

Report: Core vs Divergent Metabolism + Radiomics

Metabolic dataset – Core: 99 | Divergent: 11

Radiomic dataset – Core: 99 | Divergent: 11

Metabolic variables analyzed: 248 | Significant ($p_{\text{adj}} < 0.05$): 1

Radiomic variables analyzed: 36 | Significant ($p_{\text{adj}} < 0.05$): 0

FBA solutions: FBA, pFBA, L1w, L2, L2w

Test: two-sided Mann-Whitney U, FDR-adjusted (Benjamini-Hochberg) | Effect size: Cliff's η

η magnitude: ≥ 0.147 small · ≥ 0.330 medium · ≥ 0.474 large (Romano et al., 2006)

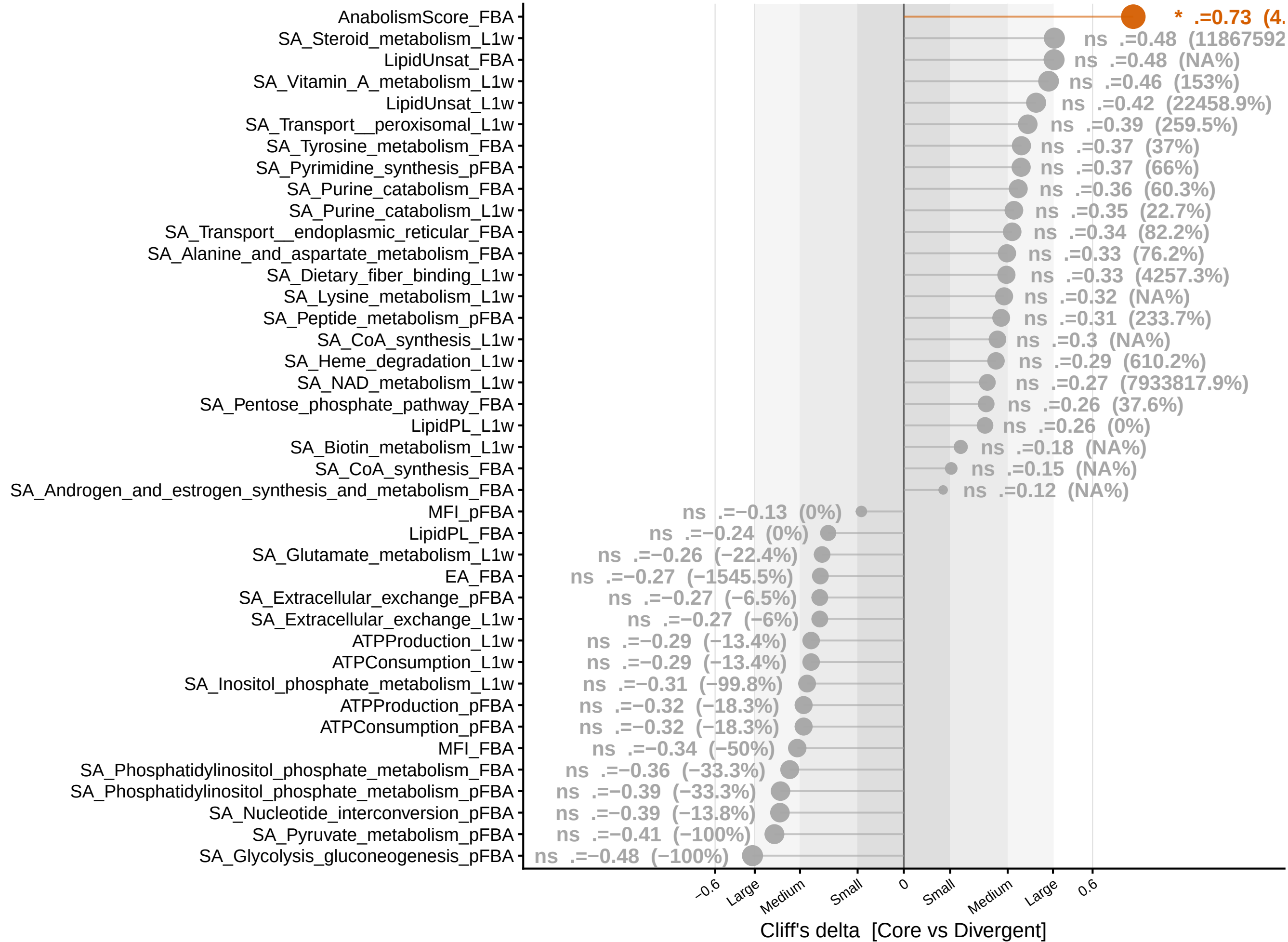
SECTION I

Metabolic Differences: Core vs Divergent

5 functional blocks: Energy | Redox | Lipids | Dependencies | Systemic Reprogramming

Top 40 Metabolic Variables – Cliff's .

Cliff's . + .% median | . higher in Core · . higher in Divergent



Cliff's .: . 1. Energy and Bioenergetics

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

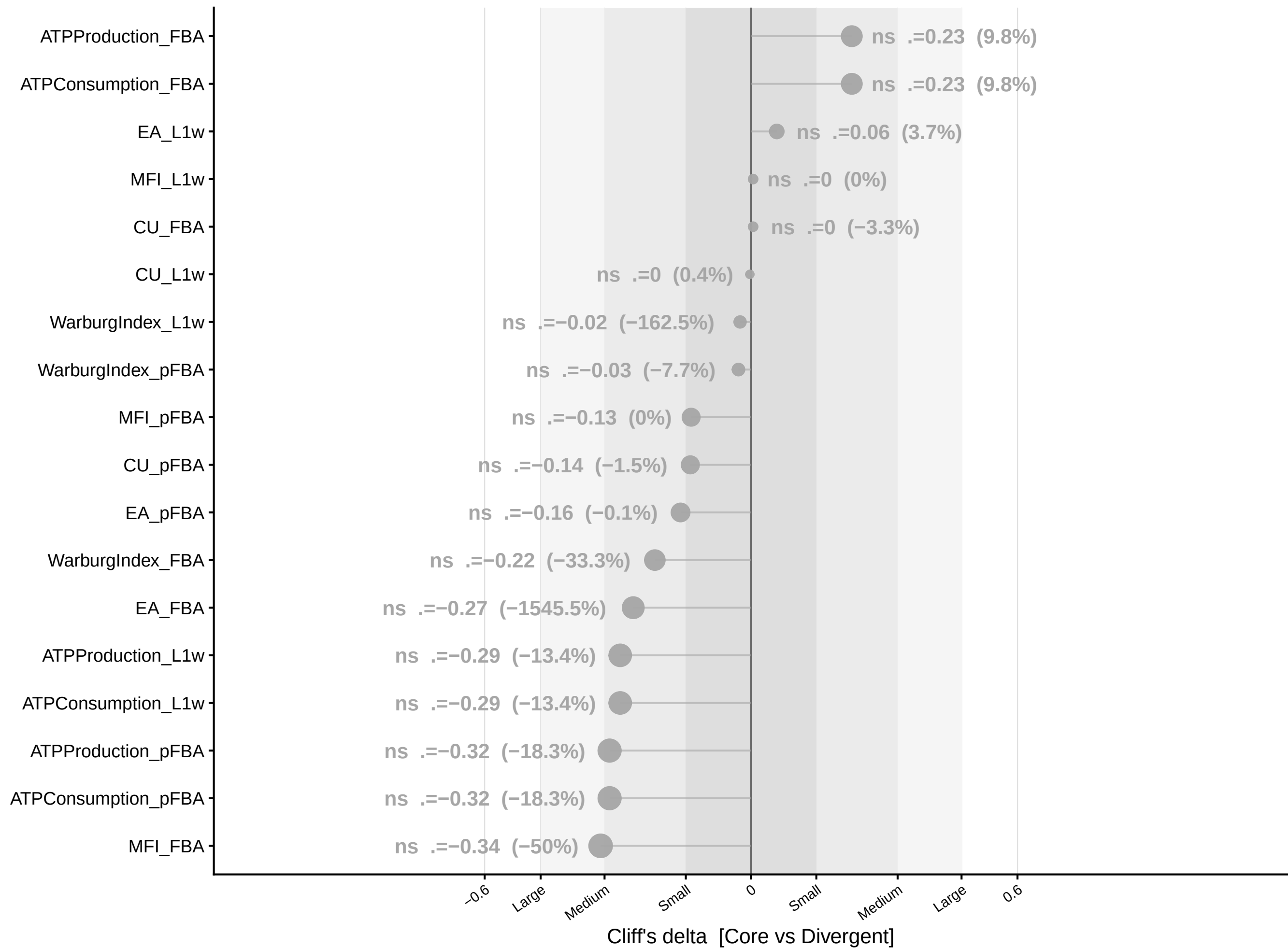
Cliff's .
Div vs Core



ATPProduction_FBA	ns .=-0.23
ATPConsumption_FBA	ns .=-0.23
EA_L1w	ns .=-0.06
MFI_L1w	ns .=-0
CU_FBA	ns .=-0
CU_L1w	ns .=-0
WarburgIndex_L1w	ns .=-0.02
WarburgIndex_pFBA	ns .=-0.03
MFI_pFBA	ns .=-0.13
CU_pFBA	ns .=-0.14
EA_pFBA	ns .=-0.16
WarburgIndex_FBA	ns .=-0.22
EA_FBA	ns .=-0.27
ATPProduction_L1w	ns .=-0.29
ATPConsumption_L1w	ns .=-0.29
ATPProduction_pFBA	ns .=-0.32
ATPConsumption_pFBA	ns .=-0.32
MFI_FBA	ns .=-0.34

Cliff's . – . 1. Energy and Bioenergetics

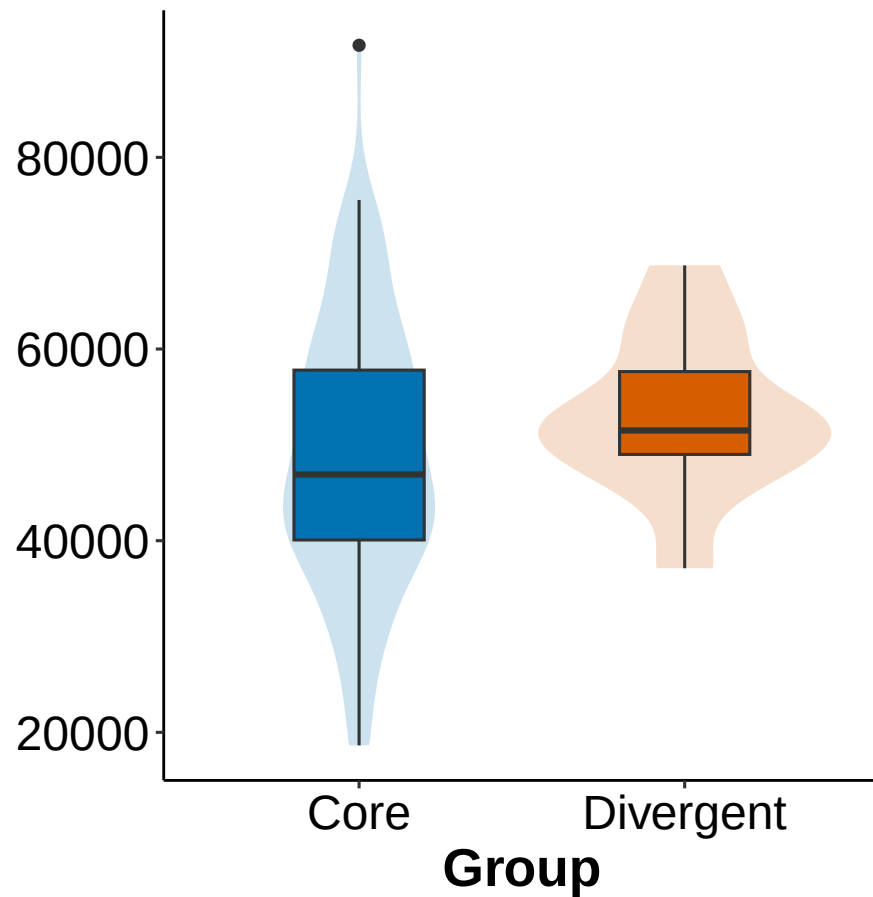
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 1. Energy and Bioenergetics

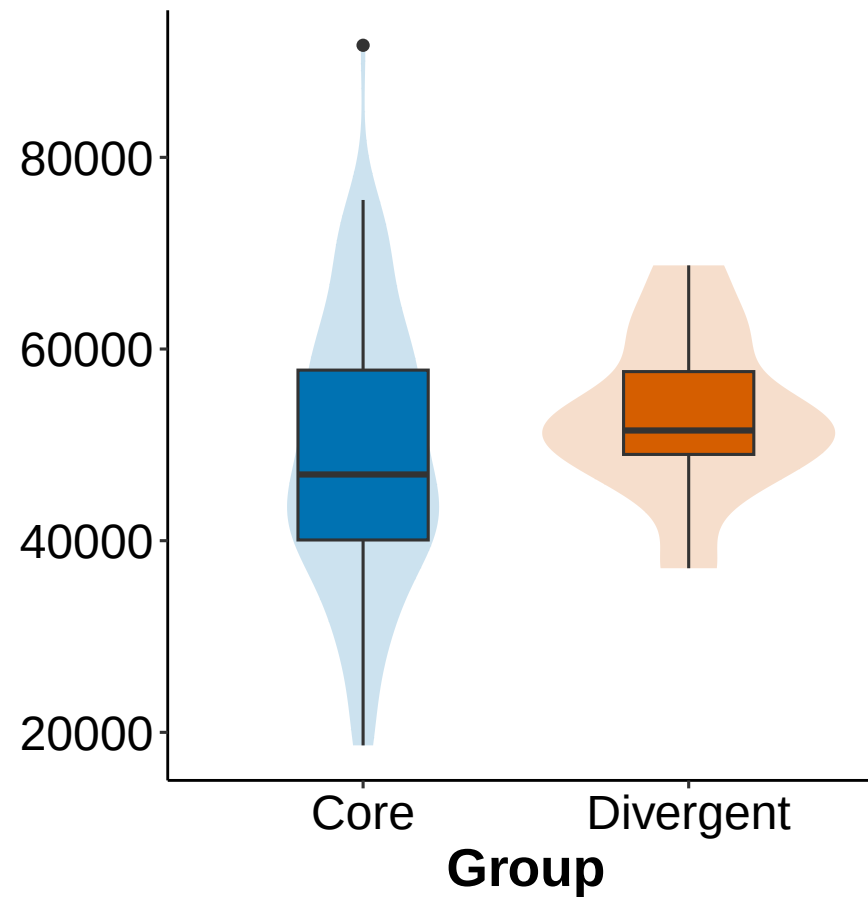
ATPConsumption_FBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 9.8



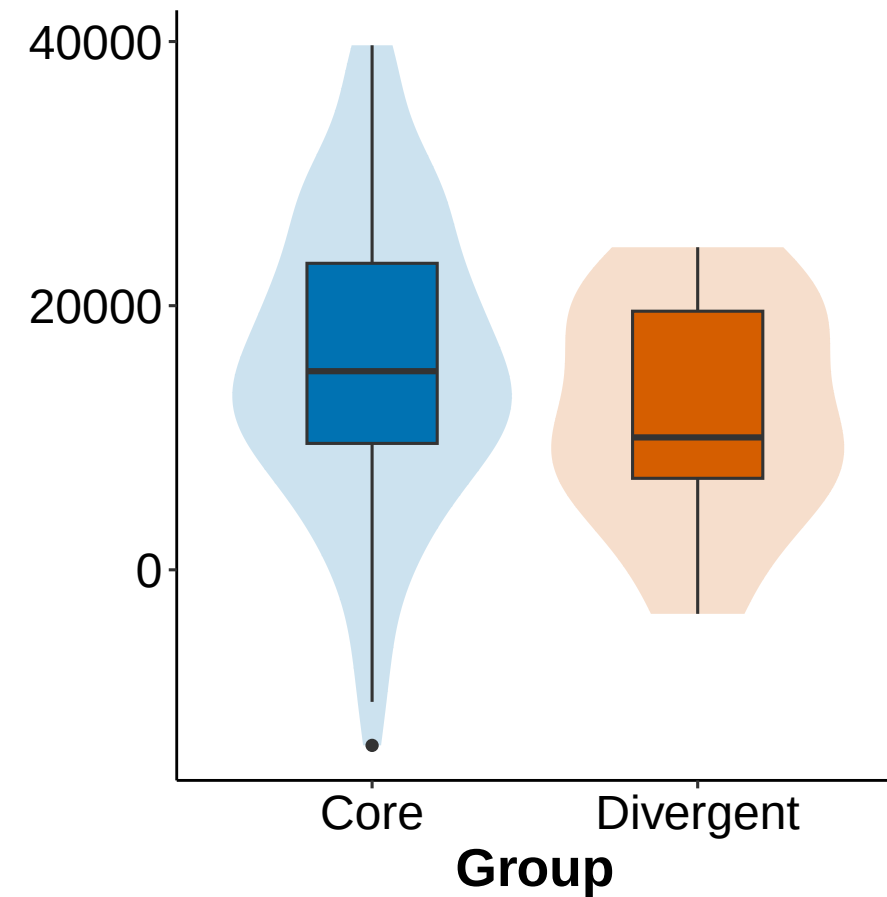
ATPProduction_FBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 9.8



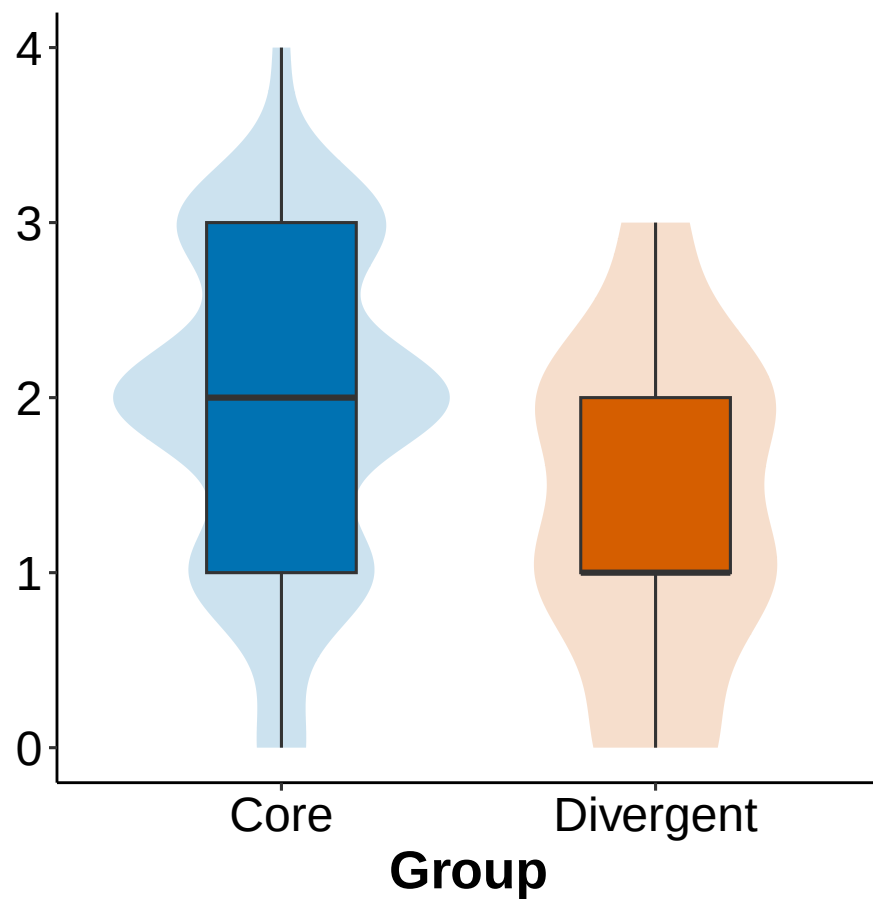
WarburgIndex_FBA

p-adj: 0.885 | .:= -0.22 (Small) | .%: -33.3



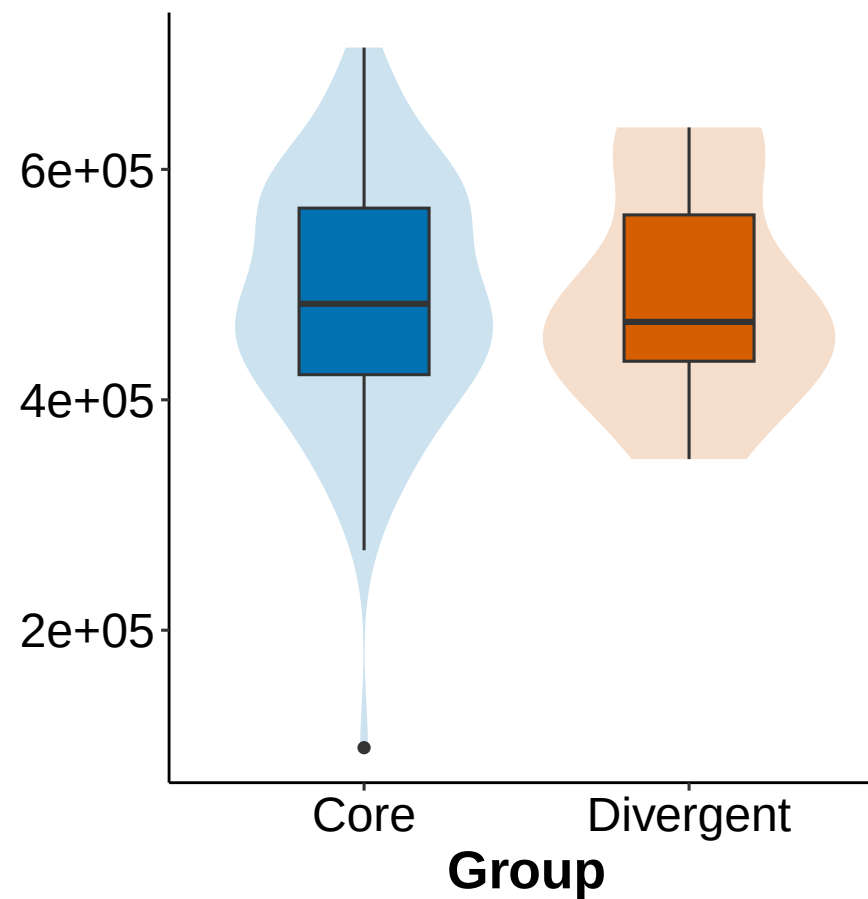
MFI_FBA

p-adj: 0.843 | .:= -0.34 (Medium) | .%: -50



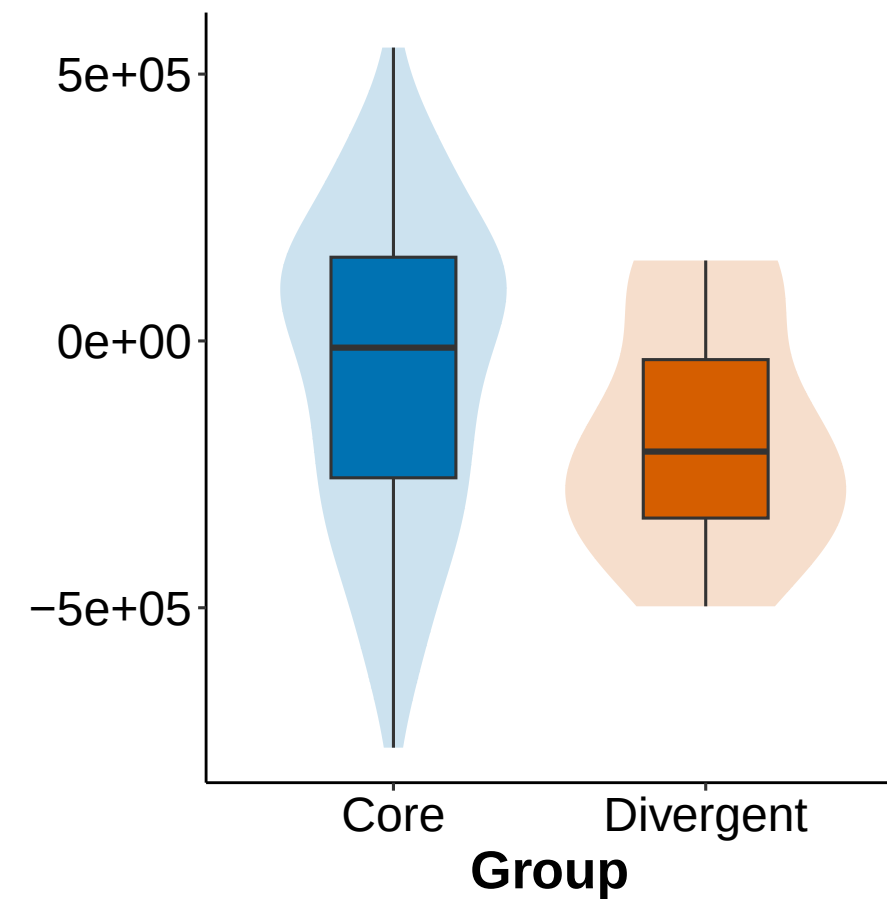
CU_FBA

p-adj: 0.992 | .:=0 (Negligible) | .%: -3.3



EA_FBA

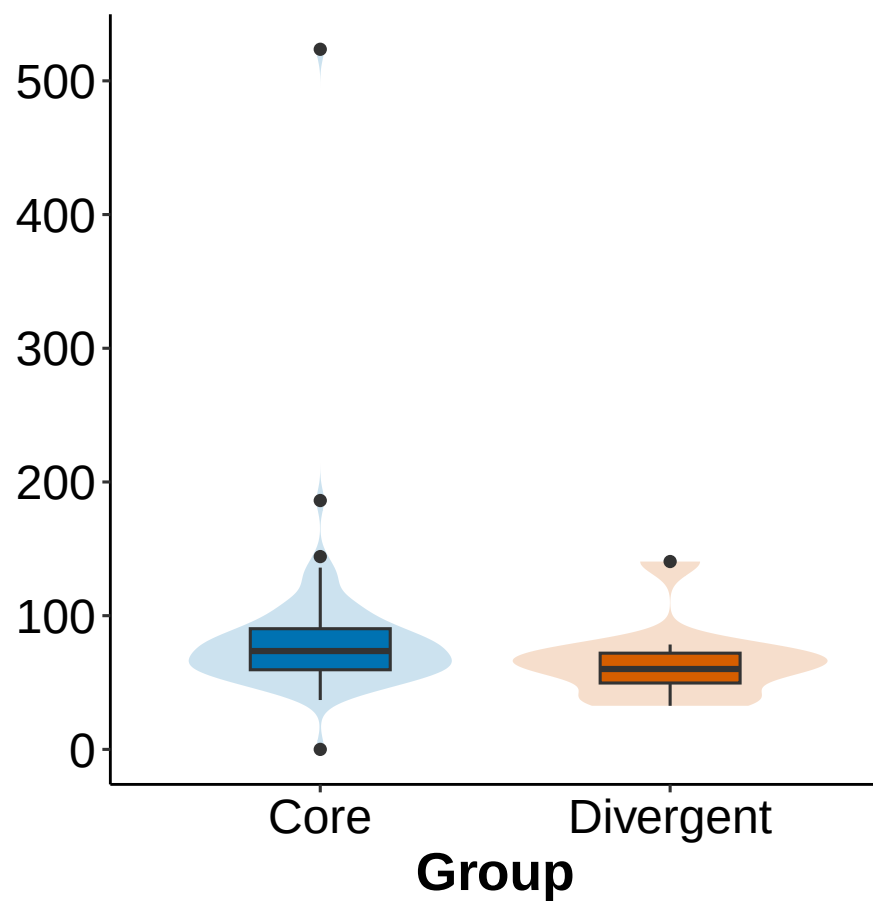
p-adj: 0.872 | .:= -0.27 (Small) | .%: -1545.5



Metabolism – . 1. Energy and Bioenergetics

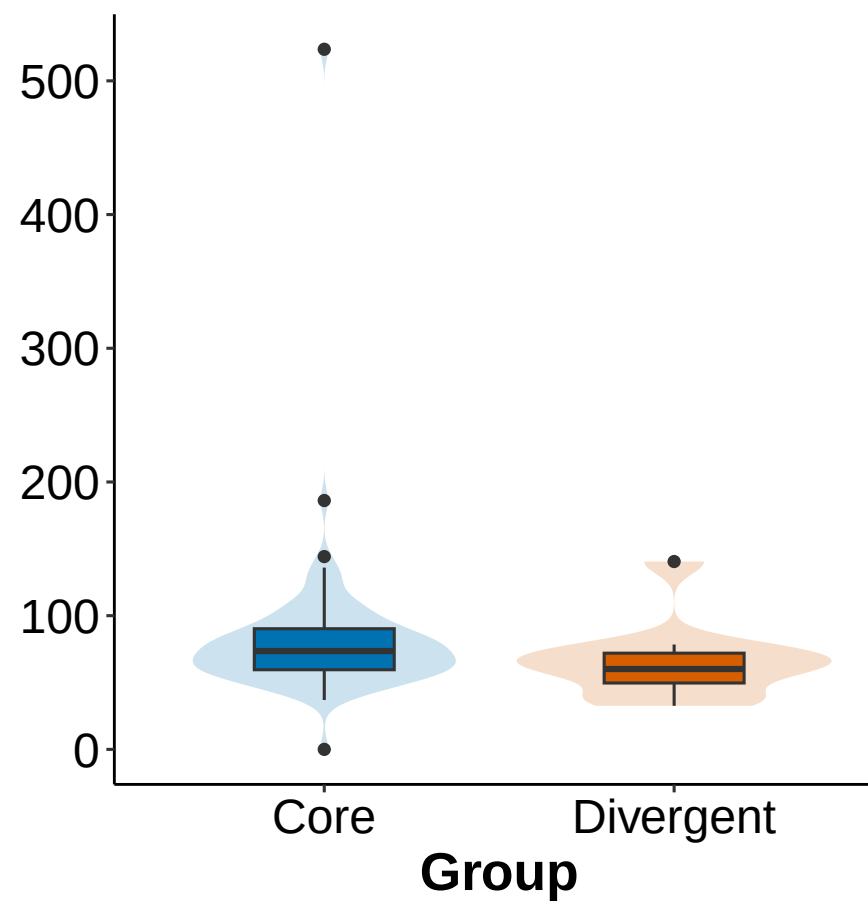
ATPConsumption_pFBA

p-adj: 0.843 | .=-0.32 (Small) | .%: -18.3



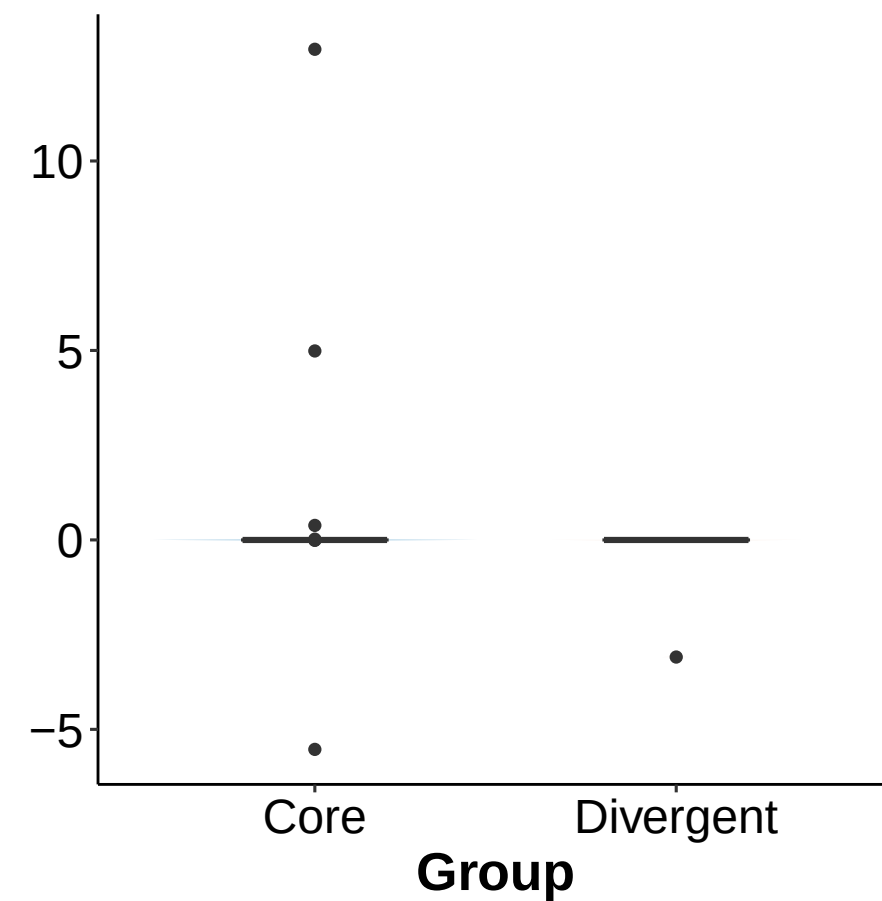
ATPProduction_pFBA

p-adj: 0.843 | .=-0.32 (Small) | .%: -18.3



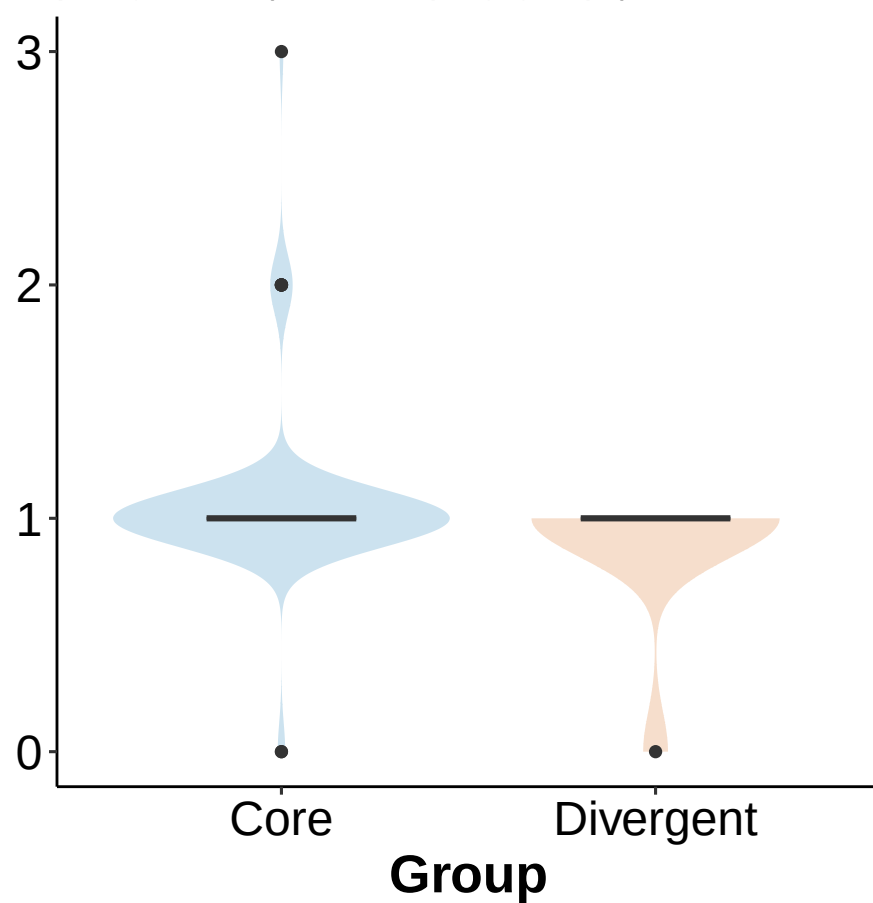
WarburgIndex_pFBA

p-adj: 0.954 | .=-0.03 (Negligible) | .%: -7.7



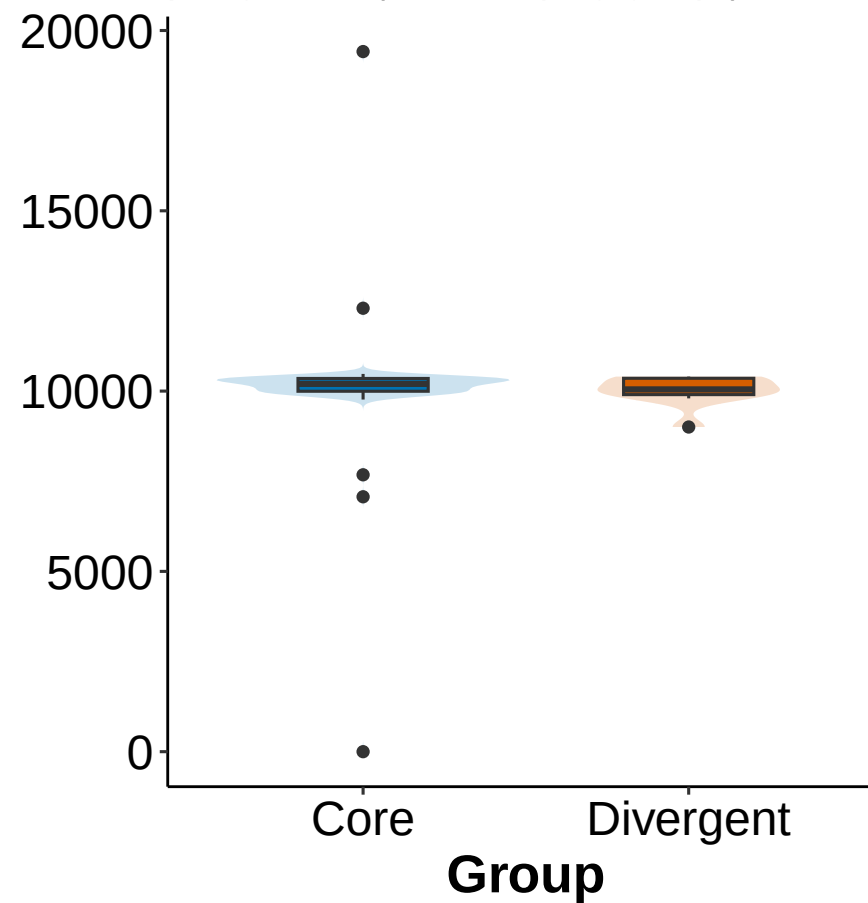
MFI_pFBA

p-adj: 0.872 | .=-0.13 (Negligible) | .%: 0



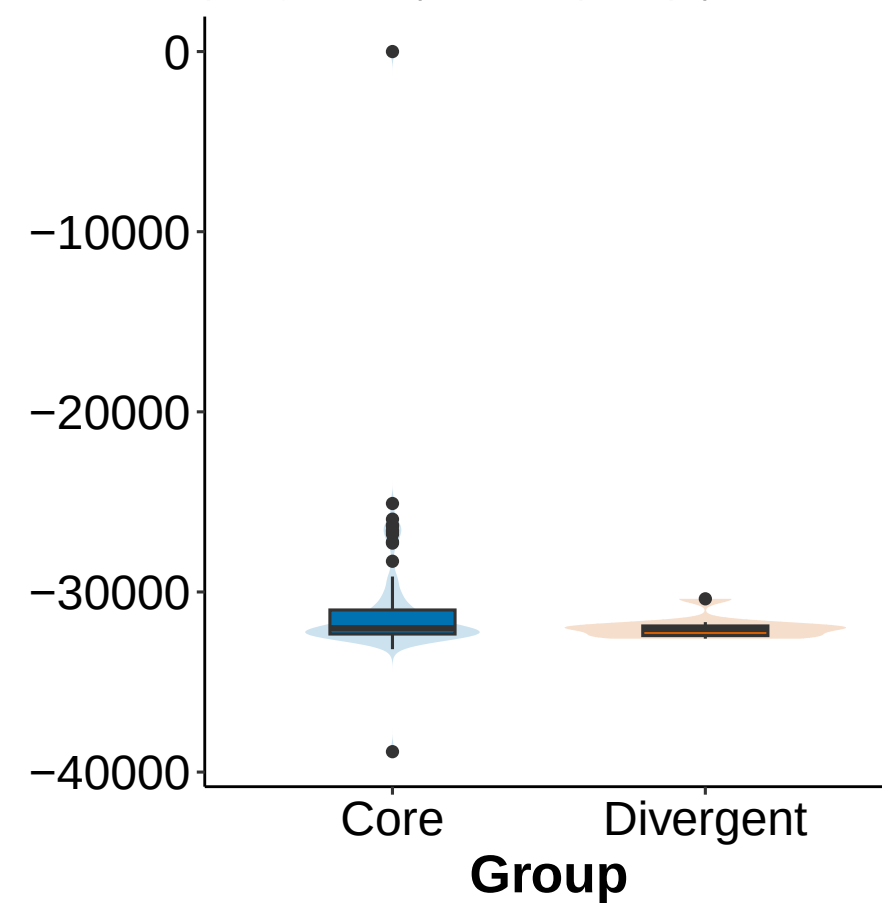
CU_pFBA

p-adj: 0.896 | .=-0.14 (Negligible) | .%: -1.5



EA_pFBA

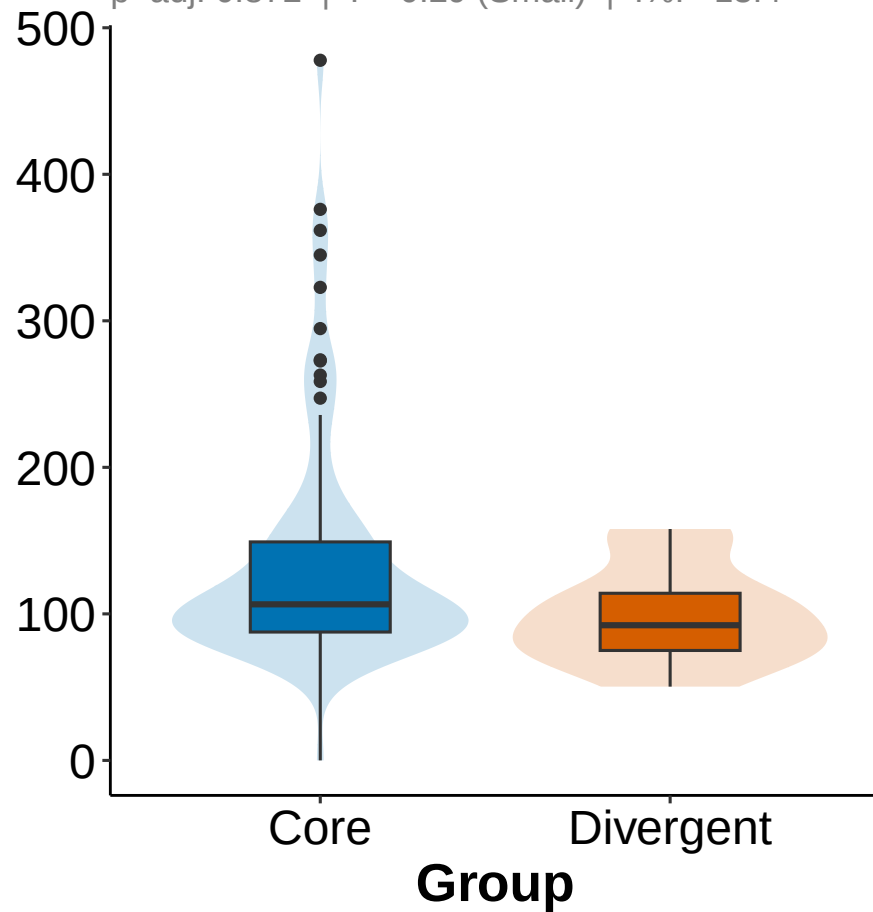
p-adj: 0.885 | .=-0.16 (Small) | .%: -0.1



Metabolism – . 1. Energy and Bioenergetics

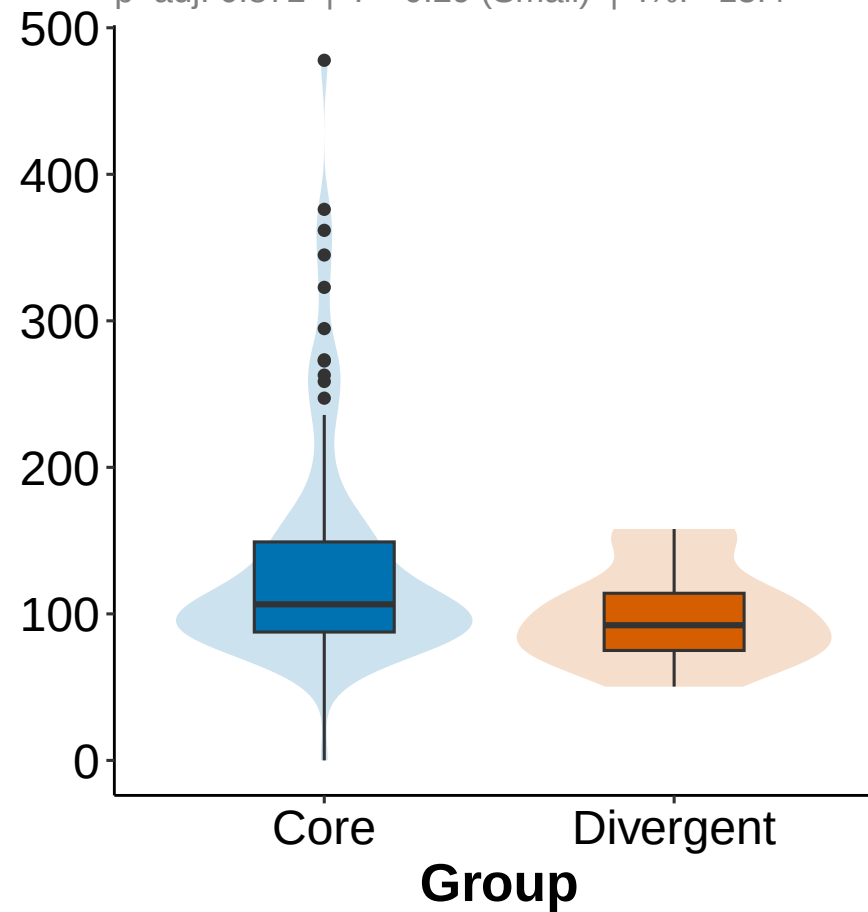
ATPConsumption_L1w

p-adj: 0.872 | .=-0.29 (Small) | .%: -13.4



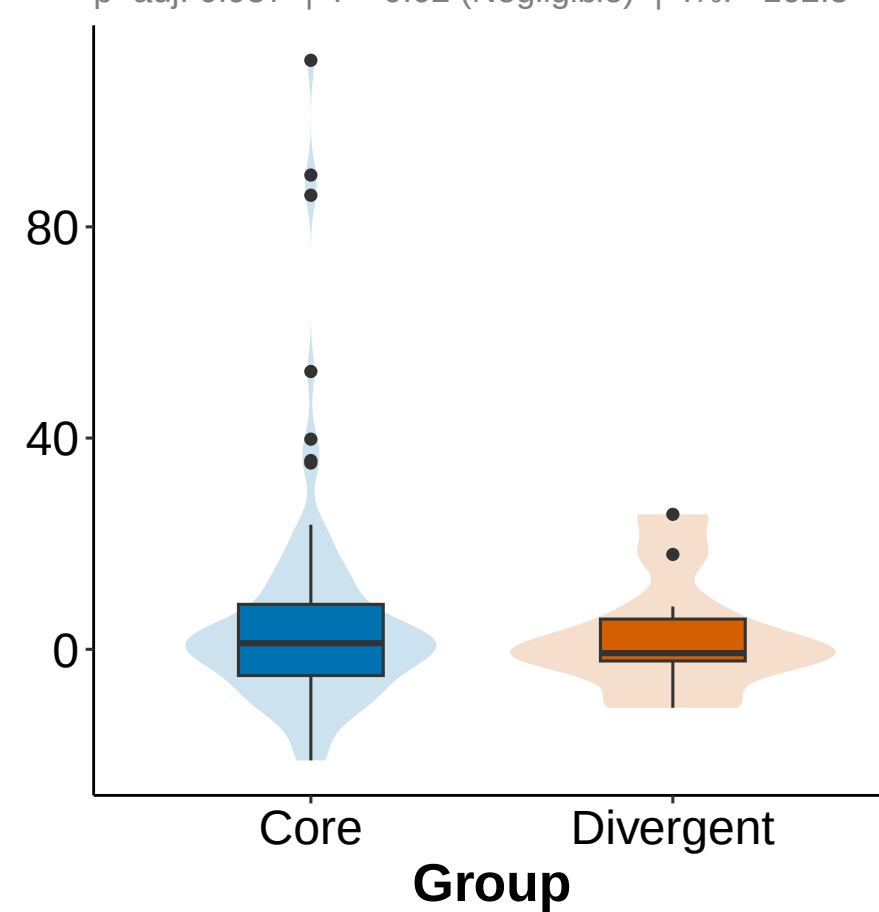
ATPProduction_L1w

p-adj: 0.872 | .=-0.29 (Small) | .%: -13.4



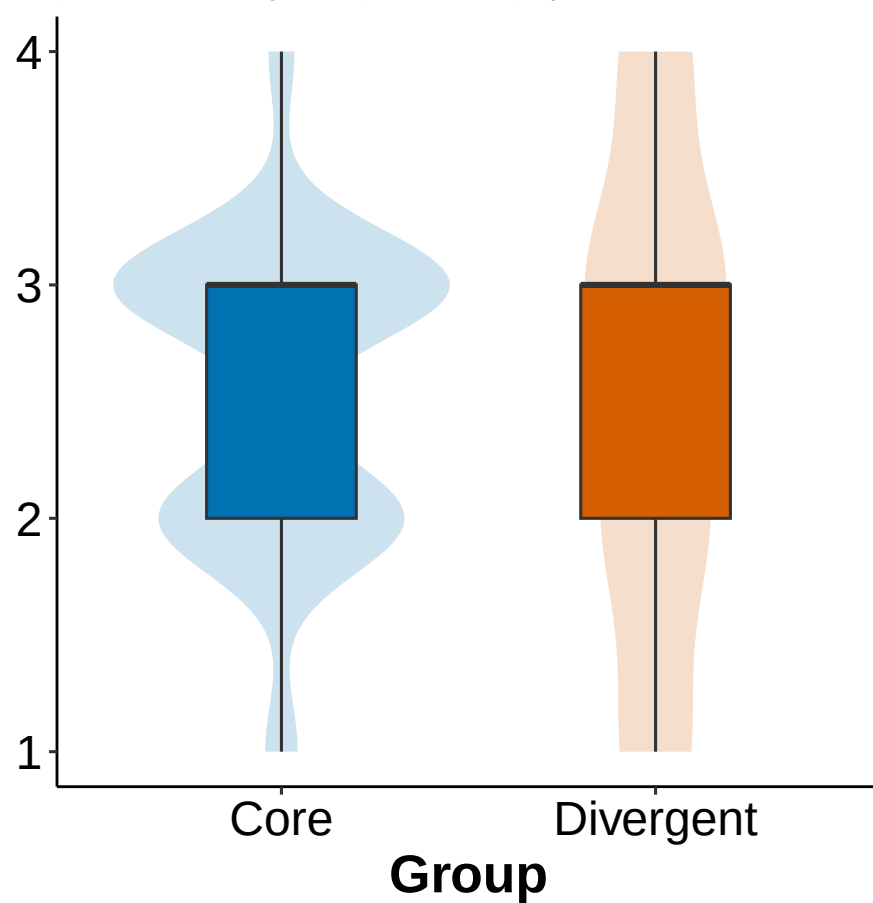
WarburgIndex_L1w

p-adj: 0.957 | .=-0.02 (Negligible) | .%: -162.5



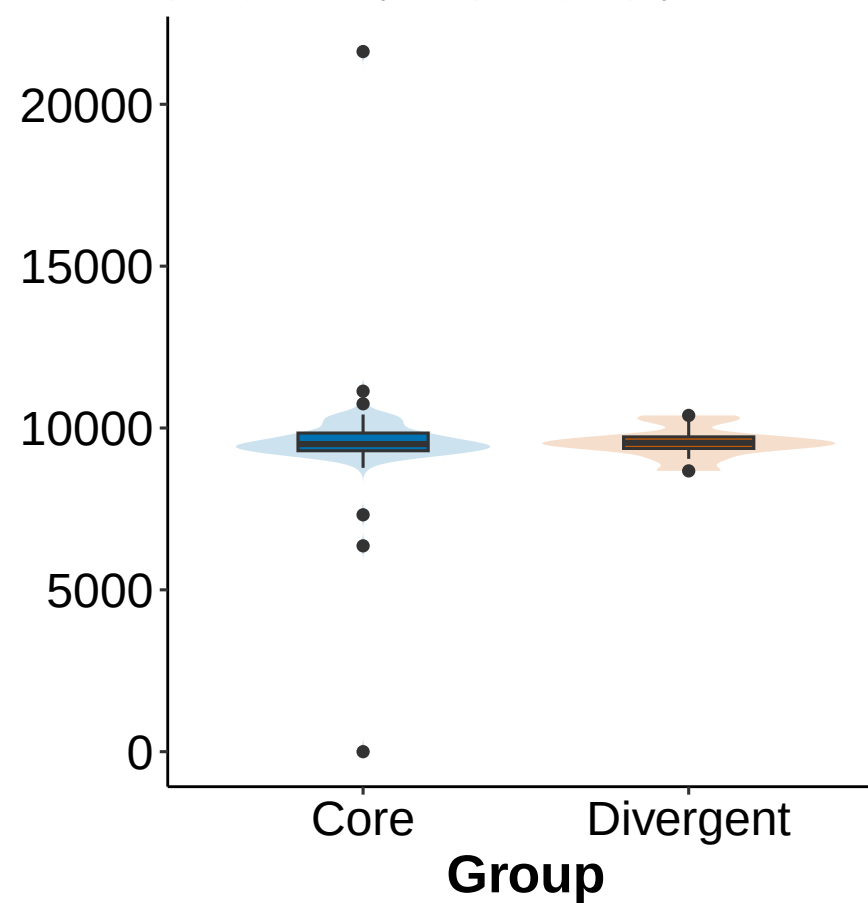
MFI_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 0



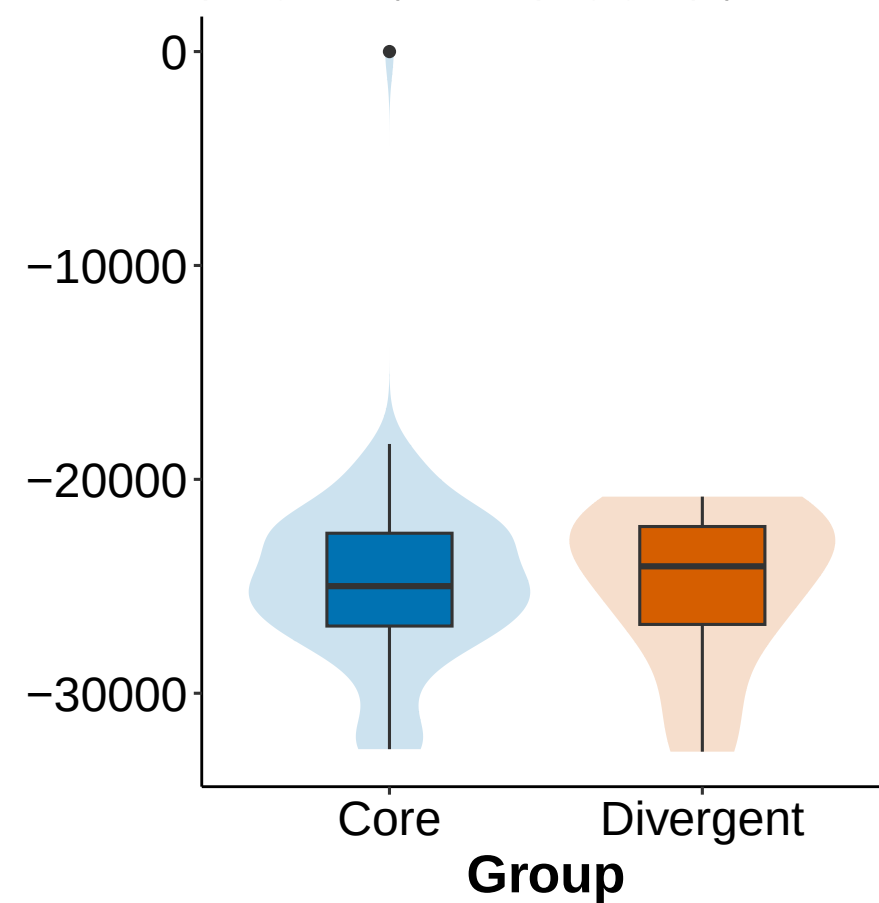
CU_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 0.4



EA_L1w

p-adj: 0.95 | .=0.06 (Negligible) | .%: 3.7



Cliff's .: . 2. Redox State and Anabolism

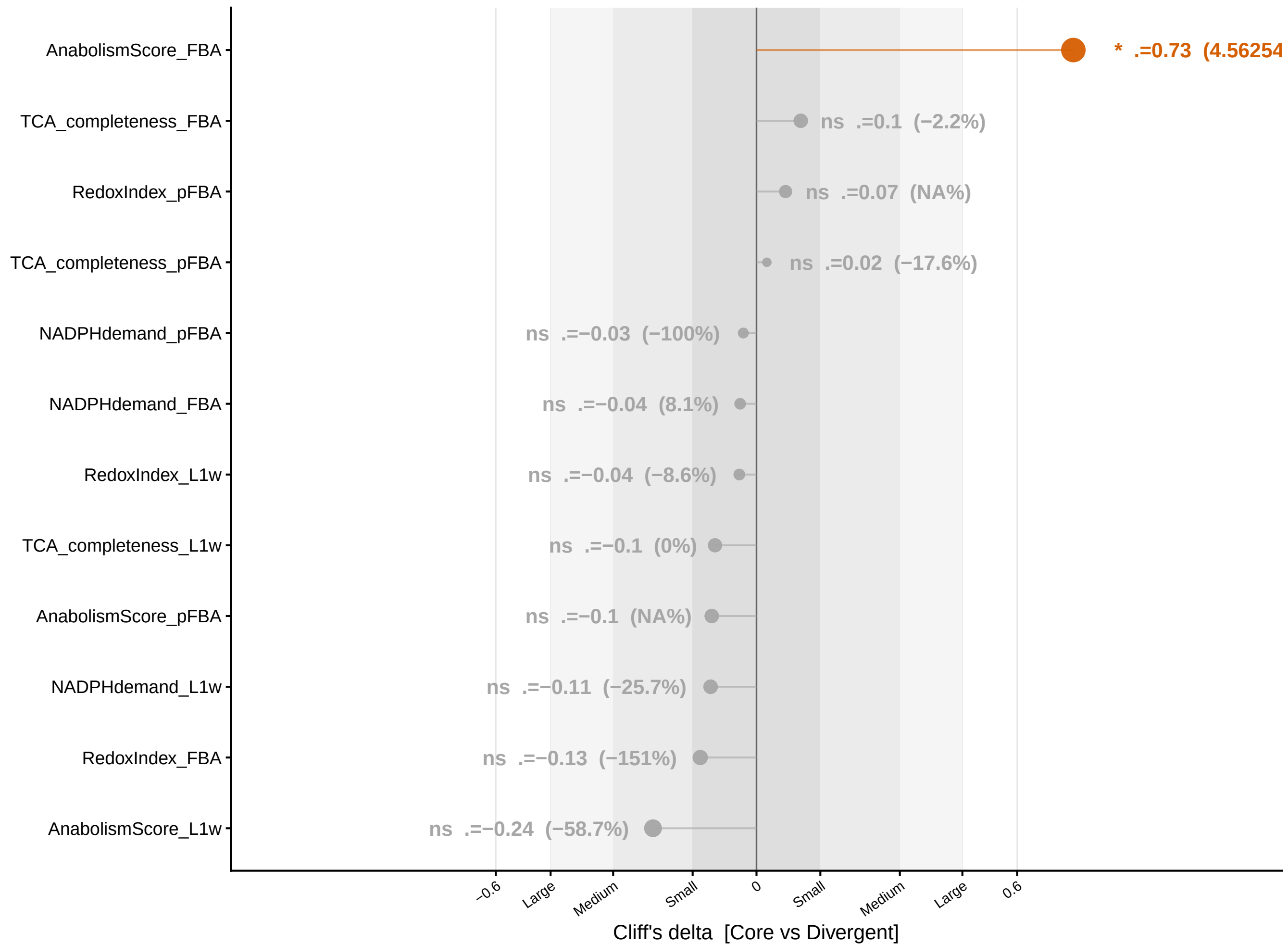
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



AnabolismScore_FBA	* .=-0.73
TCA_completeness_FBA	ns .=-0.1
RedoxIndex_pFBA	ns .=-0.07
TCA_completeness_pFBA	ns .=-0.02
NADPHdemand_pFBA	ns .=-0.03
NADPHdemand_FBA	ns .=-0.04
RedoxIndex_L1w	ns .=-0.04
TCA_completeness_L1w	ns .=-0.1
AnabolismScore_pFBA	ns .=-0.1
NADPHdemand_L1w	ns .=-0.11
RedoxIndex_FBA	ns .=-0.13
AnabolismScore_L1w	ns .=-0.24

Cliff's . – . 2. Redox State and Anabolism

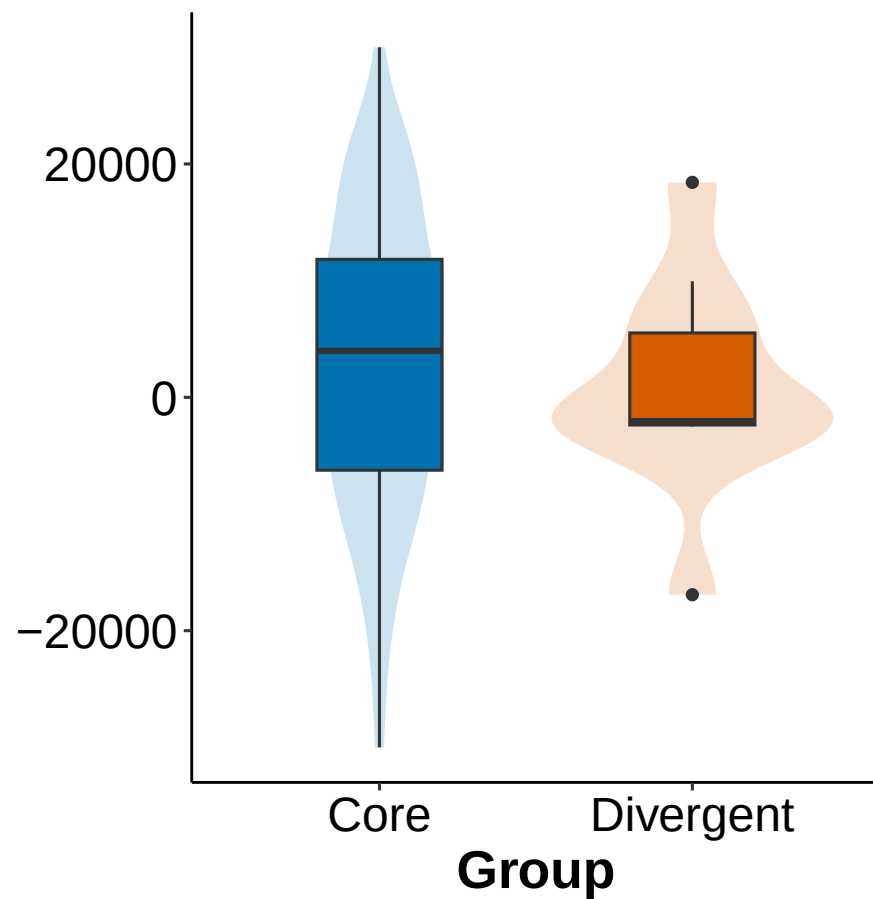
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 2. RedoxState and Anabolism

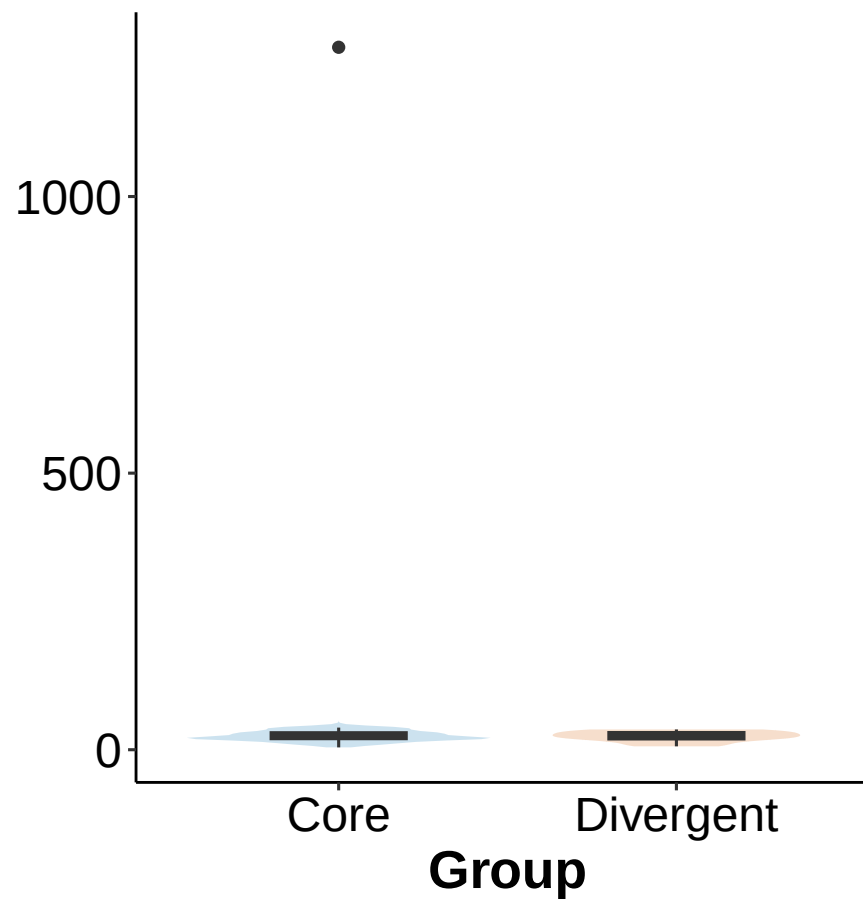
RedoxIndex_FBA

p-adj: 0.917 | .=-0.13 (Negligible) | .%: -15



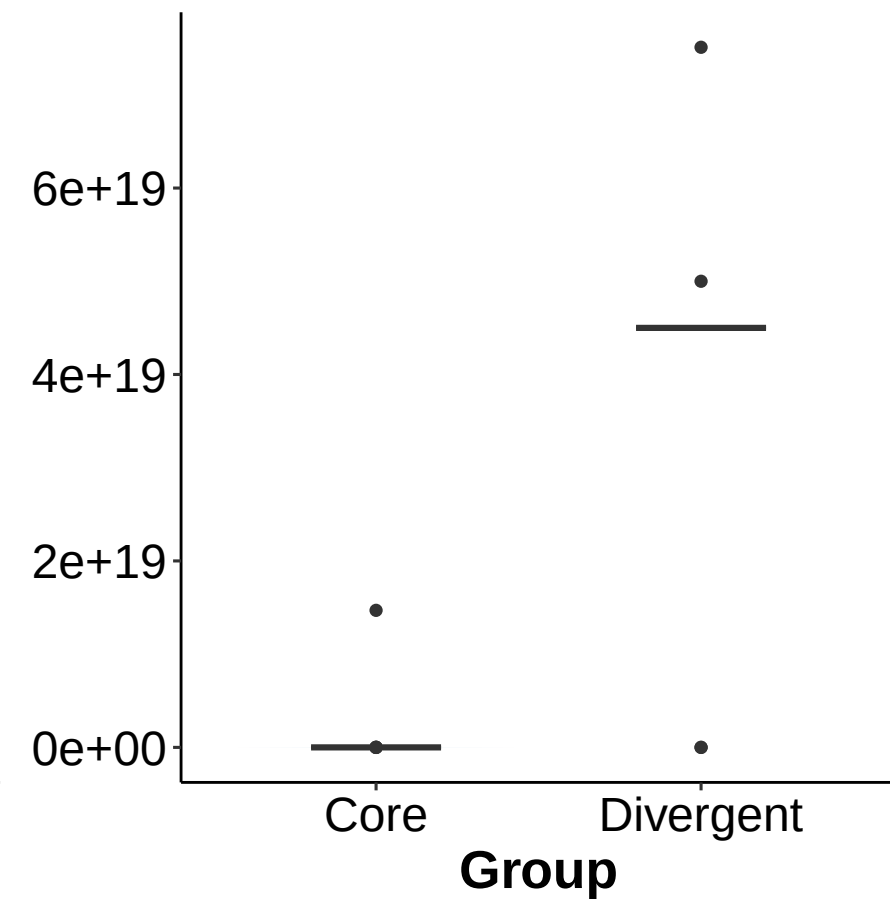
NADPHdemand_FBA

p-adj: 0.954 | .=-0.04 (Negligible) | .%: 8.1



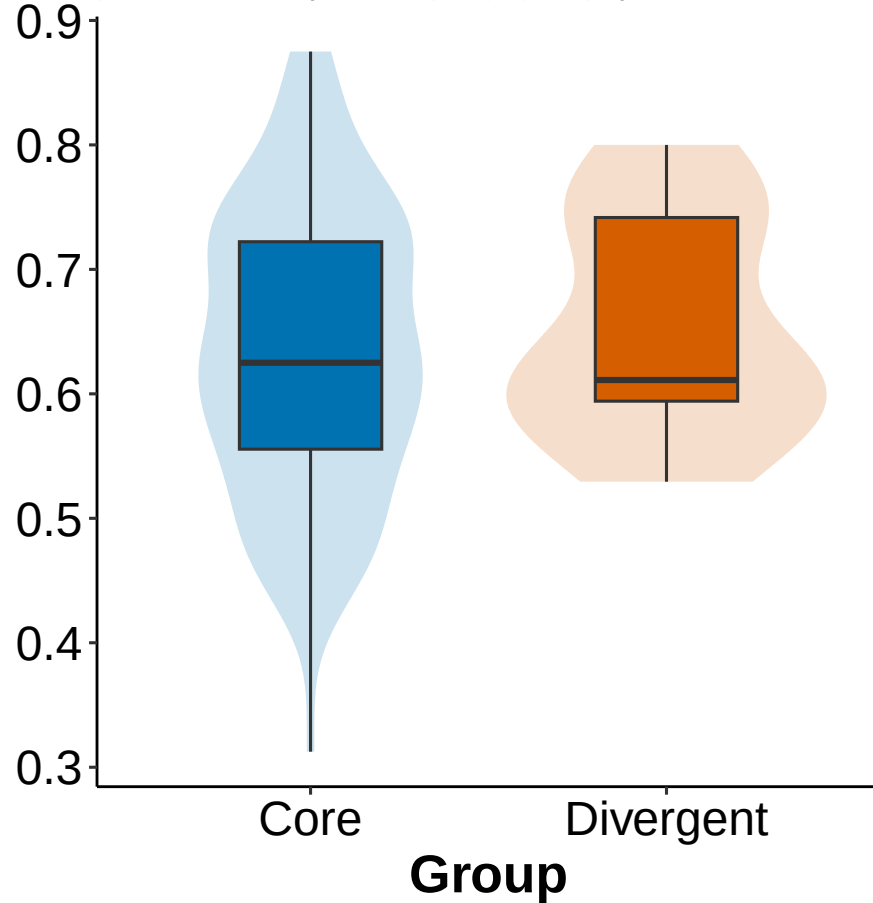
AnabolismScore_FBA

p-adj: 0.0481 | .=0.73 (Large) | .%: 4.562543



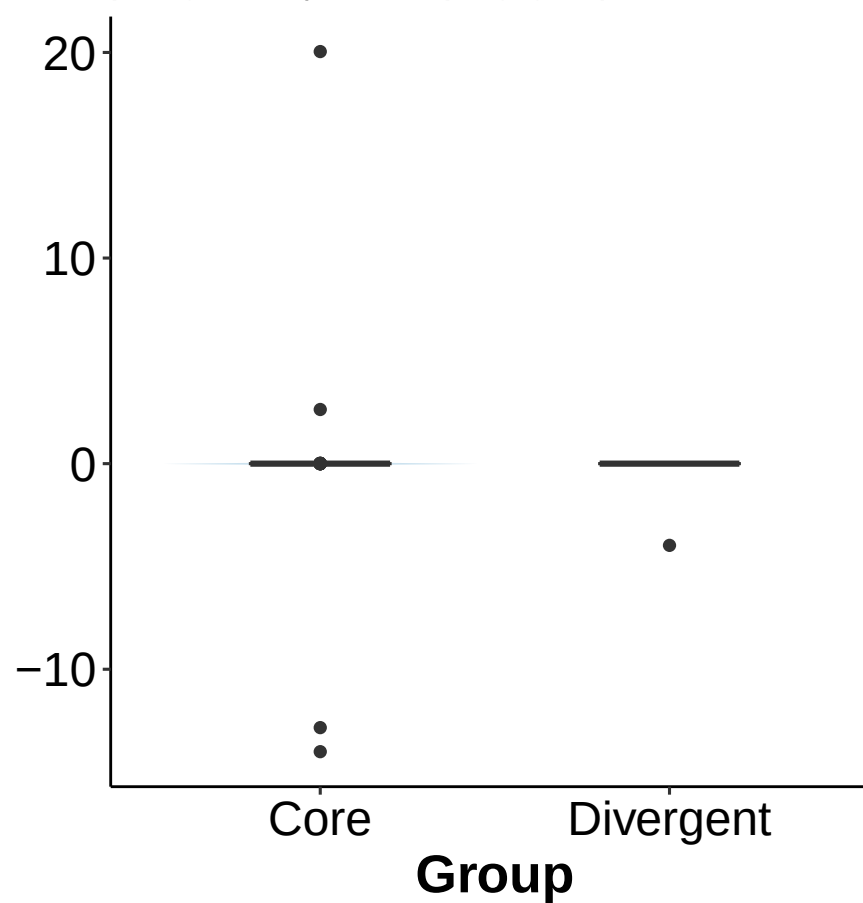
TCA_completeness_FBA

p-adj: 0.928 | .=0.1 (Negligible) | .%: -2.2



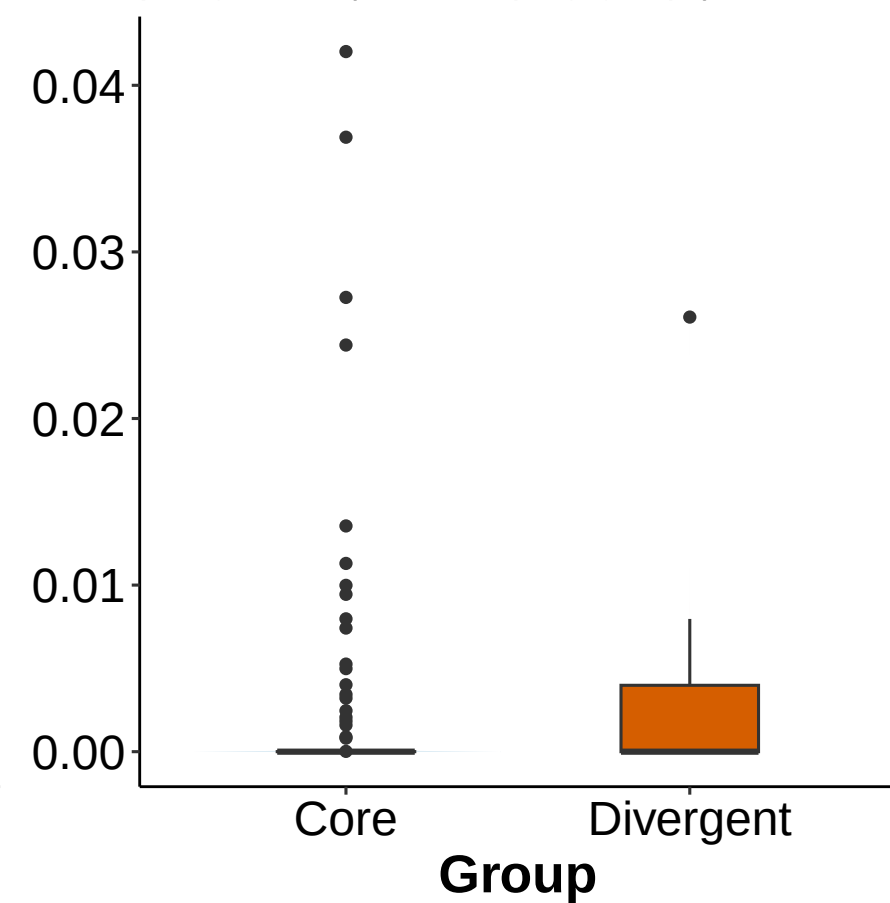
RedoxIndex_pFBA

p-adj: 0.95 | .=0.07 (Negligible)



NADPHdemand_pFBA

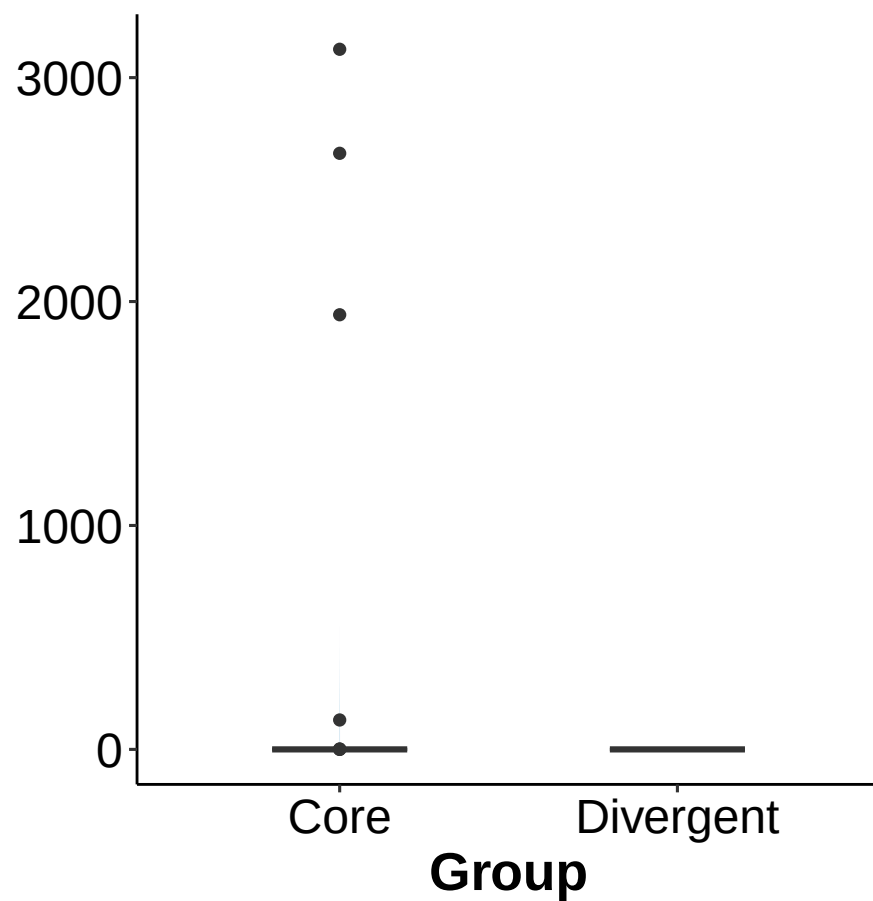
p-adj: 0.954 | .=-0.03 (Negligible) | .%: -100



Metabolism – . 2. RedoxState and Anabolism

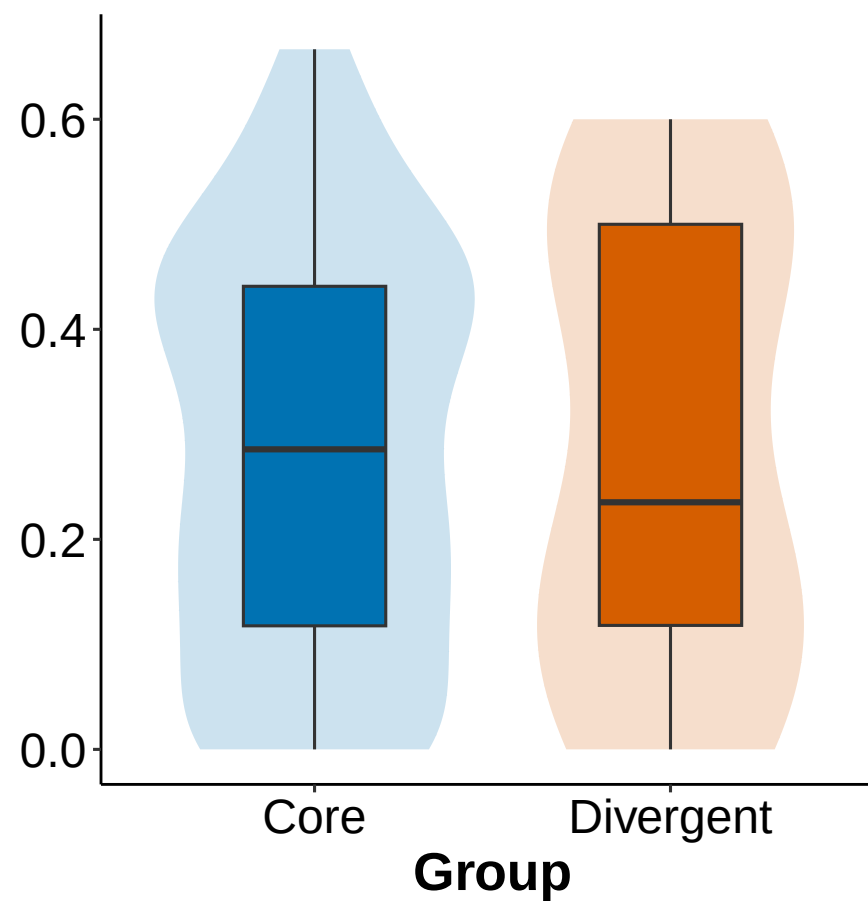
AnabolismScore_pFBA

p-adj: 0.885 | .=-0.1 (Negligible)



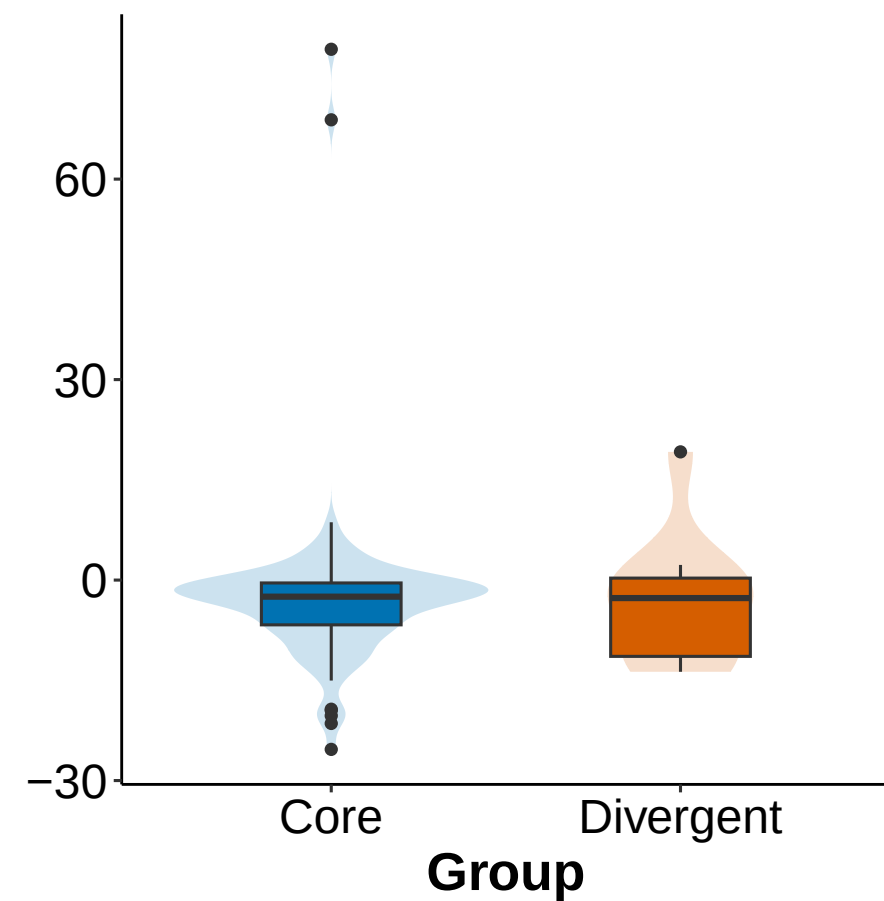
TCA_completeness_pFBA

p-adj: 0.957 | .=0.02 (Negligible) | .%: -17.6



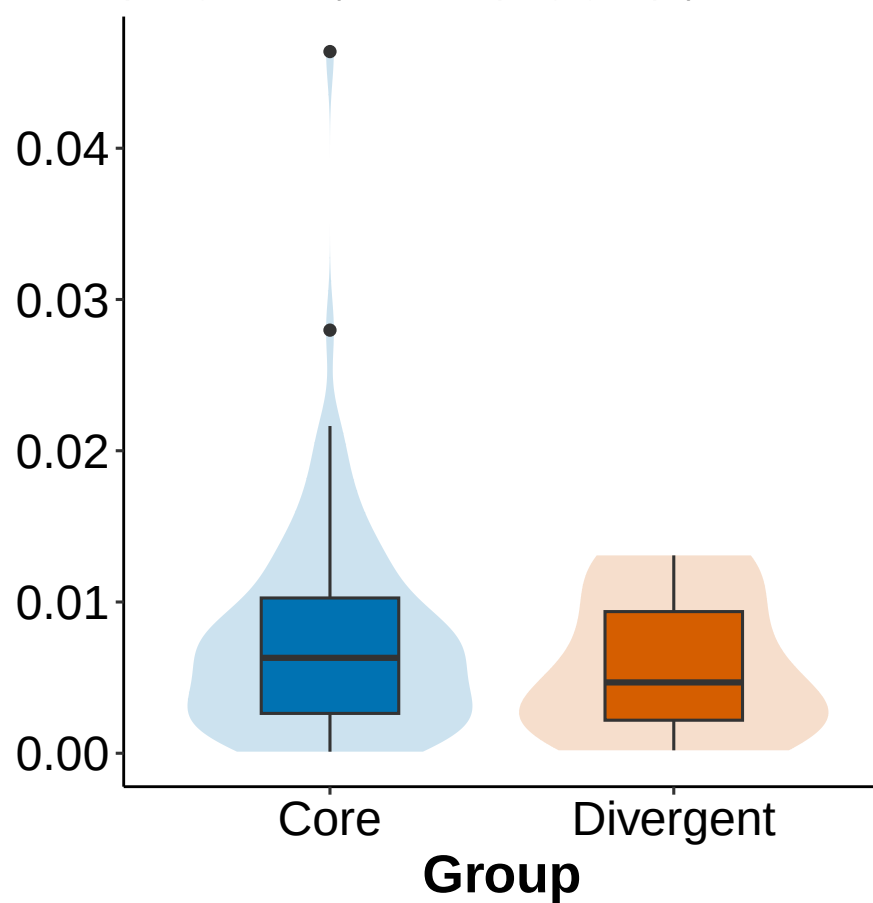
RedoxIndex_L1w

p-adj: 0.954 | .=-0.04 (Negligible) | .%: -8.6



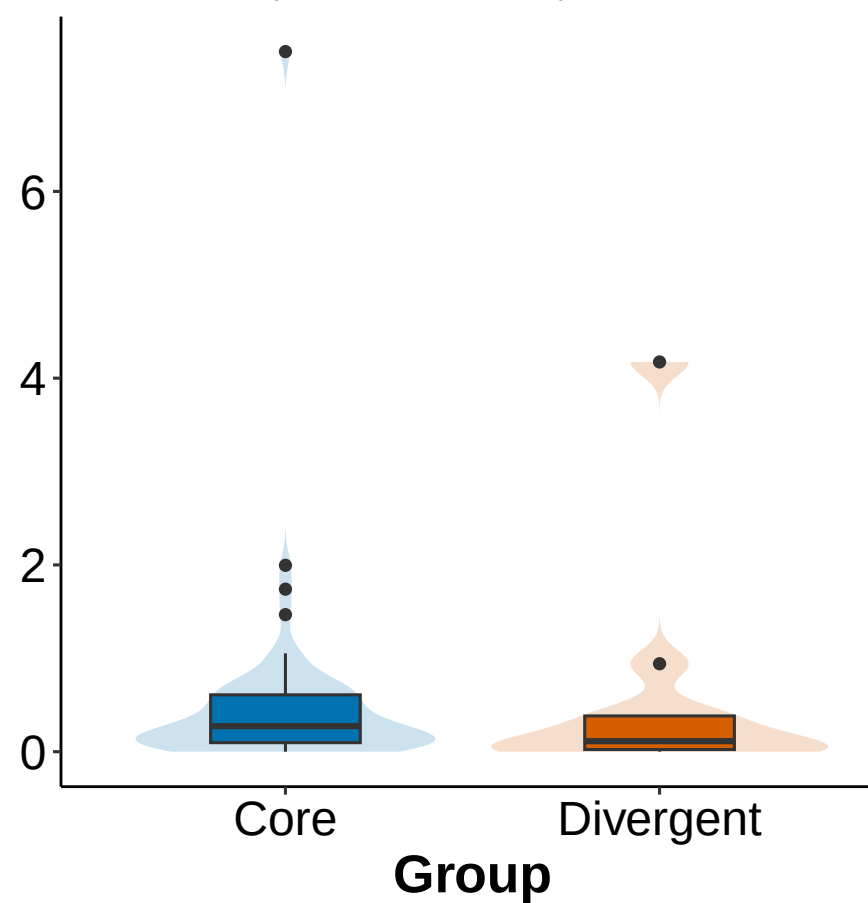
NADPHdemand_L1w

p-adj: 0.927 | .=-0.11 (Negligible) | .%: -25.7



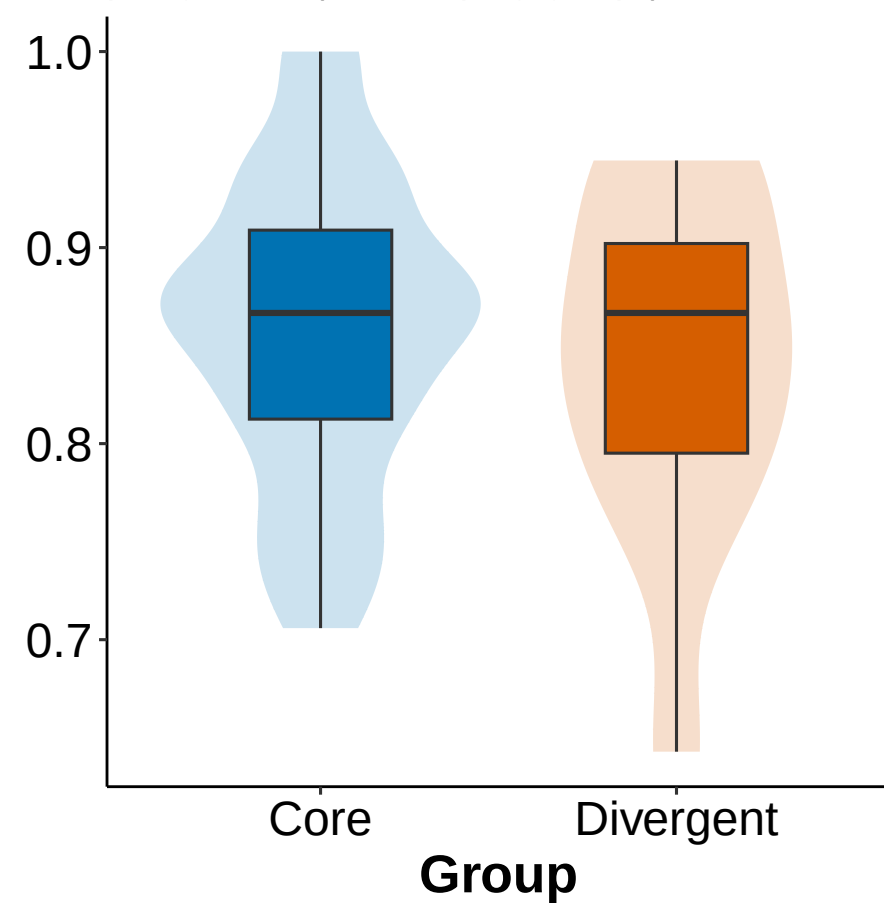
AnabolismScore_L1w

p-adj: 0.885 | .=-0.24 (Small) | .%: -58.7



TCA_completeness_L1w

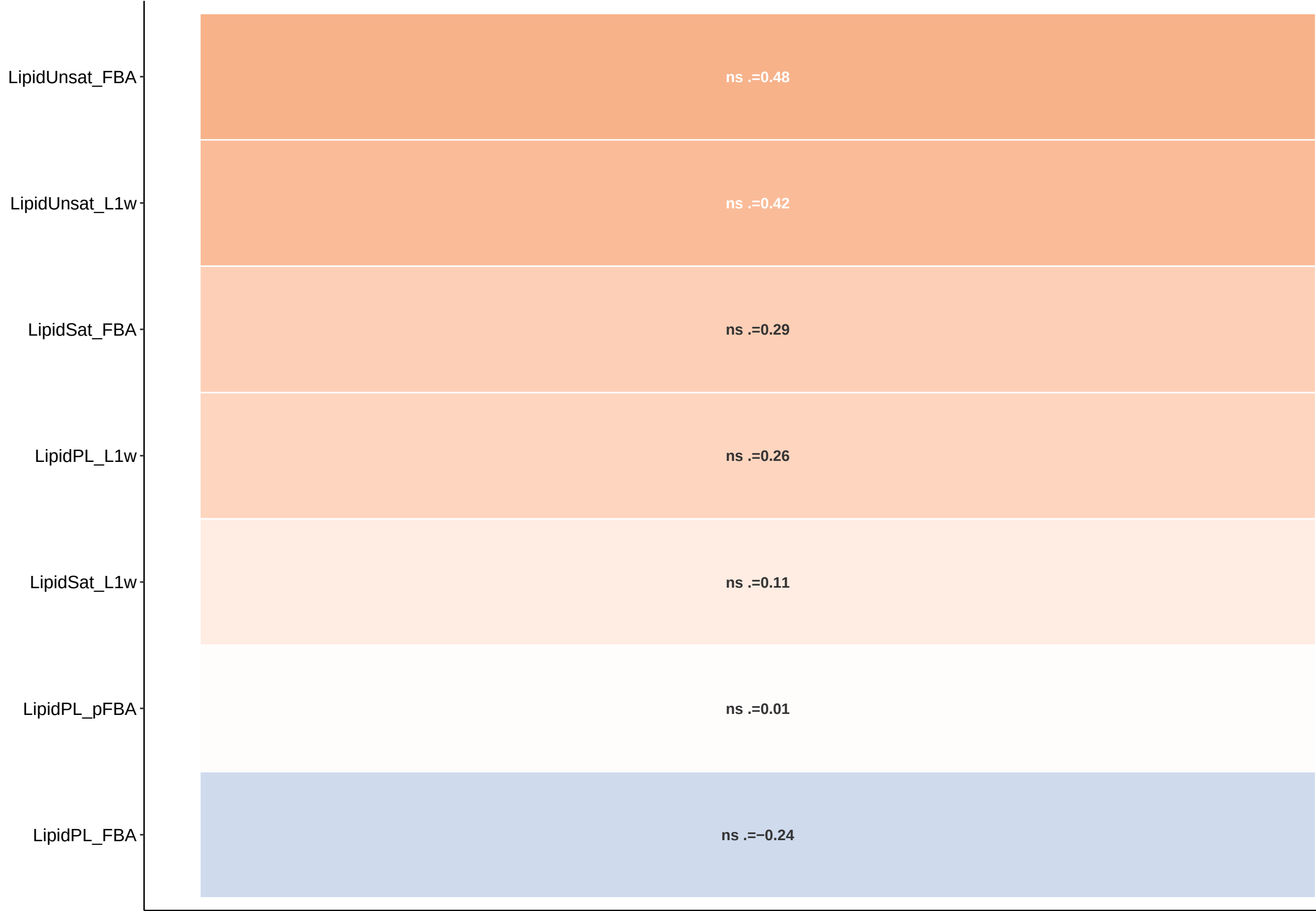
p-adj: 0.942 | .=-0.1 (Negligible) | .%: 0



Cliff's .: . 3. Lipid Metabolism

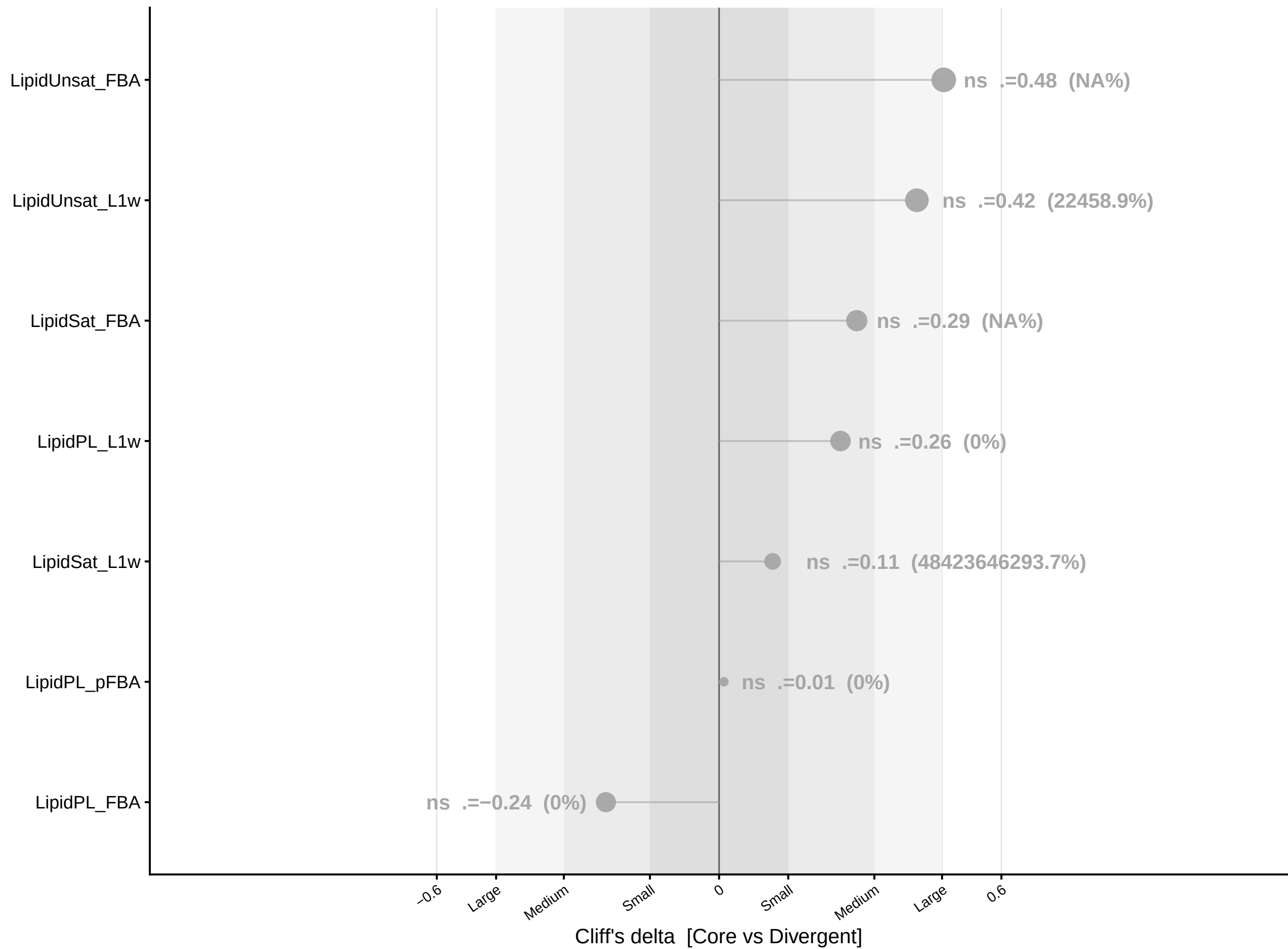
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0

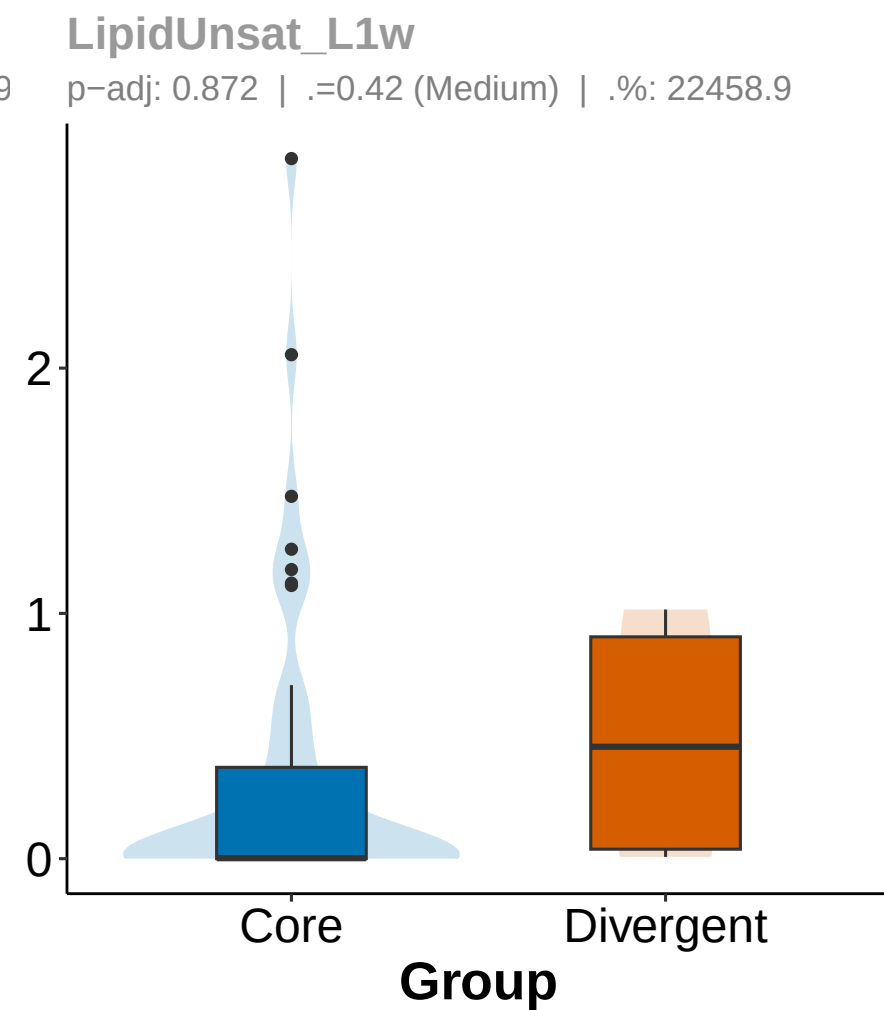
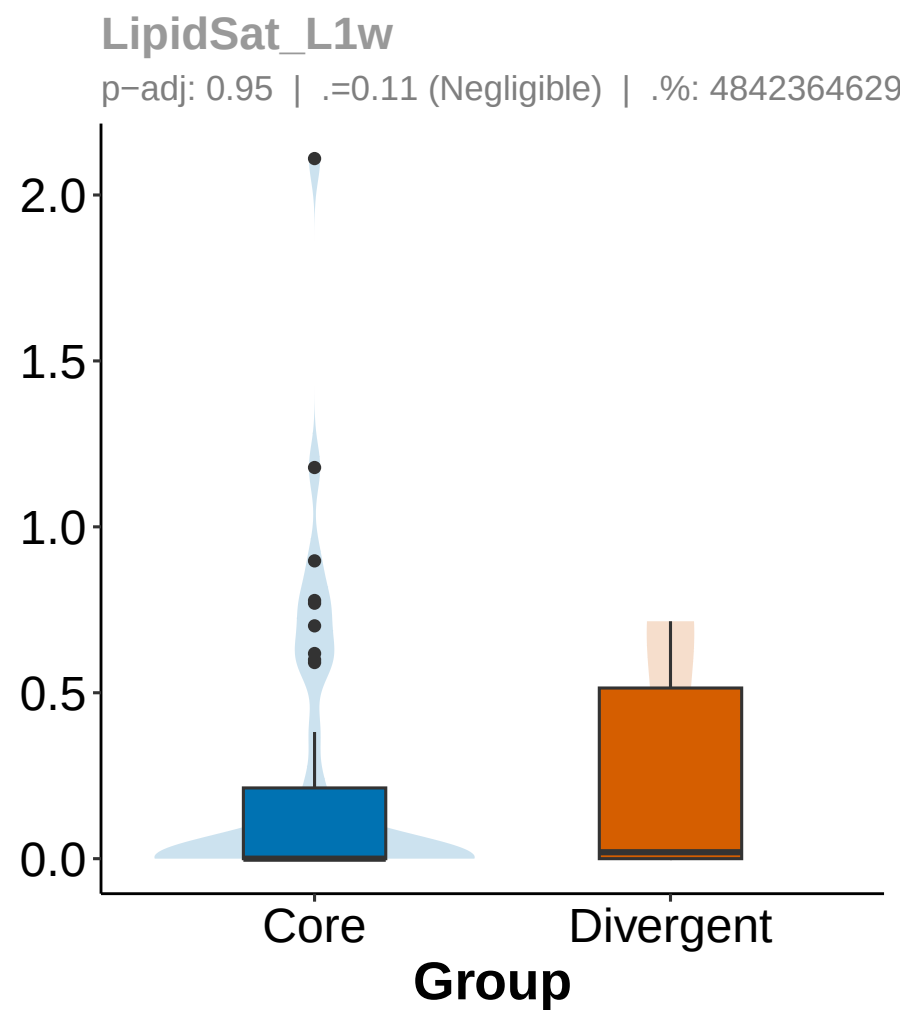
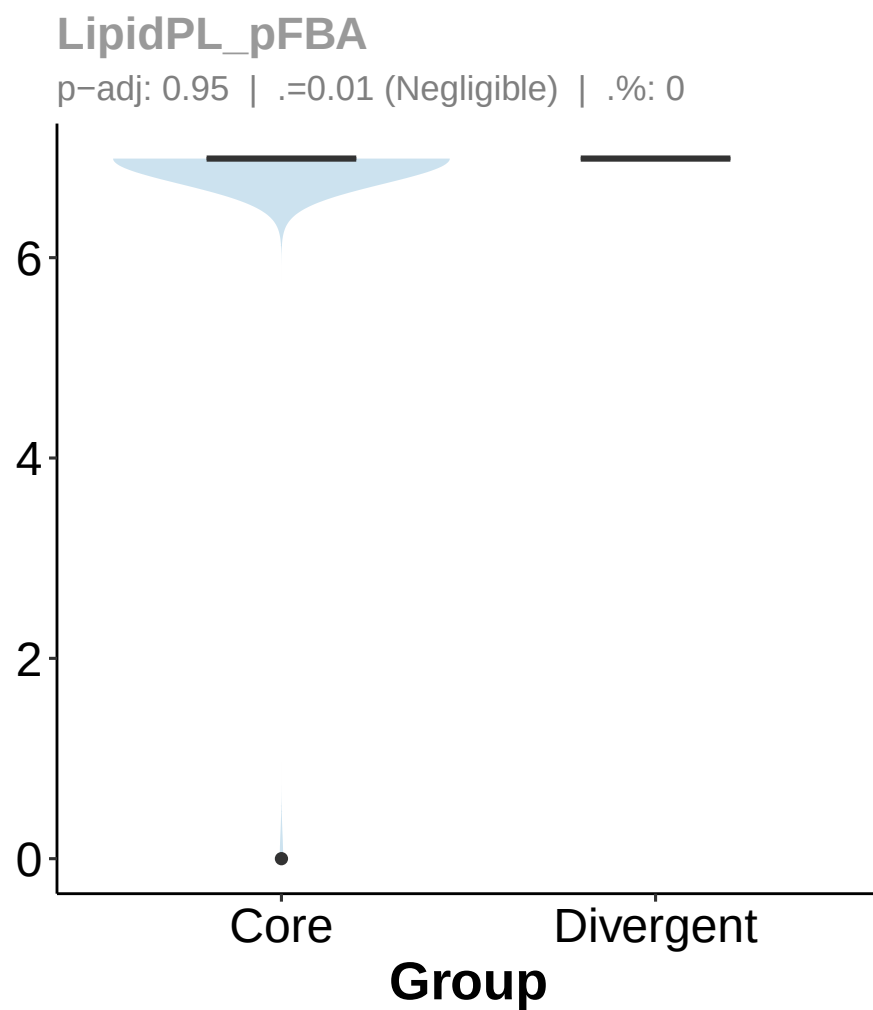
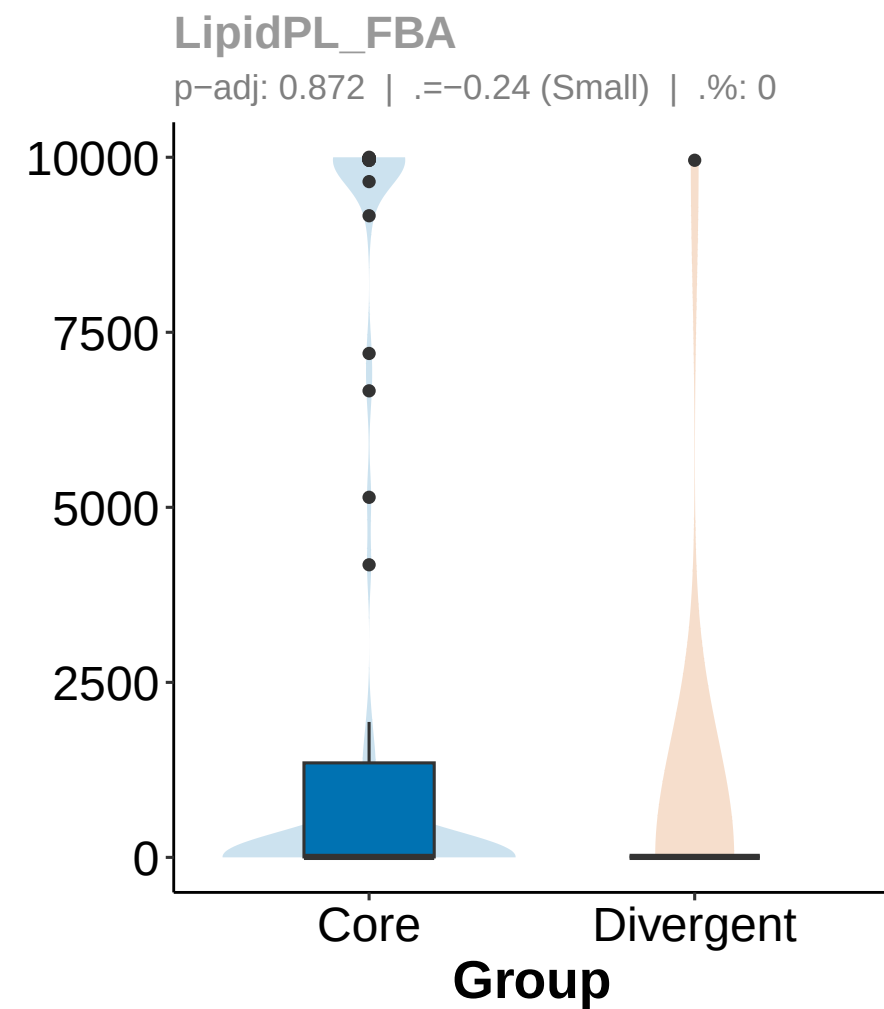
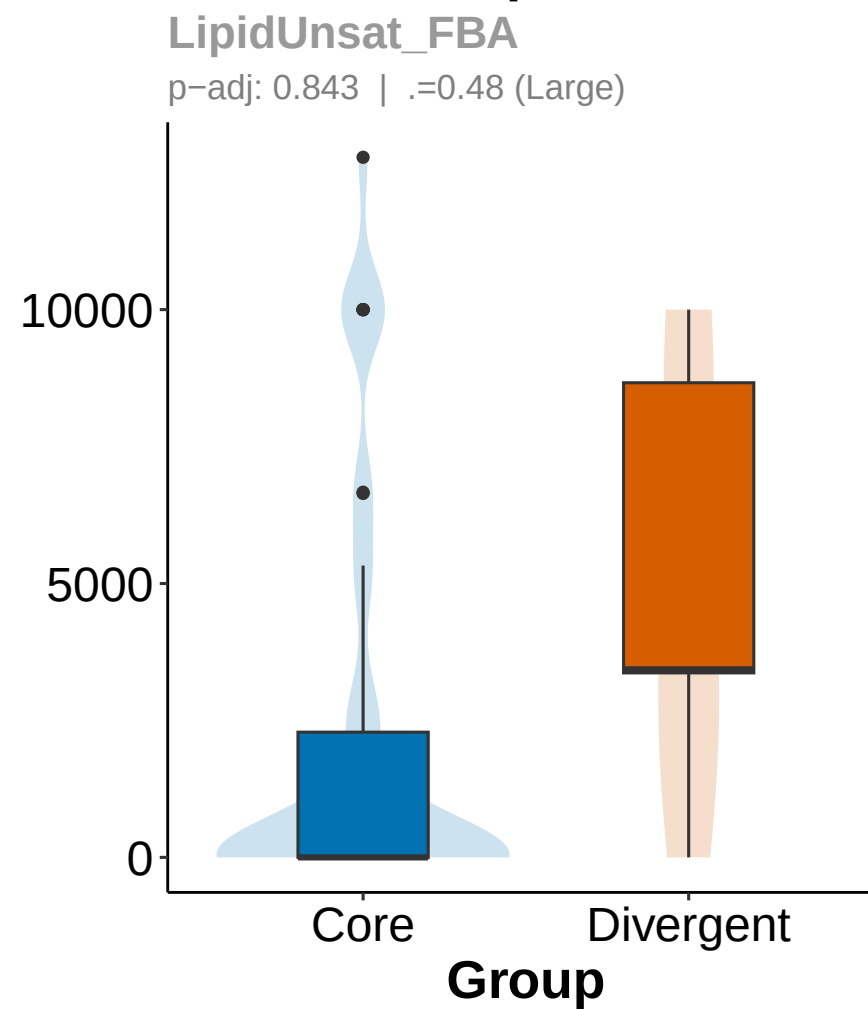
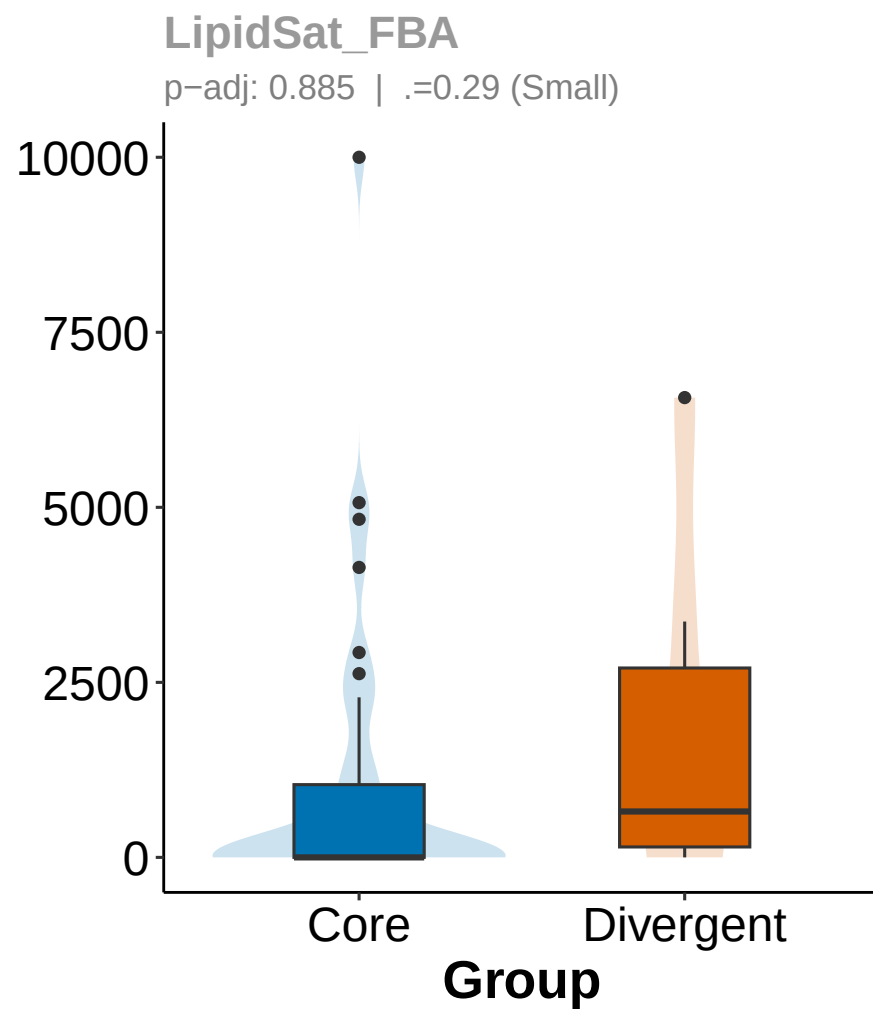


Cliff's . – . 3. Lipid Metabolism

Cliff's . + .% median | . higher in Core · . higher in Divergent



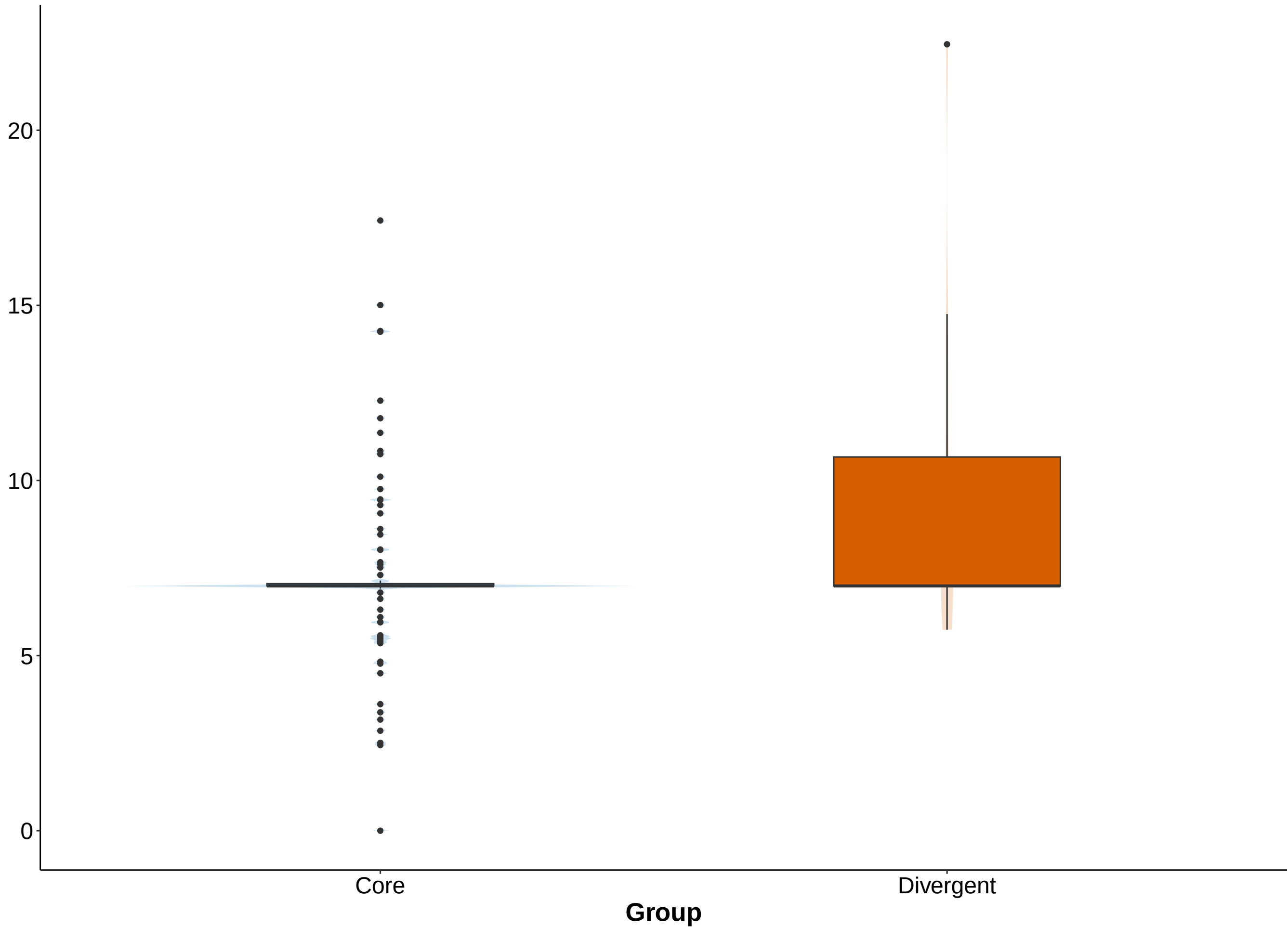
Metabolism – . 3. Lipid Metabolism



Metabolism - . 3. Lipid Metabolism

LipidPL_L1w

p-adj: 0.872 | . = 0.26 (Small) | .%: 0



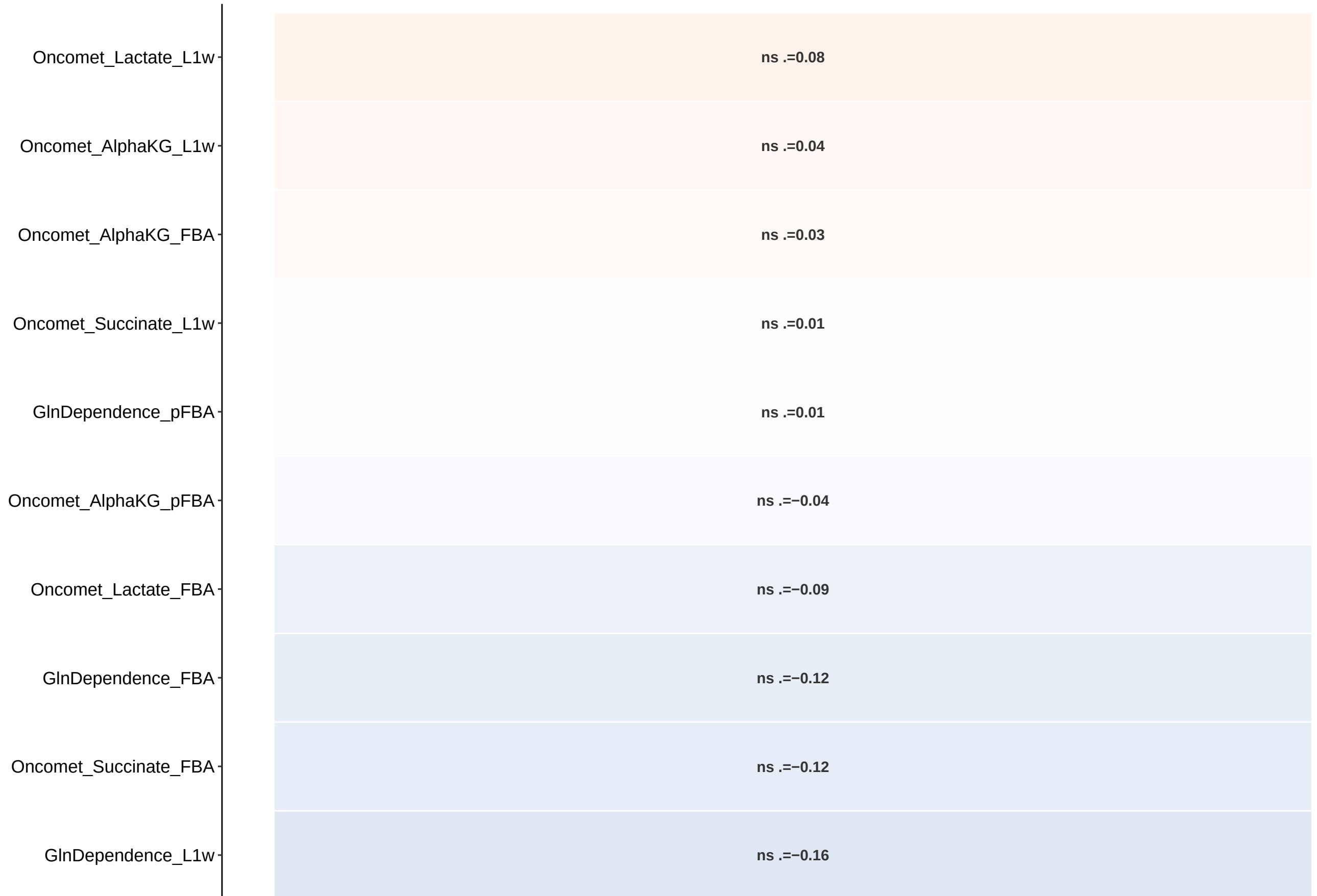
Cliff's .: . 4. Specific Metabolic Dependencies

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core

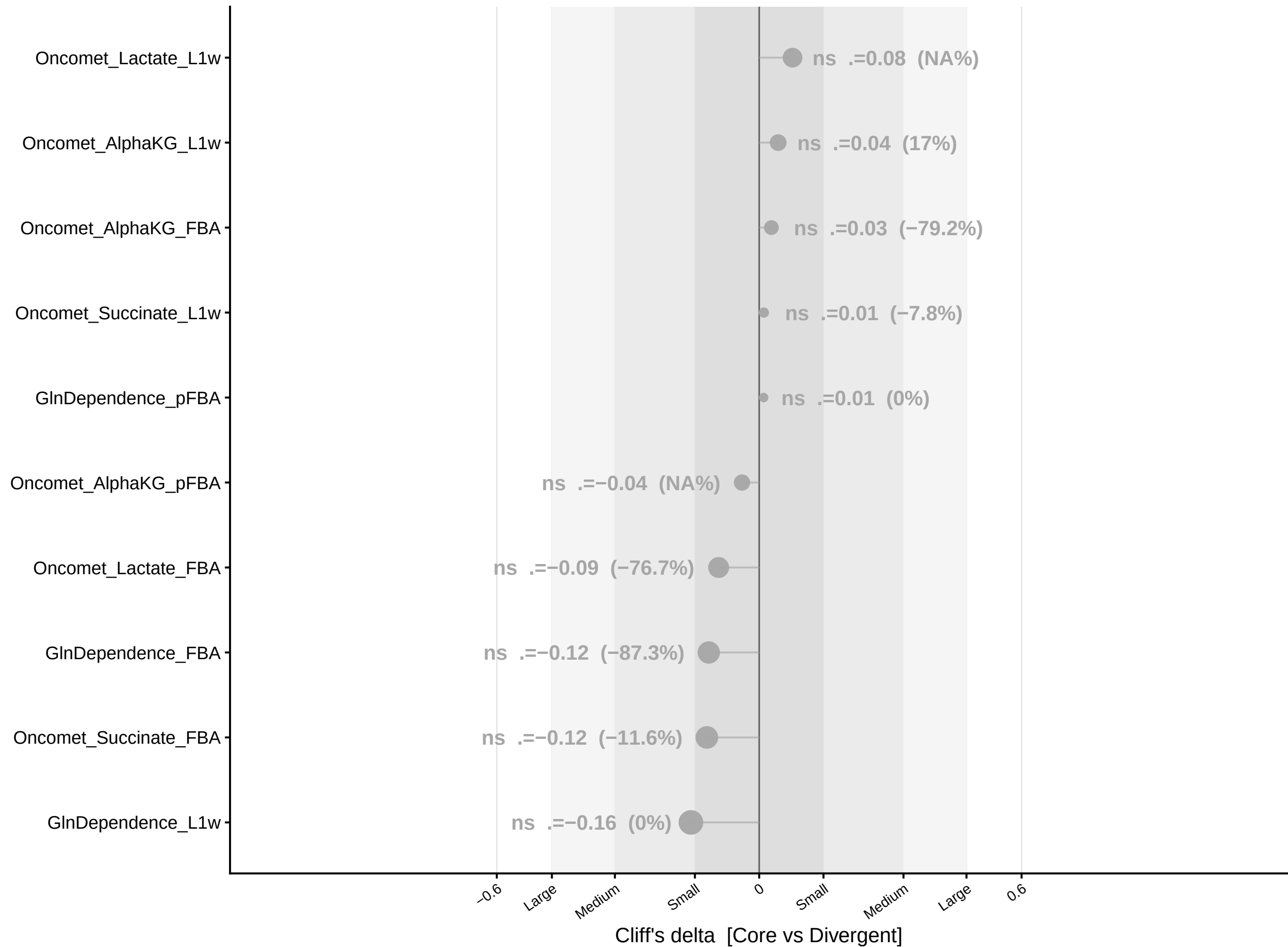


1.0 0.5 0.0 0.5 1.0

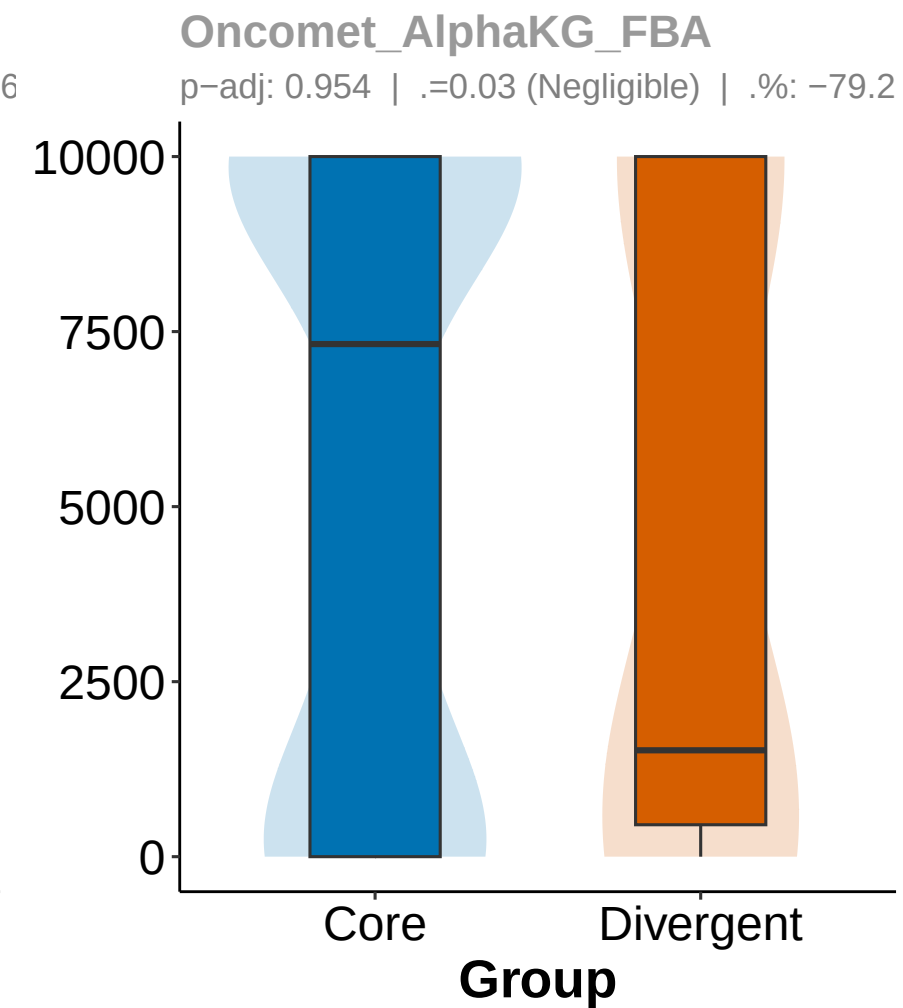
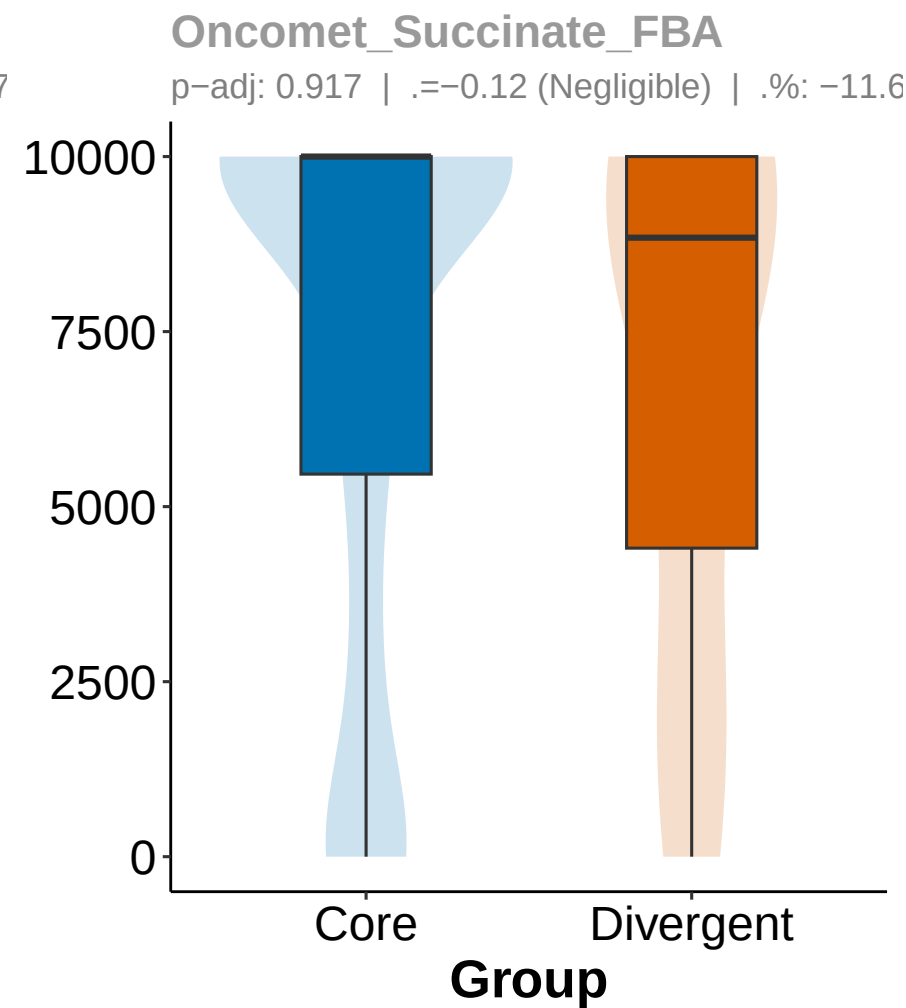
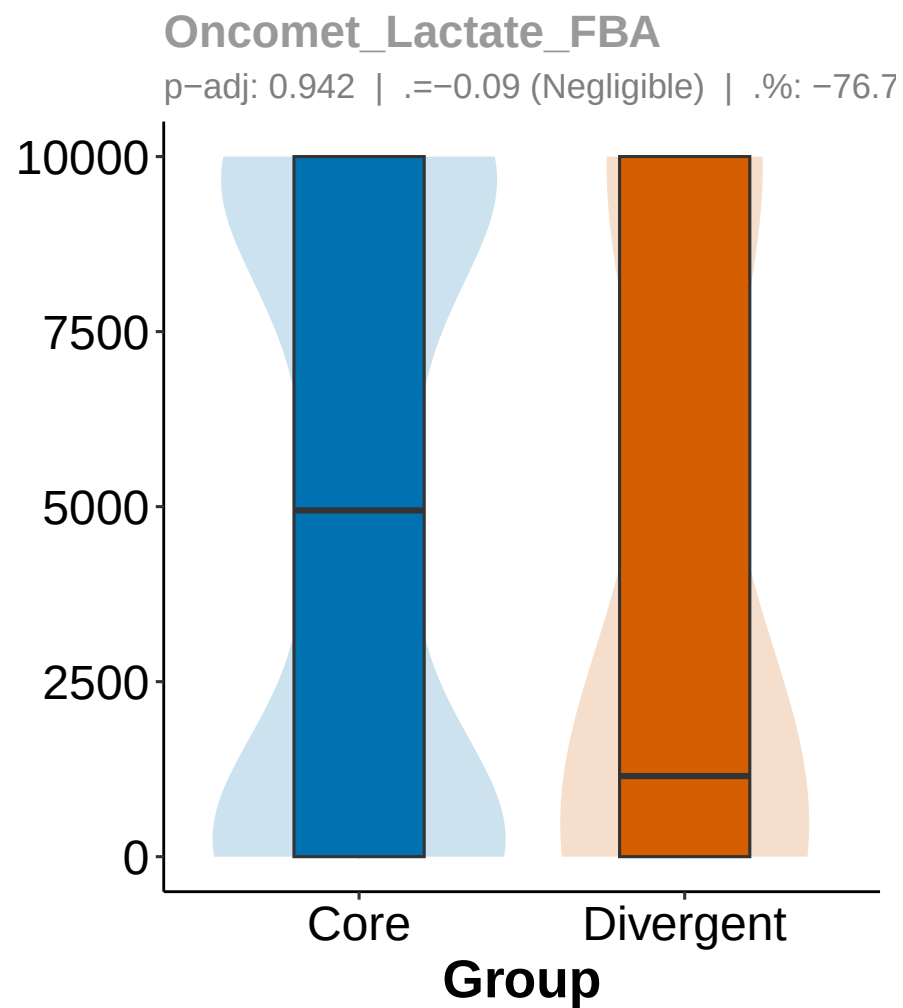
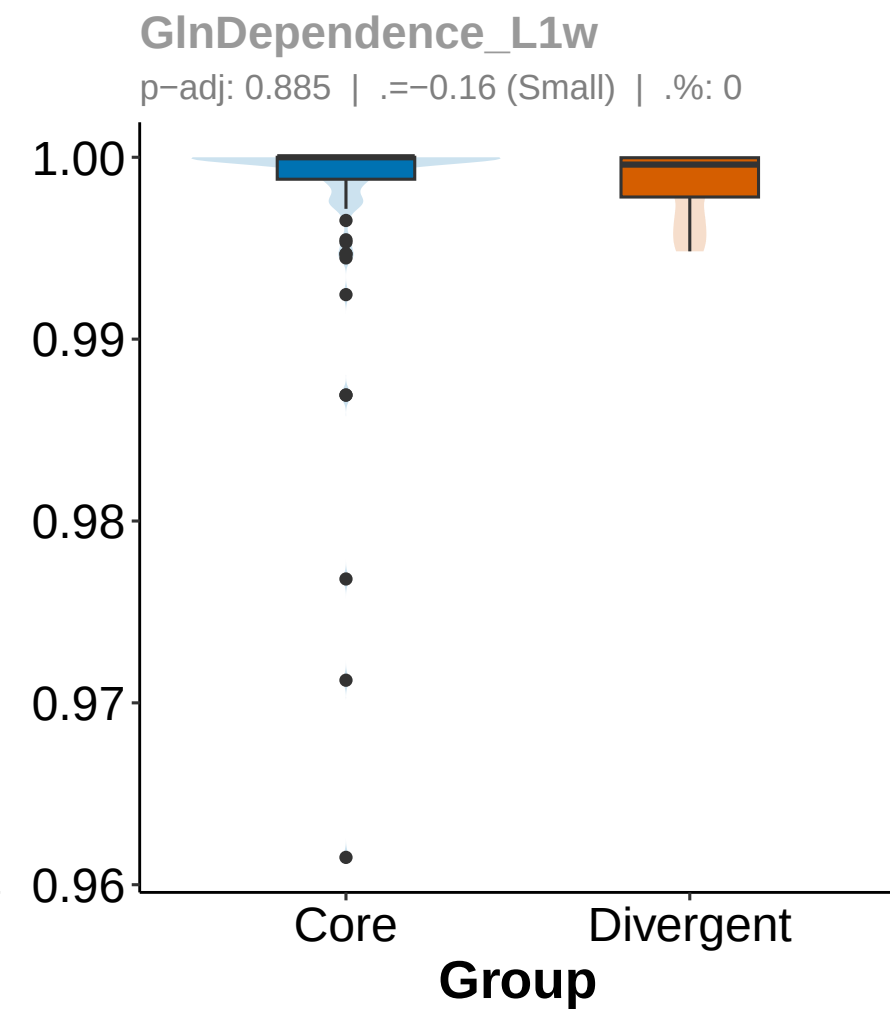
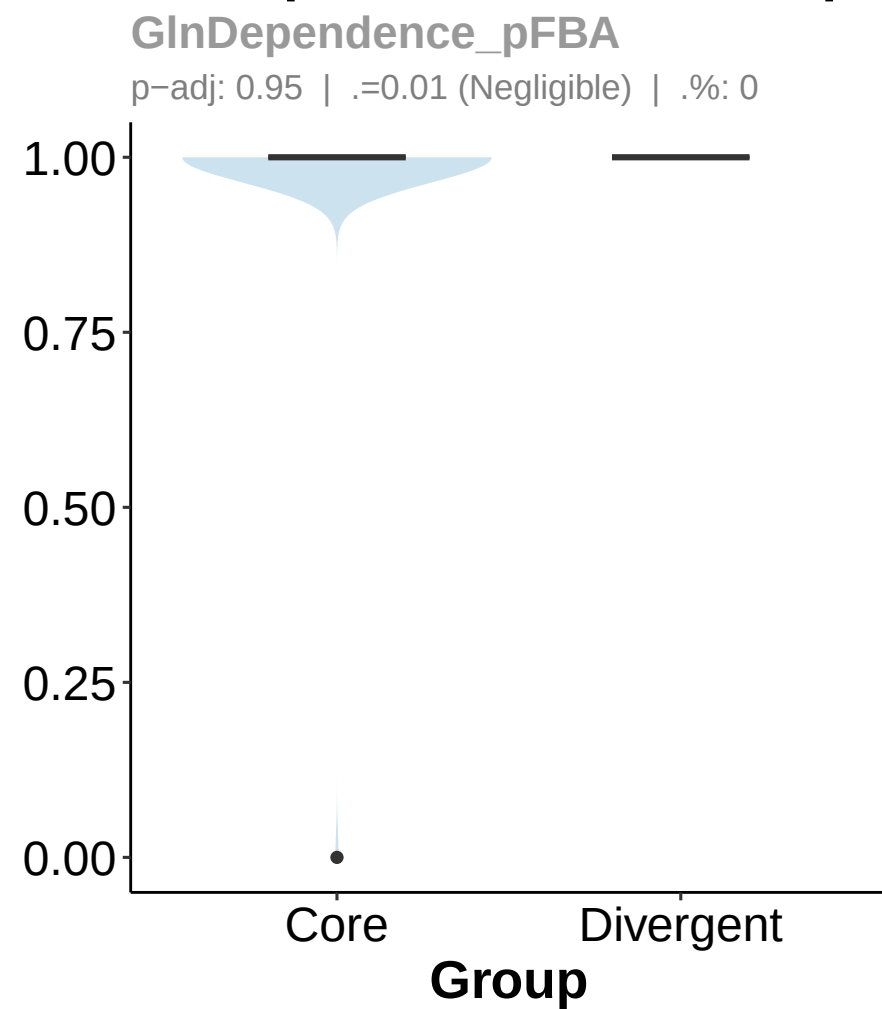
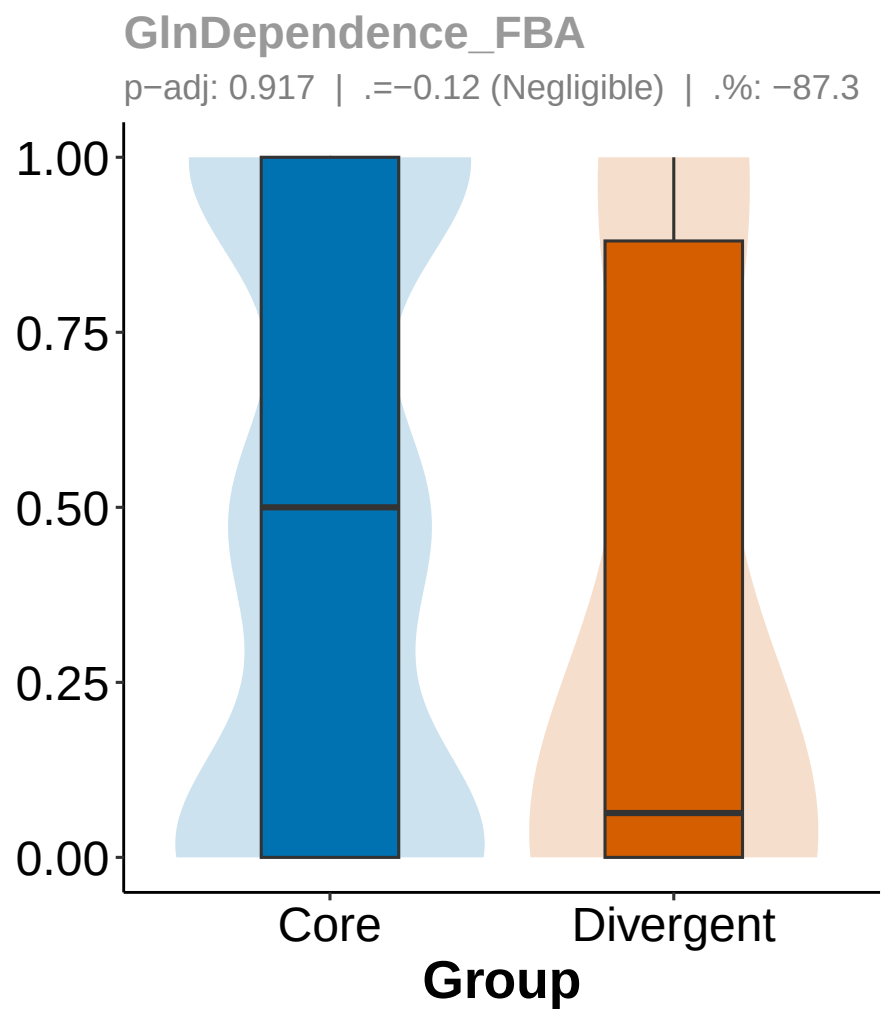


Cliff's . – . 4. Specific Metabolic Dependencies

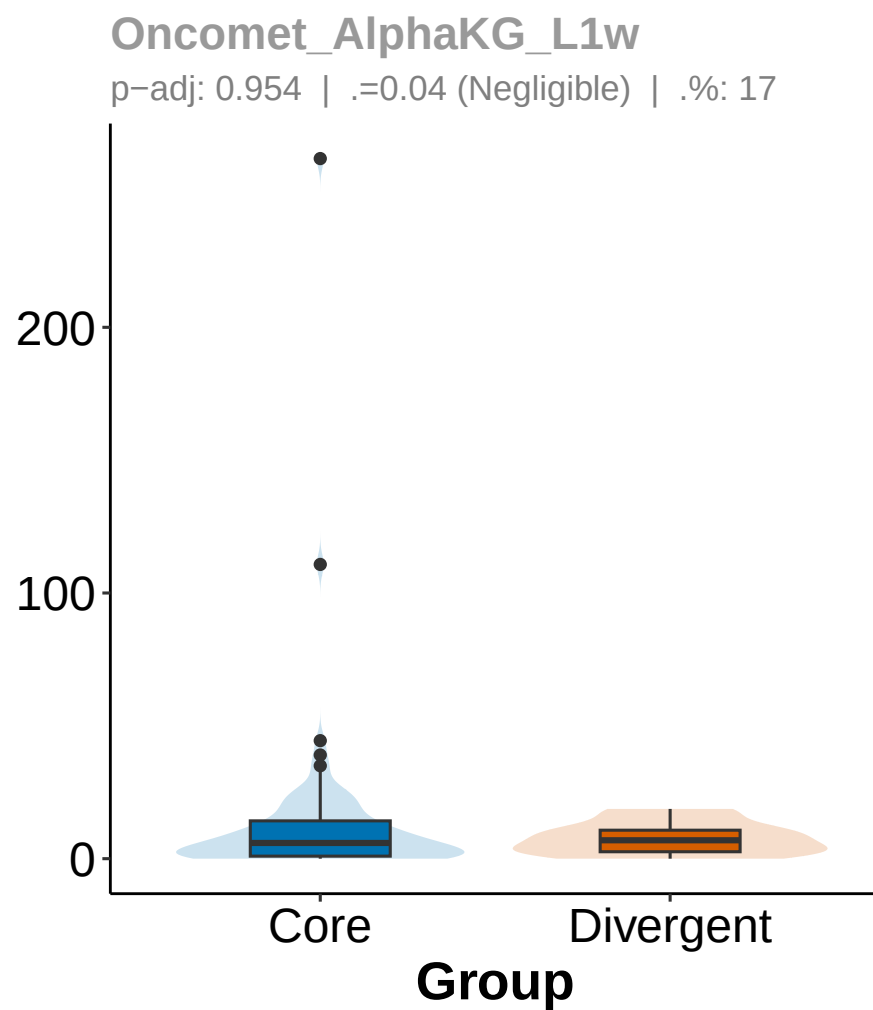
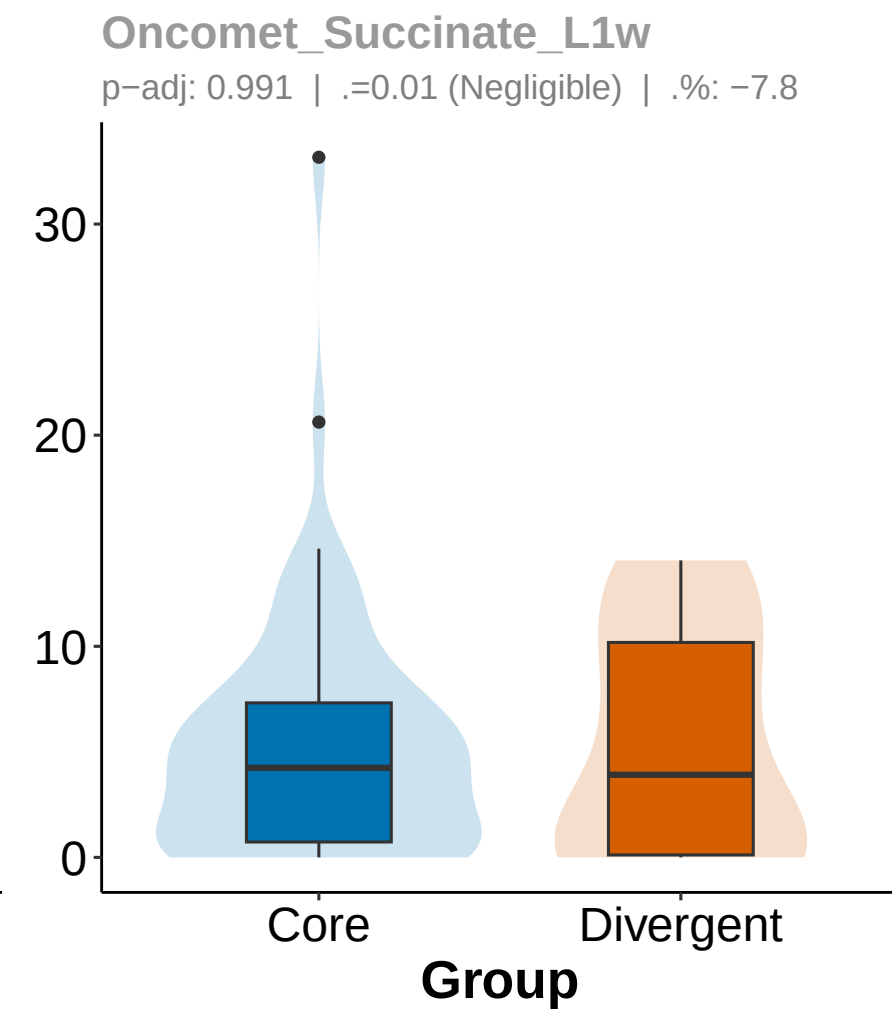
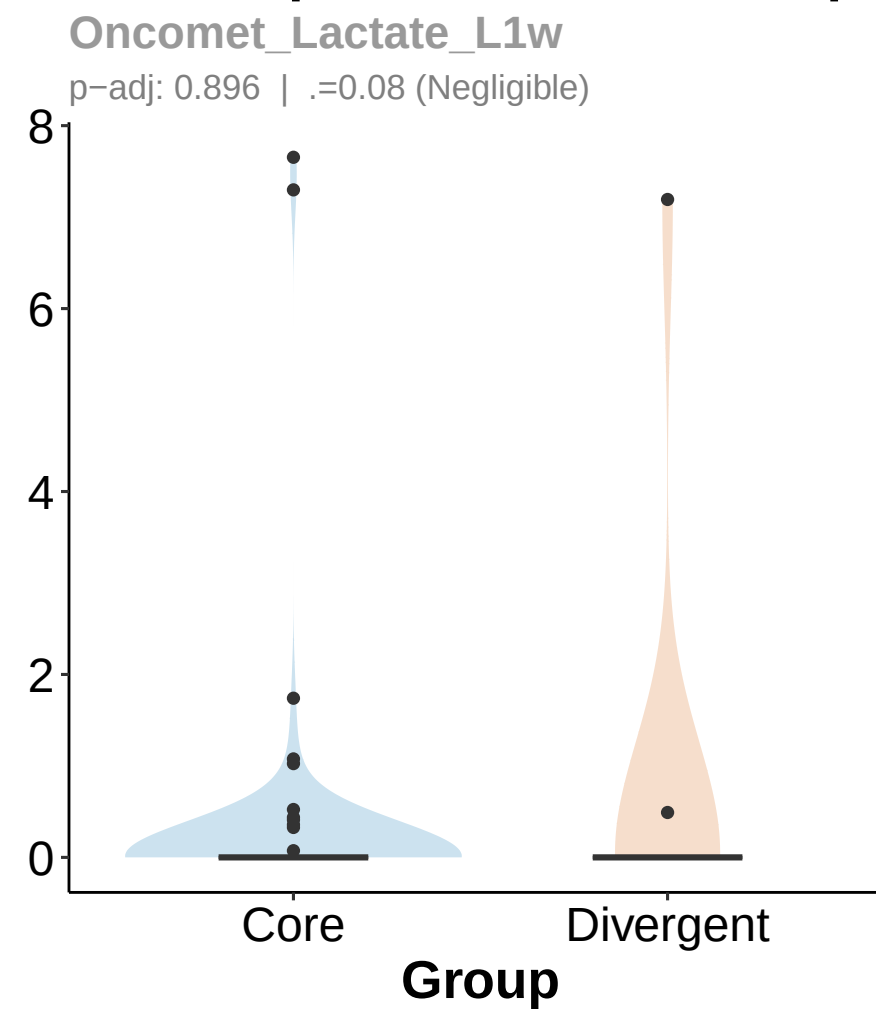
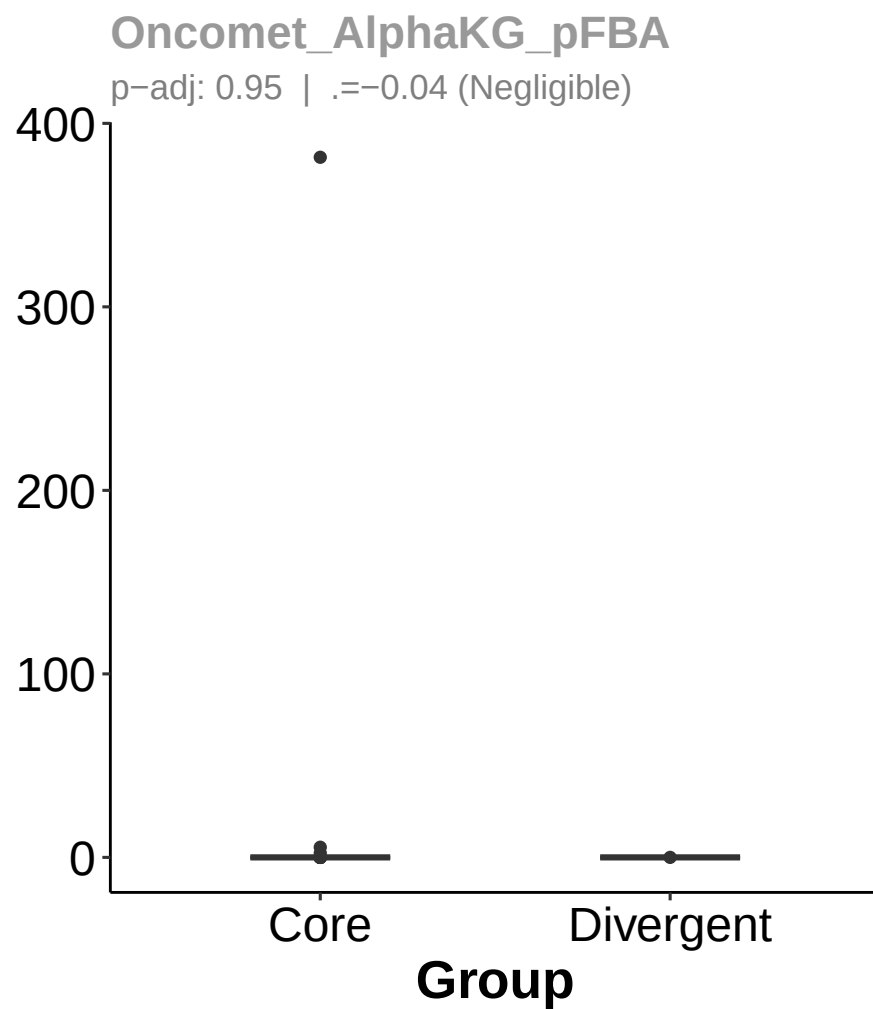
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 4. Specific Metabolic Dependencies



Metabolism – . 4. Specific Metabolic Dependencies



Cliff's .: . 5. Systemic Reprogramming (SA_*)

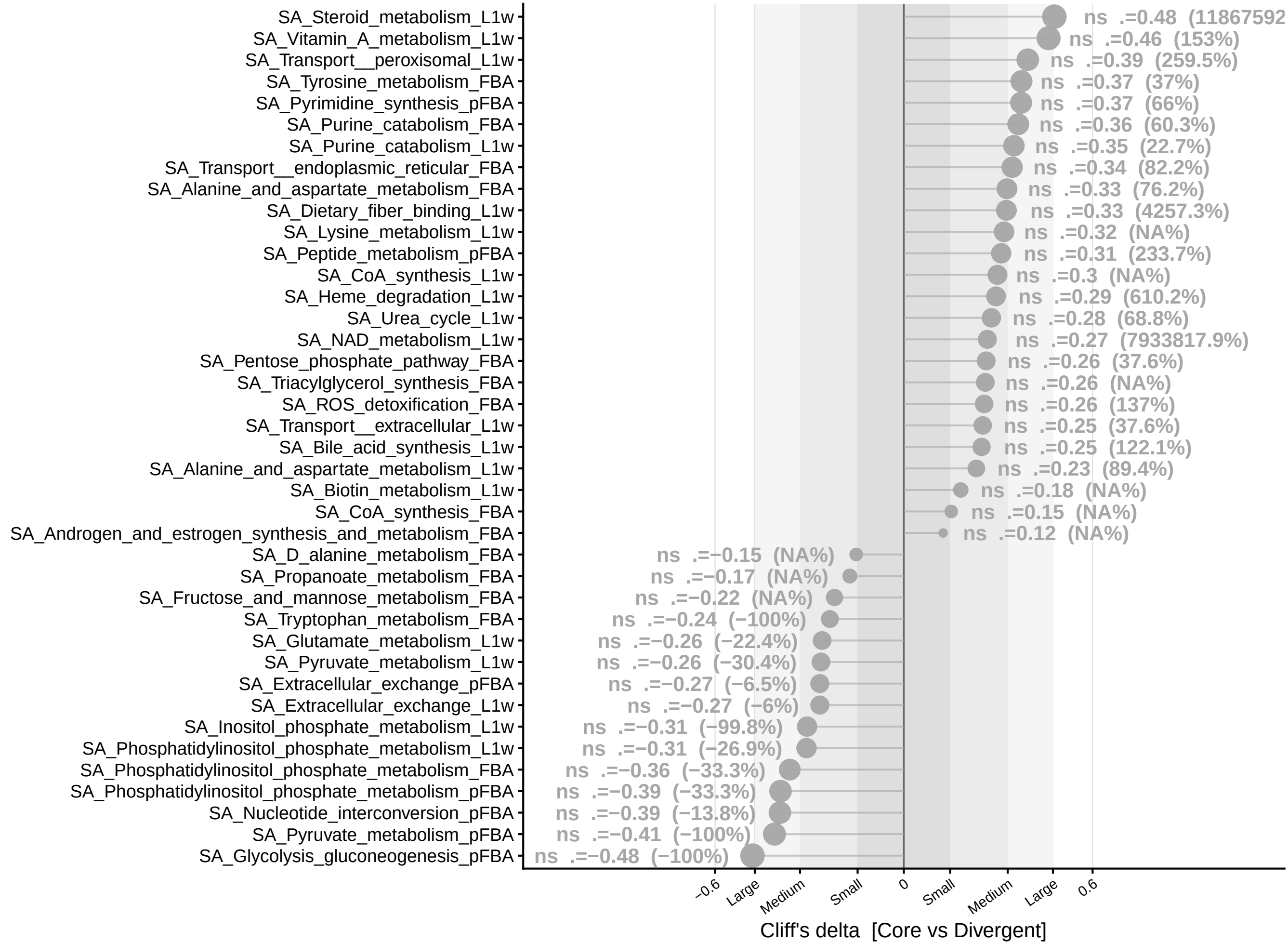
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



SA_Steroid_metabolism_L1w	ns .=0.48
SA_Vitamin_A_metabolism_L1w	ns .=0.46
SA_Transport__peroxisomal_L1w	ns .=0.39
SA_Tyrosine_metabolism_FBA	ns .=0.37
SA_Pyrimidine_synthesis_pFBA	ns .=0.37
SA_Purine_catabolism_FBA	ns .=0.36
SA_Purine_catabolism_L1w	ns .=0.35
SA_Transport__endoplasmic_reticular_FBA	ns .=0.34
SA_Alanine_and_aspartate_metabolism_FBA	ns .=0.33
SA_Dietary_fiber_binding_L1w	ns .=0.33
SA_Lysine_metabolism_L1w	ns .=0.32
SA_Peptide_metabolism_pFBA	ns .=0.31
SA_CoA_synthesis_L1w	ns .=0.3
SA_Heme_degradation_L1w	ns .=0.29
SA_Urea_cycle_L1w	ns .=0.28
SA_NAD_metabolism_L1w	ns .=0.27
SA_Pentose_phosphate_pathway_FBA	ns .=0.26
SA_Triacylglycerol_synthesis_FBA	ns .=0.26
SA_ROS_detoxification_FBA	ns .=0.26
SA_Transport__extracellular_L1w	ns .=0.25
SA_Bile_acid_synthesis_L1w	ns .=0.25
SA_Alanine_and_aspartate_metabolism_L1w	ns .=0.23
SA_Biotin_metabolism_L1w	ns .=0.18
SA_CoA_synthesis_FBA	ns .=0.15
SA_Androgen_and_estrogen_synthesis_and_metabolism_FBA	ns .=0.12
SA_D_alanine_metabolism_FBA	ns .=-0.15
SA_Propanoate_metabolism_FBA	ns .=-0.17
SA_Fructose_and_mannose_metabolism_FBA	ns .=-0.22
SA_Tryptophan_metabolism_FBA	ns .=-0.24
SA_Glutamate_metabolism_L1w	ns .=-0.26
SA_Pyruvate_metabolism_L1w	ns .=-0.26
SA_Extracellular_exchange_pFBA	ns .=-0.27
SA_Extracellular_exchange_L1w	ns .=-0.27
SA_Inositol_phosphate_metabolism_L1w	ns .=-0.31
SA_Phosphatidylinositol_phosphate_metabolism_L1w	ns .=-0.31
SA_Phosphatidylinositol_phosphate_metabolism_FBA	ns .=-0.36
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	ns .=-0.39
SA_Nucleotide_interconversion_pFBA	ns .=-0.39
SA_Pyruvate_metabolism_pFBA	ns .=-0.41
SA_Glycolysis_gluconeogenesis_pFBA	ns .=-0.48

Cliff's . – . 5. Systemic Reprogramming (SA_*)

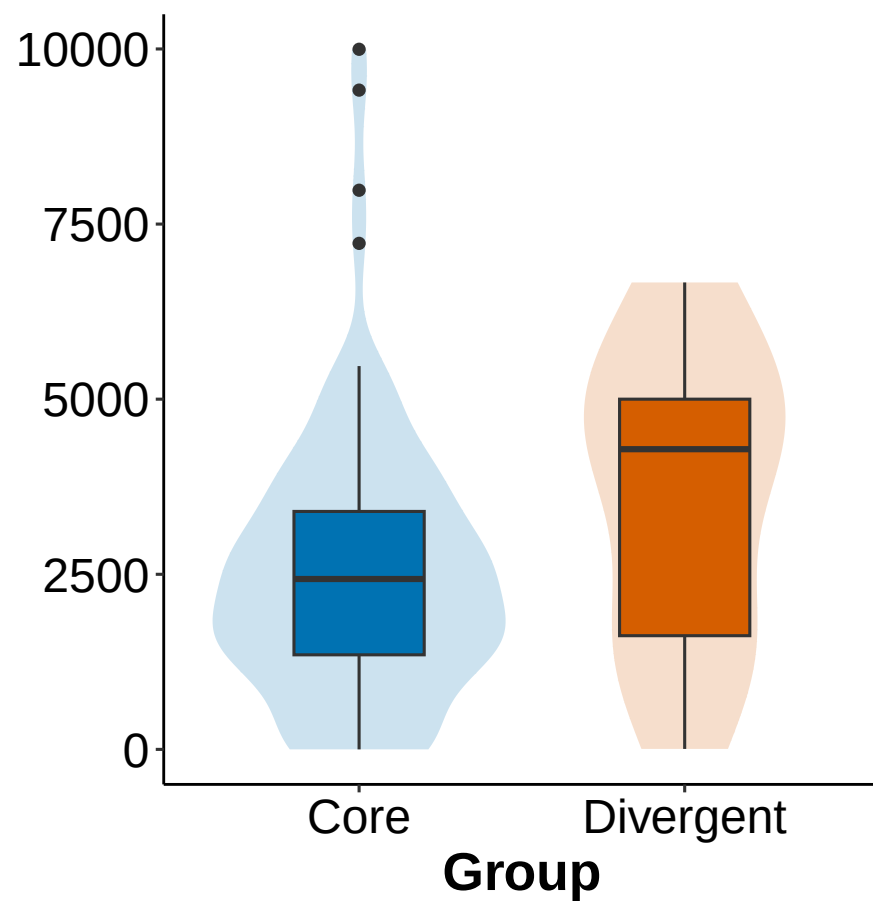
Cliff's . + .% median | . higher in Core · . higher in Divergent



Metabolism – . 5. Systemic Reprogramming (SA_*)

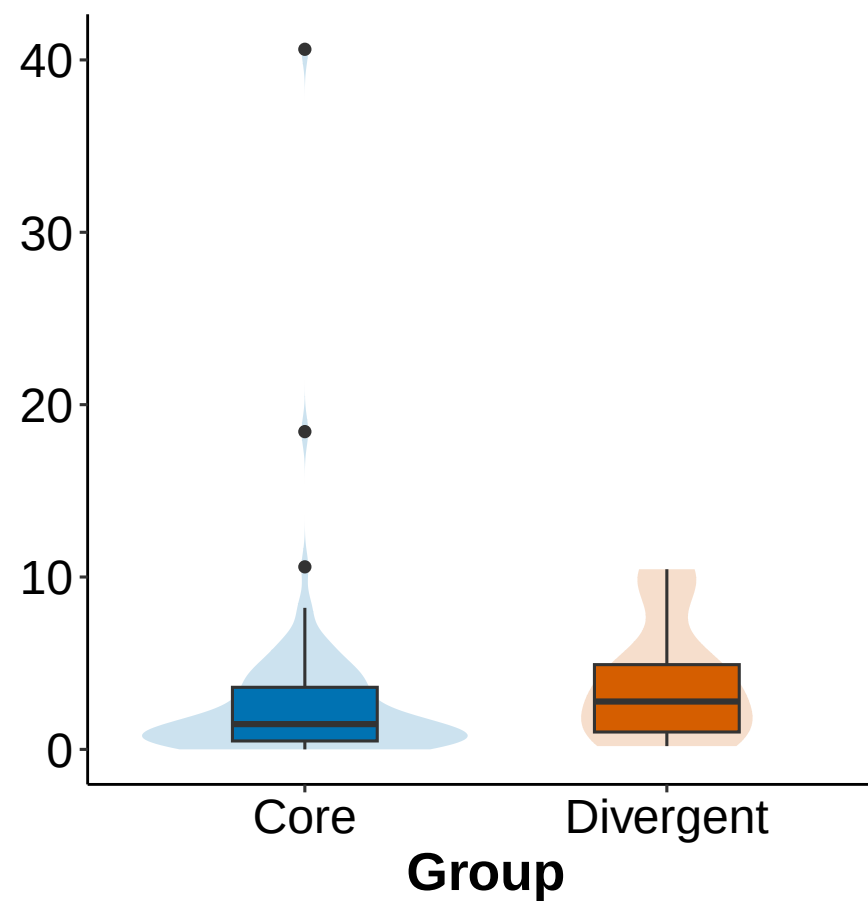
SA_Alanine_and_aspartate_metabolism

p-adj: 0.843 | .= 0.33 (Small) | .%: 76.2



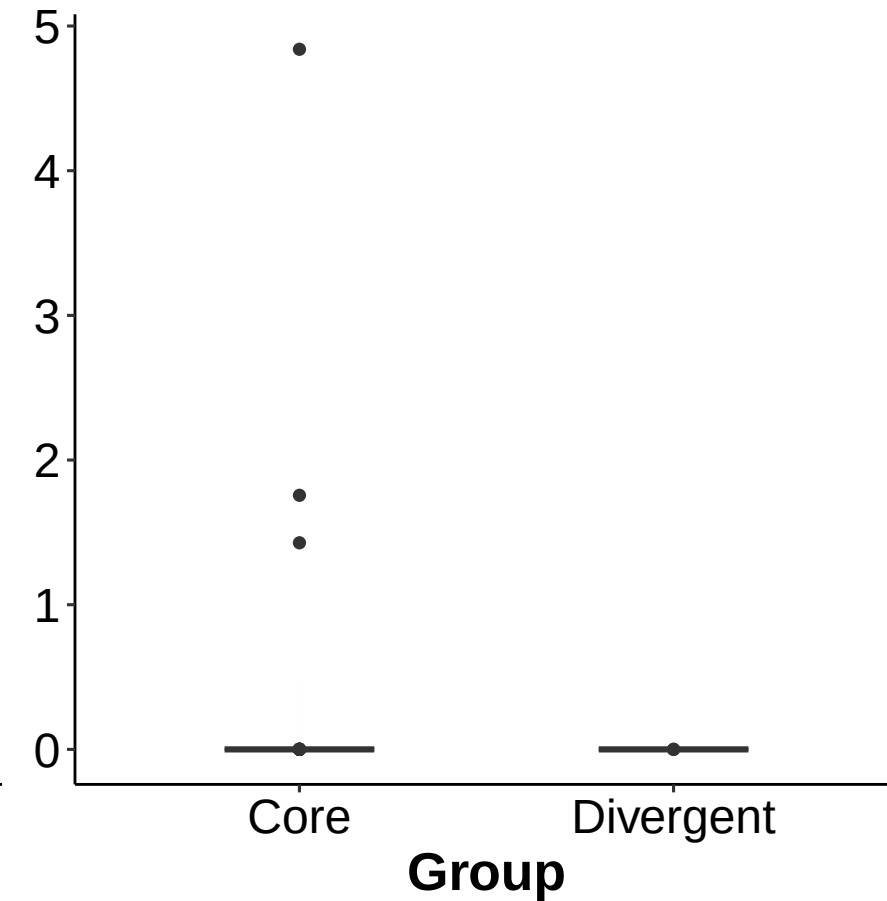
SA_Alanine_and_aspartate_metabolism

p-adj: 0.885 | .= 0.23 (Small) | .%: 89.4



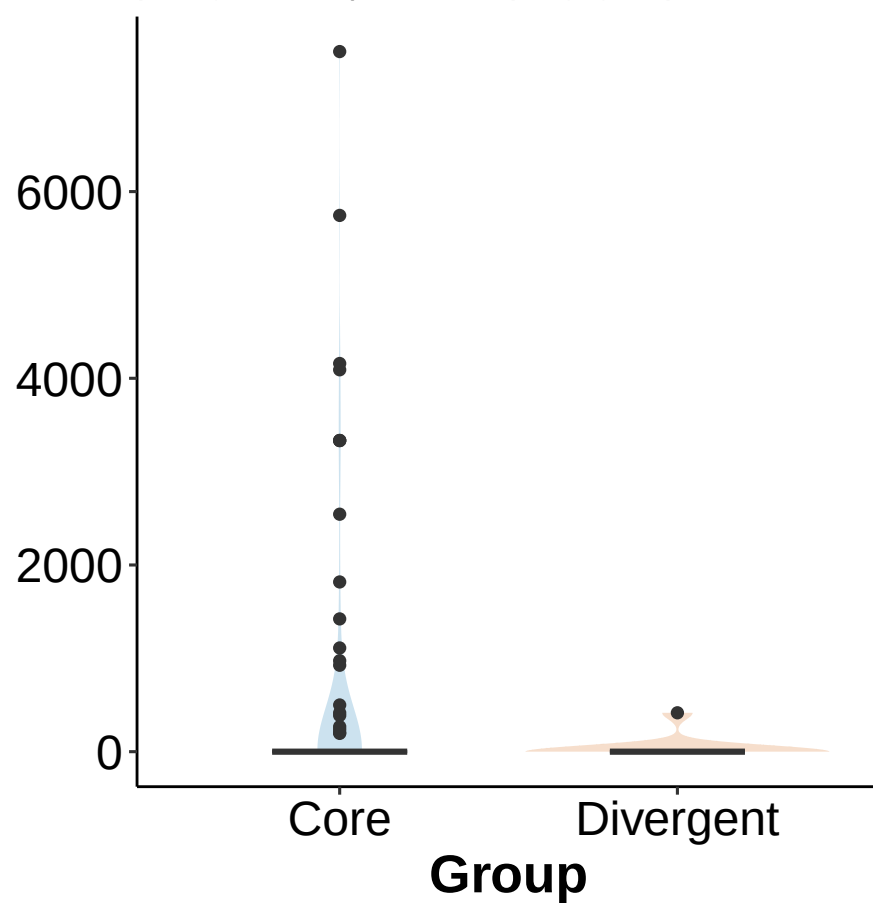
SA_Alanine_and_aspartate_metabolism

p-adj: 0.95 | .= 0.05 (Negligible)



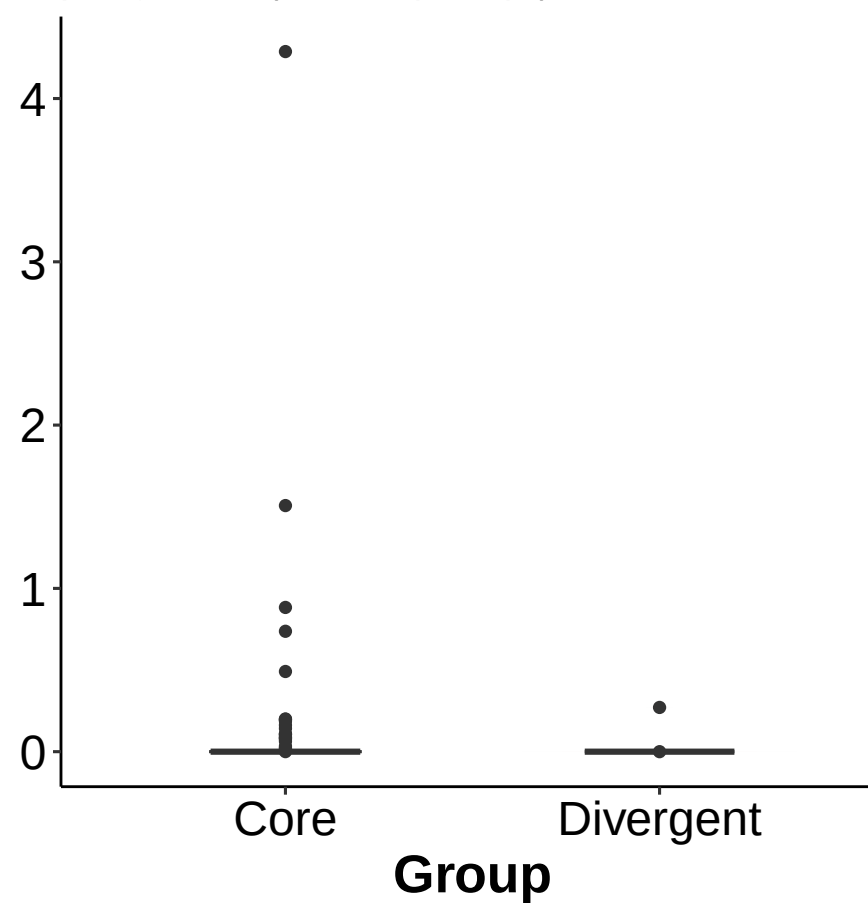
SA_Aminosugar_metabolism_FBA

p-adj: 0.885 | .= -0.12 (Negligible)



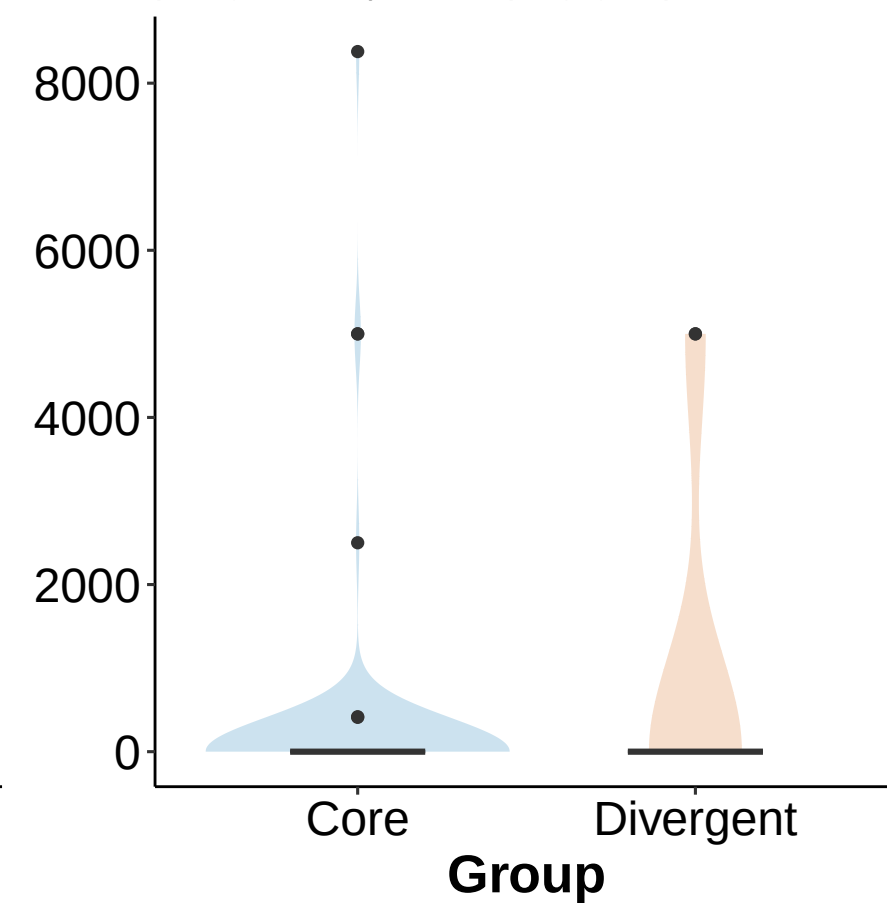
SA_Aminosugar_metabolism_L1w

p-adj: 0.885 | .= -0.2 (Small) | .%: -40.3



SA_Androgen_and_estrogen_synthesis

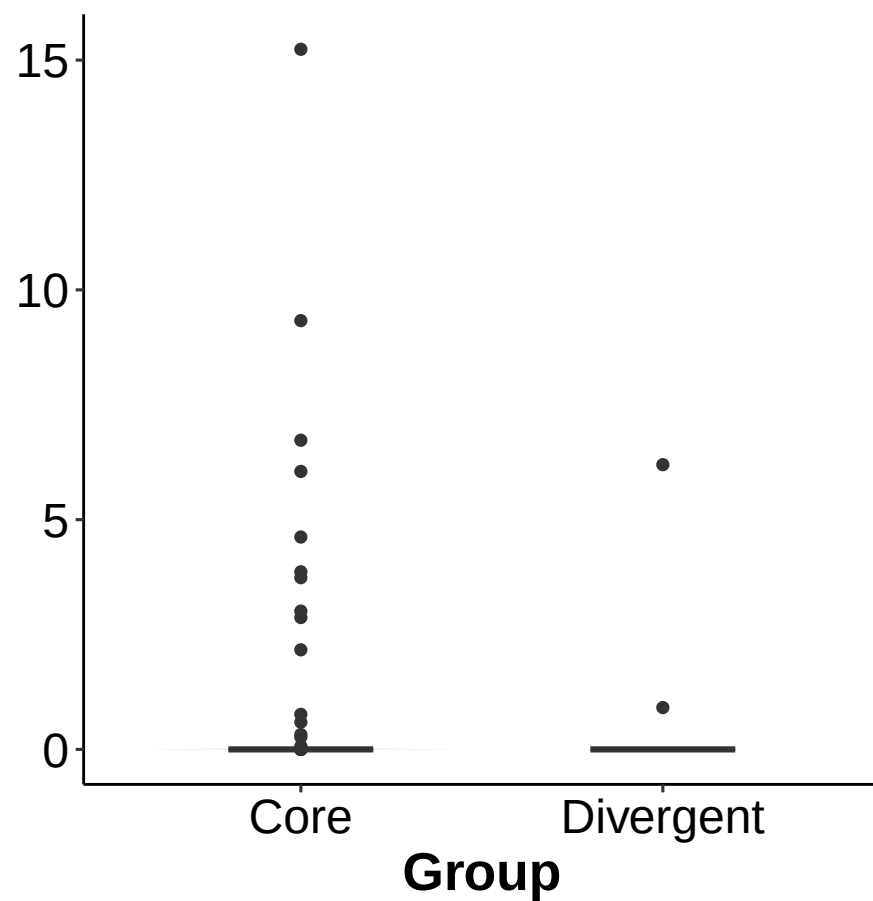
p-adj: 0.872 | .= 0.12 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

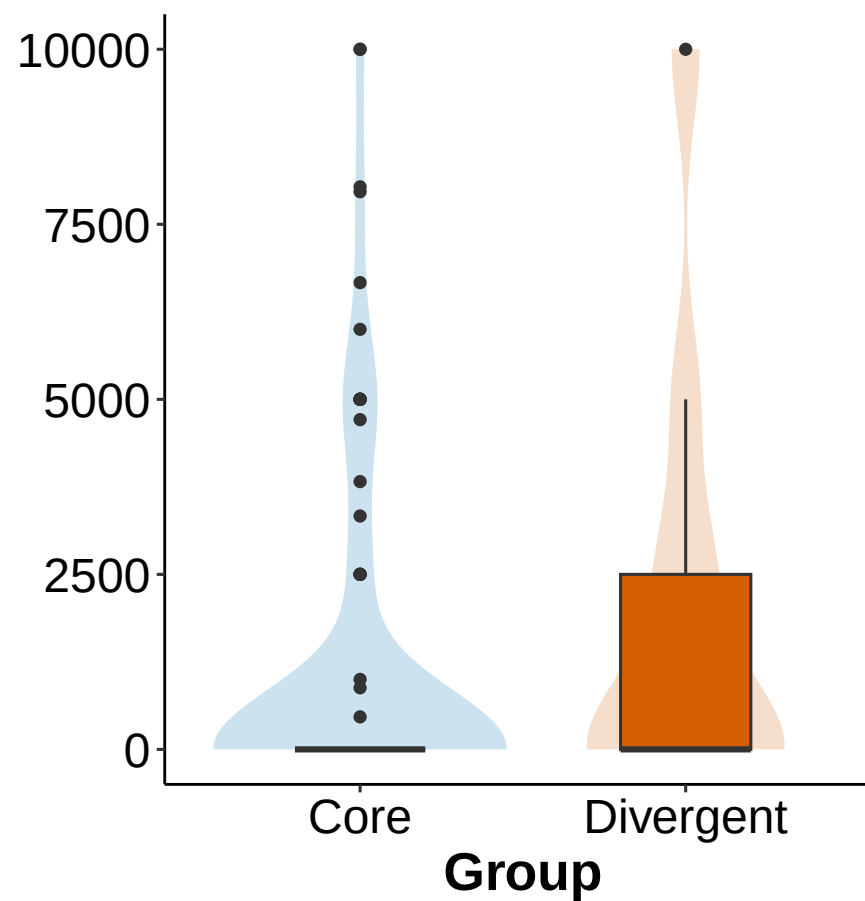
SA_Androgen_and_estrogen_synthes

p-adj: 0.885 | .:=0.15 (Small)



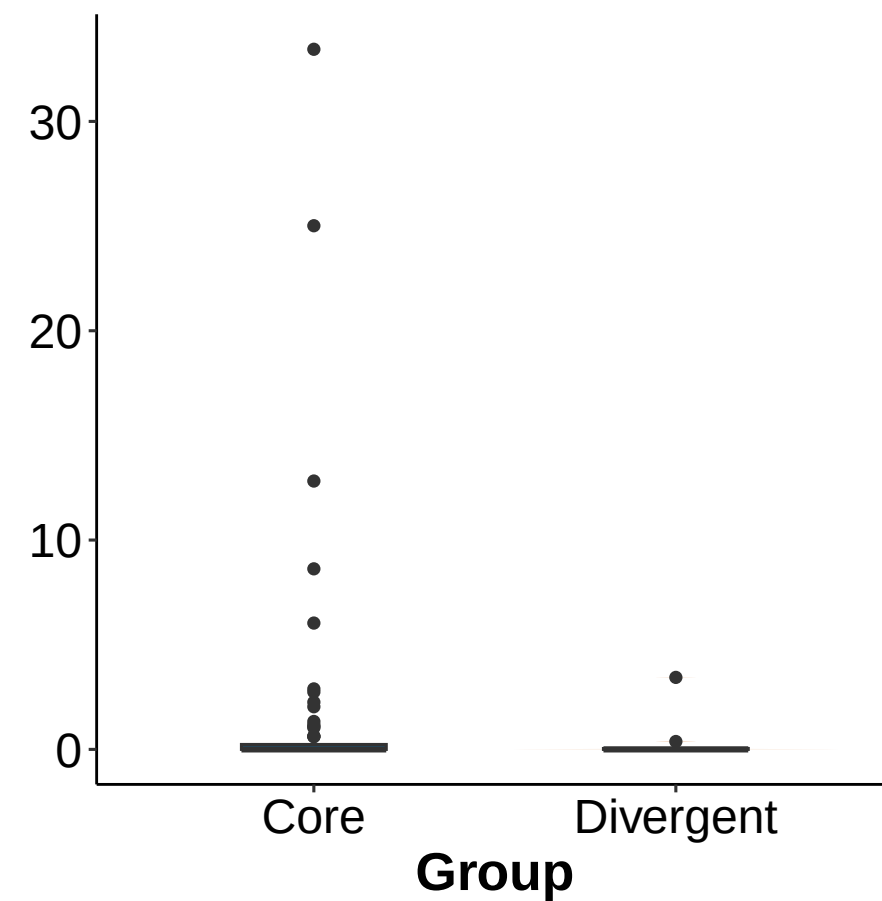
SA_Arachidonic_acid_metabolism

p-adj: 0.885 | .:=0.12 (Negligible)



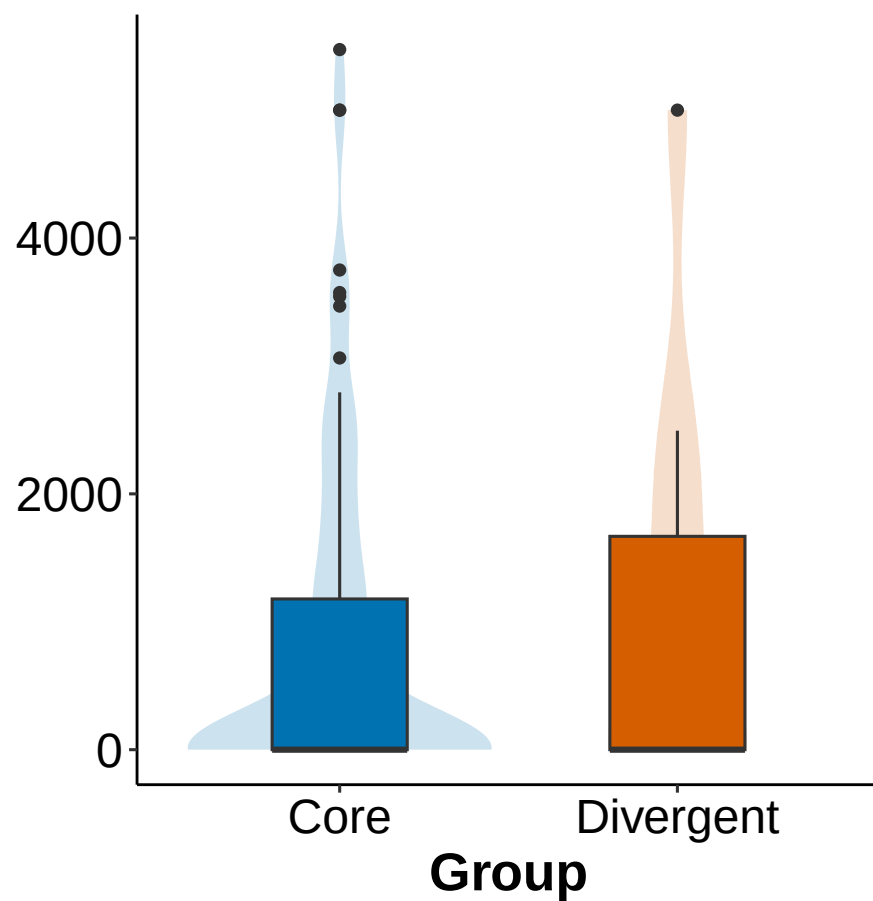
SA_Arachidonic_acid_metabolism_L1

p-adj: 0.885 | .:=−0.17 (Small) | .%: −100



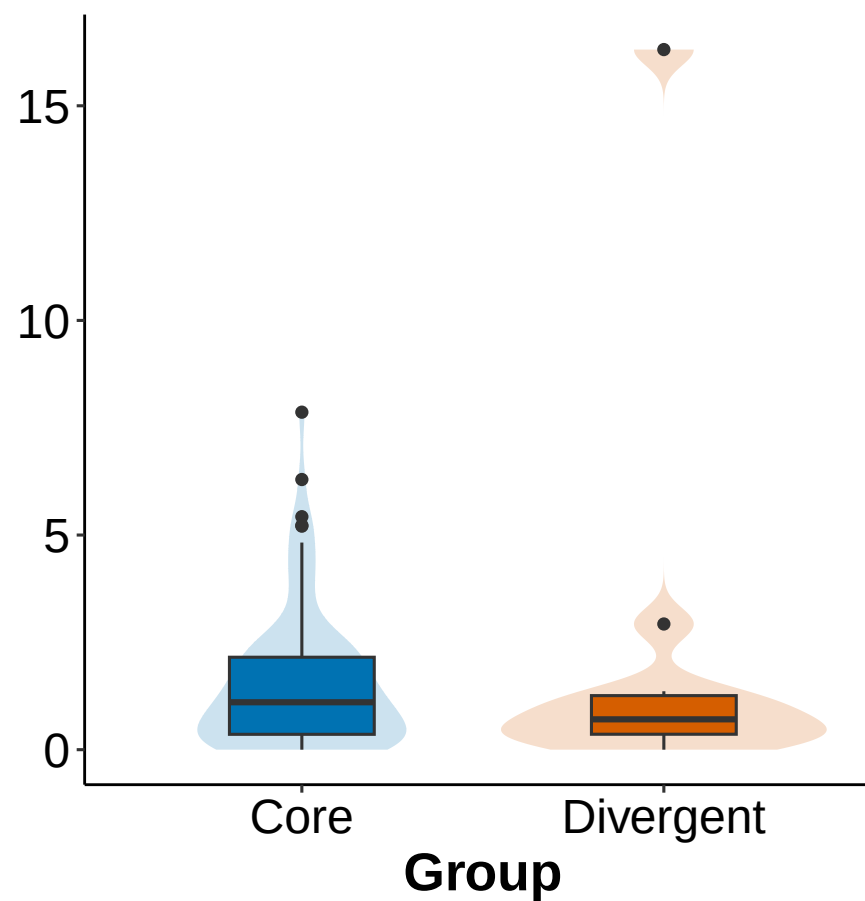
SA_Arginine_and_proline_metaboli

p-adj: 0.95 | .:=0.06 (Negligible)



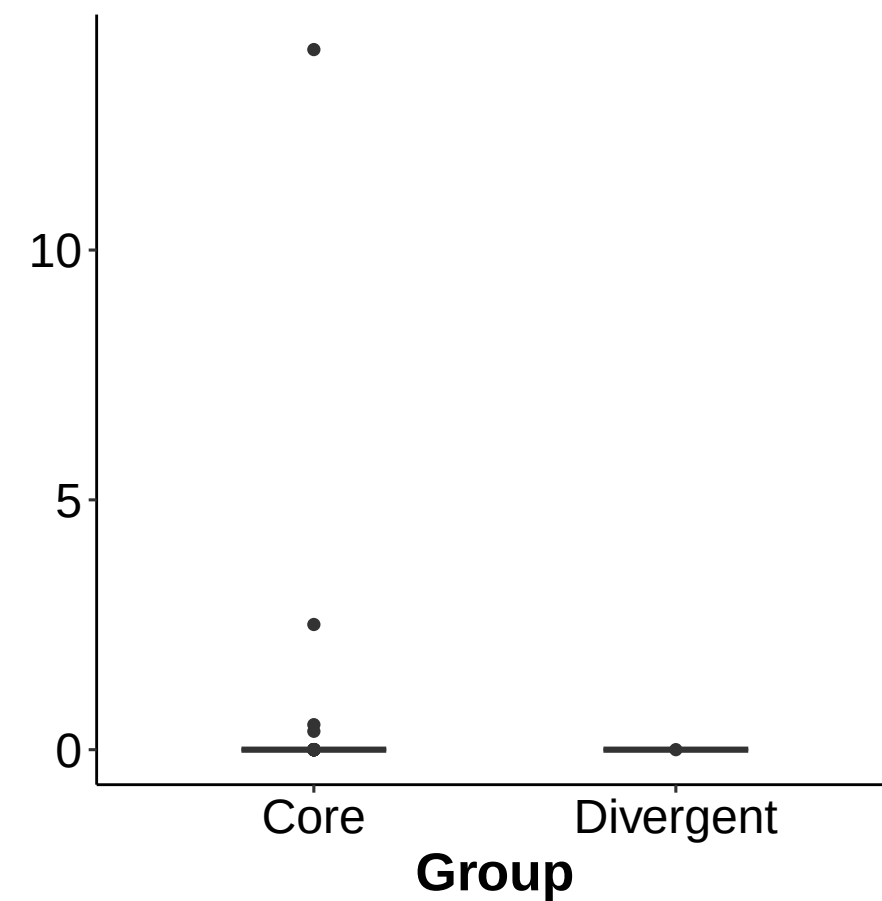
SA_Arginine_and_proline_metabolism

p-adj: 0.917 | .:=−0.12 (Negligible) | .%: −36



SA_Arginine_and_proline_metabolism

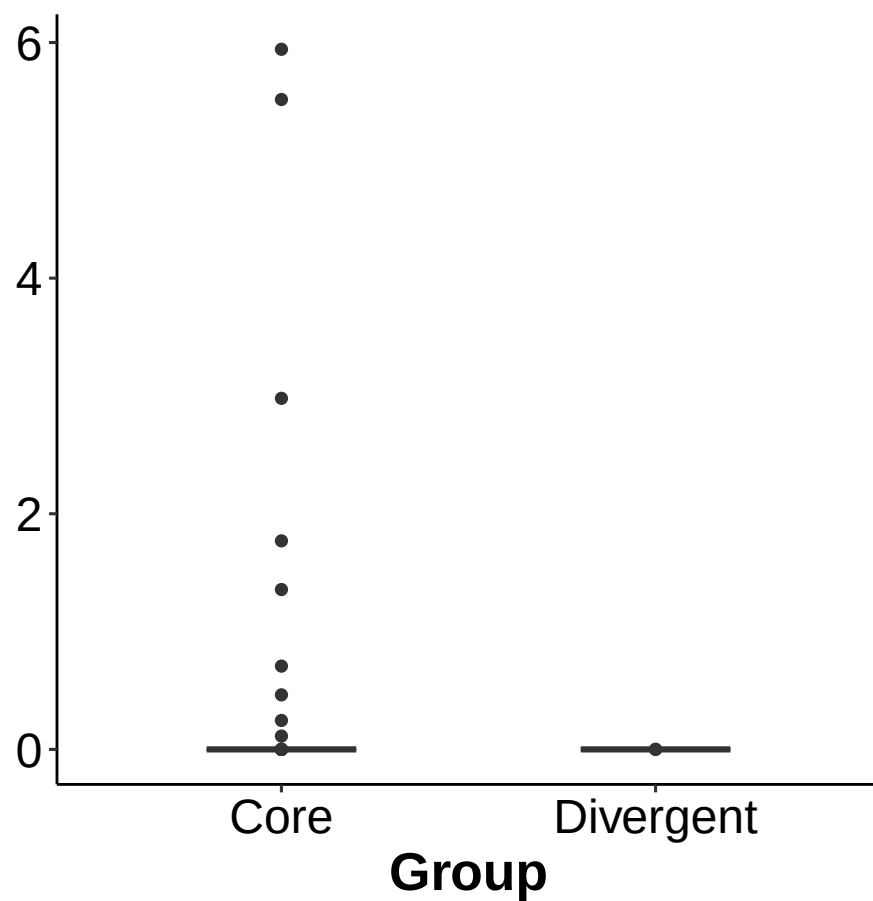
p-adj: 0.896 | .:=−0.1 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

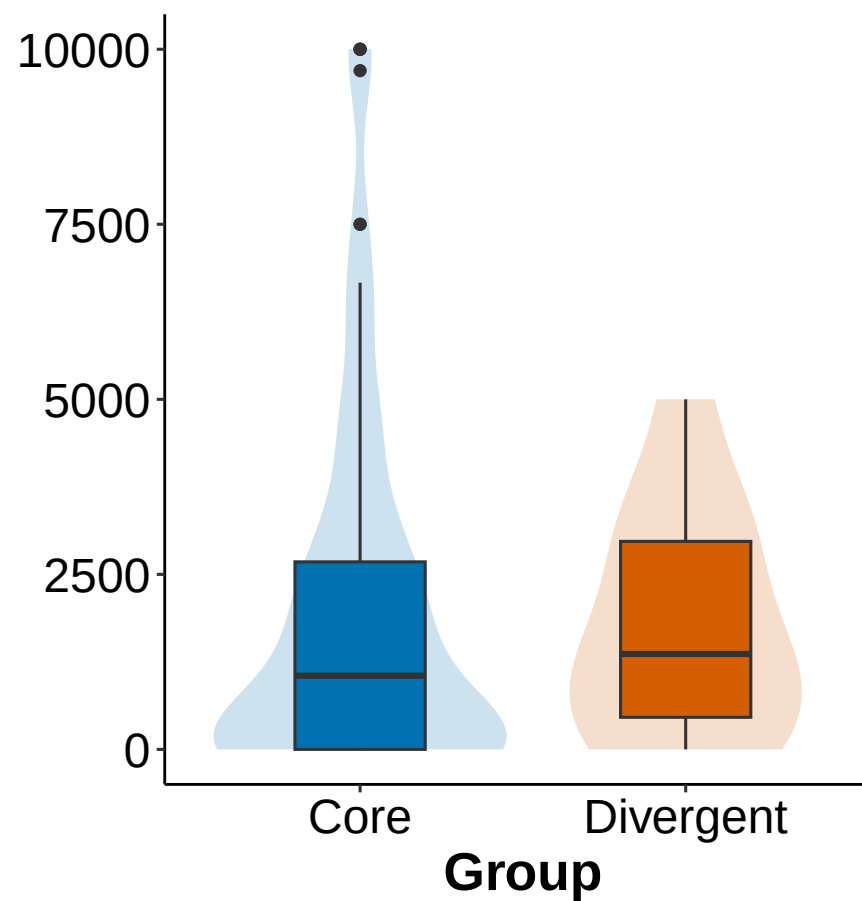
SA_Beta_Alanine_metabolism_L1w

p-adj: 0.954 | .:=0.03 (Negligible)



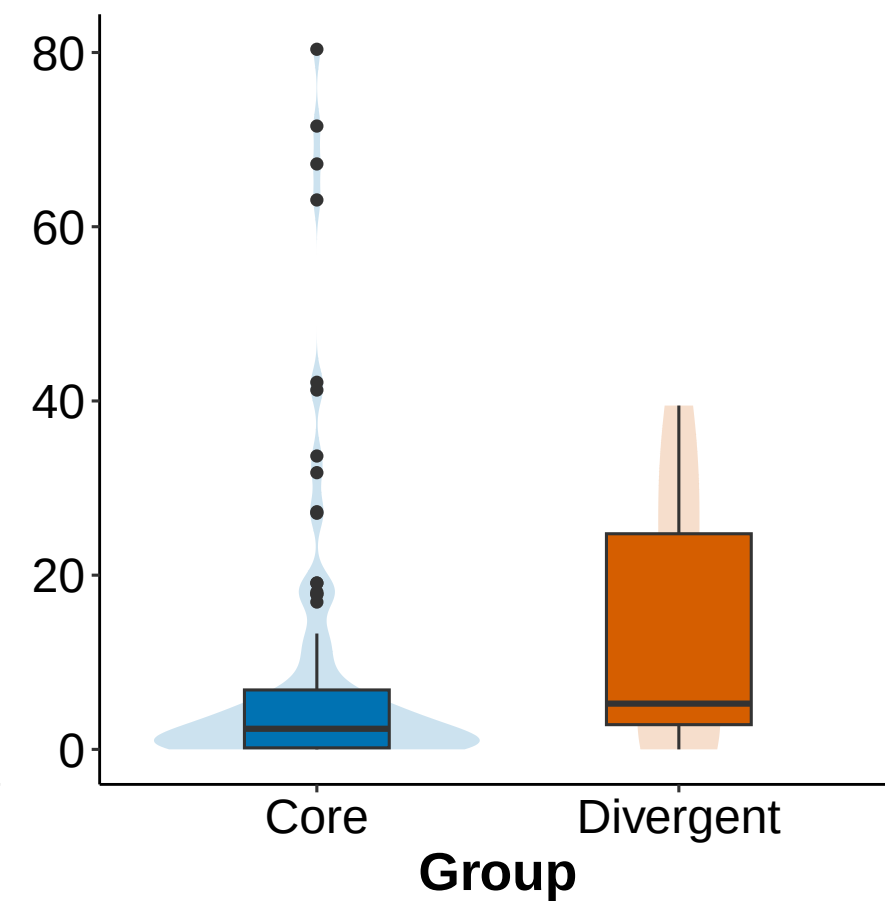
SA_Bile_acid_synthesis_FBA

p-adj: 0.917 | .:=0.12 (Negligible) | .%: 29.5



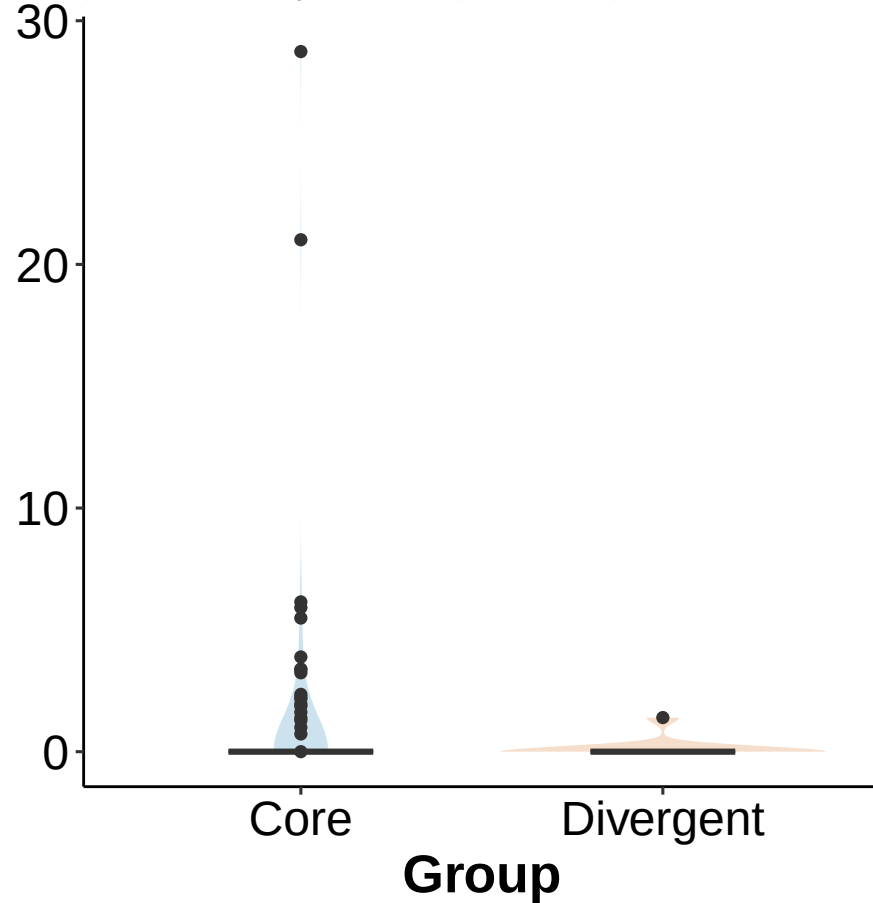
SA_Bile_acid_synthesis_L1w

p-adj: 0.885 | .:=0.25 (Small) | .%: 122.1



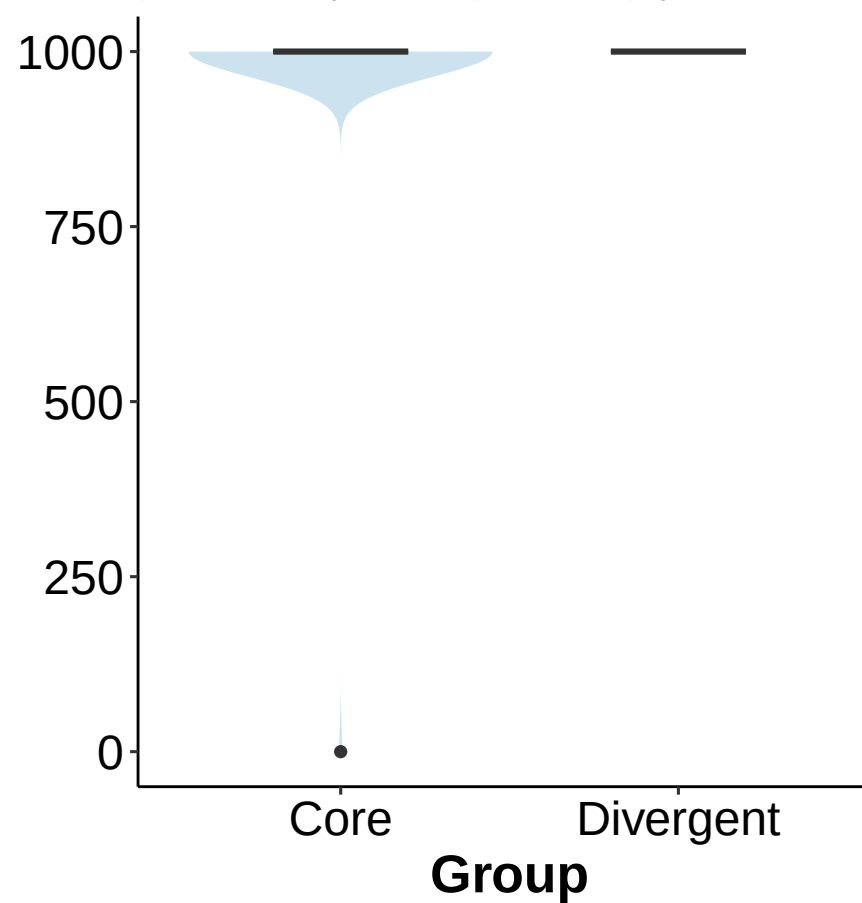
SA_Bile_acid_synthesis_pFBA

p-adj: 0.885 | .:=-0.12 (Negligible)



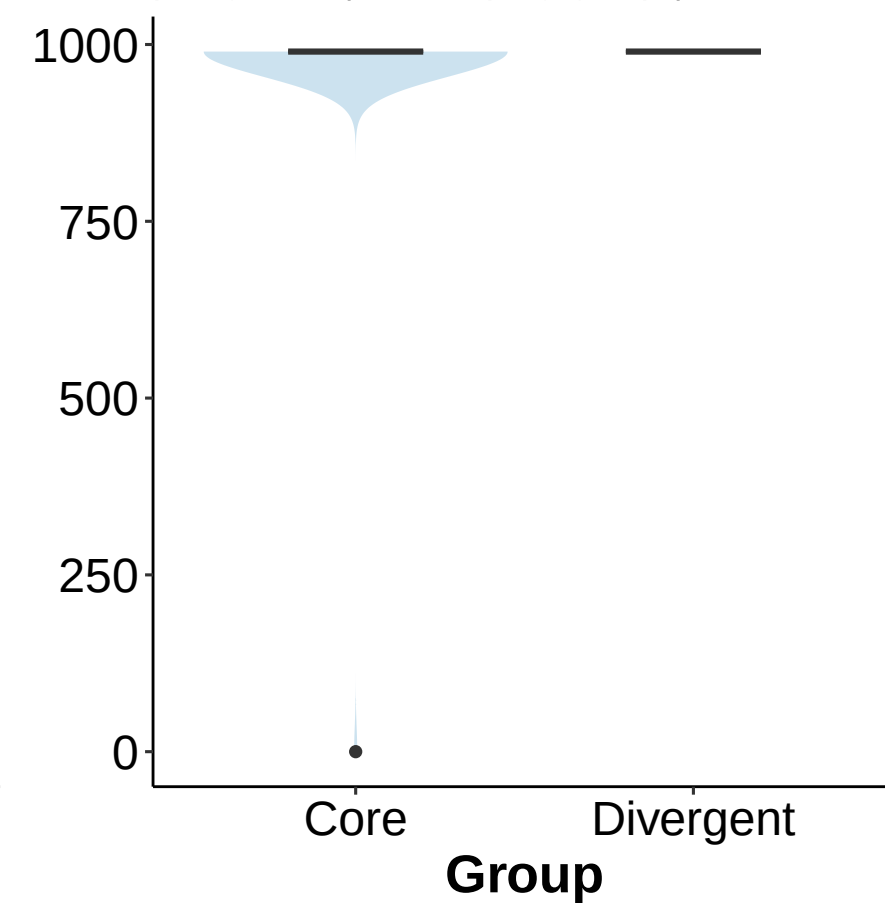
SA_Biomass_reactions_FBA

p-adj: 0.95 | .:=0.01 (Negligible) | .%: 0

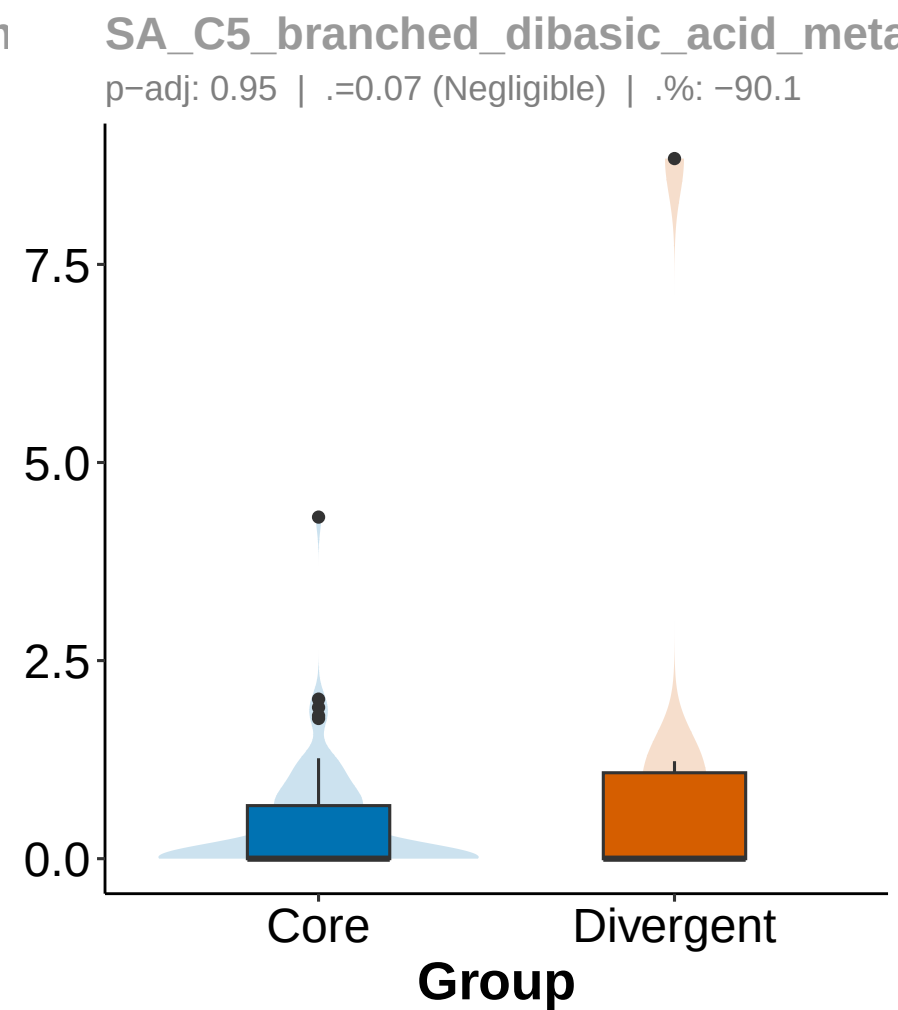
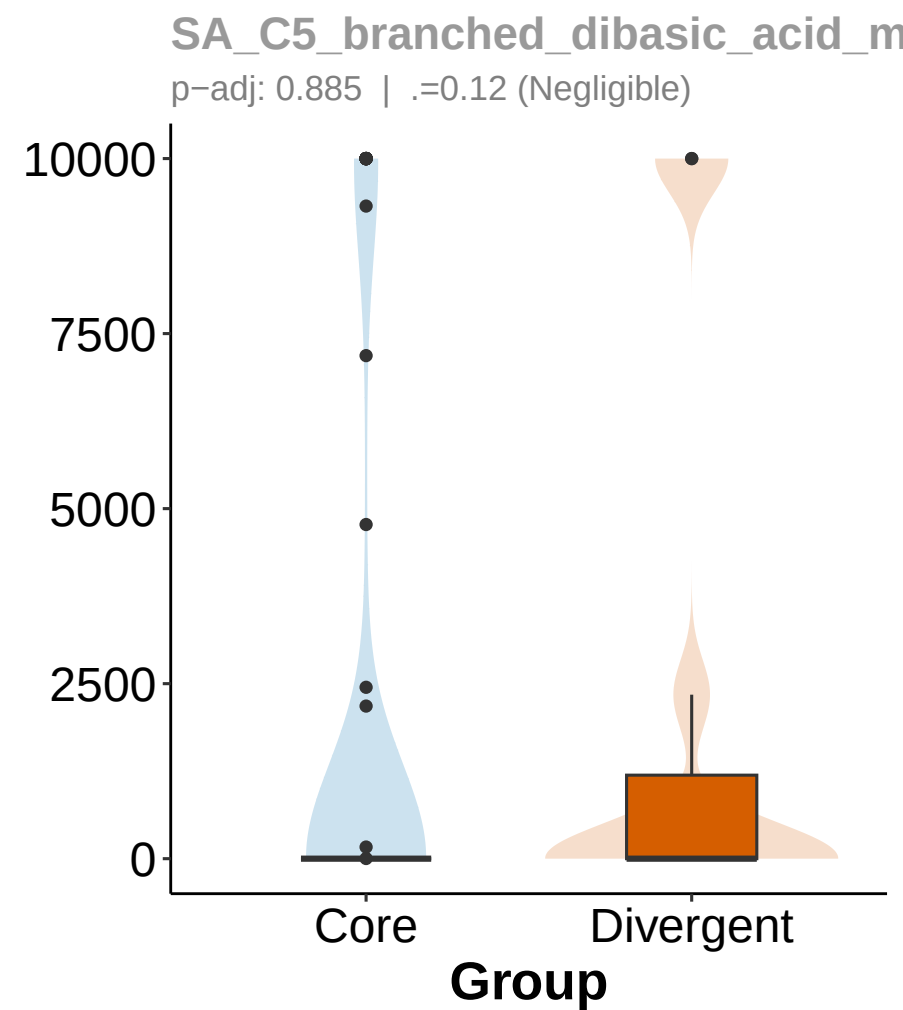
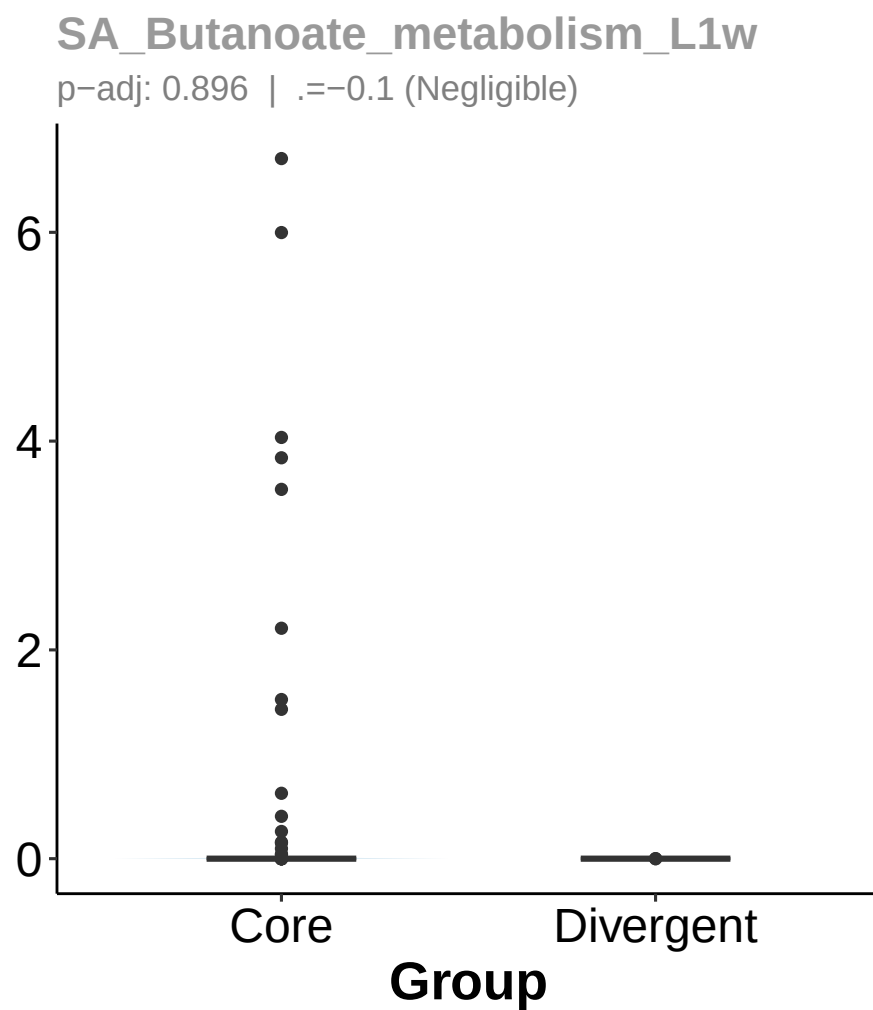
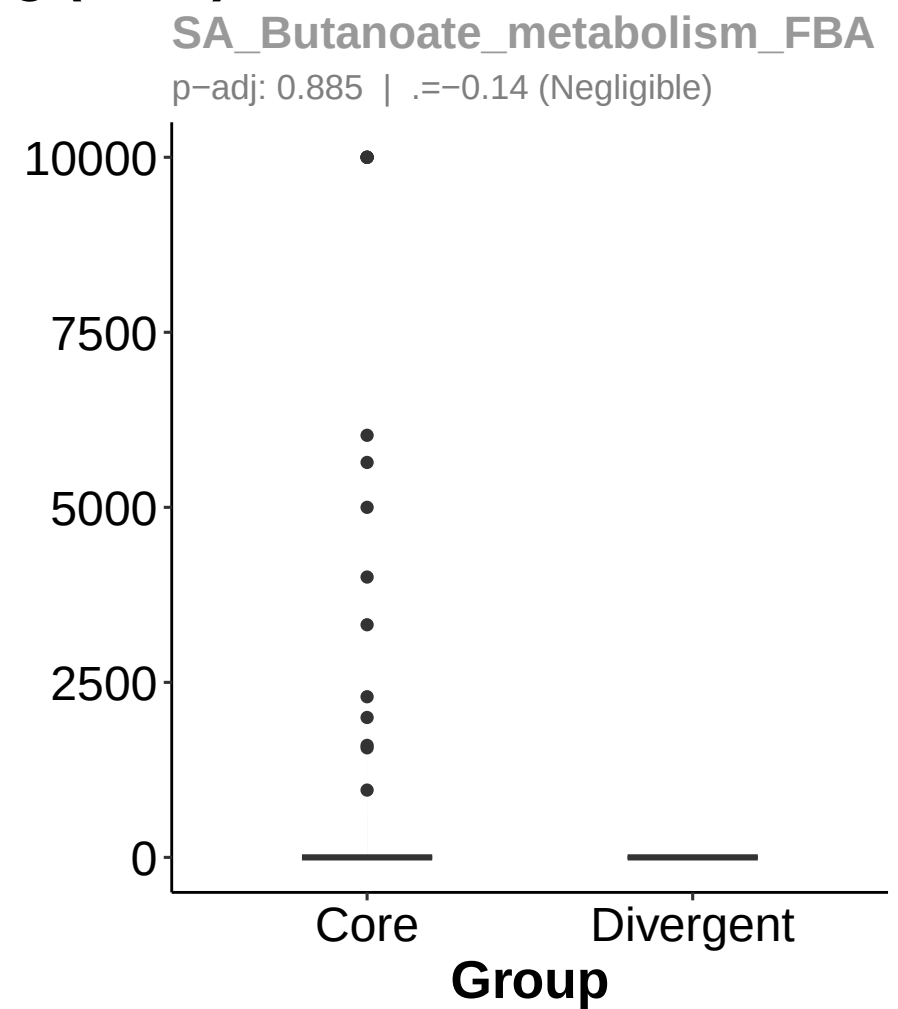
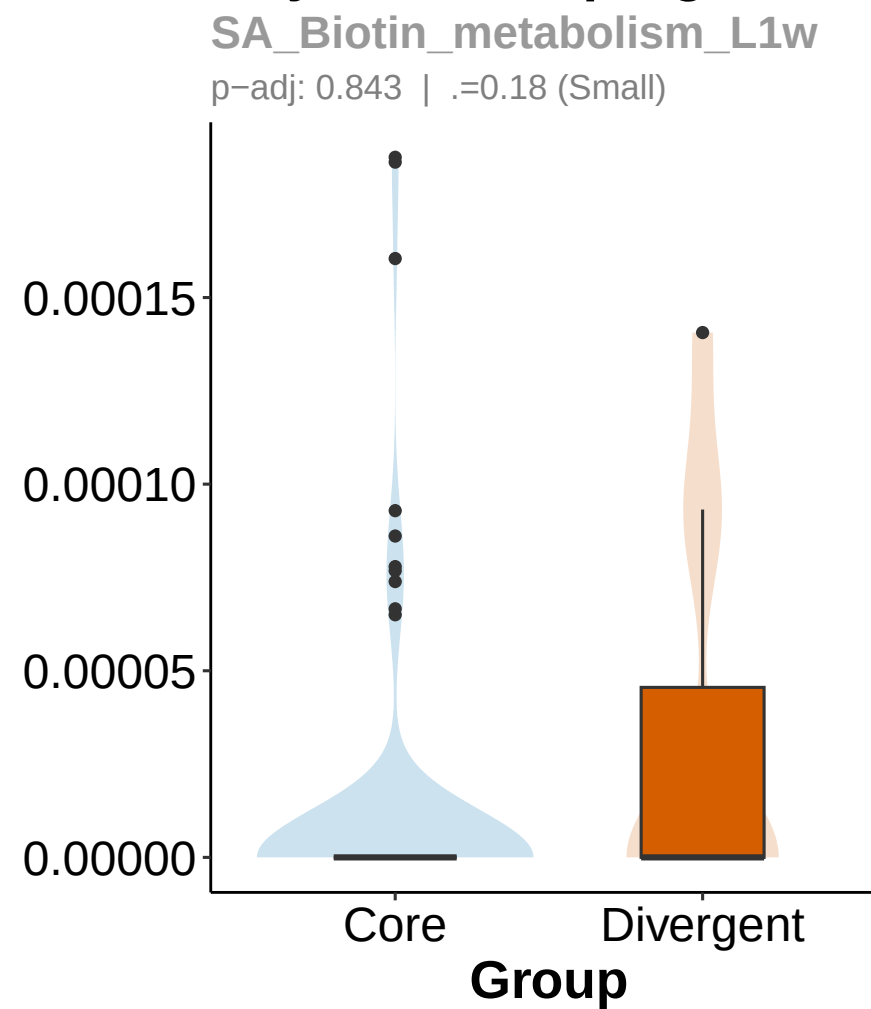
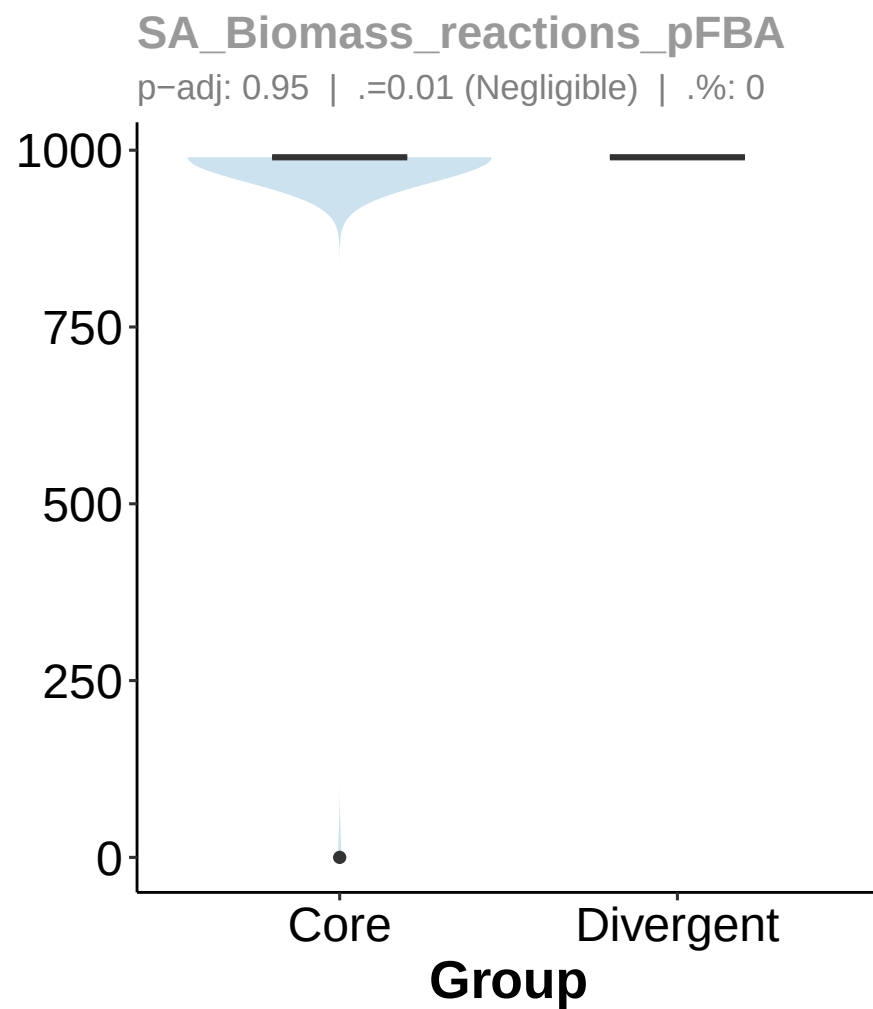


SA_Biomass_reactions_L1w

p-adj: 0.95 | .:=0.01 (Negligible) | .%: 0



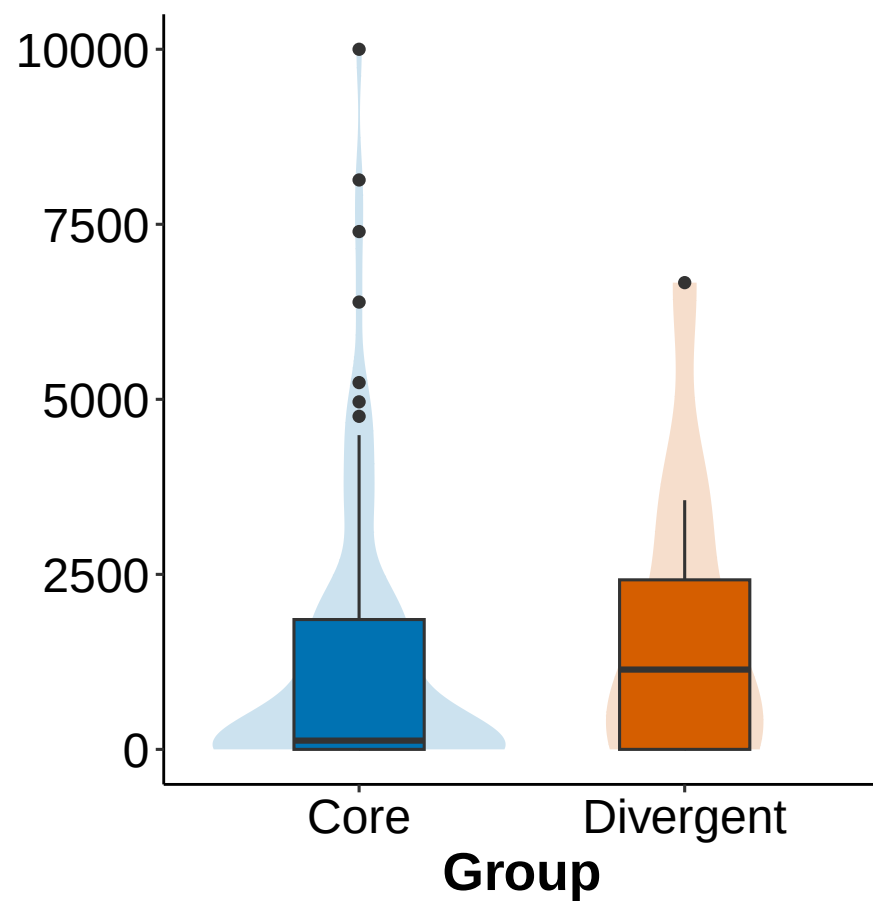
Metabolism – 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

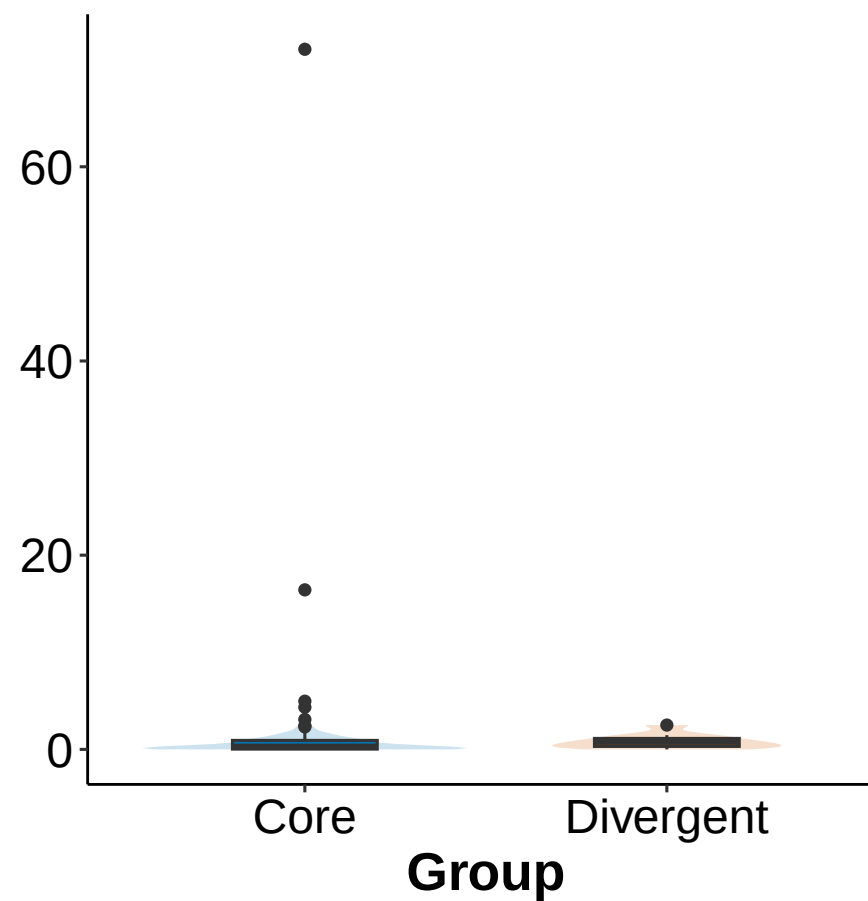
SA_Cholesterol_metabolism_FBA

p-adj: 0.933 | .=0.1 (Negligible) | .%: 808.5



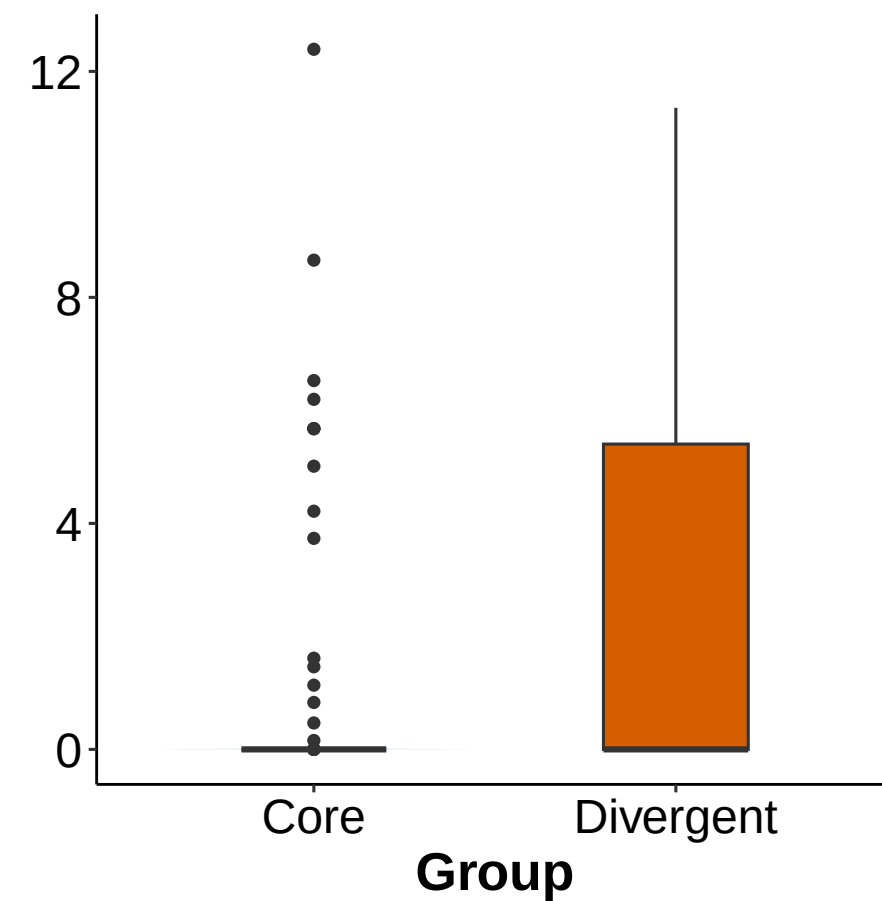
SA_Cholesterol_metabolism_L1w

p-adj: 0.885 | .=0.21 (Small) | .%: 188.7



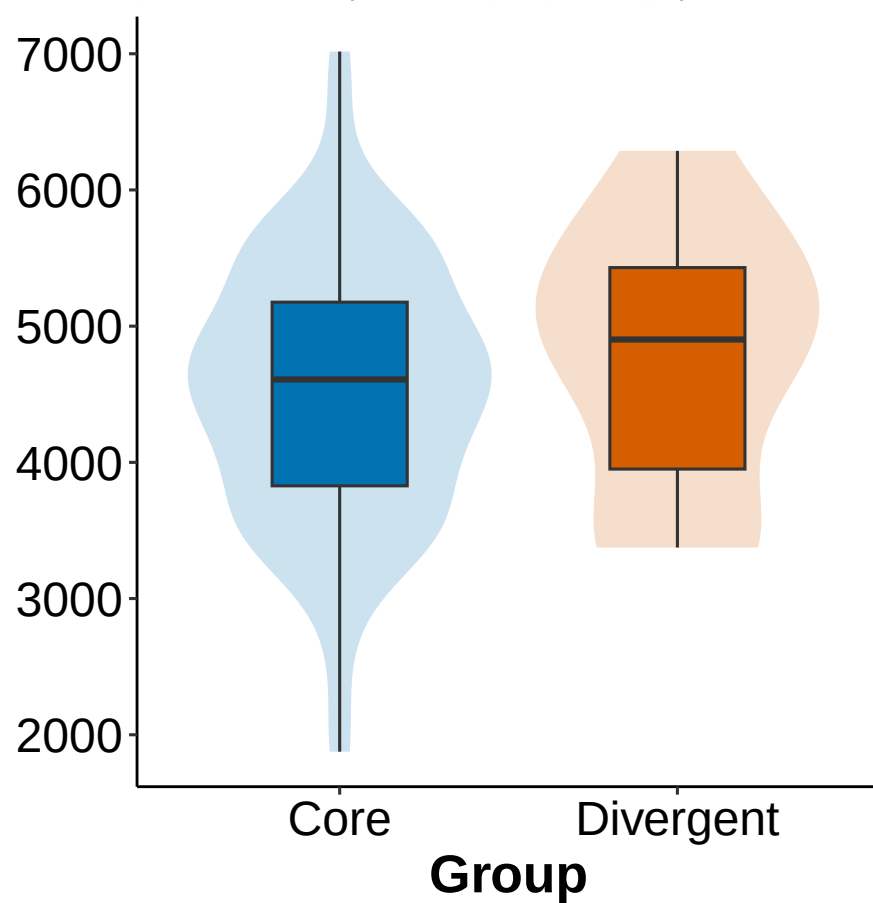
SA_Cholesterol_metabolism_pFBA

p-adj: 0.885 | .=0.15 (Small)



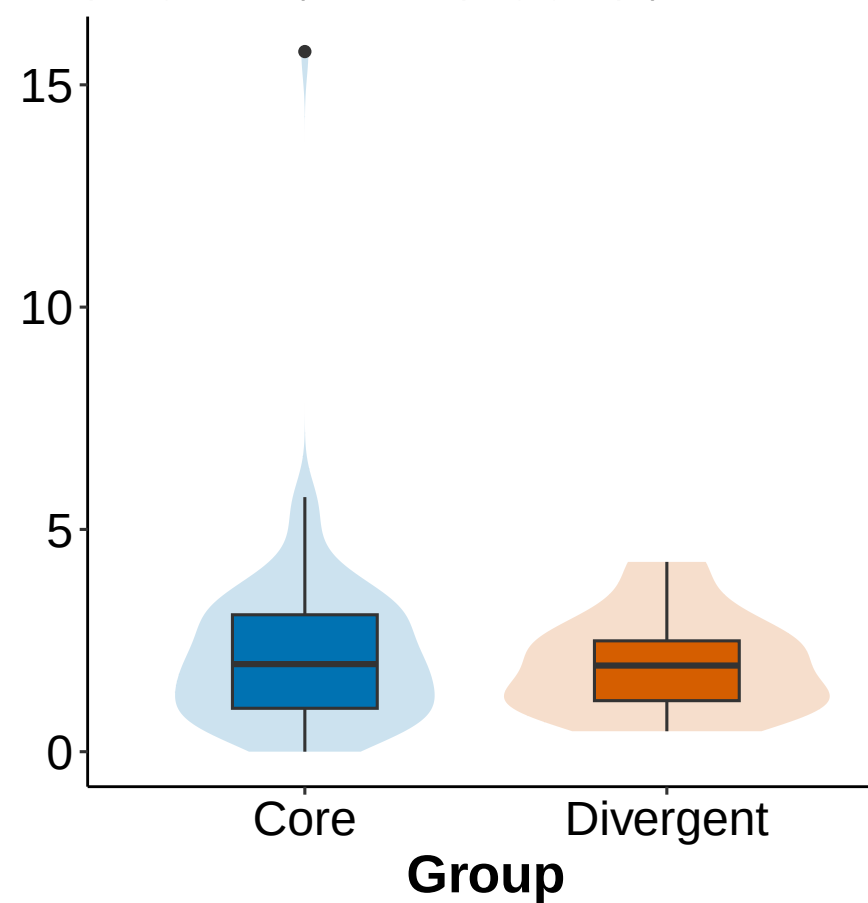
SA_Citric_acid_cycle_FBA

p-adj: 0.896 | .=0.14 (Negligible) | .%: 6.4



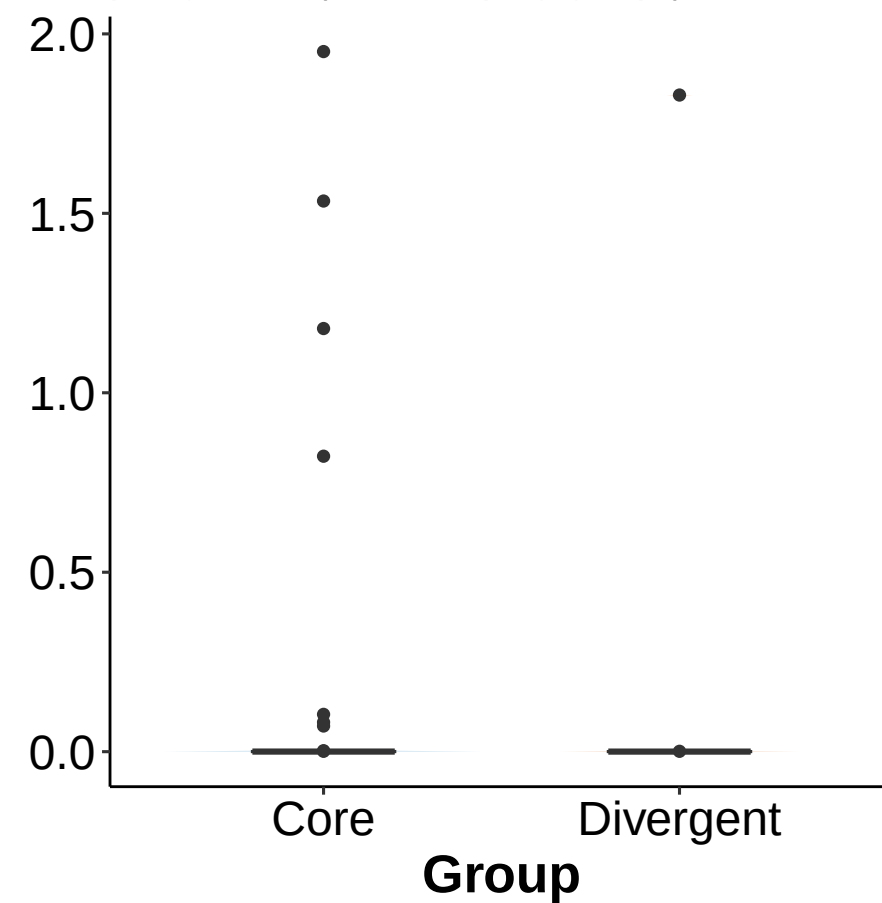
SA_Citric_acid_cycle_L1w

p-adj: 0.954 | .=-0.03 (Negligible) | .%: -1.7



SA_Citric_acid_cycle_pFBA

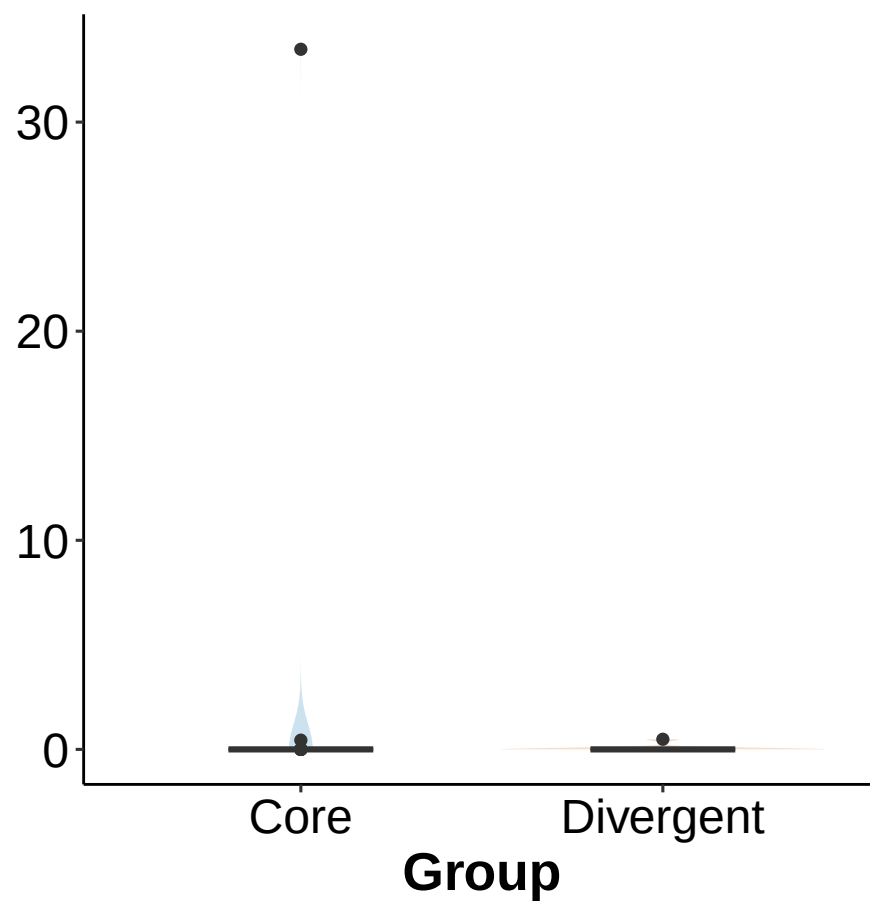
p-adj: 0.945 | .=-0.09 (Negligible) | .%: -24.4



Metabolism – . 5. Systemic Reprogramming (SA_*)

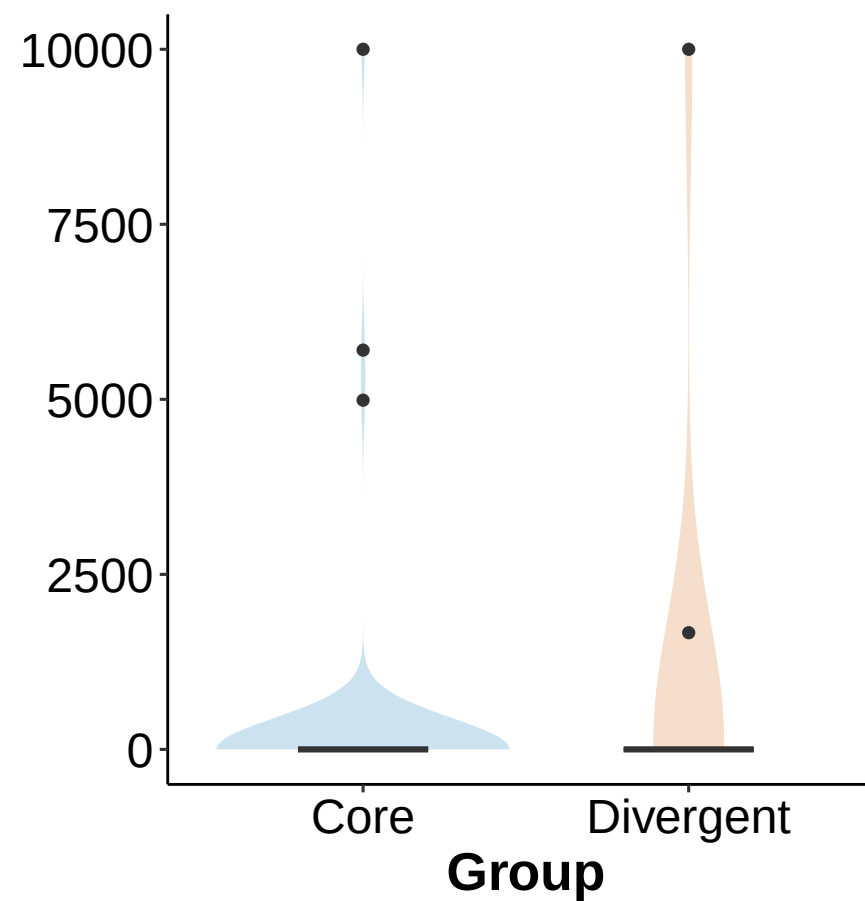
SA_CoA_catabolism_L1w

p-adj: 0.95 | . = 0.02 (Negligible)



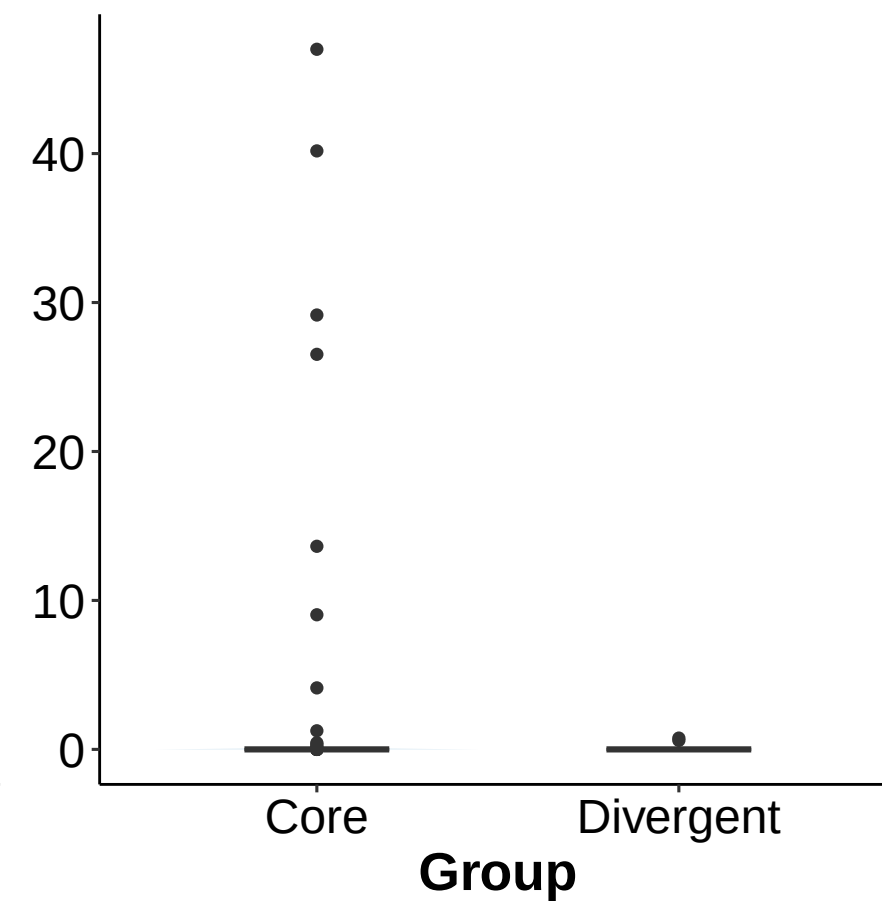
SA_CoA_synthesis_FBA

p-adj: 0.843 | . = 0.15 (Small)



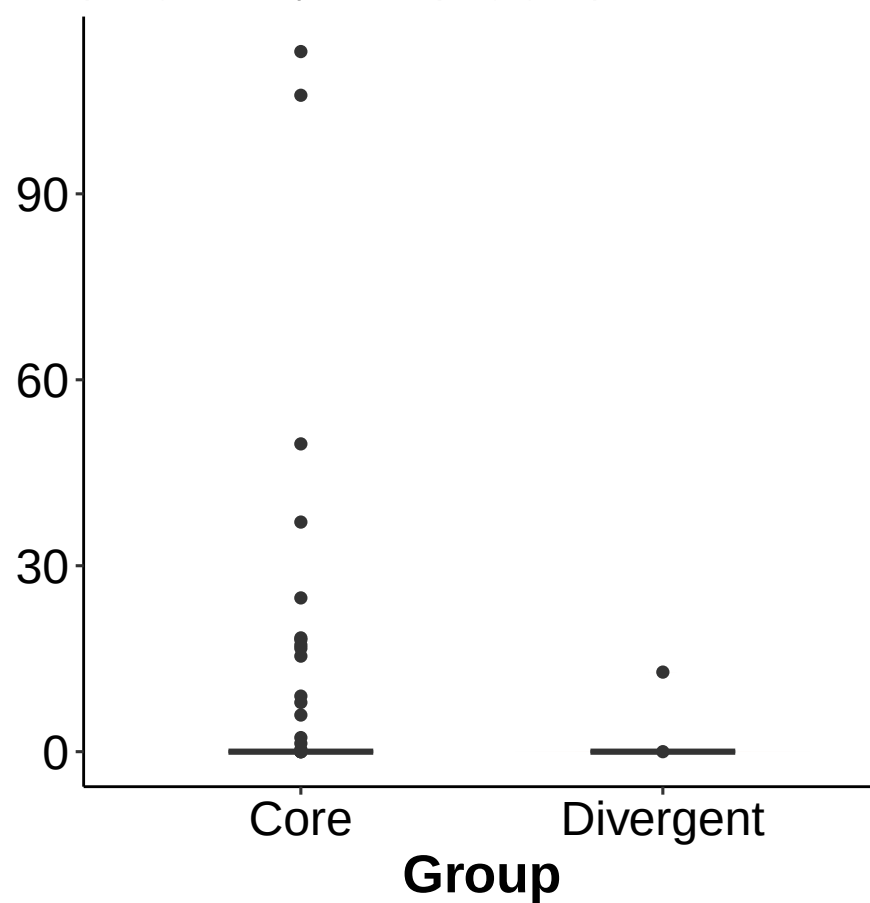
SA_CoA_synthesis_L1w

p-adj: 0.843 | . = 0.3 (Small)



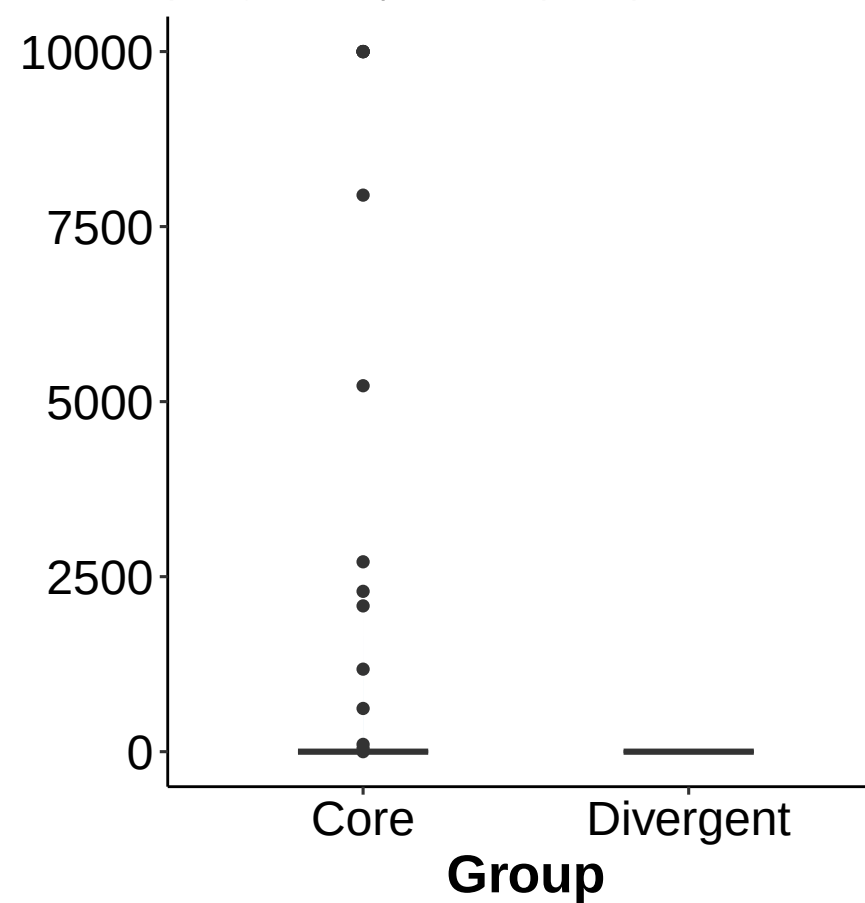
SA_Cytochrome_metabolism_L1w

p-adj: 0.954 | . = 0.02 (Negligible)



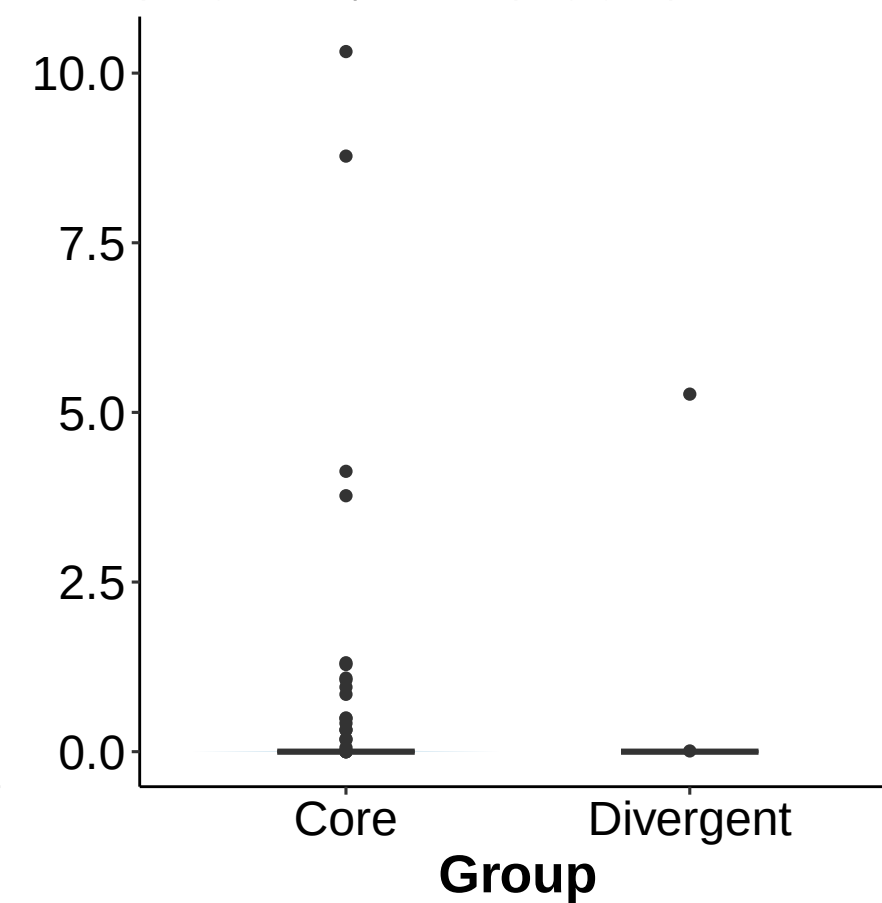
SA_D_alanine_metabolism_FBA

p-adj: 0.885 | . = -0.15 (Small)

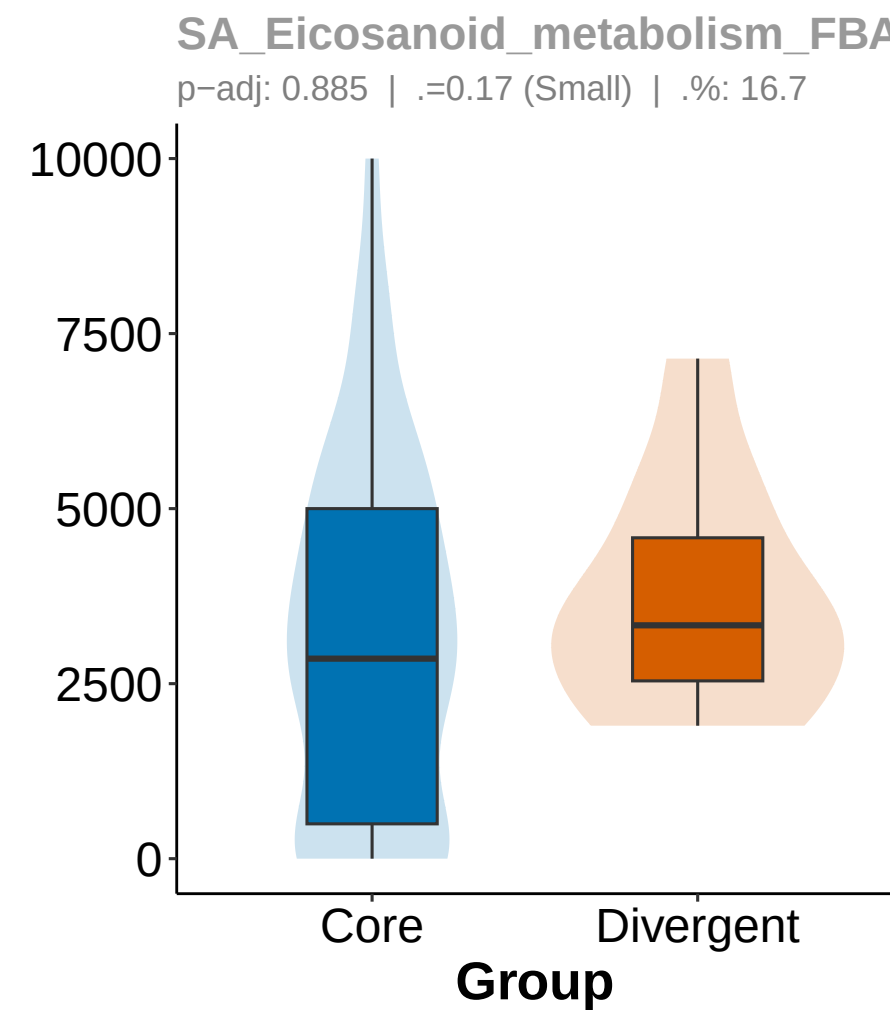
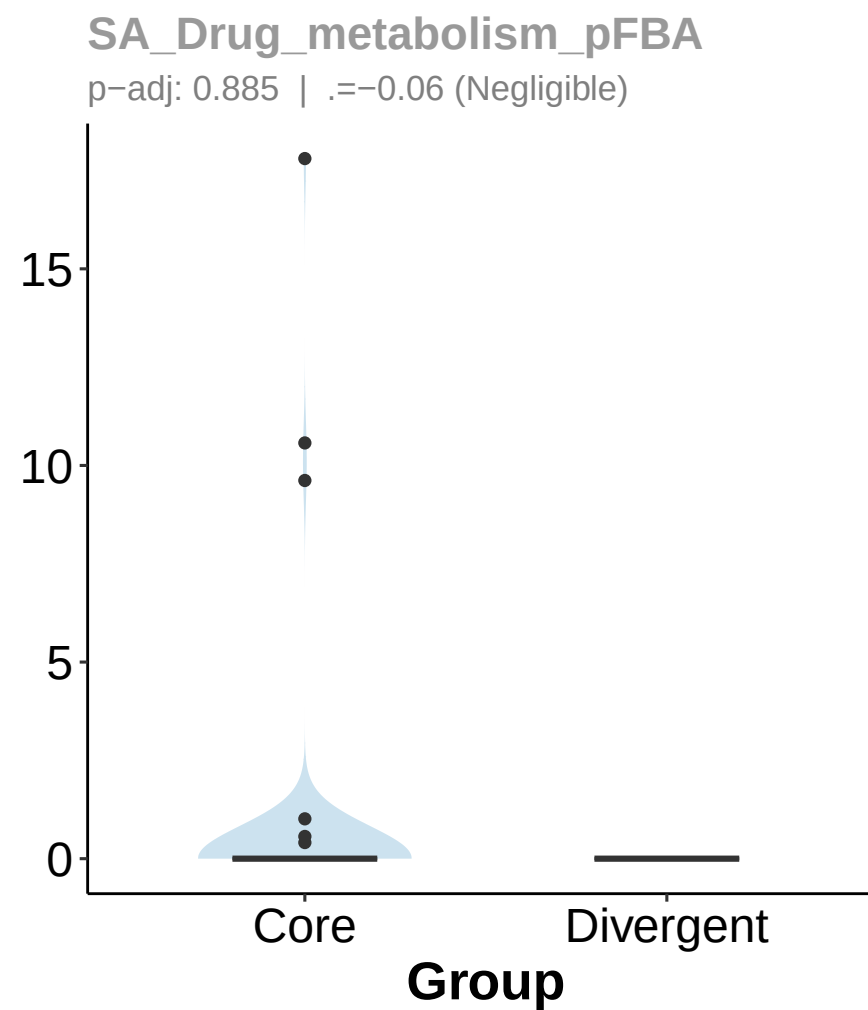
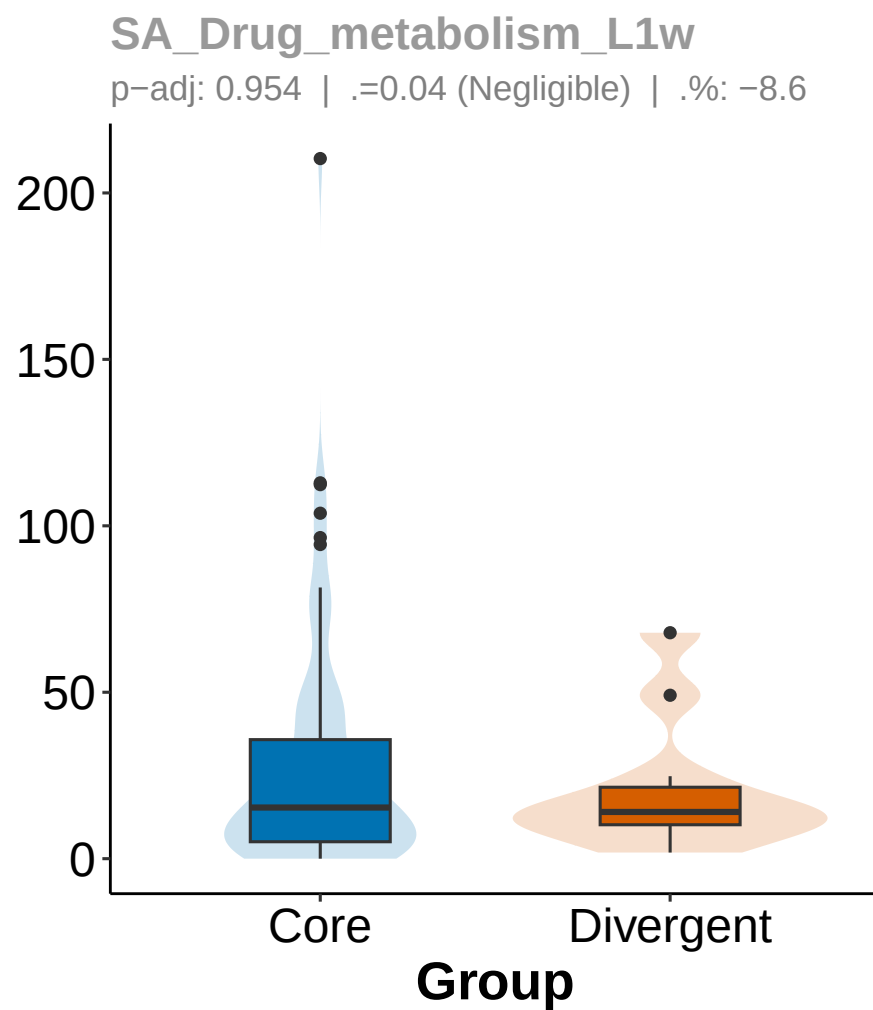
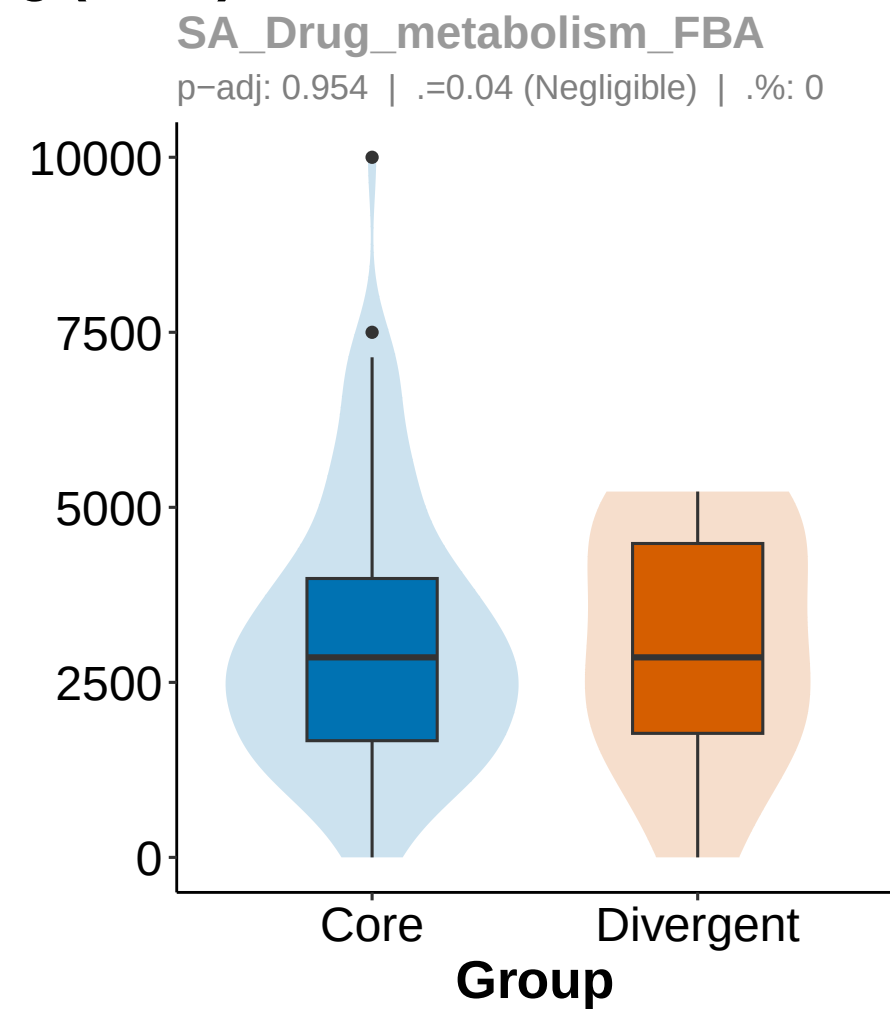
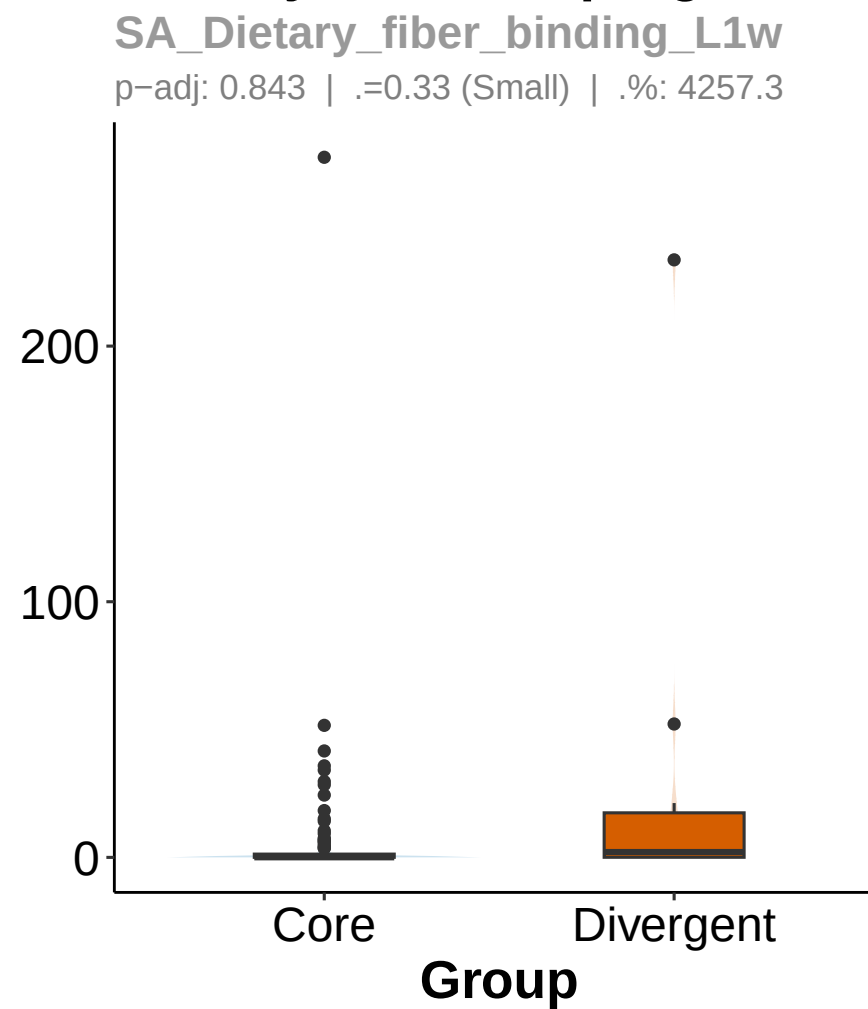
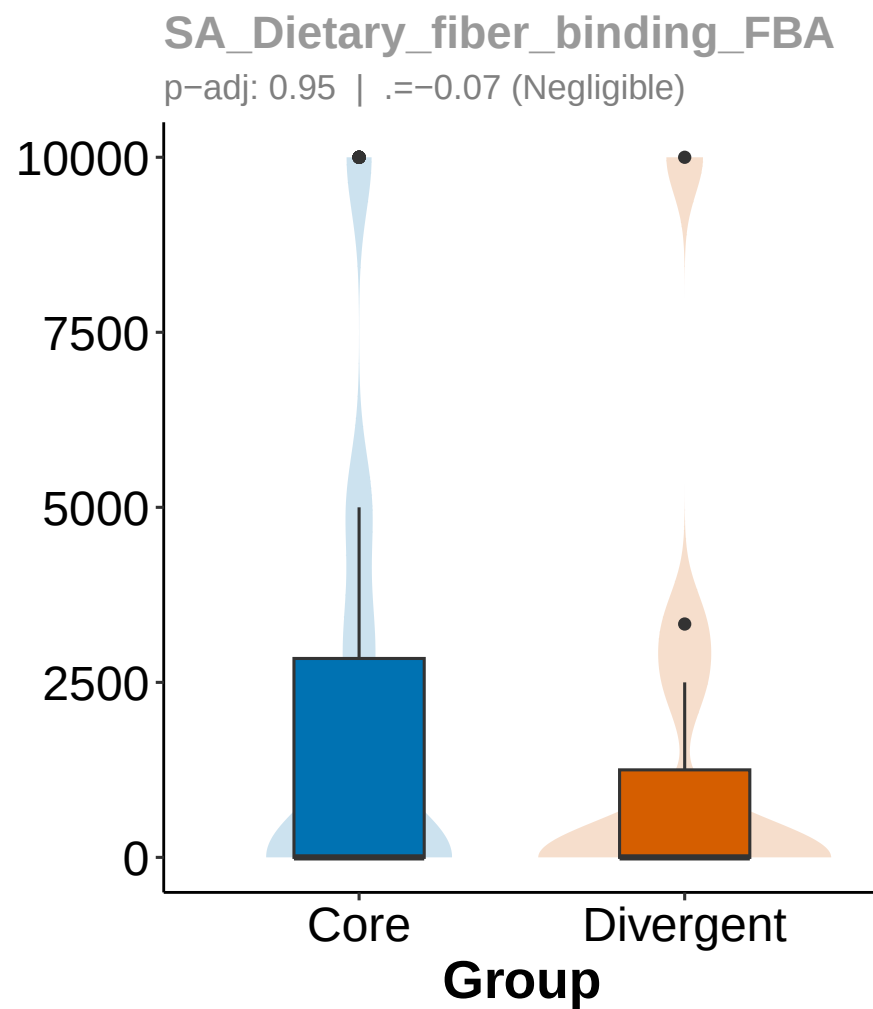


SA_D_alanine_metabolism_L1w

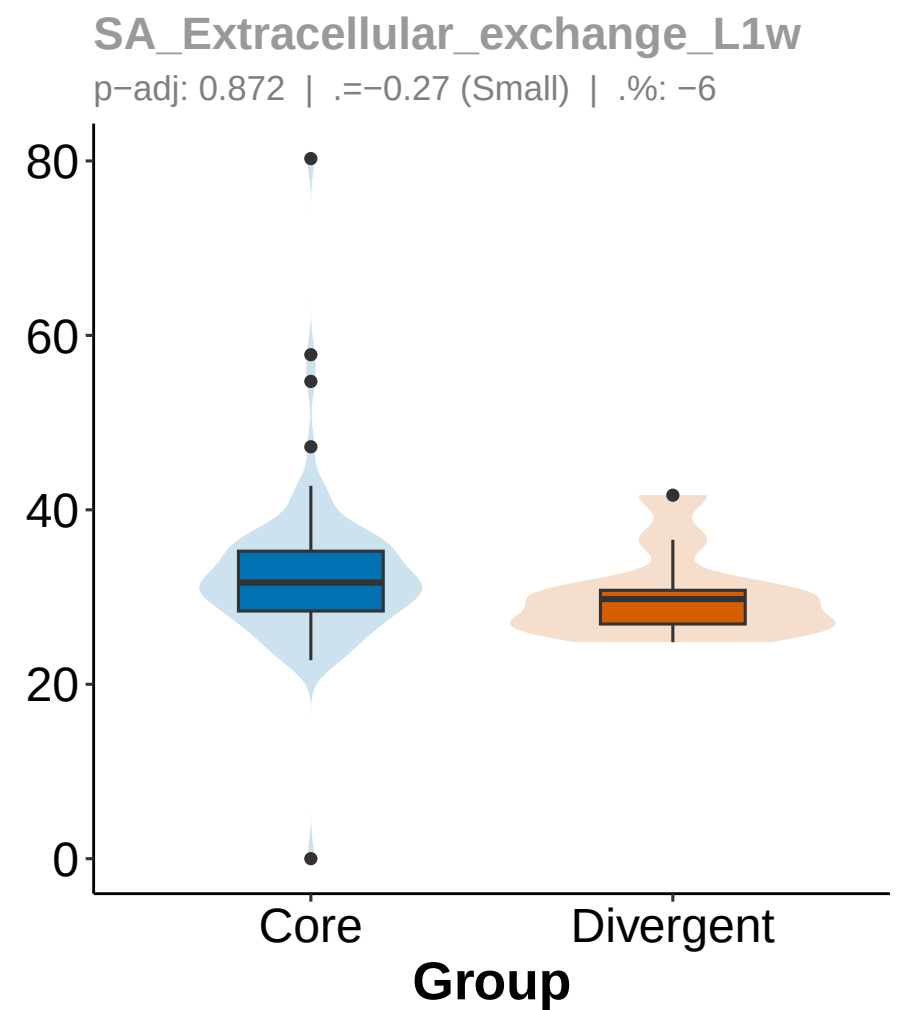
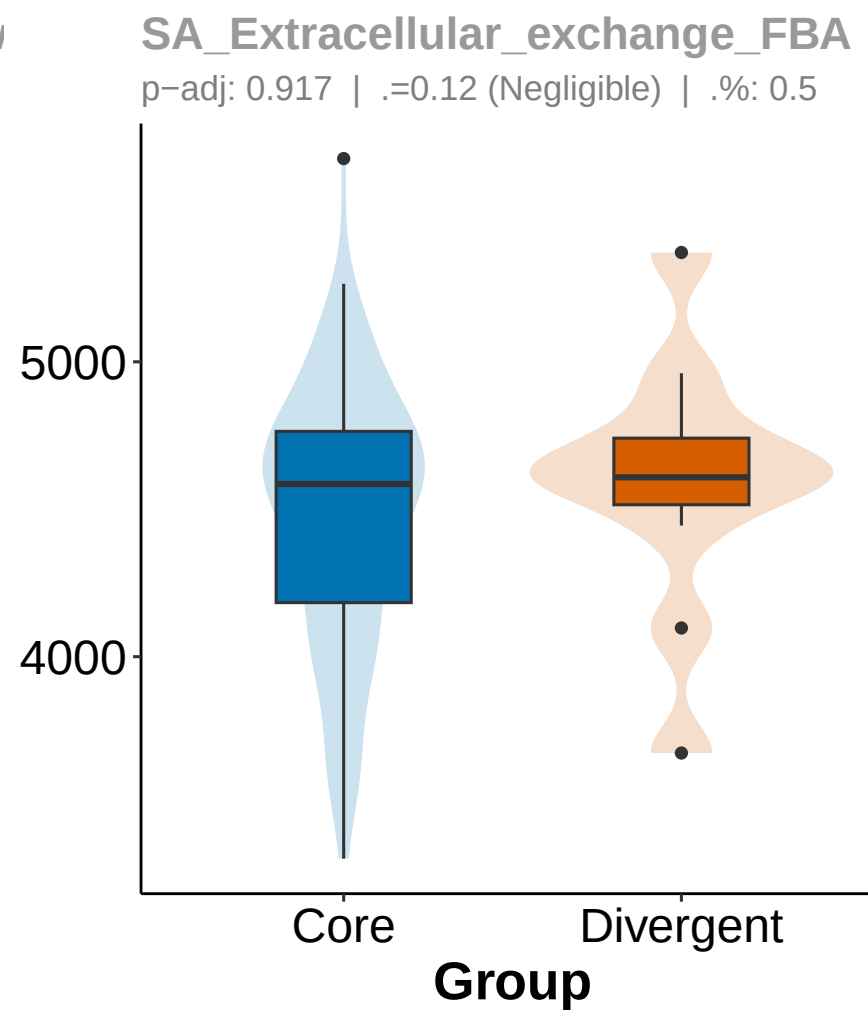
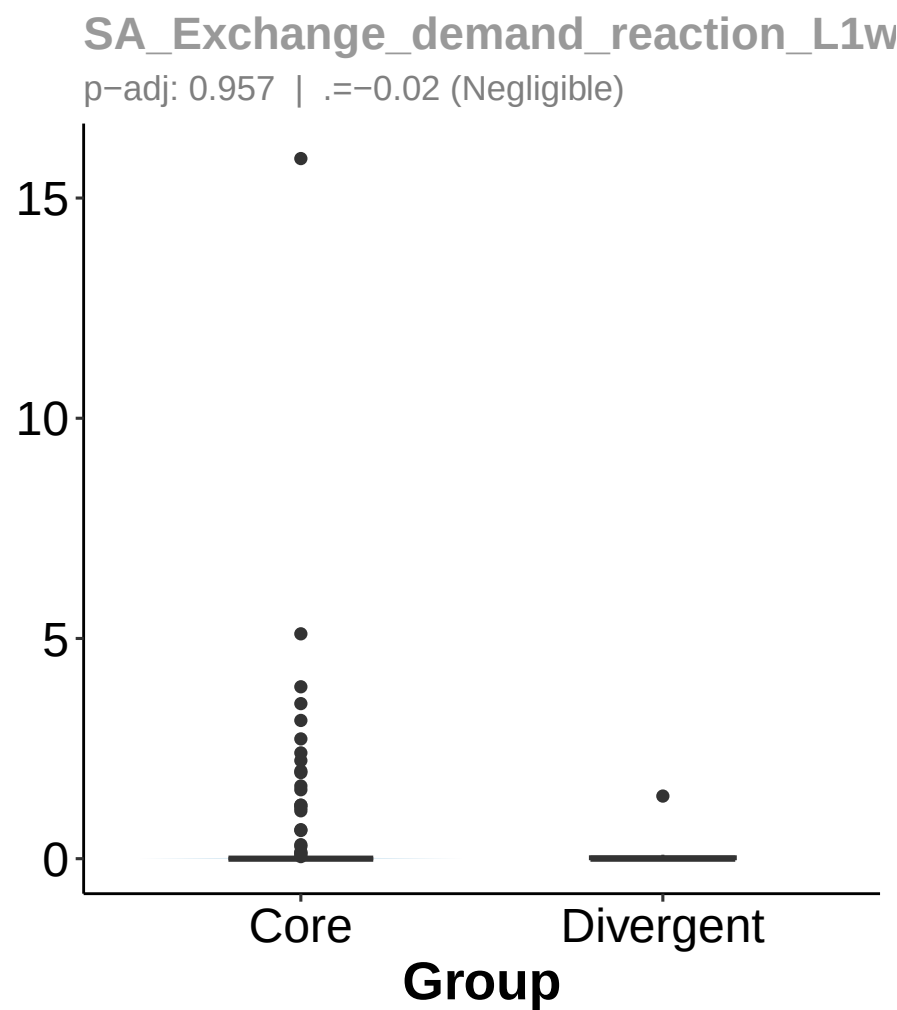
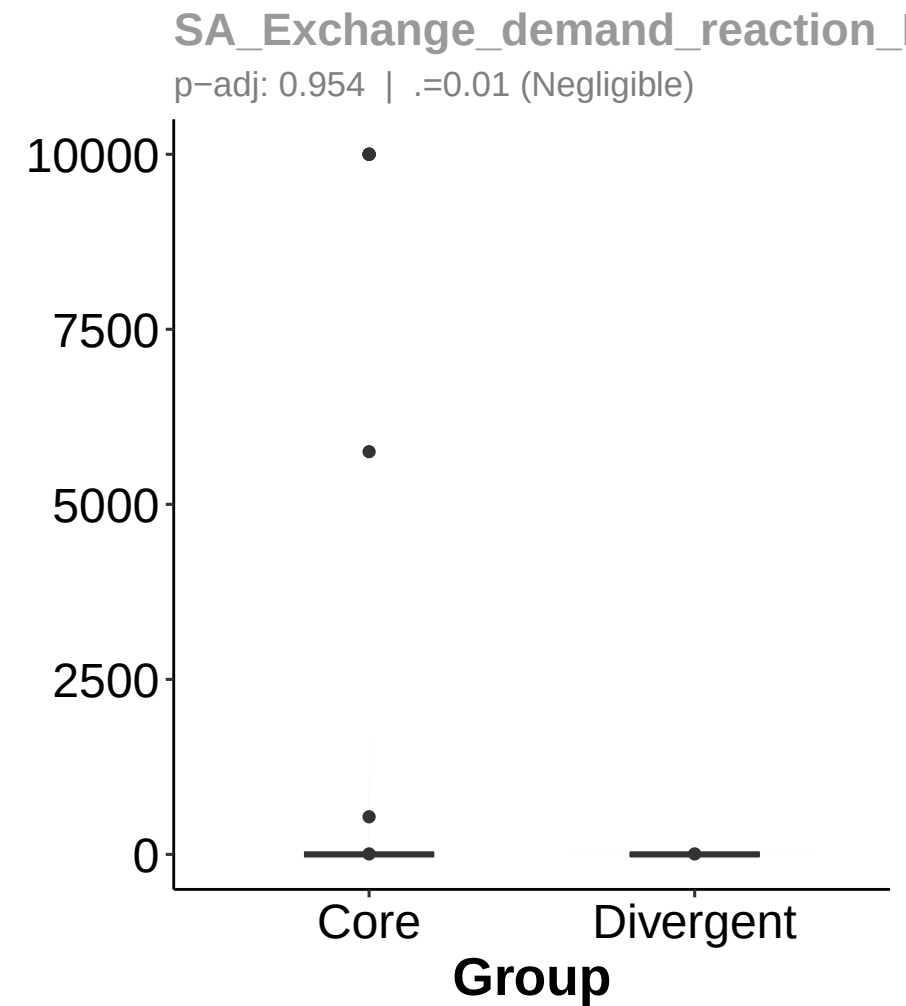
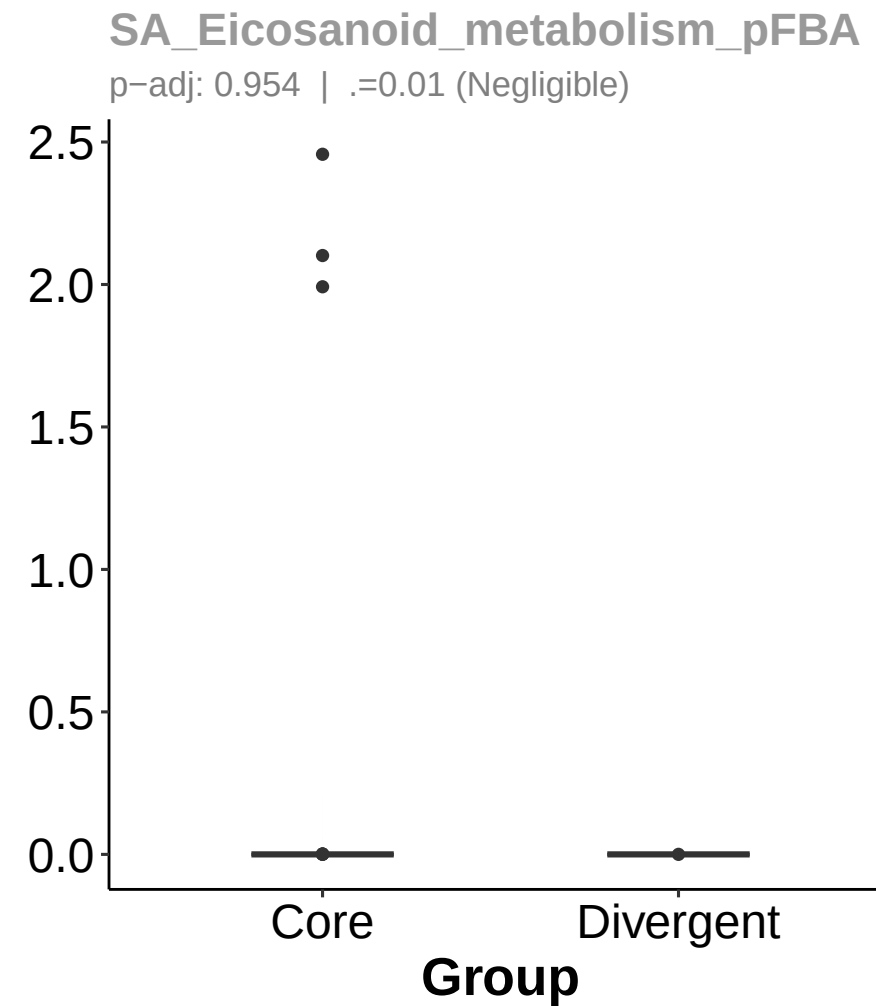
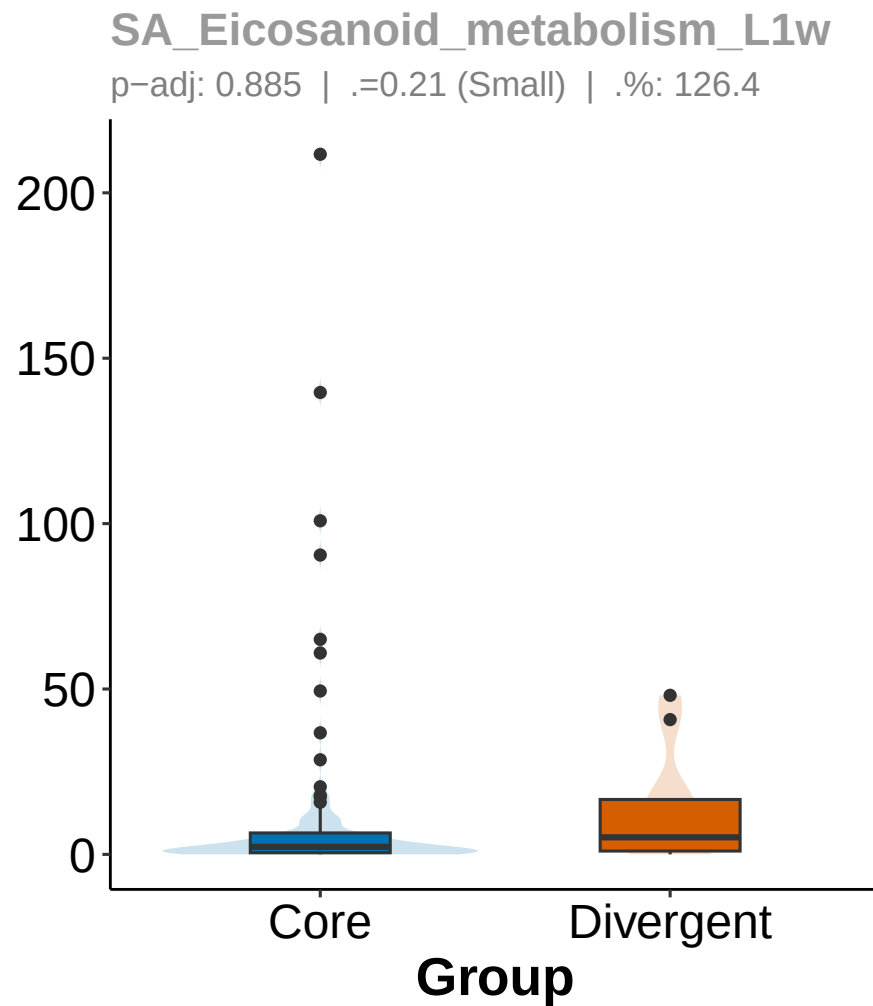
p-adj: 0.896 | . = -0.11 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)



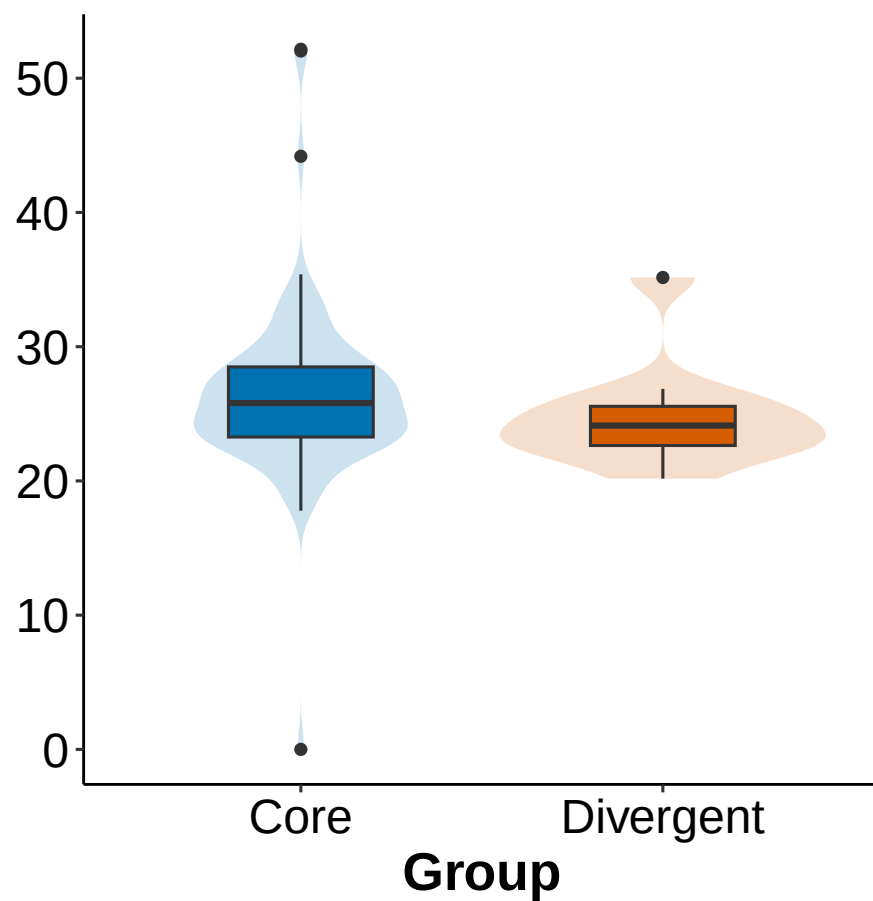
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

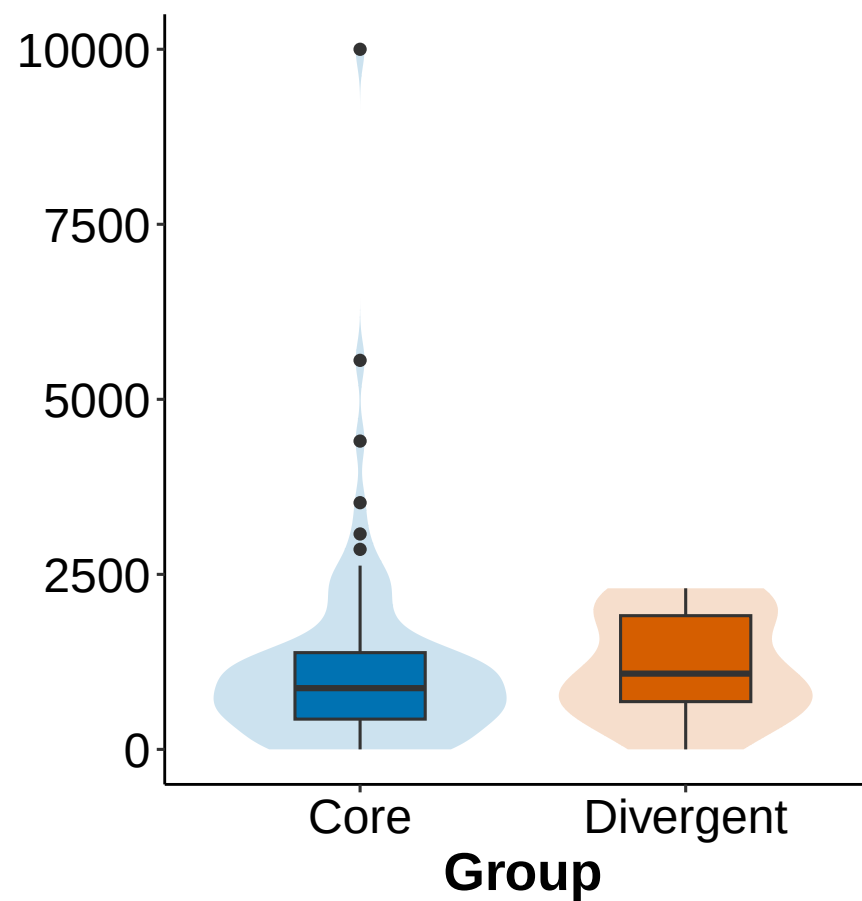
SA_Extracellular_exchange_pFBA

p-adj: 0.872 | .=-0.27 (Small) | .%: -6.5



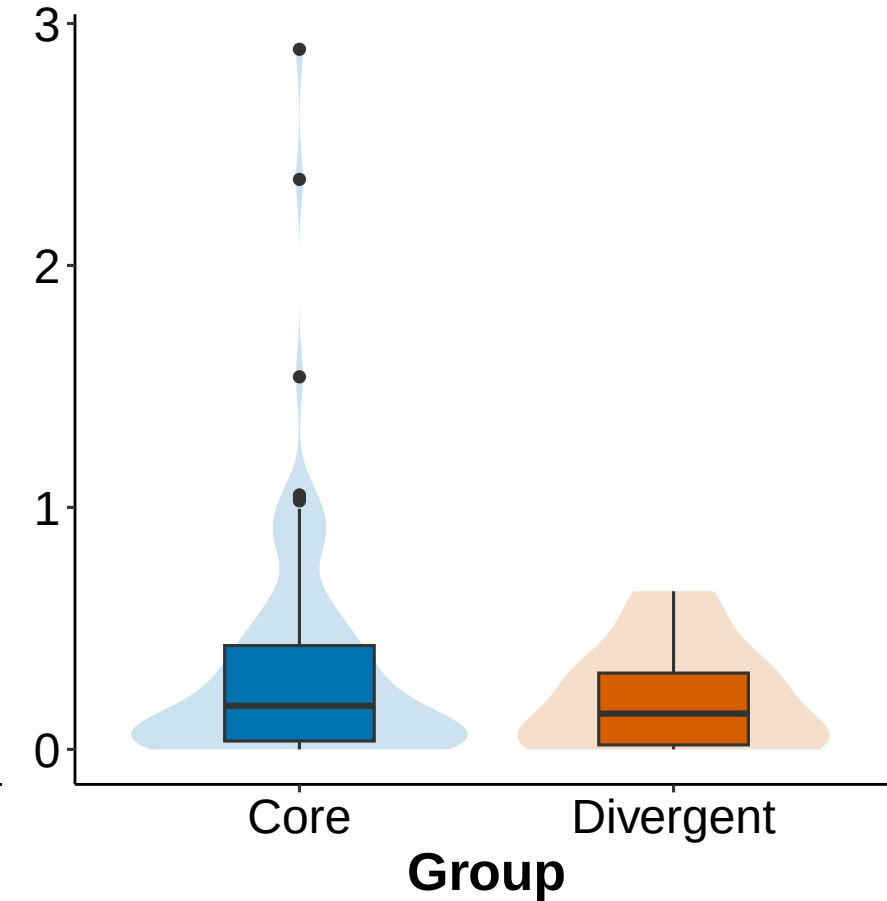
SA_Fatty_acid_oxidation_FBA

p-adj: 0.896 | .=-0.14 (Negligible) | .%: 23.7



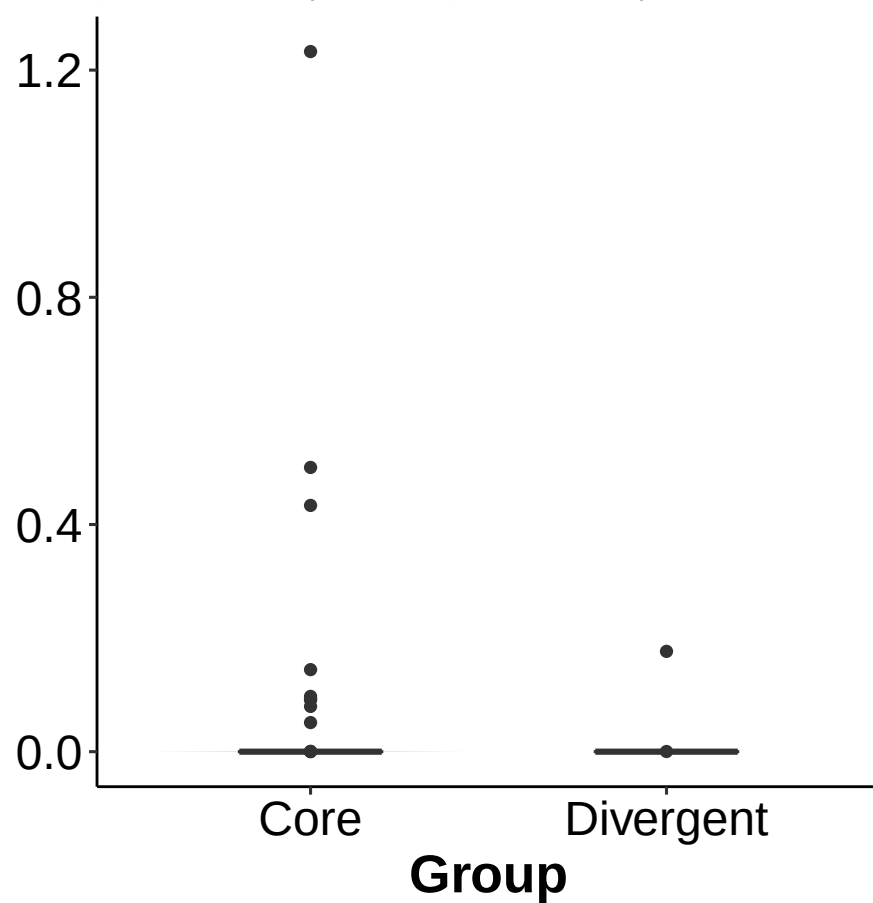
SA_Fatty_acid_oxidation_L1w

p-adj: 0.928 | .=-0.1 (Negligible) | .%: -17.8



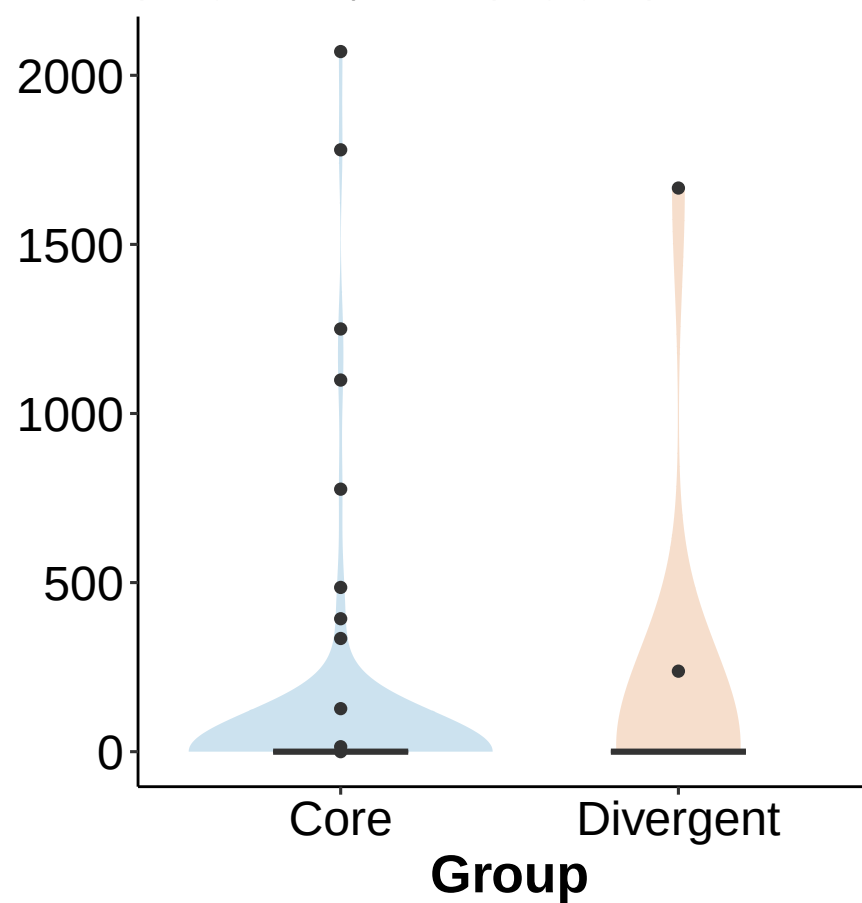
SA_Fatty_acid_oxidation_pFBA

p-adj: 0.896 | .=-0.13 (Negligible) | .%: 126



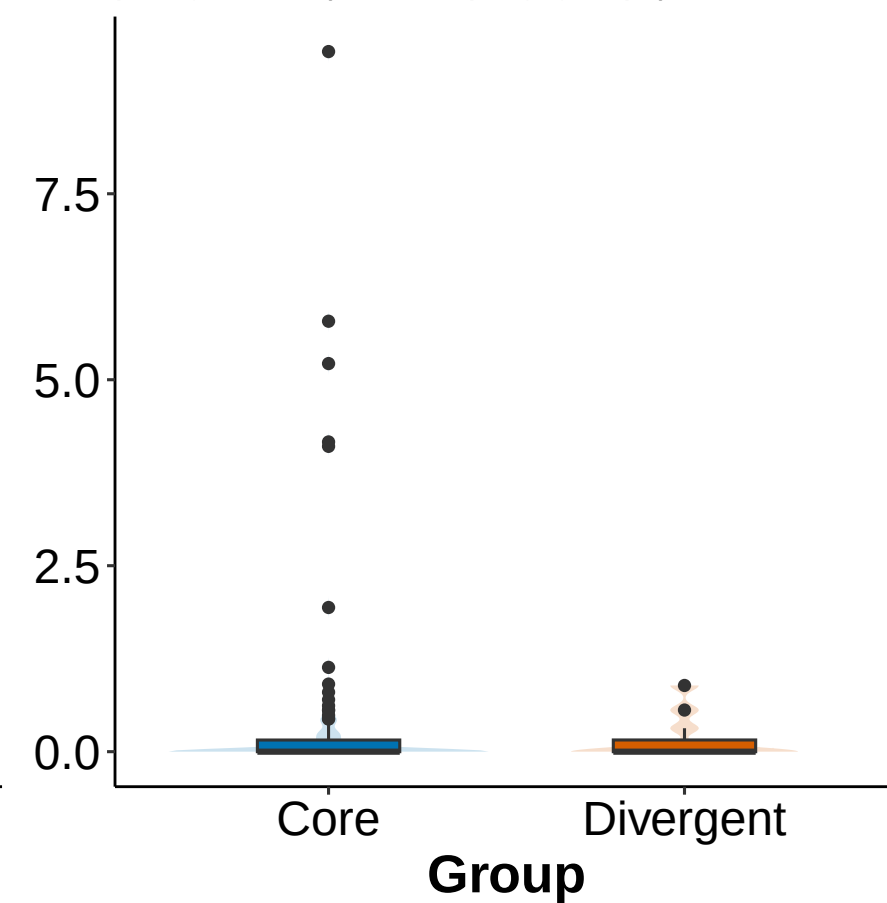
SA_Fatty_acid_synthesis_FBA

p-adj: 0.917 | .=-0.07 (Negligible)



SA_Fatty_acid_synthesis_L1w

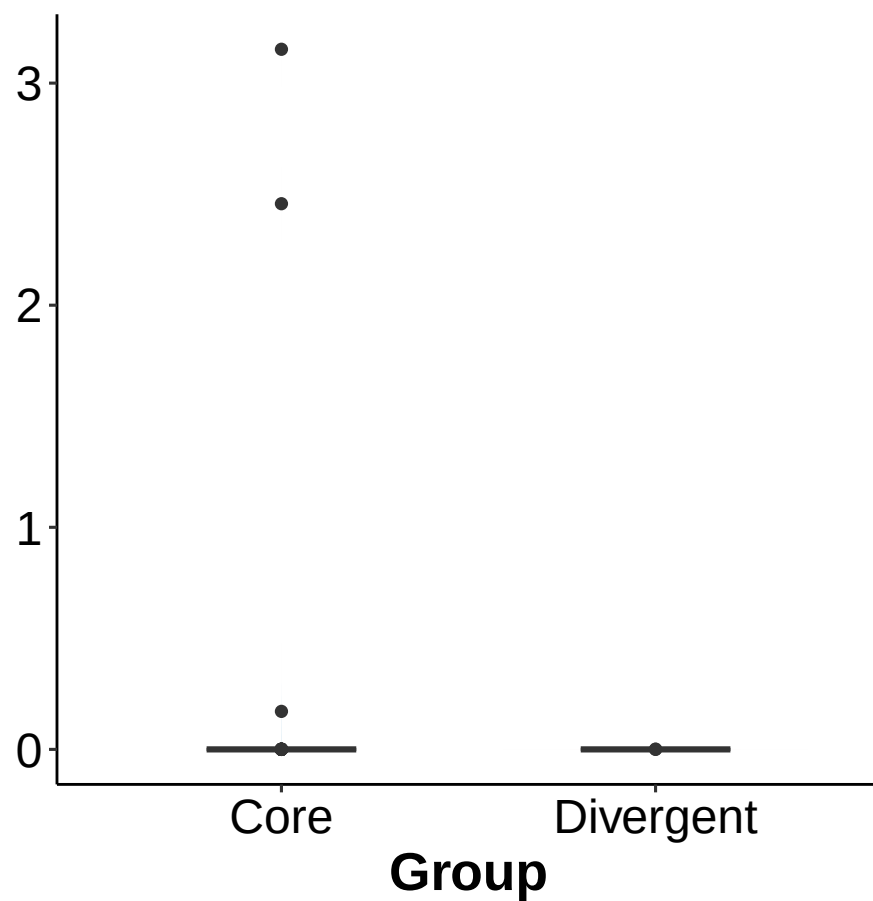
p-adj: 0.942 | .=-0.1 (Negligible) | .%: -78.5



Metabolism – . 5. Systemic Reprogramming (SA_*)

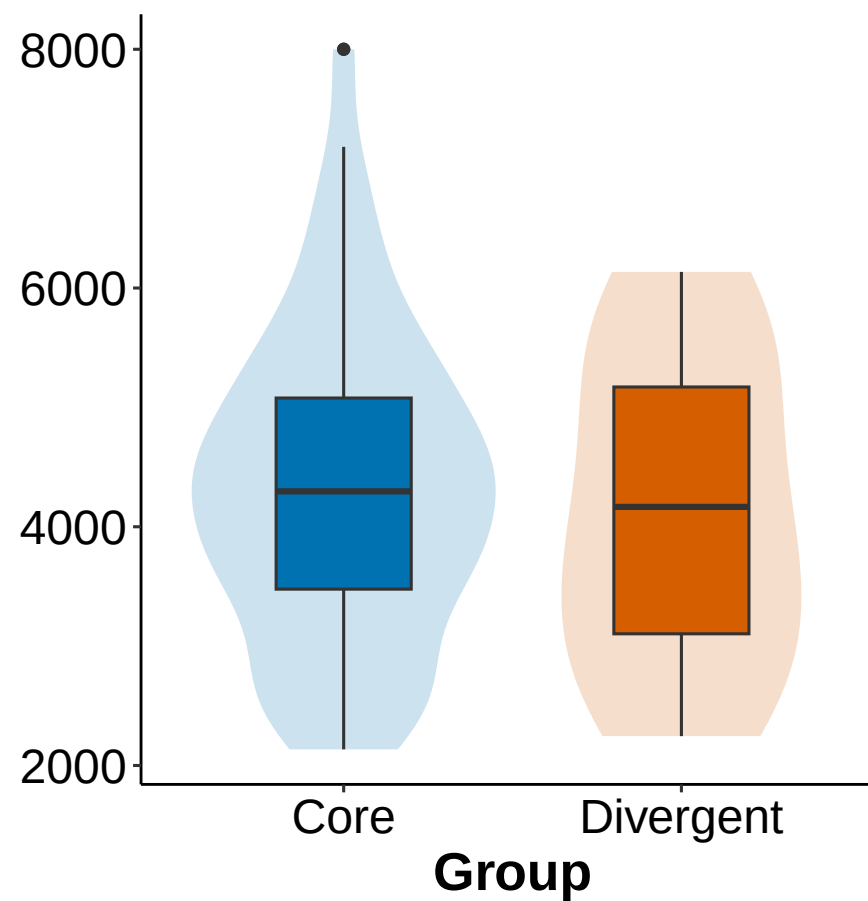
SA_Fatty_acid_synthesis_pFBA

p-adj: 0.95 | .:=0.04 (Negligible)



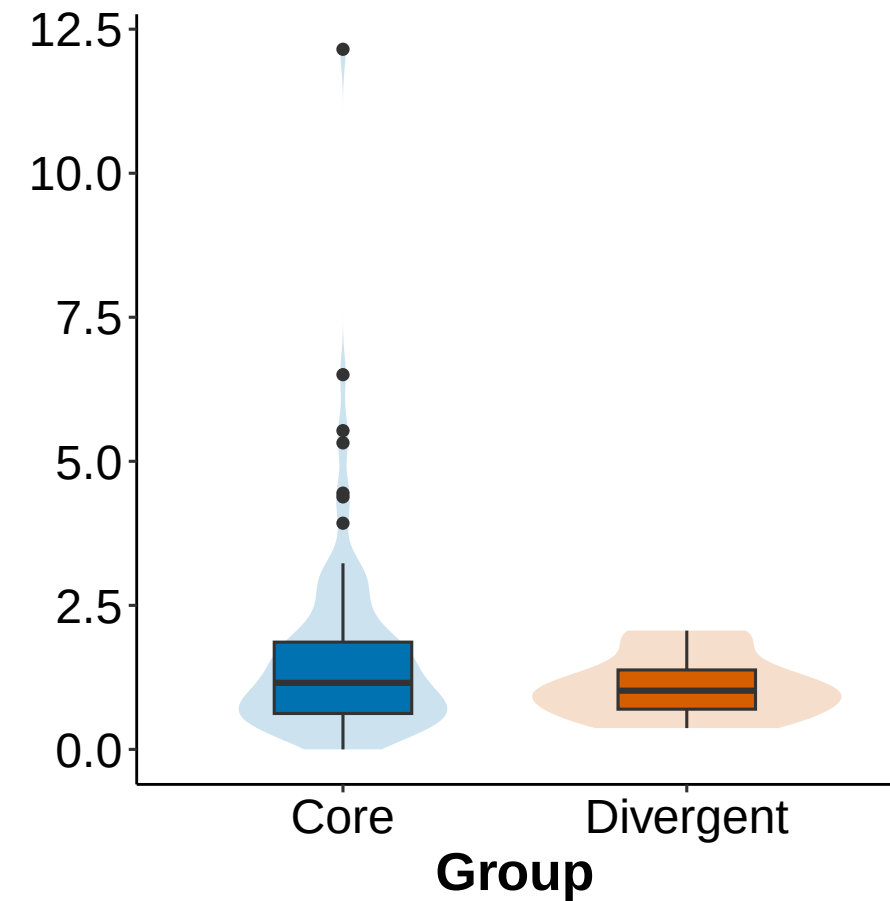
SA_Folate_metabolism_FBA

p-adj: 0.95 | .:=−0.09 (Negligible) | .%: −3



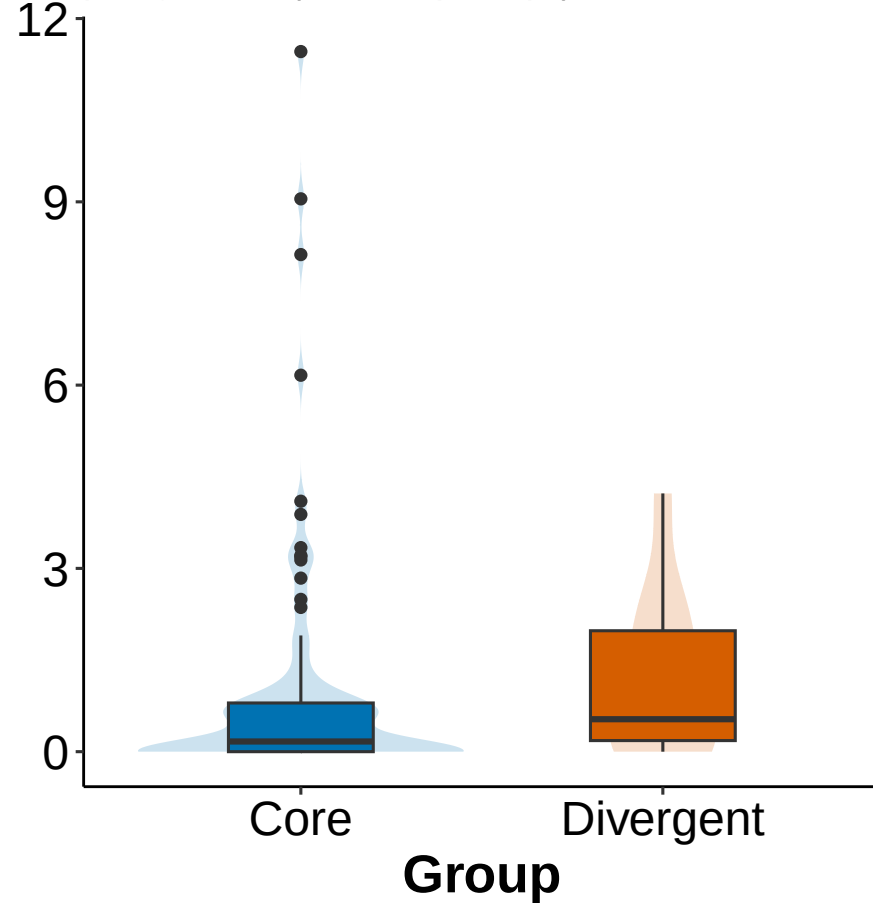
SA_Folate_metabolism_L1w

p-adj: 0.917 | .:=−0.11 (Negligible) | .%: −11.8



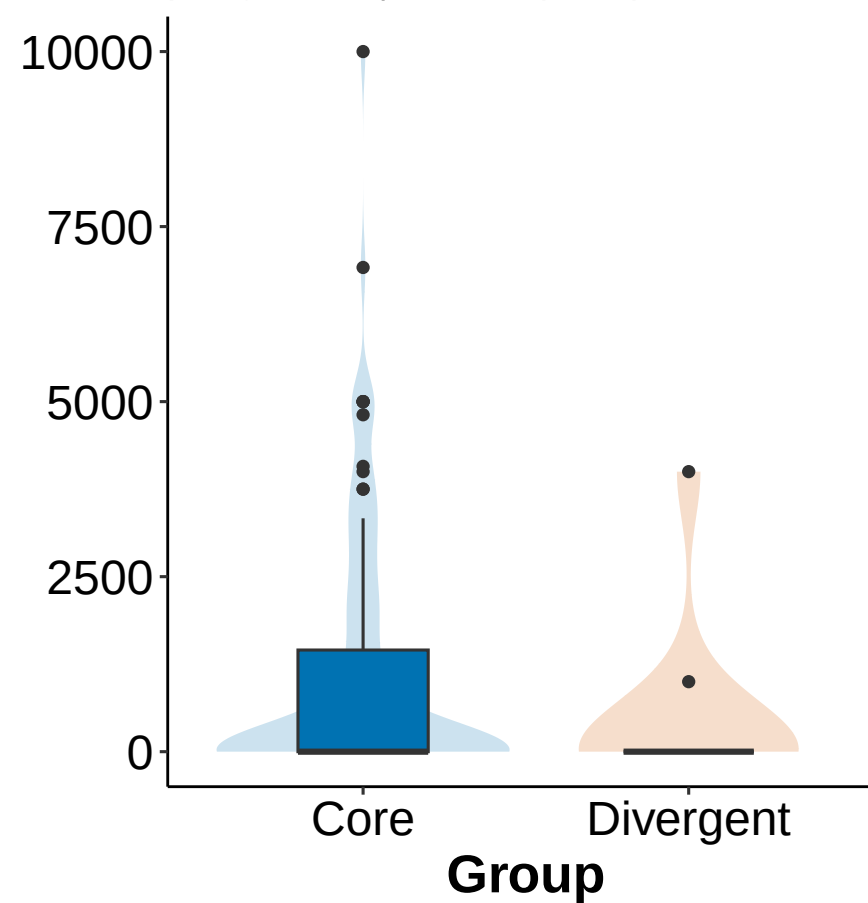
SA_Folate_metabolism_pFBA

p-adj: 0.885 | .:=0.23 (Small) | .%: 218.7



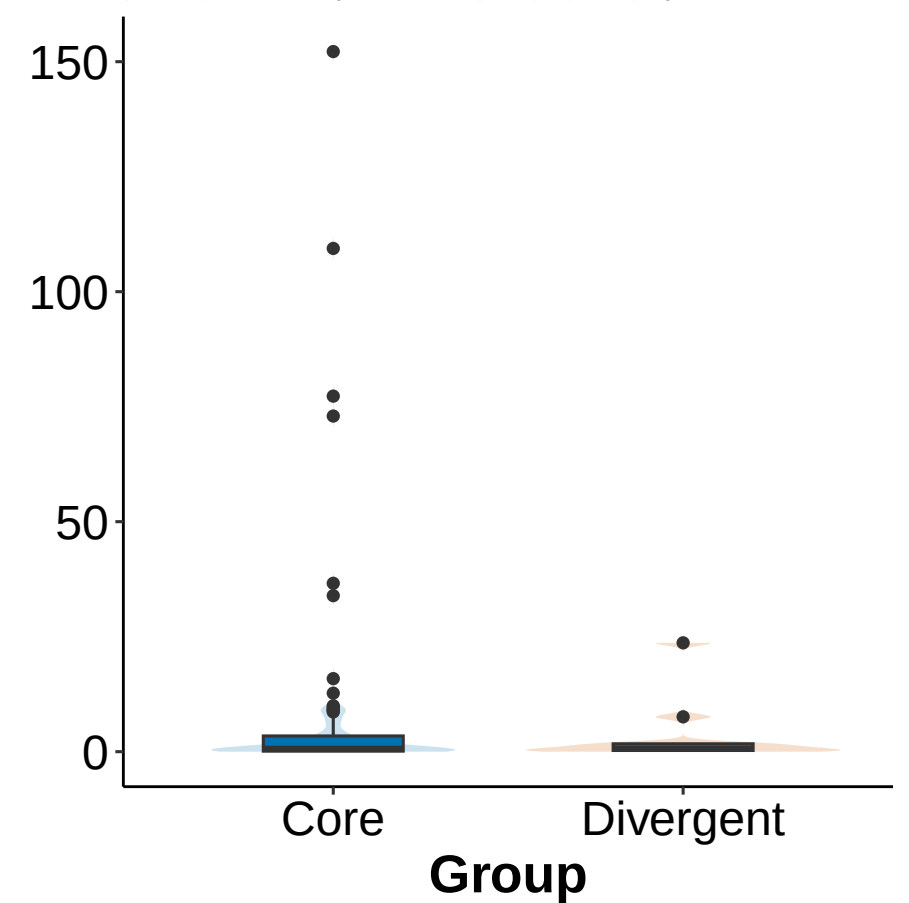
SA_Fructose_and_mannose_metabo

p-adj: 0.885 | .:=−0.22 (Small)



SA_Fructose_and_mannose_metabo

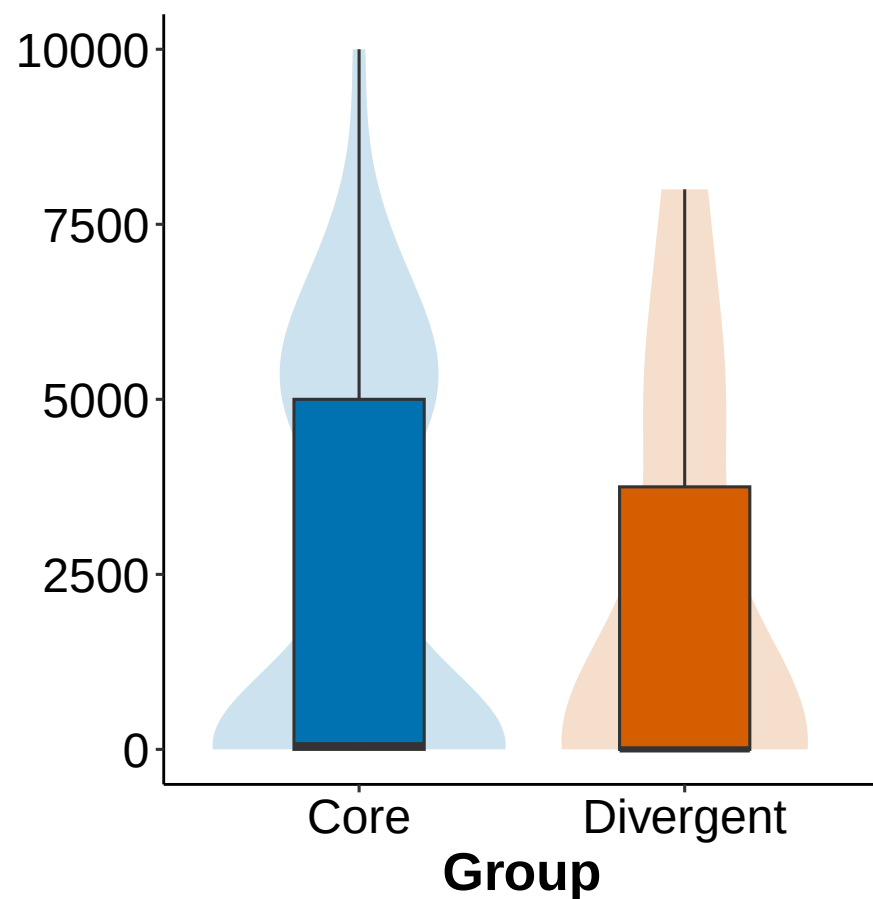
p-adj: 0.959 | .:=0.02 (Negligible) | .%: 7.9



Metabolism – . 5. Systemic Reprogramming (SA_*)

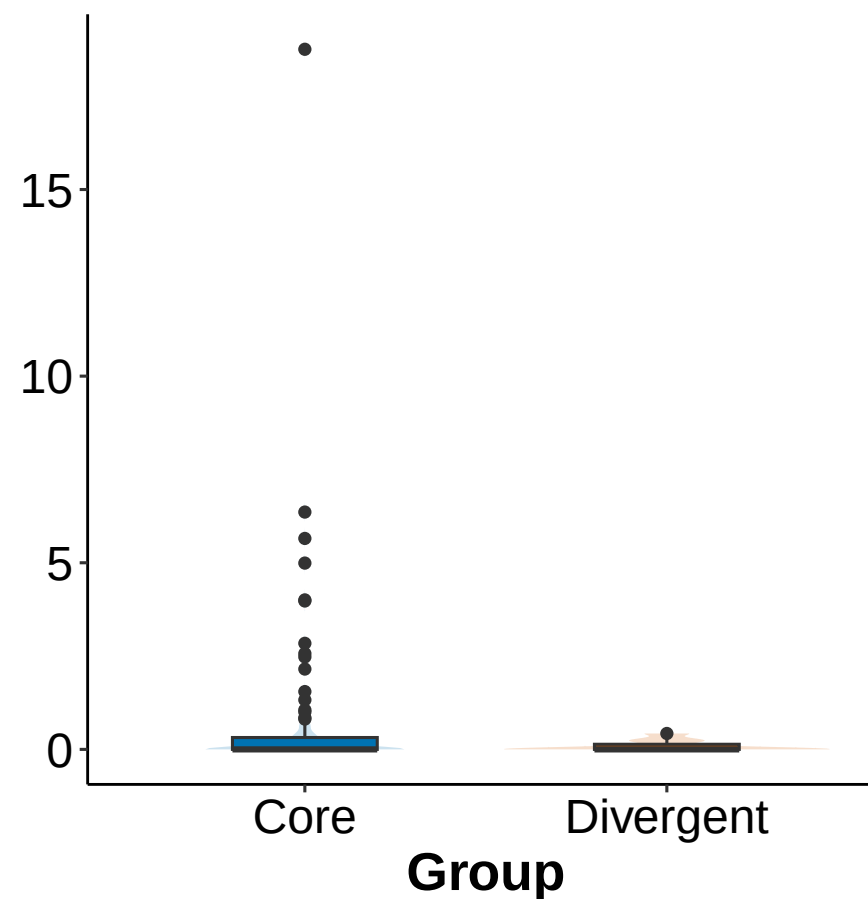
SA_Galactose_metabolism_FBA

p-adj: 0.896 | .=-0.13 (Negligible) | .%: -100



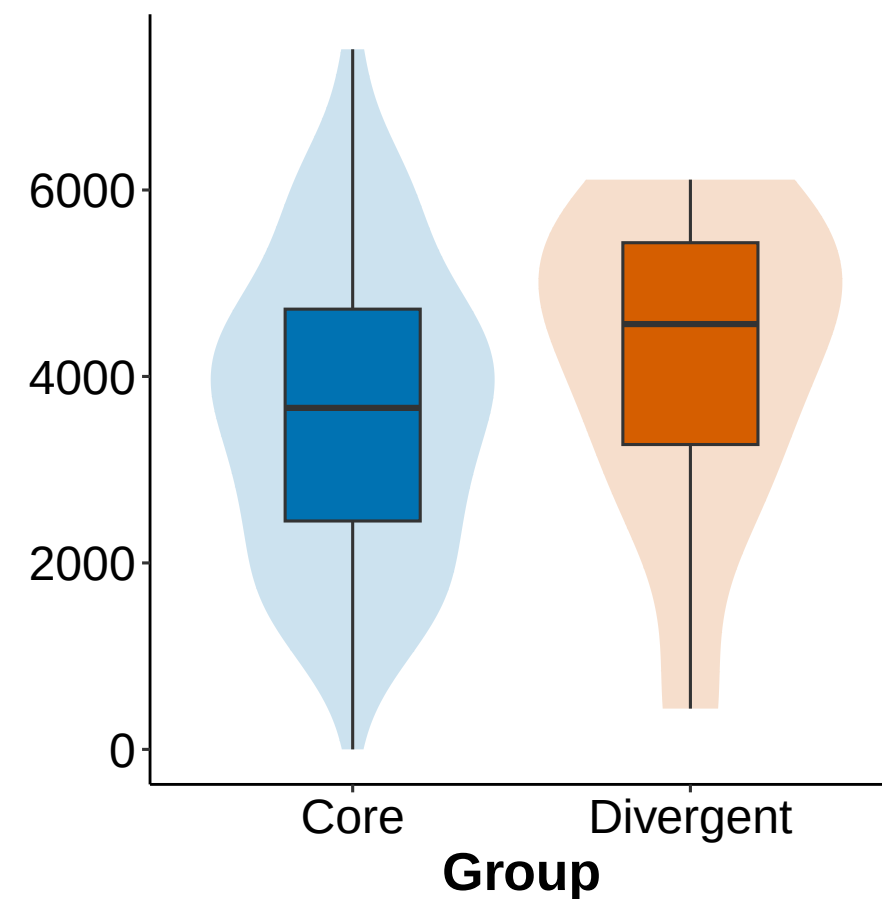
SA_Galactose_metabolism_L1w

p-adj: 0.992 | .=0 (Negligible) | .%: 69.1



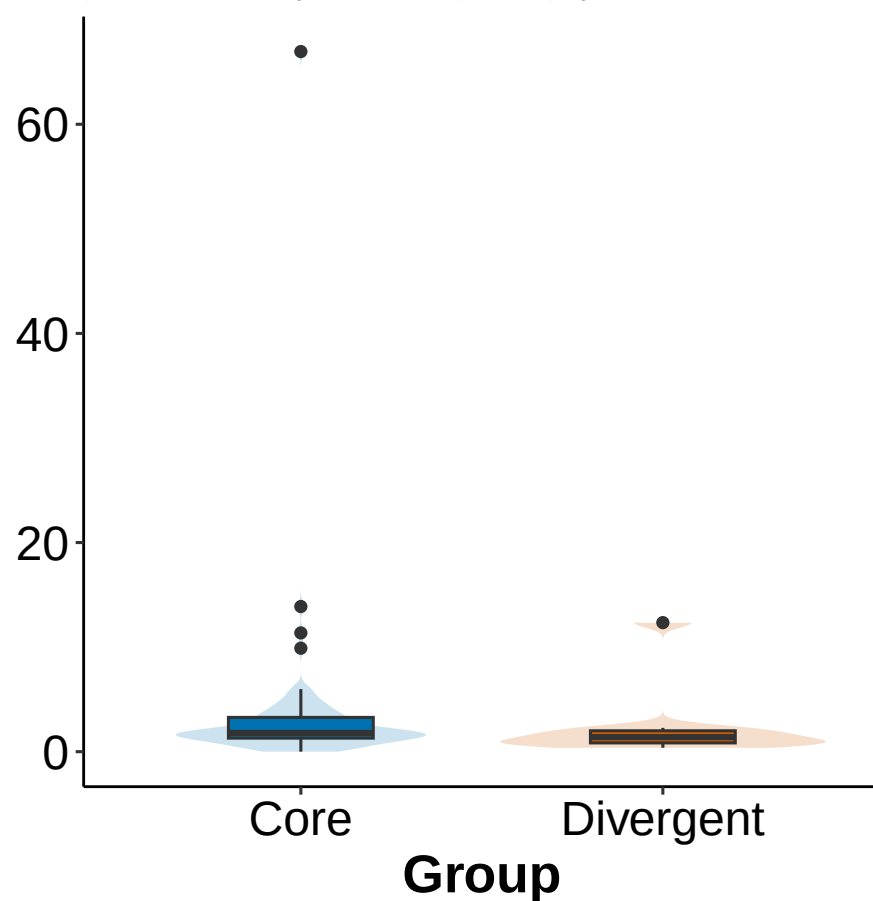
SA_Glutamate_metabolism_FBA

p-adj: 0.885 | .=0.24 (Small) | .%: 24.5



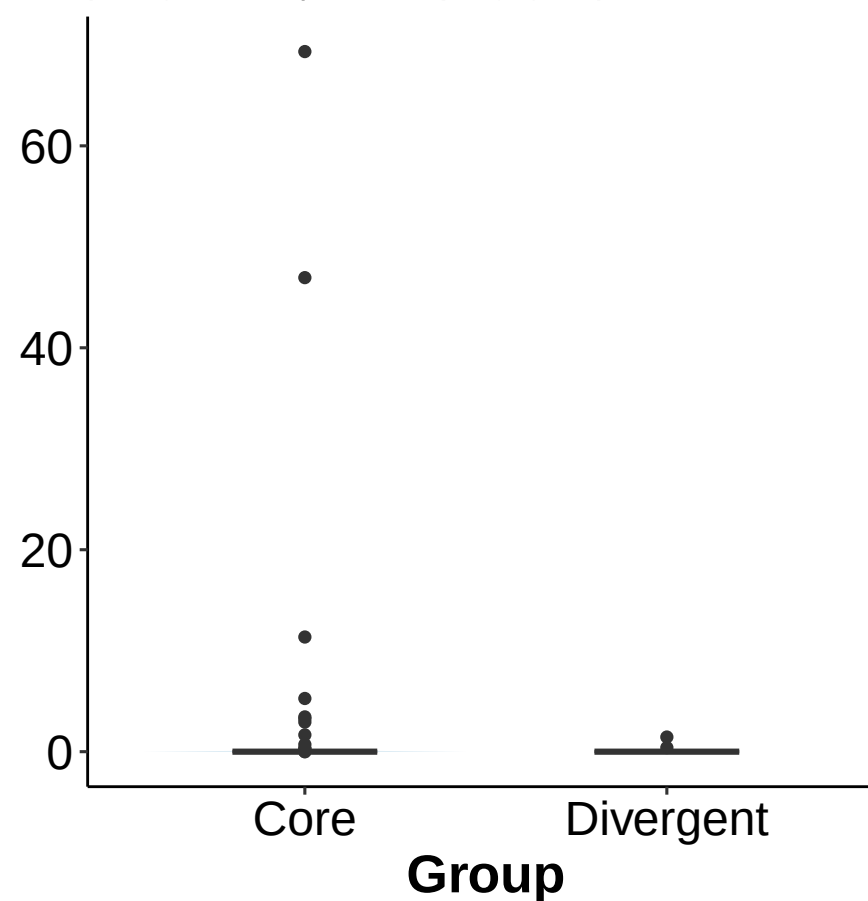
SA_Glutamate_metabolism_L1w

p-adj: 0.872 | .=-0.26 (Small) | .%: -22.4



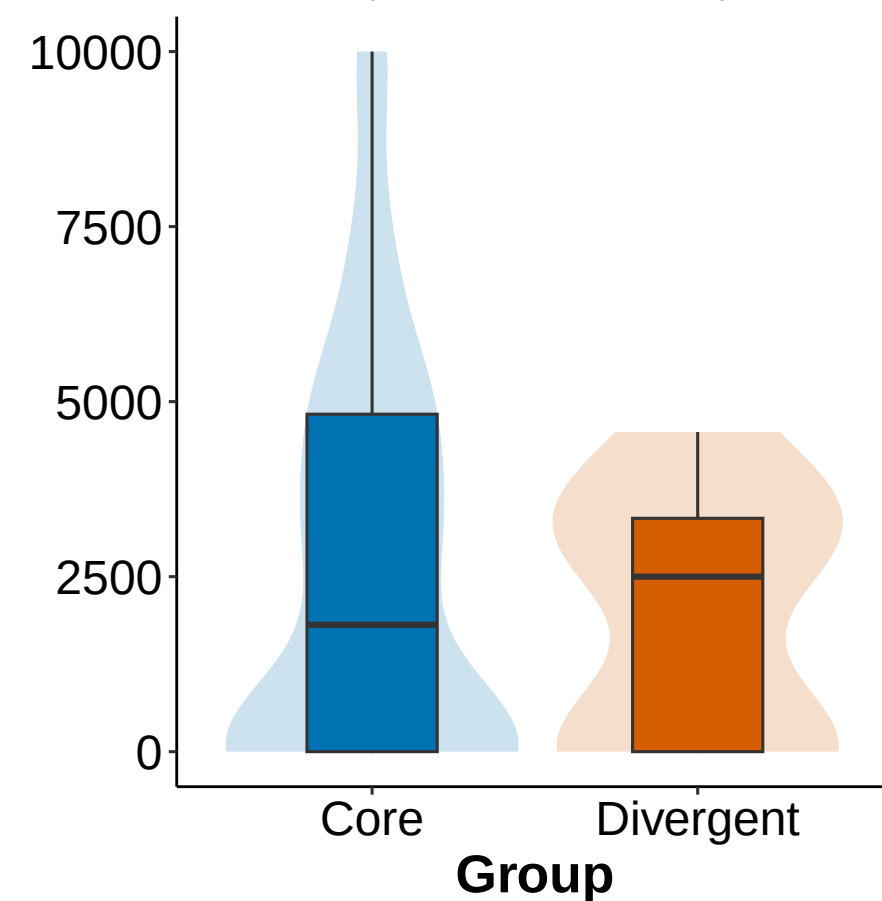
SA_Glutamate_metabolism_pFBA

p-adj: 0.885 | .=0.15 (Negligible)



SA_Glutathione_metabolism_FBA

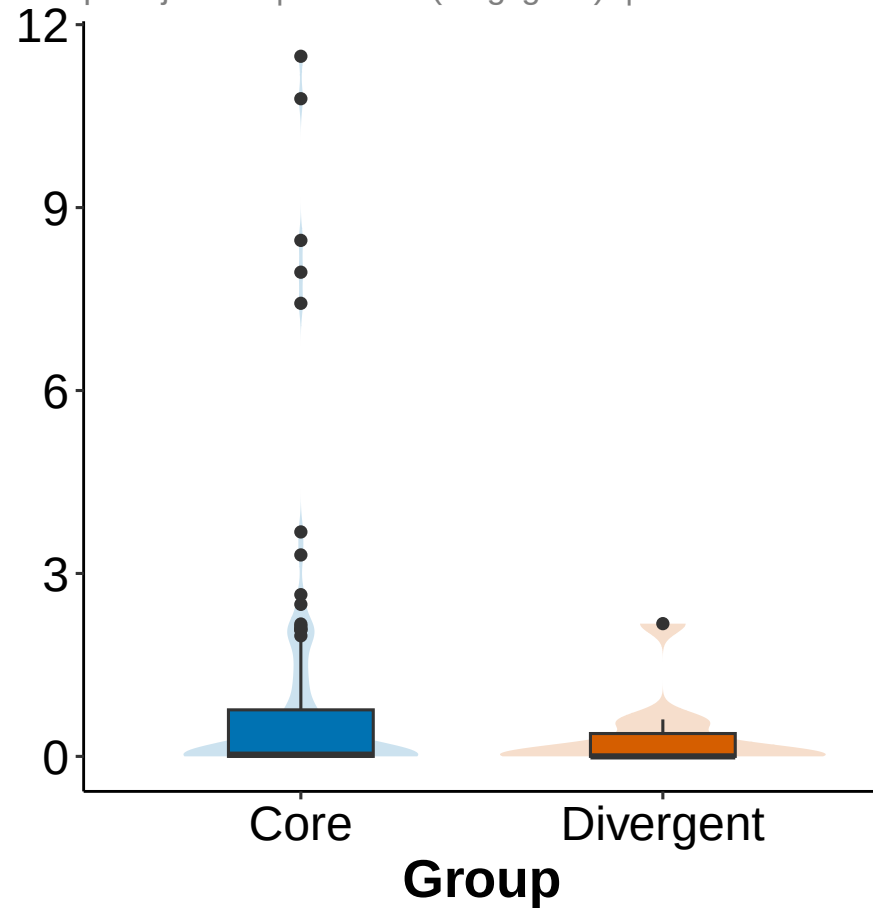
p-adj: 0.95 | .=-0.07 (Negligible) | .%: 37.9



Metabolism – . 5. Systemic Reprogramming (SA_*)

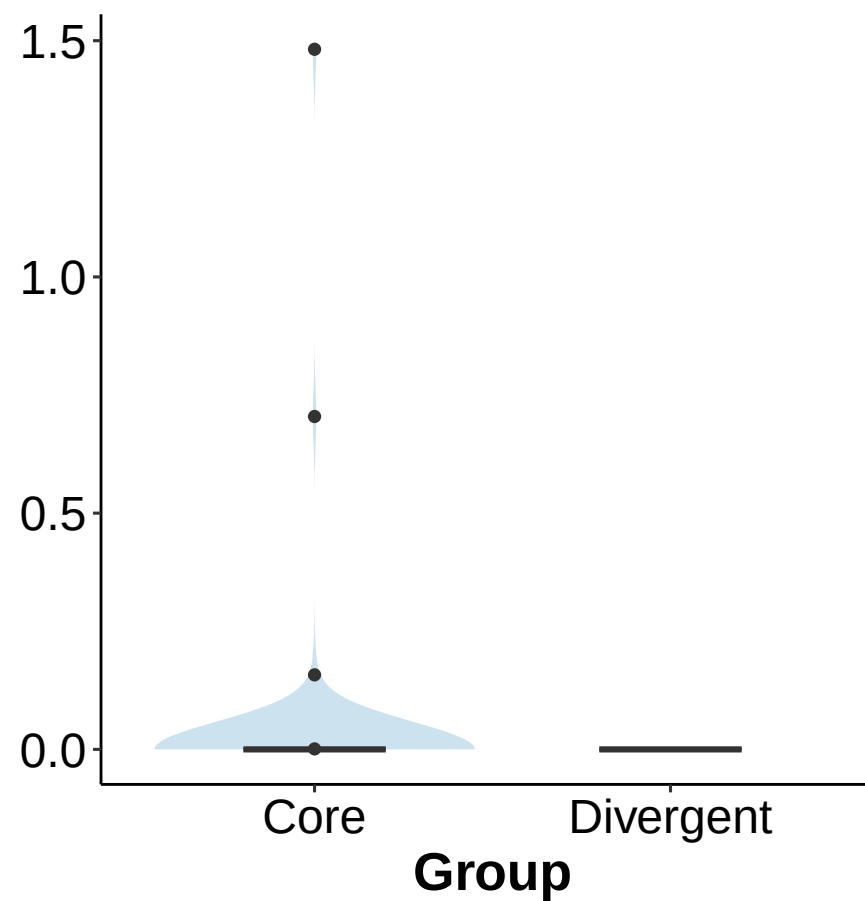
SA_Glutathione_metabolism_L1w

p-adj: 0.95 | .=-0.06 (Negligible) | .%: -99.5



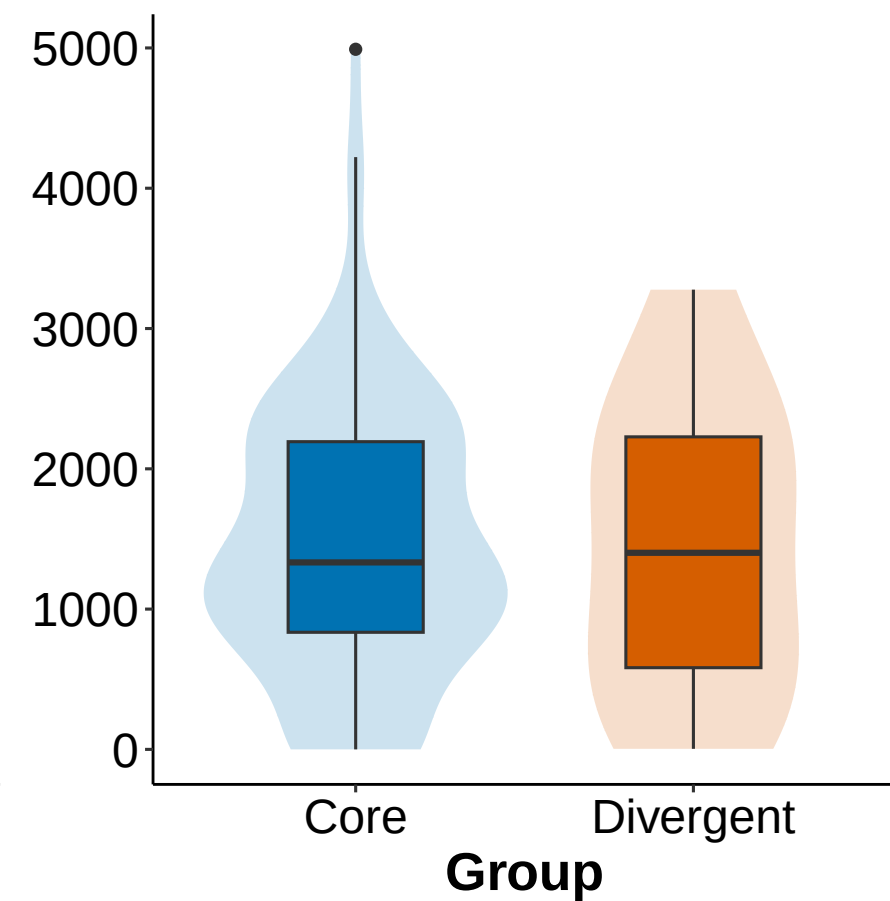
SA_Glutathione_metabolism_pFBA

p-adj: 0.917 | .=-0.04 (Negligible)



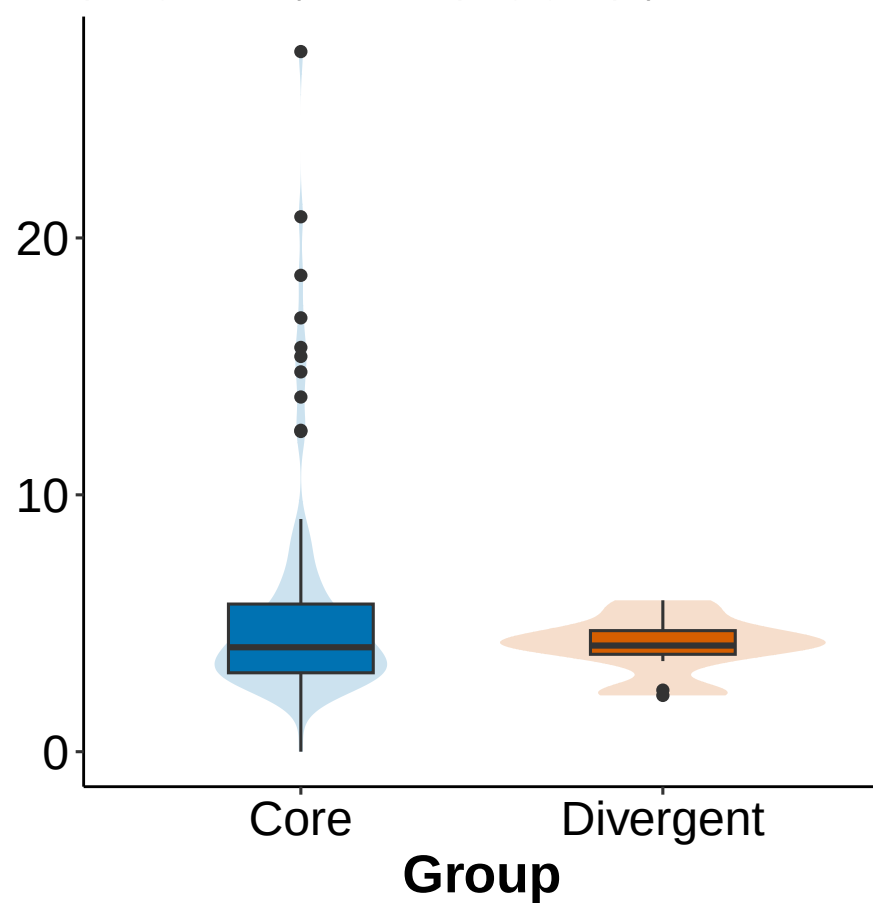
SA_Glycerophospholipid_metabolism

p-adj: 0.954 | .=-0.03 (Negligible) | .%: 5.1



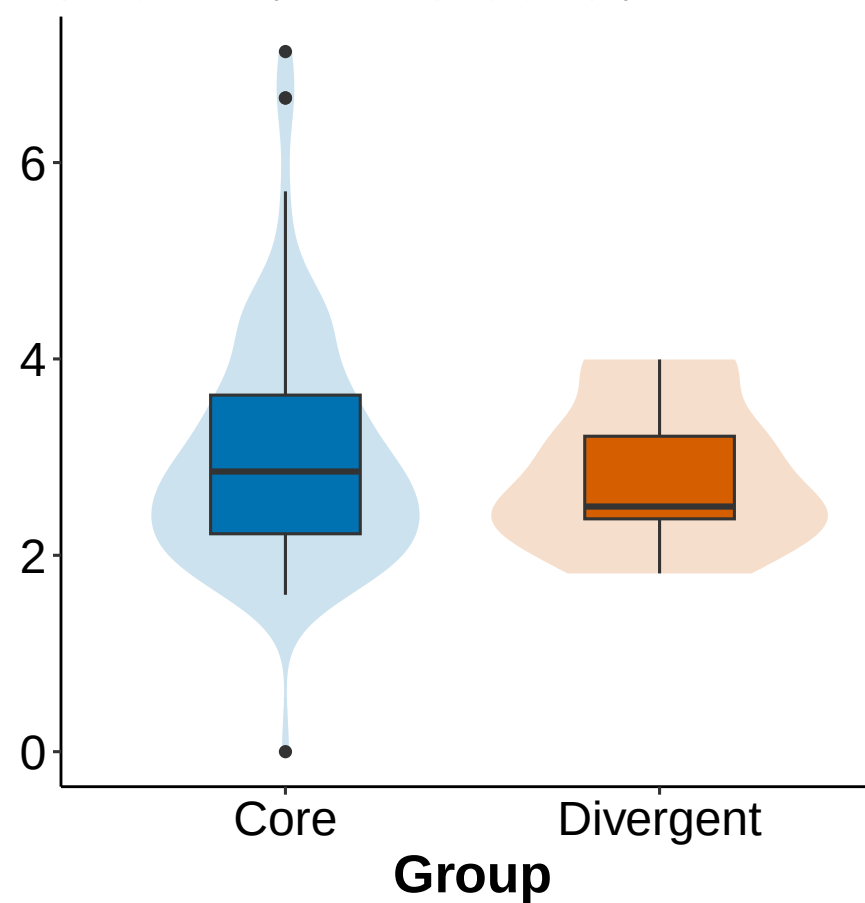
SA_Glycerophospholipid_metabolism

p-adj: 0.954 | .=-0.04 (Negligible) | .%: 1.7



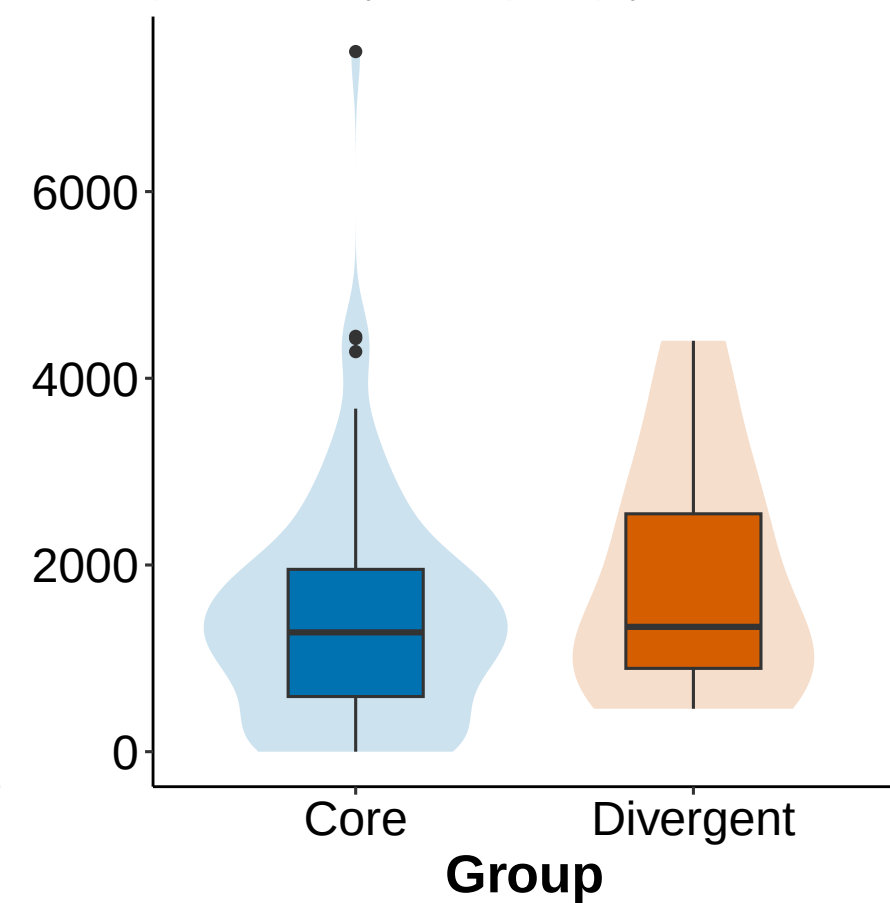
SA_Glycerophospholipid_metabolism

p-adj: 0.954 | .=-0.04 (Negligible) | .%: -12.5



SA_Glycine__serine__alanine__and

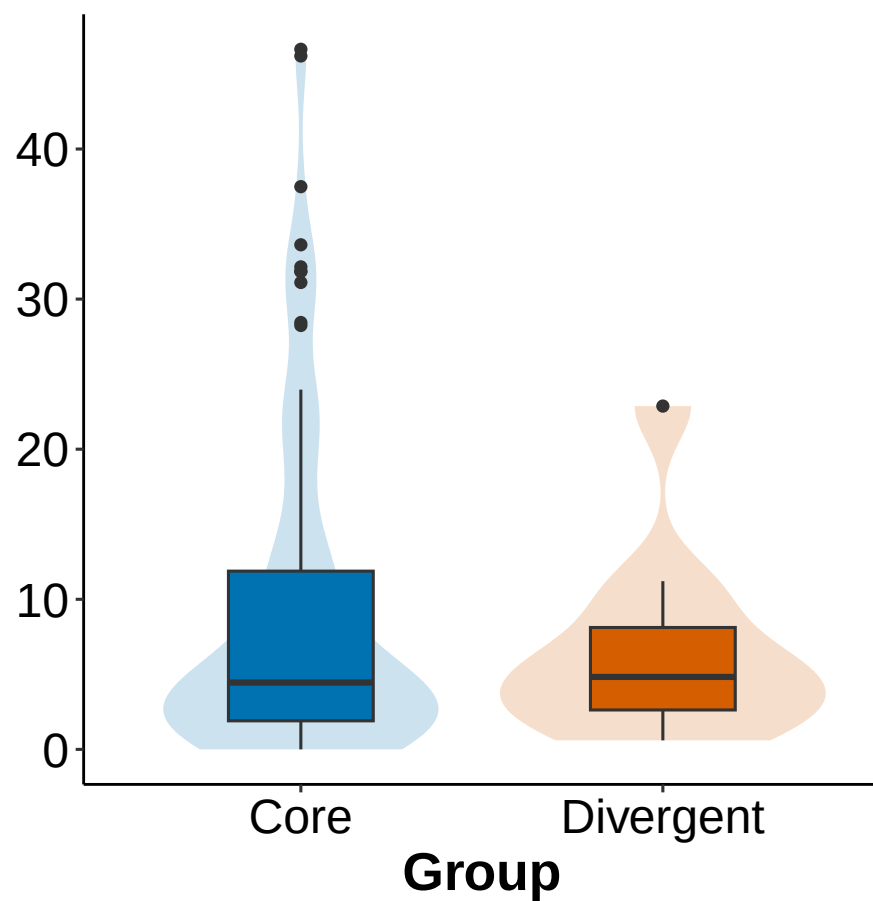
p-adj: 0.885 | .=0.19 (Small) | .%: 4.6



Metabolism – . 5. Systemic Reprogramming (SA_*)

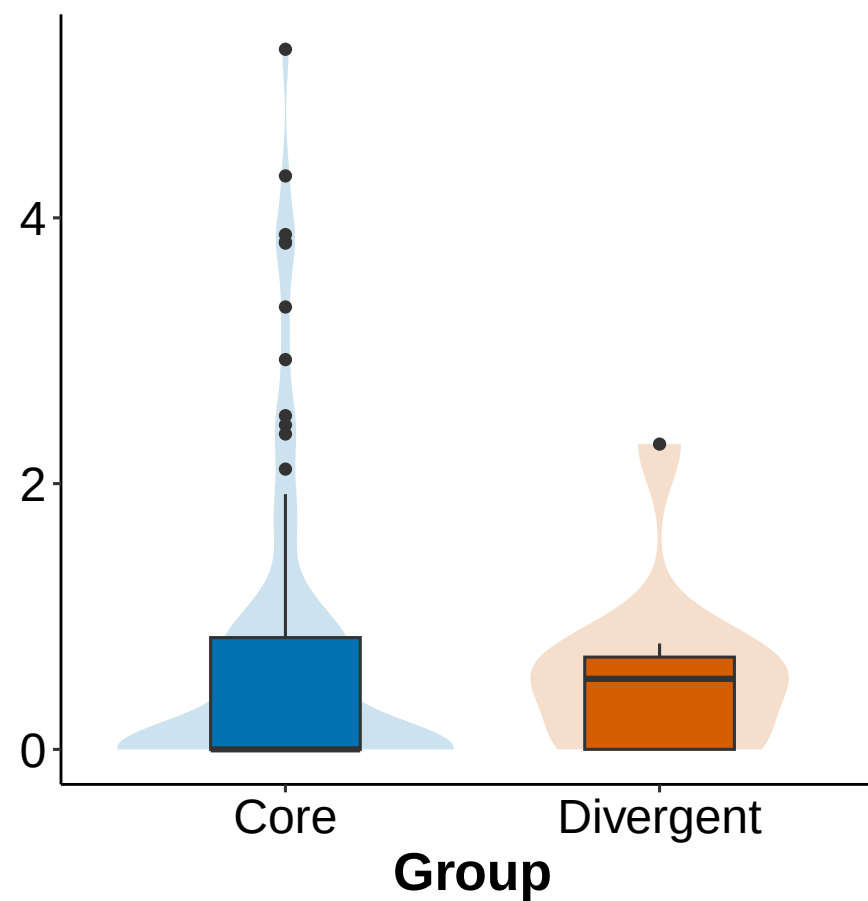
SA_Glycine__serine__alanine__and_t

p-adj: 0.992 | .=0 (Negligible) | .%: 8.4



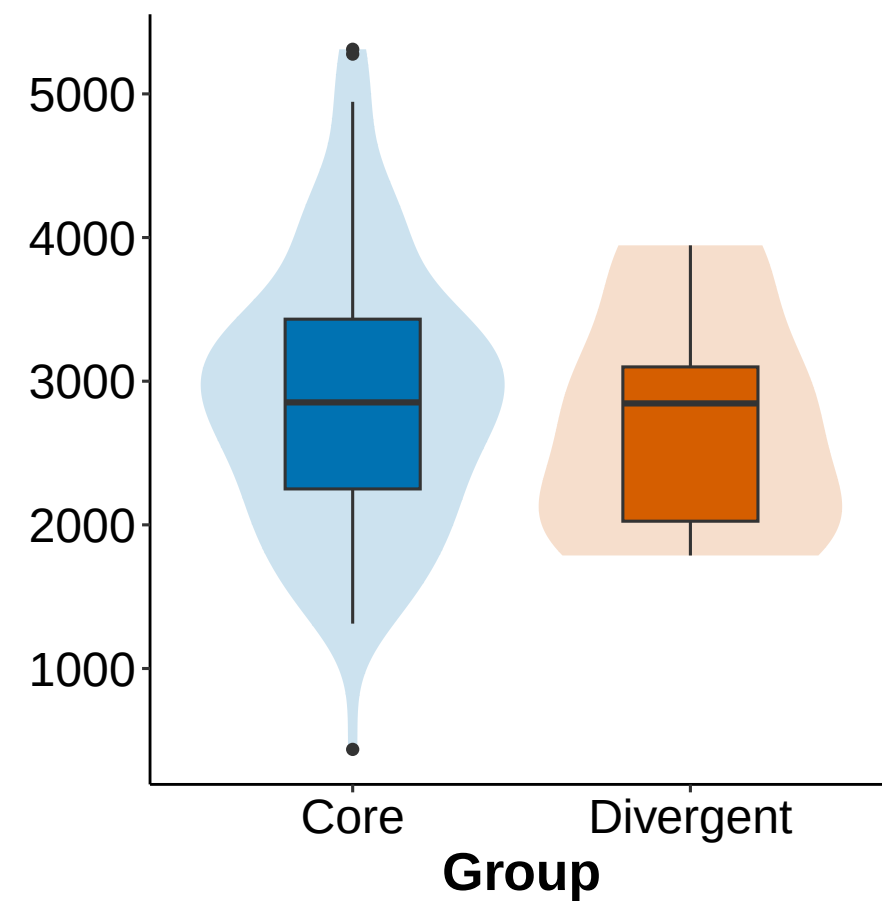
SA_Glycine__serine__alanine__and_th

p-adj: 0.95 | .=0.08 (Negligible) | .%: 140163.2



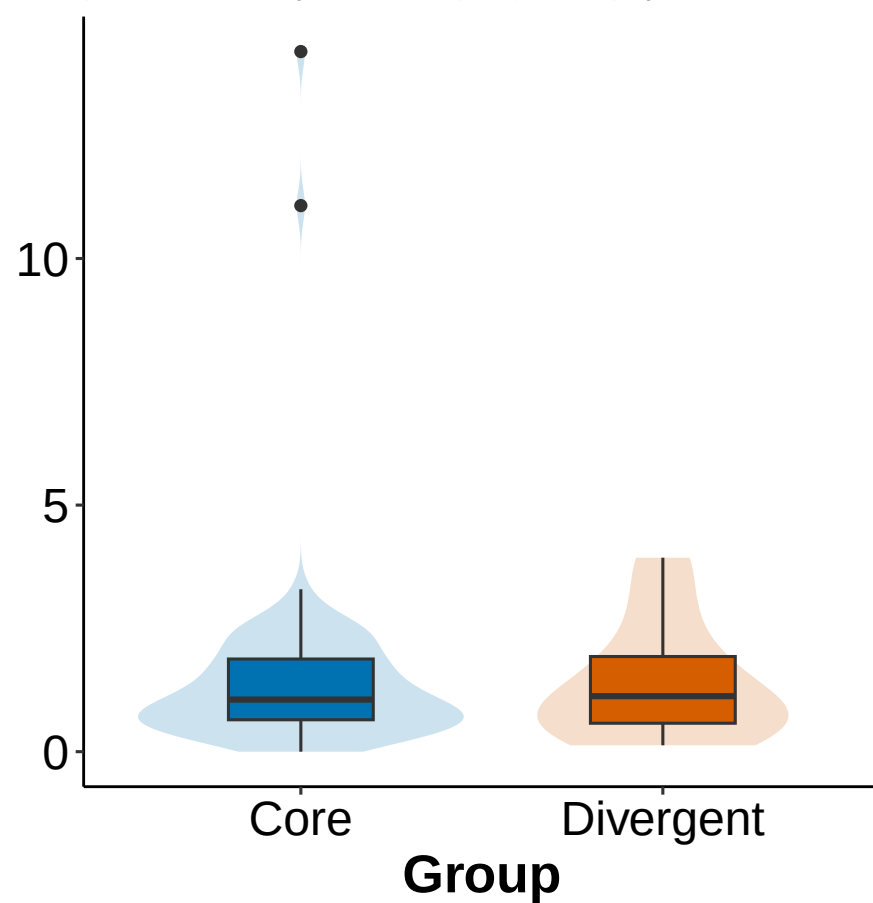
SA_Glycolysis_gluconeogenesis_F

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -0.2



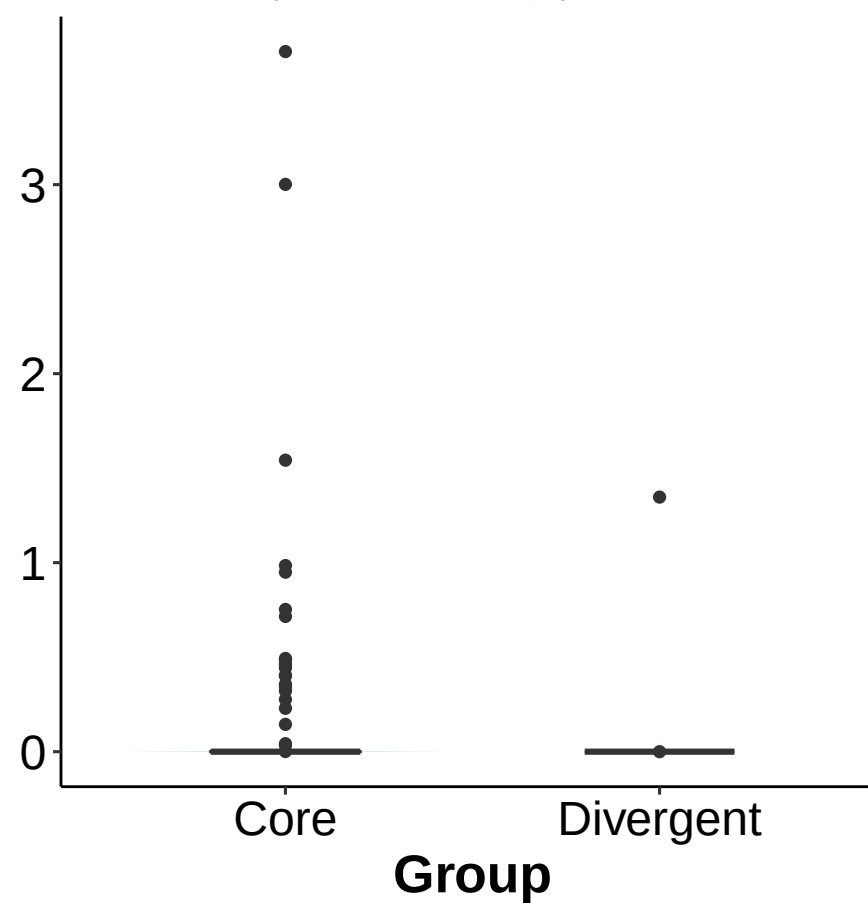
SA_Glycolysis_gluconeogenesis_L1w

p-adj: 0.992 | .=-0.01 (Negligible) | .%: 6.5



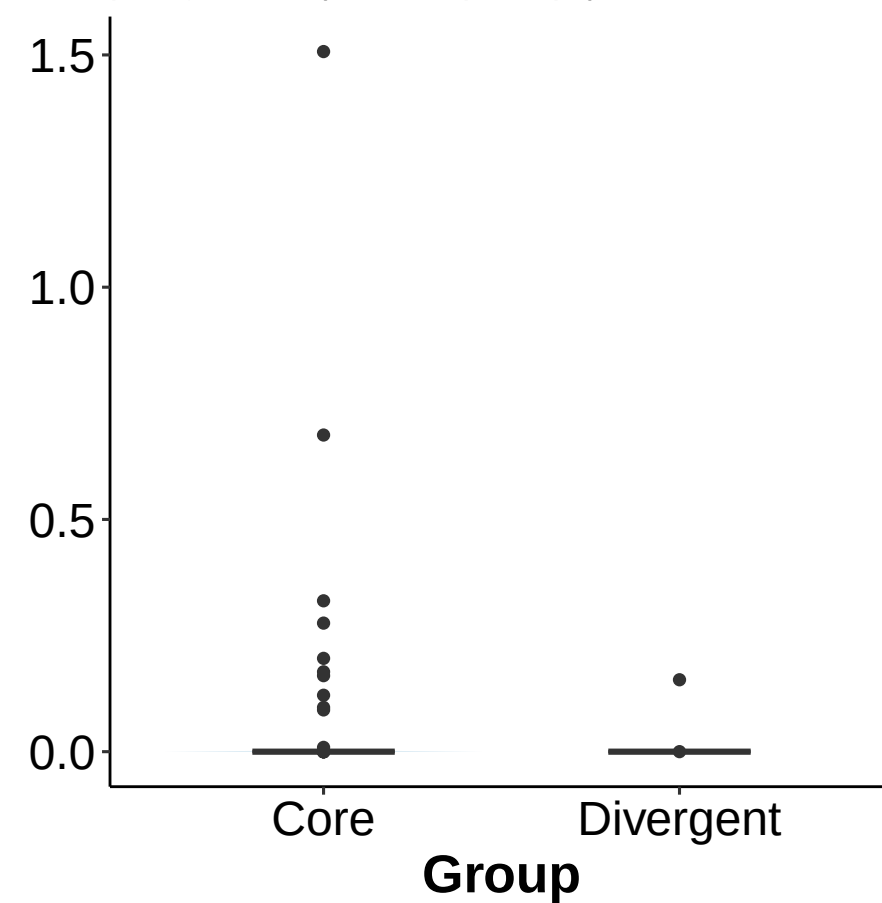
SA_Glycolysis_gluconeogenesis_pFB/

p-adj: 0.749 | .=-0.48 (Large) | .%: -100



SA_Glycosphingolipid_metabolism_L

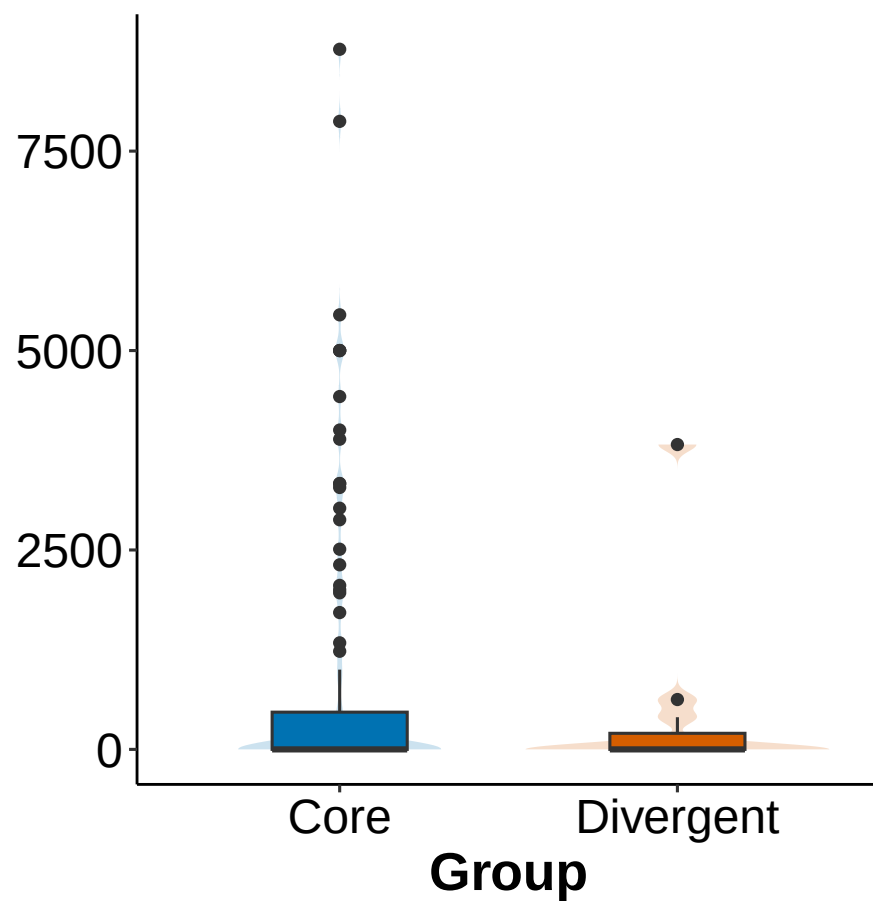
p-adj: 0.885 | .=0.15 (Small) | .%: 50.8



Metabolism – . 5. Systemic Reprogramming (SA_*)

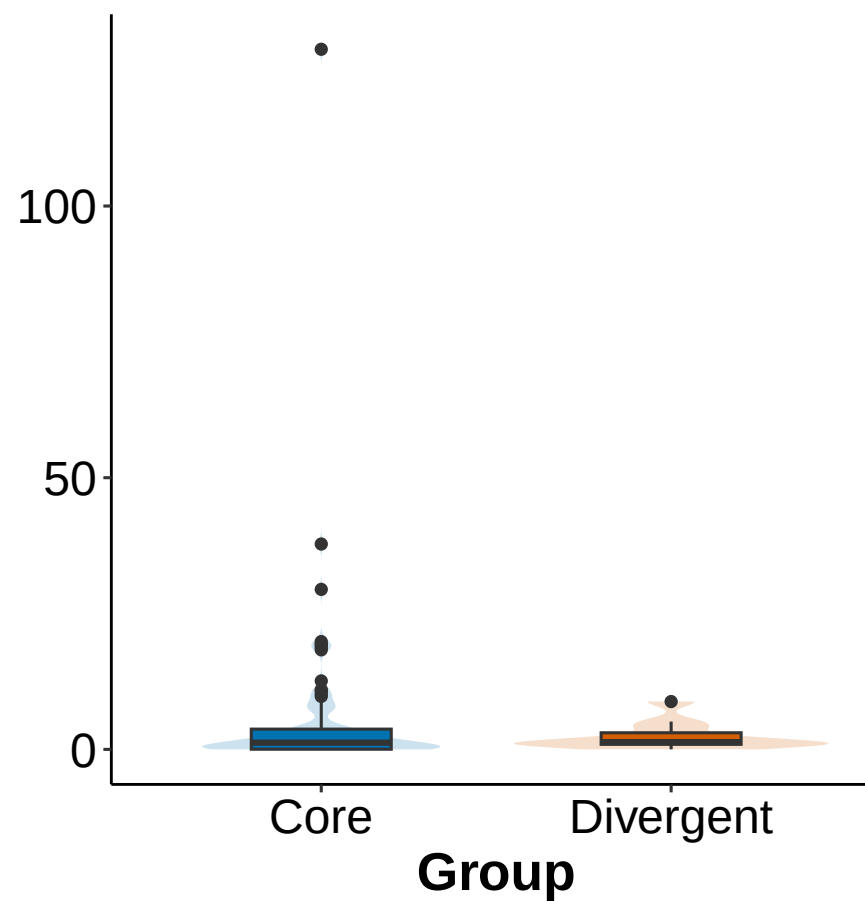
SA_Glyoxylate_and_dicarboxylate_

p-adj: 0.95 | .=-0.06 (Negligible)



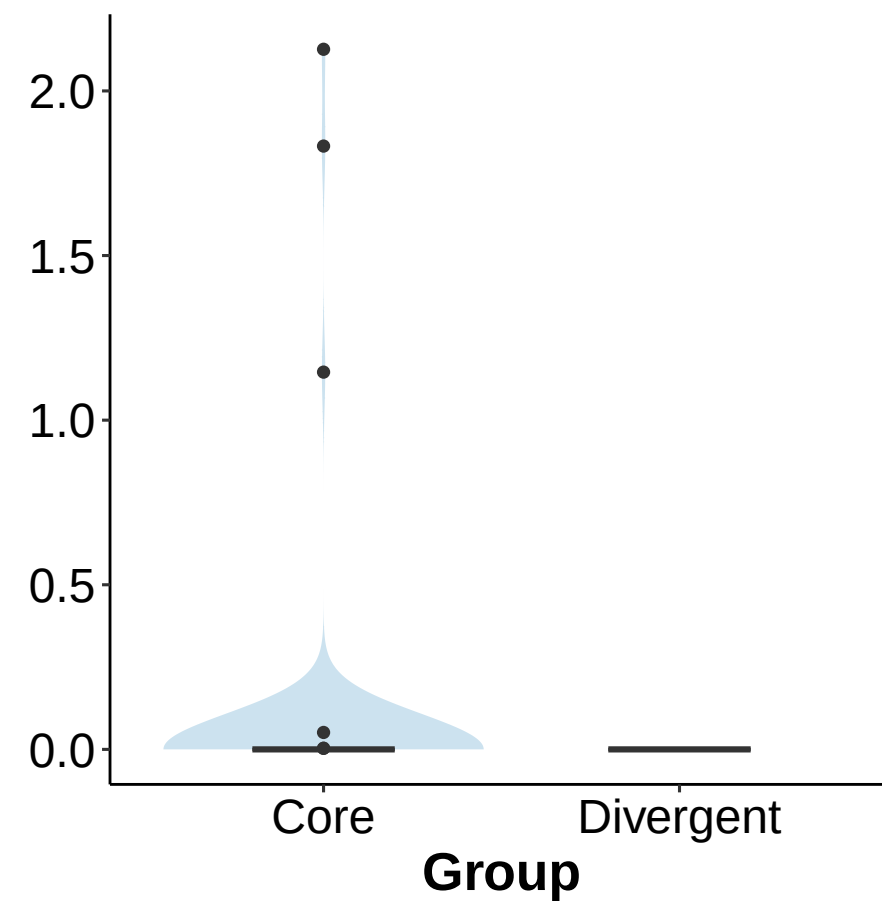
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.917 | .:=0.11 (Negligible) | .%: 8.9



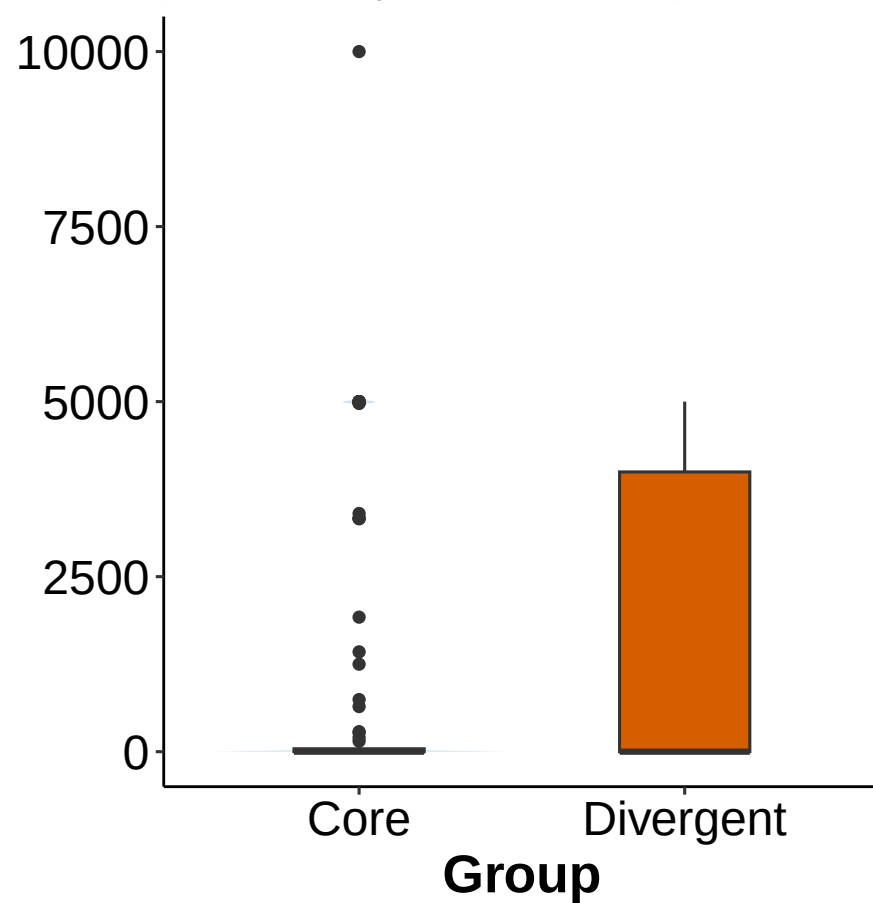
SA_Glyoxylate_and_dicarboxylate_m

p-adj: 0.885 | .=-0.06 (Negligible)



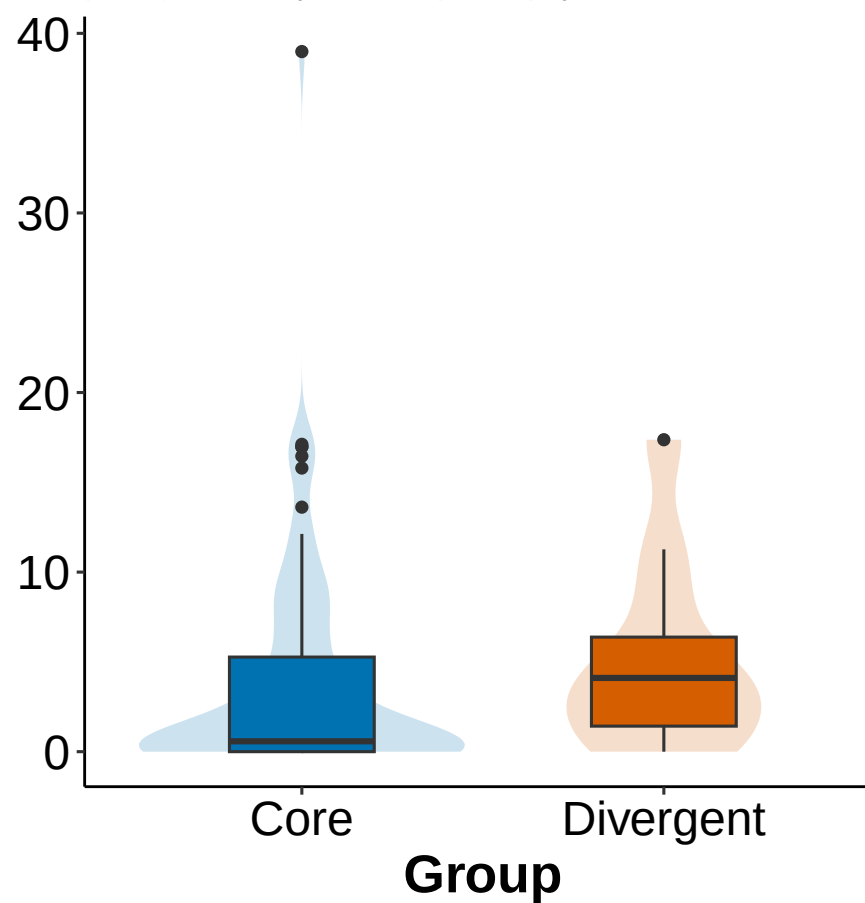
SA_Heme_degradation_FBA

p-adj: 0.885 | .:=0.14 (Negligible)



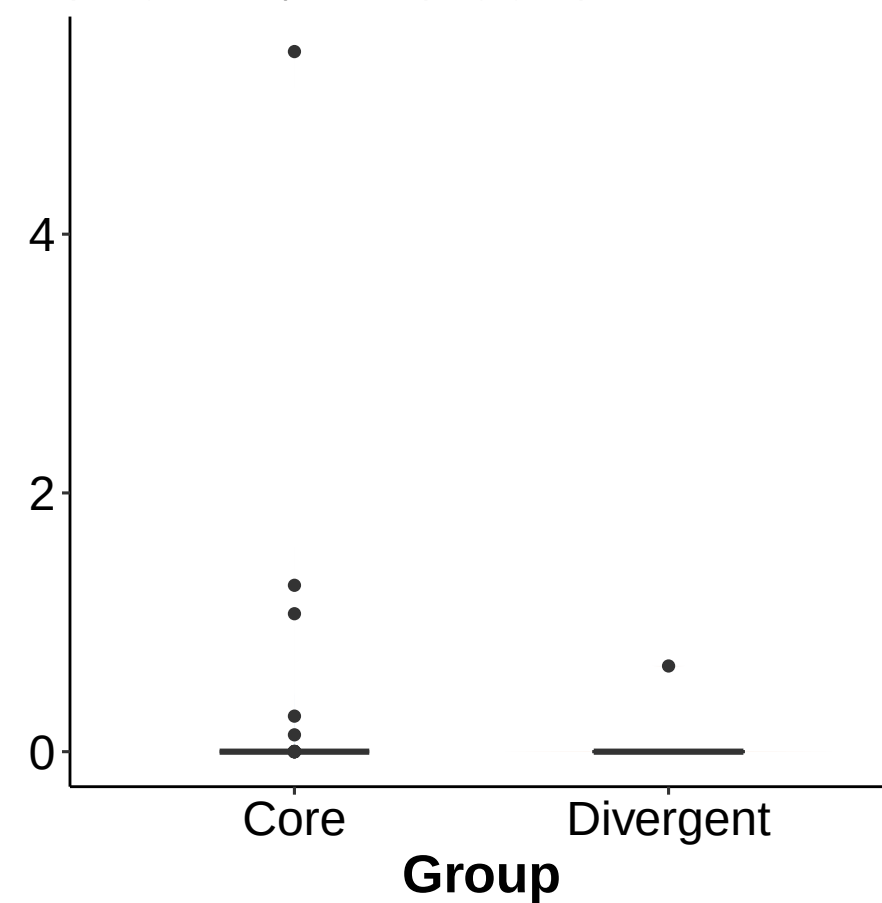
SA_Heme_degradation_L1w

p-adj: 0.872 | .:=0.29 (Small) | .%: 610.2



SA_Heme_degradation_pFBA

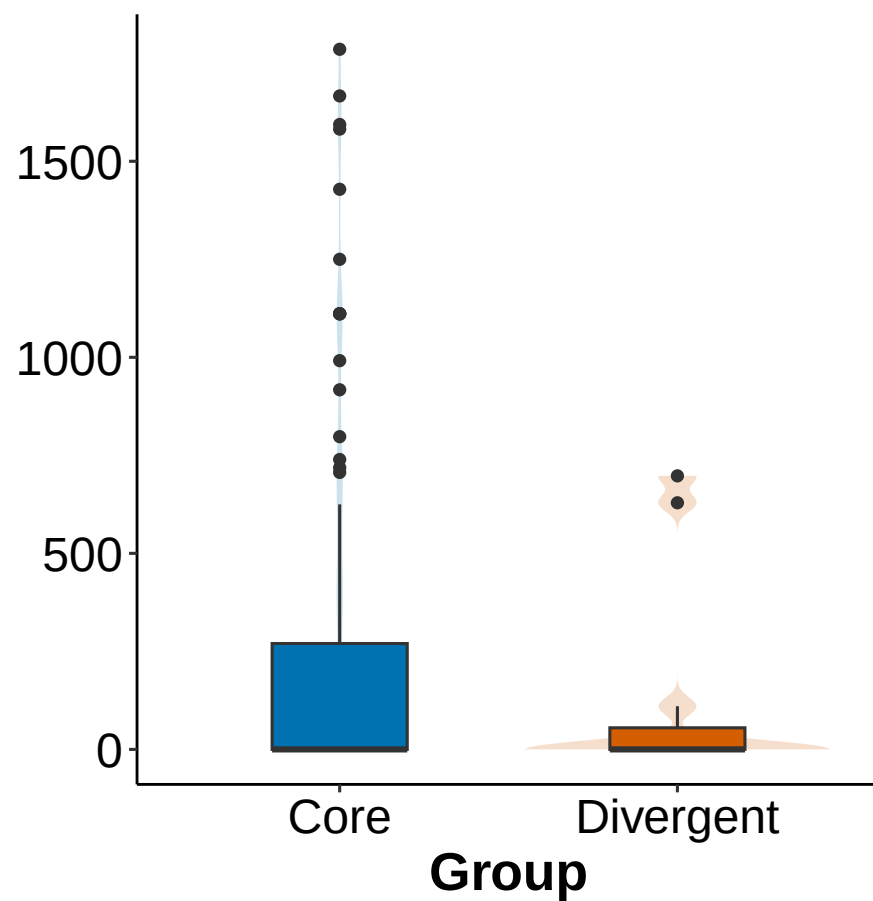
p-adj: 0.885 | .:=0.14 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

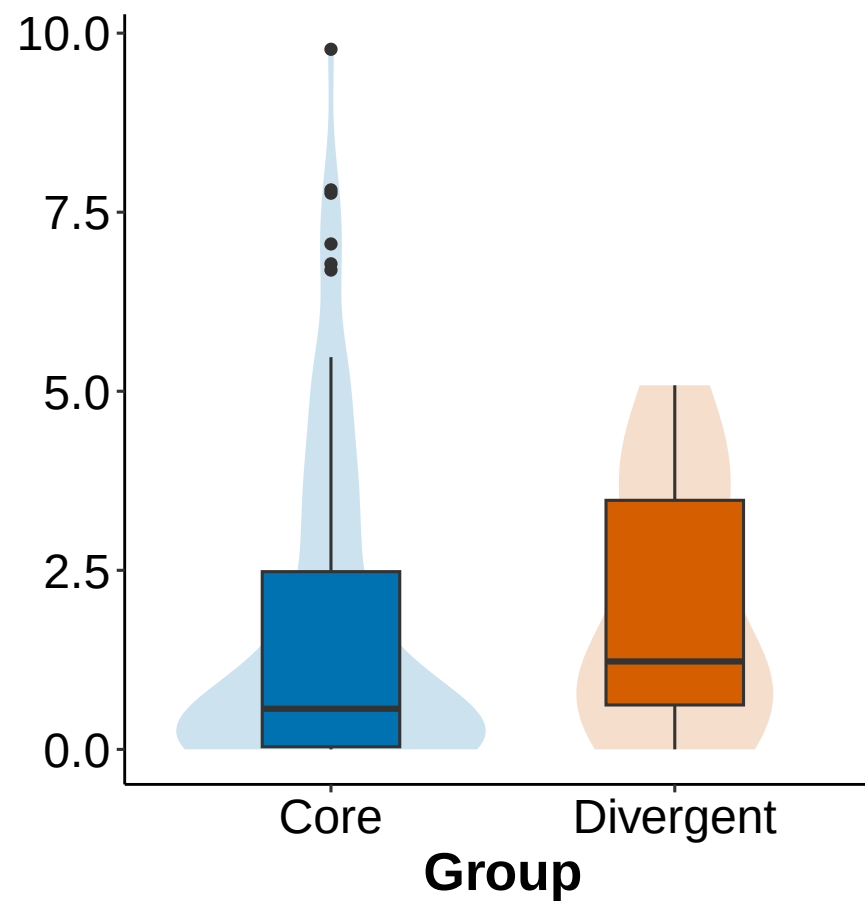
SA_Heme_synthesis_FBA

p-adj: 0.992 | . = 0 (Negligible)



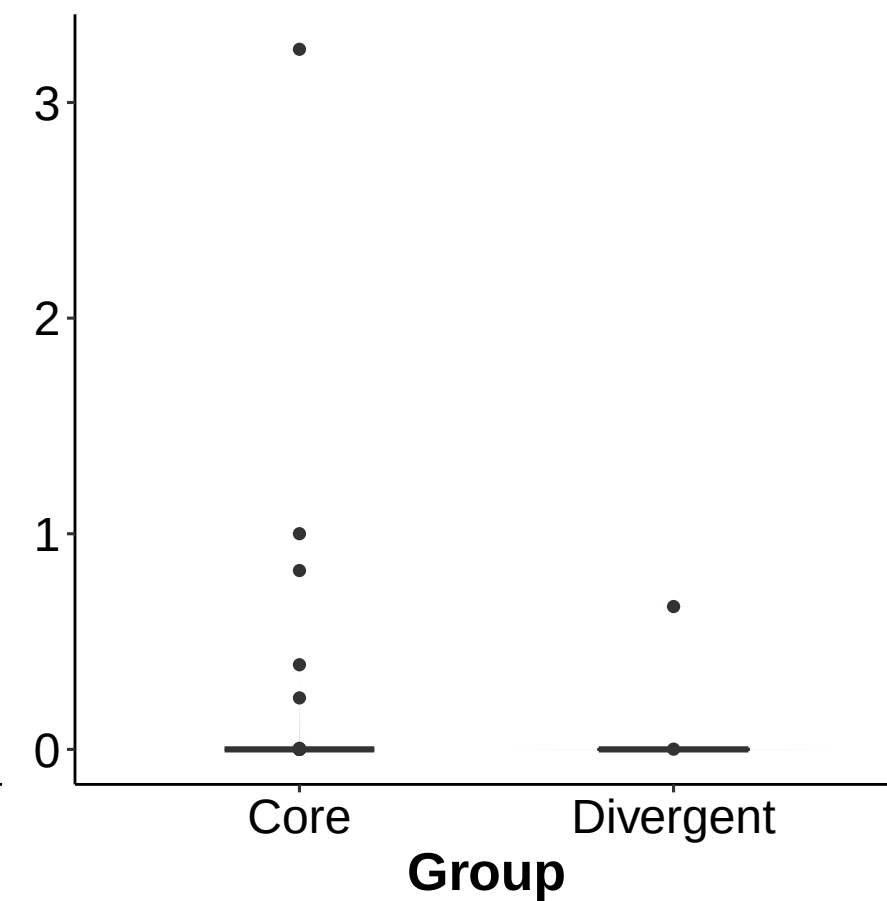
SA_Heme_synthesis_L1w

p-adj: 0.885 | . = 0.22 (Small) | .%: 116.4



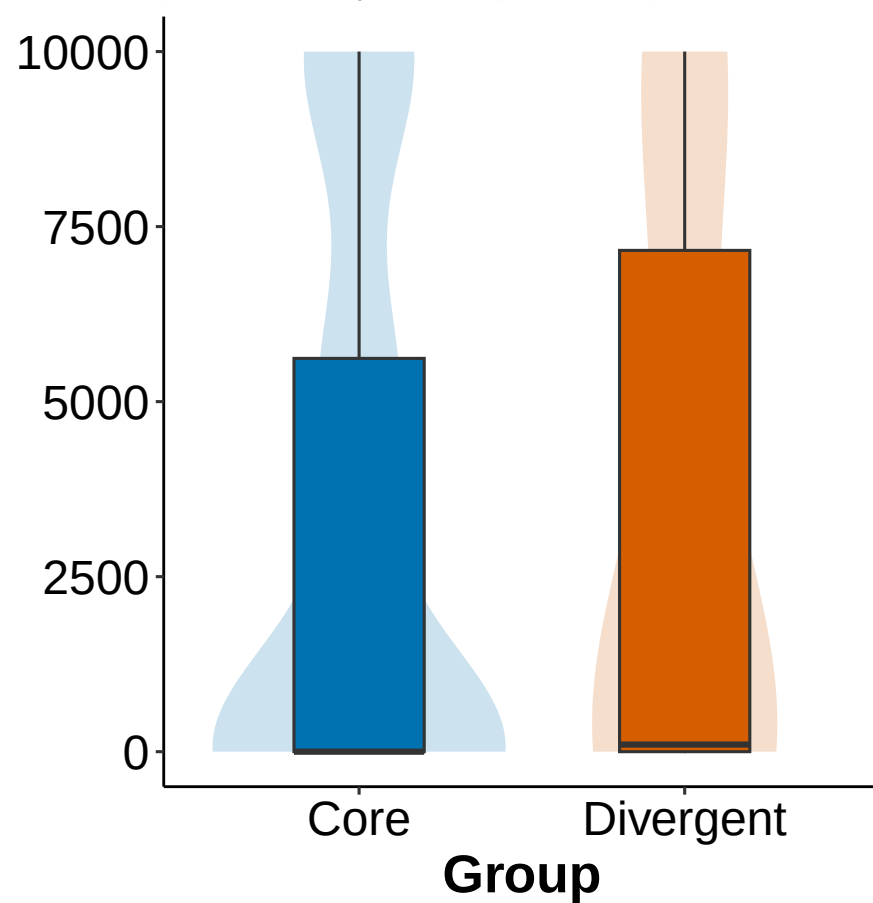
SA_Heme_synthesis_pFBA

p-adj: 0.885 | . = 0.14 (Negligible)



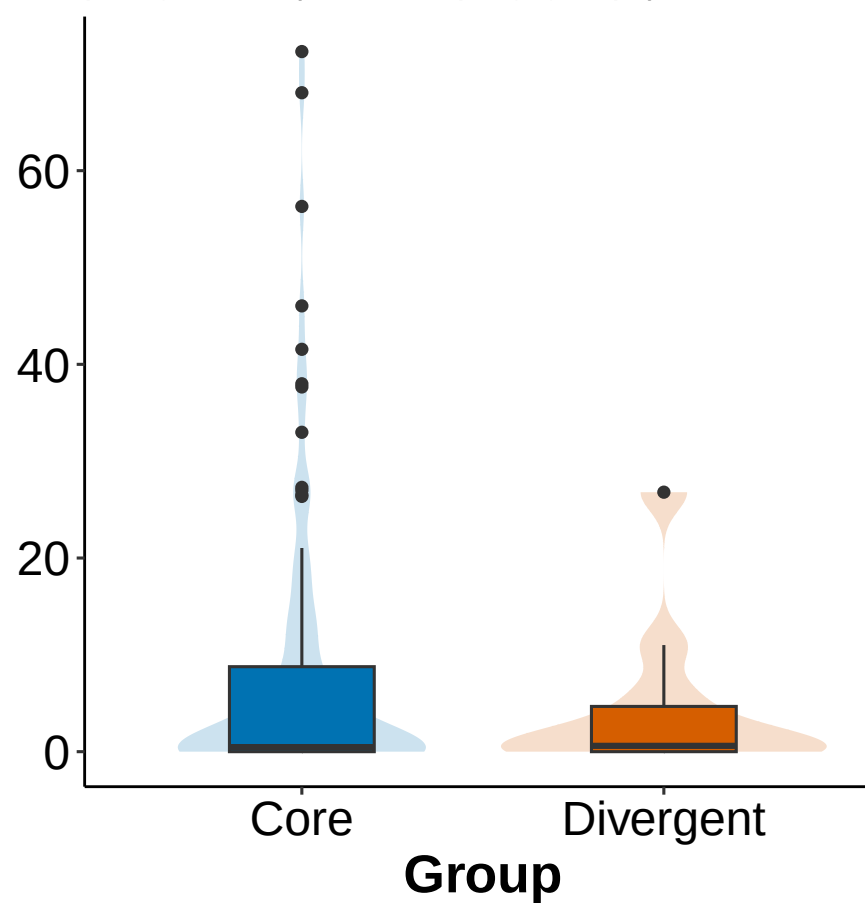
SA_Histidine_metabolism_FBA

p-adj: 0.95 | . = 0.06 (Negligible)



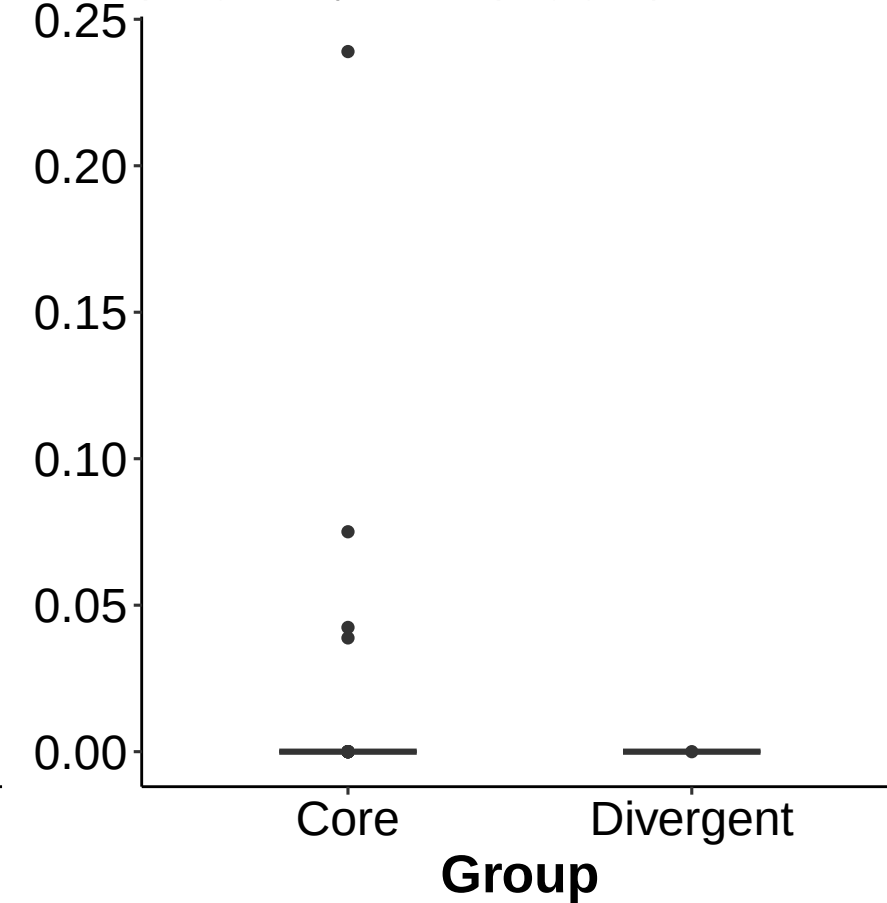
SA_Histidine_metabolism_L1w

p-adj: 0.954 | . = -0.05 (Negligible) | .%: 28.1



SA_Hyaluronan_metabolism_L1w

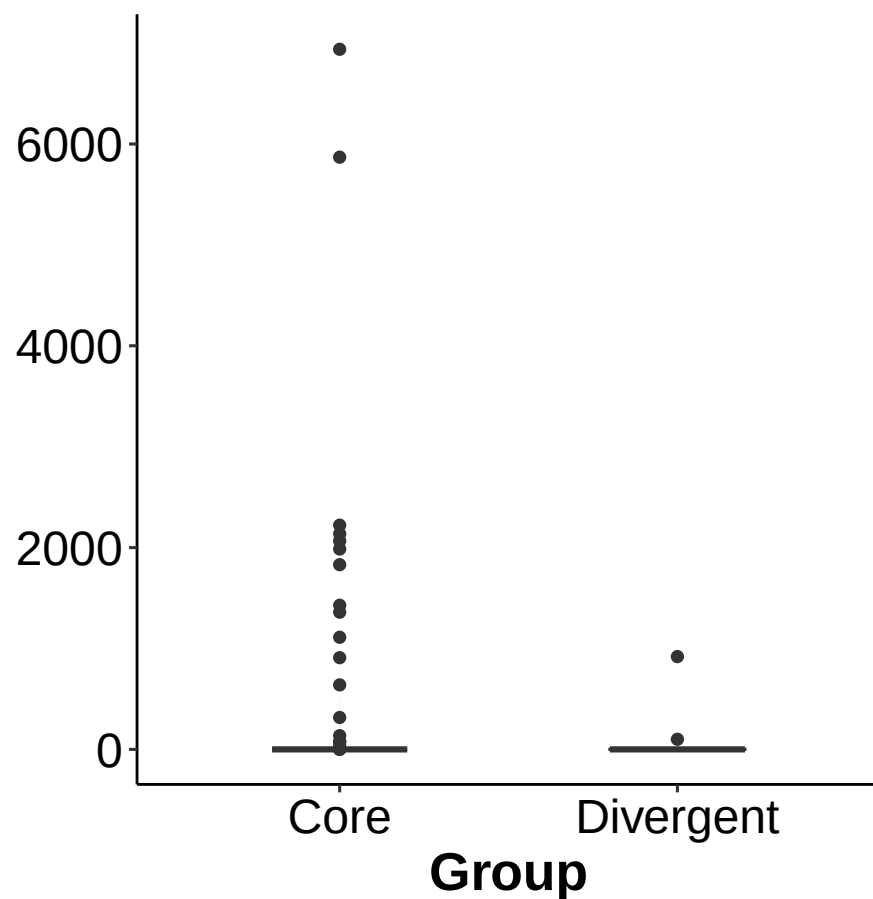
p-adj: 0.95 | . = -0.04 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

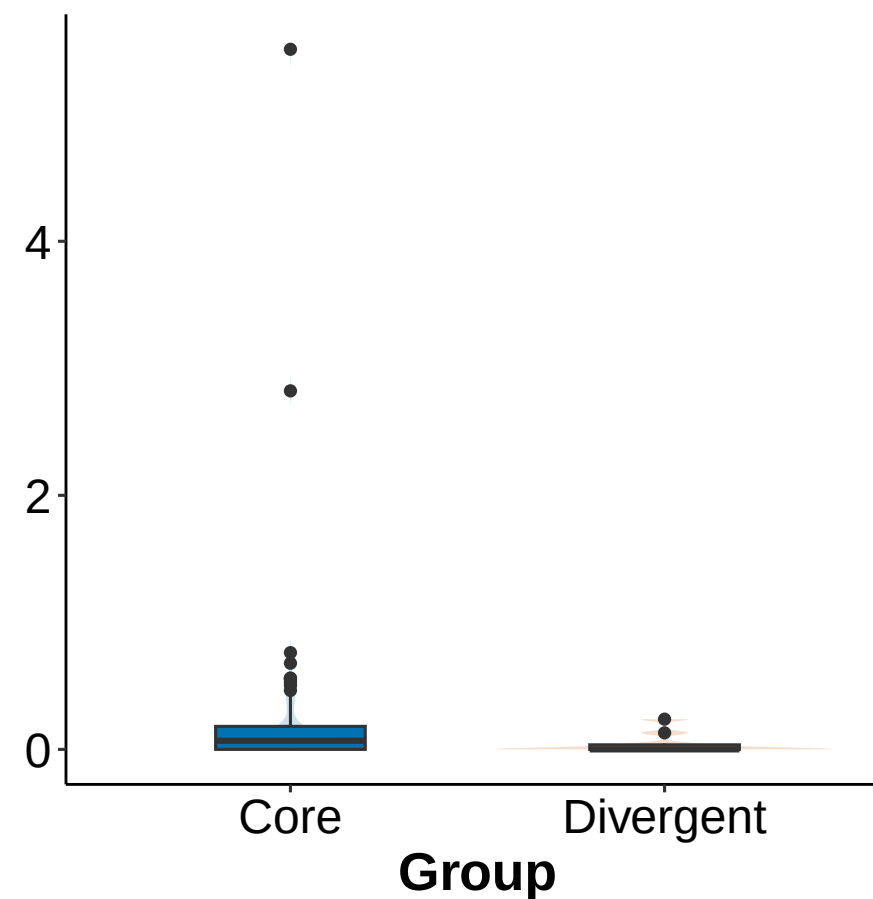
SA_Inositol_phosphate_metabolism_L1w

p-adj: 0.885 | .:=0.12 (Negligible)



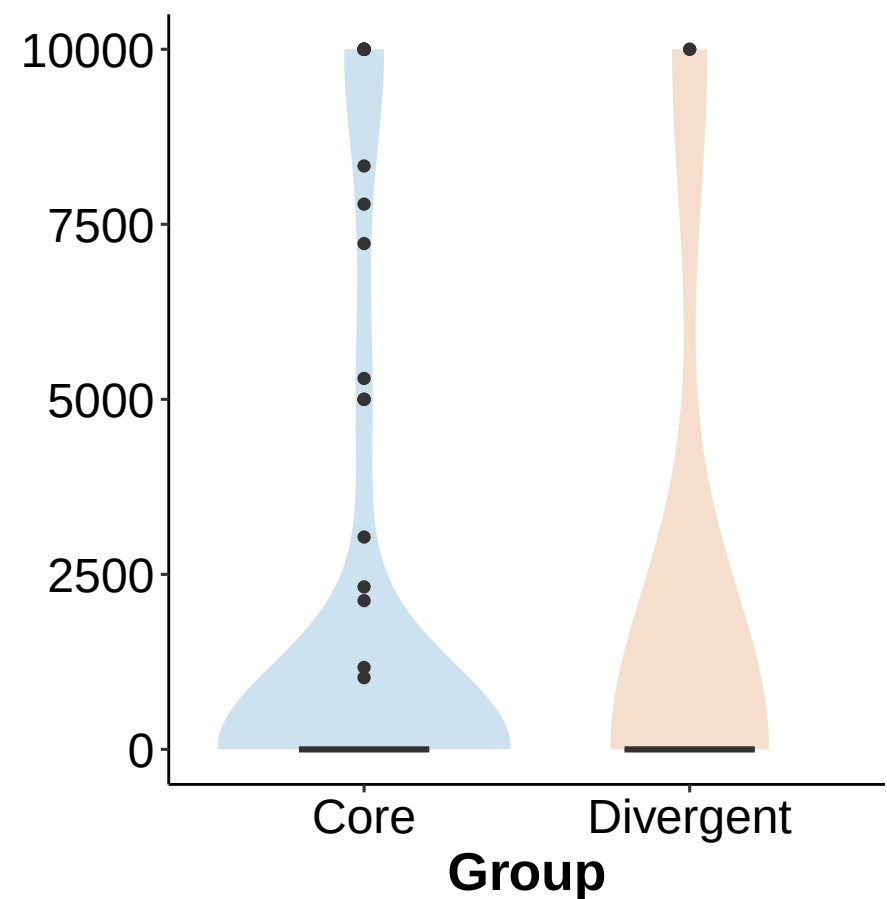
SA_Inositol_phosphate_metabolism_L1w

p-adj: 0.872 | .:=−0.31 (Small) | .%: −99.8



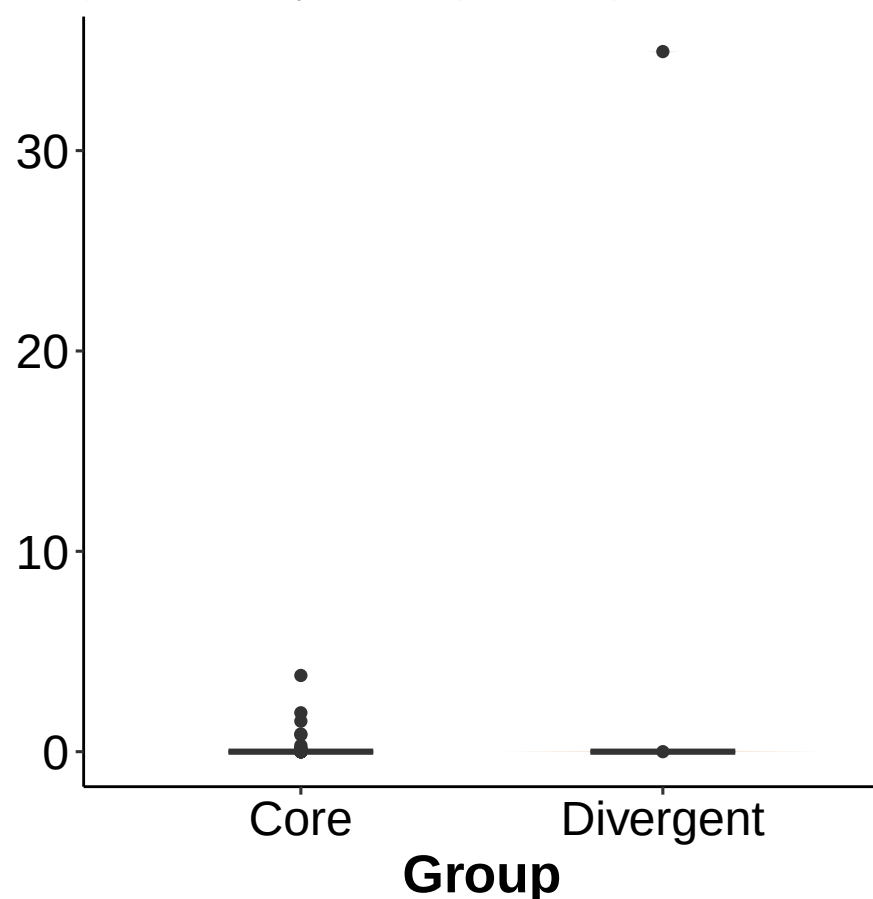
SA_Leukotriene_metabolism_FBA

p-adj: 0.957 | .:=−0.02 (Negligible)



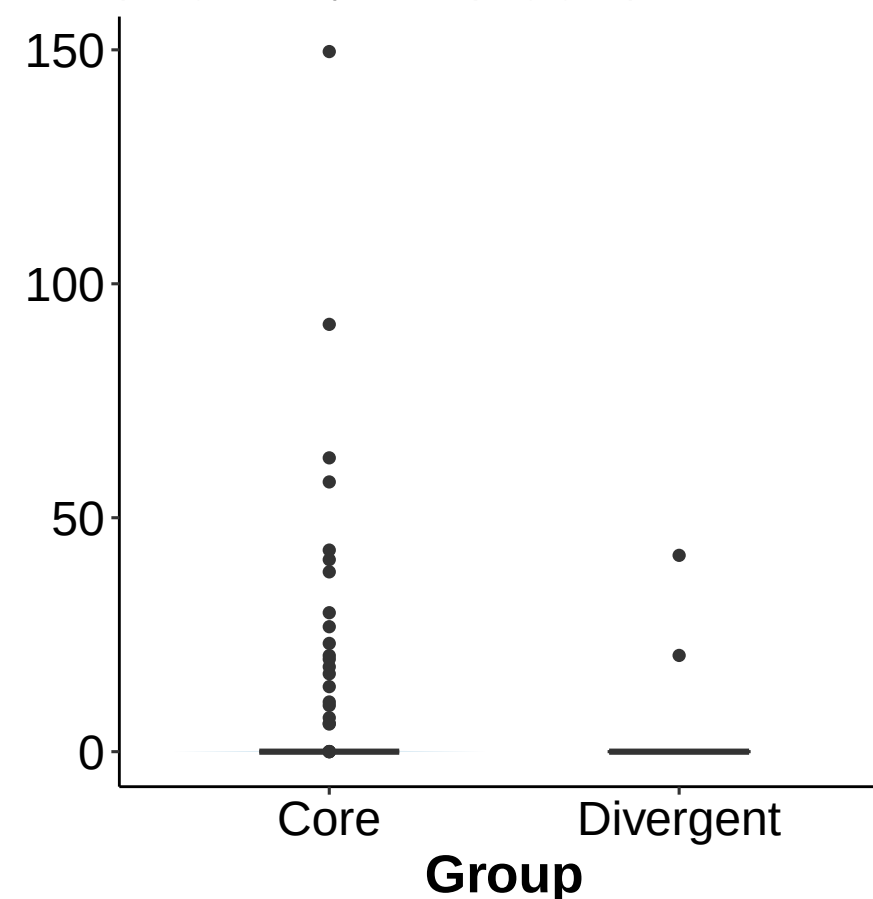
SA_Leukotriene_metabolism_L1w

p-adj: 0.917 | .:=−0.11 (Negligible)



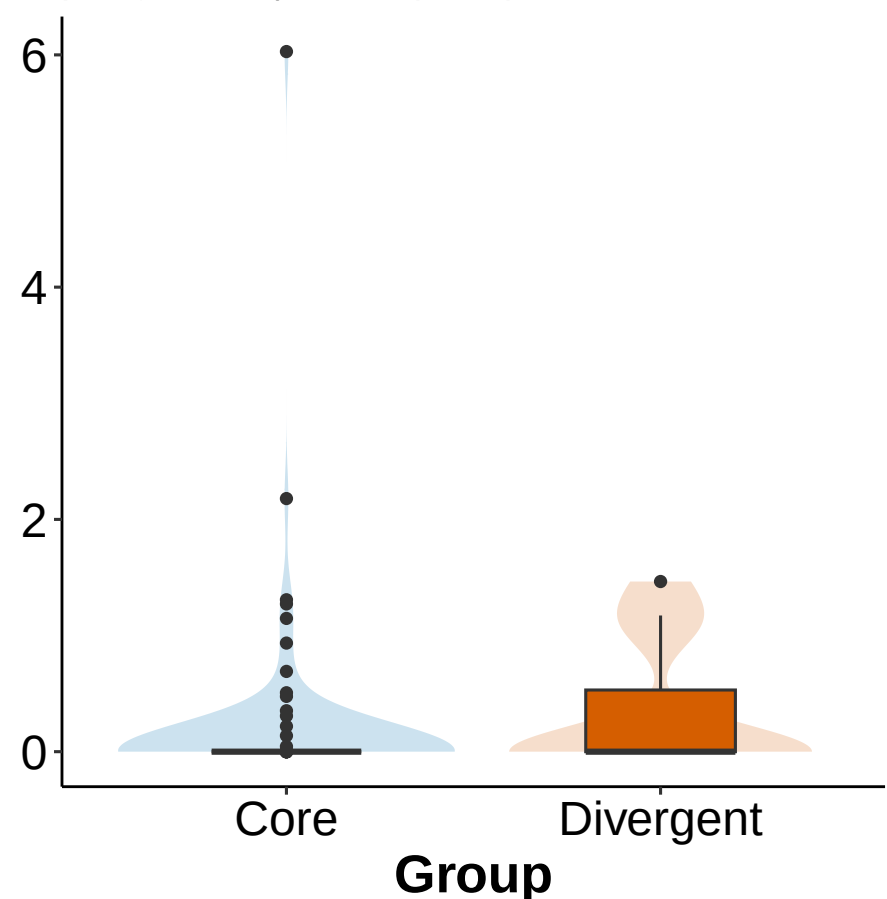
SA_Limonene_and_pinene_degradation_L1w

p-adj: 0.885 | .:=0.14 (Negligible)

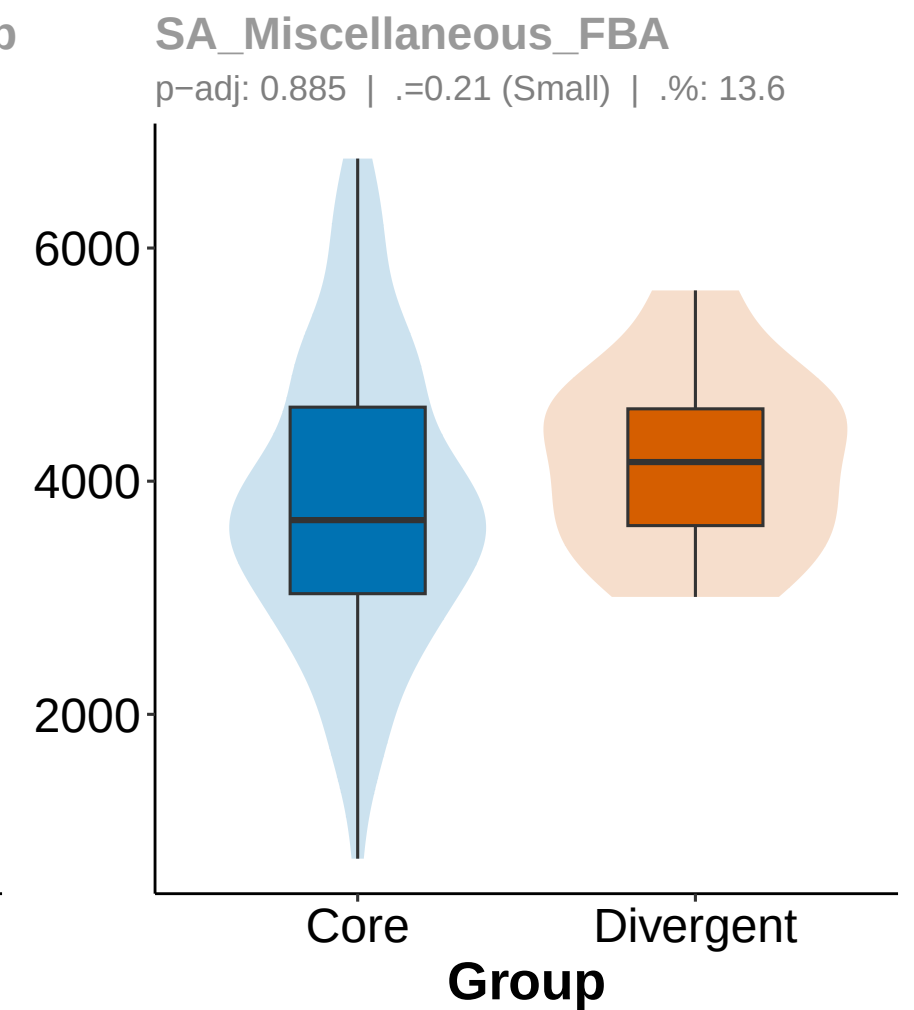
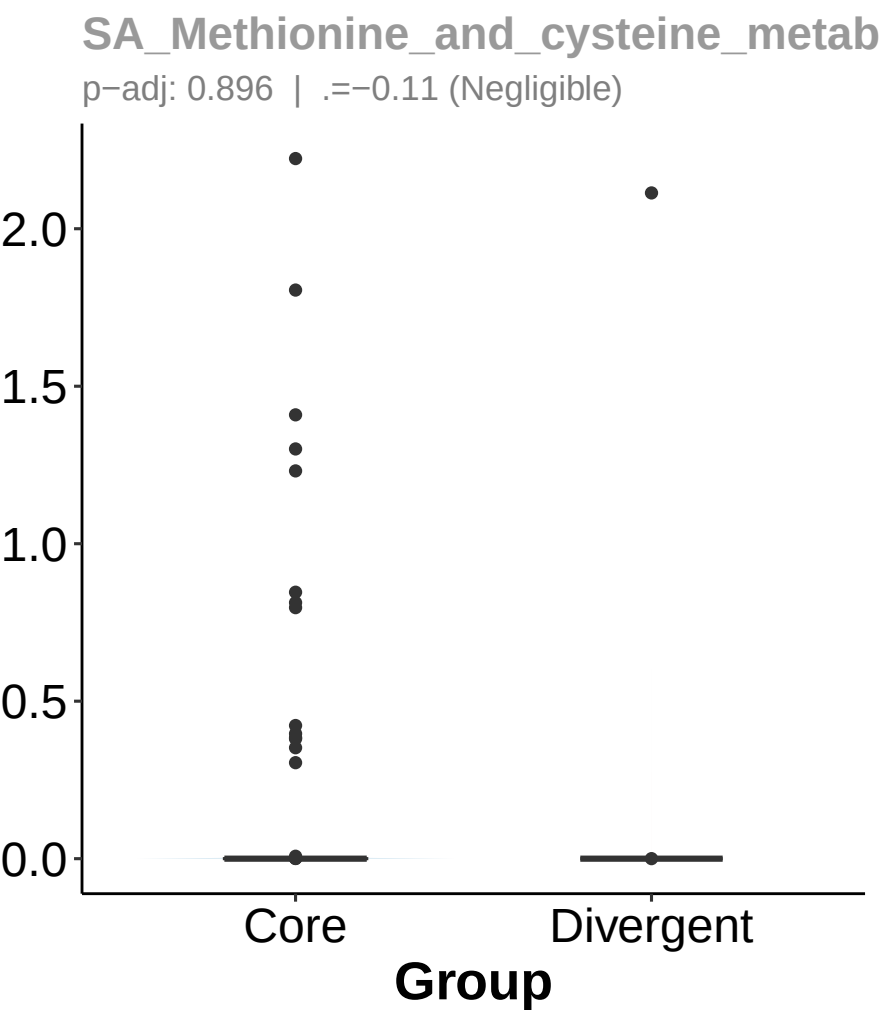
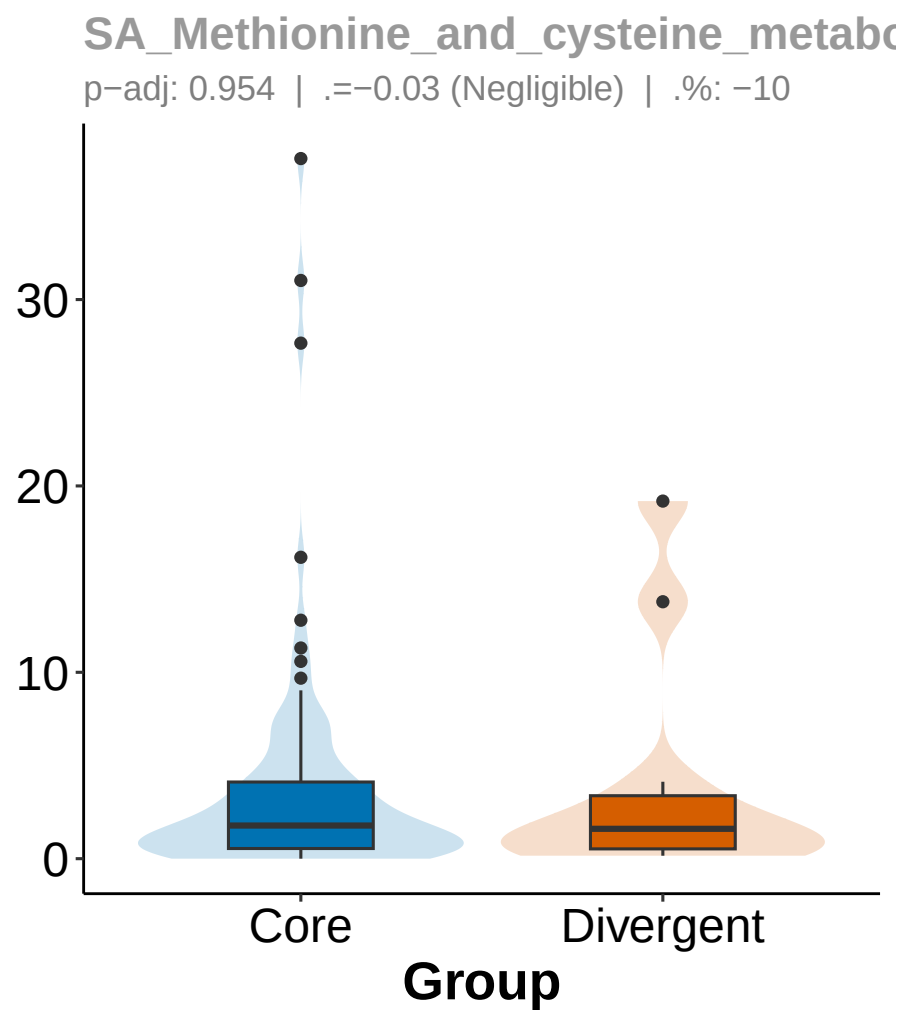
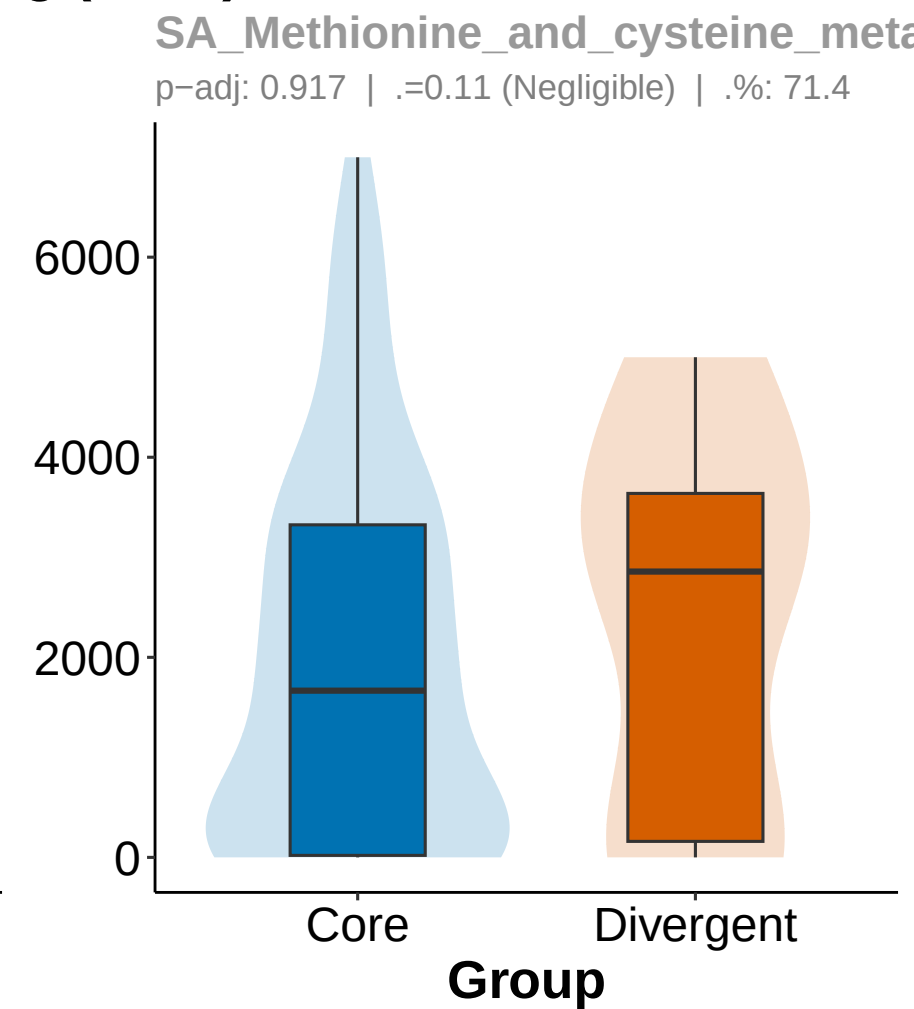
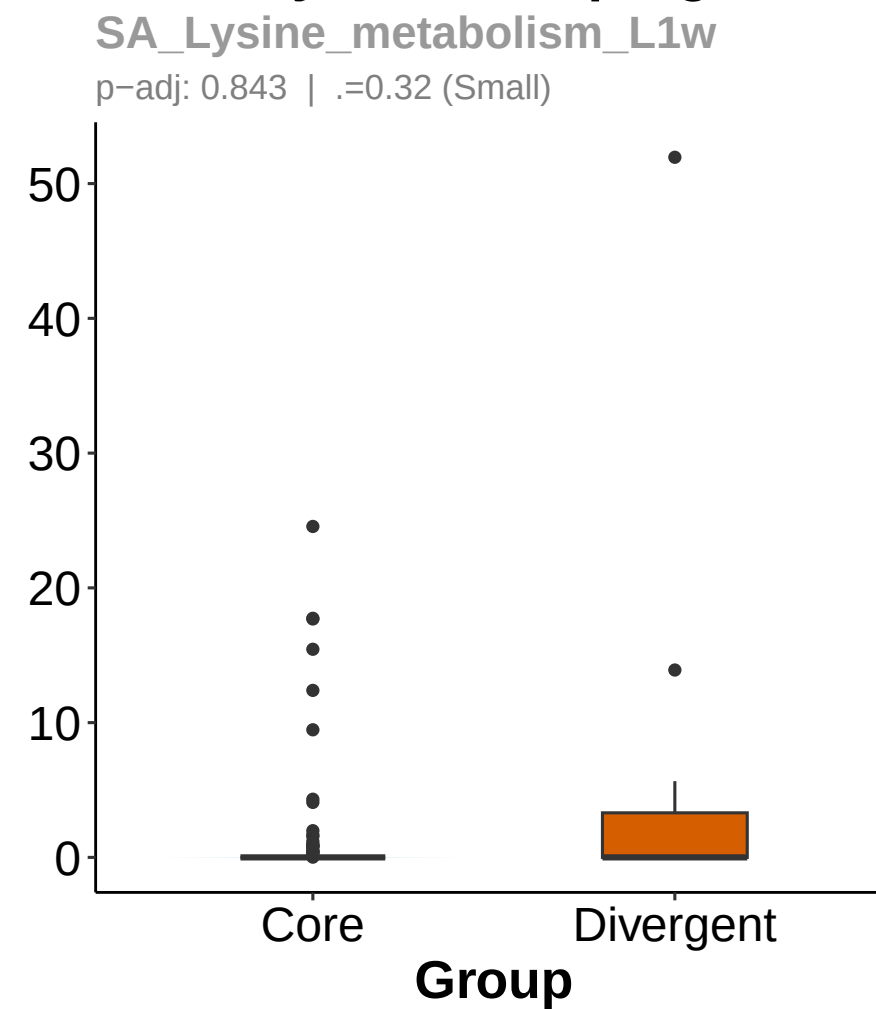
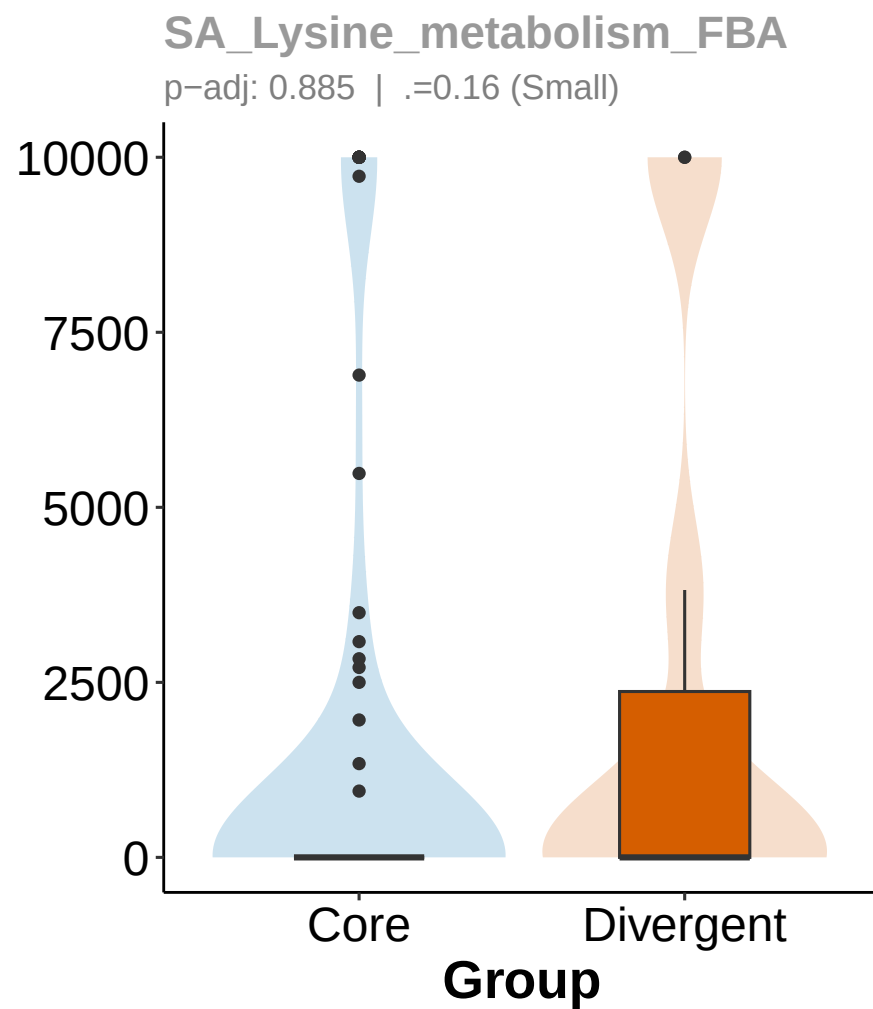


SA_Linoleate_metabolism_L1w

p-adj: 0.885 | .:=0.17 (Small)



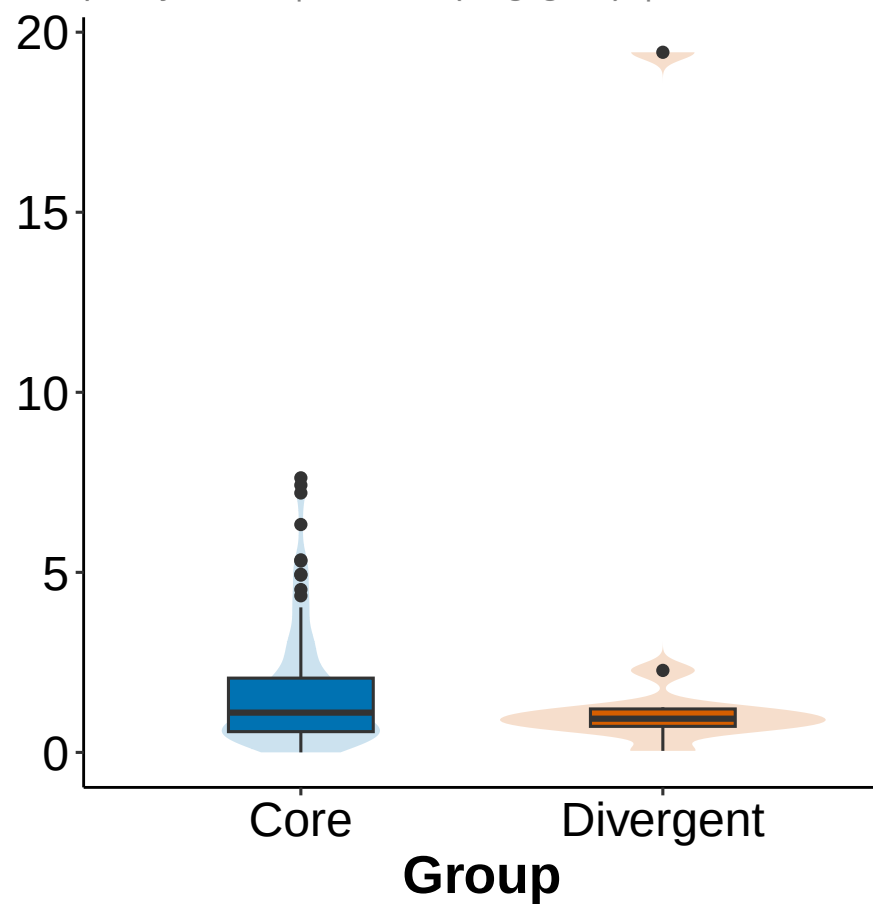
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

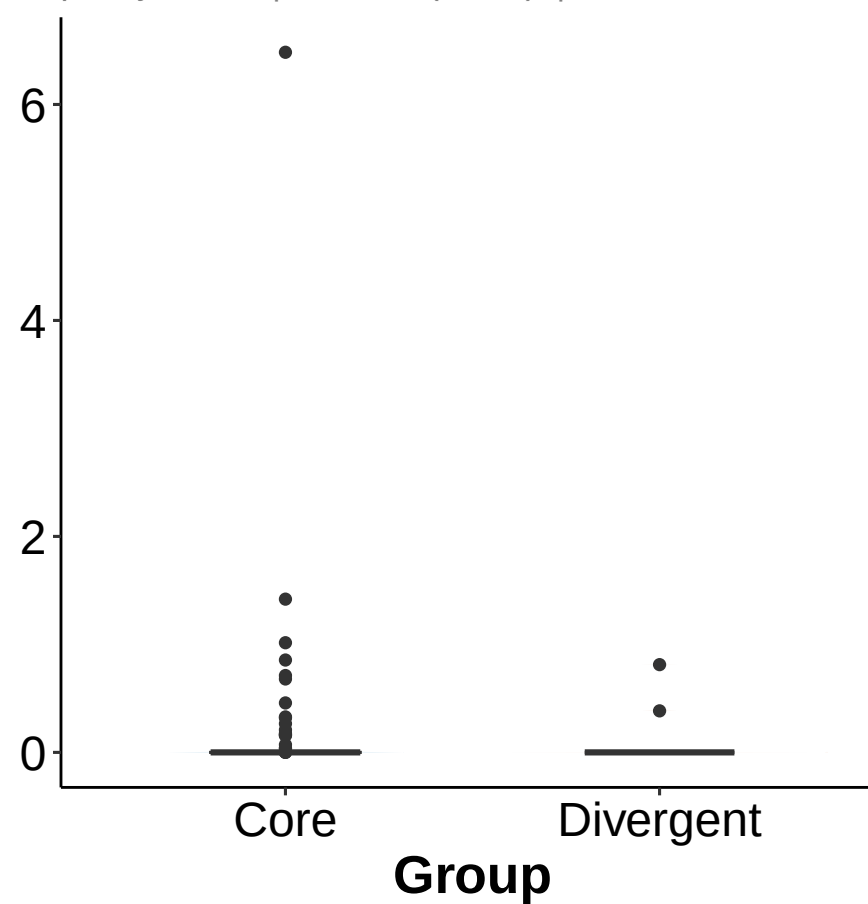
SA_Miscellaneous_L1w

p-adj: 0.954 | .=-0.04 (Negligible) | .%: -15.2



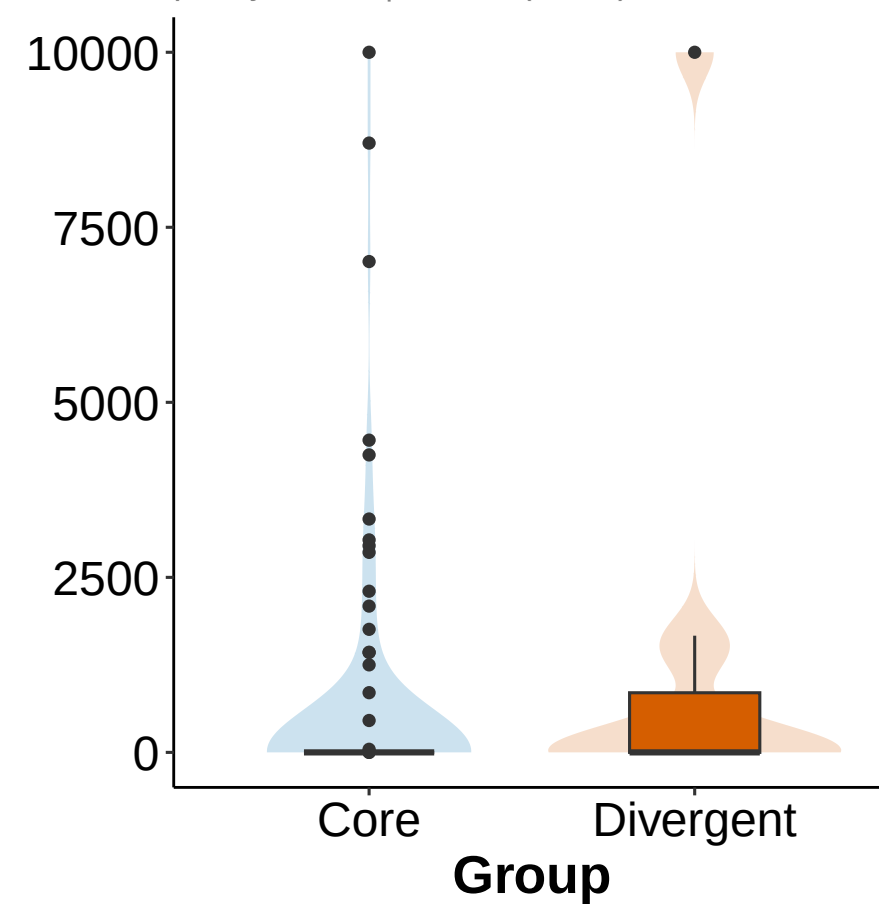
SA_Miscellaneous_pFBA

p-adj: 0.885 | .=-0.18 (Small) | .%: -57.2



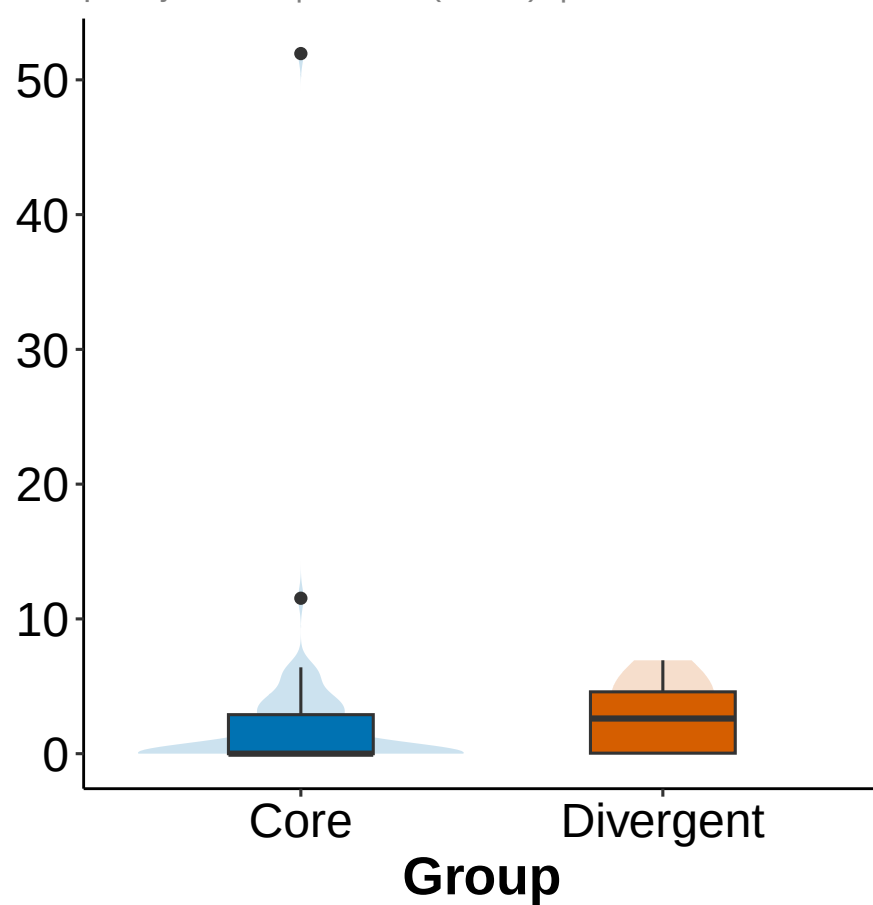
SA_NAD_metabolism_FBA

p-adj: 0.885 | .=0.15 (Small)



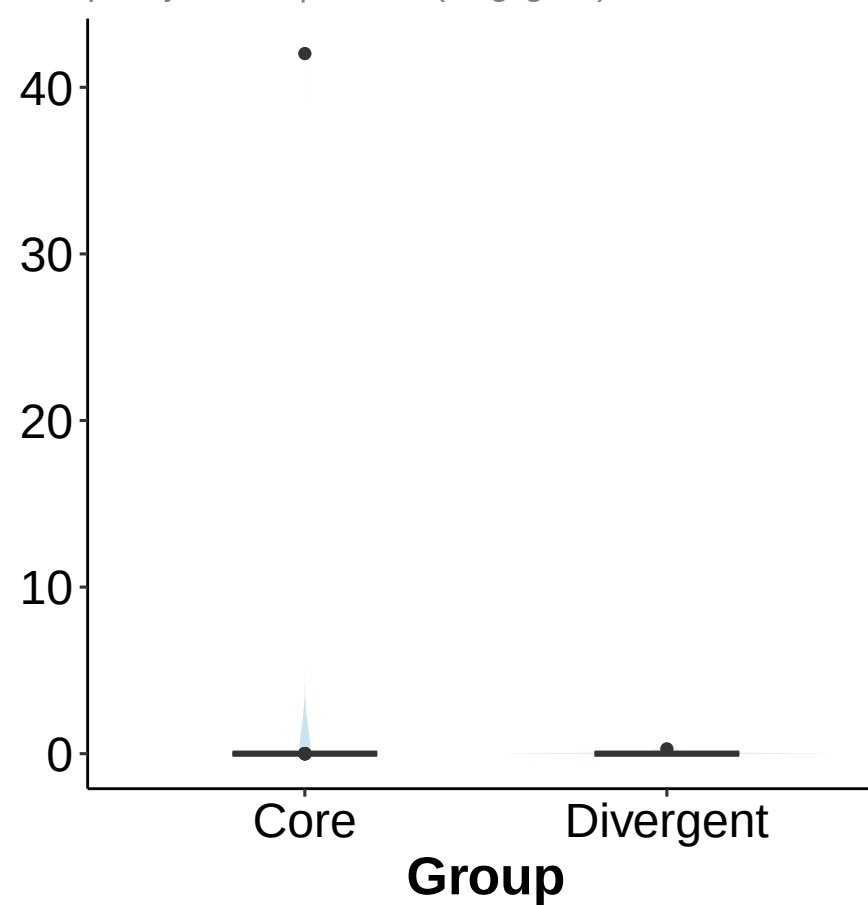
SA_NAD_metabolism_L1w

p-adj: 0.872 | .=0.27 (Small) | .%: 7933817.9



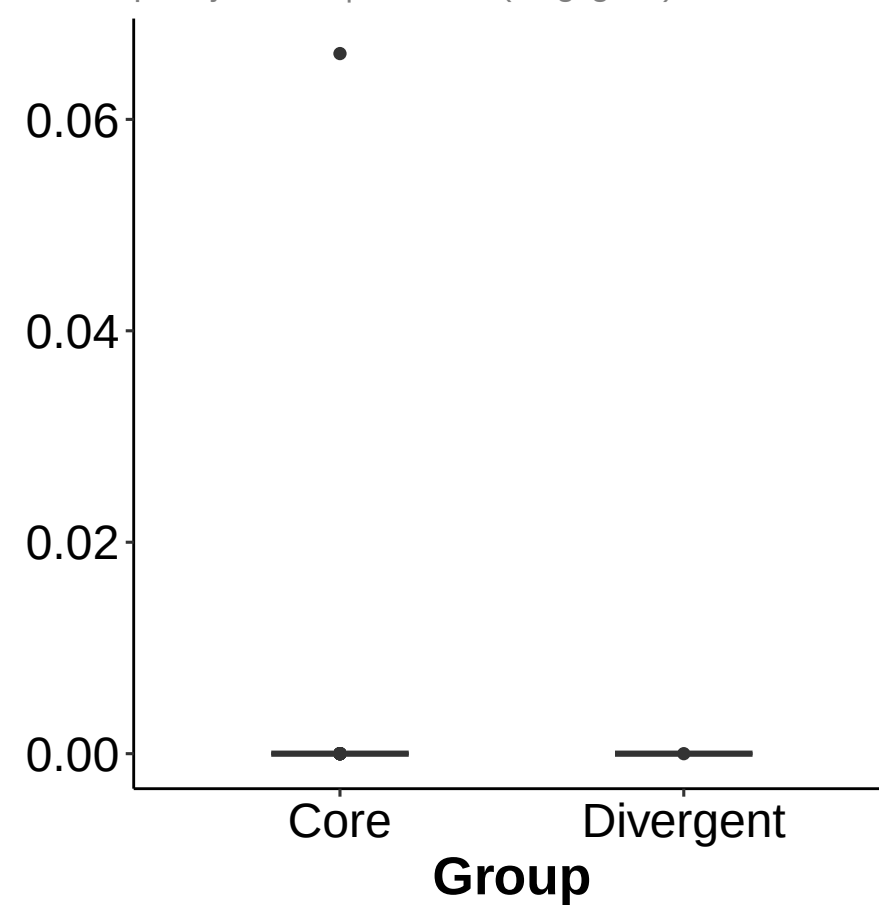
SA_NAD_metabolism_pFBA

p-adj: 0.954 | .=0.02 (Negligible)



SA_N_glycan_synthesis_L1w

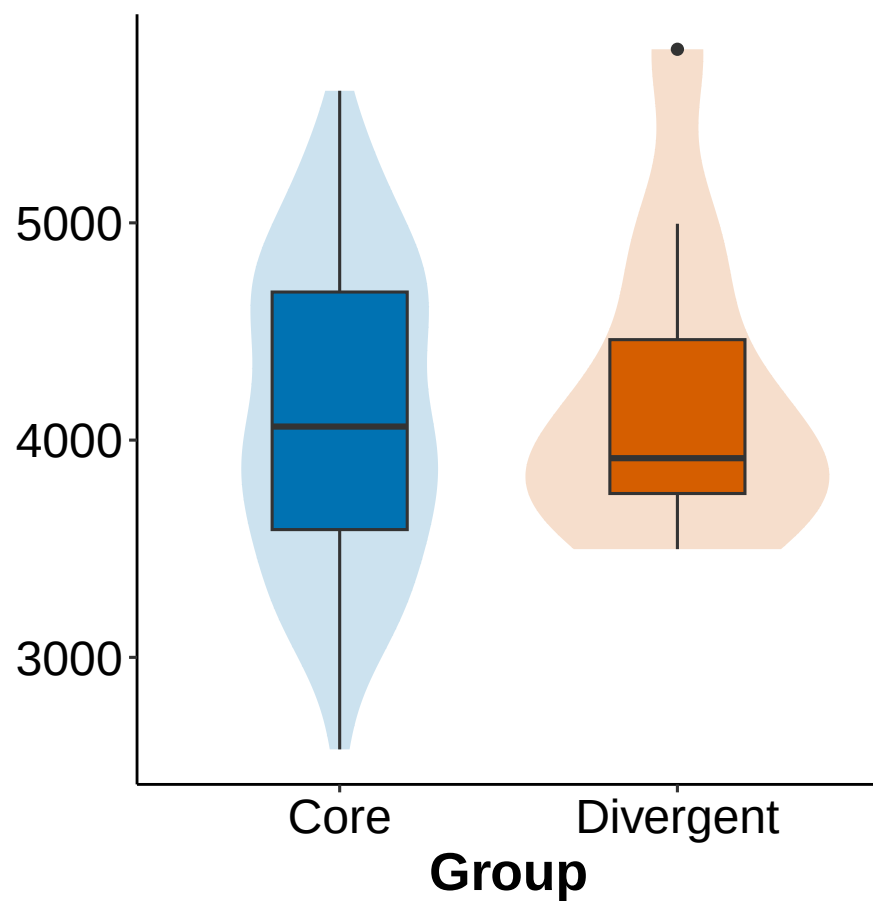
p-adj: 0.974 | .=-0.01 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

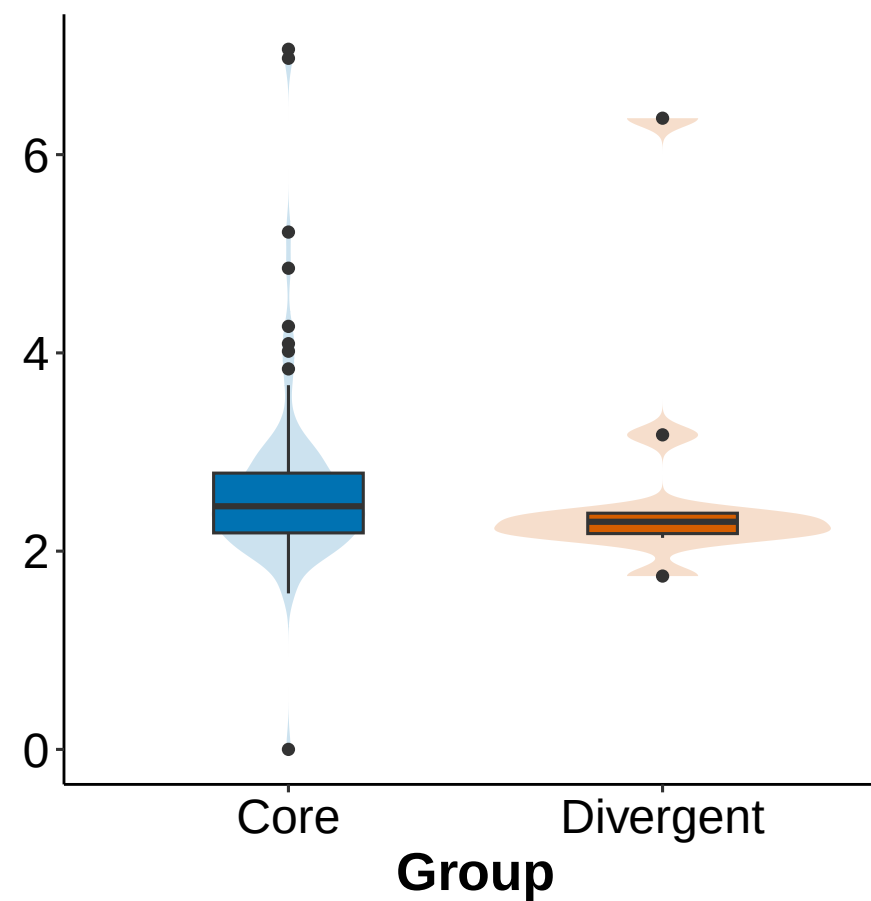
SA_Nucleotide_interconversion_FB

p-adj: 0.954 | . = 0.03 (Negligible) | .%: -3.6



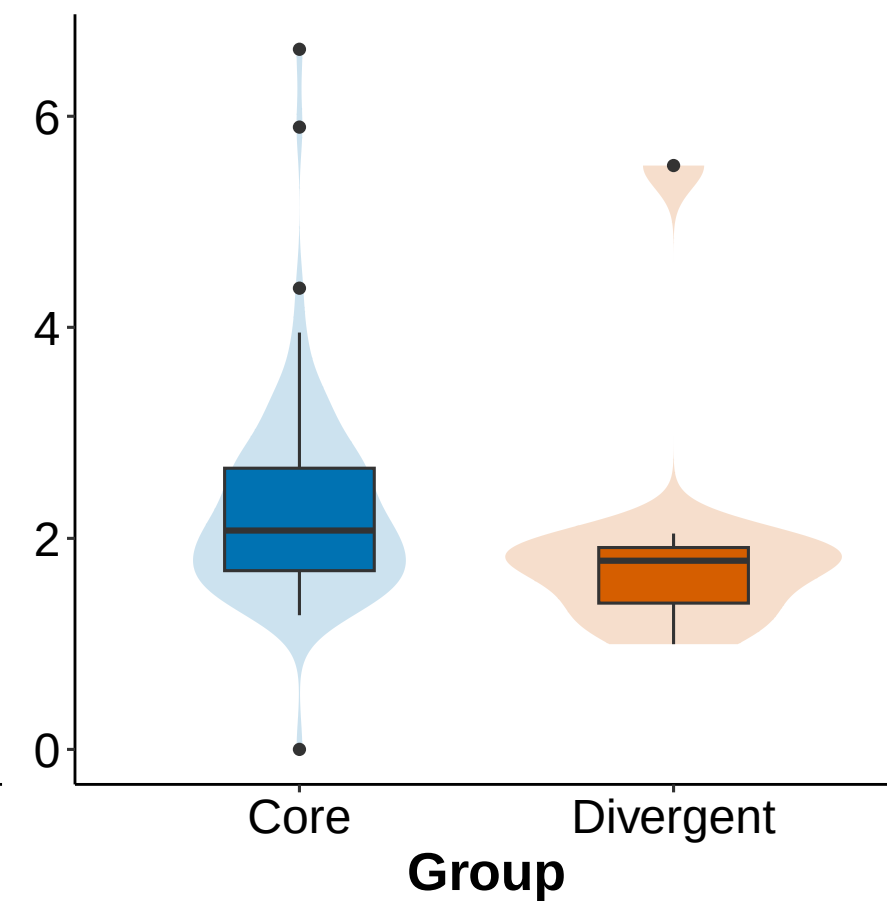
SA_Nucleotide_interconversion_L1w

p-adj: 0.885 | . = -0.18 (Small) | .%: -6.4



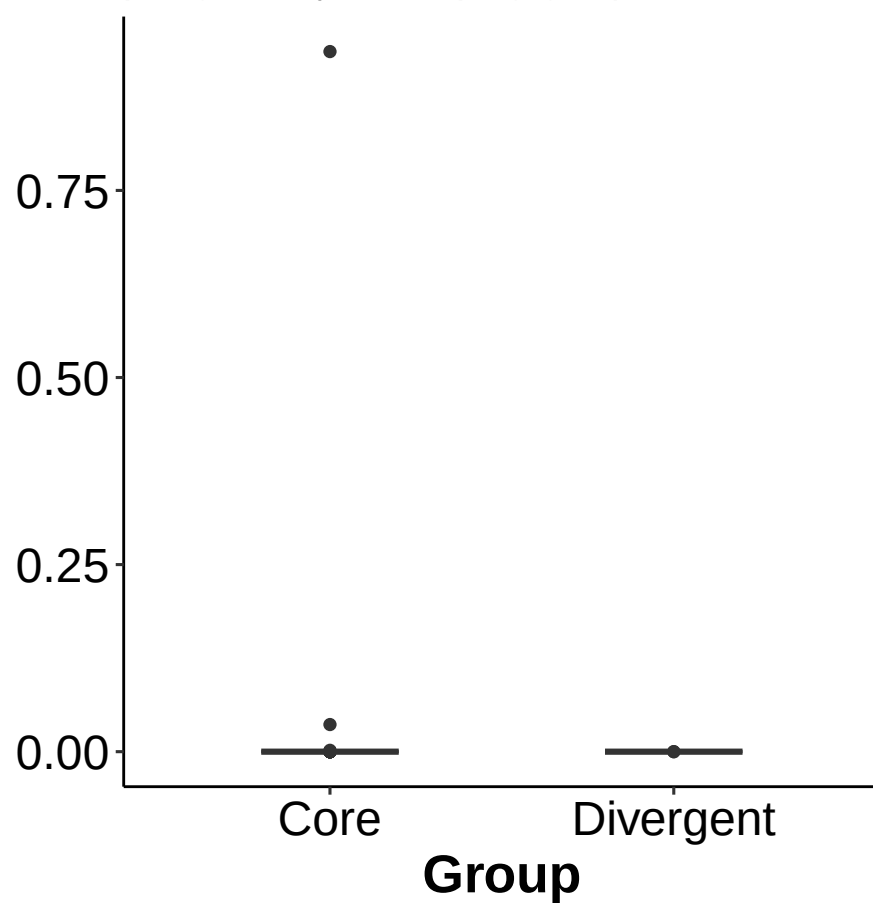
SA_Nucleotide_interconversion_pFBA

p-adj: 0.843 | . = -0.39 (Medium) | .%: -13.8



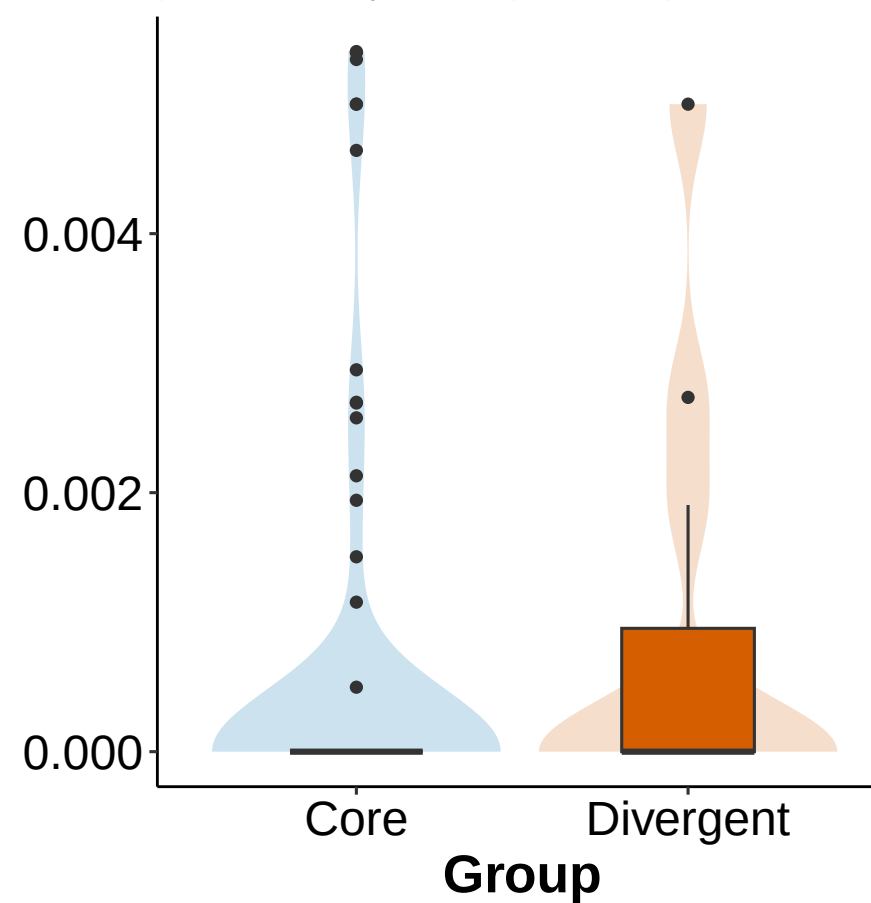
SA_Nucleotide_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible)



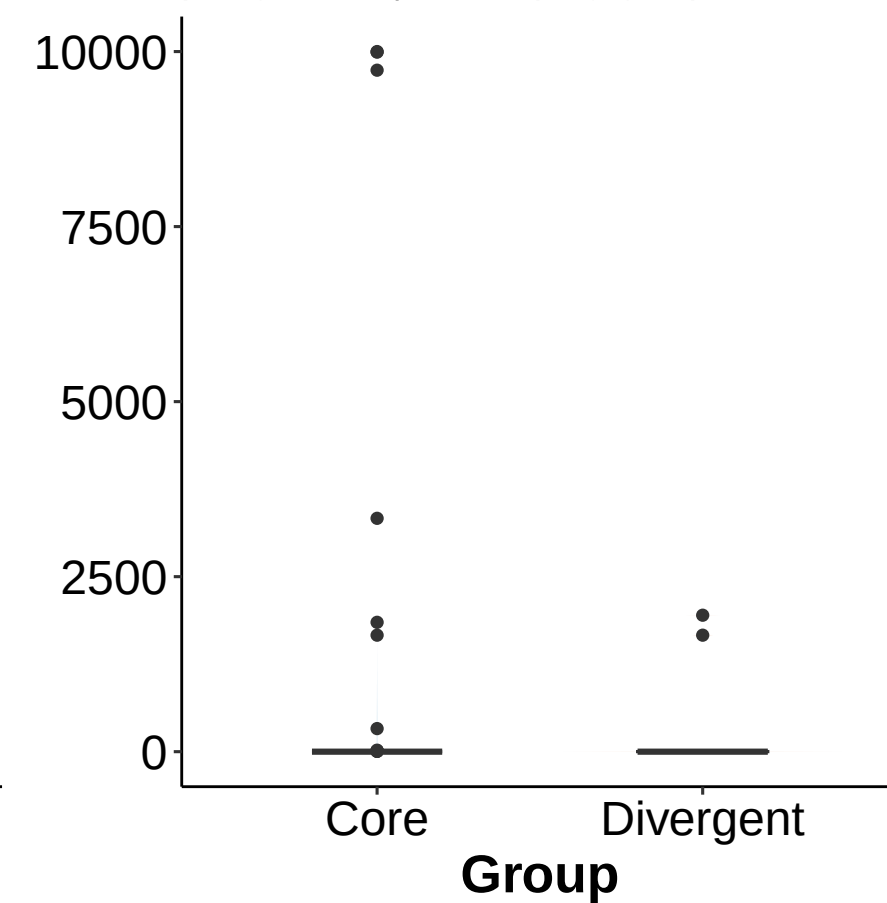
SA_Nucleotide_metabolism_pFBA

p-adj: 0.885 | . = 0.12 (Negligible)



SA_Nucleotide_salvage_pathway_pFBA

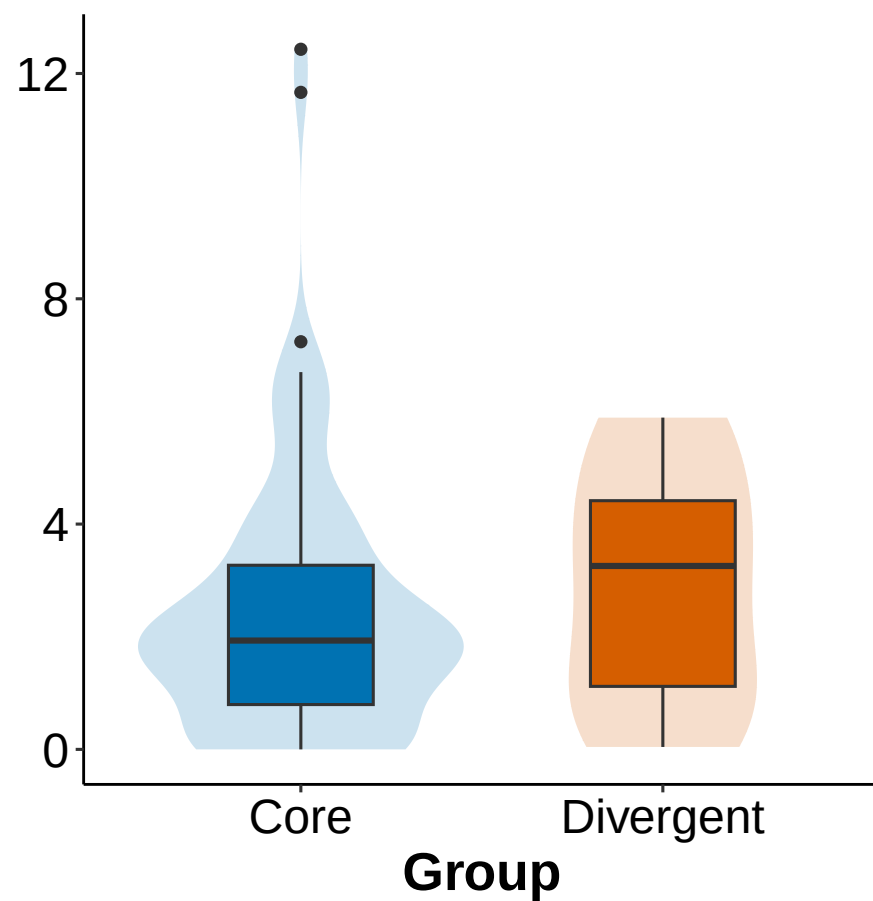
p-adj: 0.885 | . = 0.13 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

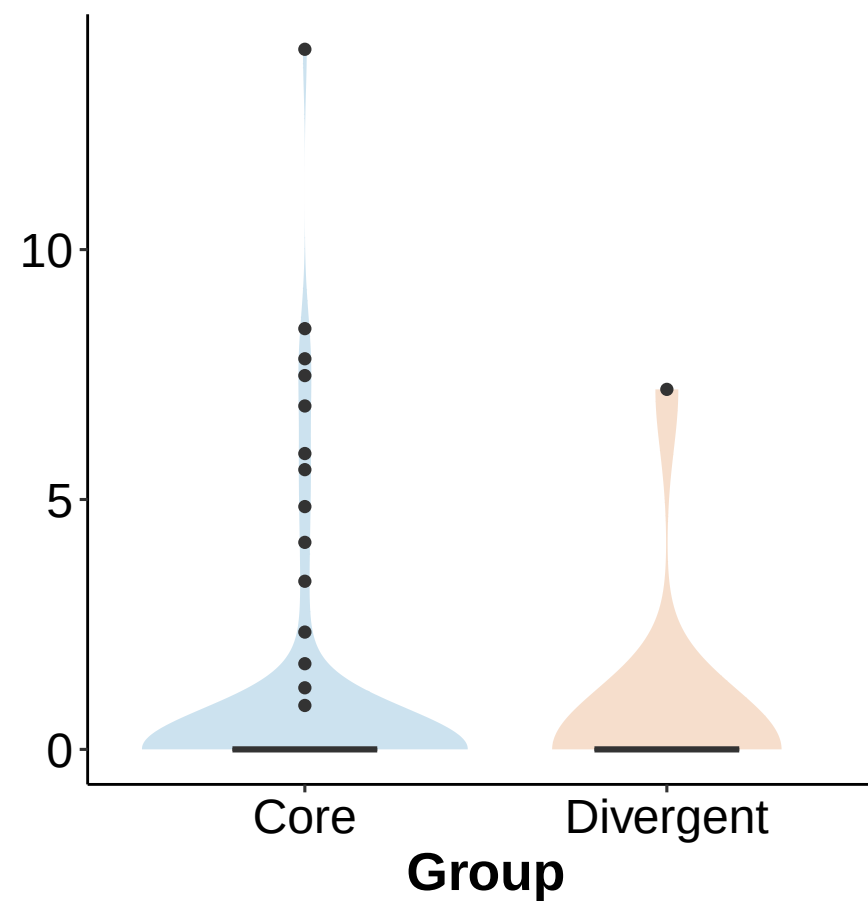
SA_Nucleotide_salvage_pathway_L1v

p-adj: 0.885 | .:=0.17 (Small) | .%: 68.5



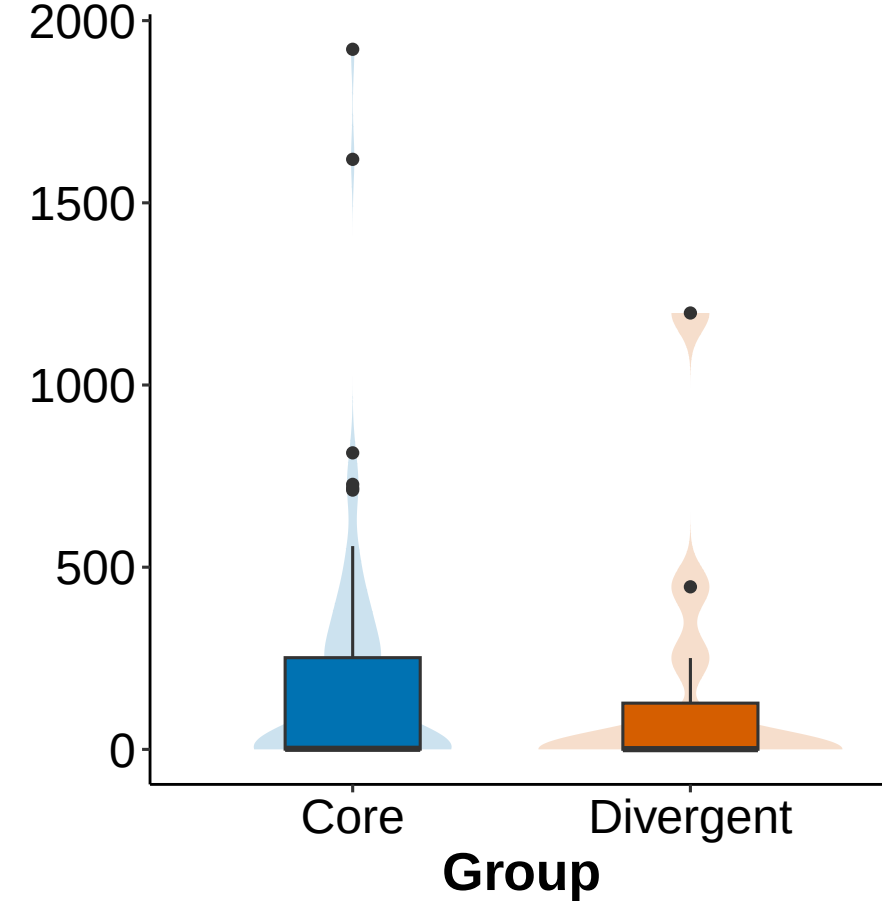
SA_Nucleotide_salvage_pathway_pFE

p-adj: 0.95 | .:=−0.04 (Negligible)



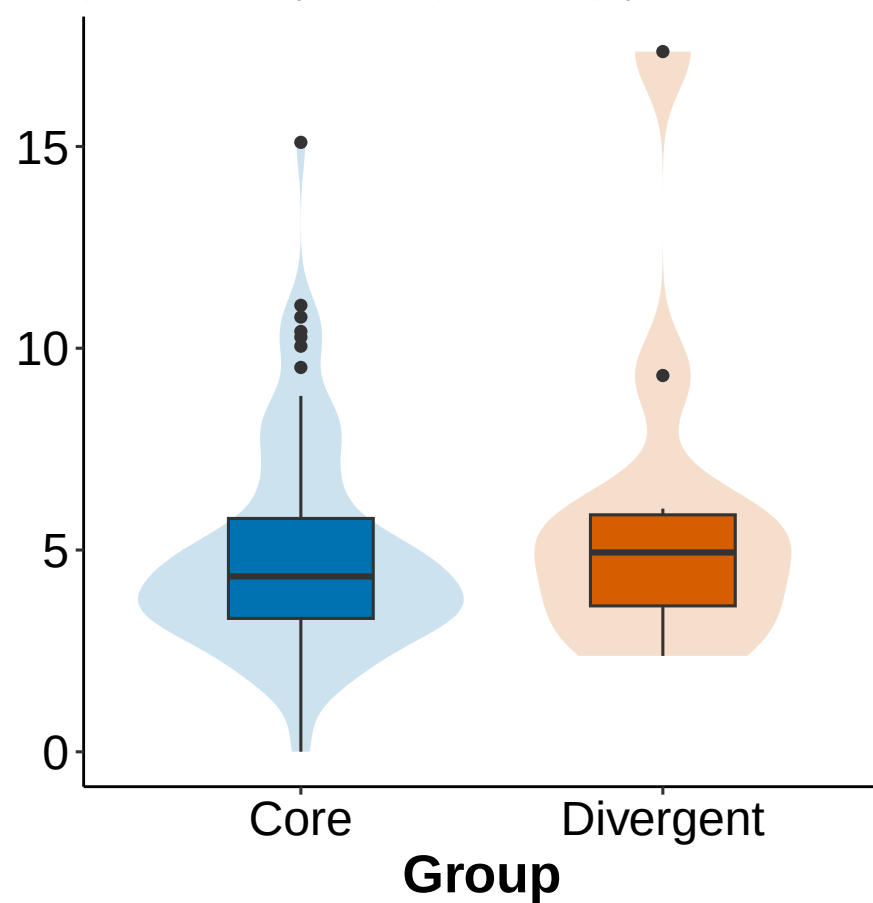
SA_Oxidative_phosphorylation_FB

p-adj: 0.885 | .:=−0.16 (Small) | .%: −59.2



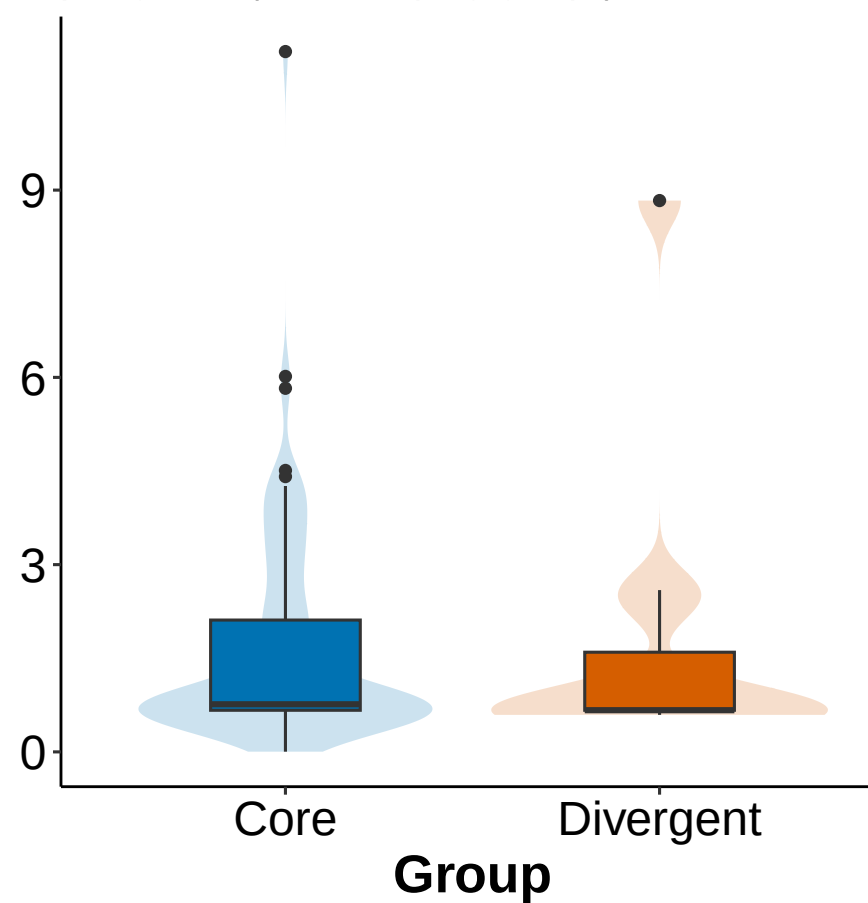
SA_Oxidative_phosphorylation_L1w

p-adj: 0.903 | .:=0.13 (Negligible) | .%: 13.7



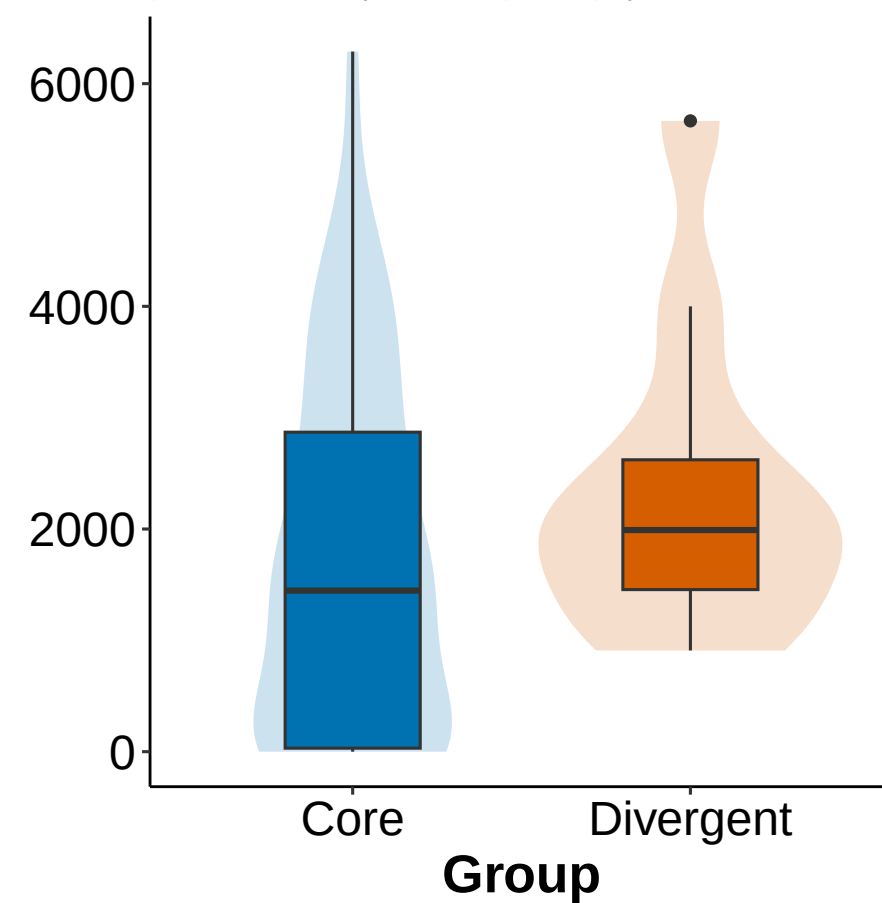
SA_Oxidative_phosphorylation_pFBA

p-adj: 0.95 | .:=−0.09 (Negligible) | .%: −12.4



SA_Pentose_phosphate_pathway_L1w

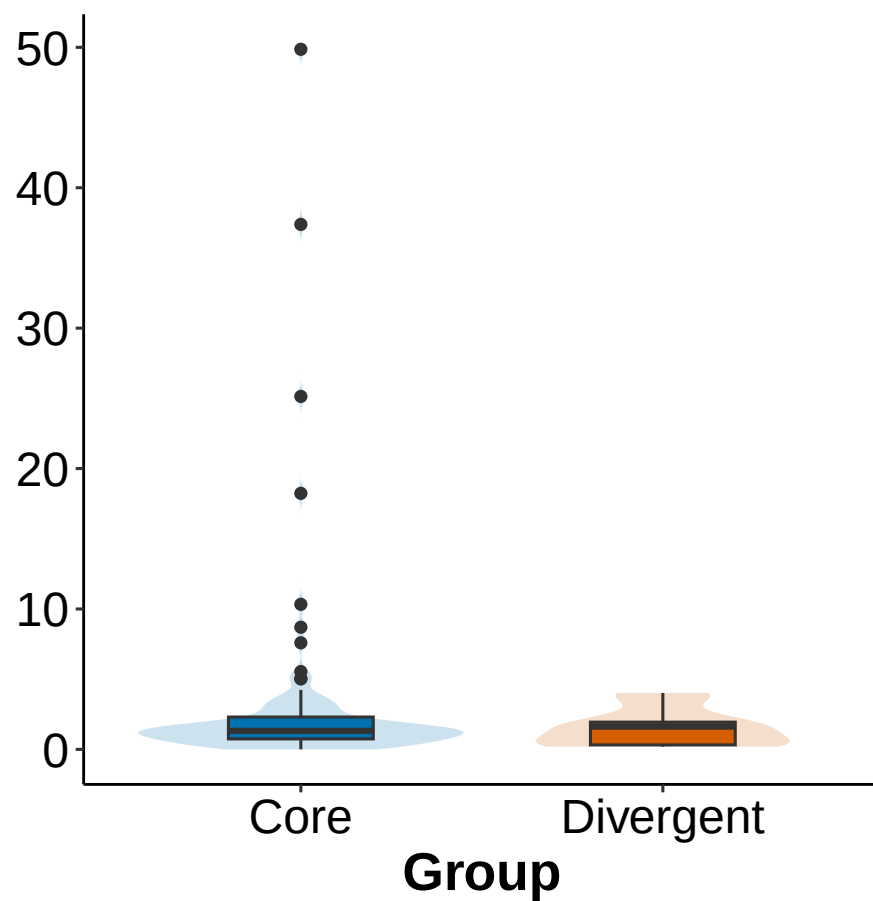
p-adj: 0.872 | .:=0.26 (Small) | .%: 37.6



Metabolism – . 5. Systemic Reprogramming (SA_*)

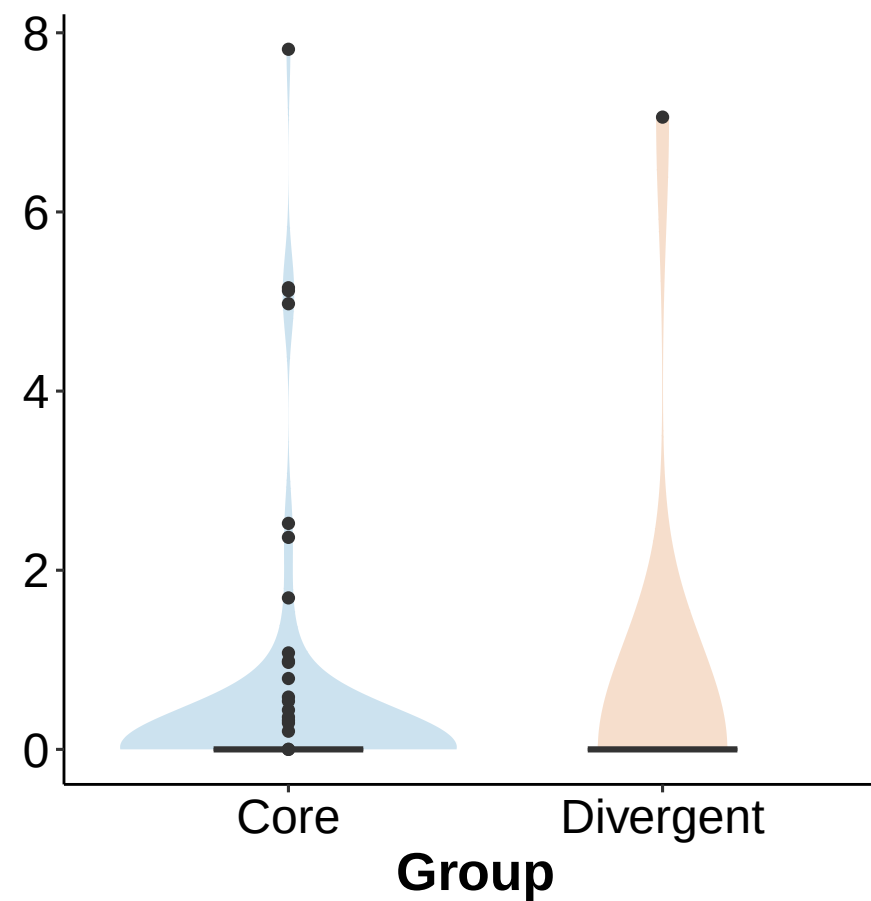
SA_Pentose_phosphate_pathway_L1w

p-adj: 0.928 | .=-0.1 (Negligible) | .%: 22.6



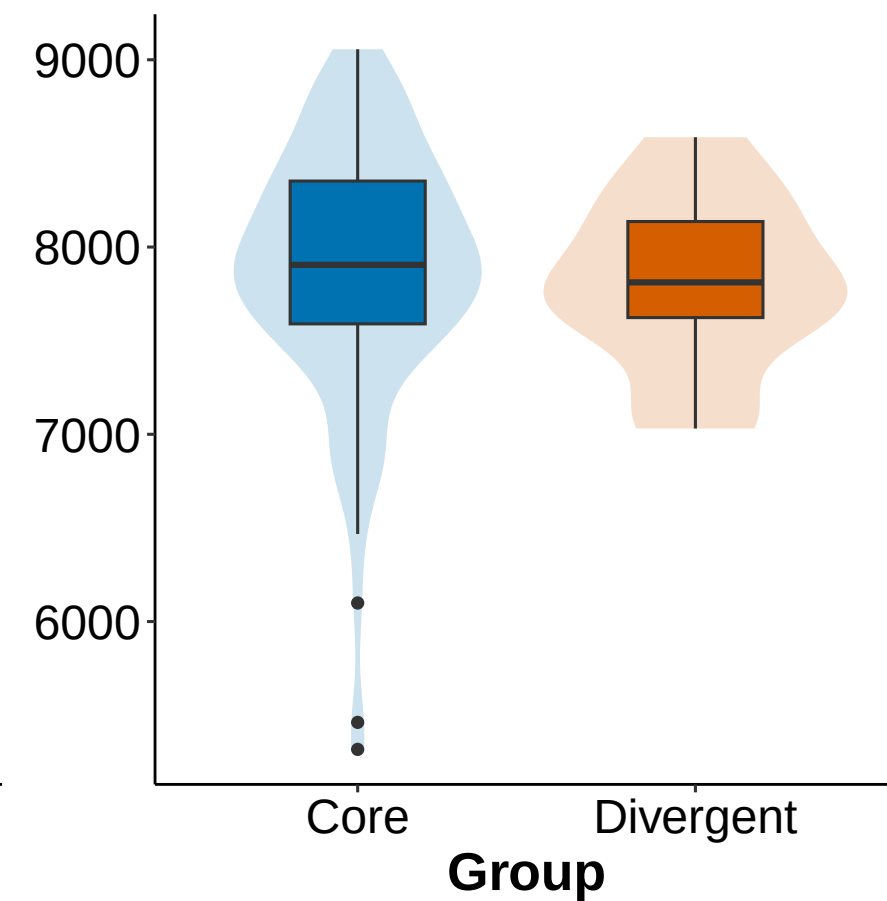
SA_Pentose_phosphate_pathway_pFB

p-adj: 0.896 | .=-0.09 (Negligible)



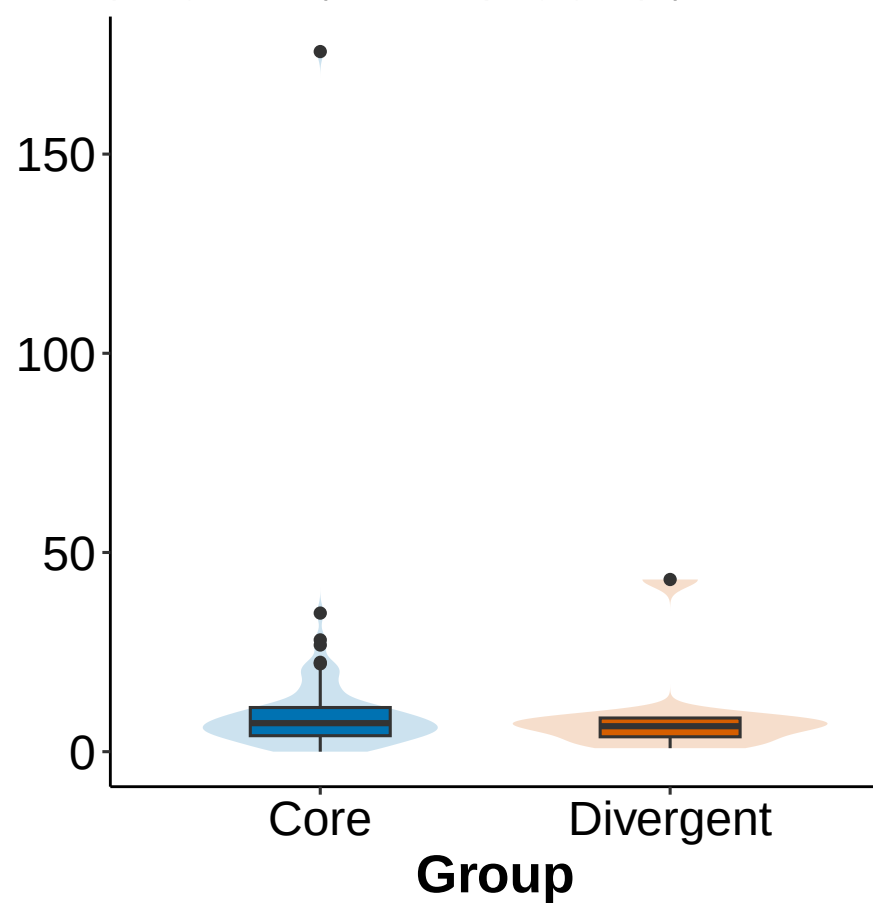
SA_Peptide_metabolism_FBA

p-adj: 0.917 | .=-0.13 (Negligible) | .%: -1.2



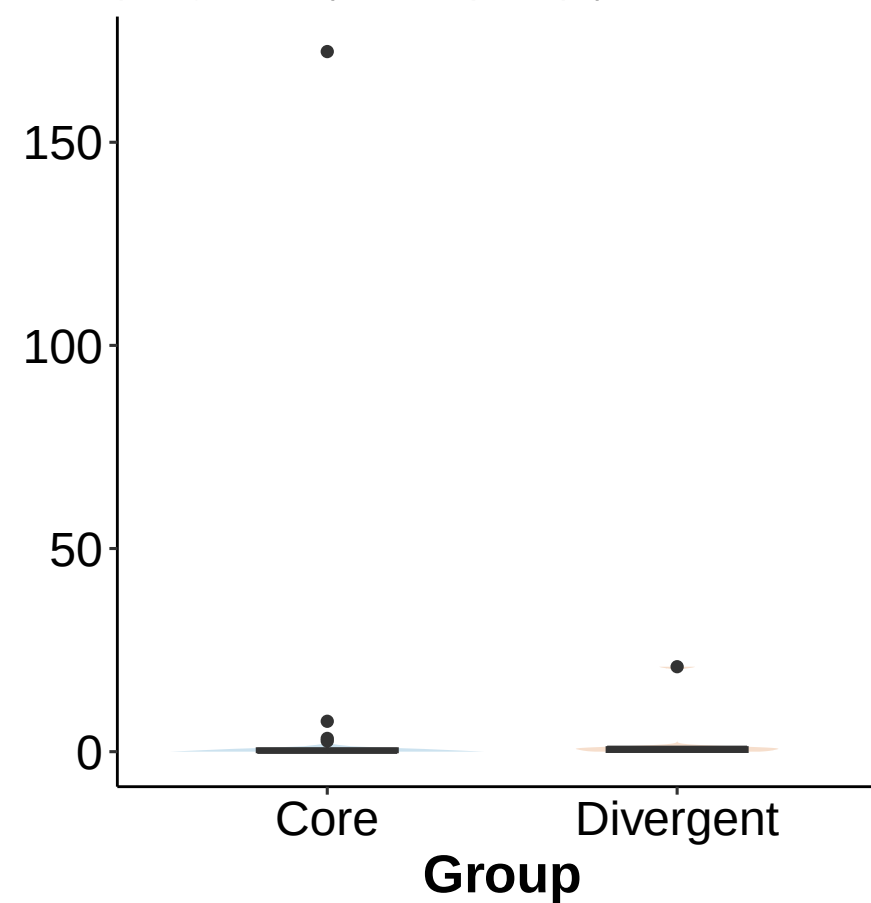
SA_Peptide_metabolism_L1w

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -10.7



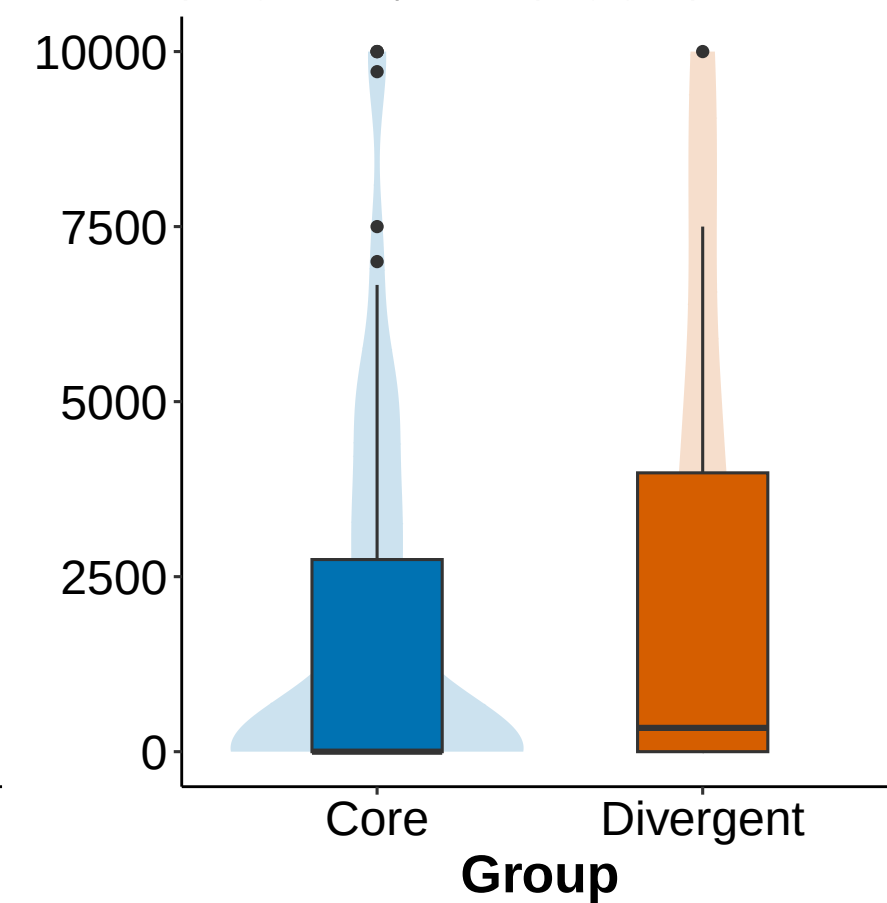
SA_Peptide_metabolism_pFBA

p-adj: 0.843 | . =0.31 (Small) | .%: 233.7



SA_Phenylalanine_metabolism_FBA

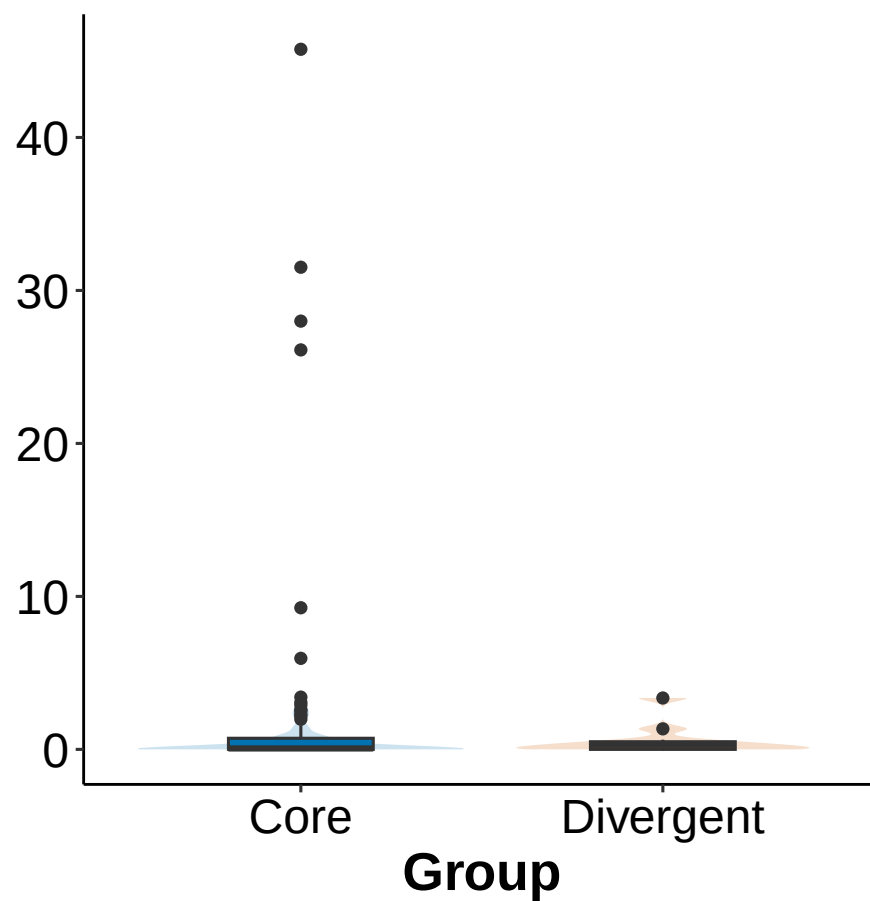
p-adj: 0.885 | . =0.14 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

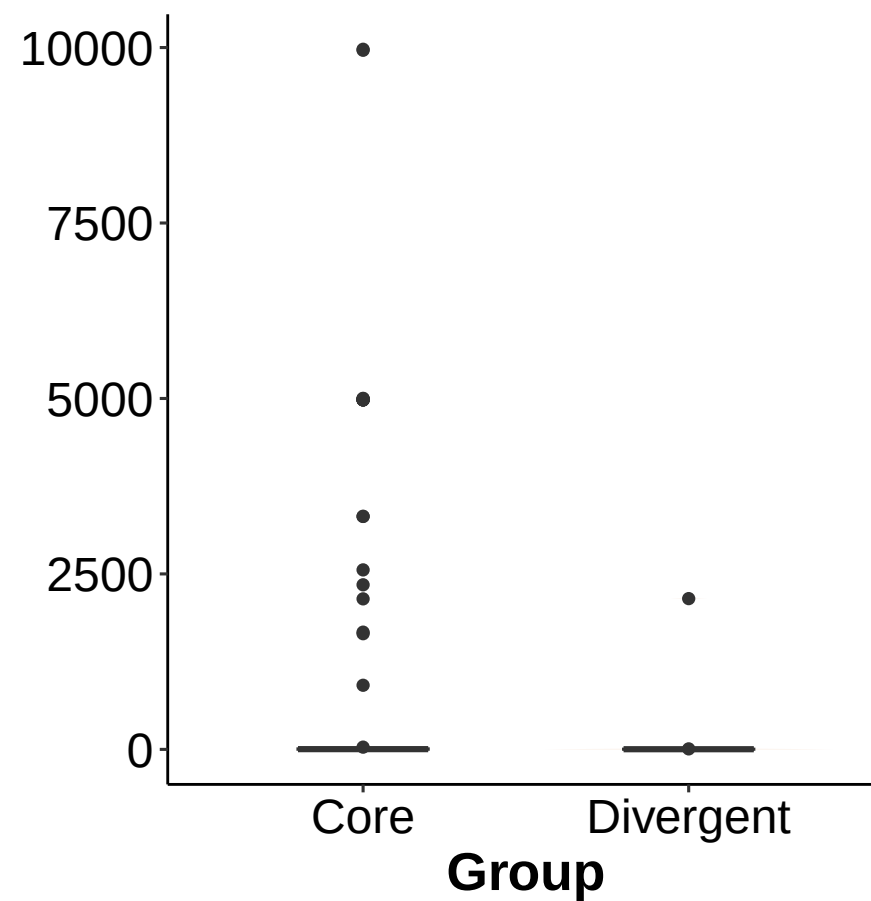
SA_Phenylalanine_metabolism_L1w

p-adj: 0.896 | .:=0.14 (Negligible) | .%: 390.3



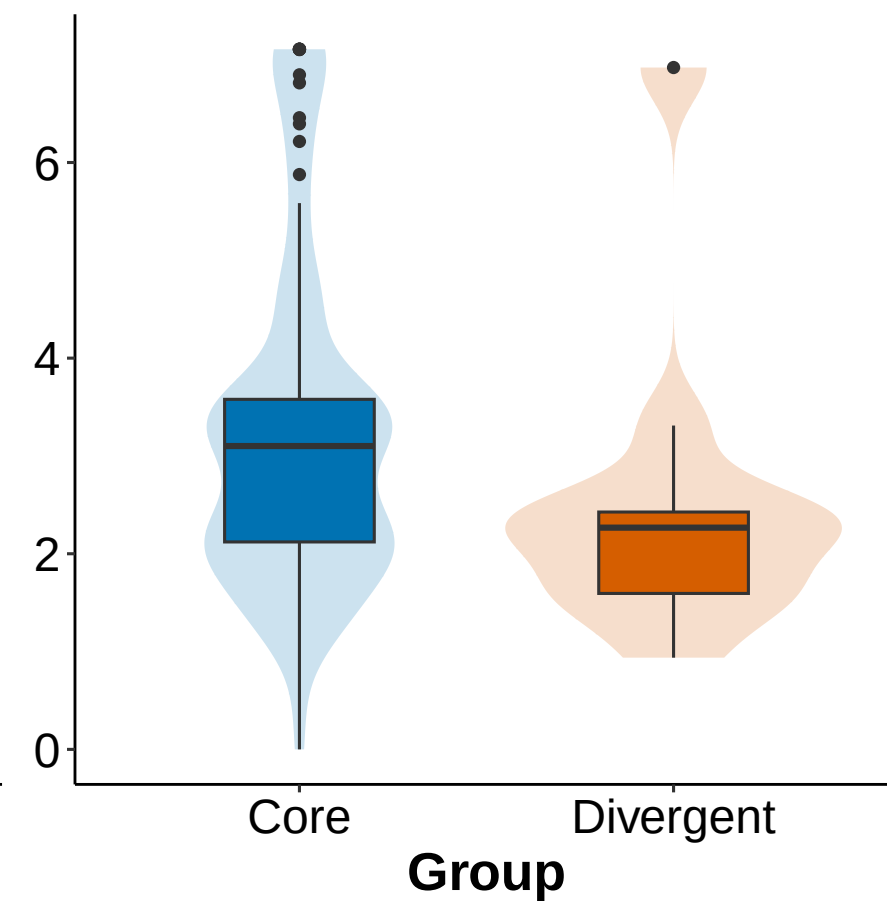
SA_Phosphatidylinositol_phosphatidylcholine_m

p-adj: 0.843 | .:=−0.36 (Medium) | .%: −33.3



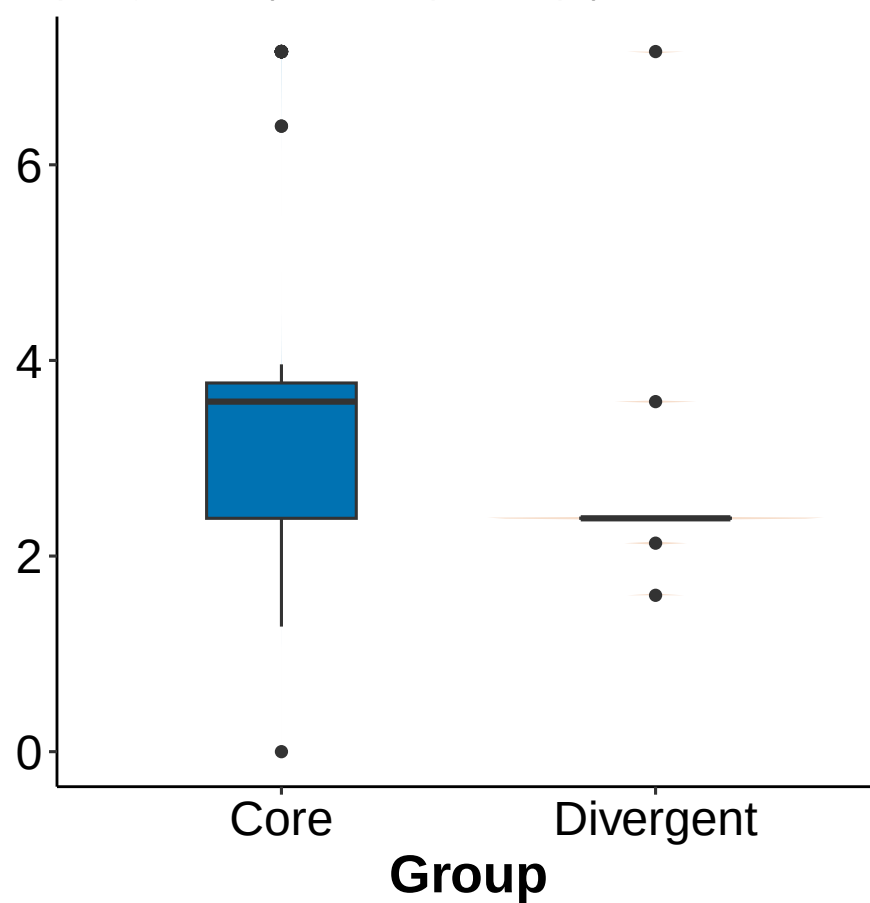
SA_Phosphatidylinositol_phosphate_m

p-adj: 0.872 | .:=−0.31 (Small) | .%: −26.9



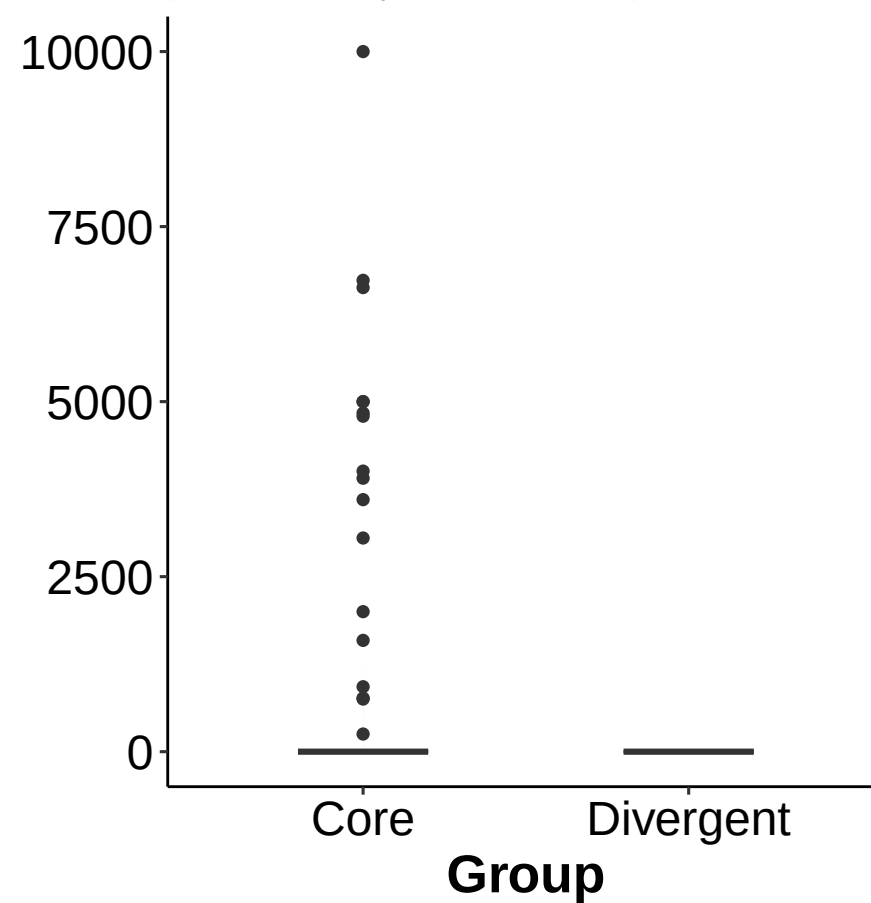
SA_Phosphatidylinositol_phosphate_n

p-adj: 0.843 | .:=−0.39 (Medium) | .%: −33.3



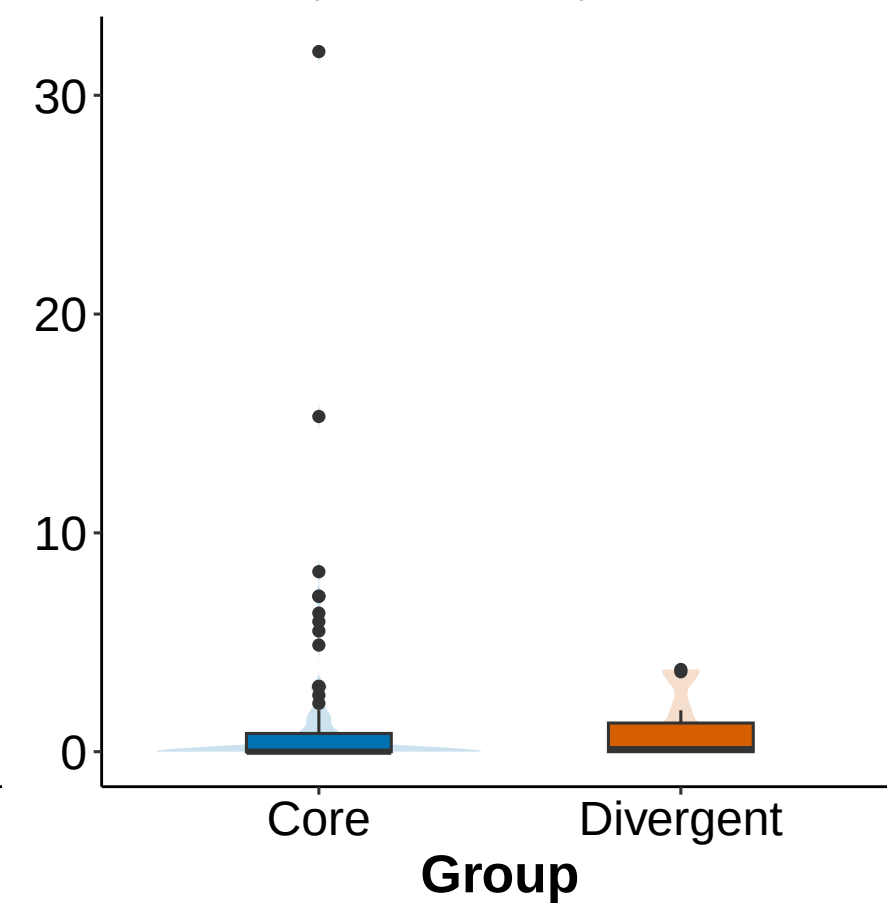
SA_Propanoate_metabolism_FBA

p-adj: 0.872 | .:=−0.17 (Small)



SA_Propanoate_metabolism_L1w

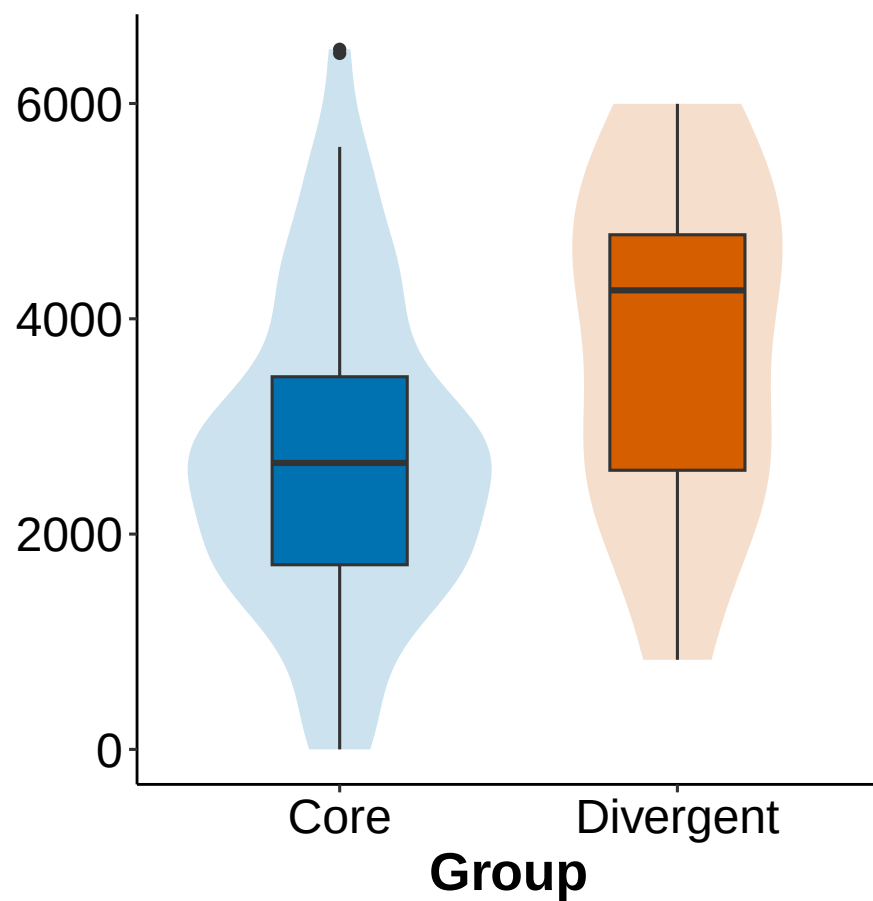
p-adj: 0.885 | .:=0.15 (Small) | .%: 1578181497152



Metabolism – . 5. Systemic Reprogramming (SA_*)

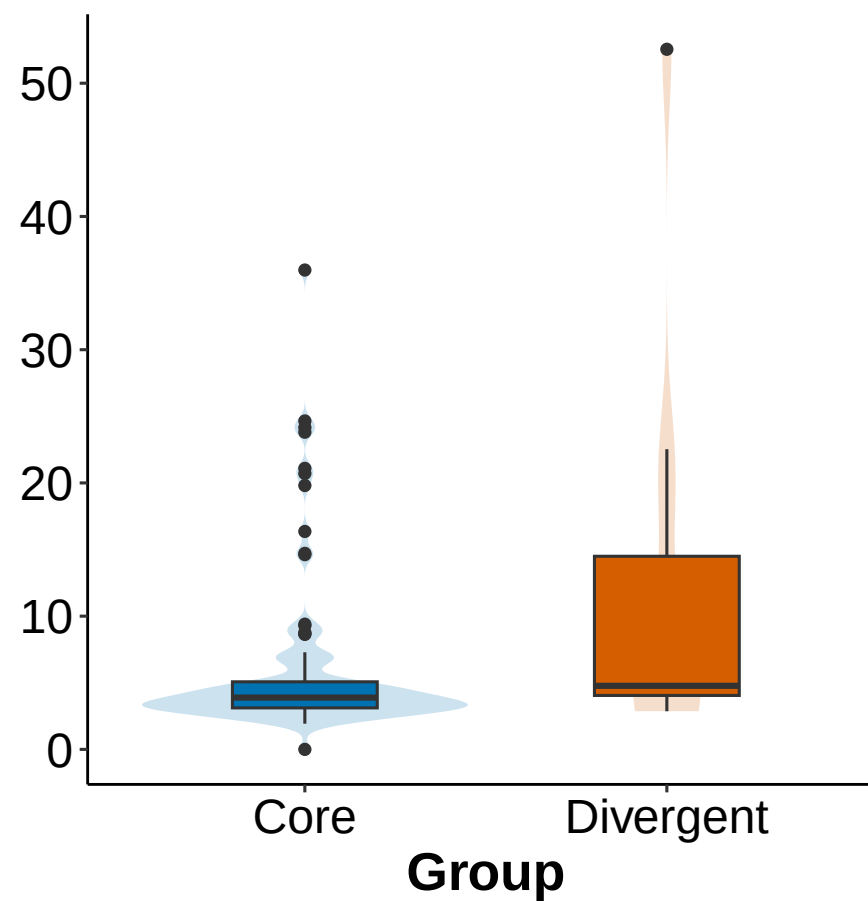
SA_Purine_catabolism_FBA

p-adj: 0.843 | .=0.36 (Medium) | .%: 60.3



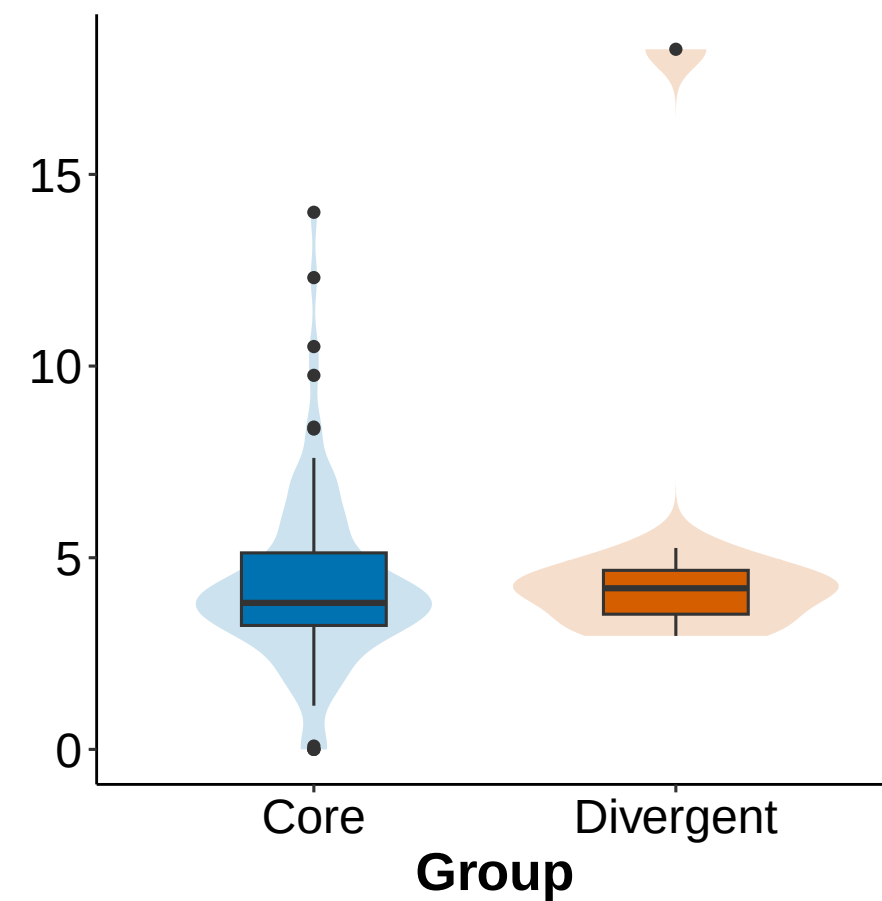
SA_Purine_catabolism_L1w

p-adj: 0.843 | .=0.35 (Medium) | .%: 22.7



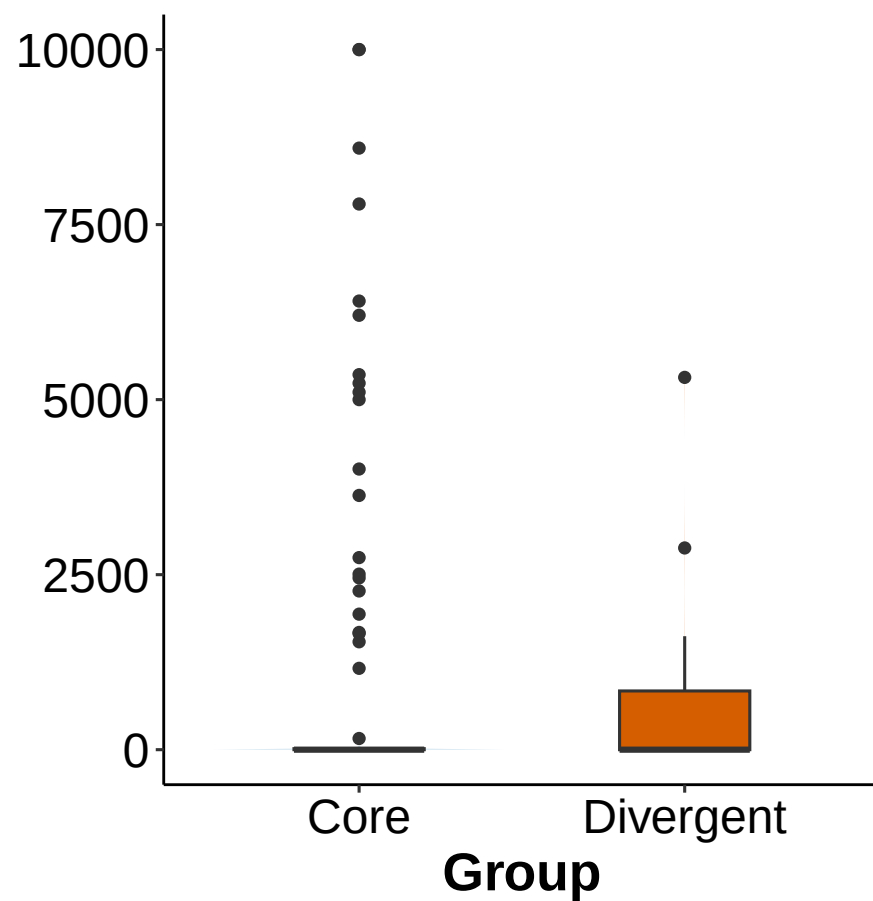
SA_Purine_catabolism_pFBA

p-adj: 0.917 | .=0.12 (Negligible) | .%: 10



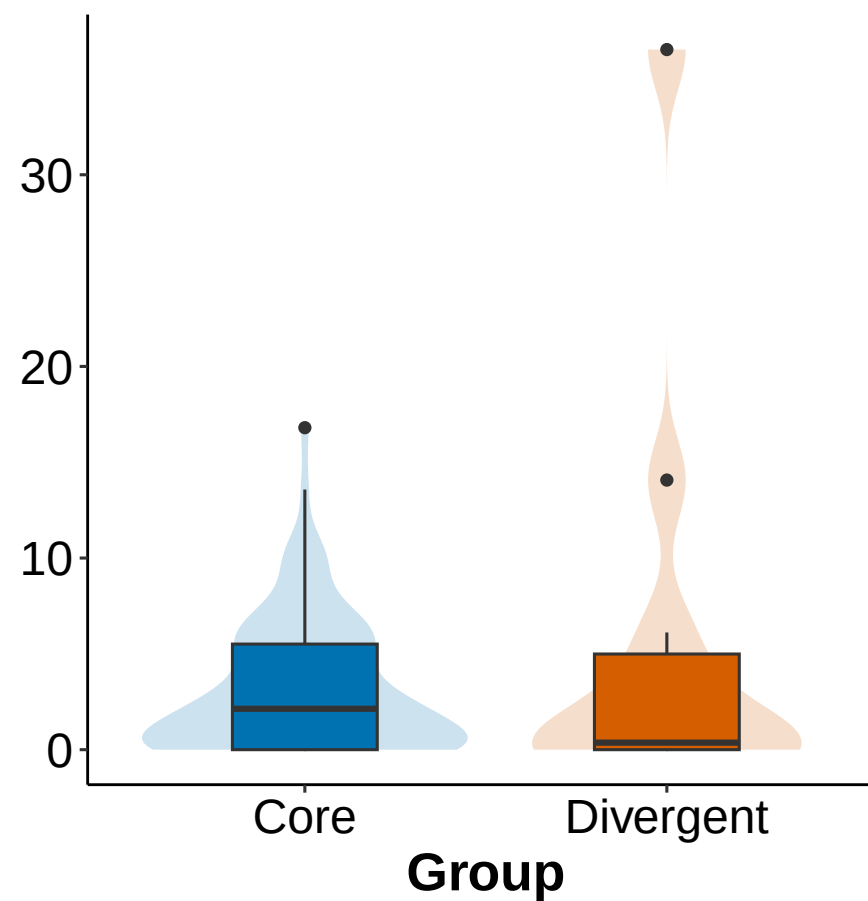
SA_Purine_synthesis_FBA

p-adj: 0.945 | .=0.07 (Negligible)



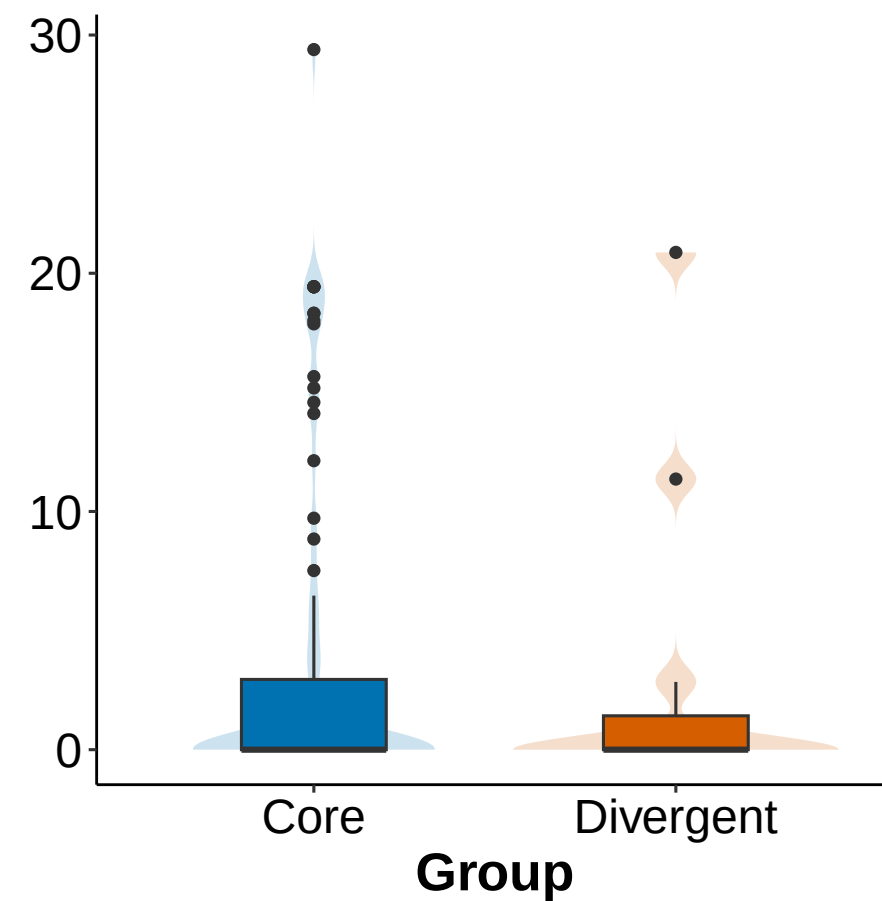
SA_Purine_synthesis_L1w

p-adj: 0.927 | .=-0.11 (Negligible) | .%: -82.8



SA_Purine_synthesis_pFBA

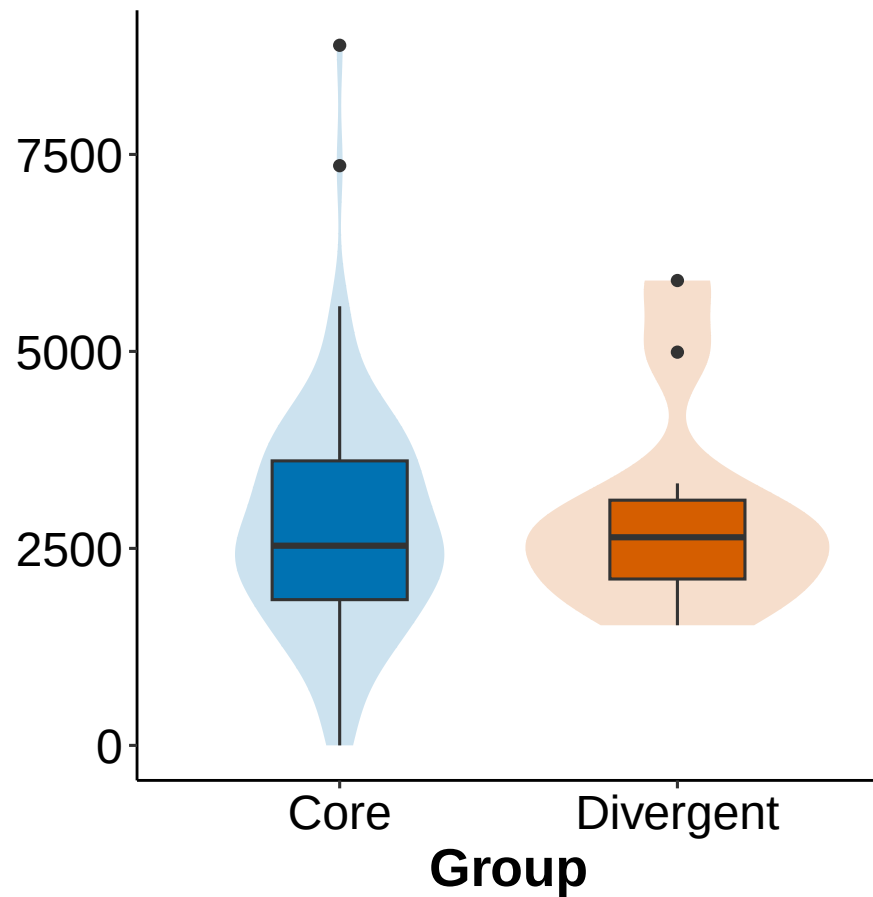
p-adj: 0.954 | .=-0.04 (Negligible)



Metabolism – . 5. Systemic Reprogramming (SA_*)

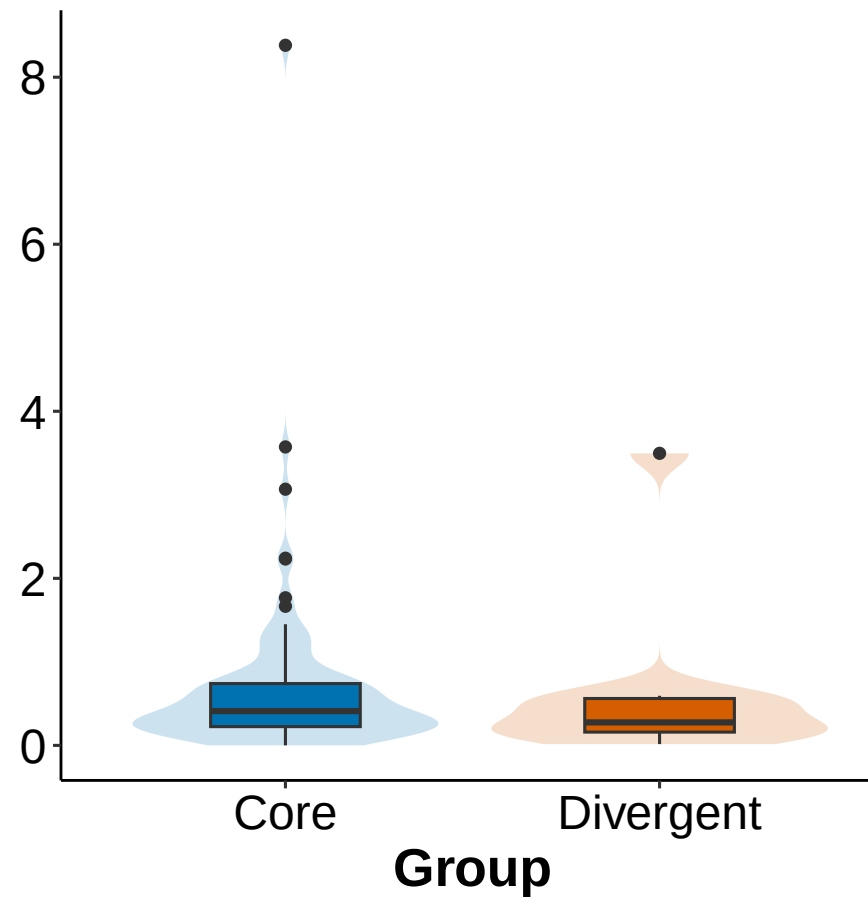
SA_Pyrimidine_catabolism_FBA

p-adj: 0.954 | .>=0.04 (Negligible) | .%: 4.2



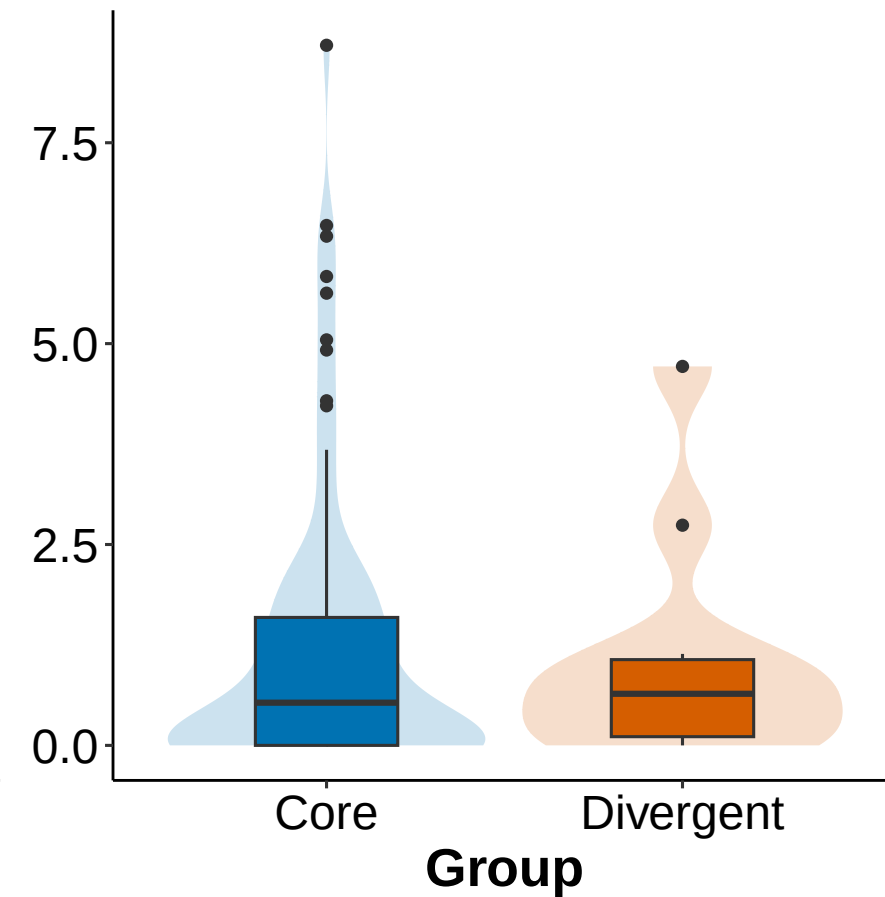
SA_Pyrimidine_catabolism_L1w

p-adj: 0.885 | .=-0.19 (Small) | .%: -32.9



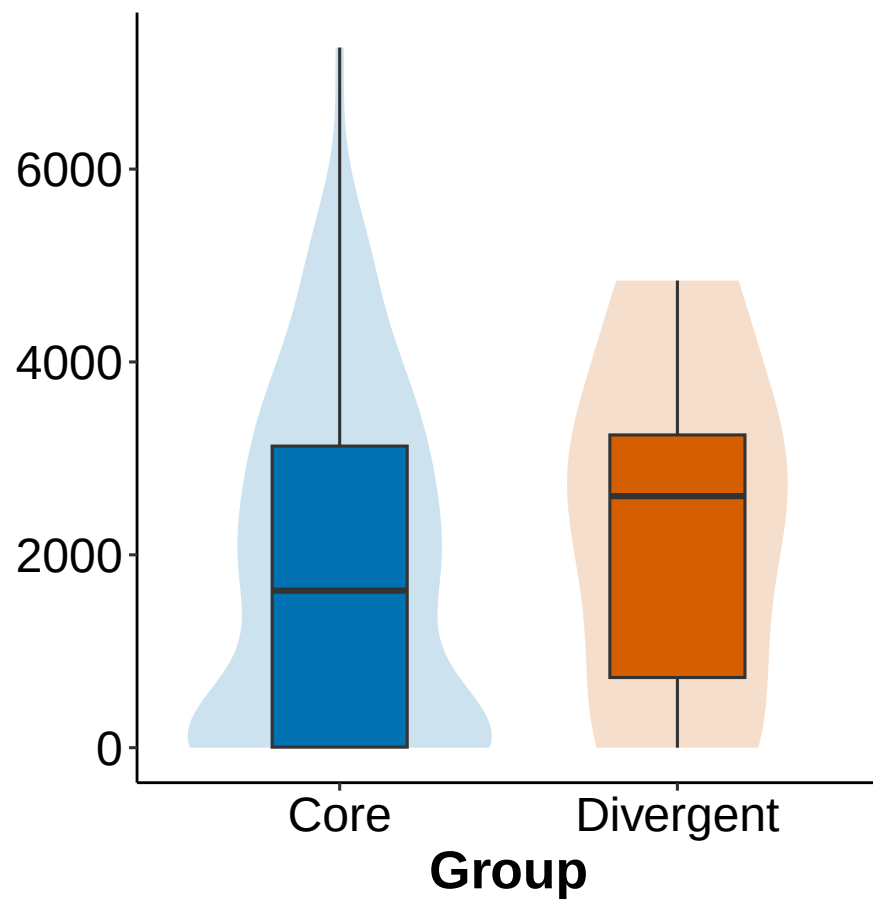
SA_Pyrimidine_catabolism_pFBA

p-adj: 0.95 | .>=0.06 (Negligible) | .%: 20.9



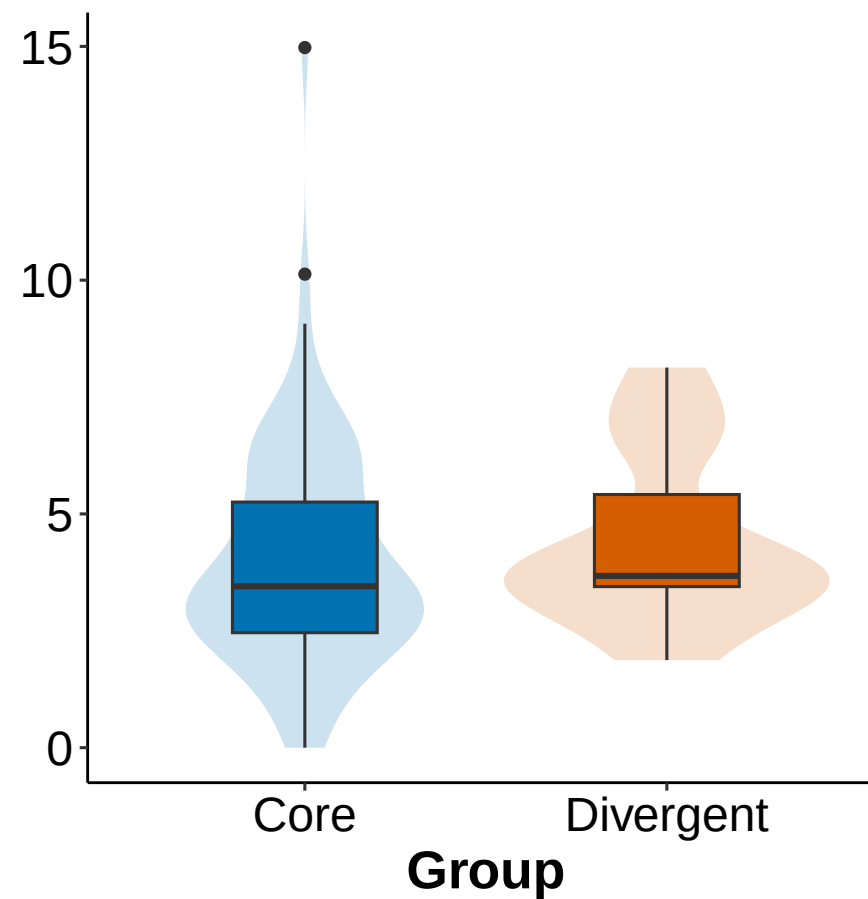
SA_Pyrimidine_synthesis_FBA

p-adj: 0.885 | .>=0.18 (Small) | .%: 60.2



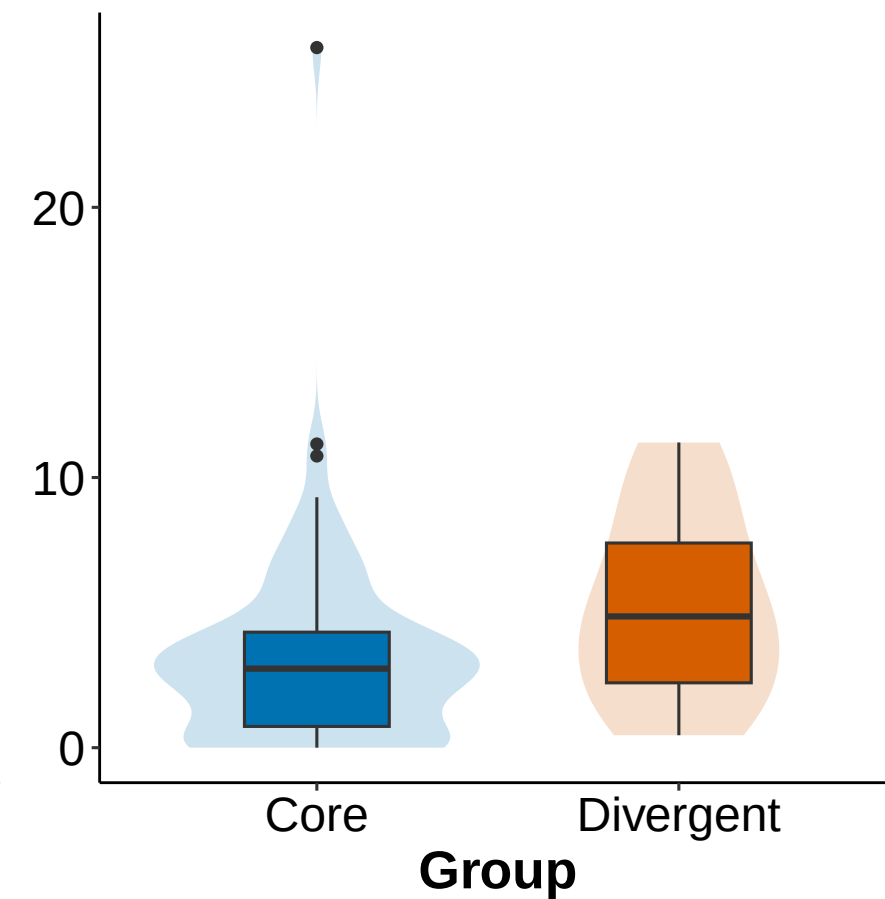
SA_Pyrimidine_synthesis_L1w

p-adj: 0.885 | .>=0.2 (Small) | .%: 6.3



SA_Pyrimidine_synthesis_pFBA

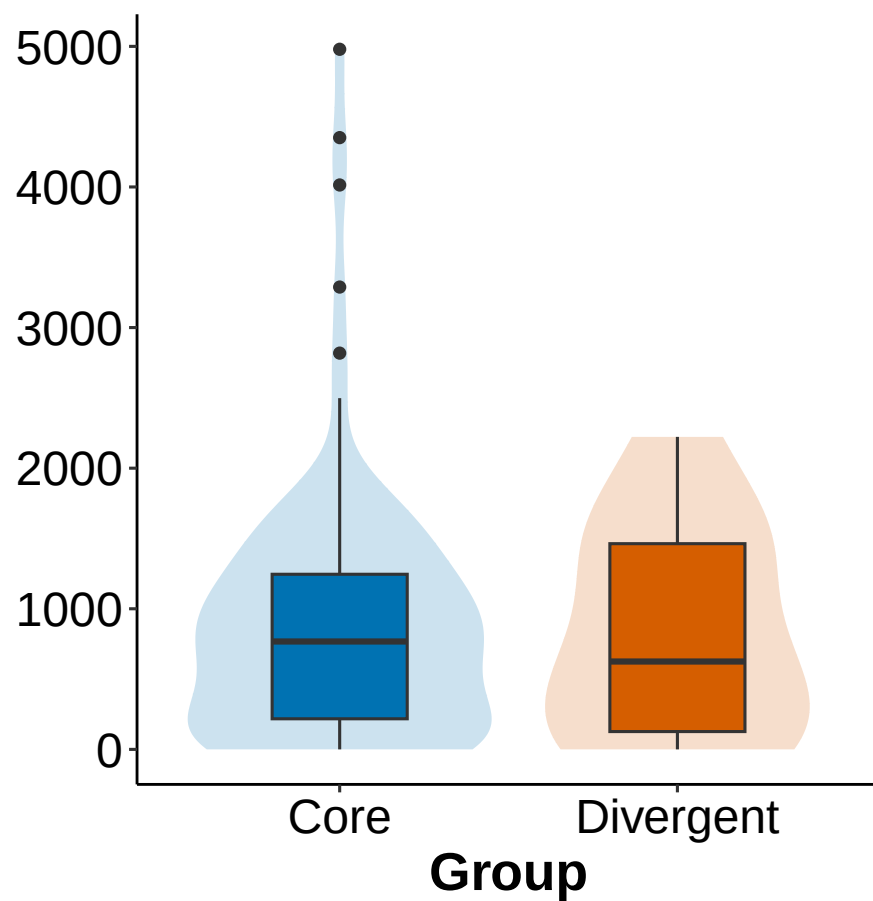
p-adj: 0.843 | .>=0.37 (Medium) | .%: 66



Metabolism – . 5. Systemic Reprogramming (SA_*)

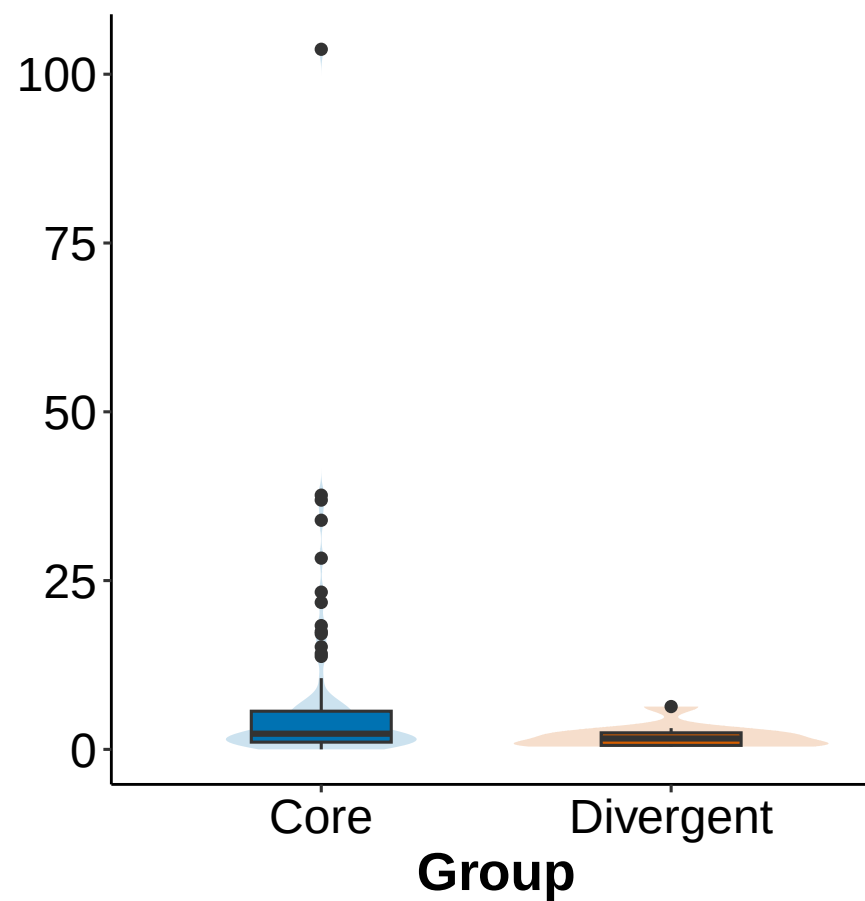
SA_Pyruvate_metabolism_FBA

p-adj: 0.957 | .=-0.02 (Negligible) | .%: -18.5



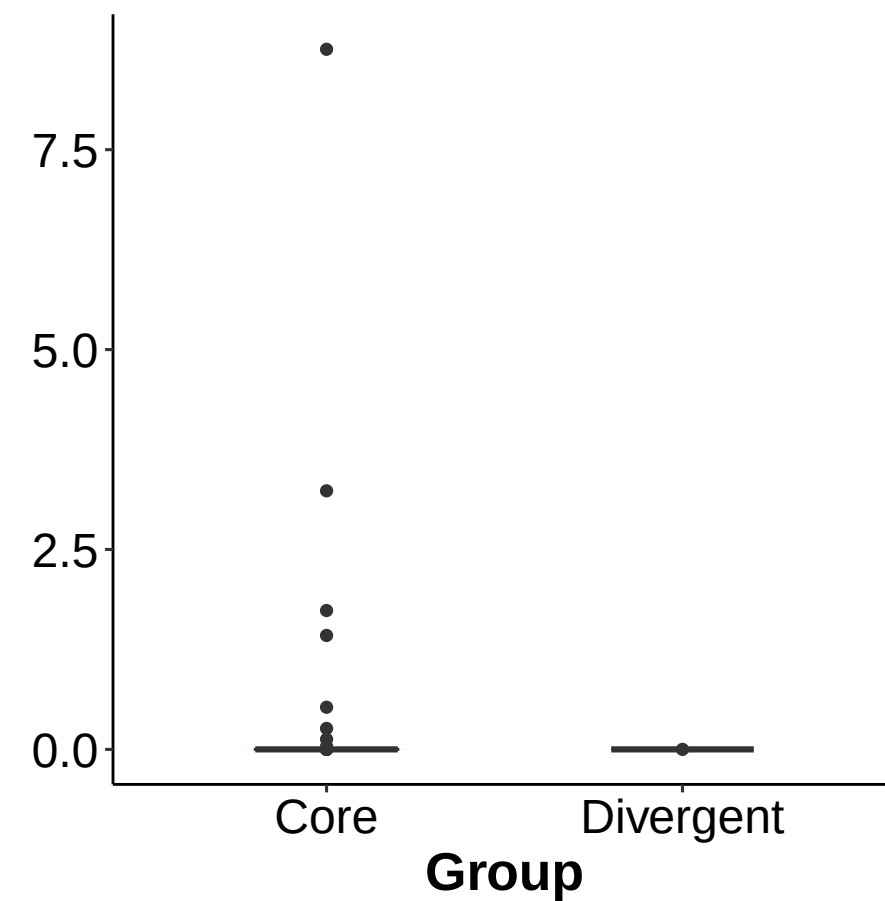
SA_Pyruvate_metabolism_L1w

p-adj: 0.872 | .=-0.26 (Small) | .%: -30.4



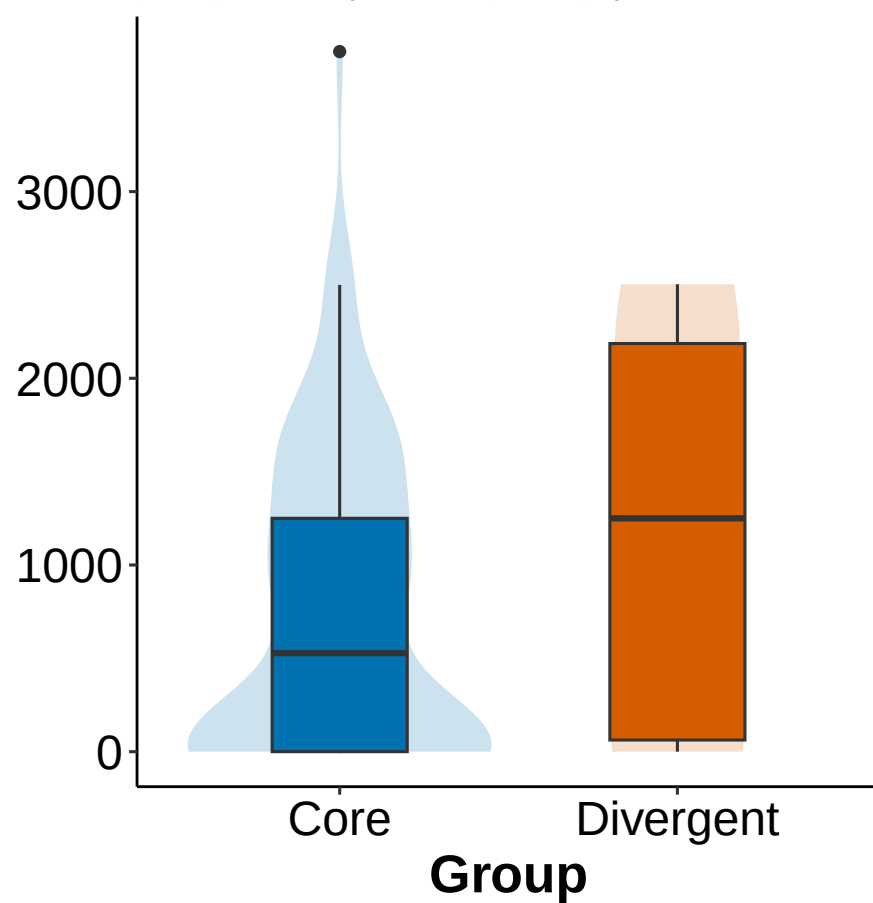
SA_Pyruvate_metabolism_pFBA

p-adj: 0.843 | .=-0.41 (Medium) | .%: -100



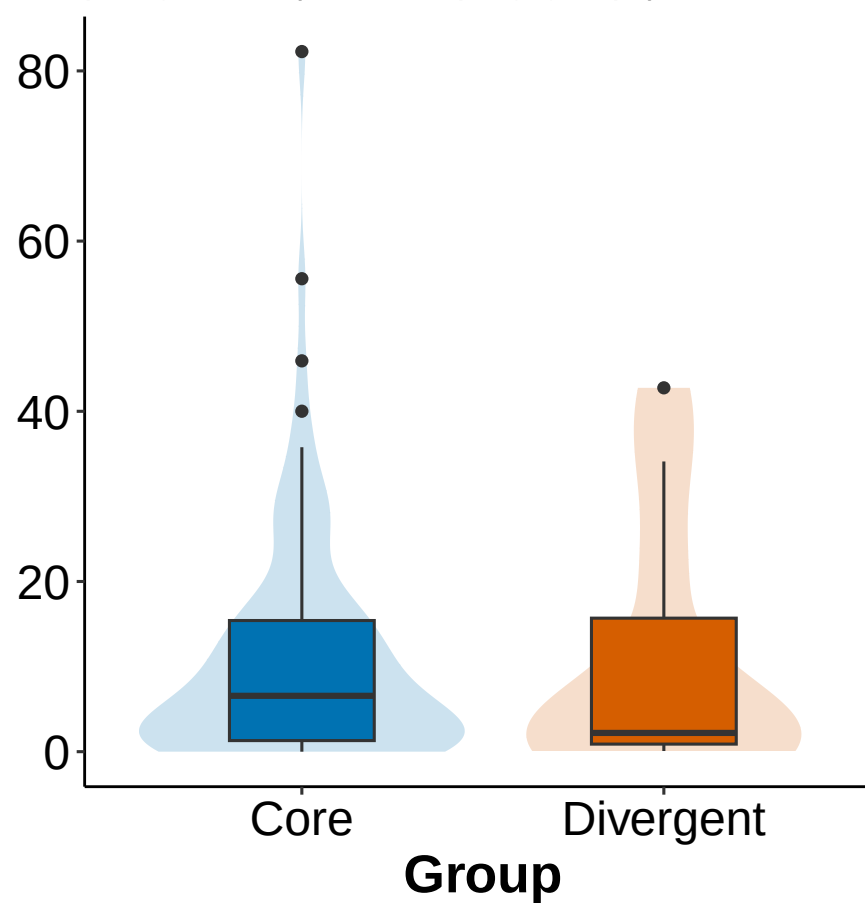
SA_ROS_detoxification_FBA

p-adj: 0.872 | .=-0.26 (Small) | .%: 137



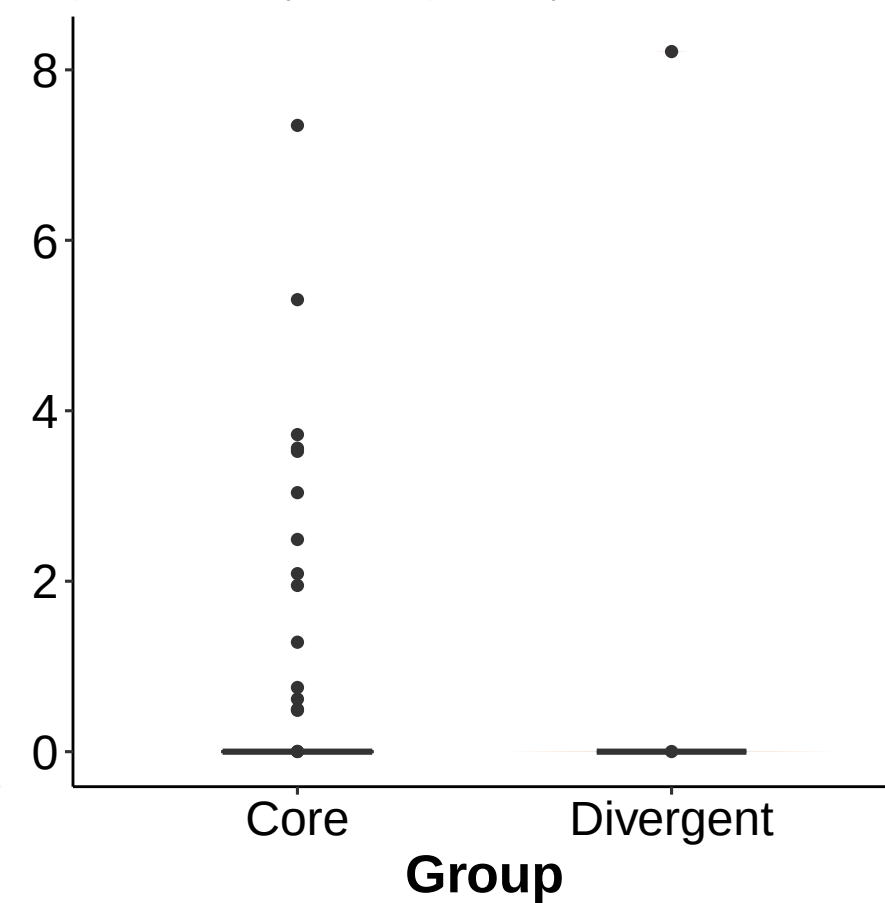
SA_ROS_detoxification_L1w

p-adj: 0.945 | .=-0.09 (Negligible) | .%: -66.4



SA_ROS_detoxification_pFBA

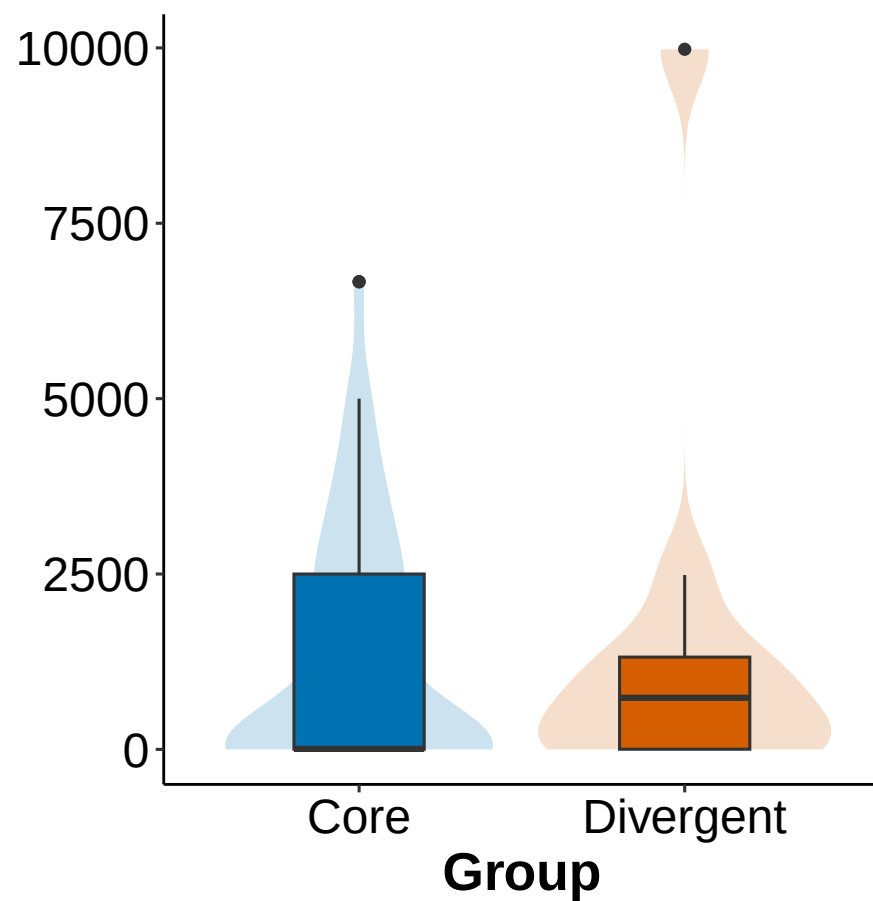
p-adj: 0.885 | .=-0.2 (Small) | .%: -100



Metabolism – . 5. Systemic Reprogramming (SA_*)

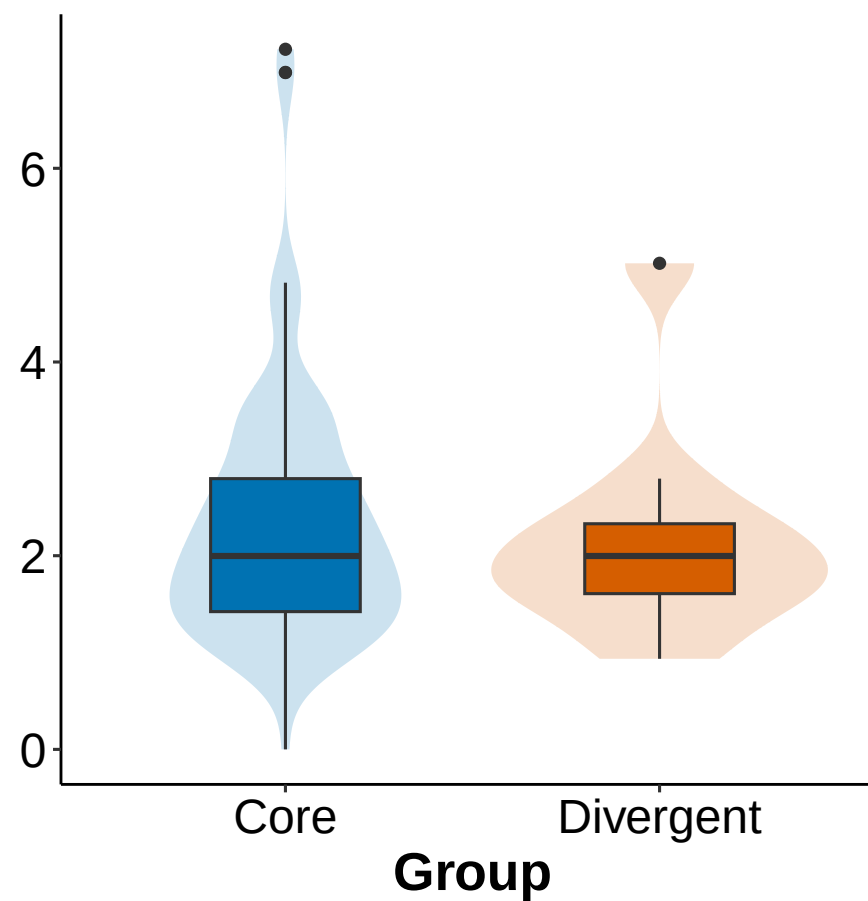
SA_Sphingolipid_metabolism_FB

p-adj: 0.954 | .=-0.03 (Negligible) | .%: 1030



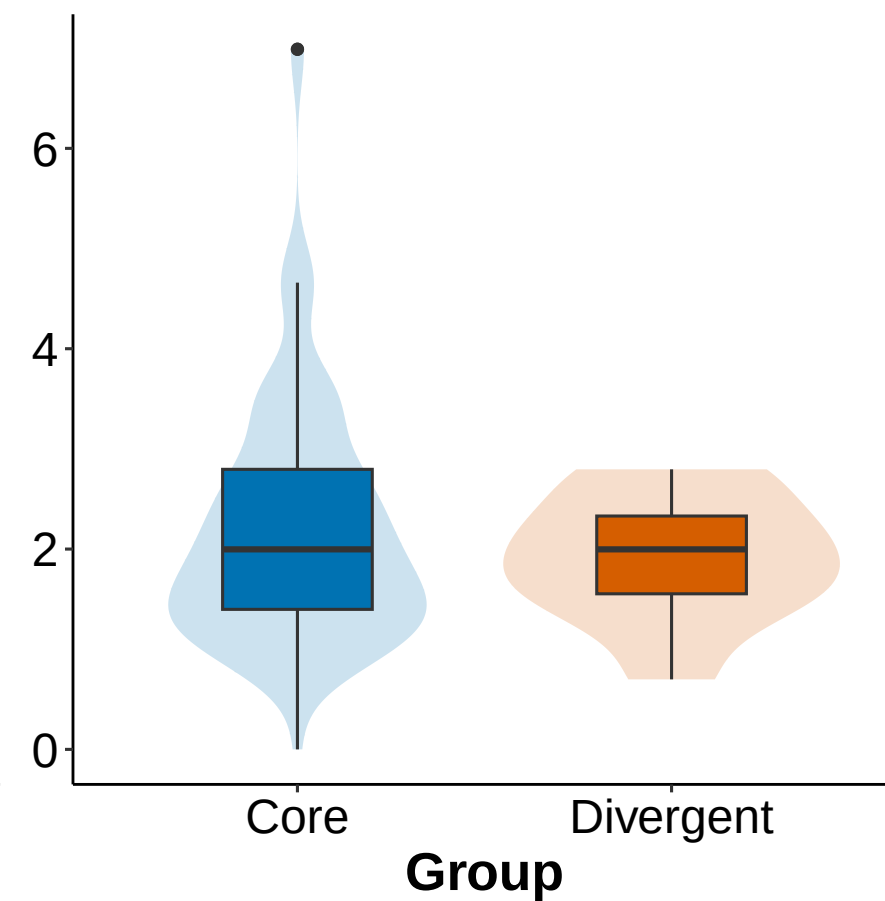
SA_Sphingolipid_metabolism_L1w

p-adj: 0.95 | .=-0.08 (Negligible) | .%: 0



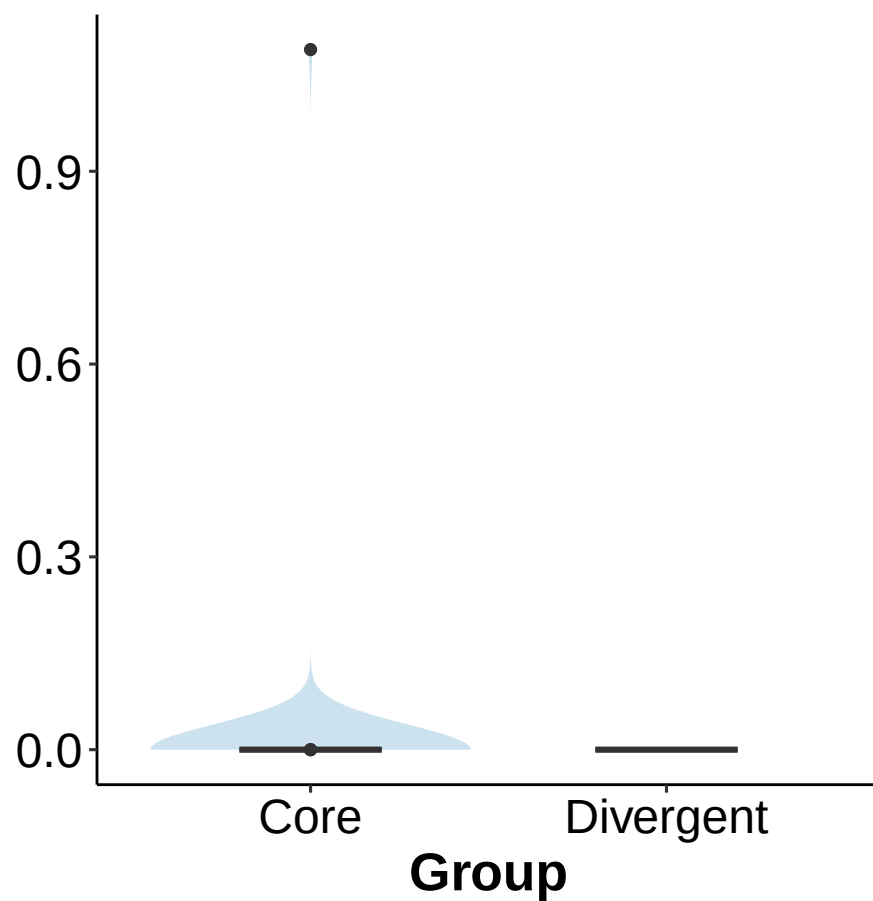
SA_Sphingolipid_metabolism_pFBA

p-adj: 0.954 | .=-0.05 (Negligible) | .%: 0



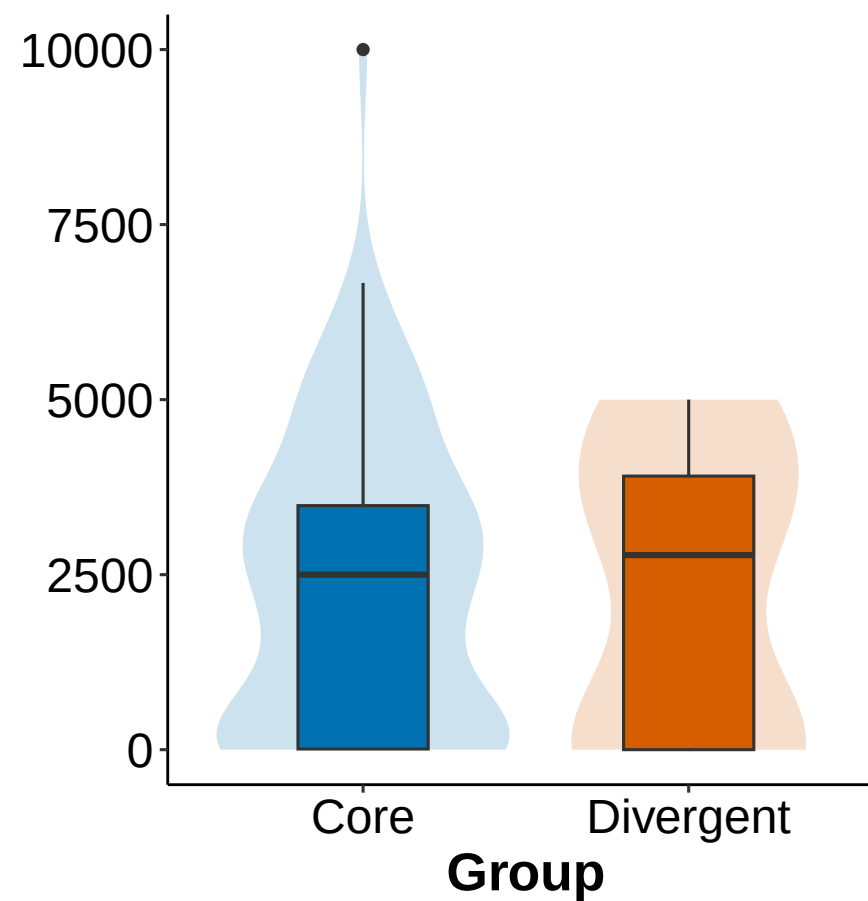
SA_Squalene_and_cholesterol_synth

p-adj: 0.927 | .=-0.03 (Negligible)



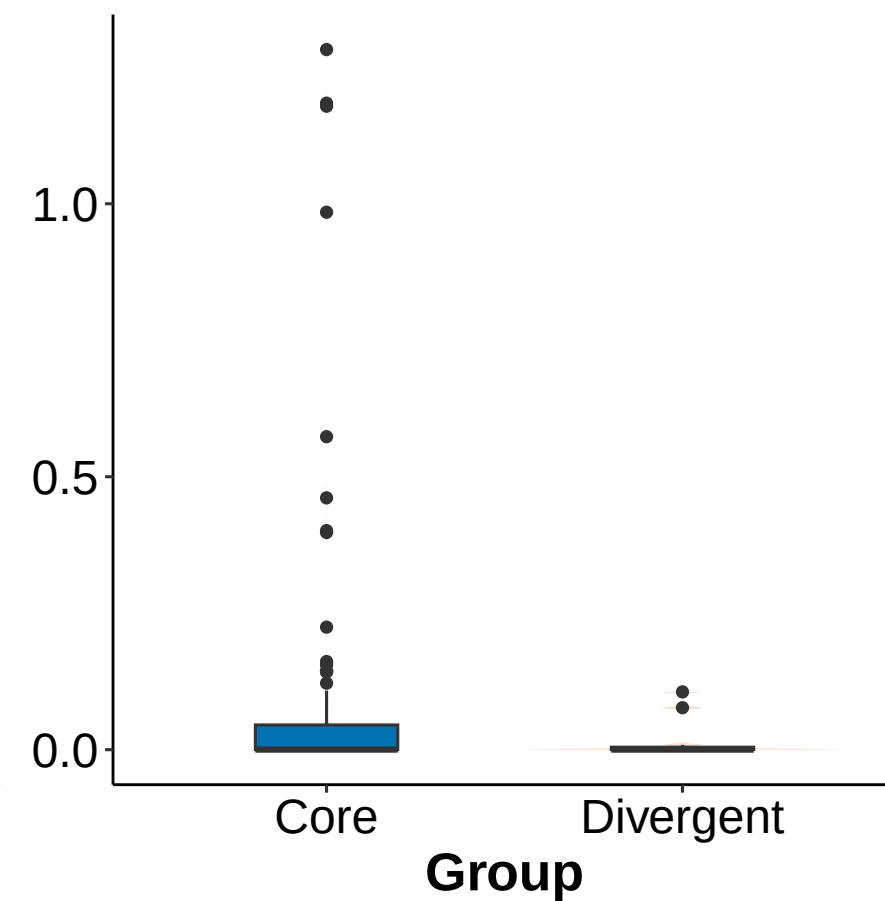
SA_Starch_and_sucrose_metabol

p-adj: 0.979 | .=-0.01 (Negligible) | .%: 11.2

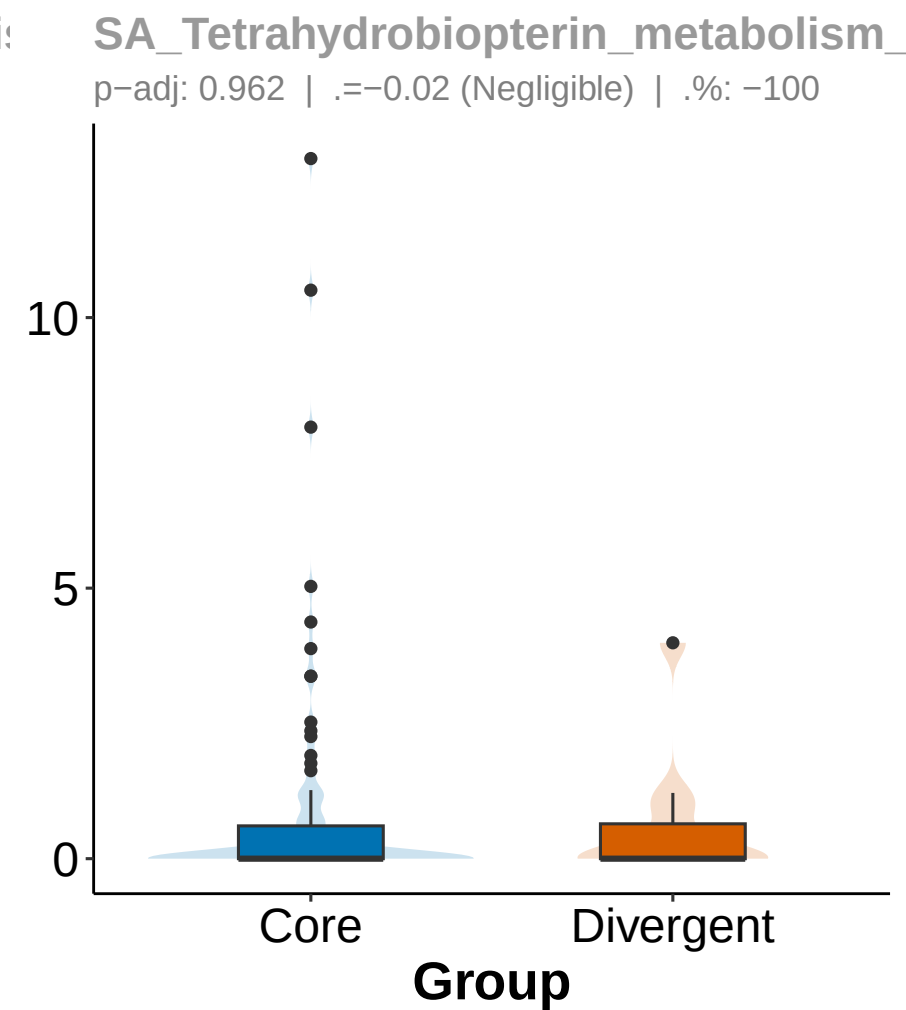
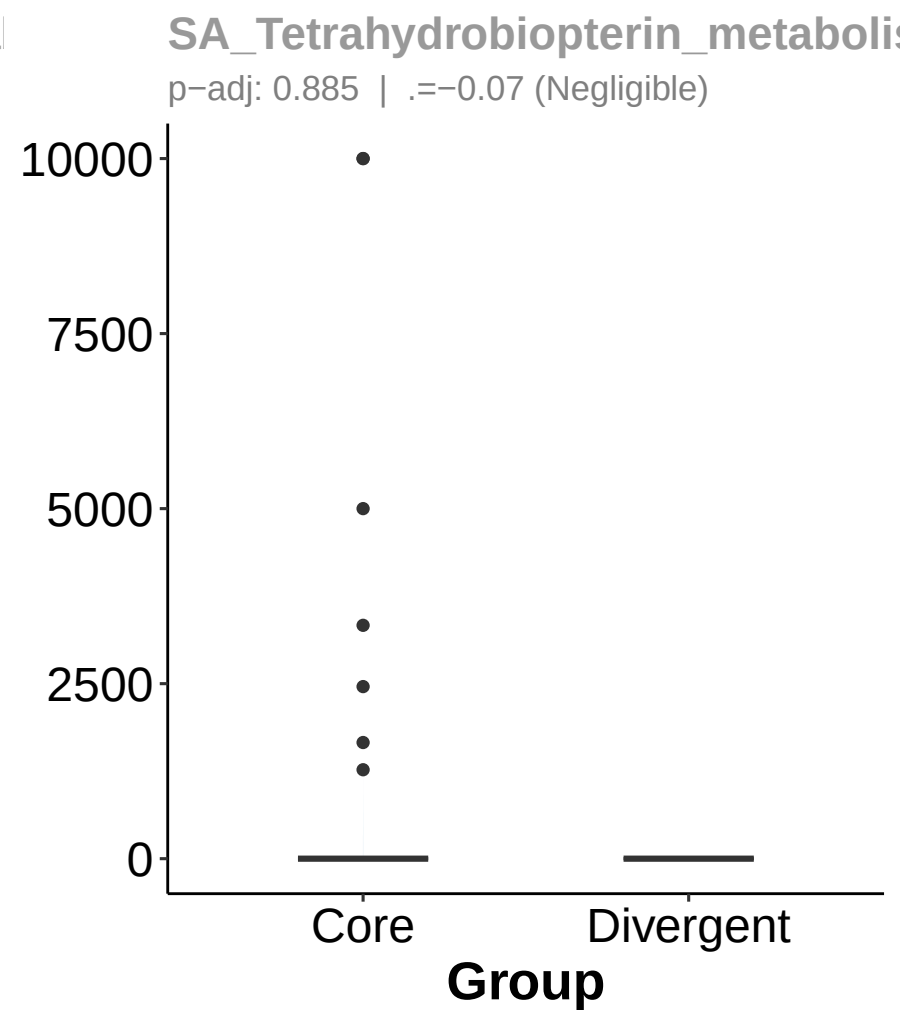
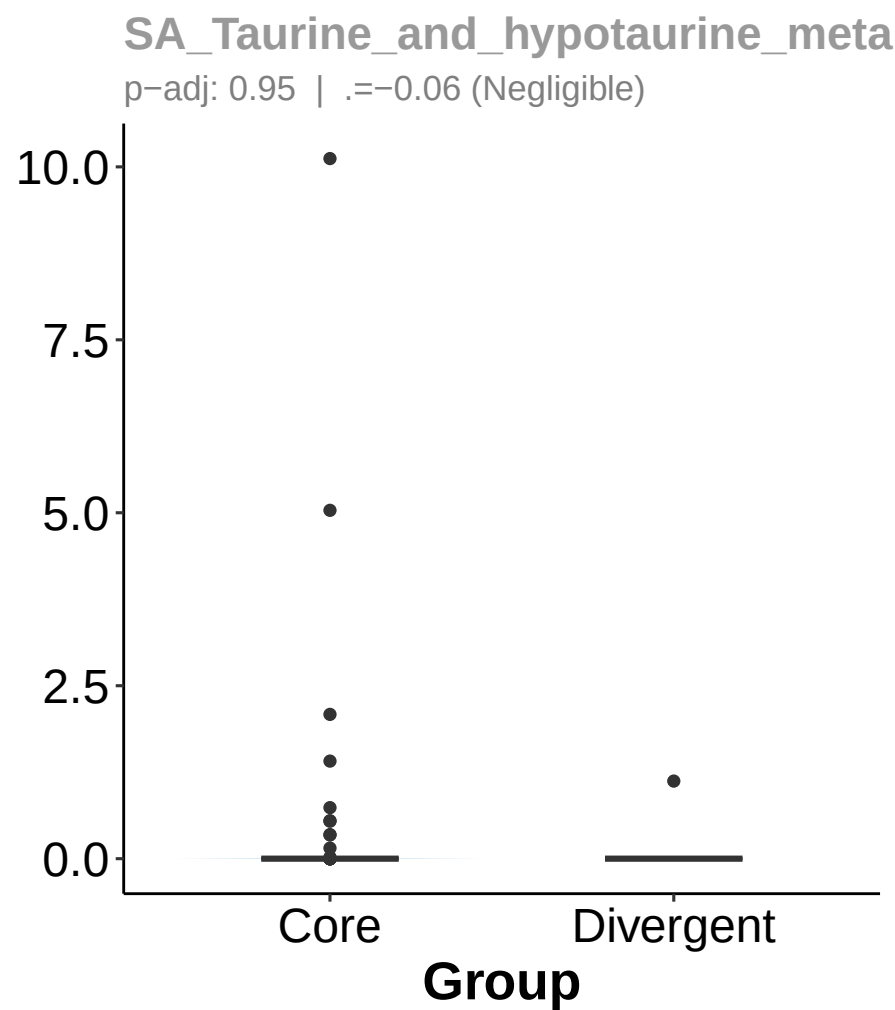
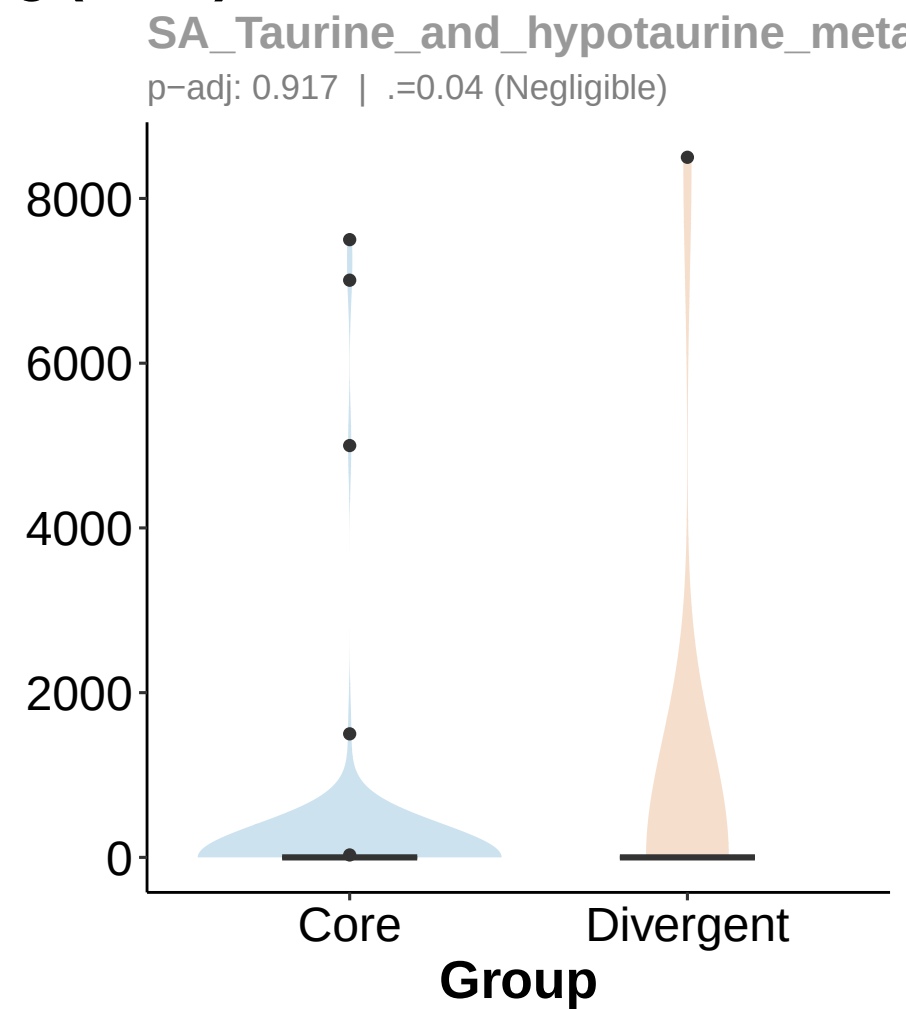
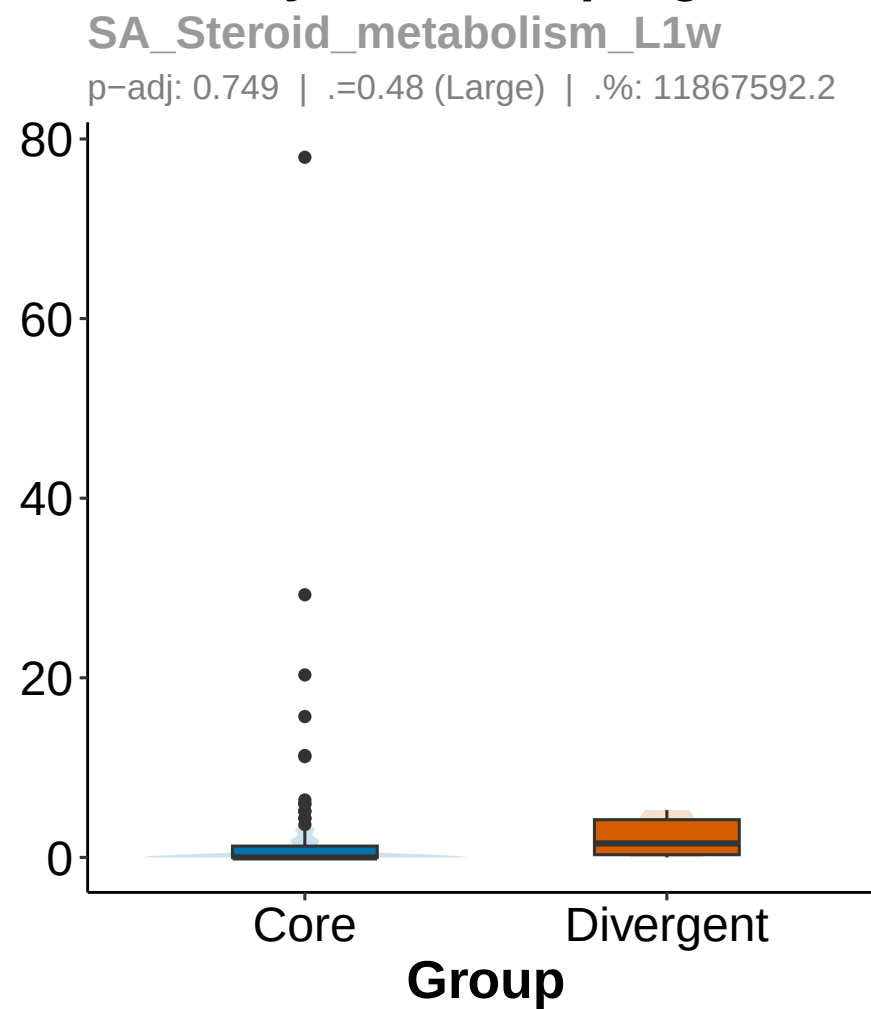
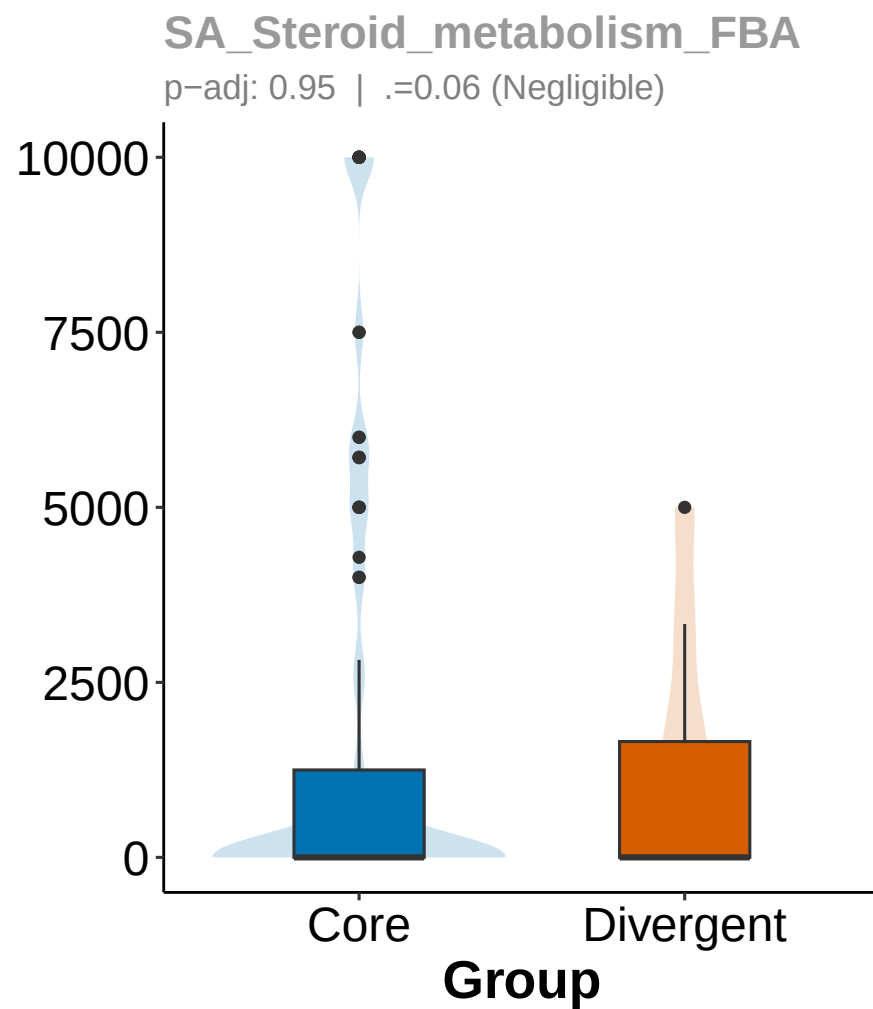


SA_Starch_and_sucrose_metabolism

p-adj: 0.917 | .=-0.11 (Negligible) | .%: -30.9



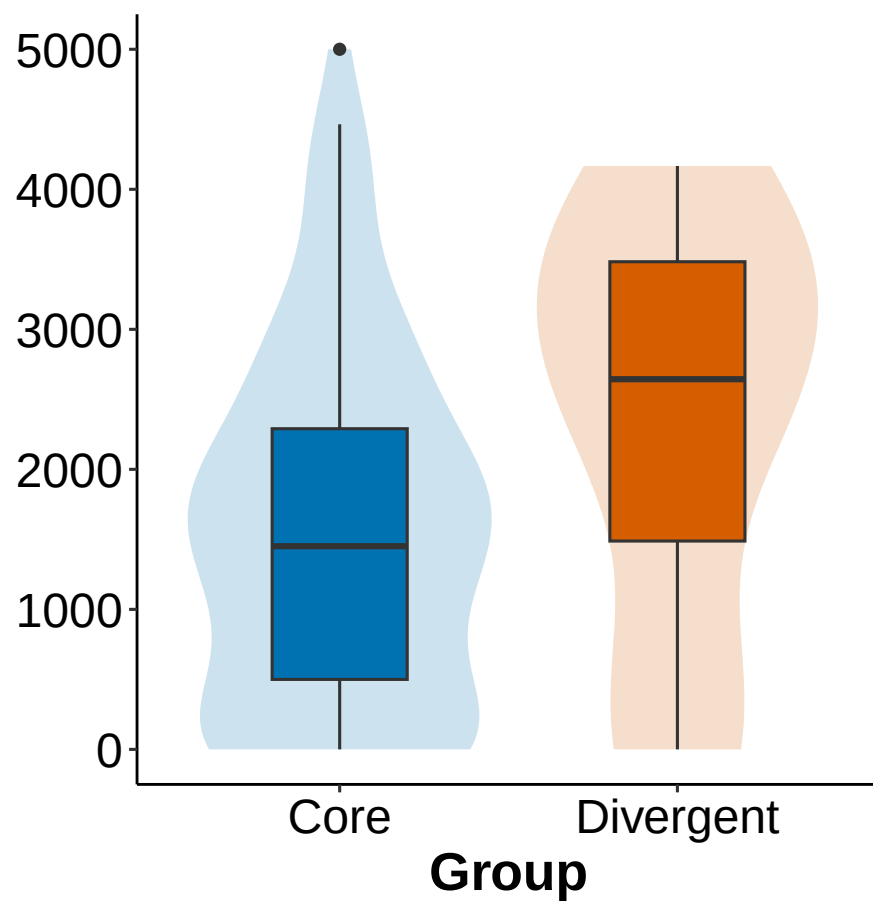
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

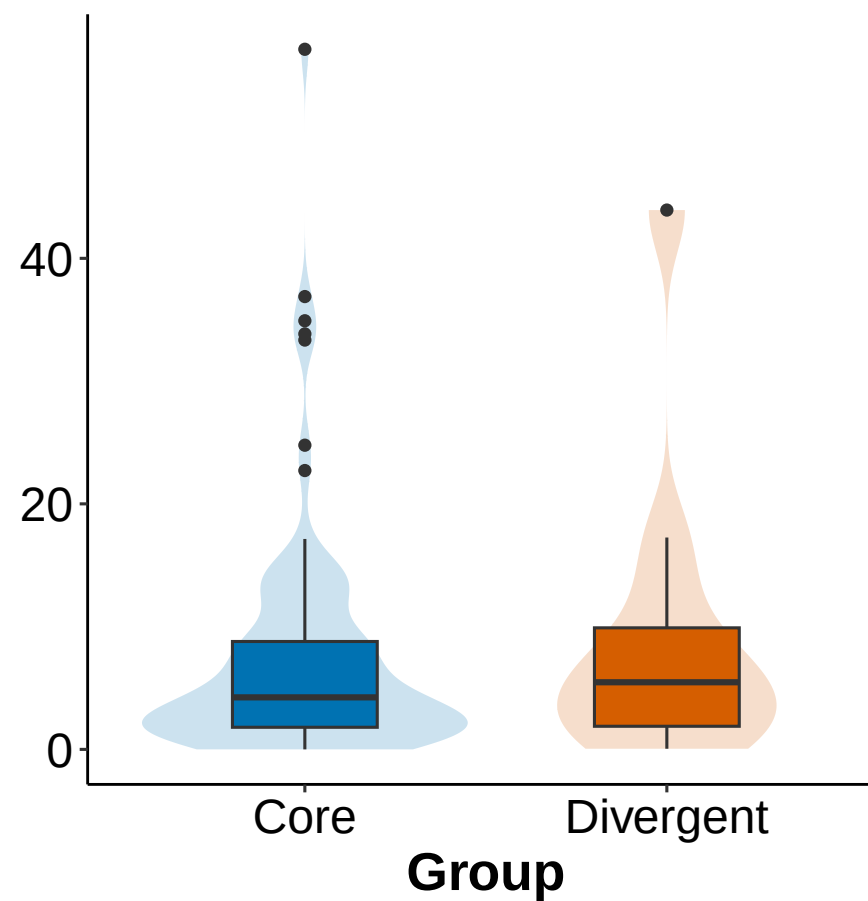
SA_Transport__endoplasmic_reticu

p-adj: 0.843 | .>=0.34 (Medium) | .%: 82.2



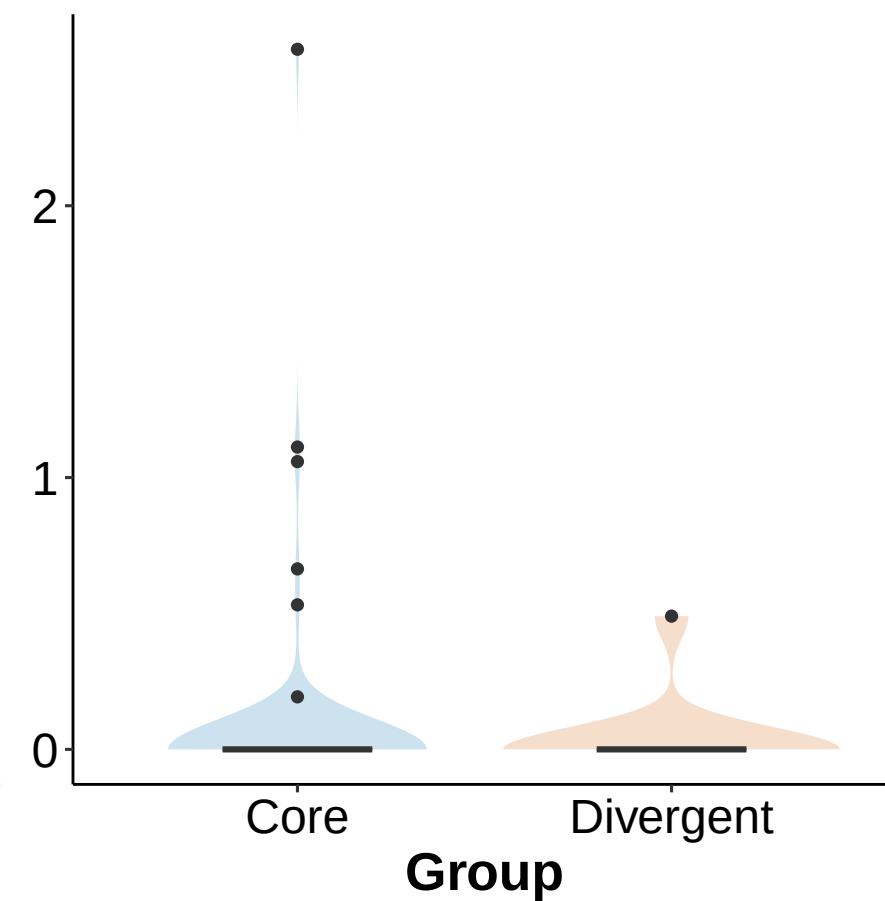
SA_Transport__endoplasmic_reticula

p-adj: 0.95 | .>=0.06 (Negligible) | .%: 29



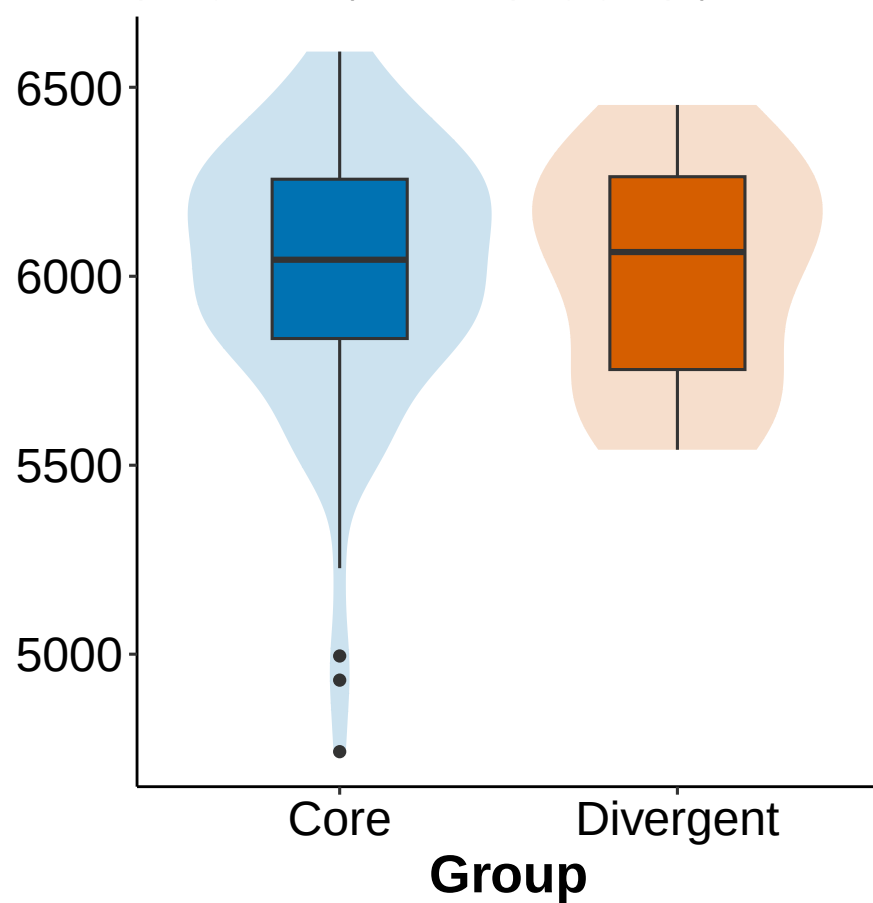
SA_Transport__endoplasmic_reticular

p-adj: 0.95 | .>=0.03 (Negligible)



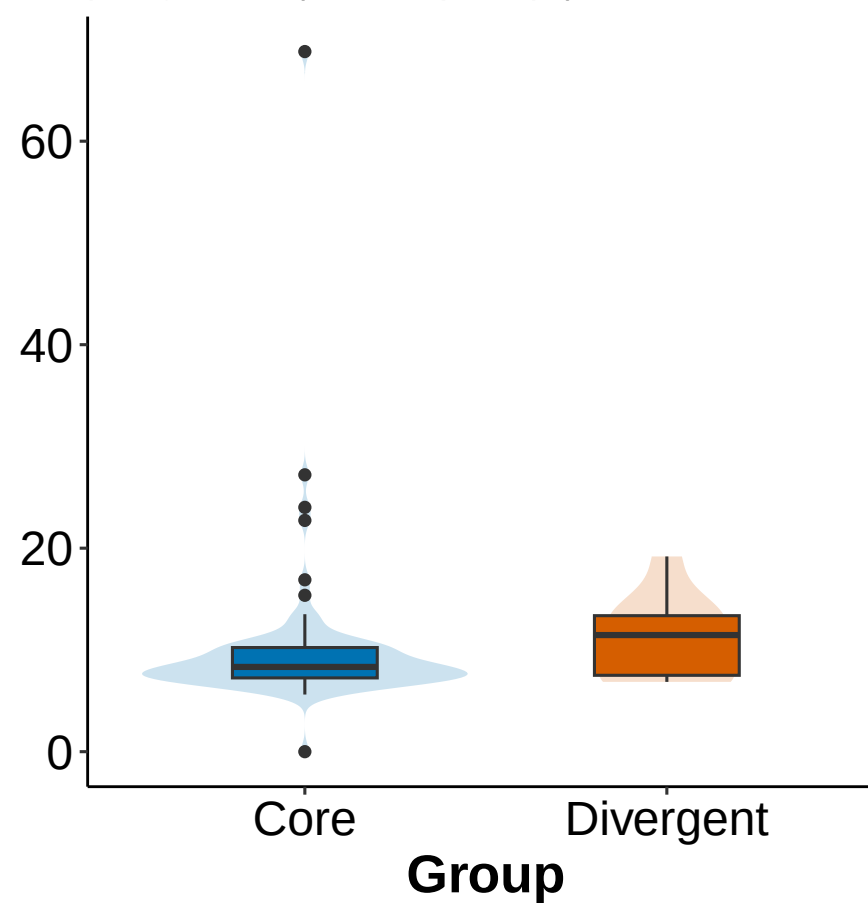
SA_Transport__extracellular_FBA

p-adj: 0.954 | .>=-0.03 (Negligible) | .%: 0.3



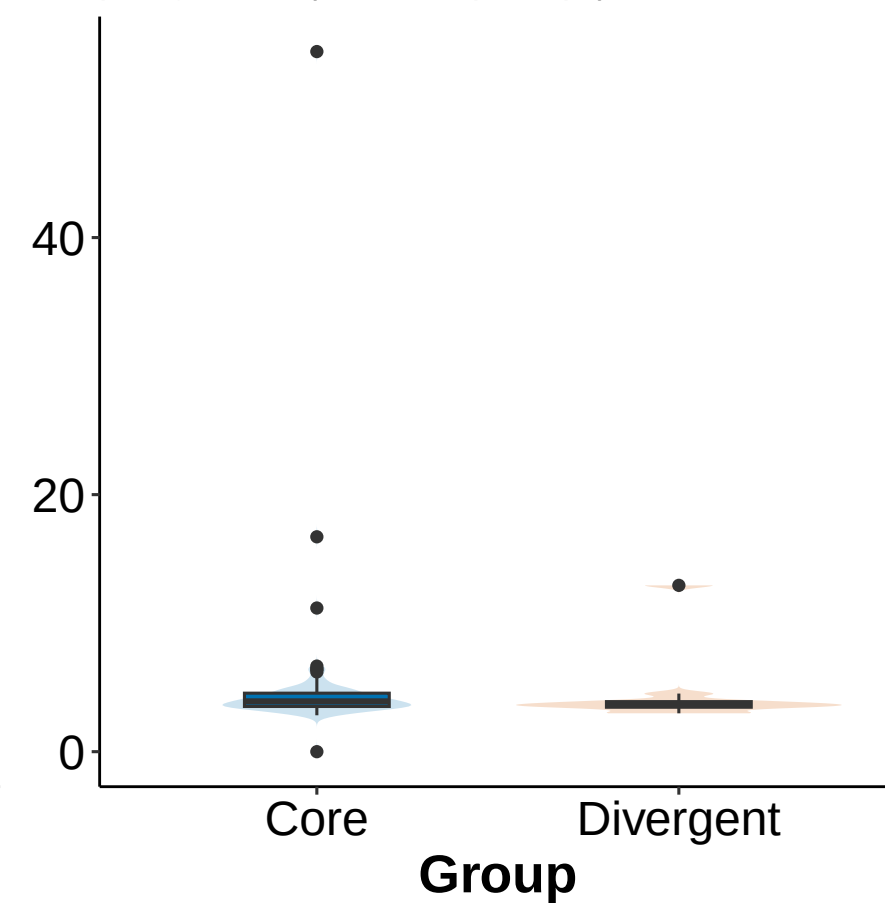
SA_Transport__extracellular_L1w

p-adj: 0.885 | .>=0.25 (Small) | .%: 37.6



SA_Transport__extracellular_pFBA

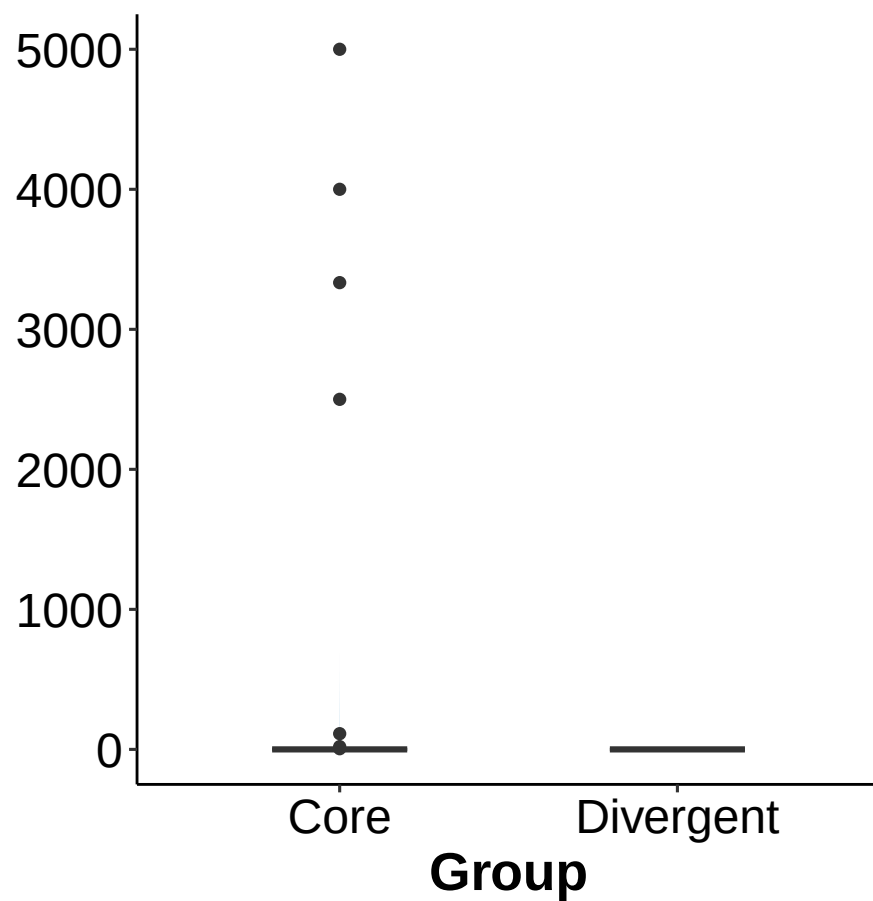
p-adj: 0.885 | .>=-0.21 (Small) | .%: -7



Metabolism – . 5. Systemic Reprogramming (SA_*)

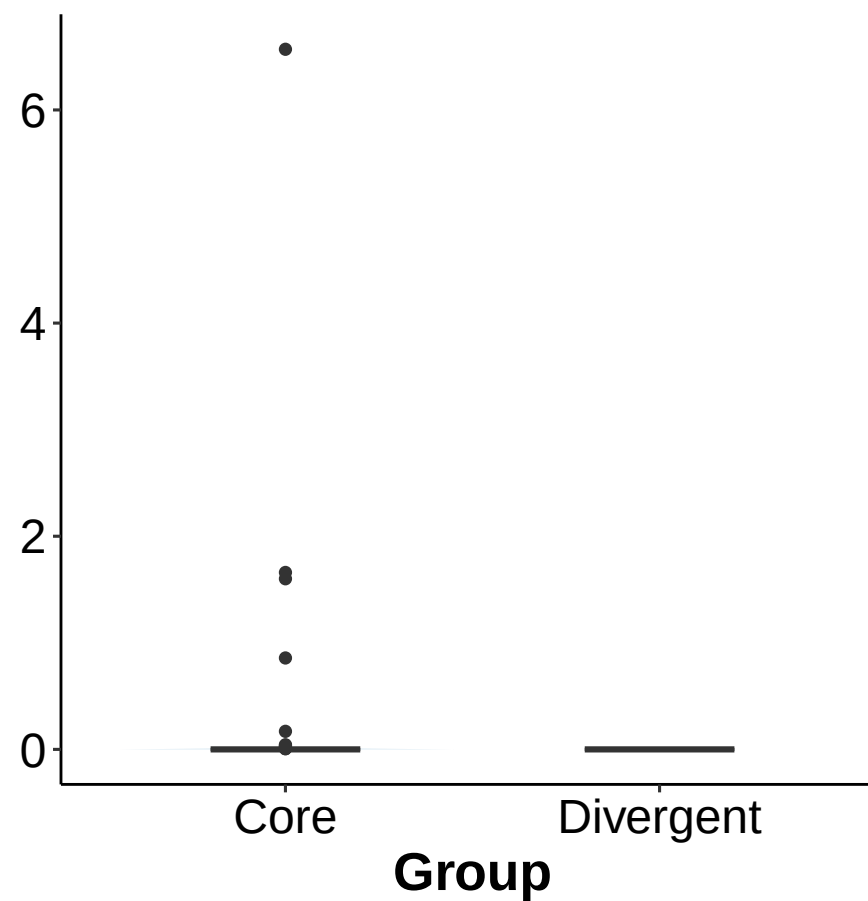
SA_Transport__golgi_apparatus_Fl

p-adj: 0.885 | .=-0.07 (Negligible)



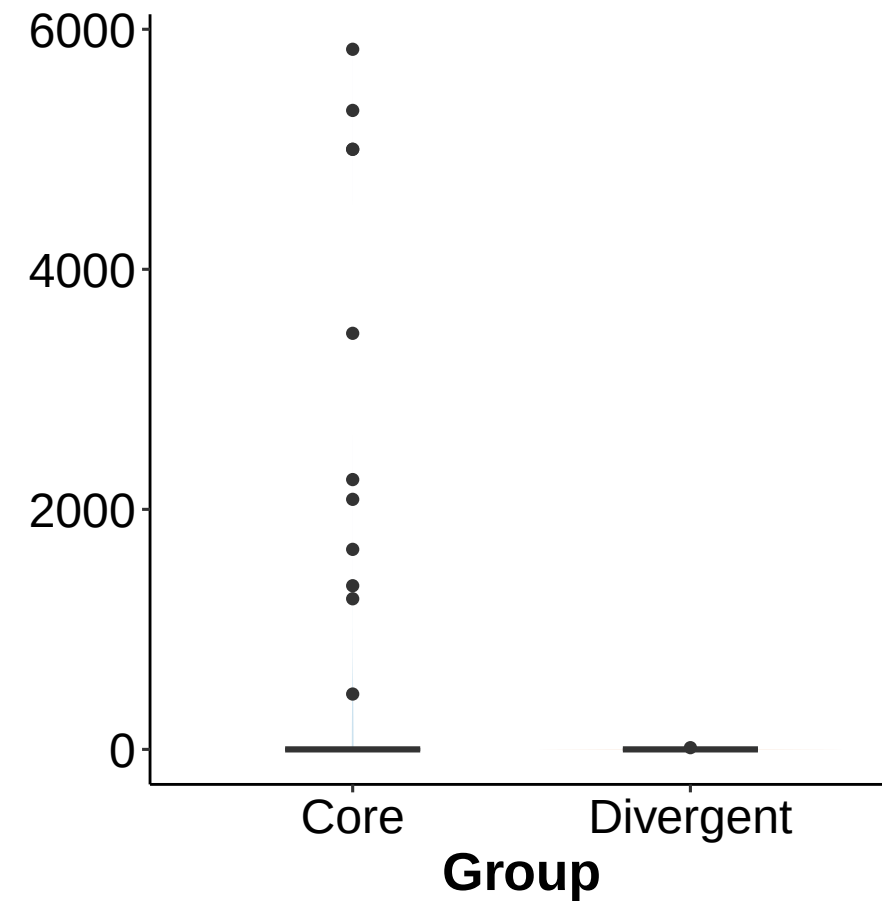
SA_Transport__golgi_apparatus_L1w

p-adj: 0.885 | .=-0.15 (Small) | .%: -28.4



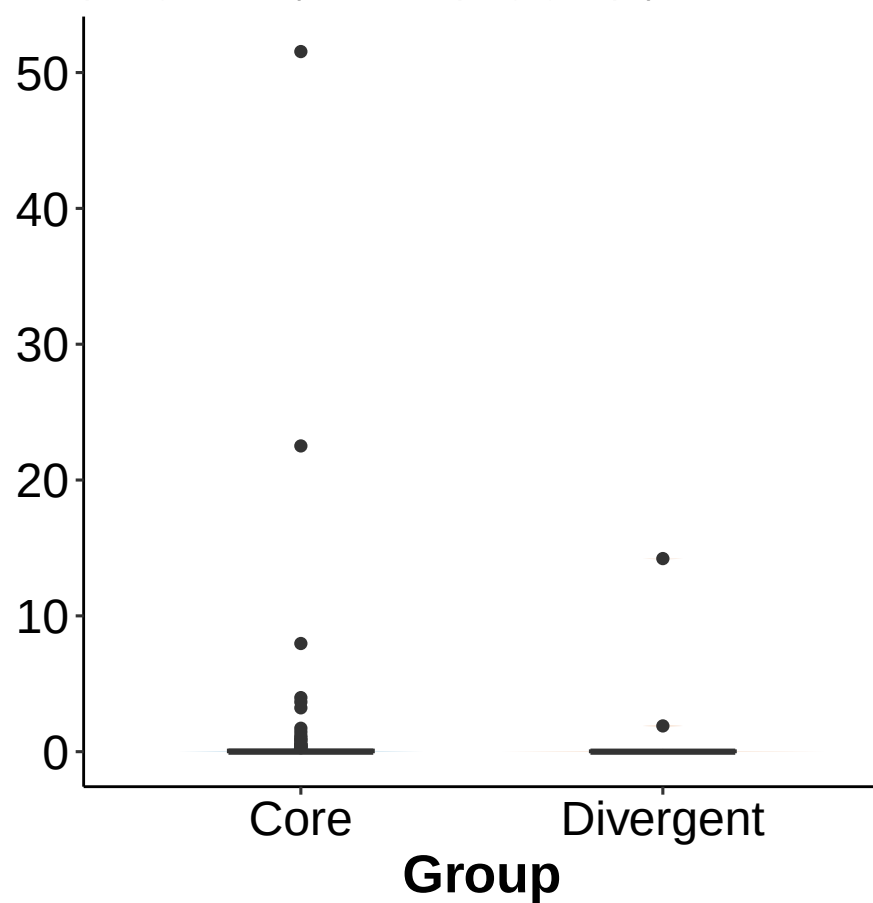
SA_Transport__lysosomal_FBA

p-adj: 0.95 | .=-0.03 (Negligible)



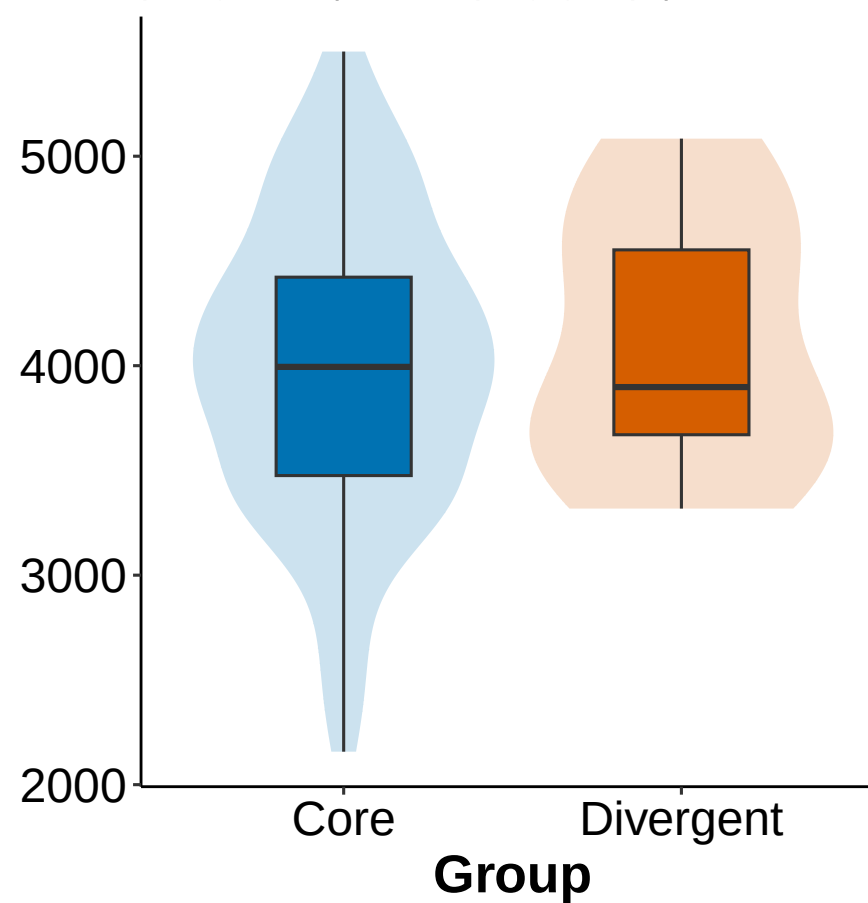
SA_Transport__lysosomal_L1w

p-adj: 0.896 | .=-0.14 (Negligible) | .%: -99.6



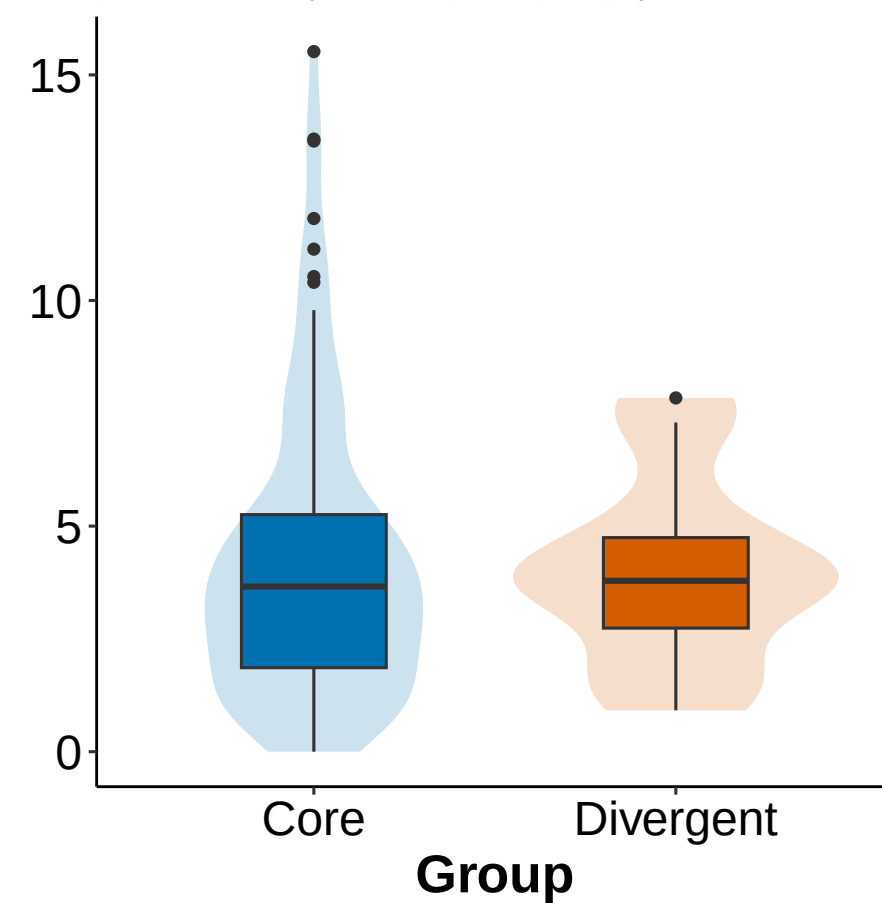
SA_Transport__mitochondrial_FBA

p-adj: 0.95 | .=0.08 (Negligible) | .%: -2.4



SA_Transport__mitochondrial_L1w

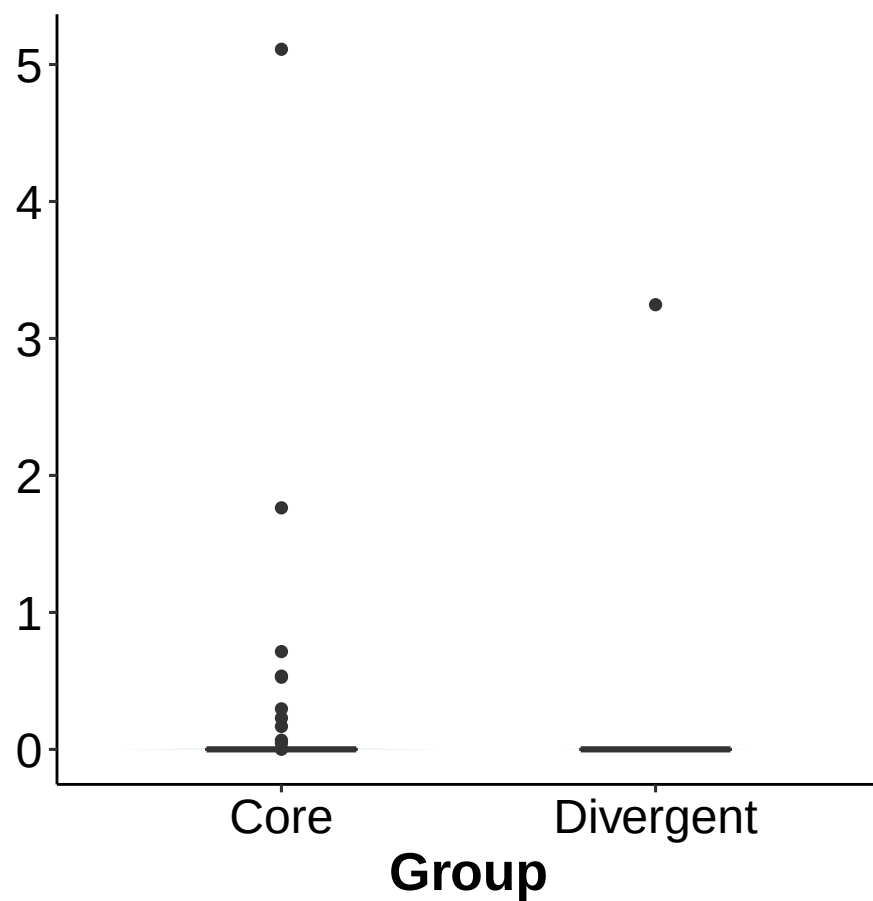
p-adj: 0.954 | .=0.05 (Negligible) | .%: 3.5



Metabolism – . 5. Systemic Reprogramming (SA_*)

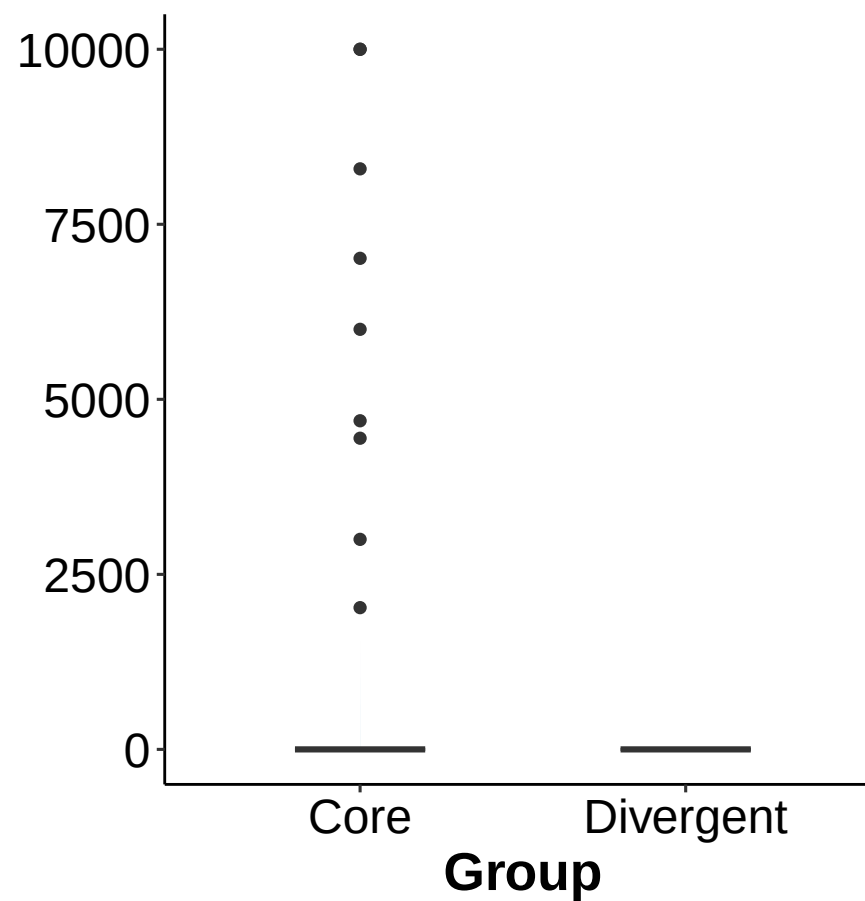
SA_Transport__mitochondrial_pFBA

p-adj: 0.885 | .=-0.16 (Small) | .%: -9.9



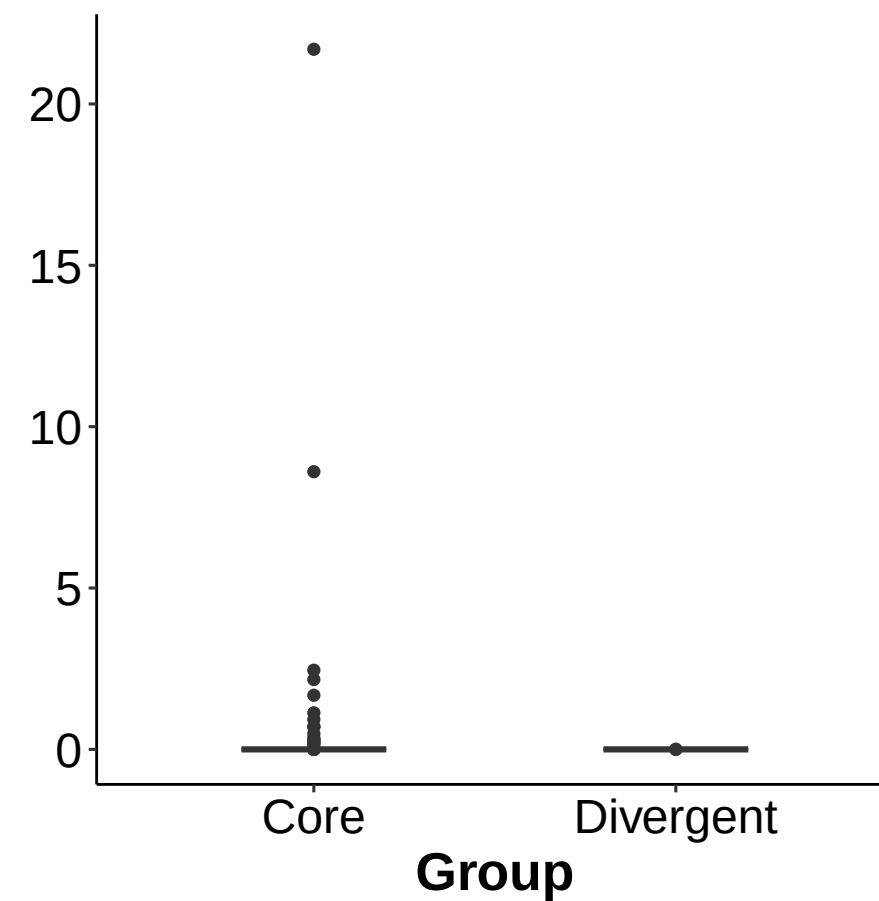
SA_Transport__nuclear_FBA

p-adj: 0.885 | .=-0.09 (Negligible)



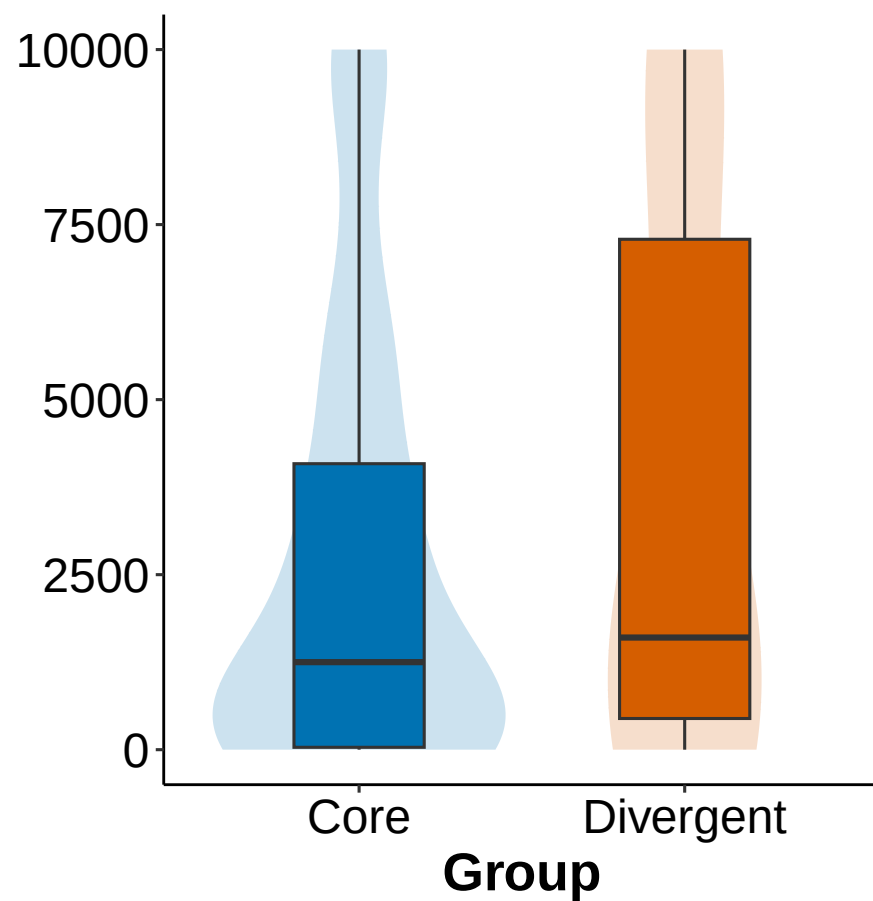
SA_Transport__nuclear_L1w

p-adj: 0.885 | .=-0.18 (Small)



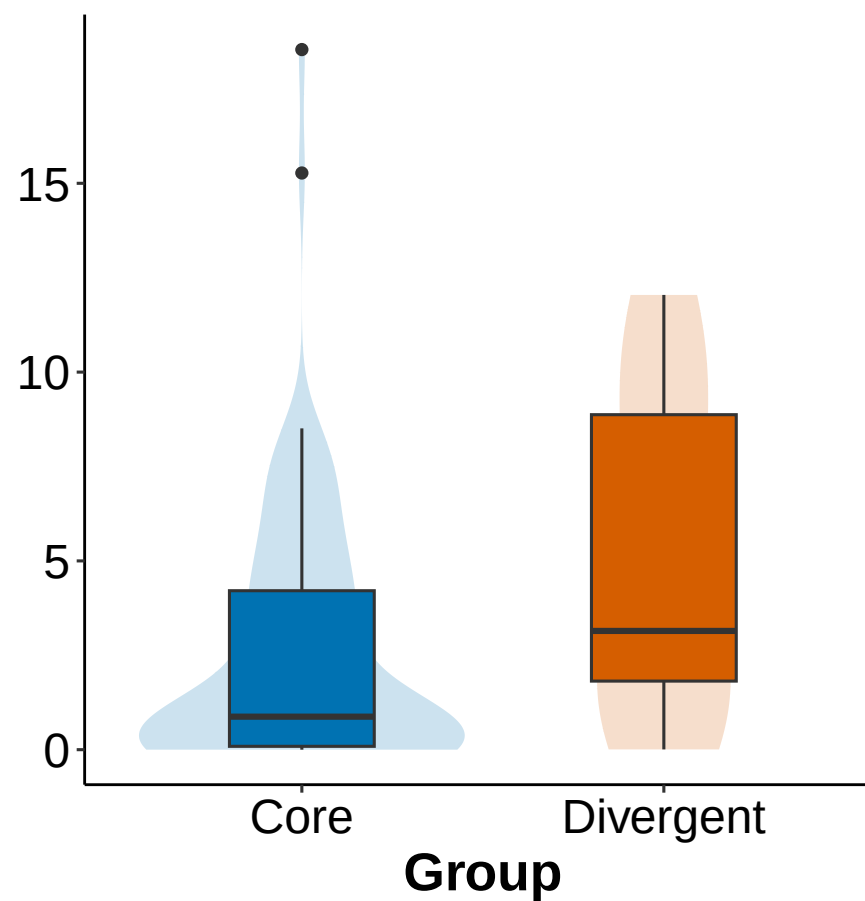
SA_Transport__peroxisomal_FBA

p-adj: 0.917 | .=0.12 (Negligible) | .%: 28.2



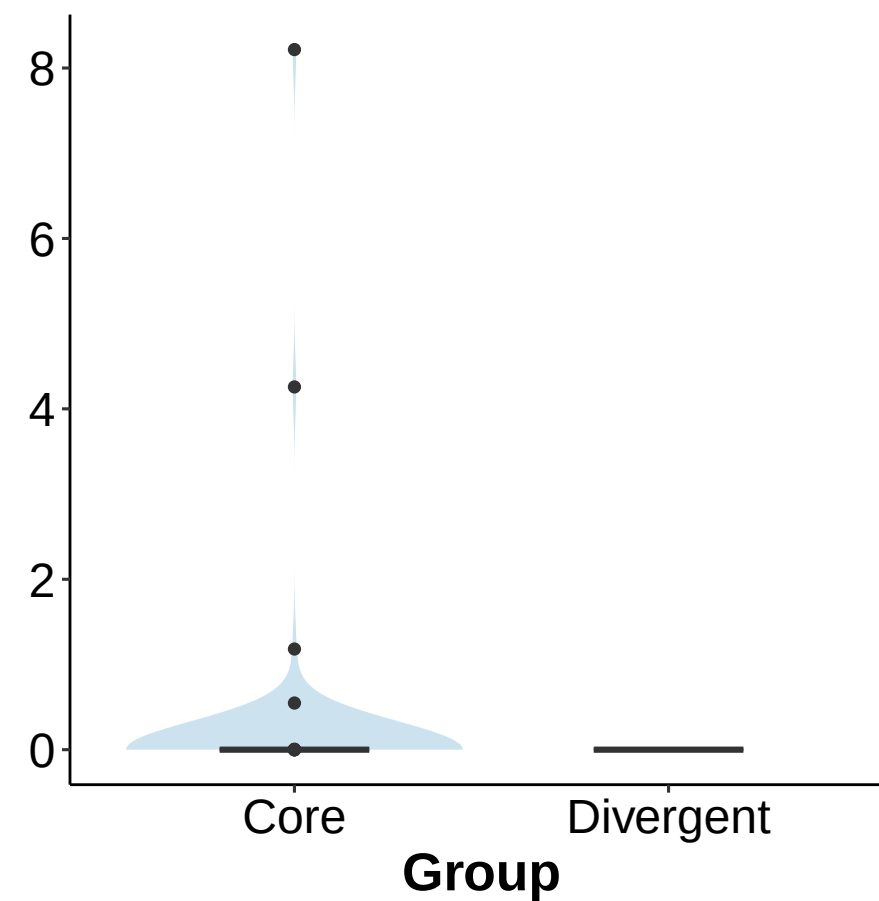
SA_Transport__peroxisomal_L1w

p-adj: 0.843 | .=0.39 (Medium) | .%: 259.5

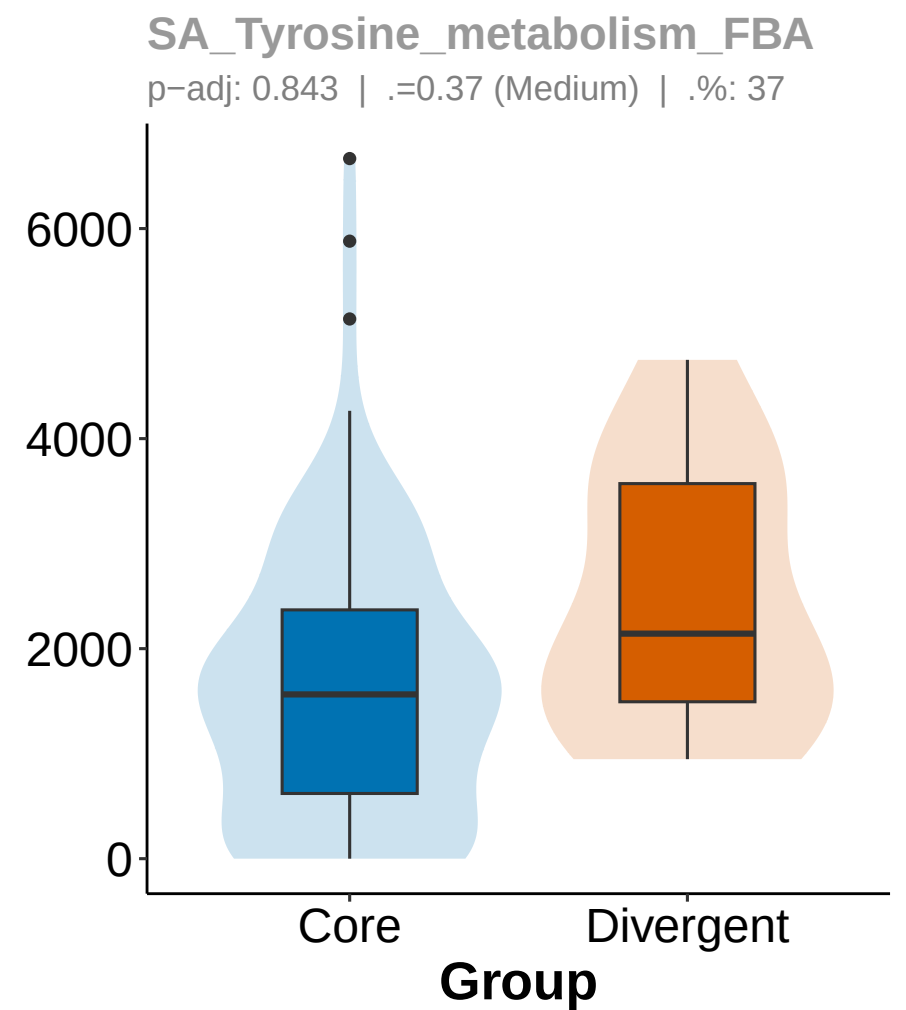
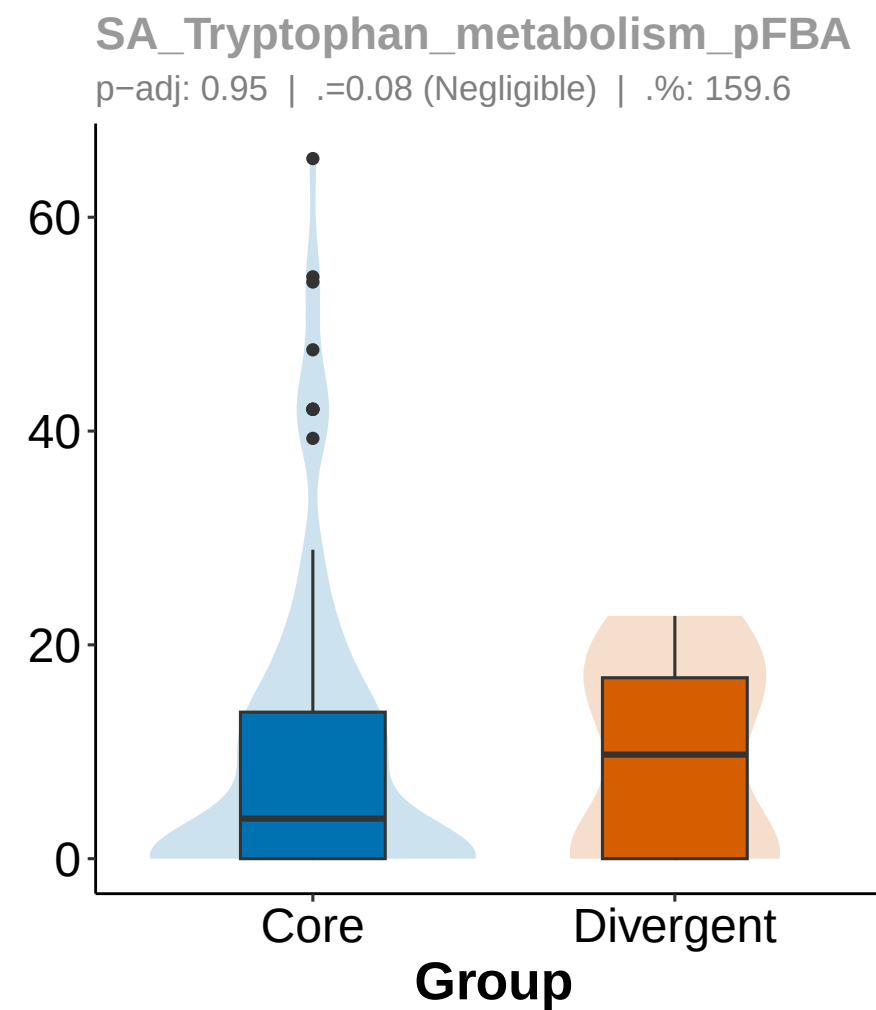
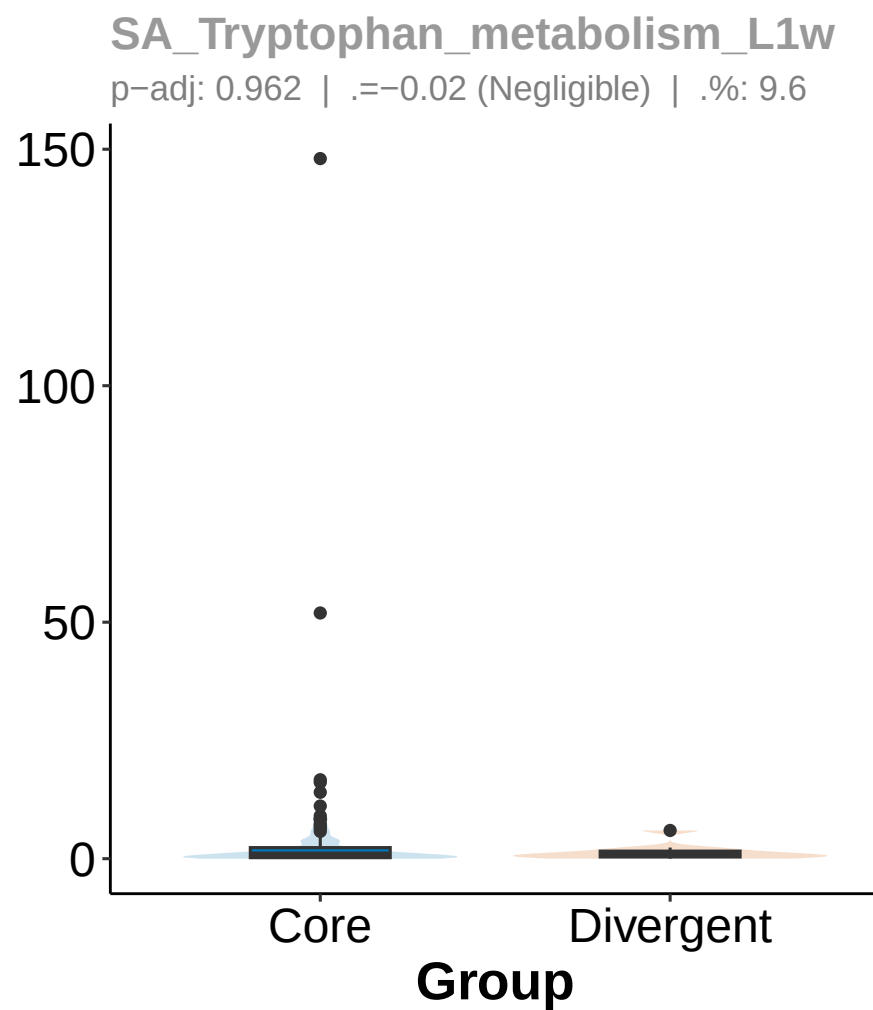
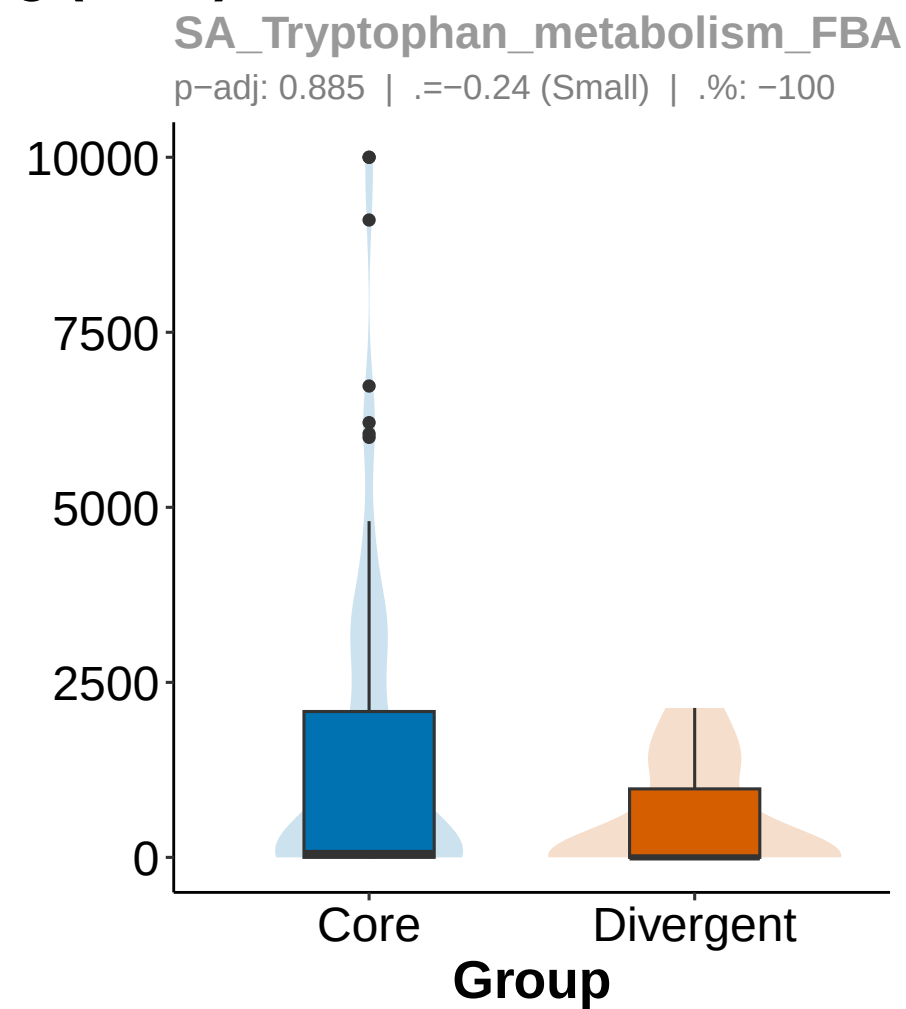
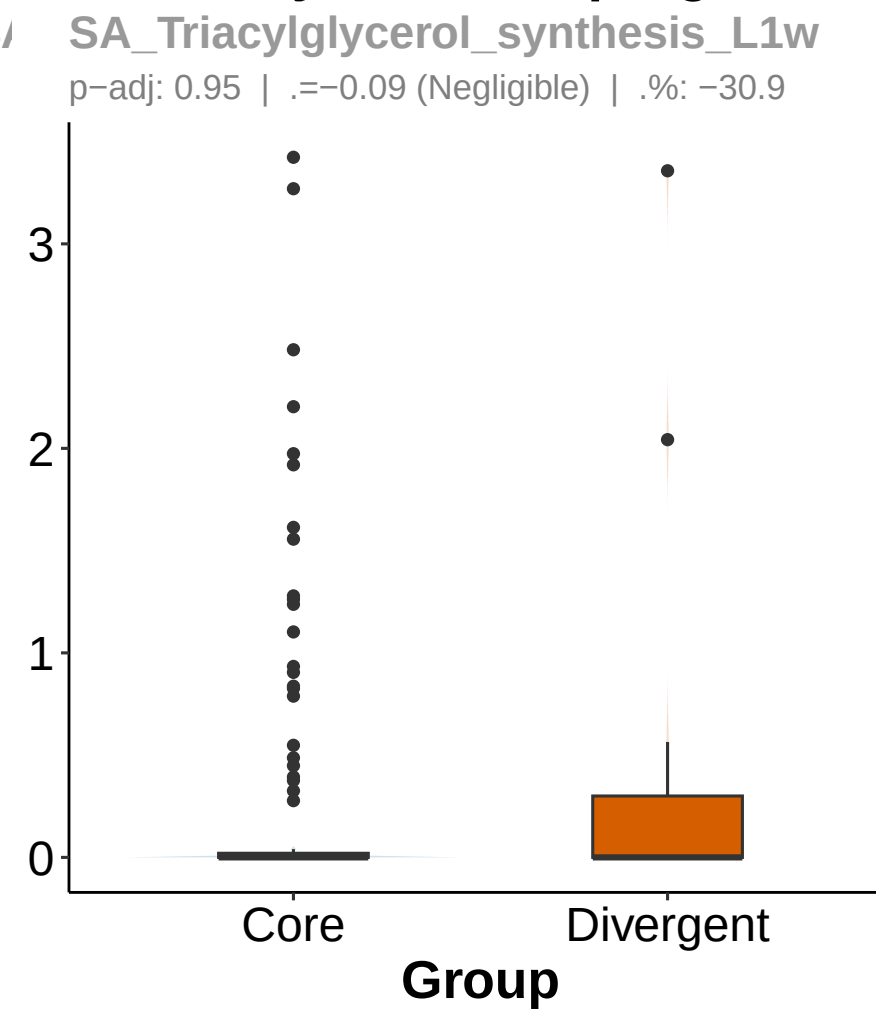
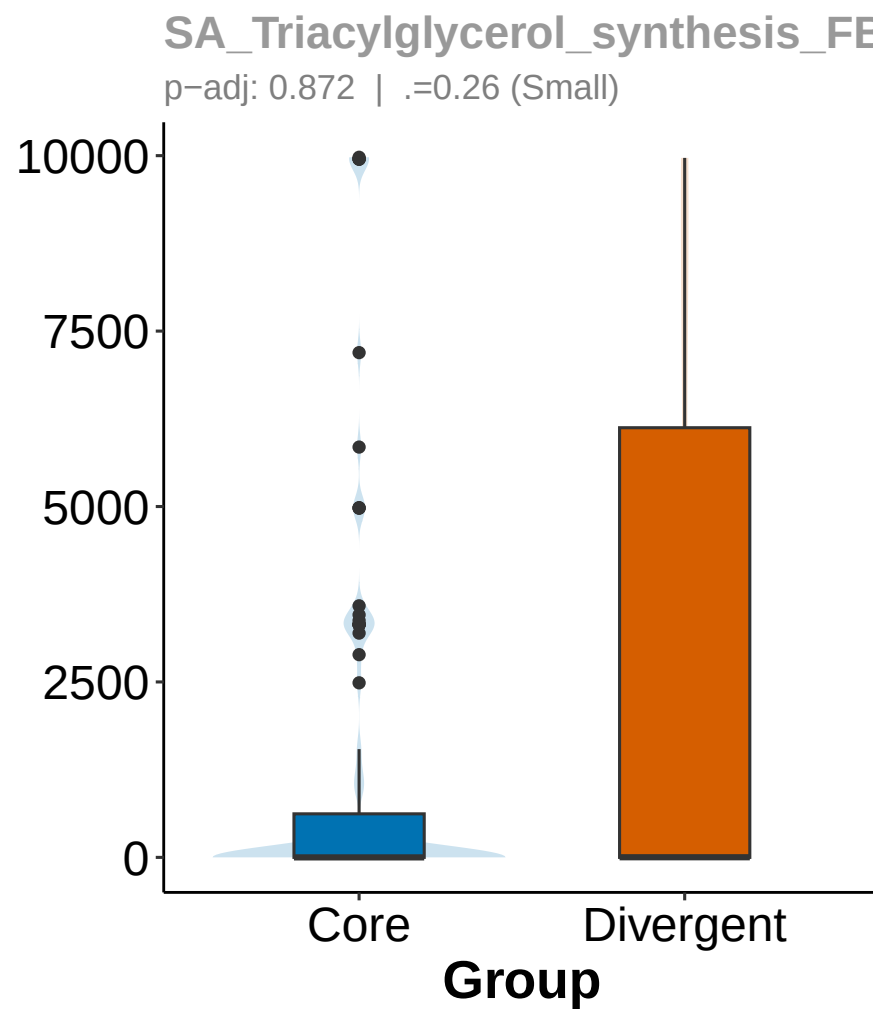


SA_Transport__peroxisomal_pFBA

p-adj: 0.885 | .=-0.1 (Negligible)



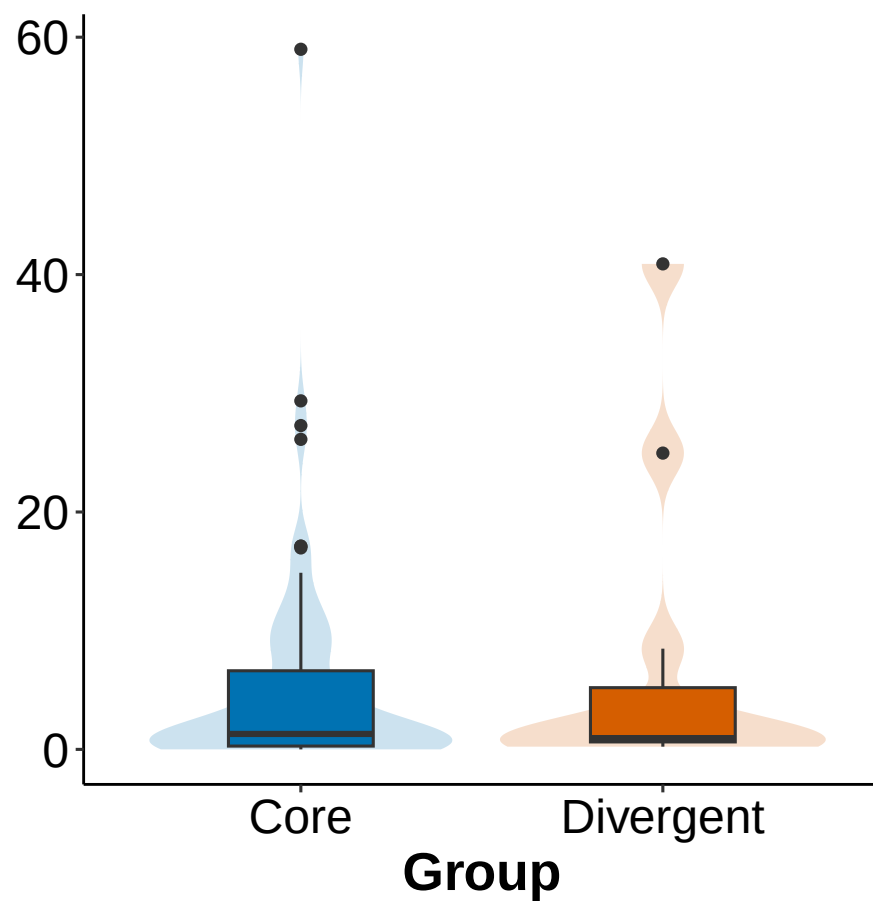
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

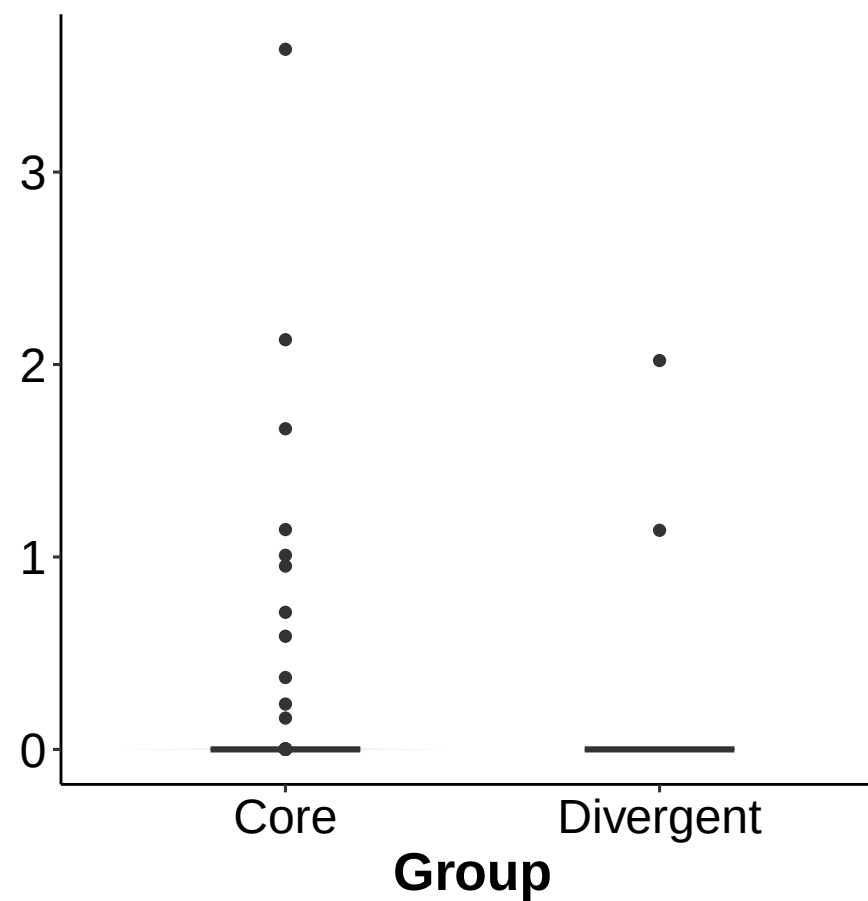
SA_Tyrosine_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible) | .%: -26.9



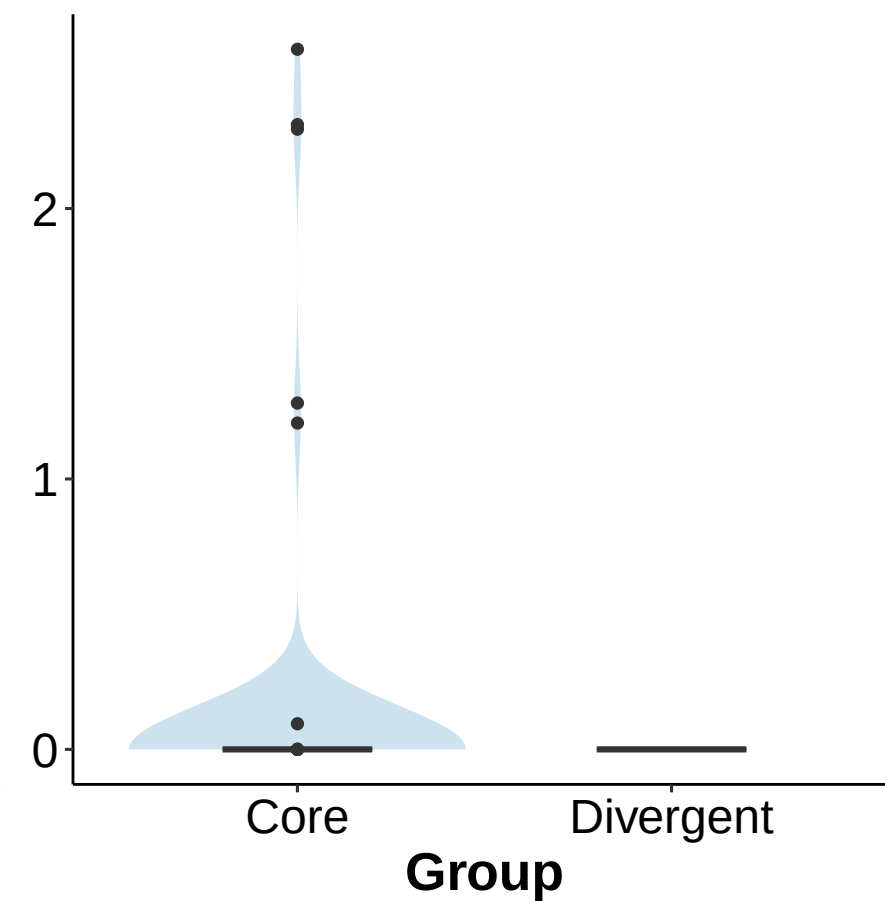
SA_Tyrosine_metabolism_pFBA

p-adj: 0.954 | . = -0.04 (Negligible)



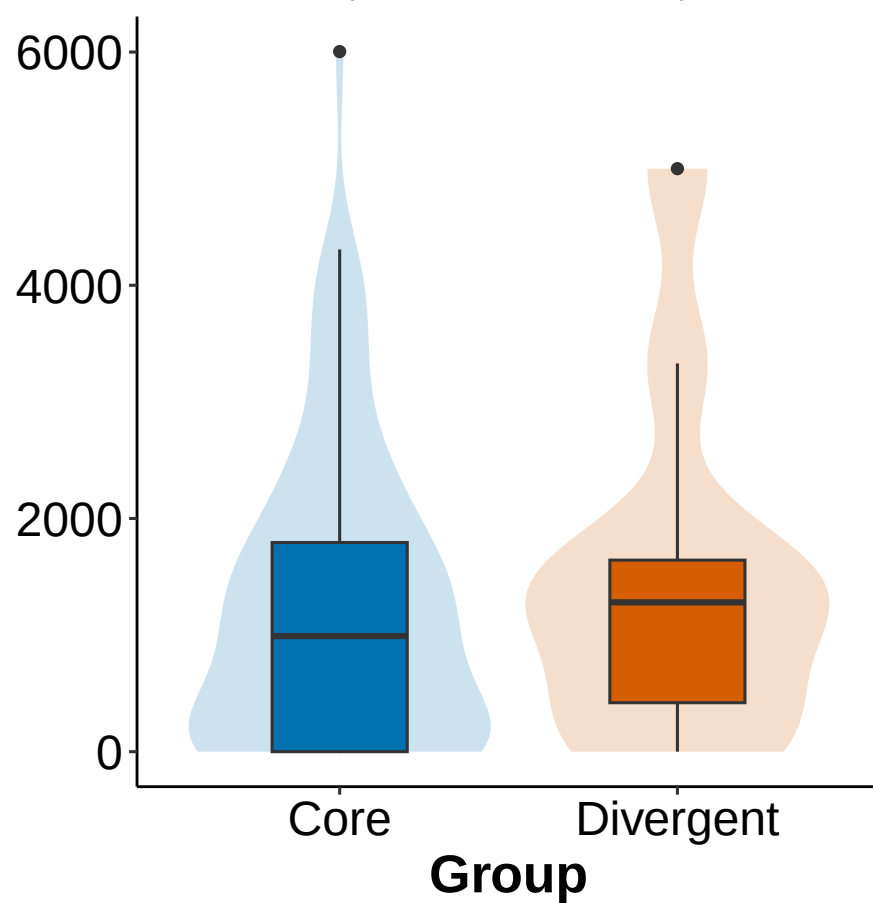
SA_Ubiquinone_synthesis_L1w

p-adj: 0.885 | . = -0.1 (Negligible)



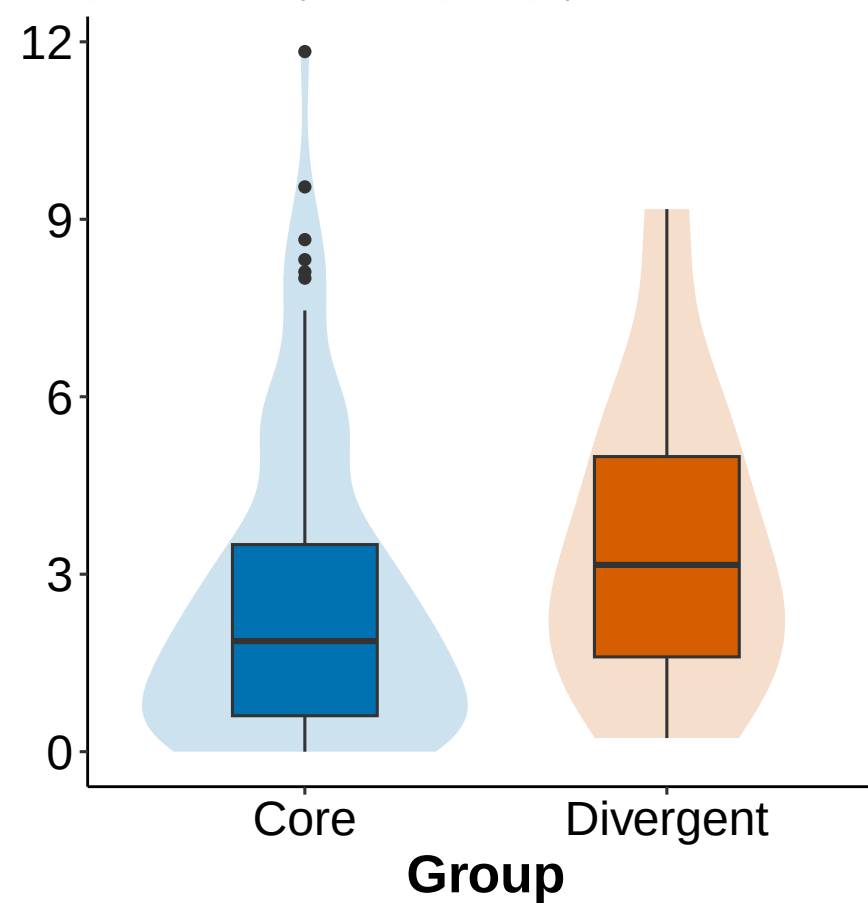
SA_Urea_cycle_FBA

p-adj: 0.945 | . = 0.09 (Negligible) | .%: 29.2



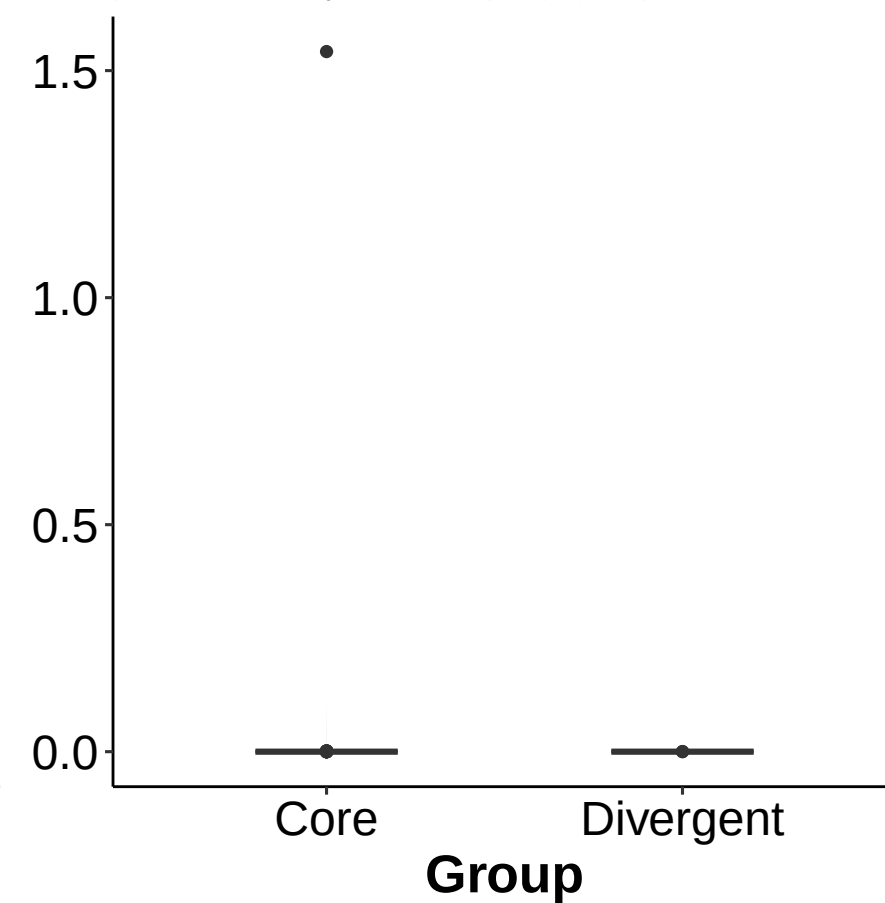
SA_Urea_cycle_L1w

p-adj: 0.872 | . = 0.28 (Small) | .%: 68.8

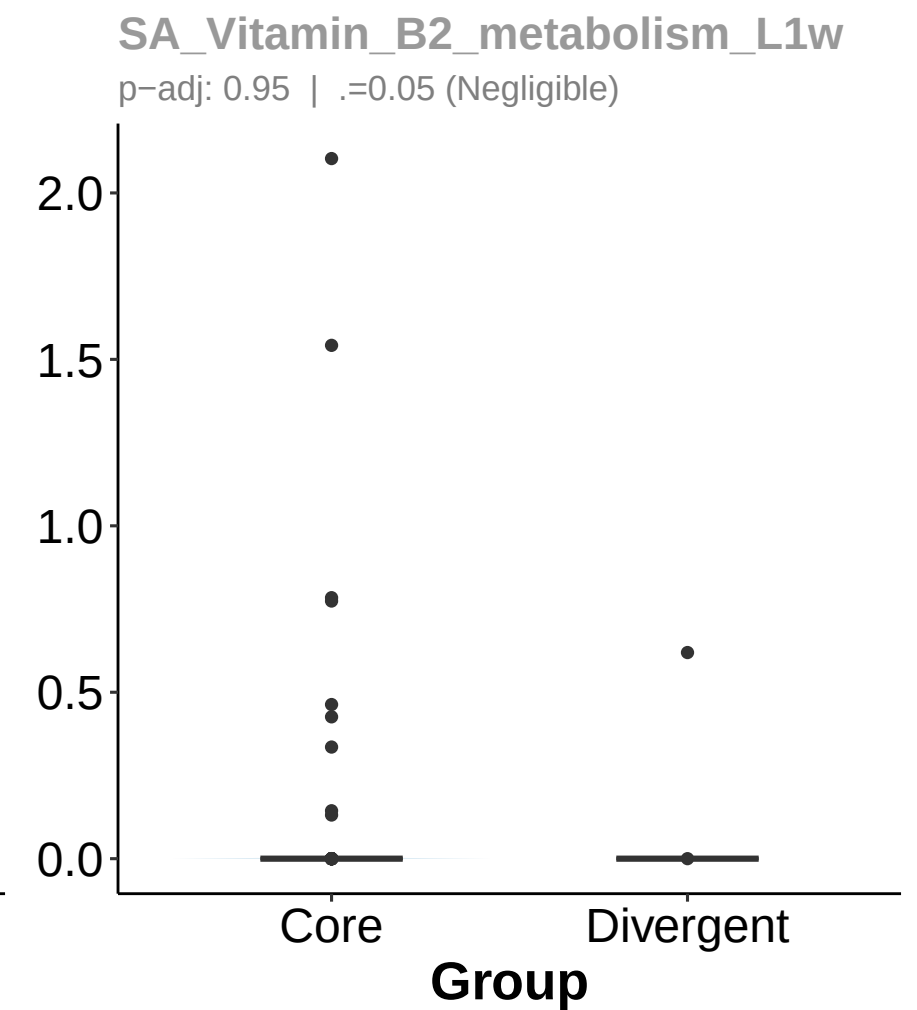
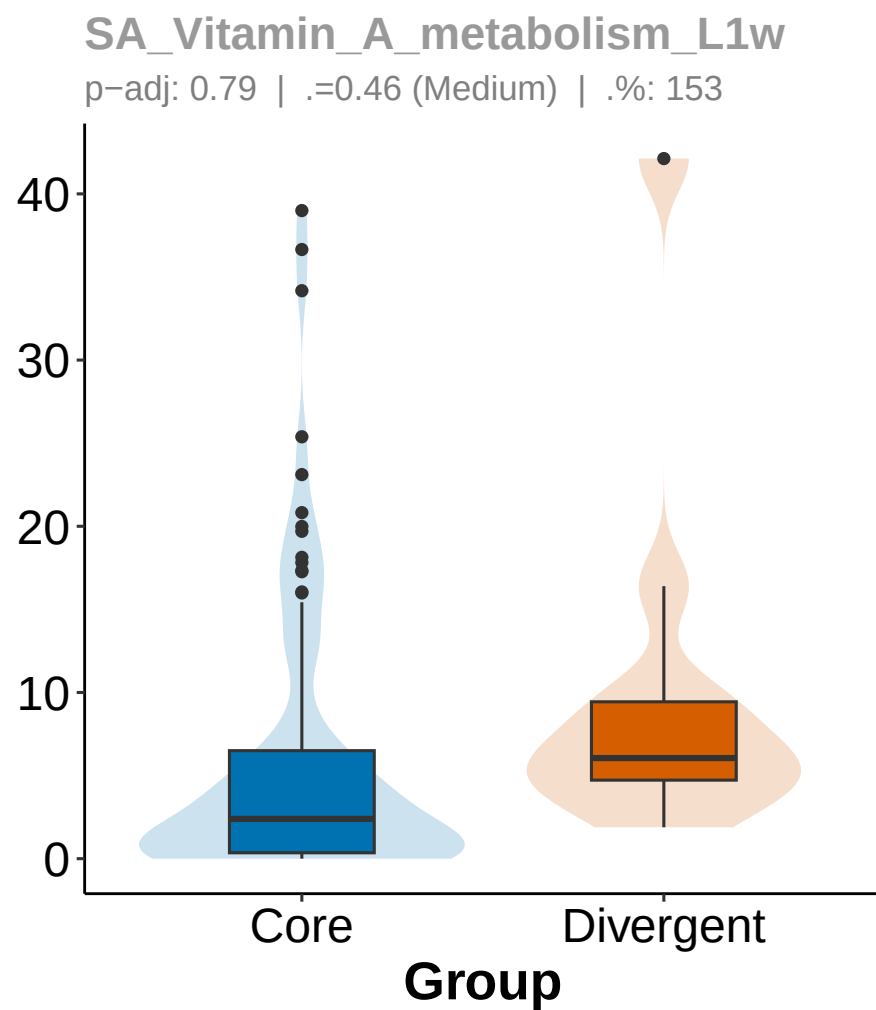
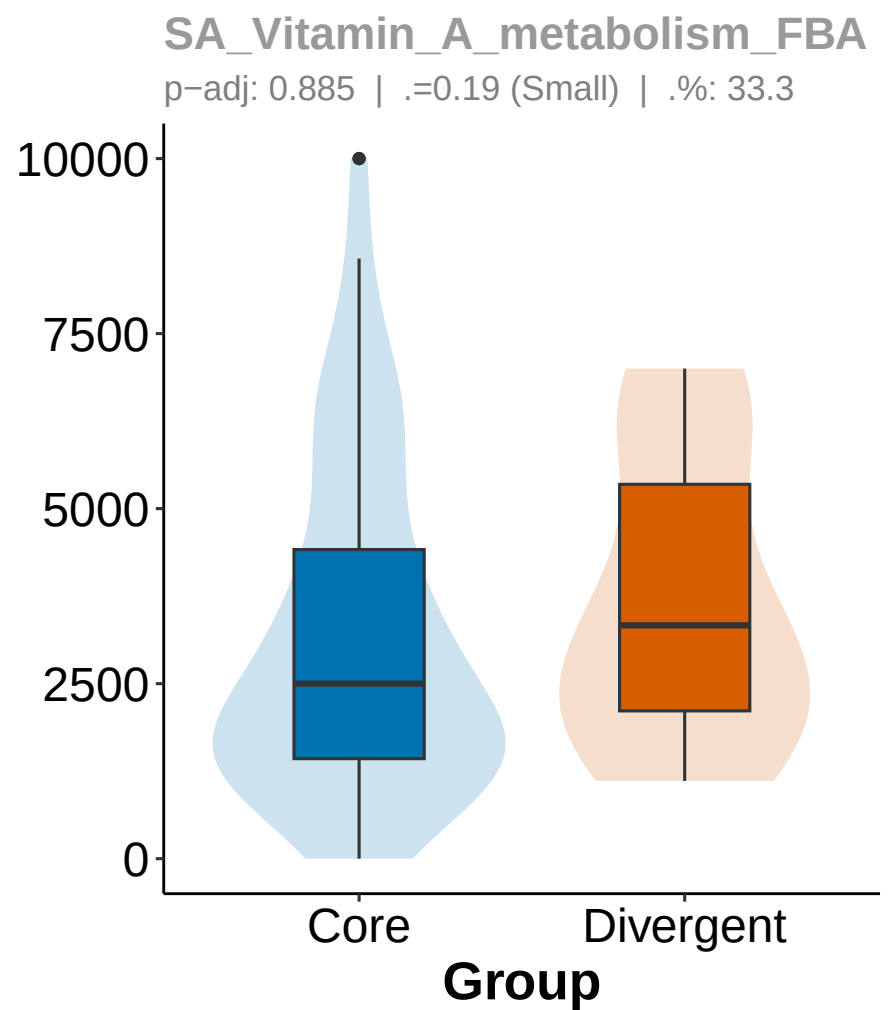
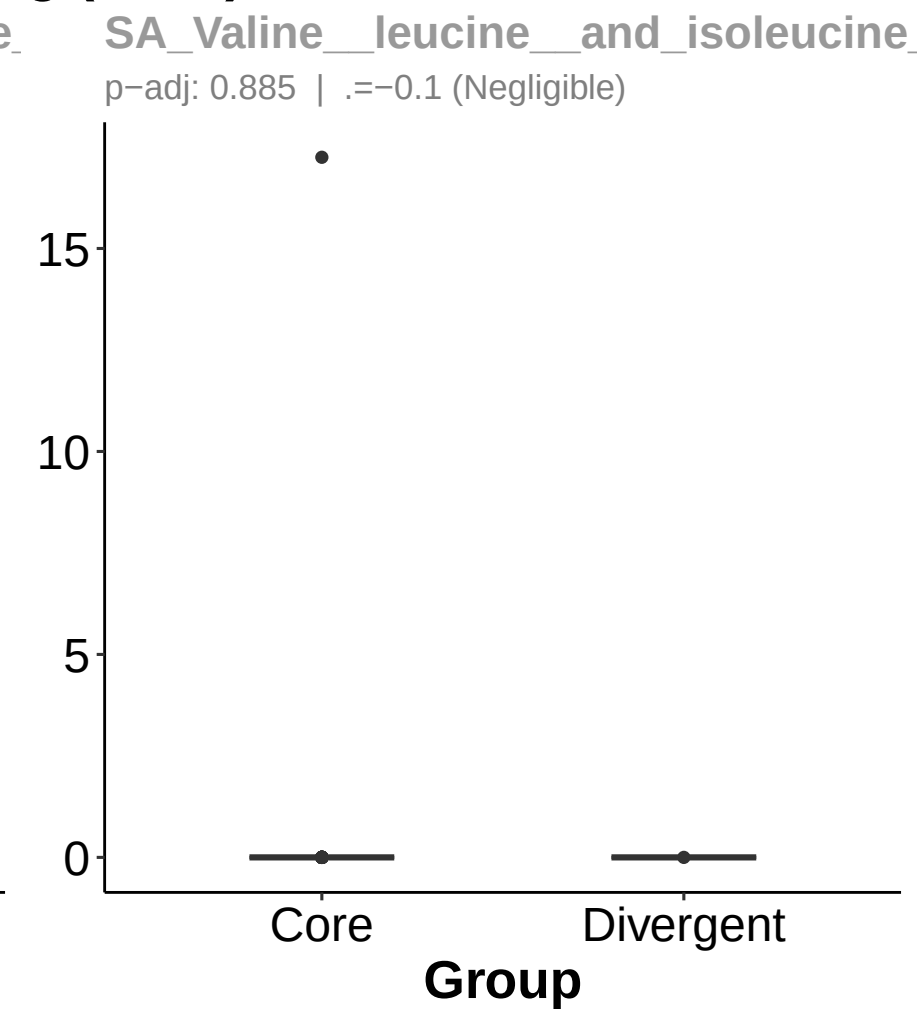
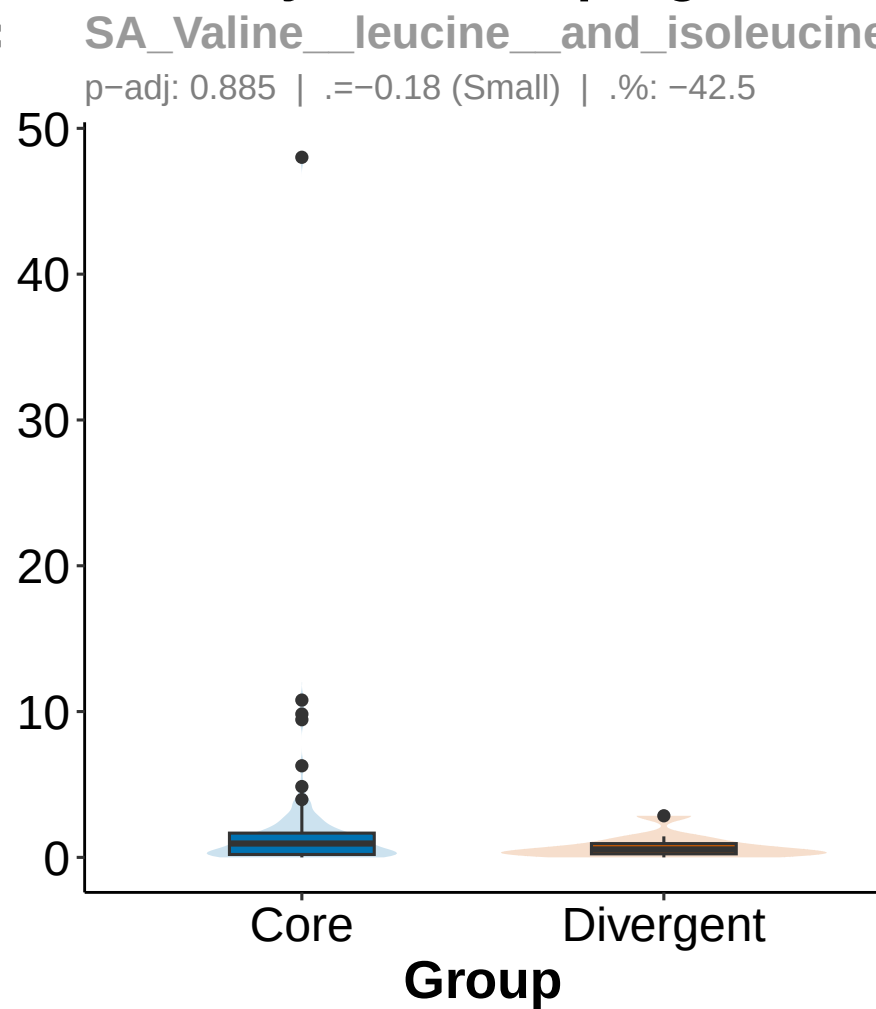
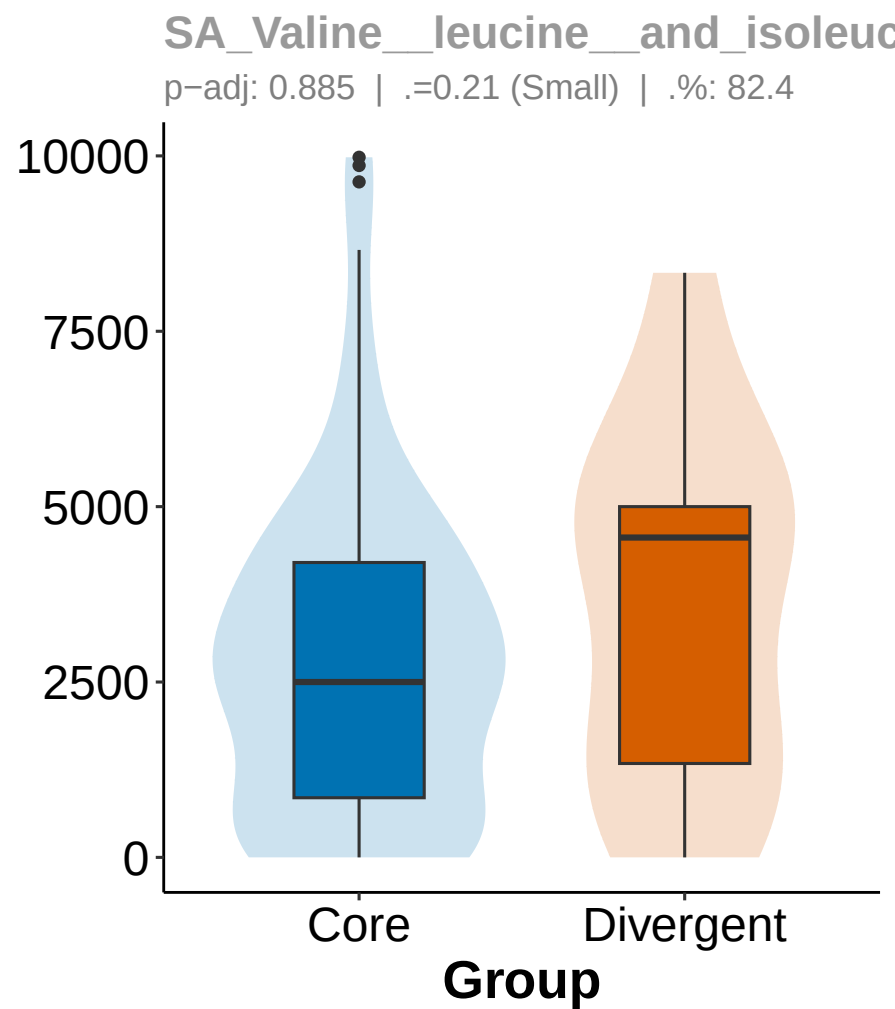


SA_Urea_cycle_pFBA

p-adj: 0.917 | . = -0.08 (Negligible)



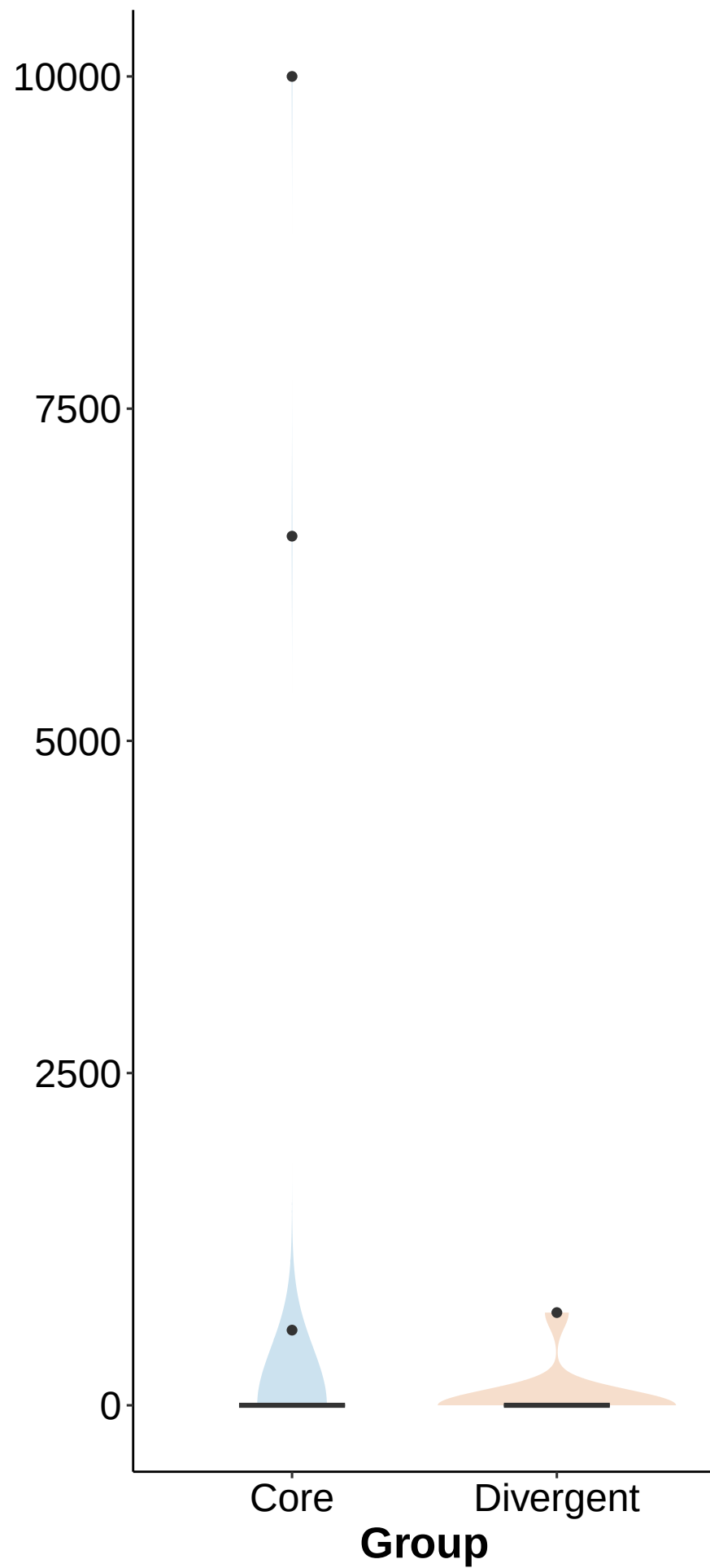
Metabolism – . 5. Systemic Reprogramming (SA_*)



Metabolism – . 5. Systemic Reprogramming (SA_*)

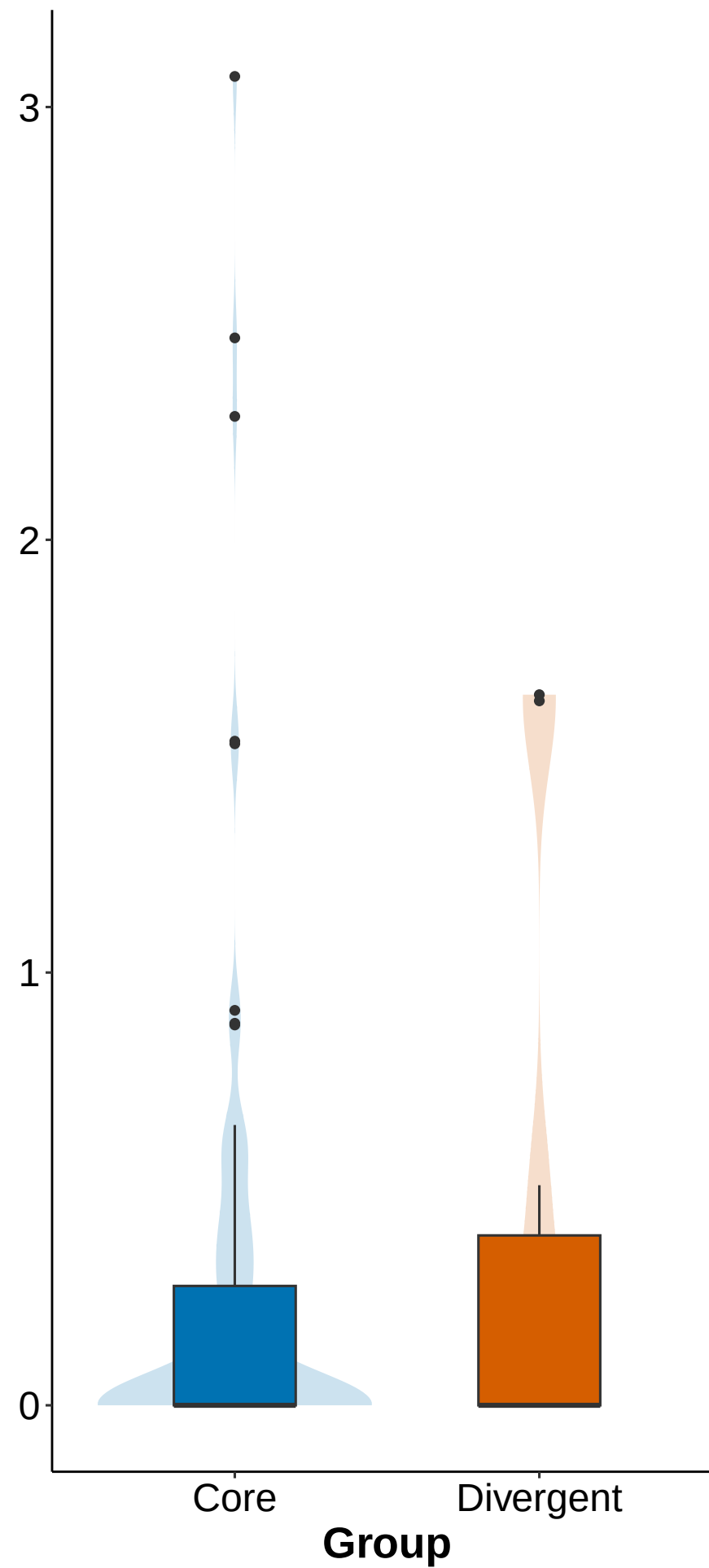
SA_Vitamin_C_metabolism_FBA

p-adj: 0.885 | . = 0.06 (Negligible)



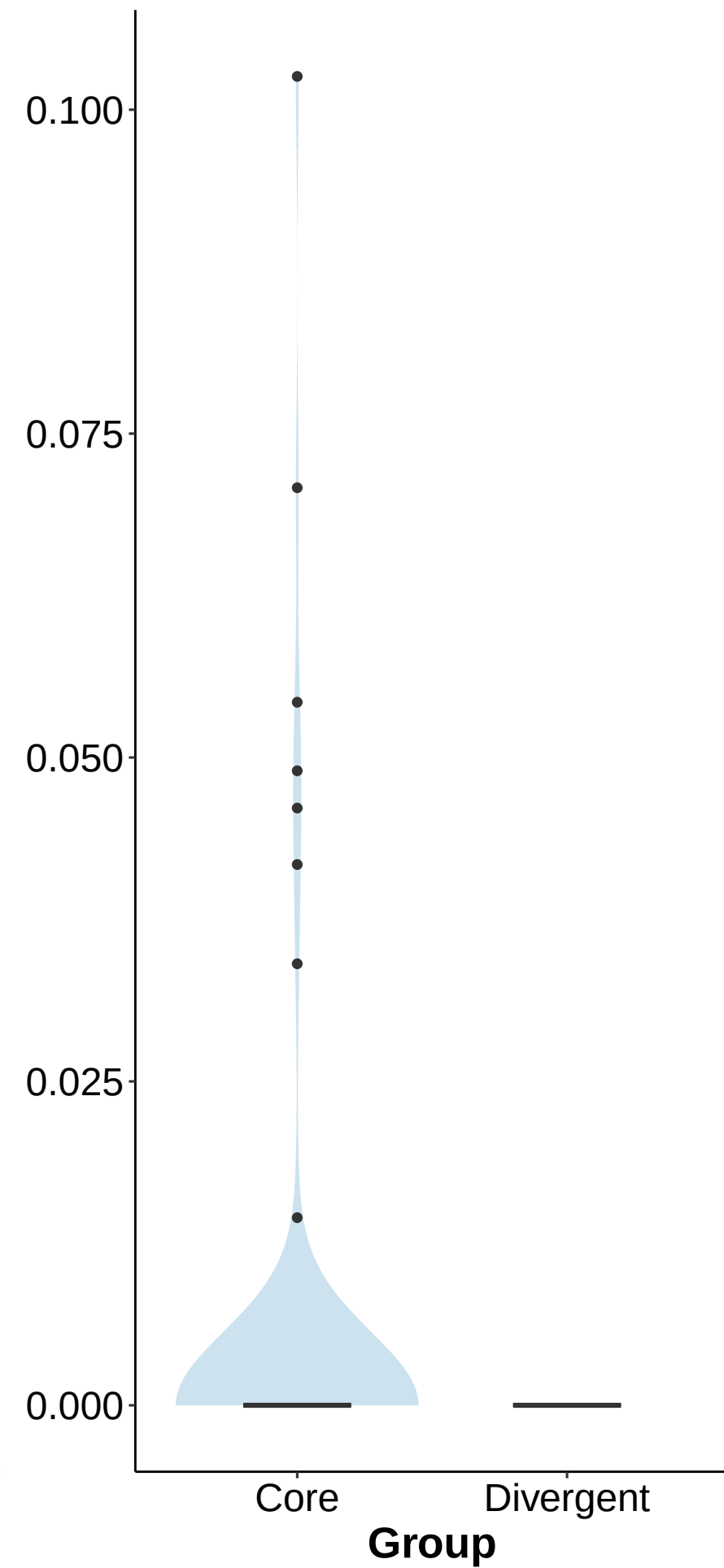
SA_Vitamin_C_metabolism_L1w

p-adj: 0.95 | . = 0.06 (Negligible) | .%: -22.6



SA_Vitamin_D_metabolism_L1w

p-adj: 0.885 | . = -0.08 (Negligible)



Cliff's .: Top 12 Metabolic Variables

Blue = Core > Divergent | Orange = Divergent > Core | * p_{adj}<0.05

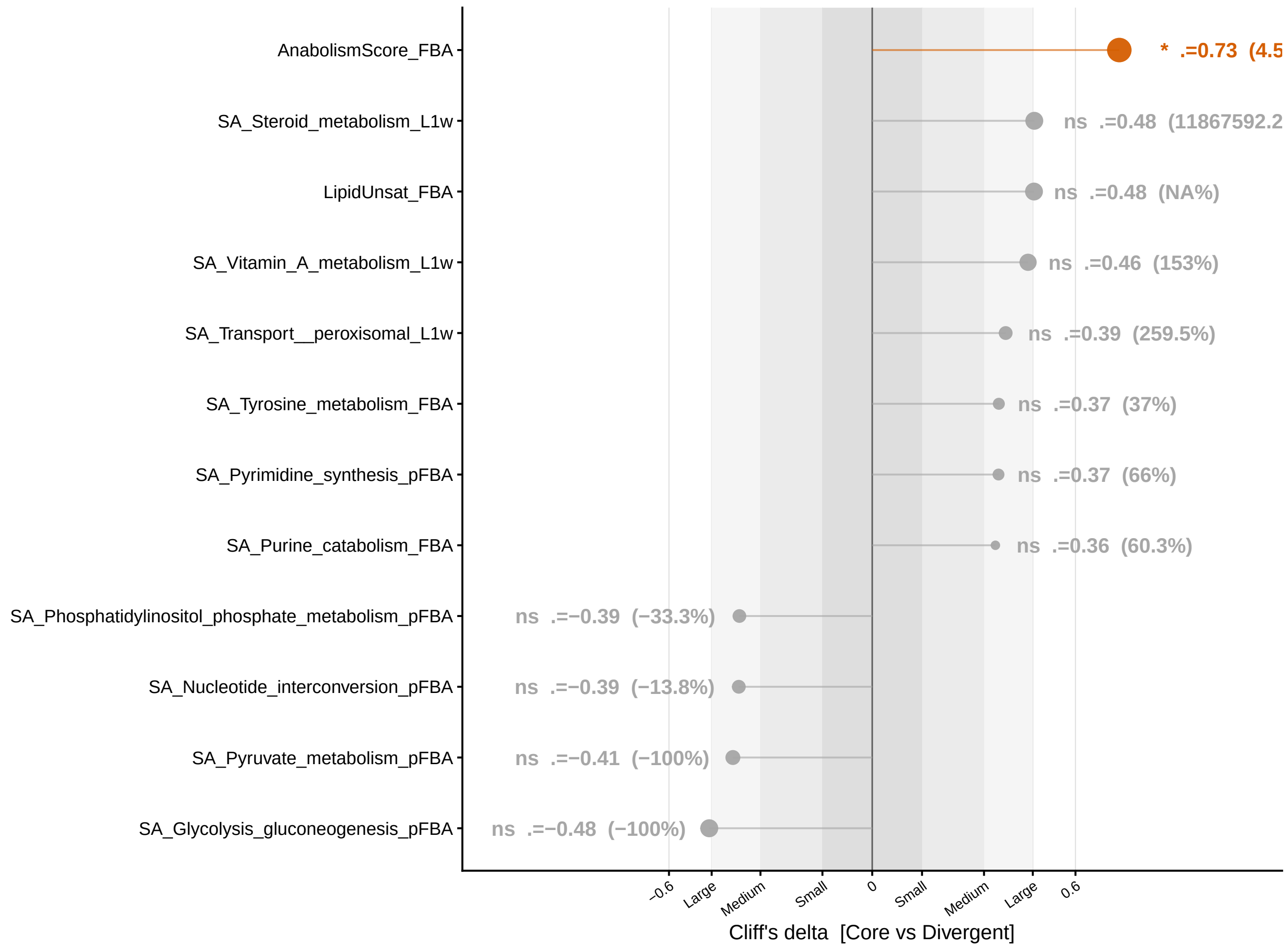
Cliff's .
Div vs Core



AnabolismScore_FBA	* .=0.73
SA_Steroid_metabolism_L1w	ns .=0.48
LipidUnsat_FBA	ns .=0.48
SA_Vitamin_A_metabolism_L1w	ns .=0.46
SA_Transport__peroxisomal_L1w	ns .=0.39
SA_Tyrosine_metabolism_FBA	ns .=0.37
SA_Pyrimidine_synthesis_pFBA	ns .=0.37
SA_Purine_catabolism_FBA	ns .=0.36
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	ns .=-0.39
SA_Nucleotide_interconversion_pFBA	ns .=-0.39
SA_Pyruvate_metabolism_pFBA	ns .=-0.41
SA_Glycolysis_gluconeogenesis_pFBA	ns .=-0.48

. Top 12 Metabolic Variables – Cliff's.

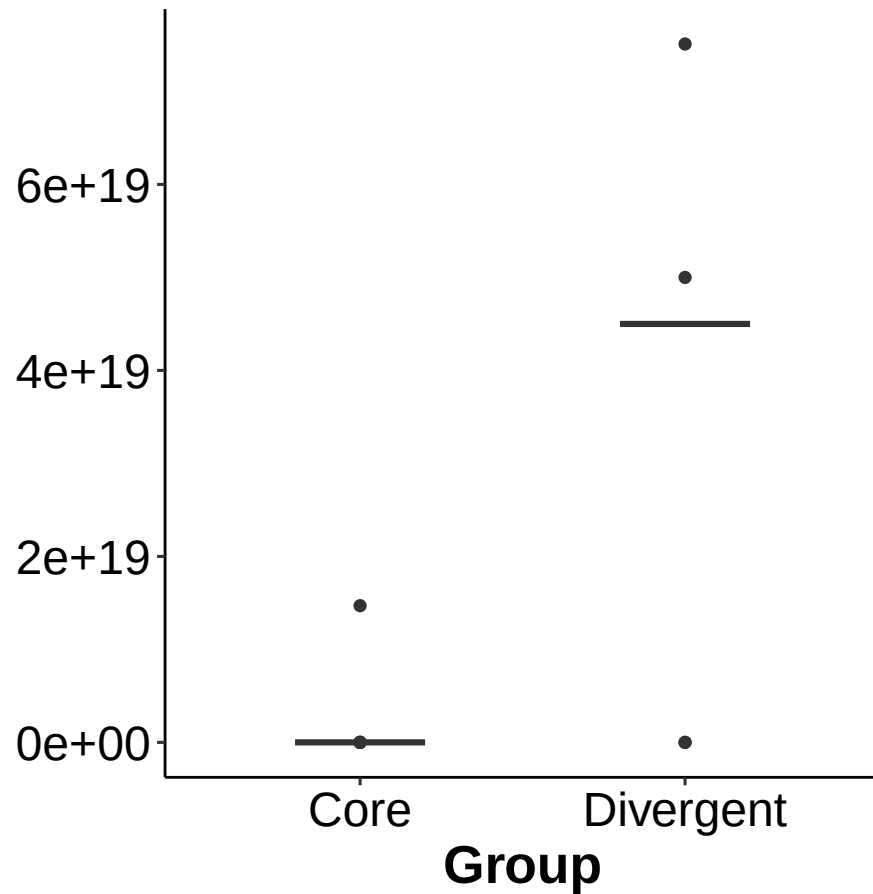
Cliff's . + .% median | . higher in Core . higher in Divergent



. Top 12 Metabolic Variables (p_adj + Cliff's .)

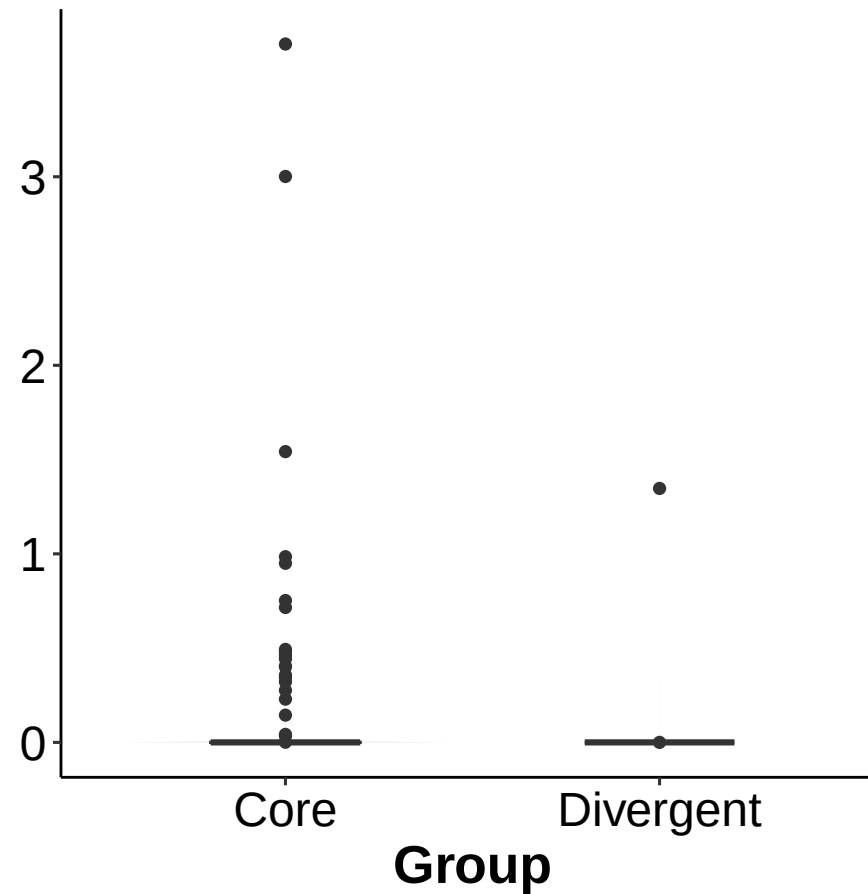
AnabolismScore_FBA

p-adj: 0.0481 | . = 0.73 (Large) | .%: 4.562543



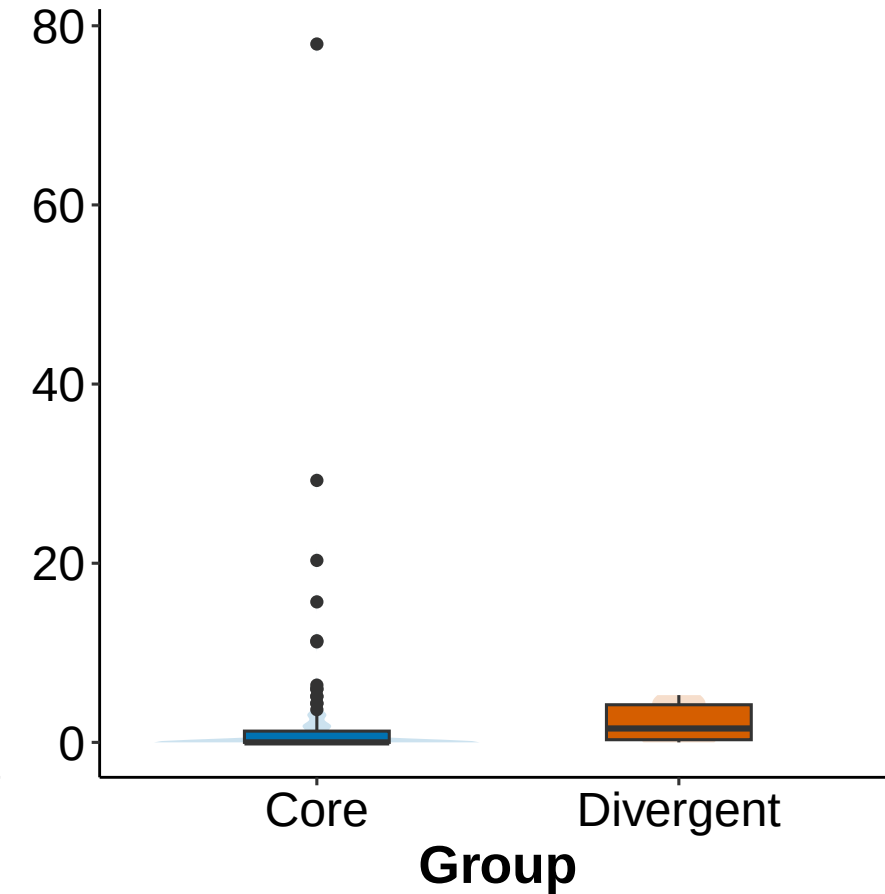
SA_Glycolysis_gluconeogenesis_pFBA

p-adj: 0.749 | . = -0.48 (Large) | .%: -100



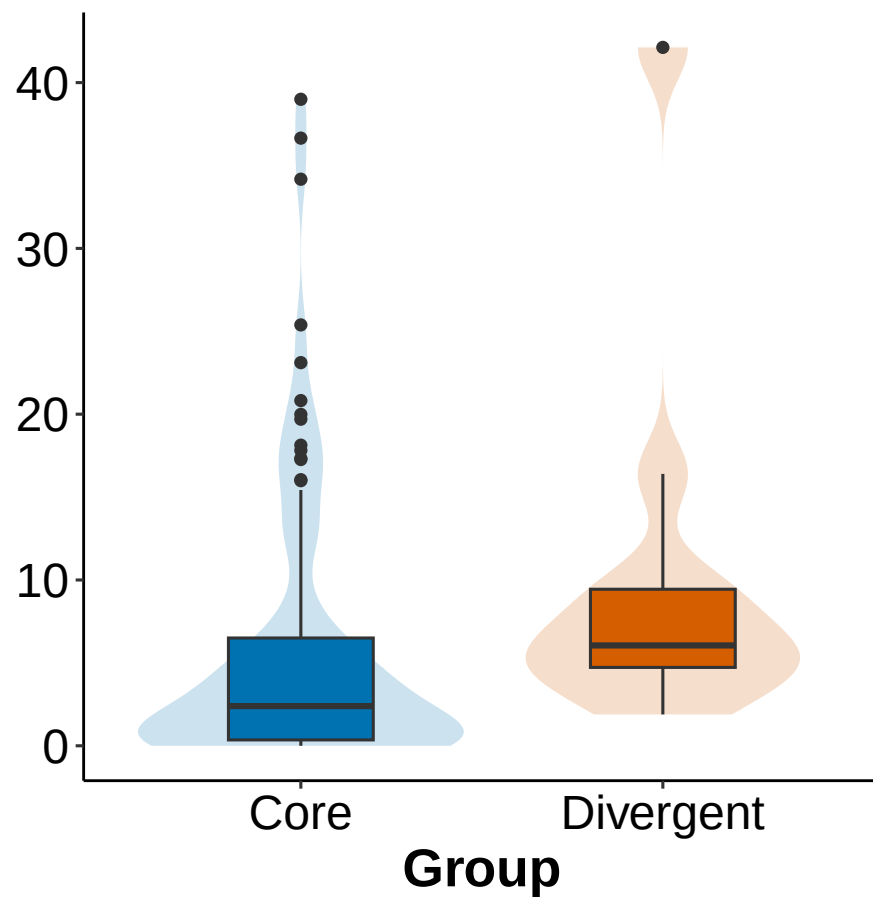
SA_Steroid_metabolism_L1w

p-adj: 0.749 | . = 0.48 (Large) | .%: 11867592.2



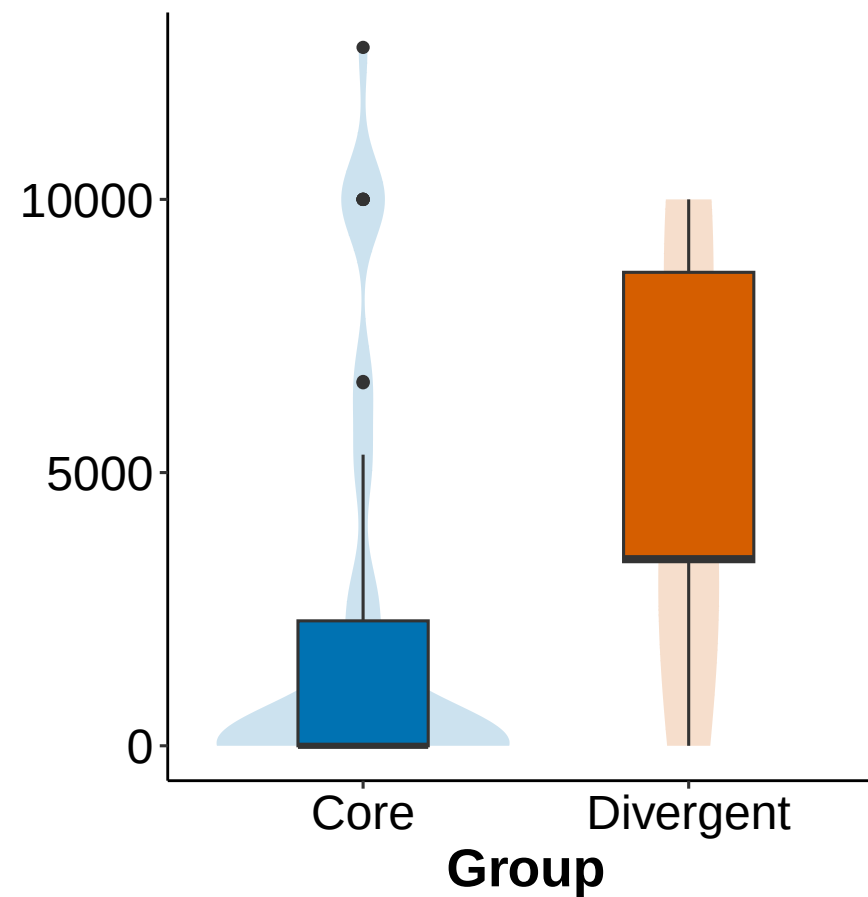
SA_Vitamin_A_metabolism_L1w

p-adj: 0.79 | . = 0.46 (Medium) | .%: 153



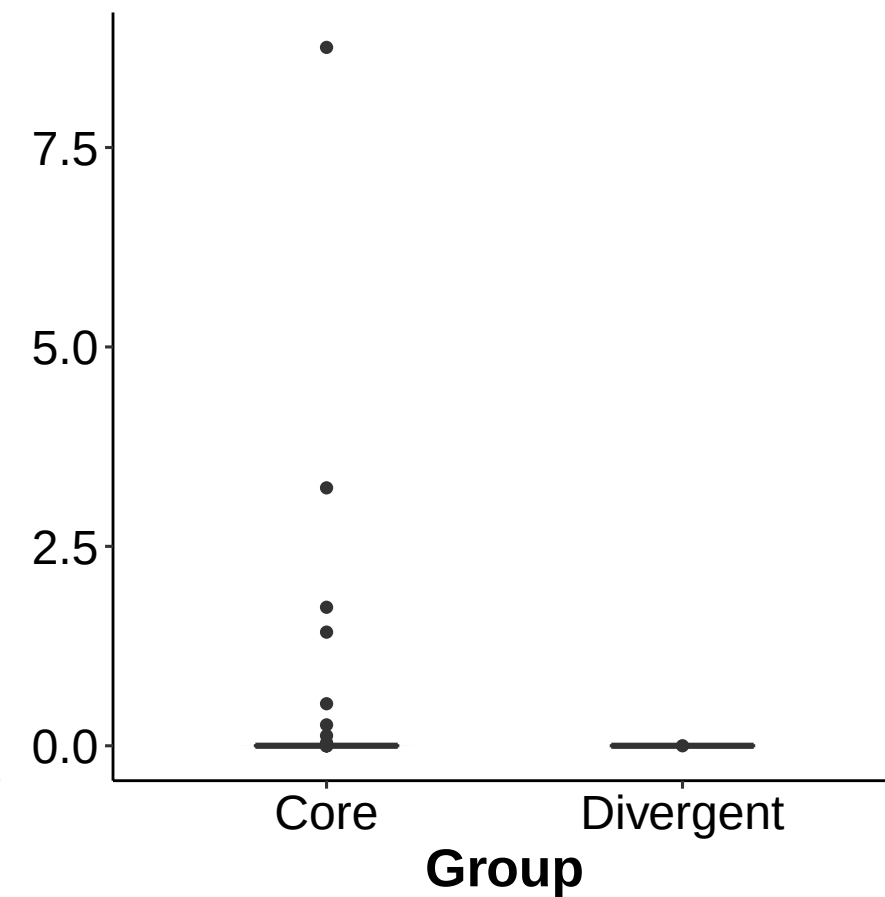
LipidUnsat_FBA

p-adj: 0.843 | . = 0.48 (Large)



SA_Pyruvate_metabolism_pFBA

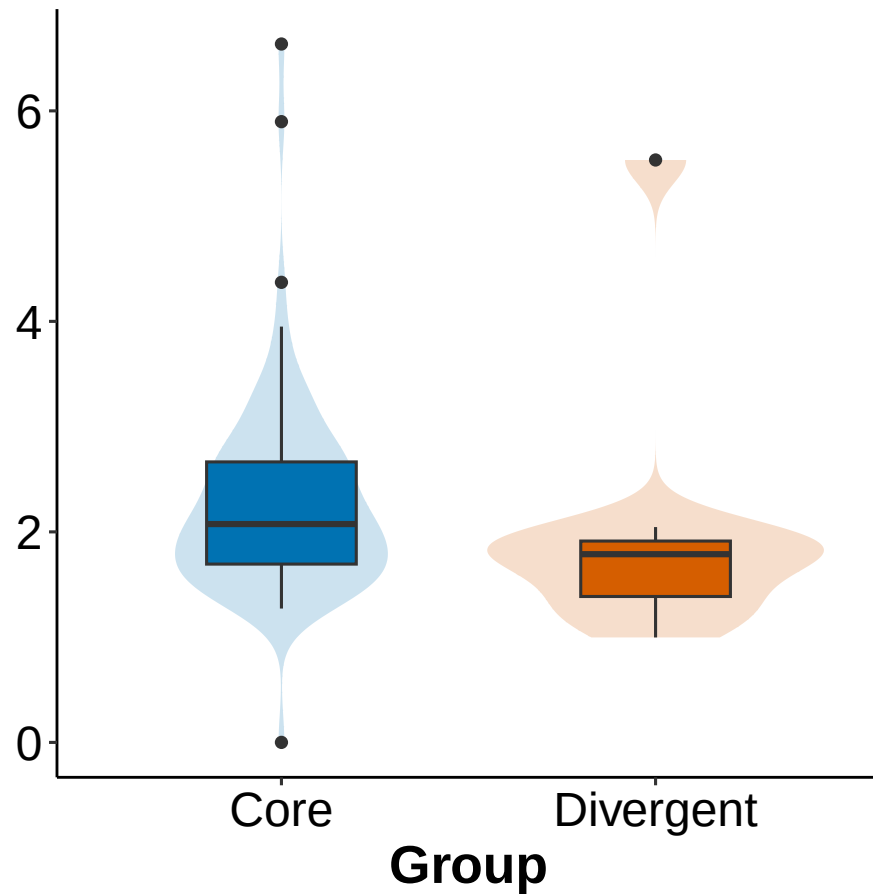
p-adj: 0.843 | . = -0.41 (Medium) | .%: -100



. Top 12 Metabolic Variables (p_adj + Cliff's .)

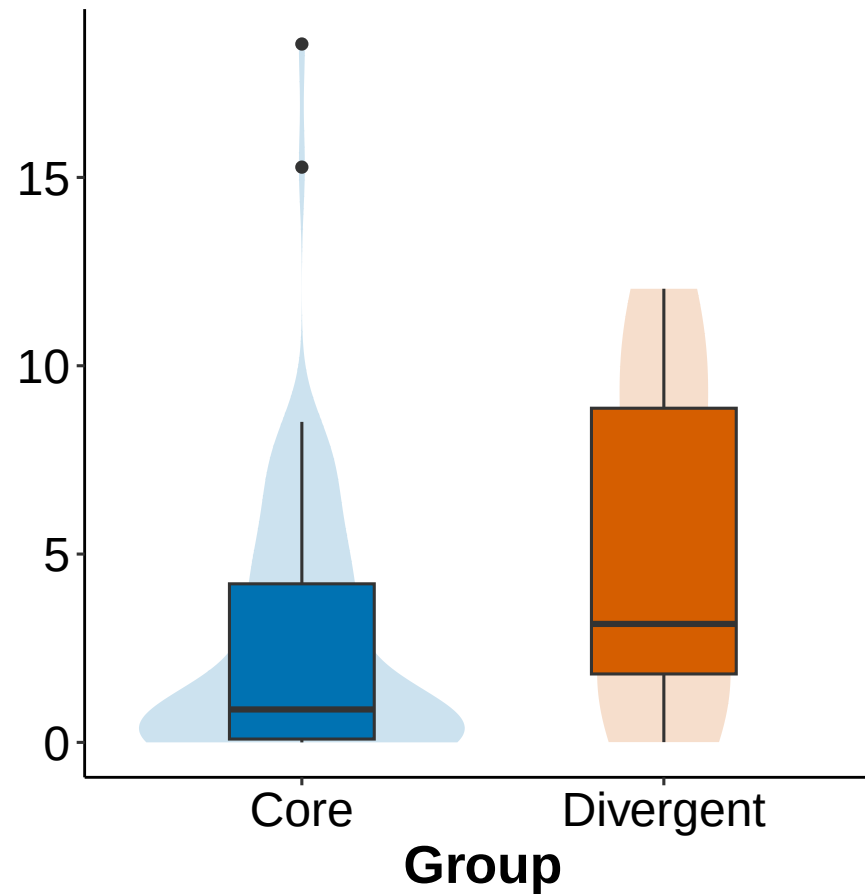
SA_Nucleotide_interconversion_pFBA

p-adj: 0.843 | .=-0.39 (Medium) | .%: -13.8



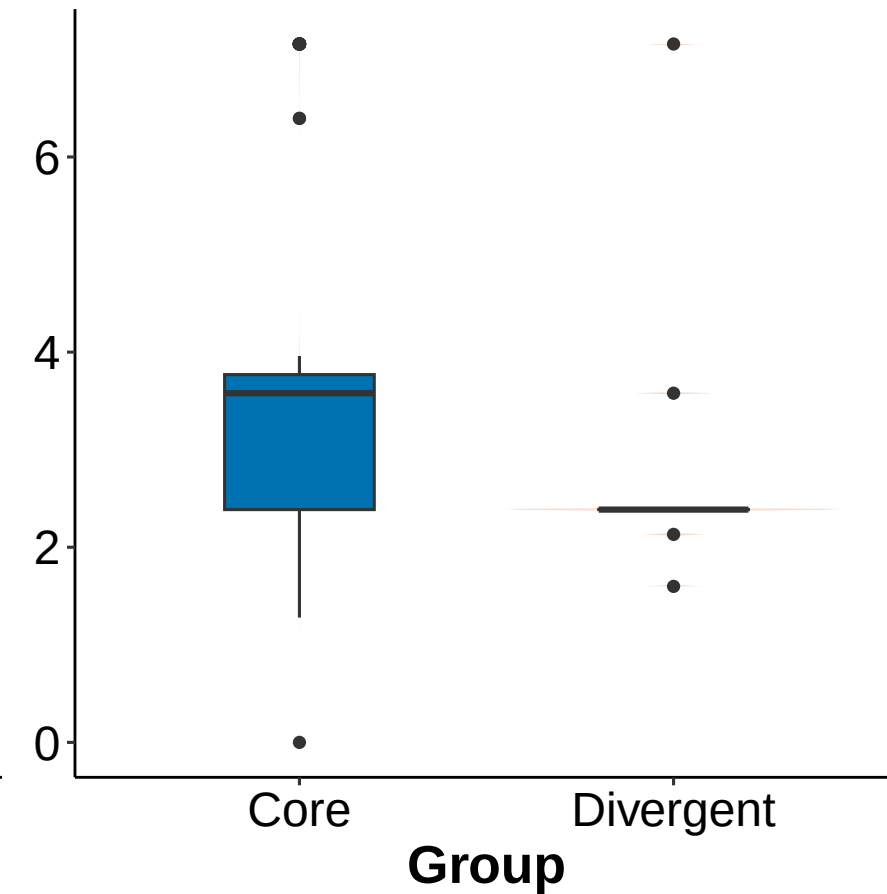
SA_Transport__peroxisomal_L1w

p-adj: 0.843 | .=0.39 (Medium) | .%: 259.5



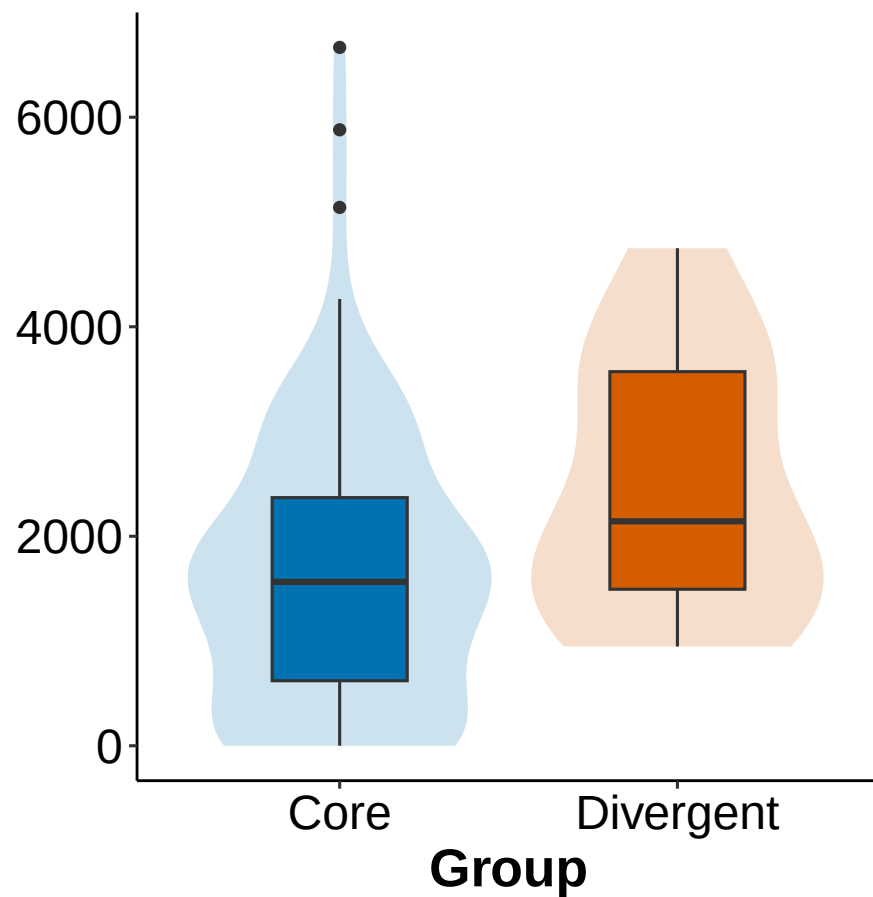
SA_Phosphatidylinositol_phosphate_m

p-adj: 0.843 | .=-0.39 (Medium) | .%: -33.3



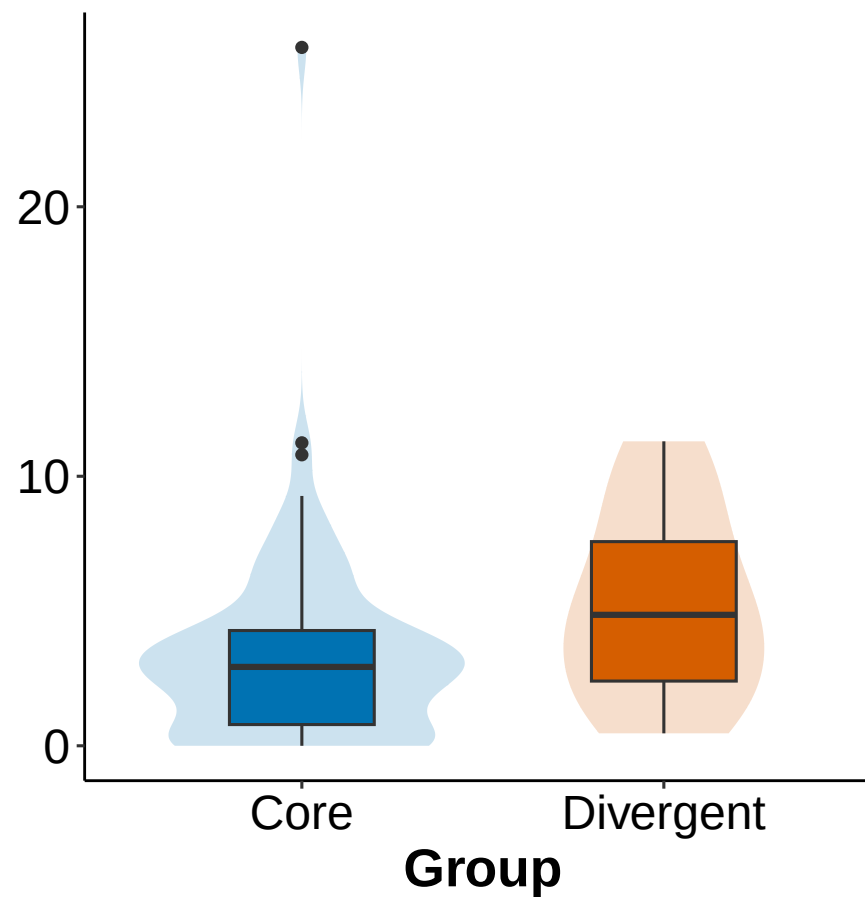
SA_Tyrosine_metabolism_FBA

p-adj: 0.843 | .=0.37 (Medium) | .%: 37



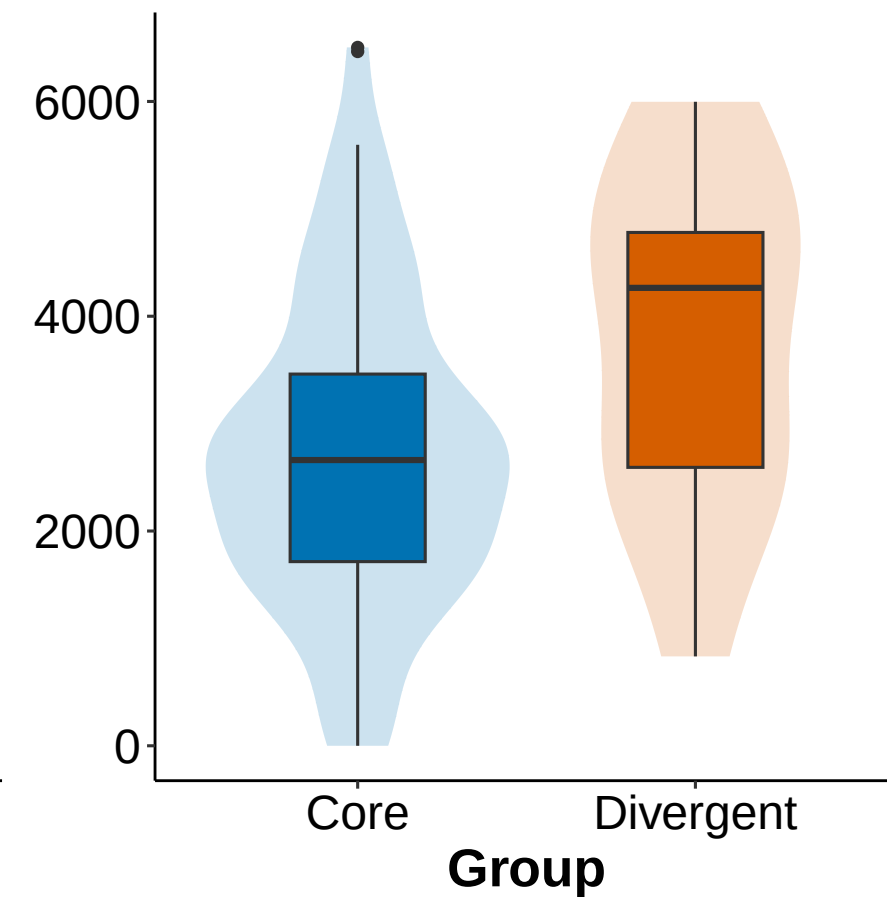
SA_Pyrimidine_synthesis_pFBA

p-adj: 0.843 | .=0.37 (Medium) | .%: 66



SA_Purine_catabolism_FBA

p-adj: 0.843 | .=0.36 (Medium) | .%: 60.3



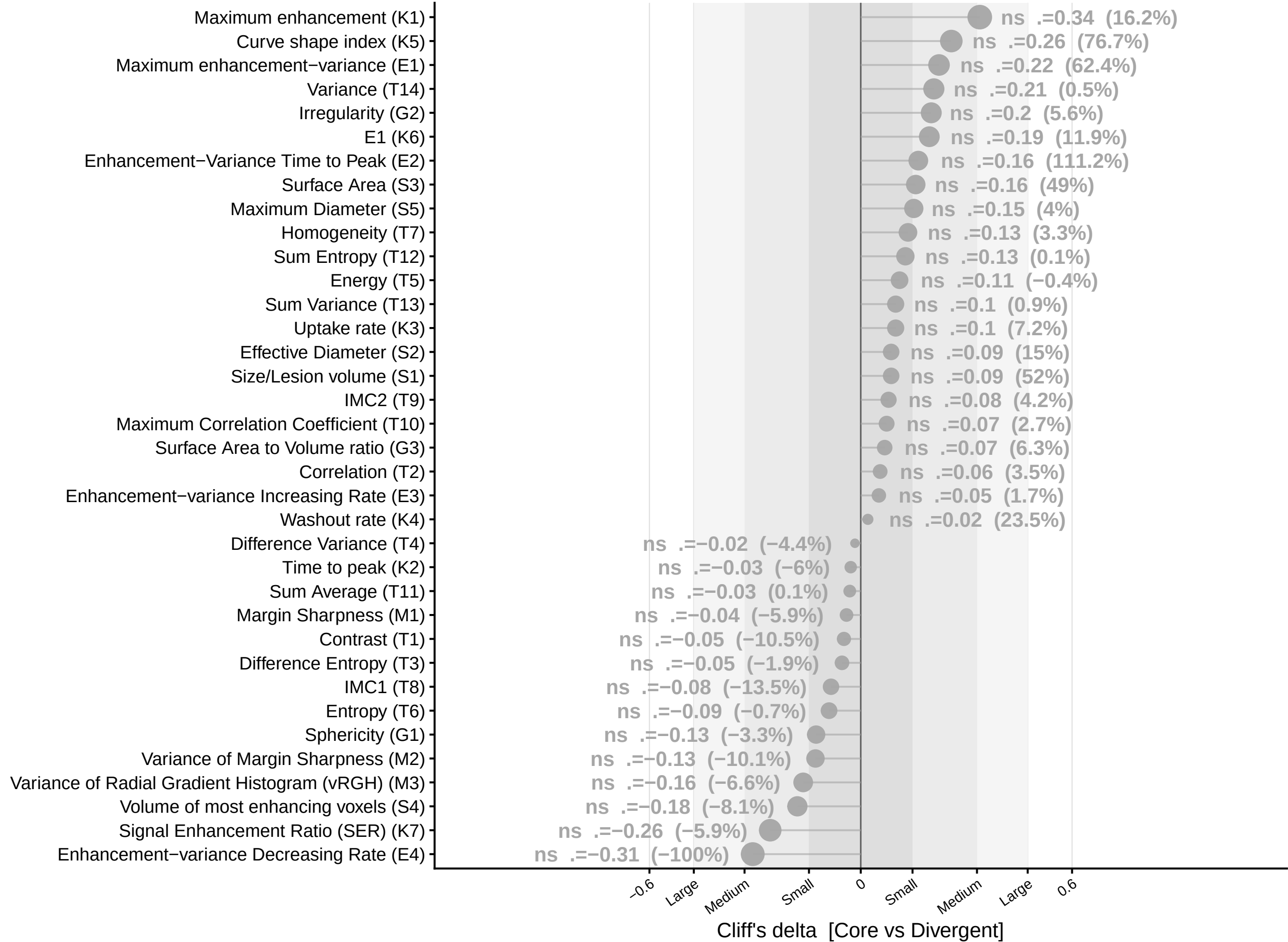
SECTION II

Radiomic Differences: Core vs Divergent

6 domains: Kinetic curve | Enhancement variance | Texture | Geometry | Margin | Size

All Radiomic Variables – Cliff's .

Cliff's . + .% median | . higher in Core . higher in Divergent



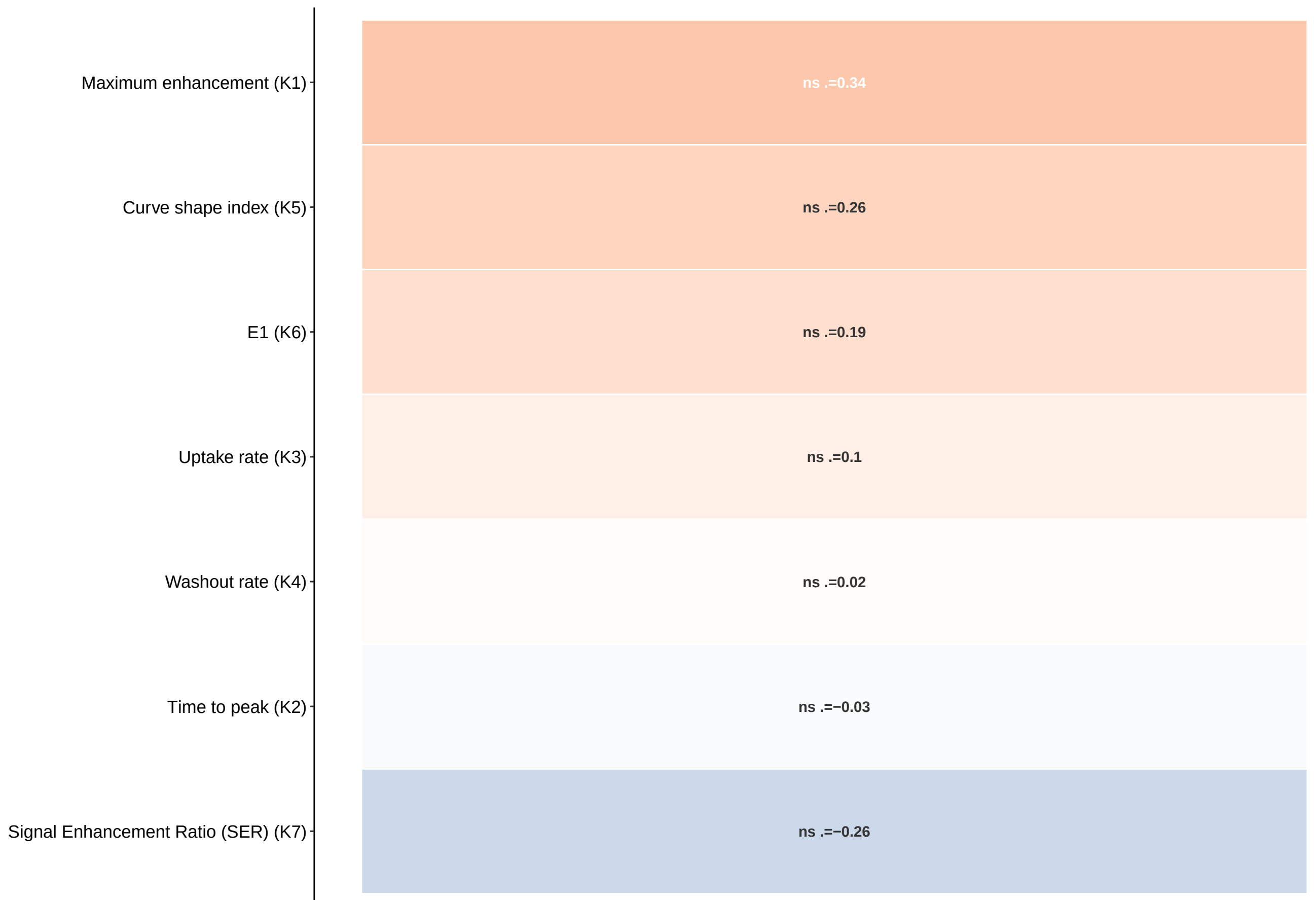
Cliff's .: K – Kinetic curve (enhancement kinetics)

Blue = Core > Divergent | Orange = Divergent > Core | * p_{adj}<0.05

Cliff's .
Div vs Core

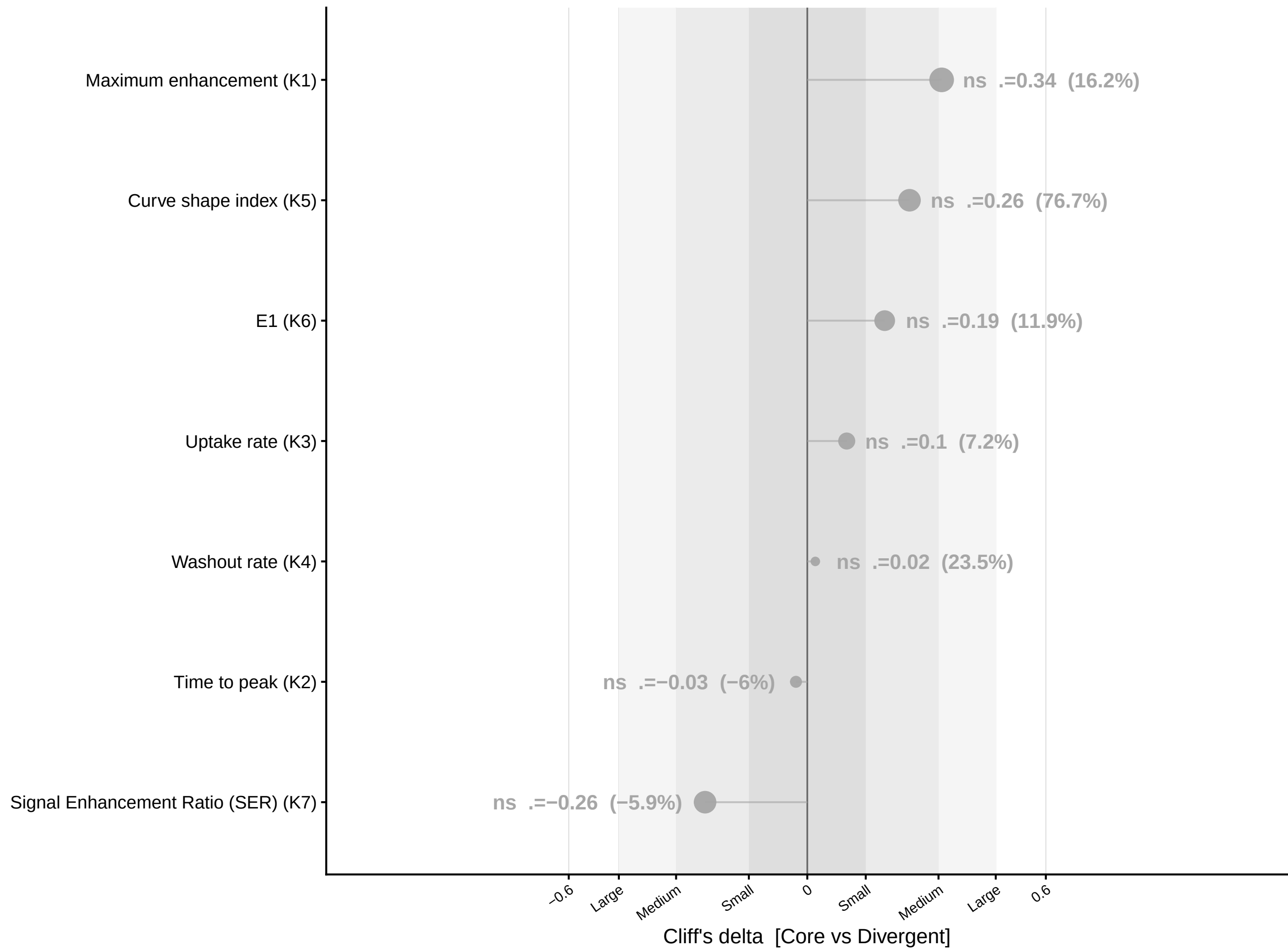


1.00.50.00.51.0



Cliff's Δ - K = Kinetic curve (enhancement kinetics)

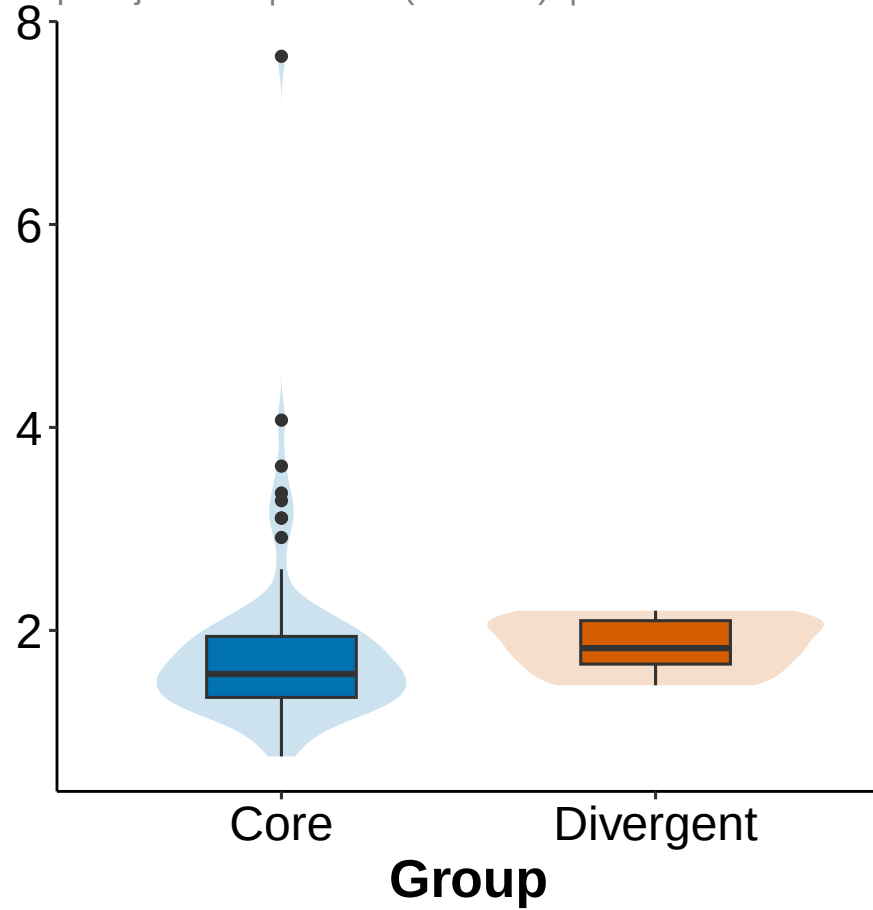
Cliff's Δ + .% median | . higher in Core · . higher in Divergent



Radiomics – K – Kinetic curve (enhancement kinetics)

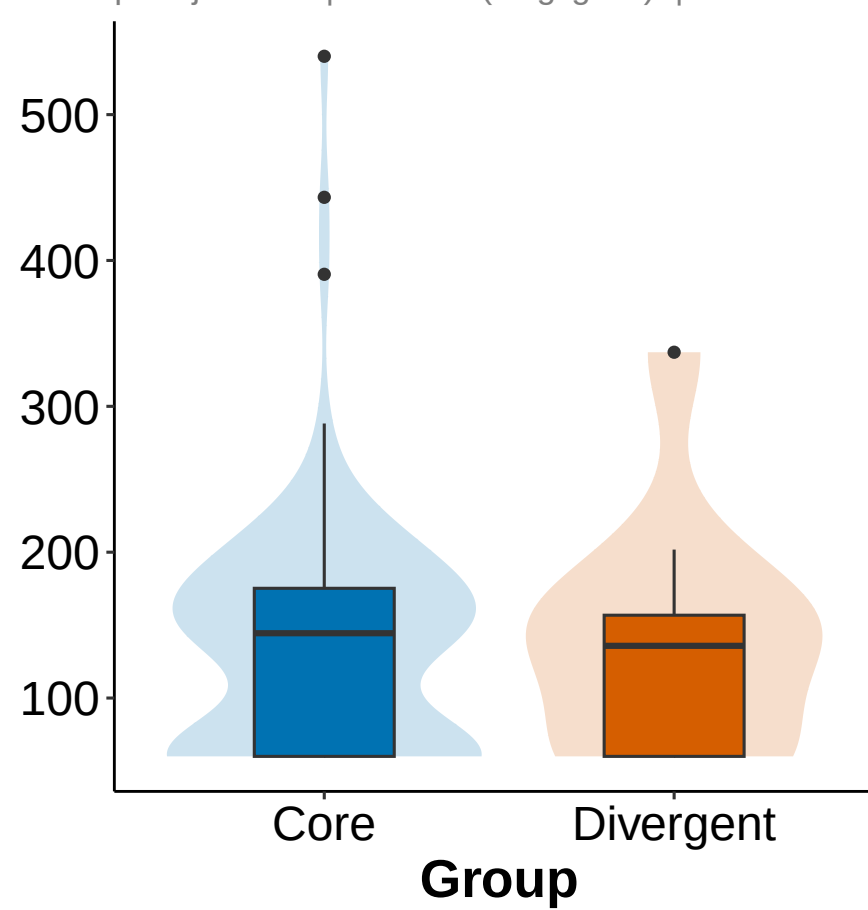
Maximum enhancement (K1)

p-adj: 0.928 | .:=0.34 (Medium) | .%: 16.2



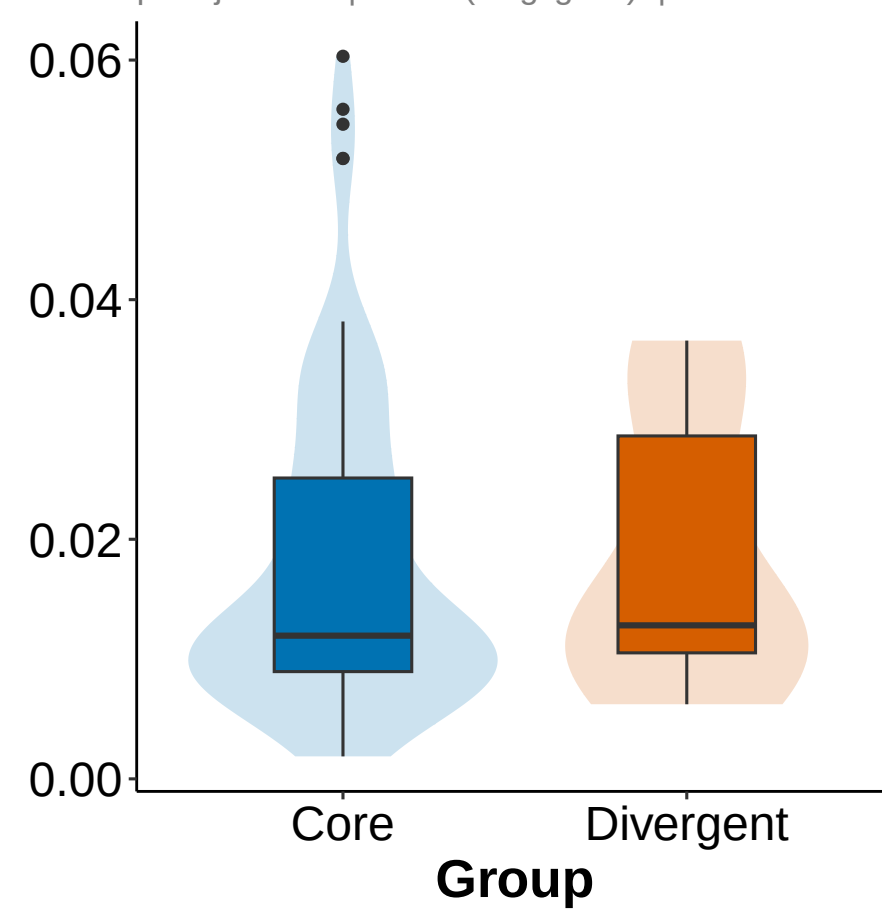
Time to peak (K2)

p-adj: 0.929 | .:=−0.03 (Negligible) | .%: −6



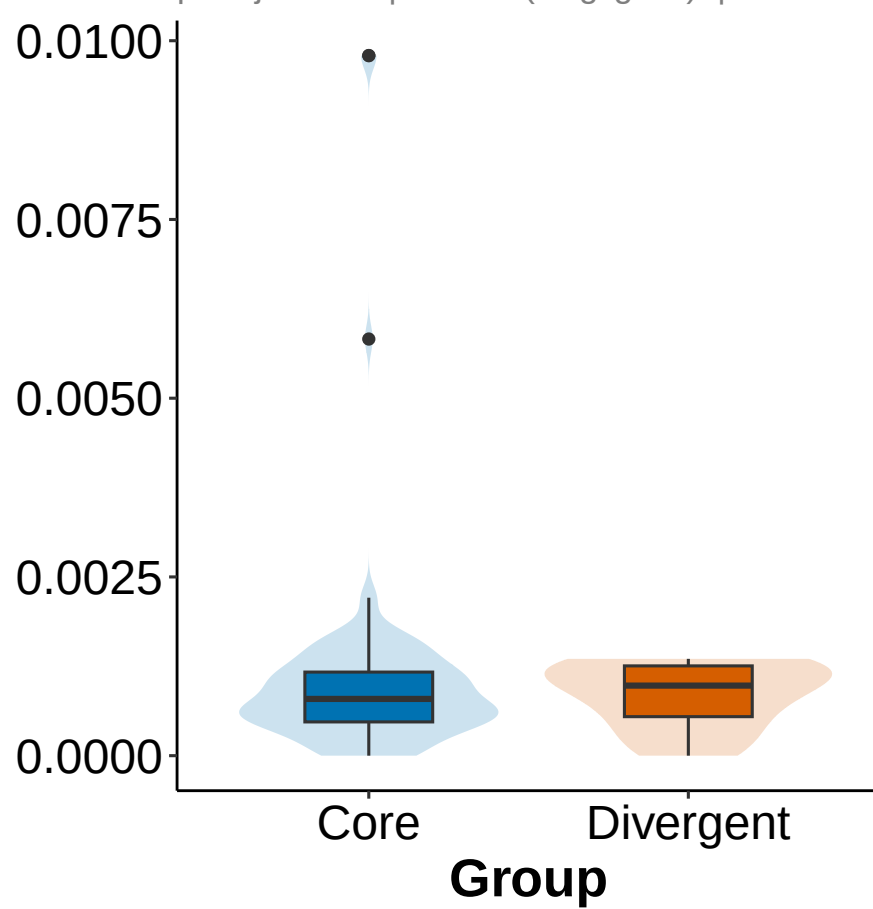
Uptake rate (K3)

p-adj: 0.928 | .:=0.1 (Negligible) | .%: 7.2



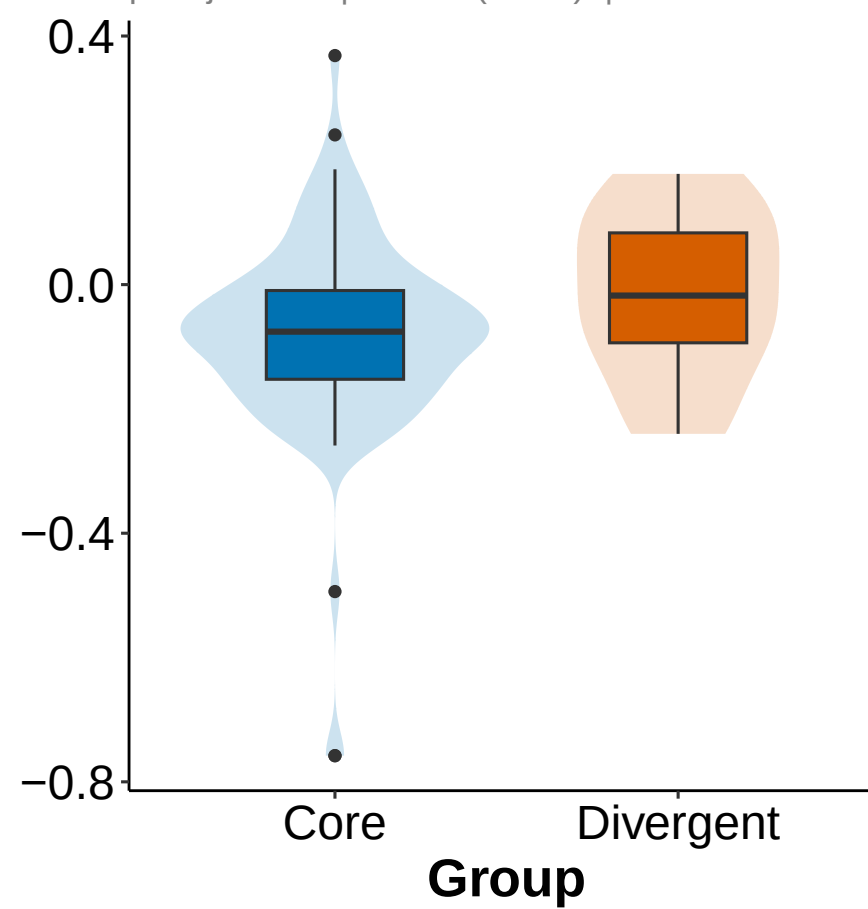
Washout rate (K4)

p-adj: 0.933 | .:=0.02 (Negligible) | .%: 23.5



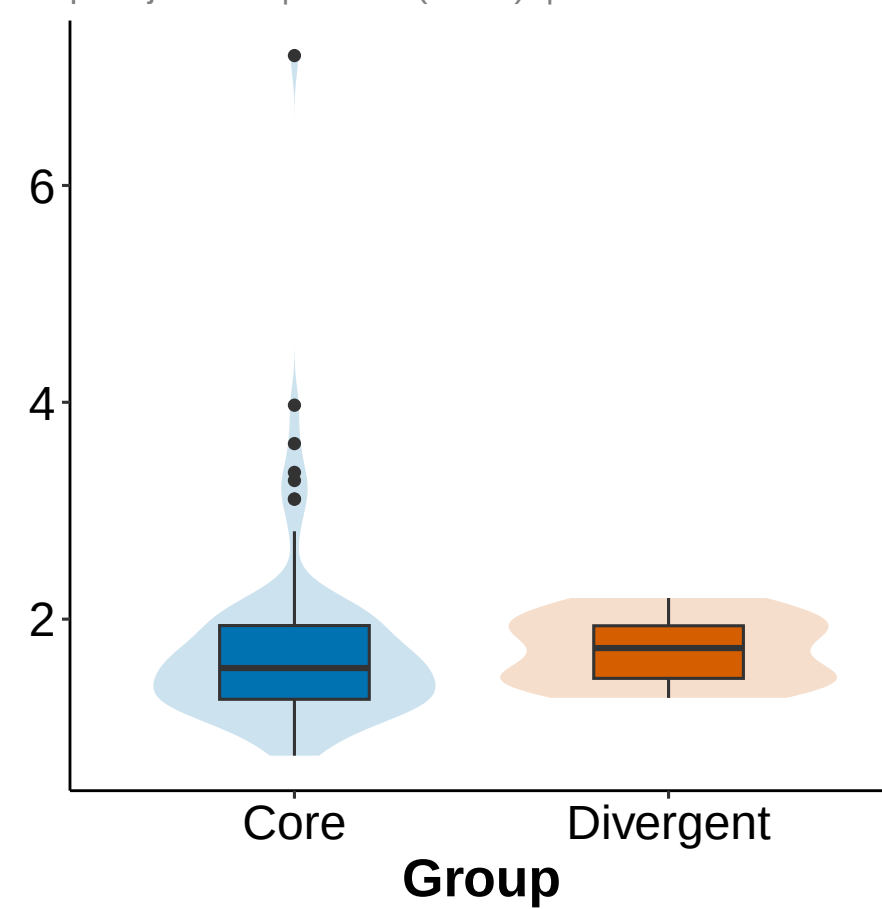
Curve shape index (K5)

p-adj: 0.928 | .:=0.26 (Small) | .%: 76.7



E1 (K6)

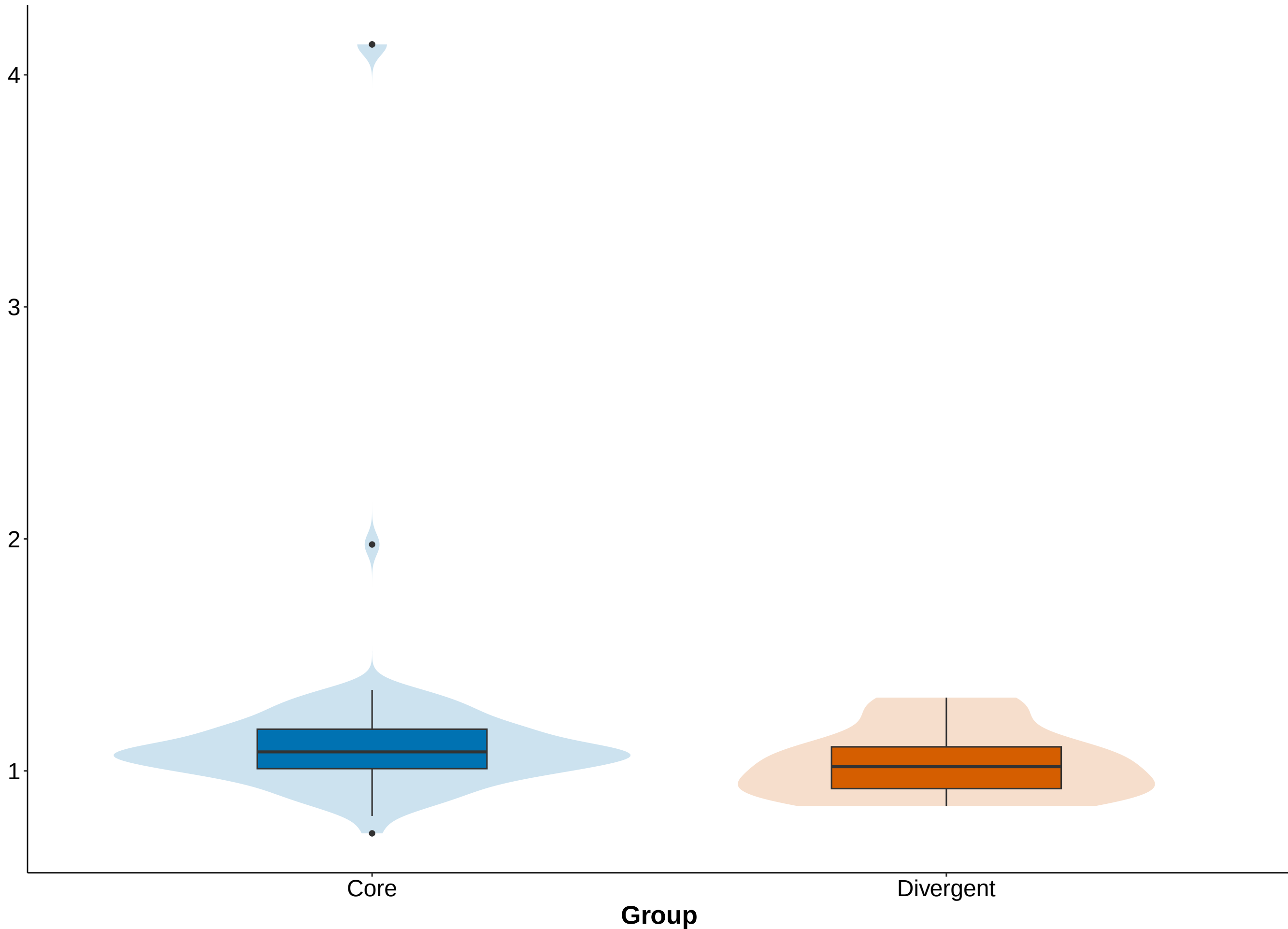
p-adj: 0.928 | .:=0.19 (Small) | .%: 11.9



Radiomics – K – Kinetic curve (enhancement kinetics)

Signal Enhancement Ratio (SER) (K7)

p-adj: 0.928 | .=-0.26 (Small) | .%: -5.9



Cliff's .: E – Enhancement variance

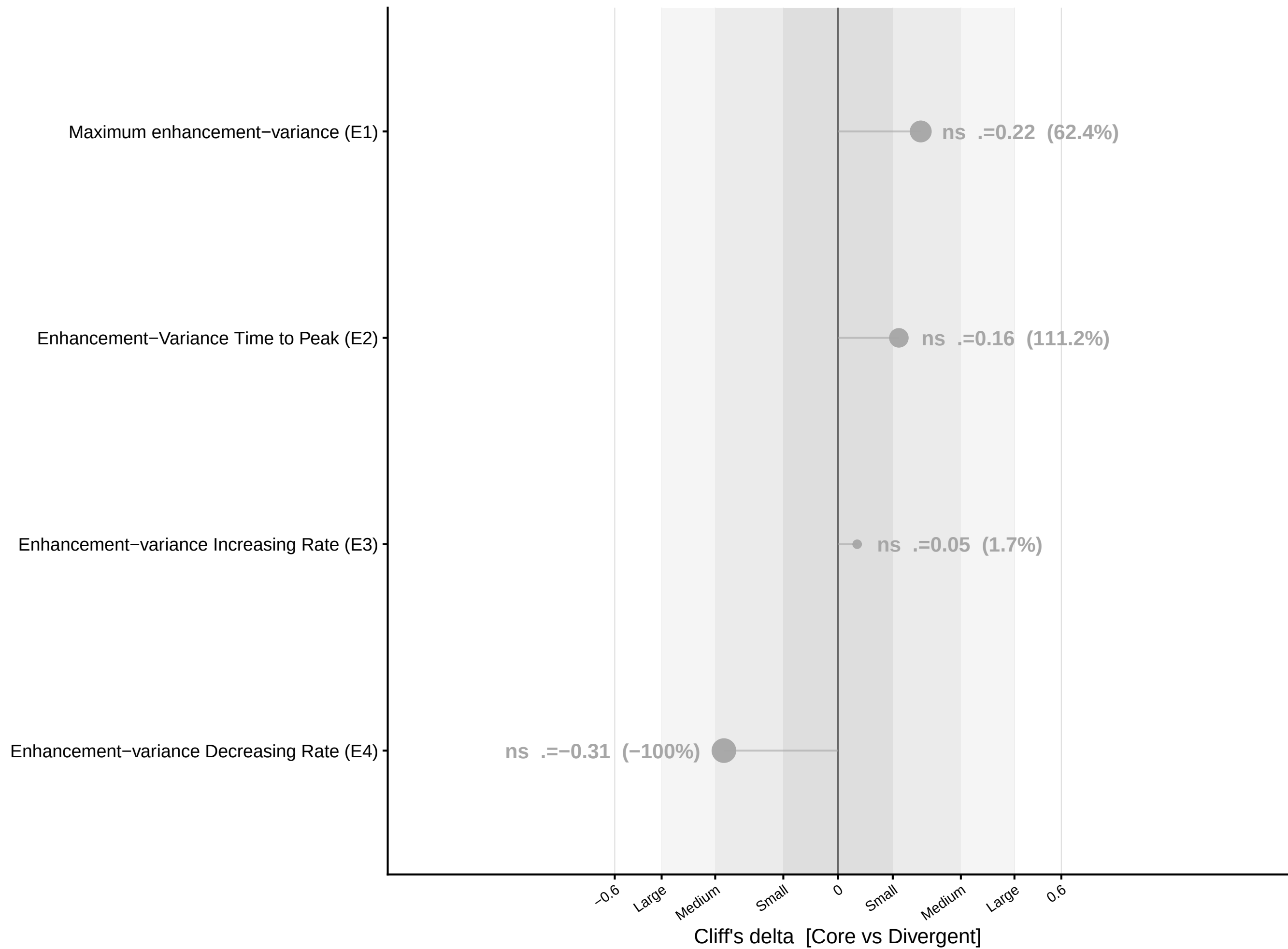
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.0.00.51.0



Cliff's . – E – Enhancement variance

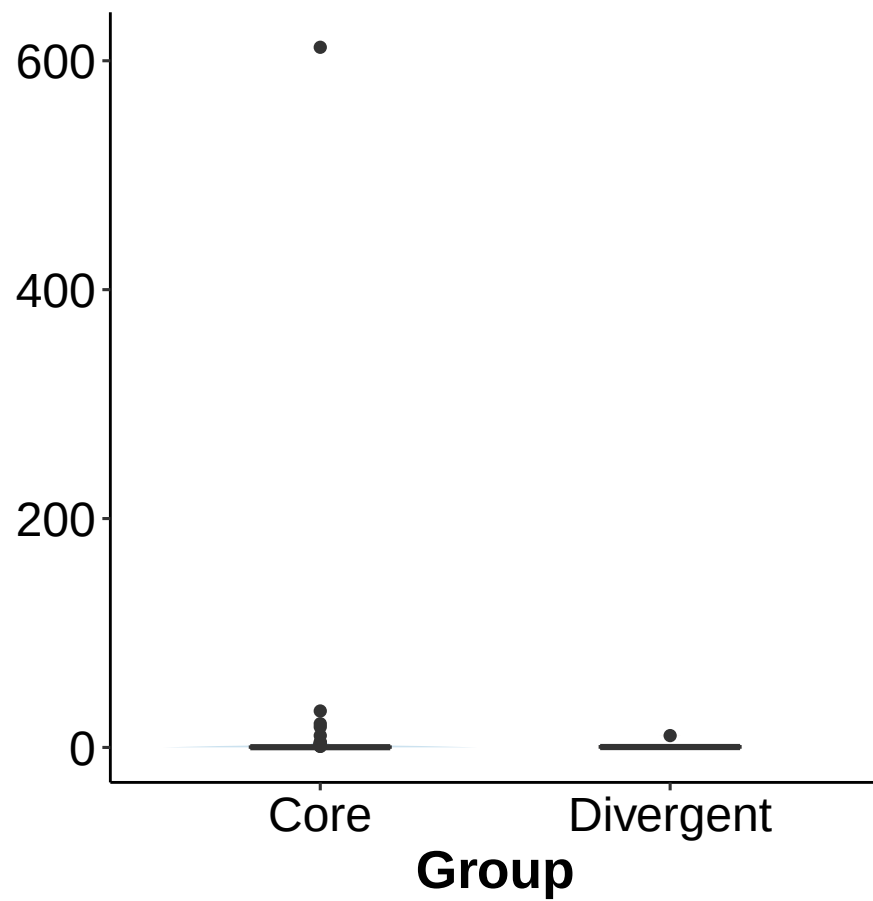
Cliff's . + .% median | . higher in Core · . higher in Divergent



Radiomics – E – Enhancement variance

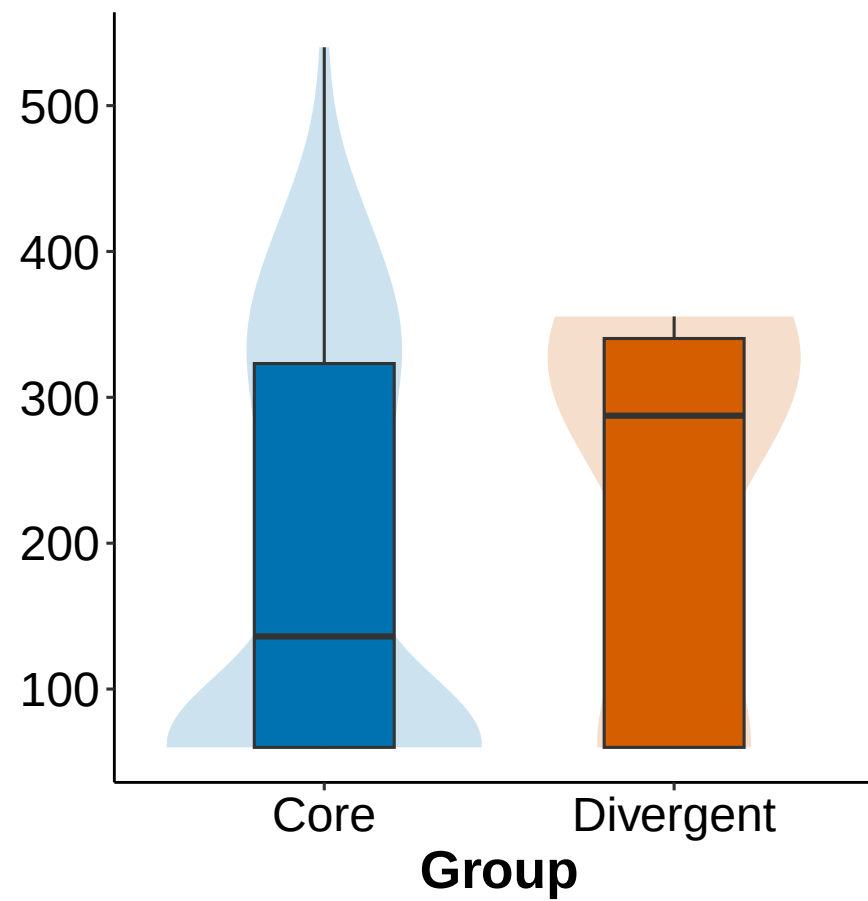
Maximum enhancement–variance (E1)

p-adj: 0.928 | .:=0.22 (Small) | .%: 62.4



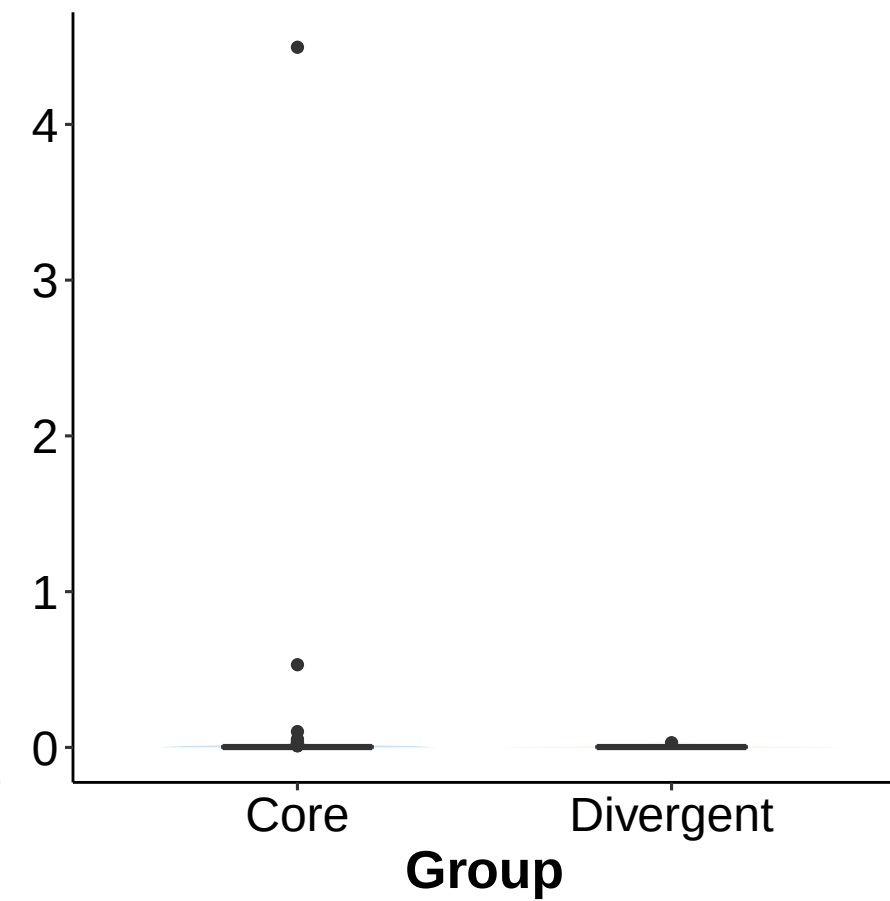
Enhancement–Variance Time to Peak (E2)

p-adj: 0.928 | .:=0.16 (Small) | .%: 111.2



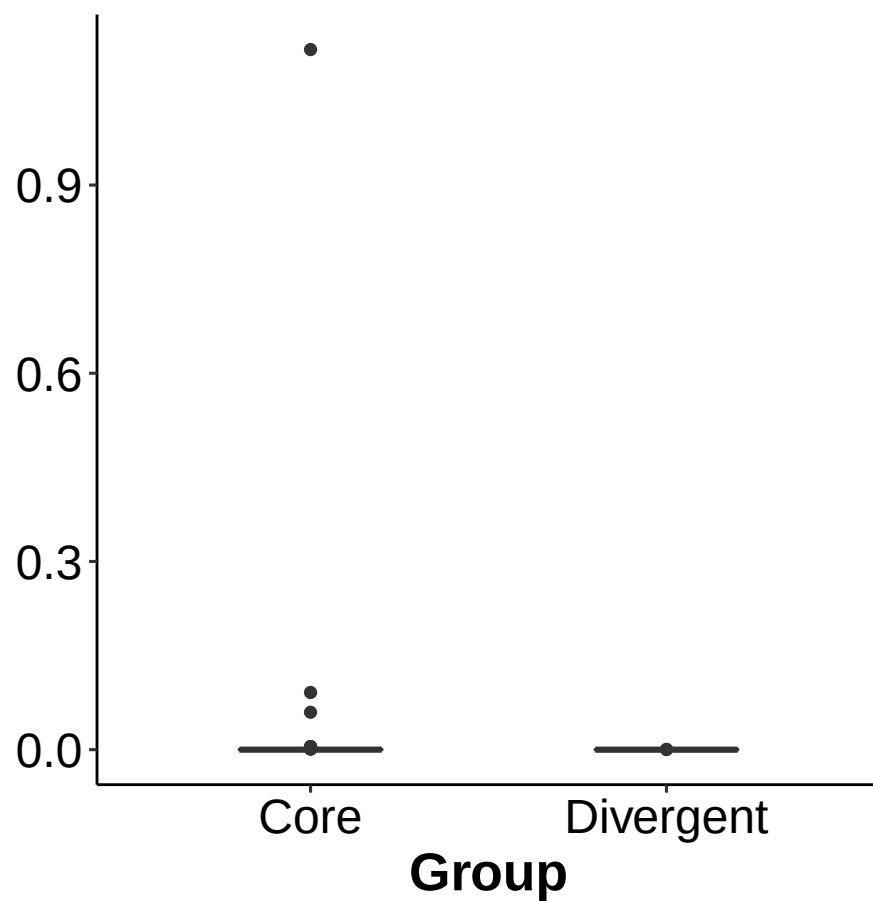
Enhancement–variance Increasing Rate (E3)

p-adj: 0.928 | .:=0.05 (Negligible) | .%: 1.7



Enhancement–variance Decreasing Rate (E4)

p-adj: 0.928 | .:=−0.31 (Small) | .%: −100



Cliff's .: T – Texture (GLCM)

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core

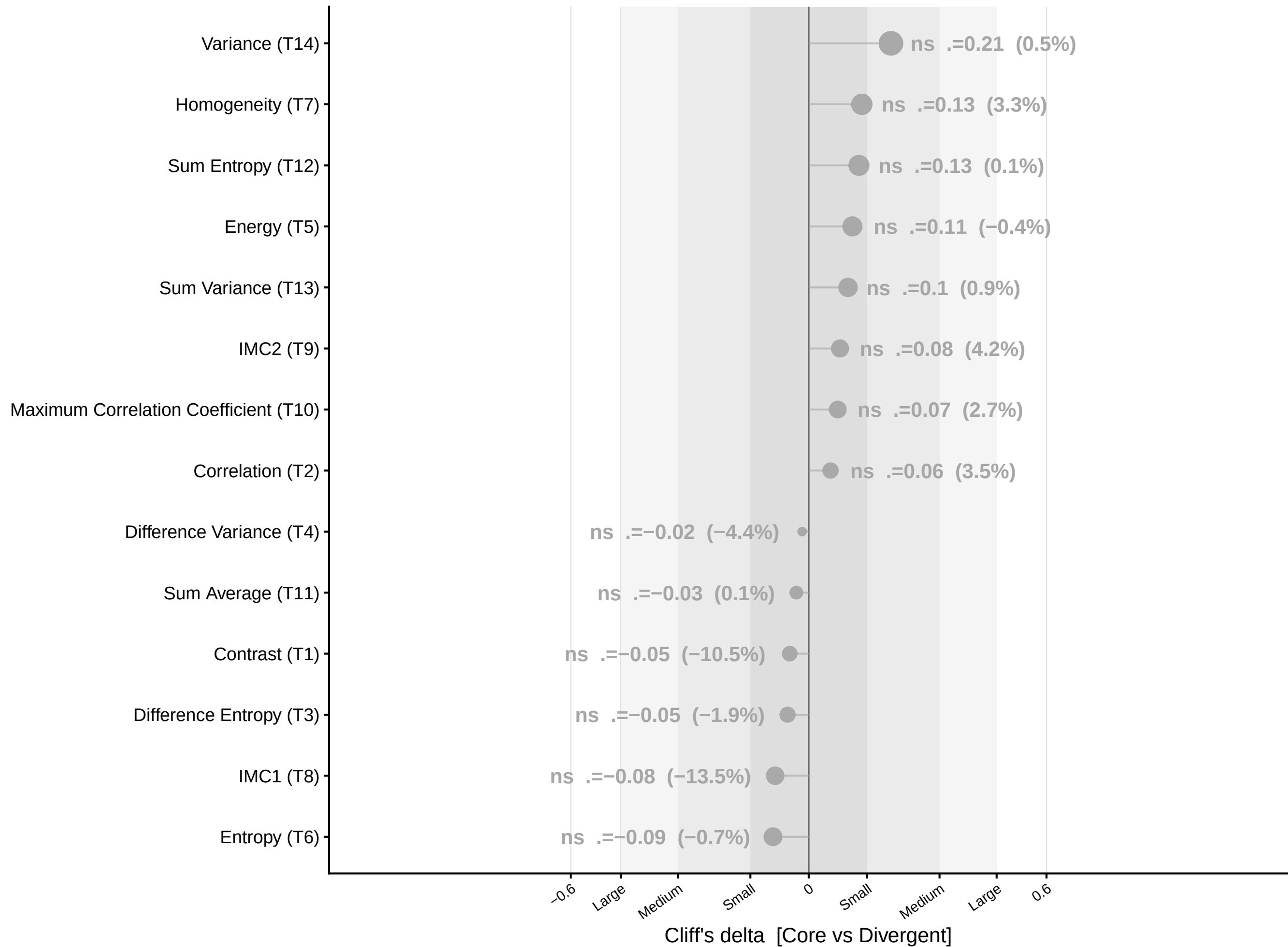


1.0 0.5 0.0 0.5 1.0

Variance (T14)	ns .=-0.21
Homogeneity (T7)	ns .=-0.13
Sum Entropy (T12)	ns .=-0.13
Energy (T5)	ns .=-0.11
Sum Variance (T13)	ns .=-0.1
IMC2 (T9)	ns .=-0.08
Maximum Correlation Coefficient (T10)	ns .=-0.07
Correlation (T2)	ns .=-0.06
Difference Variance (T4)	ns .=-0.02
Sum Average (T11)	ns .=-0.03
Contrast (T1)	ns .=-0.05
Difference Entropy (T3)	ns .=-0.05
IMC1 (T8)	ns .=-0.08
Entropy (T6)	ns .=-0.09

Cliff's δ - T - Texture (GLCM)

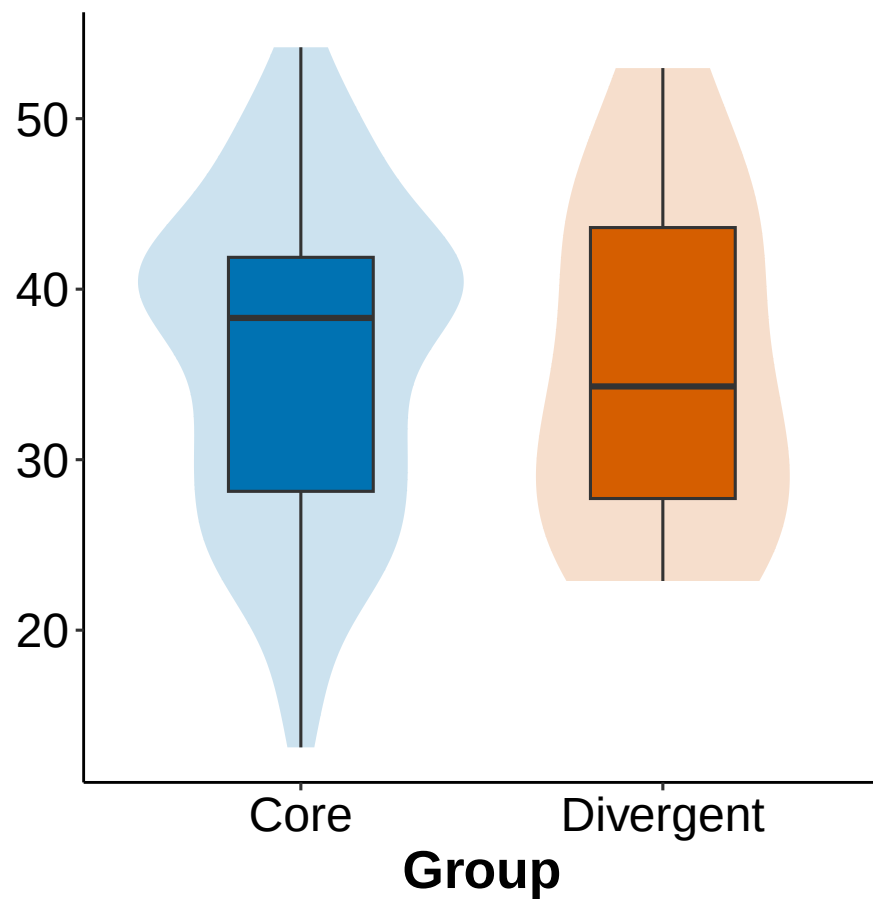
Cliff's δ + .% median | . higher in Core · . higher in Divergent



Radiomics – T – Texture (GLCM)

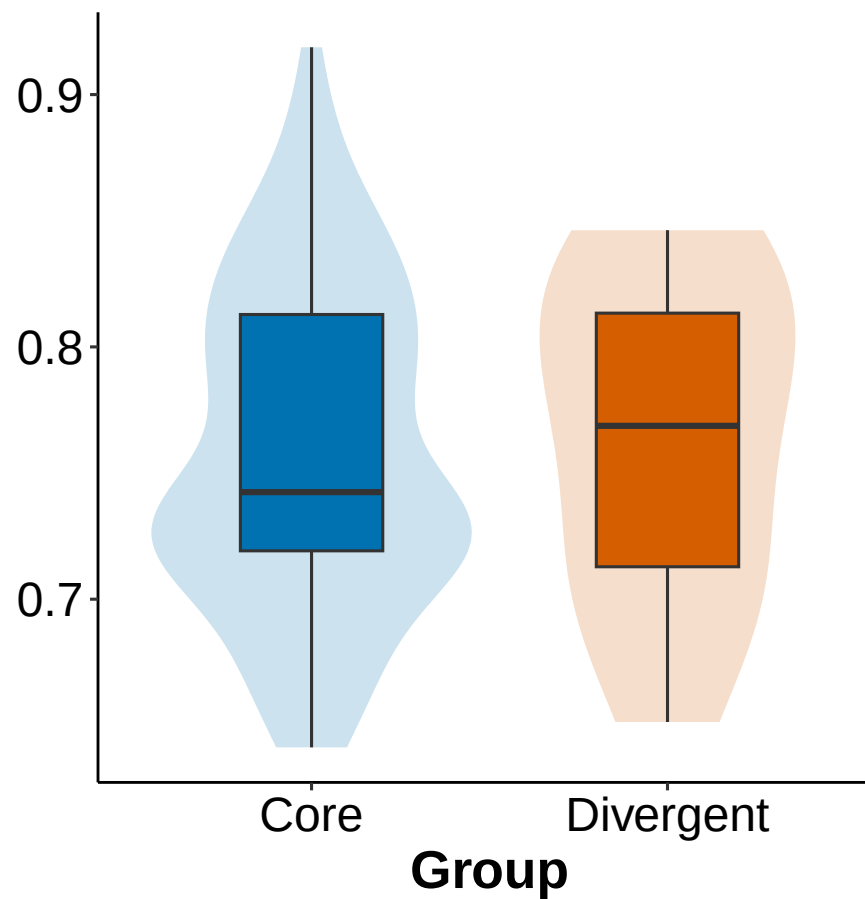
Contrast (T1)

p-adj: 0.928 | .=-0.05 (Negligible) | .%: -10.5



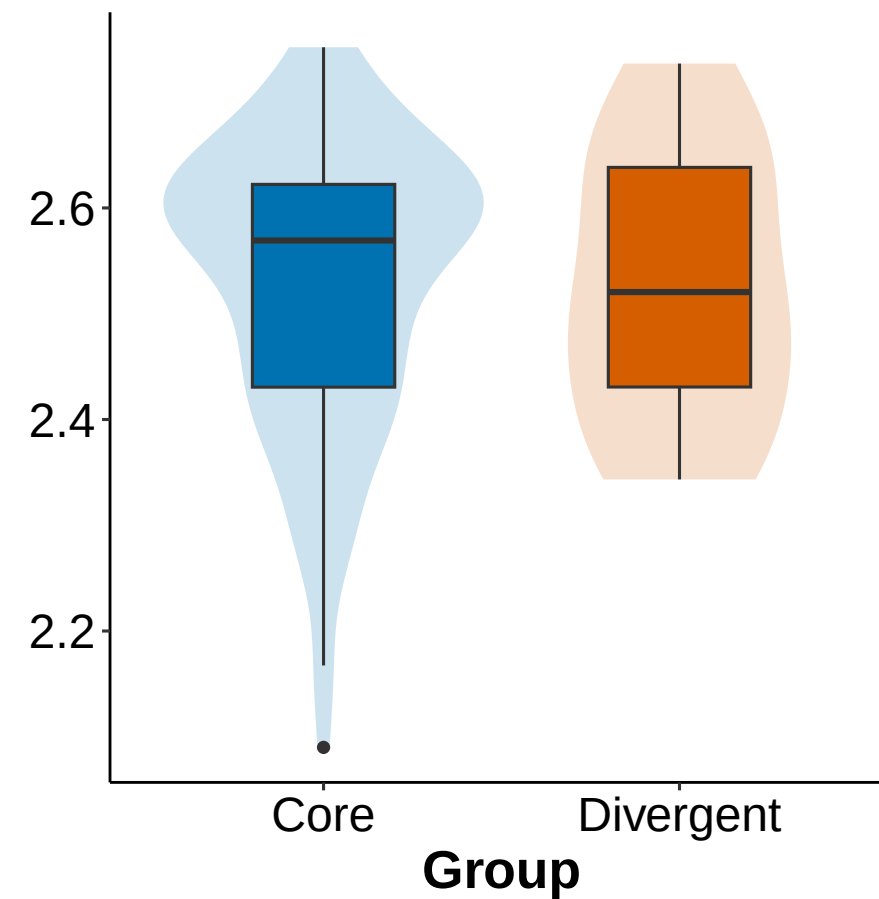
Correlation (T2)

p-adj: 0.928 | .=-0.06 (Negligible) | .%: 3.5



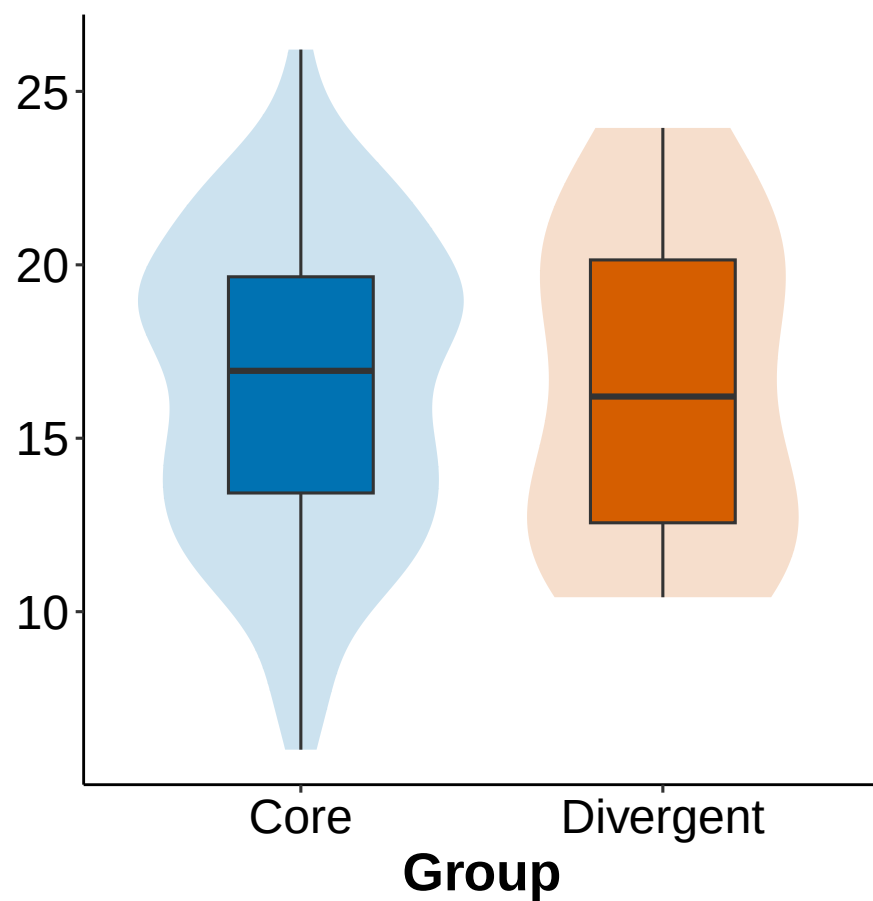
Difference Entropy (T3)

p-adj: 0.928 | .=-0.05 (Negligible) | .%: -1.9



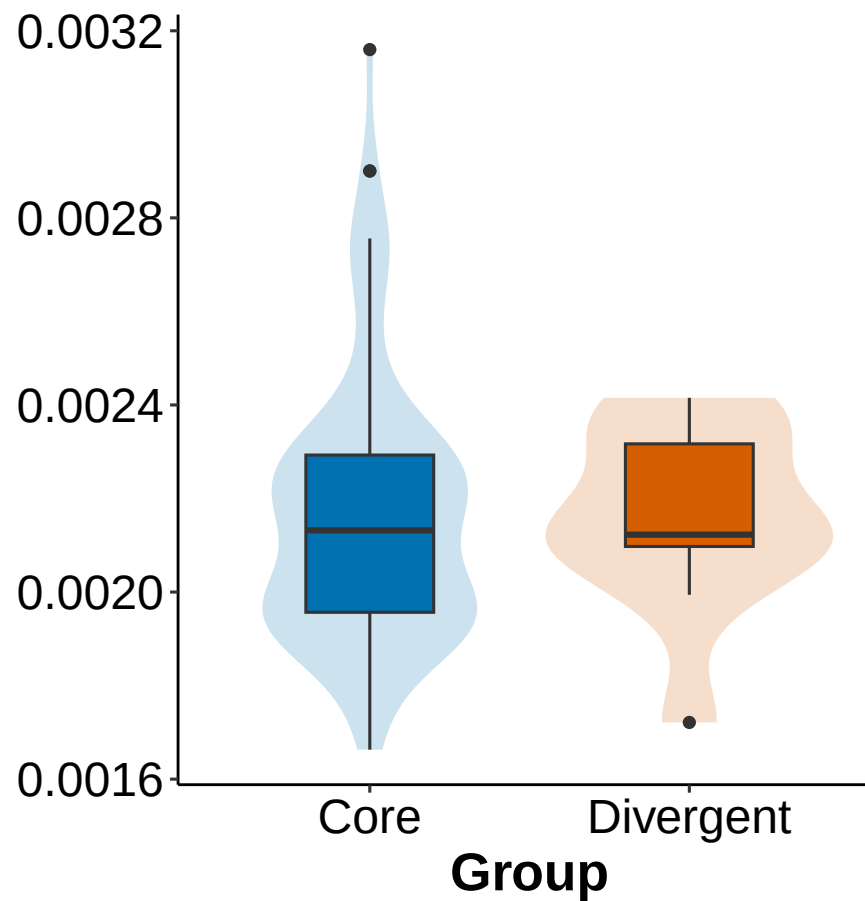
Difference Variance (T4)

p-adj: 0.933 | .=-0.02 (Negligible) | .%: -4.4



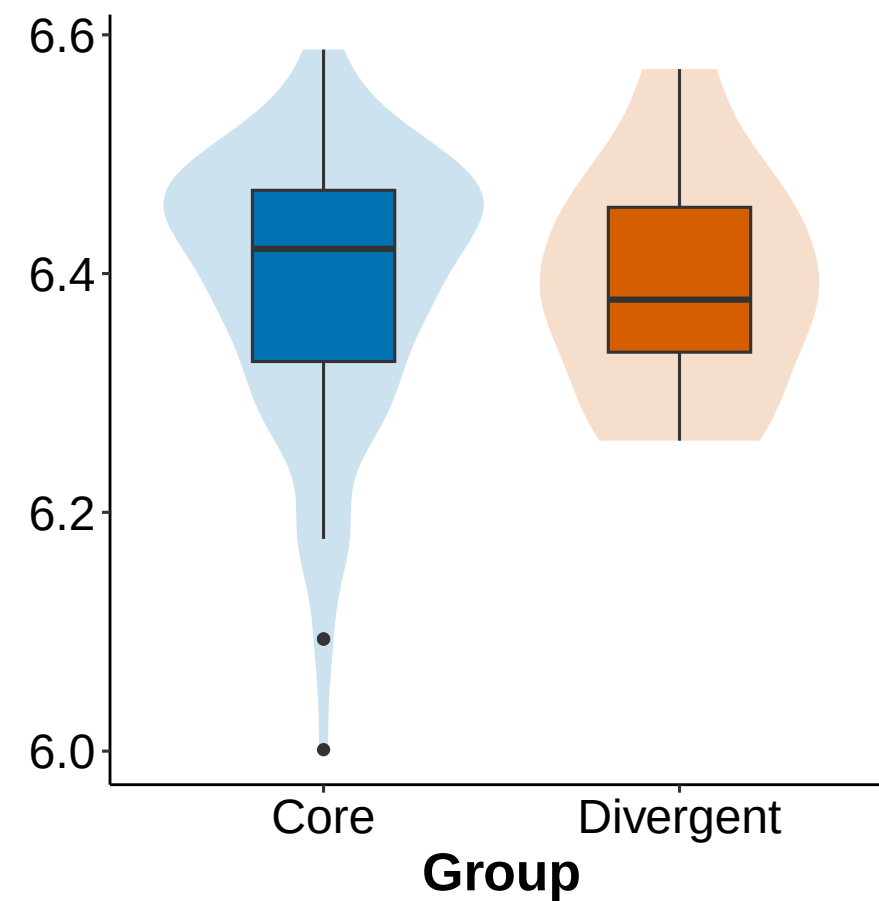
Energy (T5)

p-adj: 0.928 | .=-0.11 (Negligible) | .%: -0.4



Entropy (T6)

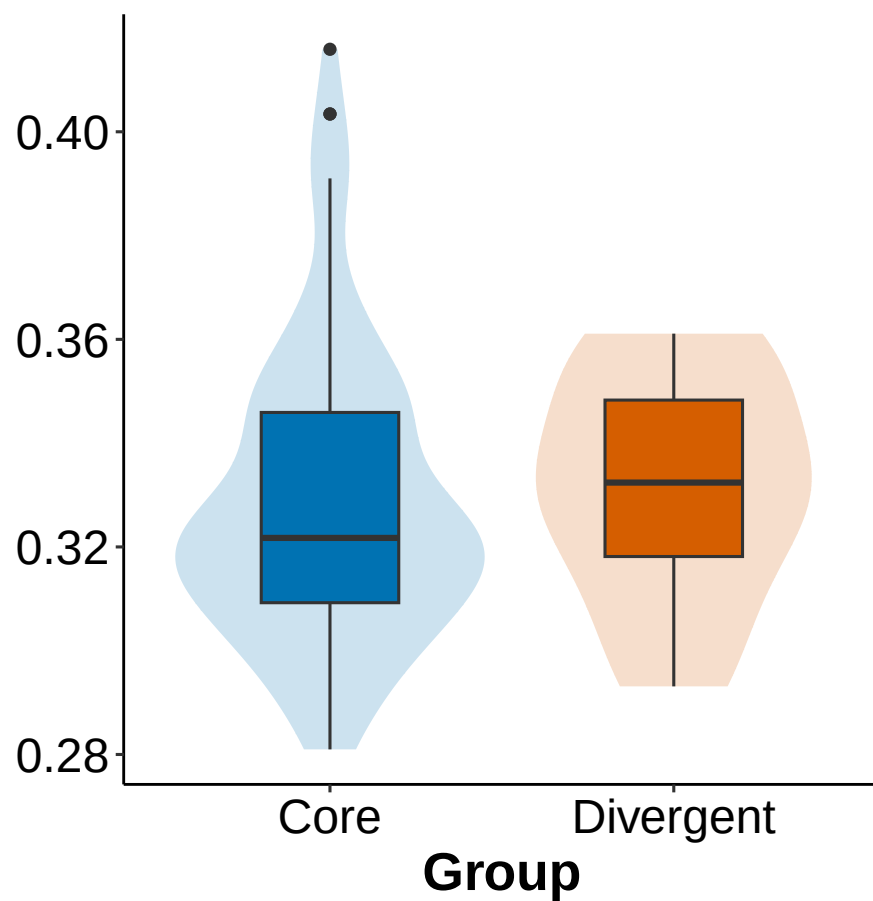
p-adj: 0.928 | .=-0.09 (Negligible) | .%: -0.7



Radiomics – T – Texture (GLCM)

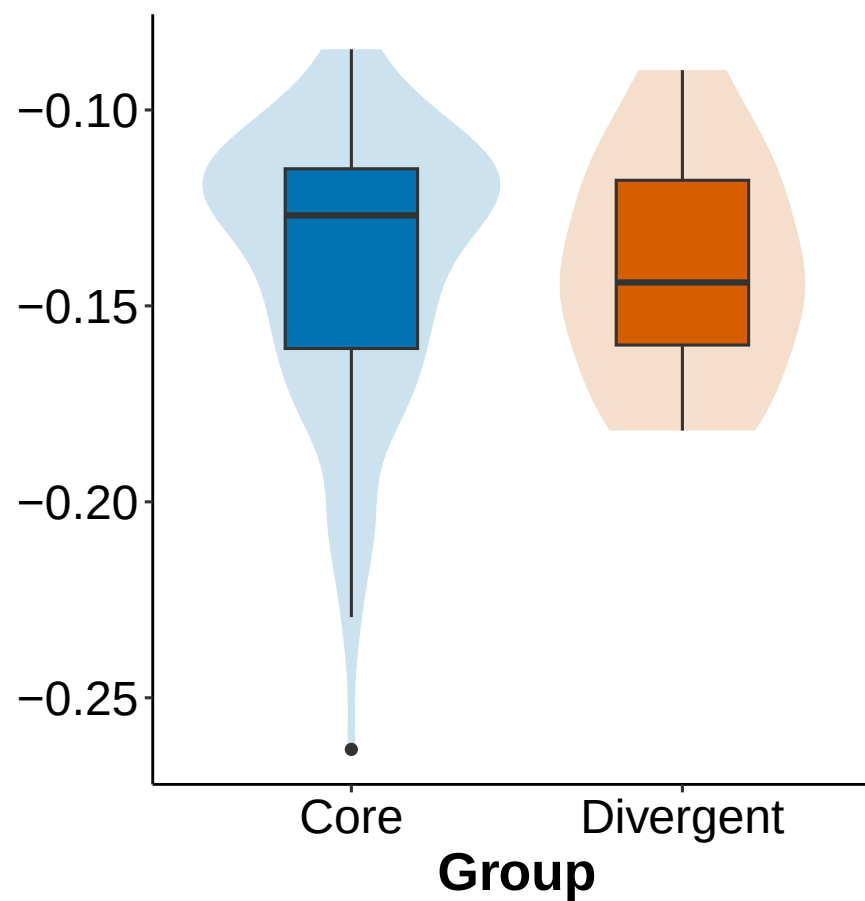
Homogeneity (T7)

p-adj: 0.928 | .:=0.13 (Negligible) | .%: 3.3



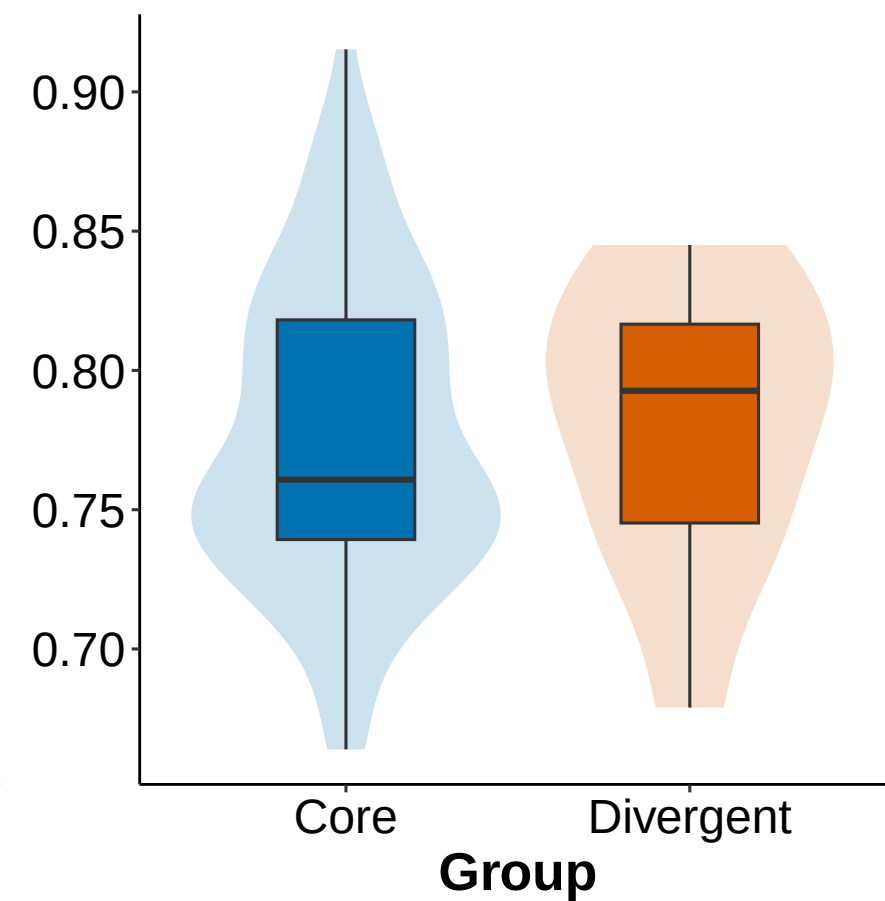
IMC1 (T8)

p-adj: 0.928 | .:=−0.08 (Negligible) | .%: −13.5



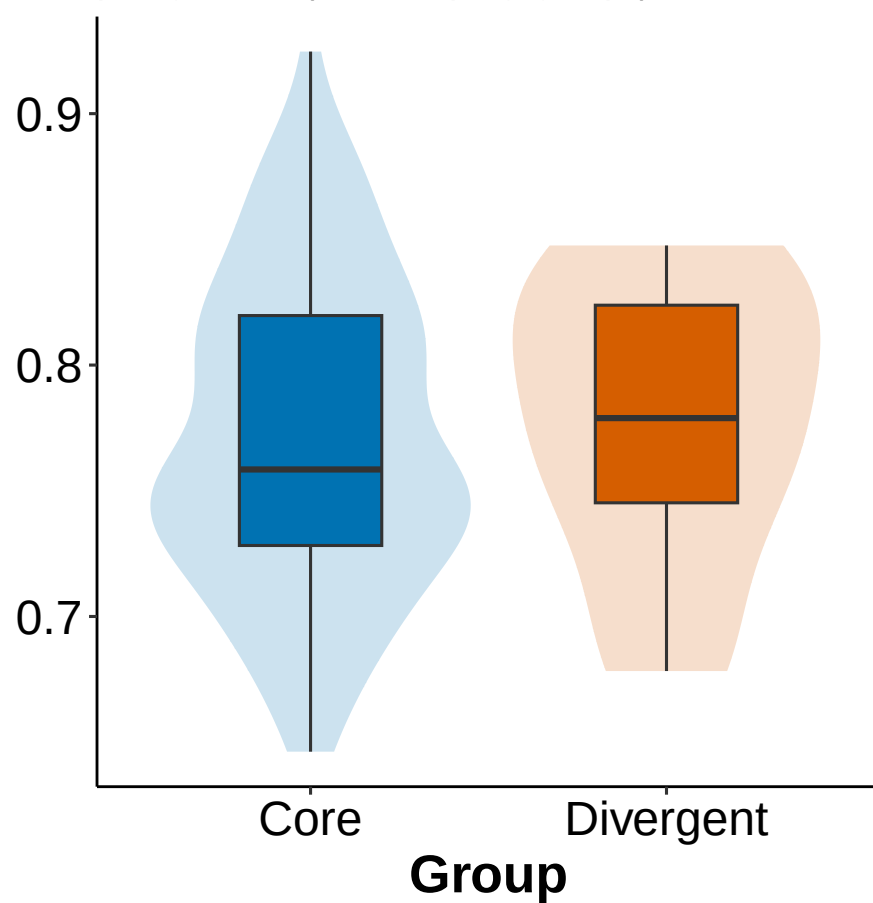
IMC2 (T9)

p-adj: 0.928 | .:=0.08 (Negligible) | .%: 4.2



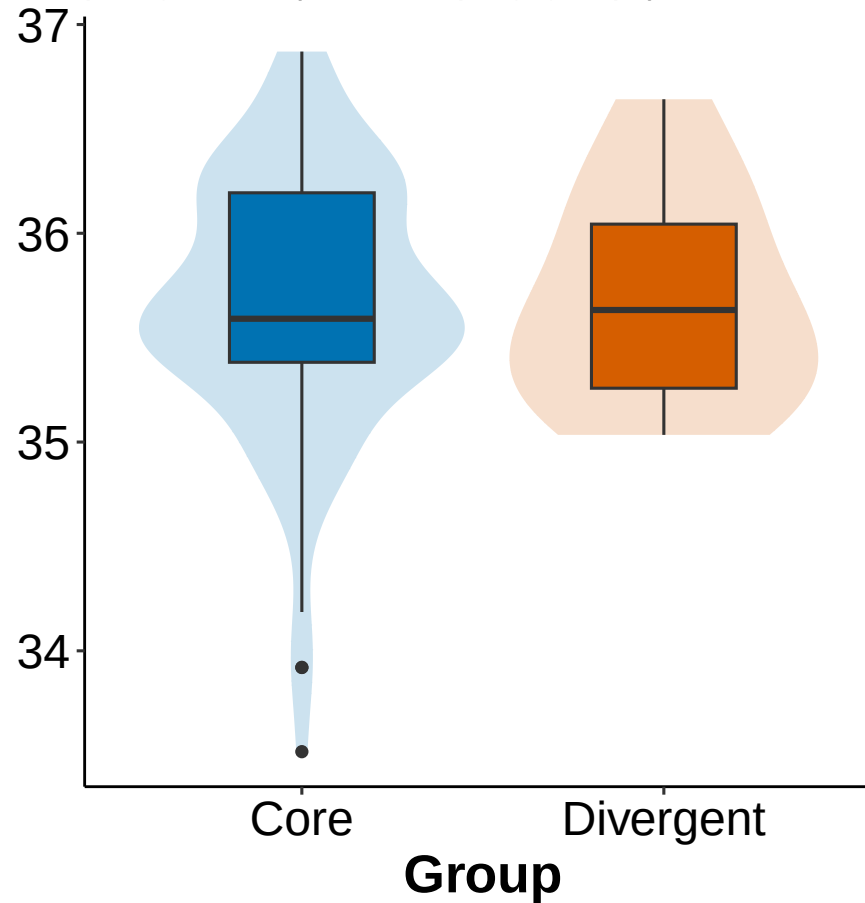
Maximum Correlation Coefficient (T10)

p-adj: 0.928 | .:=0.07 (Negligible) | .%: 2.7



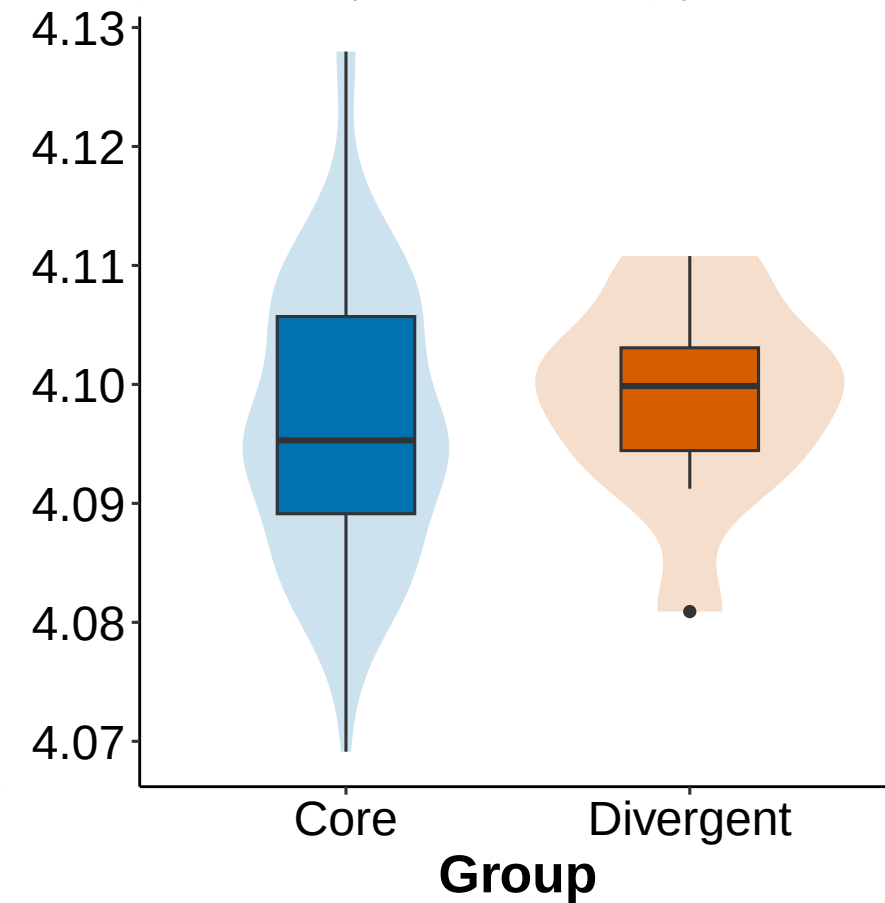
Sum Average (T11)

p-adj: 0.929 | .:=−0.03 (Negligible) | .%: 0.1



Sum Entropy (T12)

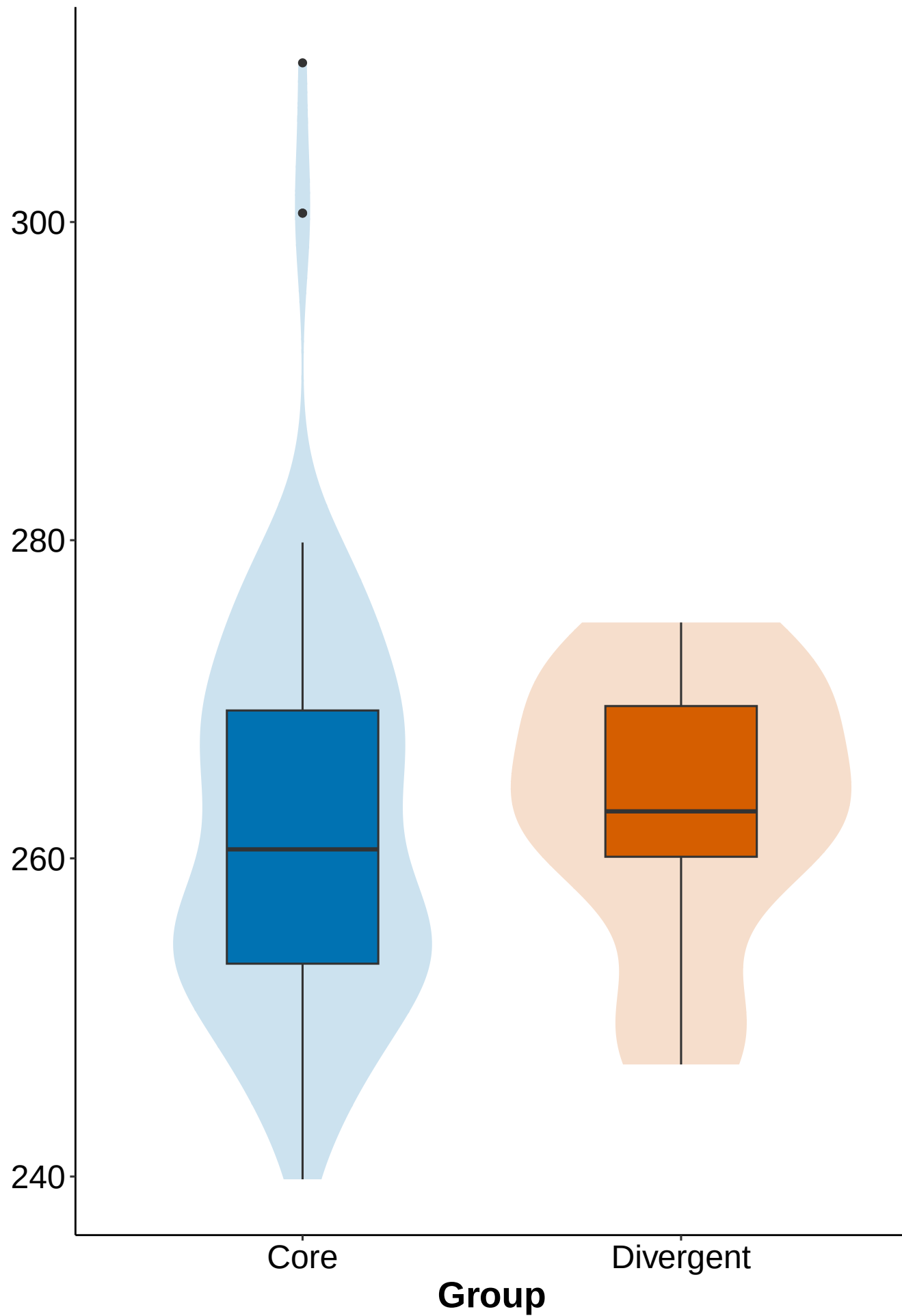
p-adj: 0.928 | .:=0.13 (Negligible) | .%: 0.1



Radiomics – T – Texture (GLCM)

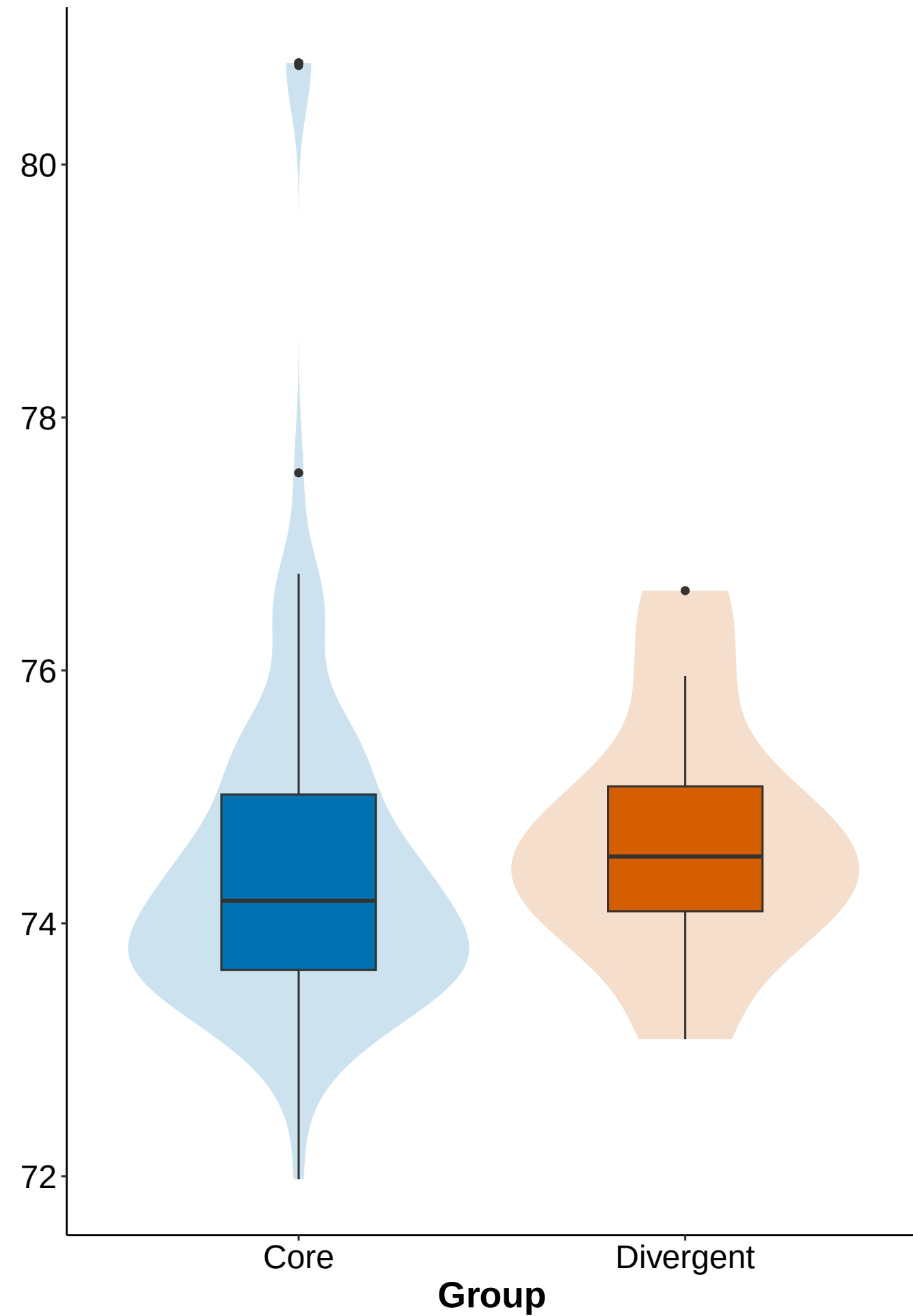
Sum Variance (T13)

p-adj: 0.928 | .:=0.1 (Negligible) | .%: 0.9



Variance (T14)

p-adj: 0.928 | .:=0.21 (Small) | .%: 0.5



Cliff's .: G – Geometry / Shape

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



1.0 0.5 0.0 0.5 1.0

Irregularity (G2)

ns .=0.2

Surface Area to Volume ratio (G3)

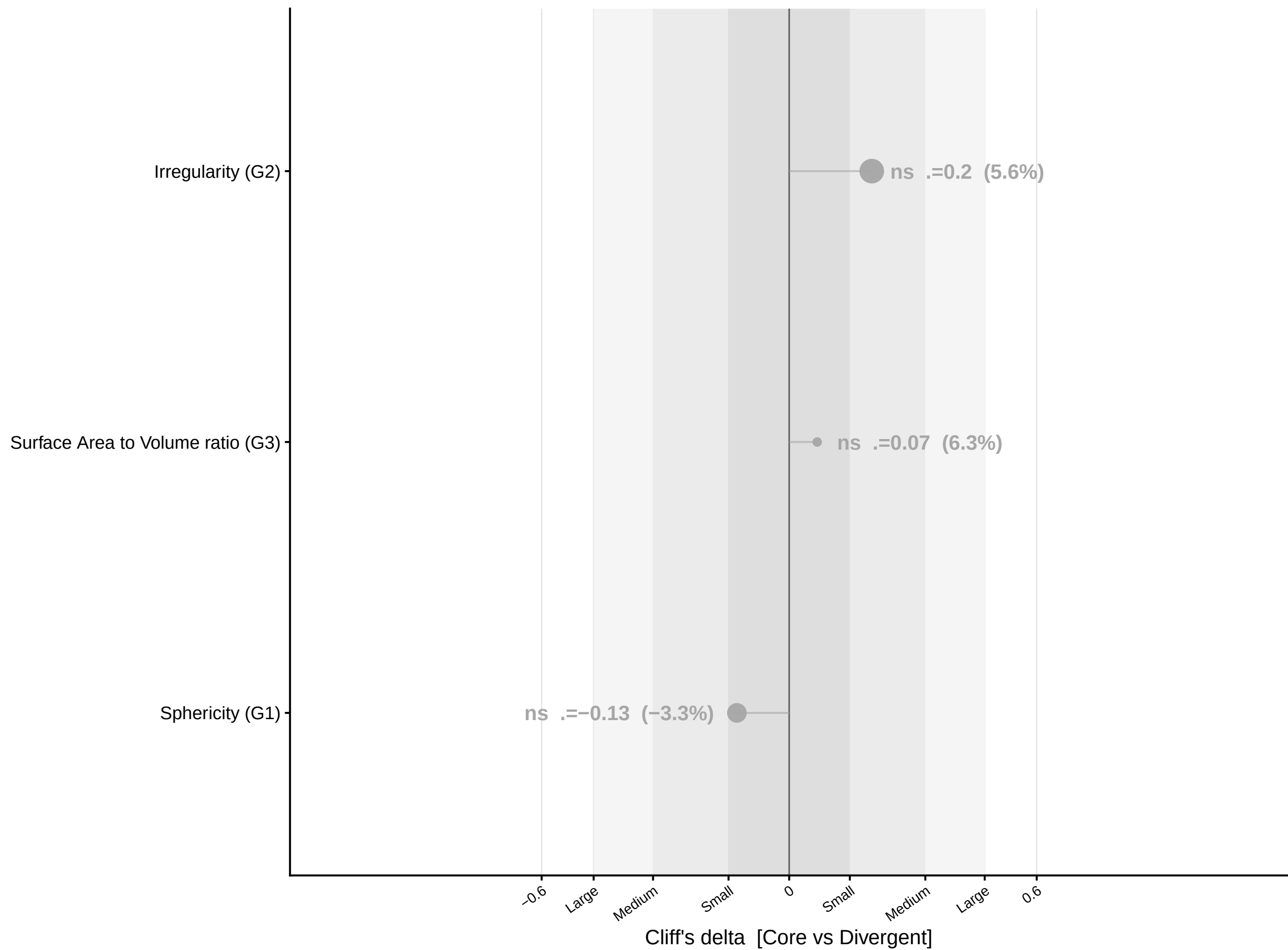
ns .=0.07

Sphericity (G1)

ns .=-0.13

Cliff's . – G – Geometry / Shape

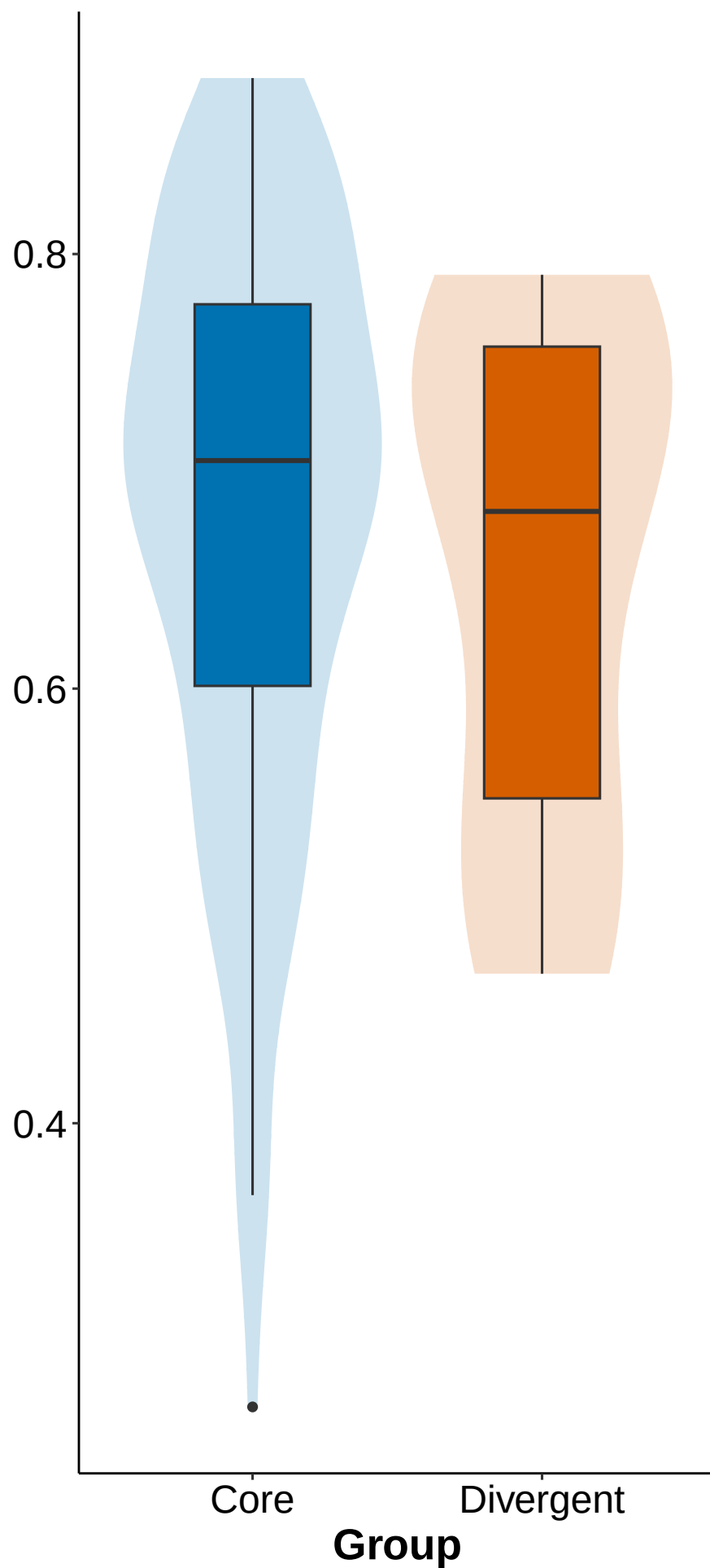
Cliff's . + .% median | . higher in Core · . higher in Divergent



Radiomics – G – Geometry / Shape

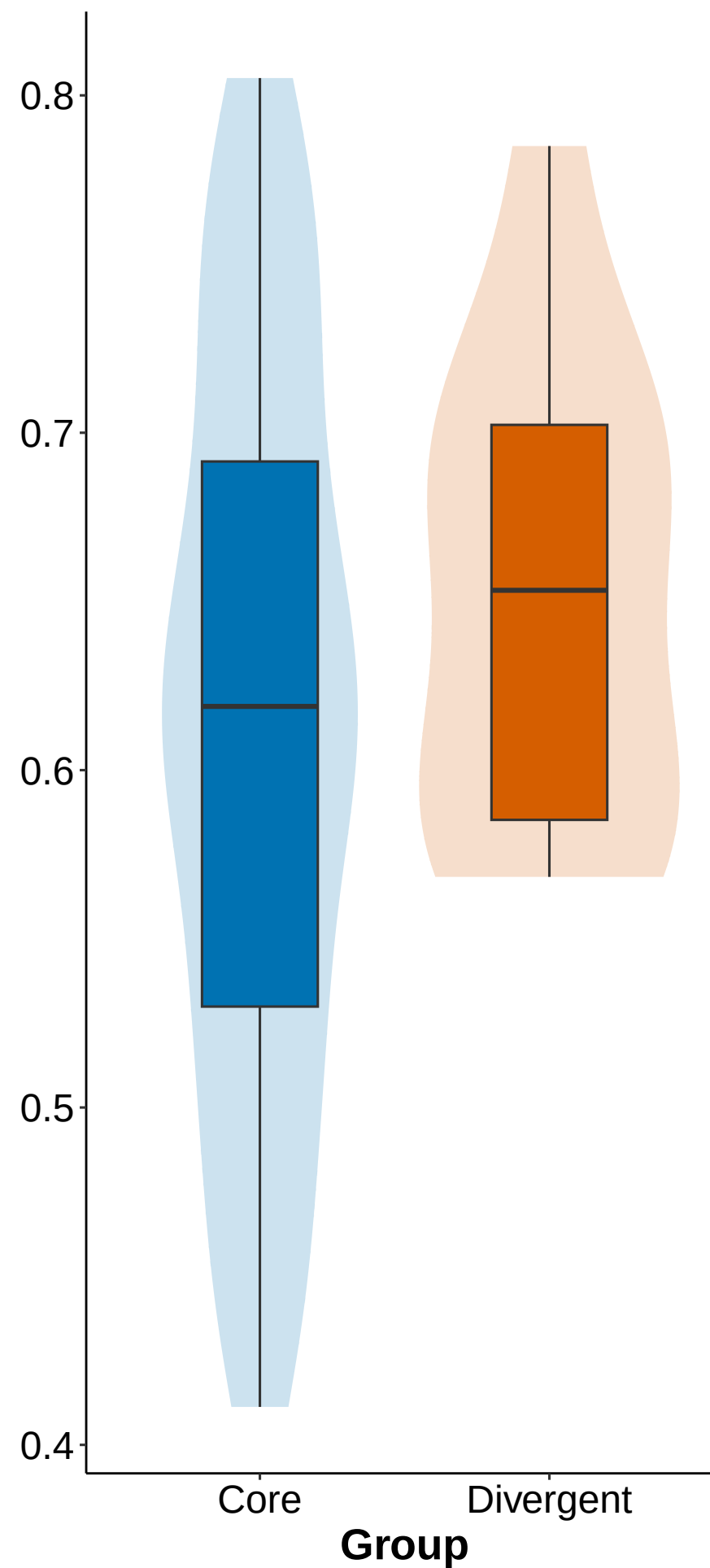
Sphericity (G1)

p-adj: 0.928 | .=-0.13 (Negligible) | .%: -3.3



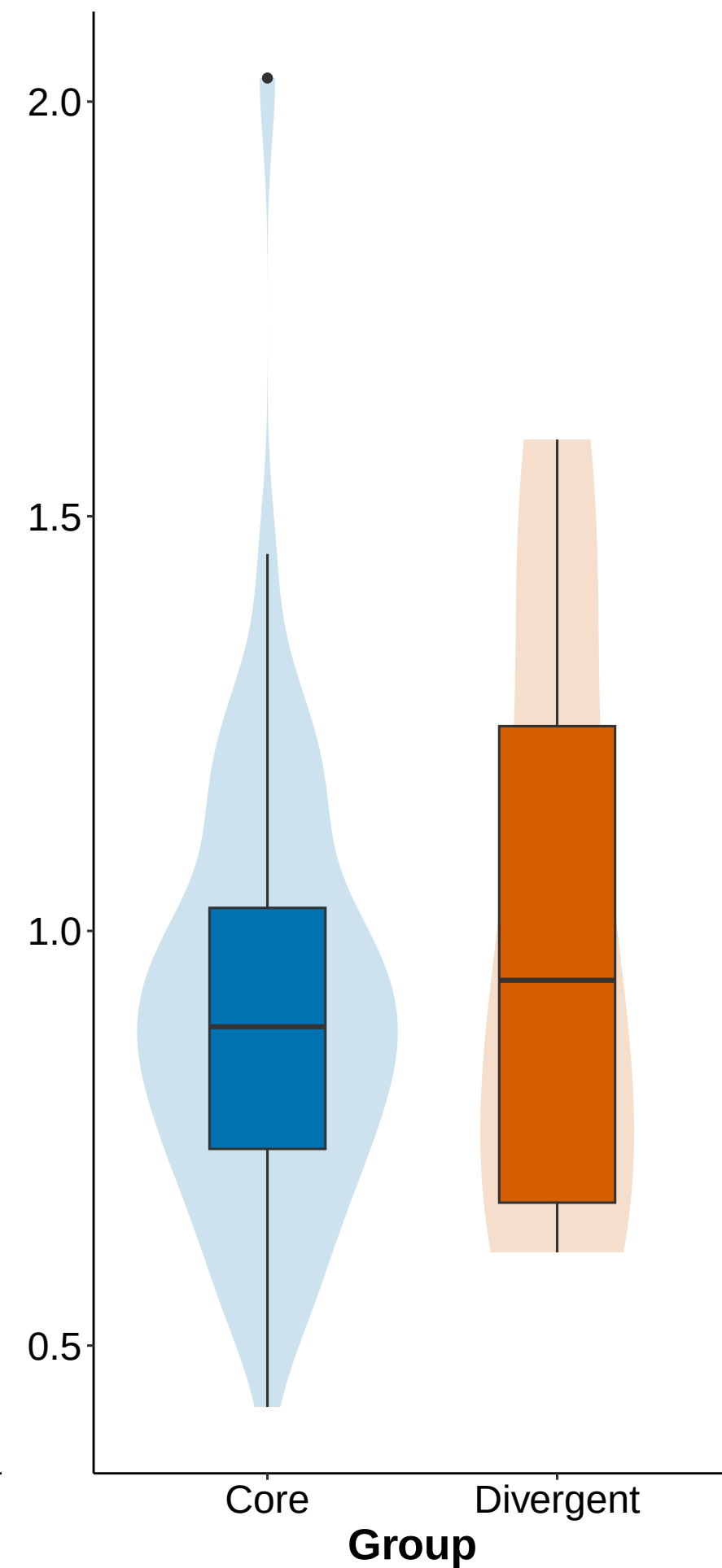
Irregularity (G2)

p-adj: 0.928 | .=0.2 (Small) | .%: 5.6



Surface Area to Volume ratio (G3)

p-adj: 0.928 | .=0.07 (Negligible) | .%: 6.3



Cliff's .: M – Margin

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



Margin Sharpness (M1)

ns .=-0.04

Variance of Margin Sharpness (M2)

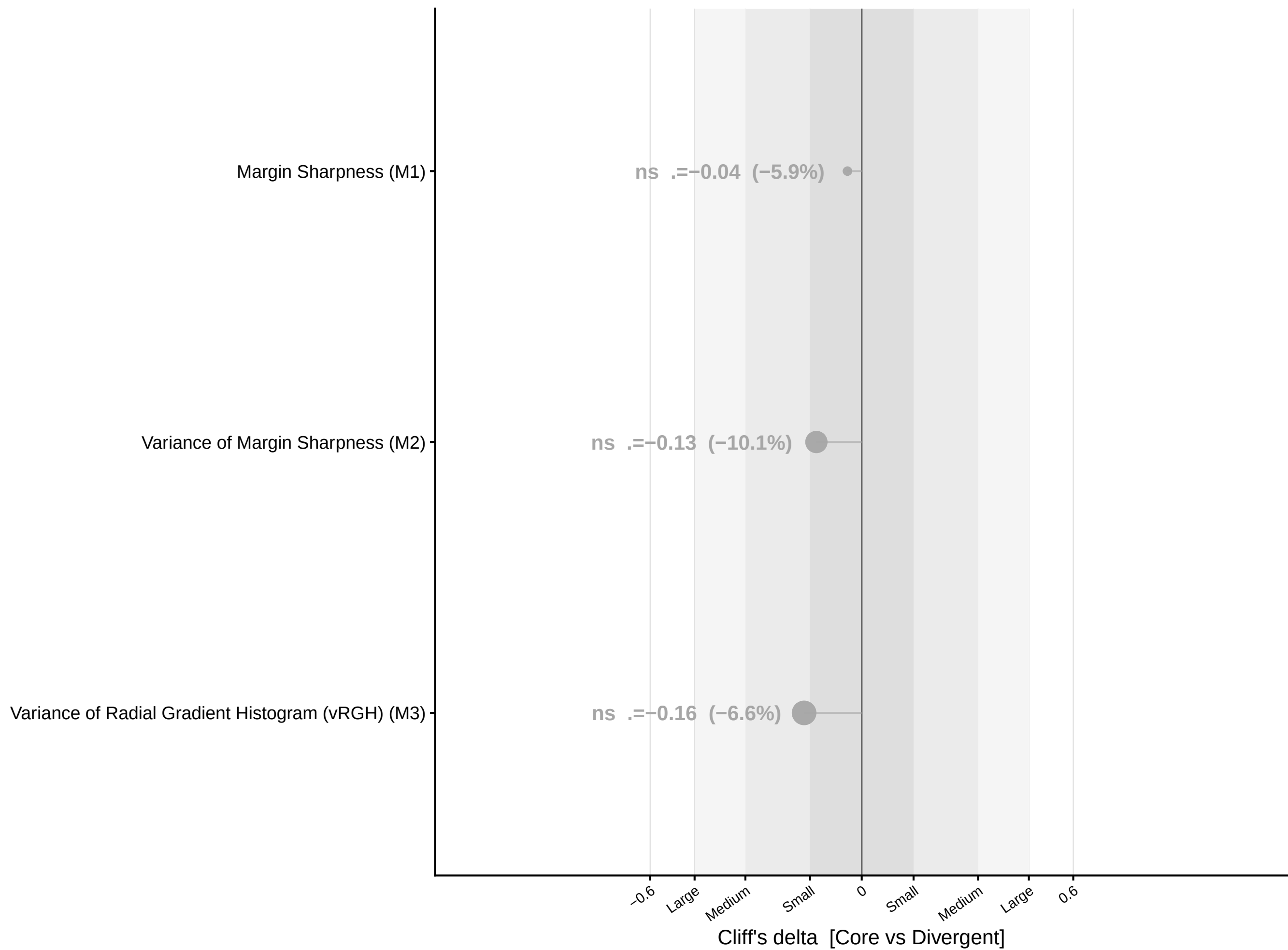
ns .=-0.13

Variance of Radial Gradient Histogram (vRGH) (M3)

ns .=-0.16

Cliff's . – M – Margin

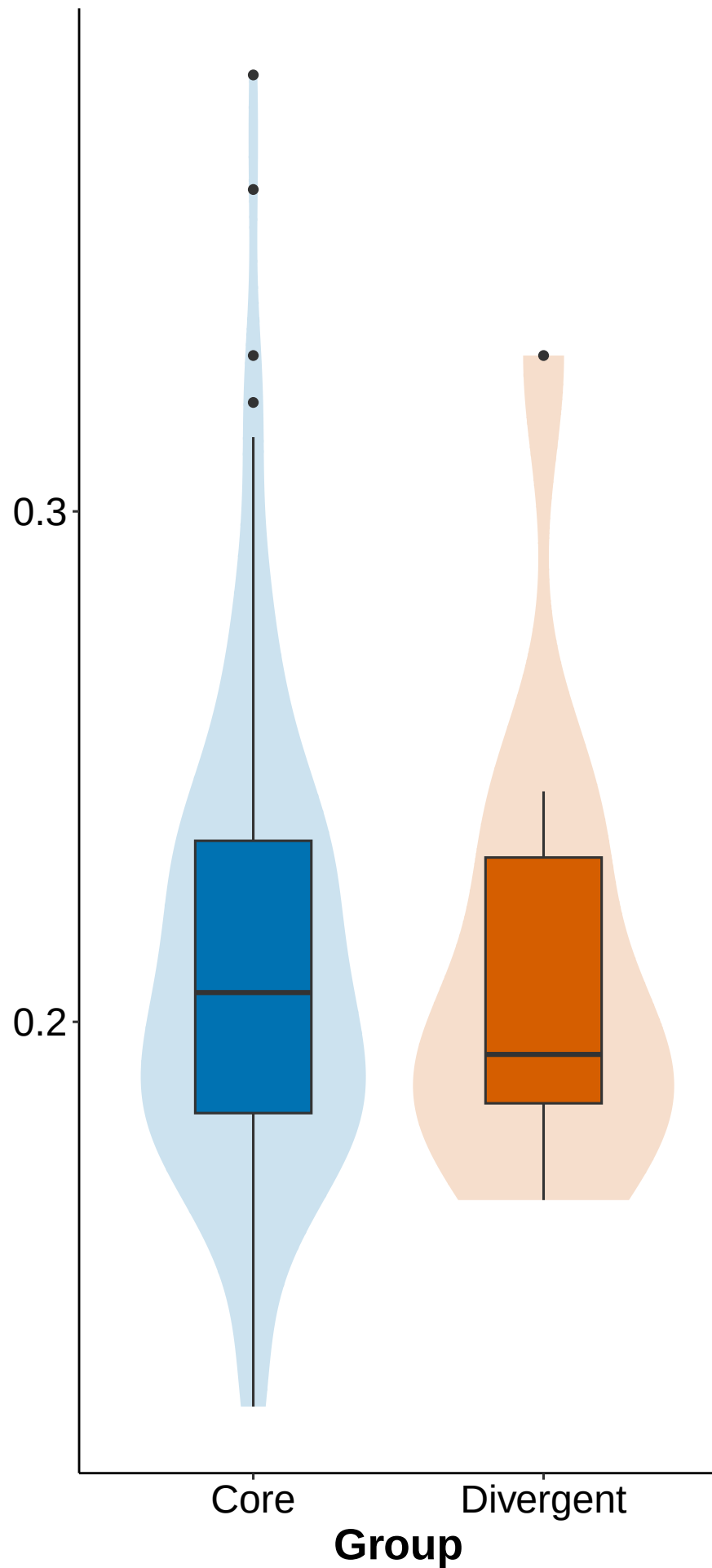
Cliff's . + .% median | . higher in Core · . higher in Divergent



Radiomics – M – Margin

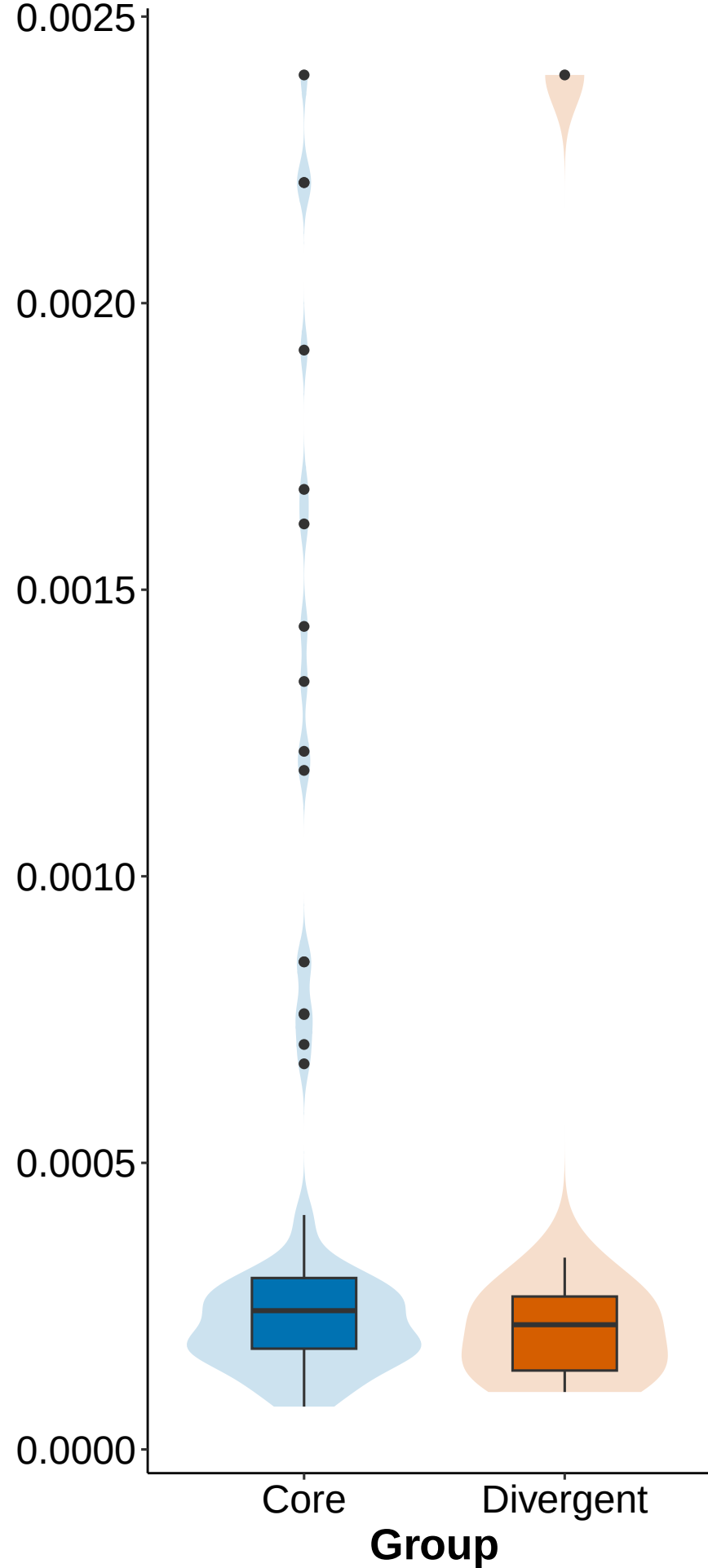
Margin Sharpness (M1)

p-adj: 0.929 | .=-0.04 (Negligible) | .%: -5.9



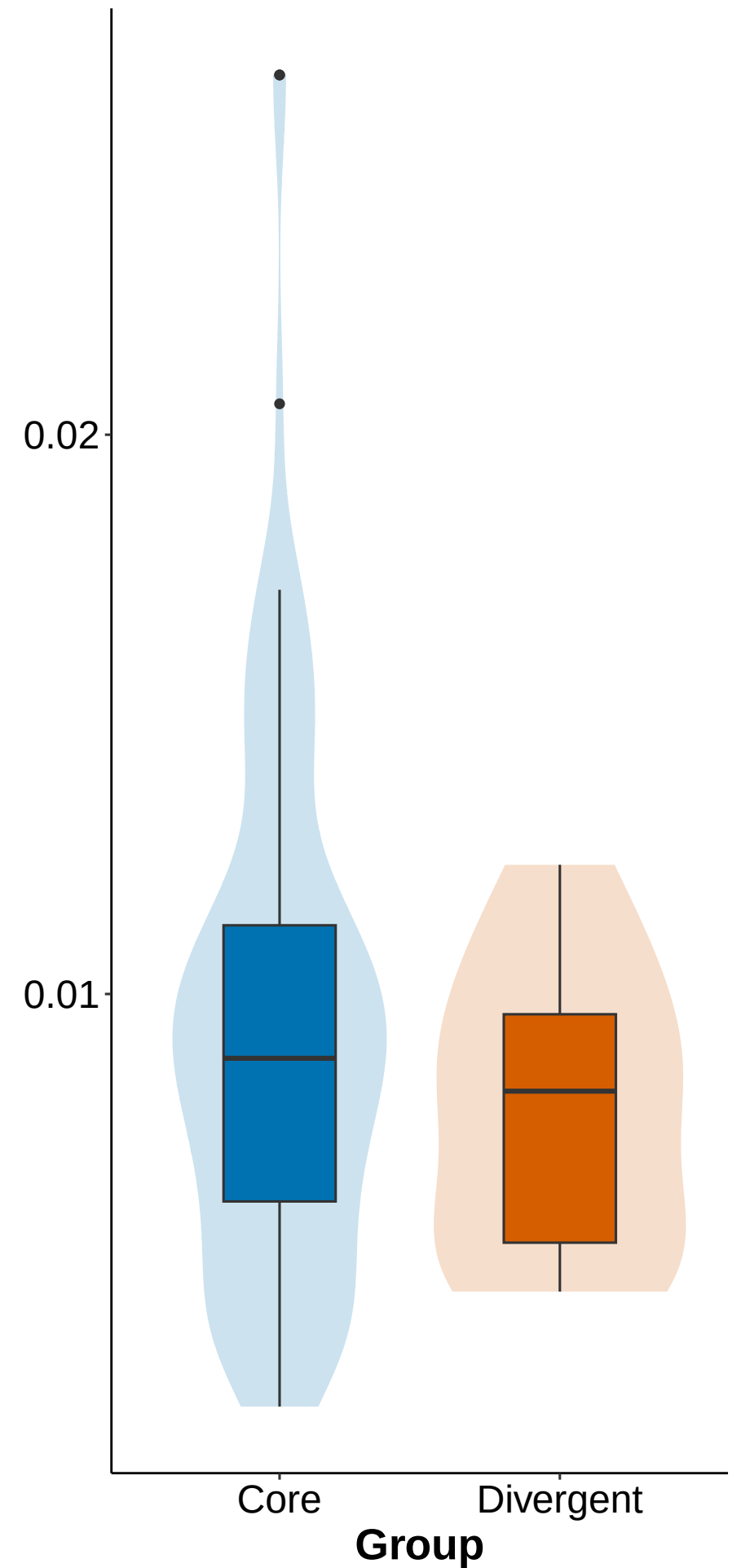
Variance of Margin Sharpness (M1)

p-adj: 0.928 | .=-0.13 (Negligible) | .%: -10.



Variance of Radial Gradient Histogram

p-adj: 0.928 | .=-0.16 (Small) | .%: -6.6



Cliff's .: S – Size

Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05



Surface Area (S3)

ns .=
0.16

Maximum Diameter (S5)

ns .=
0.15

Effective Diameter (S2)

ns .=
0.09

Size/Lesion volume (S1)

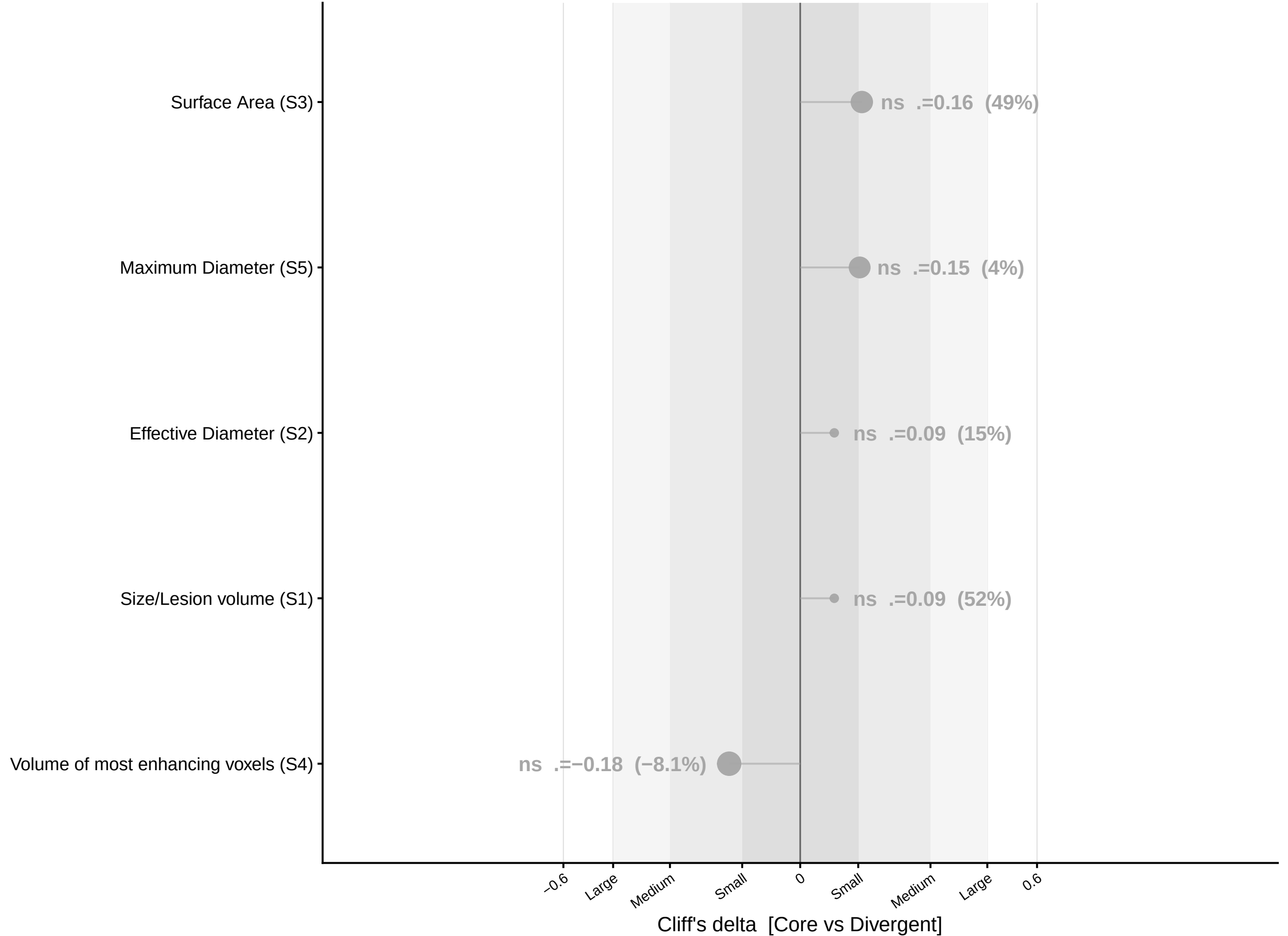
ns .=
0.09

Volume of most enhancing voxels (S4)

ns .=-
0.18

Cliff's . – S – Size

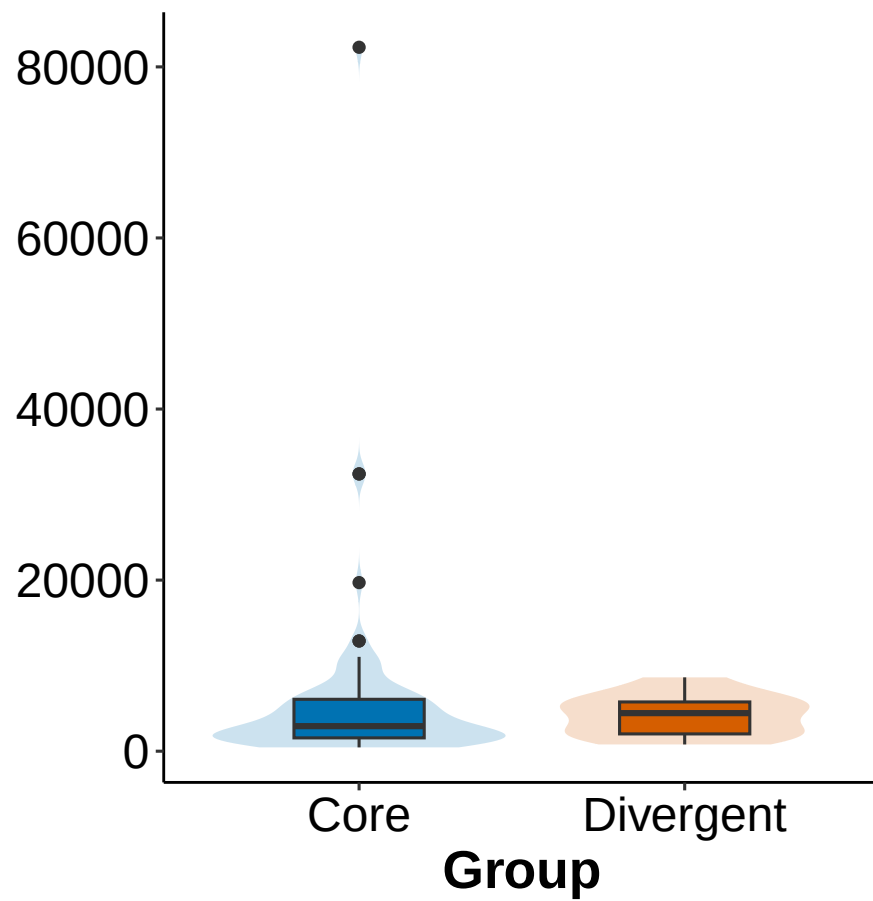
Cliff's . + .% median | . higher in Core · . higher in Divergent



Radiomics – S – Size

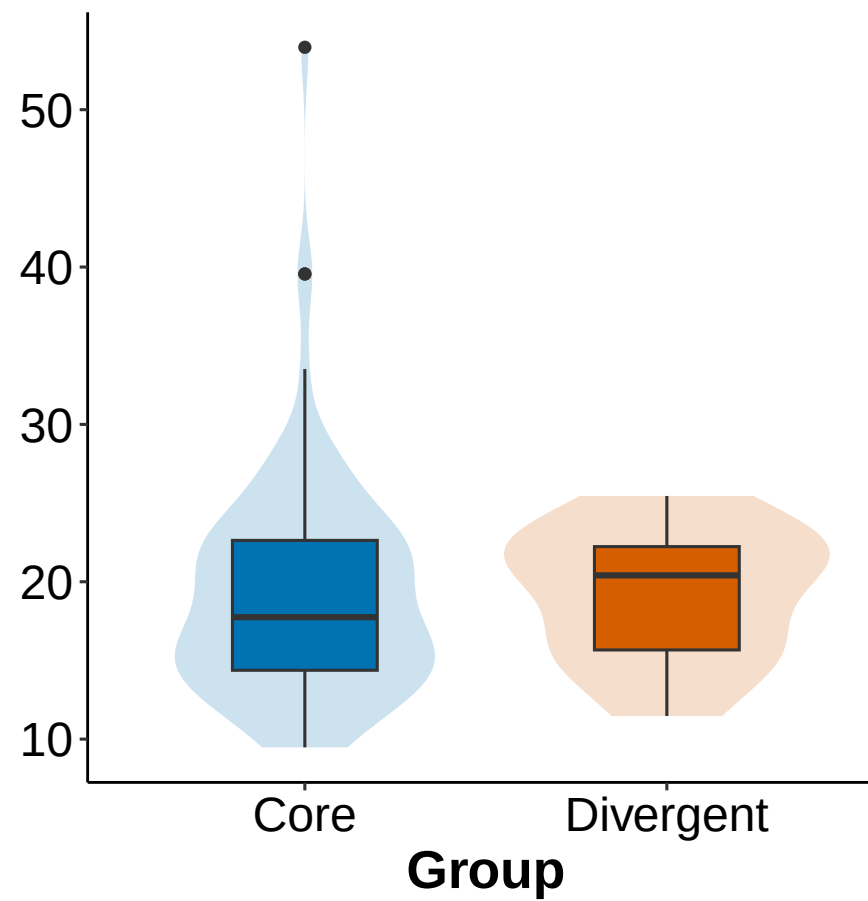
Size/Lesion volume (S1)

p-adj: 0.928 | .:=0.09 (Negligible) | .%: 52



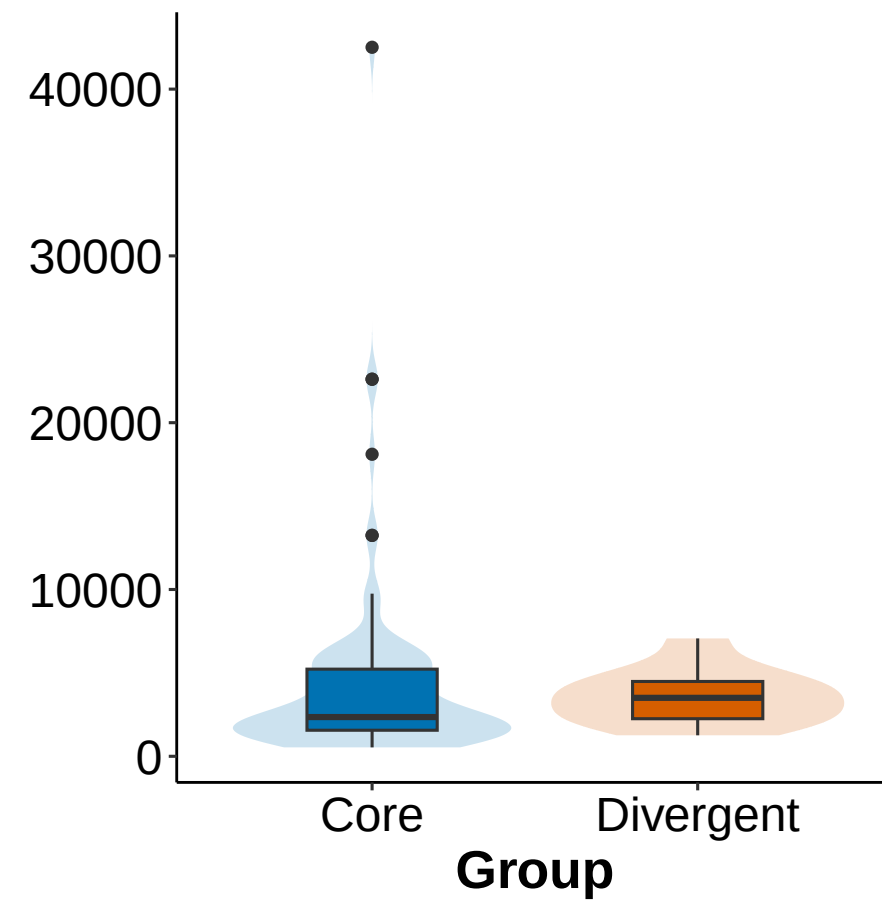
Effective Diameter (S2)

p-adj: 0.928 | .:=0.09 (Negligible) | .%: 15



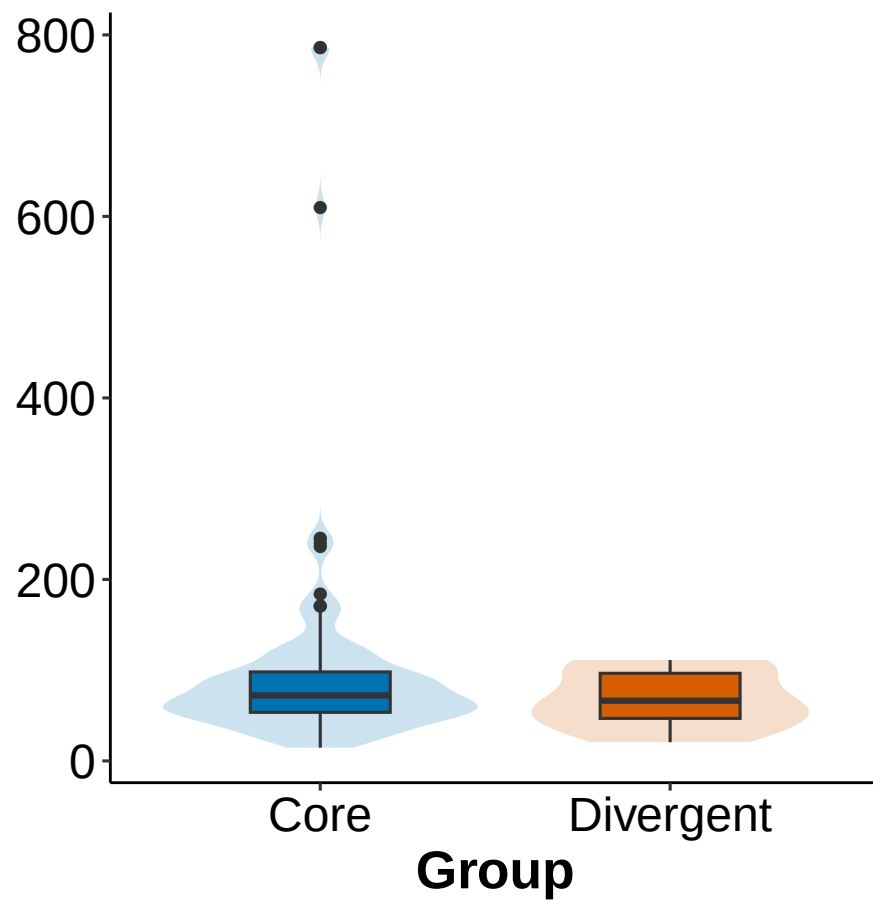
Surface Area (S3)

p-adj: 0.928 | .:=0.16 (Small) | .%: 49



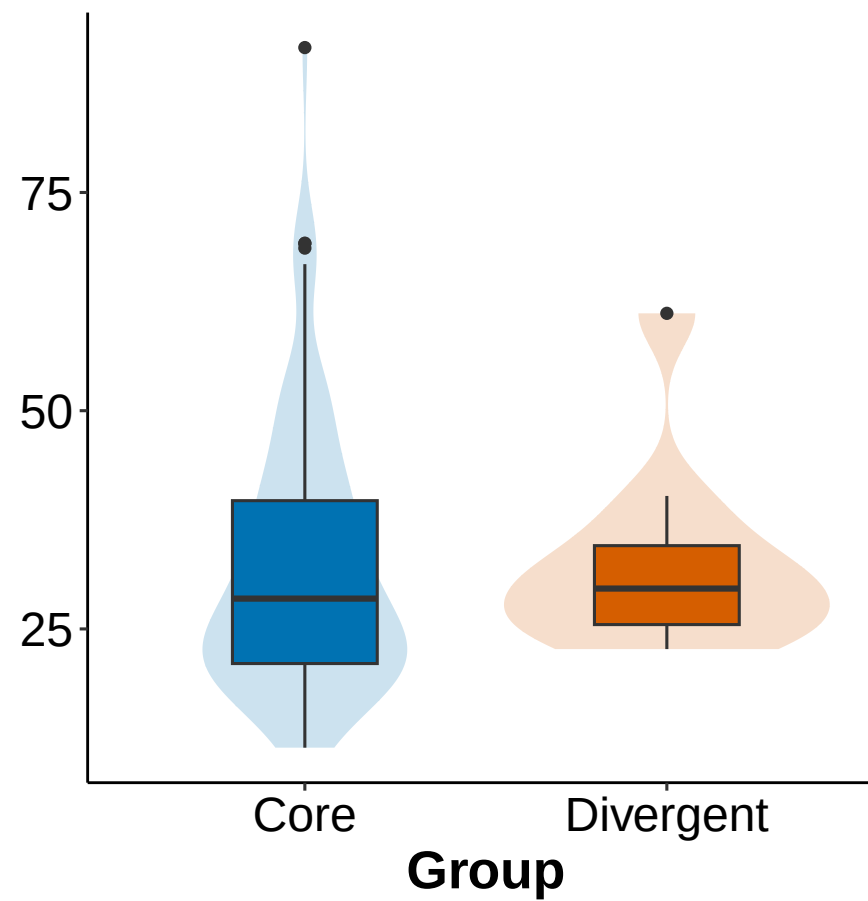
Volume of most enhancing voxels (S4)

p-adj: 0.928 | .:= -0.18 (Small) | .%: -8.1



Maximum Diameter (S5)

p-adj: 0.928 | .:=0.15 (Small) | .%: 4



Cliff's .: Top 12 Radiomic Variables

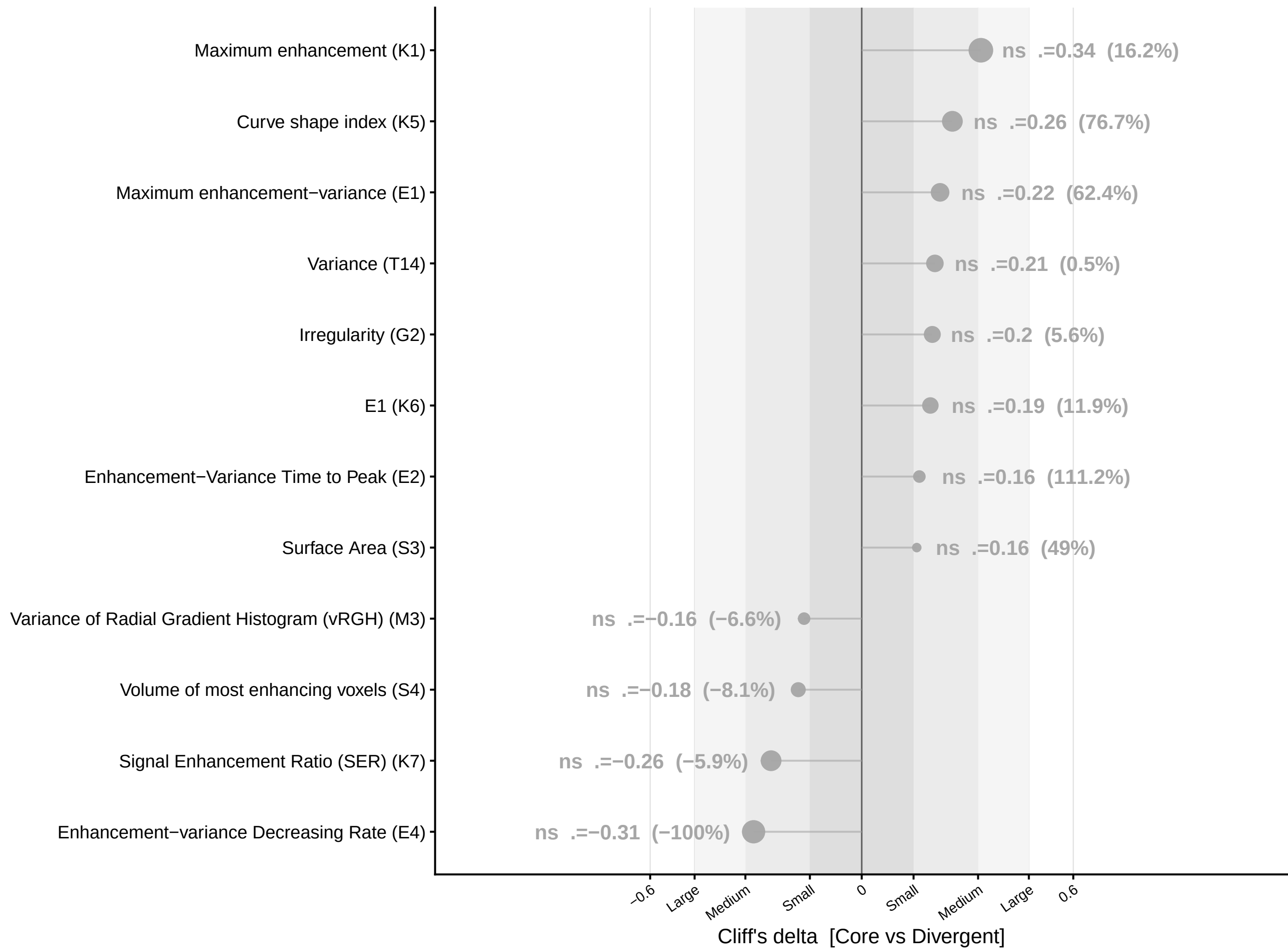
Blue = Core > Divergent | Orange = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



. Top 12 Radiomic Variables – Cliff's .

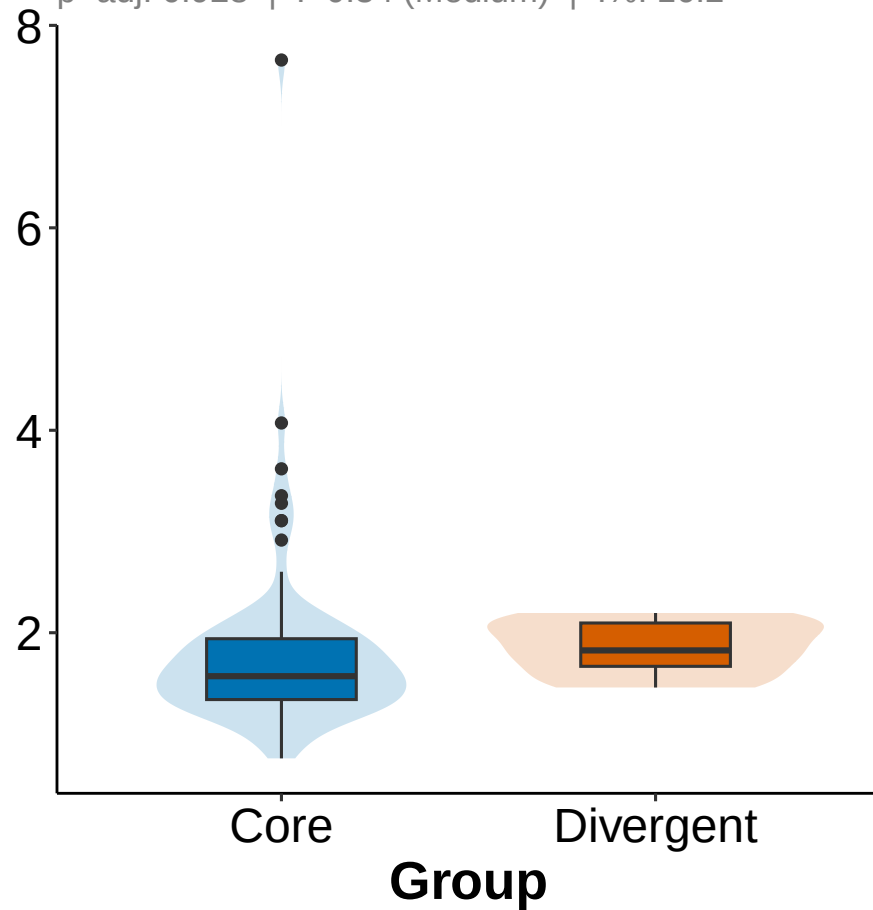
Cliff's . + .% median | . higher in Core . higher in Divergent



. Top 12 Radiomic Variables (p_adj + Cliff's .)

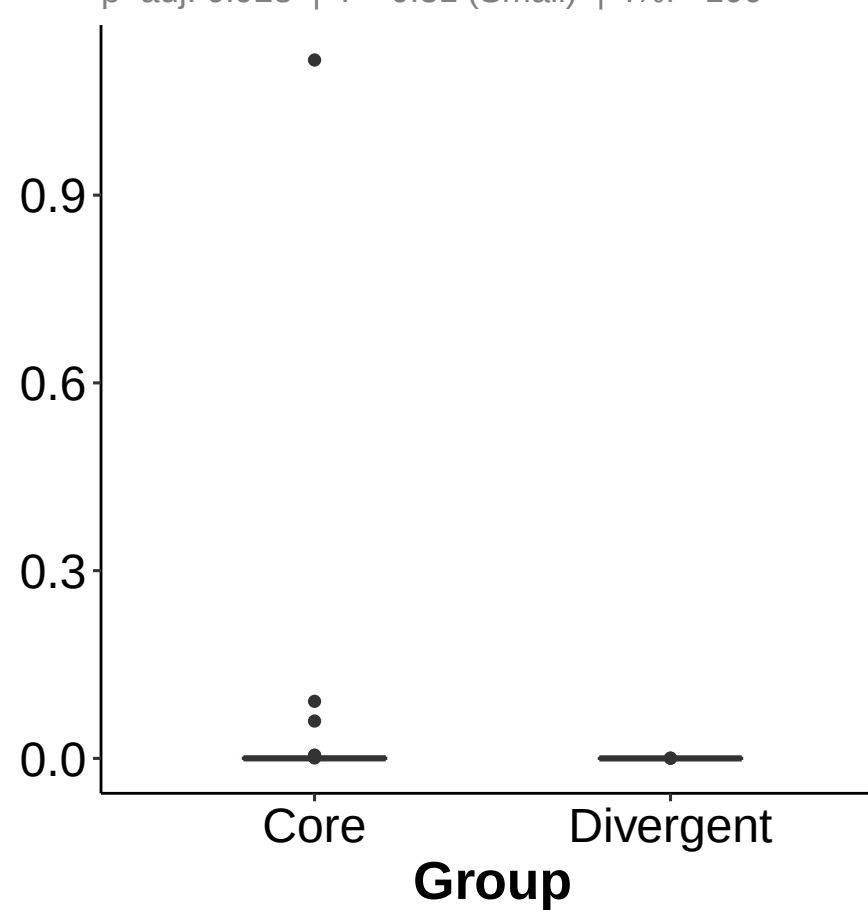
Maximum enhancement (K1)

p-adj: 0.928 | .:=0.34 (Medium) | .%: 16.2



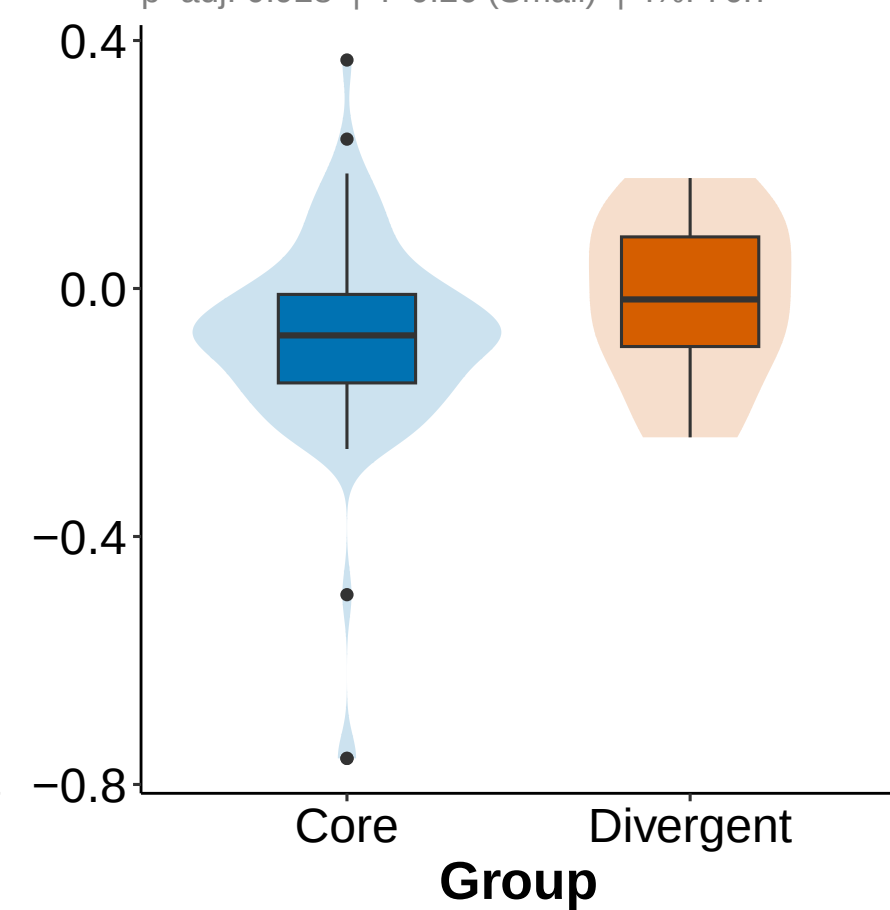
Enhancement-variance Decreasing F

p-adj: 0.928 | .:=−0.31 (Small) | .%: −100



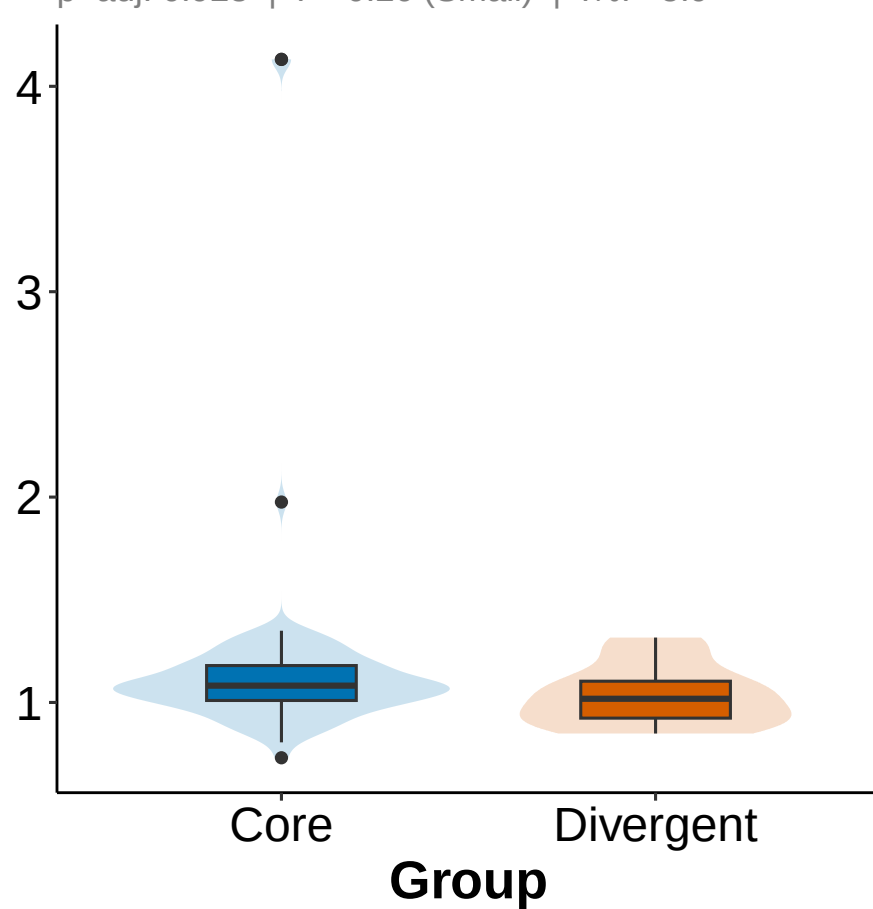
Curve shape index (K5)

p-adj: 0.928 | .:=0.26 (Small) | .%: 76.7



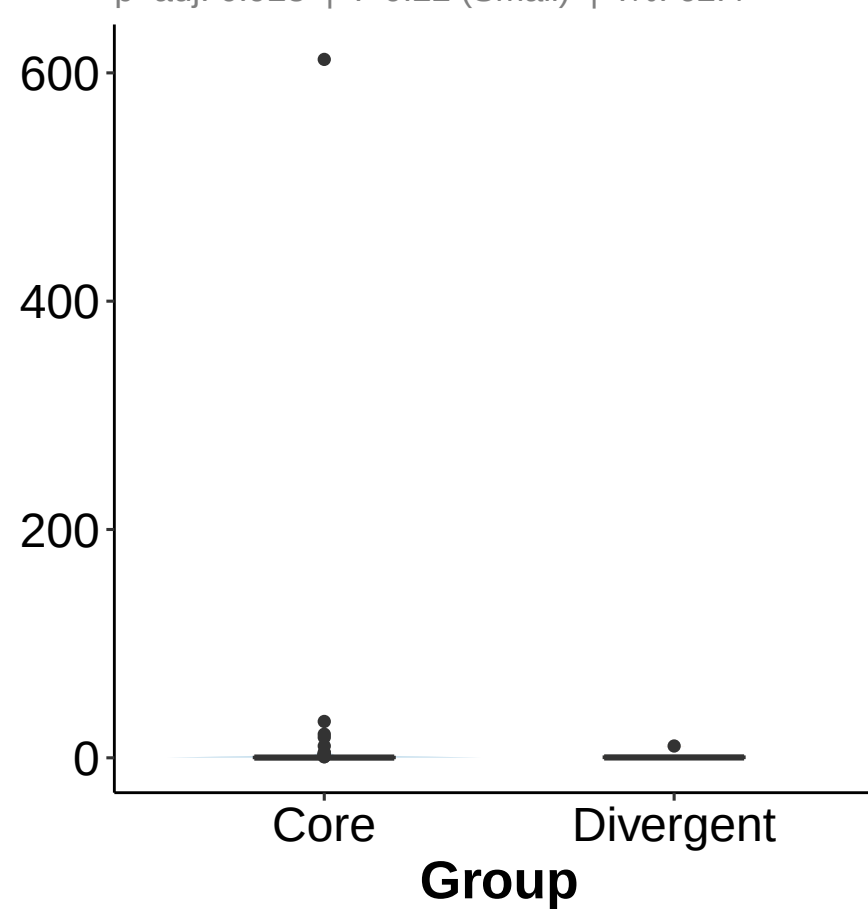
Signal Enhancement Ratio (SER) (K7)

p-adj: 0.928 | .:=−0.26 (Small) | .%: −5.9



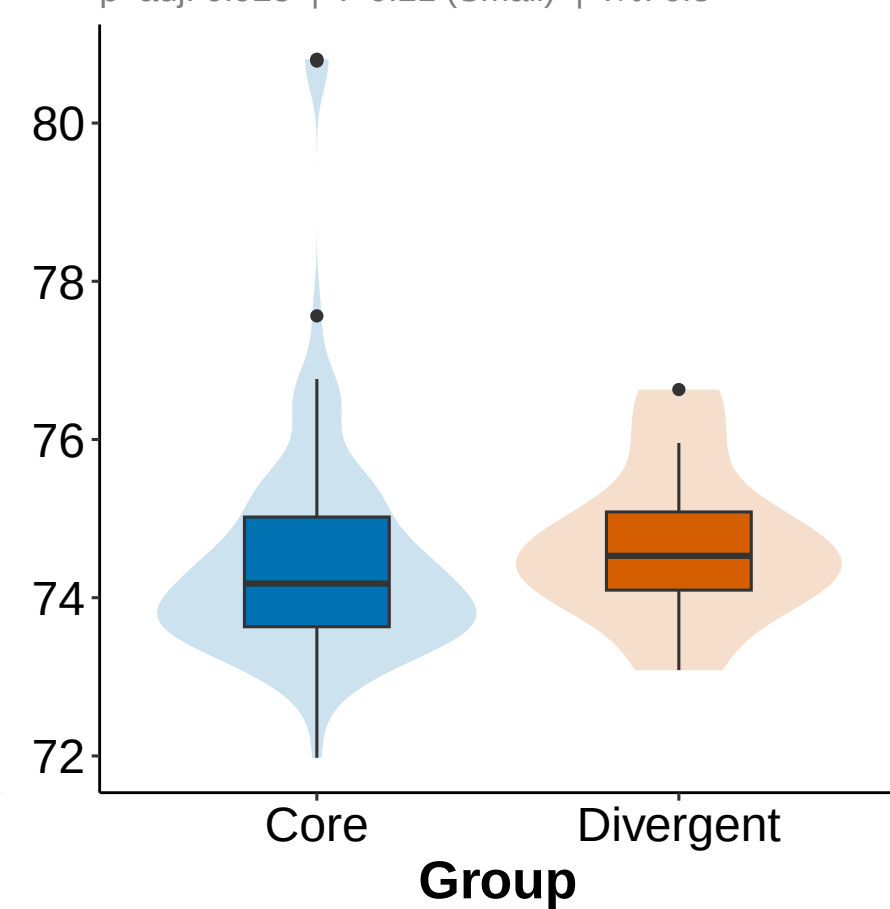
Maximum enhancement-variance (E:)

p-adj: 0.928 | .:=0.22 (Small) | .%: 62.4



Variance (T14)

p-adj: 0.928 | .:=0.21 (Small) | .%: 0.5



. Top 12 Radiomic Variables (p_adj + Cliff's .)

